

CHAPTER 4

ENGINE SYSTEM

CHAPTER OVERVIEW

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4-5	Maintenance Procedures (BENCH)

4-1/(4-2 Blank)

SECTION 4-1

EQUIPMENT DESCRIPTION AND DATA

SECTION OVERVIEW

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4-1-1 POWERPLANT DESCRIPTION AND DATA.

4-1-1.1 Powerplant System.

- The Powerplant System consists of:
- a. Two Demountable Power Packages:
 - (1) Engines
 - (2) Engine Anti-Ice System
 - (3) Engine Control System
 - (4) Engine Overspeed Protection System
 - (5) Engine Speed Trim System
 - (6) Engine Starting System
 - (7) Indicating Systems

4-1-1.2 Demountable Power Package.

Two Demountable Power Packages are installed, one on each side of the main transmission (Figure 4-1-1-1). Each power package consists of an engine, a pneumatic (air) starter, drive shaft assembly, fuel and lubrication lines, and a wiring harness. A drive shaft assembly and forward support

tube connect the engine to the input module of the main transmission. A power turbine shaft extending through the engine drives the input module. Each power package allows a quick change of engines with all components installed, for ease of maintenance and handling. The demountable power packages, less starters, may be ordered already built up as quick engine change kits, or assembled and disassembled on a buildup stand. The demountable power packages are then installed or removed as complete units.

4-1-1.3 Engine.

Two General Electric turboshaft engines, T700-GE-700, T700-GE-701A, or T700-GE-701C, with front drive are used as the primary powerplant (Figures 4-1-1-2 and 4-1-1-3). Refer to TABLE NO. 4-1-1-1 and TABLE NO. 4-1-1-2 for basic engine data. Each engine has four modules: cold section, hot section, power turbine section, and accessory section. The engine, with bleed-air capability at the compressor, provides heated air for engine inlet anti-icing and cockpit/cabin heating, and crossbleed engine starting. For a more detailed description of the engine, refer to SEI 671 and TM 1-2840-248-23.

TABLE NO. 4-1-1-1. Basic Engine Data, T700-GE-700 and T700-GE-701A

Models	T700-GE-700, T700-GE-701A
Type of Engine	Turboshaft
Output Power (Intermediate)	1,622 SHP at sea level, standard day conditions at 20,900 RPM
Type of Compressor	Combined axial/centrifugal
Number of Compressor Stages	6 stages: 5 axial and 1 centrifugal
Variable Geometry	Inlet guide vanes, and stage 1 and 2 stator vanes
Type of Combustion Chamber	Single annular chamber with axial flow

TABLE NO. 4-1-1-1. Basic Engine Data, T700-GE-700 and T700-GE-701A (Cont)

Gas Generator Turbine Stages	2 cooled
Power Turbine Stages	2 uncooled
Direction of Engine Rotation (from rear looking to front)	Clockwise
Engine Weight (Dry)	437 lb max.
Max. Engine Length	47 in.
Max. Engine Diameter	25 in.
Fuel	MIL-DTL-5624, Grade JP-4, JP-5, or MIL-DTL-83133, Grade JP-8
Lubricating Oil	MIL-PRF-23699 or MIL-PRF-7808
HELICOPTER POWER REQUIREMENTS:	
History Recorder, and Np overspeed protection	40w, 115 VAC, 400 Hz
Anti-icing valve, fuel filter bypass indication, oil filter bypass indication, and magnetic chip detector	1 amp, 28 VDC, each

TABLE NO. 4-1-1-2. Basic Engine Data, T700-GE-701C

Model	T700-GE-701C
Type of Engine	Turboshaft
Output Power (Intermediate)	1,800 SHP at sea level, standard day conditions at 20,900 RPM
Type of Compressor	Combined axial/centrifugal
Number of Compressor Stages	6 stages: 5 axial and 1 centrifugal
Variable Geometry	Inlet guide vanes, and stage 1 and 2 stator vanes
Type of Combustion Chamber	Single annular chamber with axial flow

TABLE NO. 4-1-1-2. Basic Engine Data, T700-GE-701C (Cont)

Gas Generator Turbine Stages	2 cooled
Power Turbine Stages	2 uncooled
Direction of Engine Rotation (from rear looking to front)	Clockwise
Engine Weight (Dry)	456 lb max.
Max. Engine Length	47 in.
Max. Engine Diameter	25 in.
Fuel	MIL-DTL-5624, Grade JP-4, JP-5, or MIL-DTL-83133, Grade JP-8
Lubricating Oil	MIL-PRF-23699 or MIL-PRF-7808
HELICOPTER POWER REQUIREMENTS:	
History Counter, and Np overspeed protection	40w, 115 VAC, 400 Hz
Anti-icing valve, fuel filter bypass indication, oil filter bypass indication, and magnetic chip detector	1 amp, 28 VDC each

4-1-1.4 Engine Mounts.

Each engine is held in place by front and rear engine mounts. The mounts, parts of the demountable power package, hold the demountable power package to the airframe. Vibrations transmitted

from the engine to the fuselage are reduced by the shock-absorbing mounts. The forward mount is bolted to the swirlframe on the engine and the input module of the main transmission. The rear mounts are bolted to the combustion chamber midframe and to the airframe (Figure 4-1-1-4).

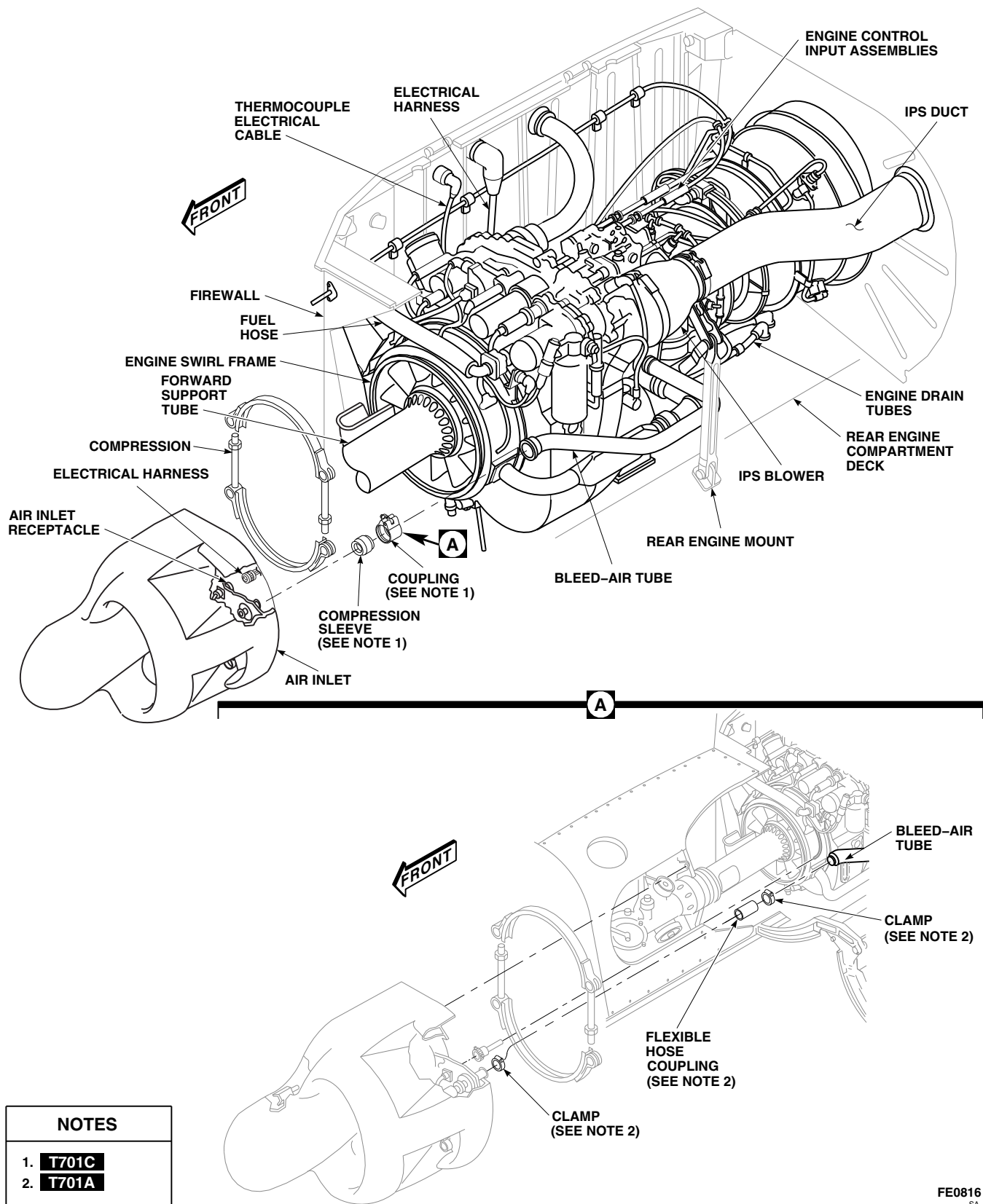


FIGURE 4-1-1-1. DEMOUNTABLE POWER PACKAGE

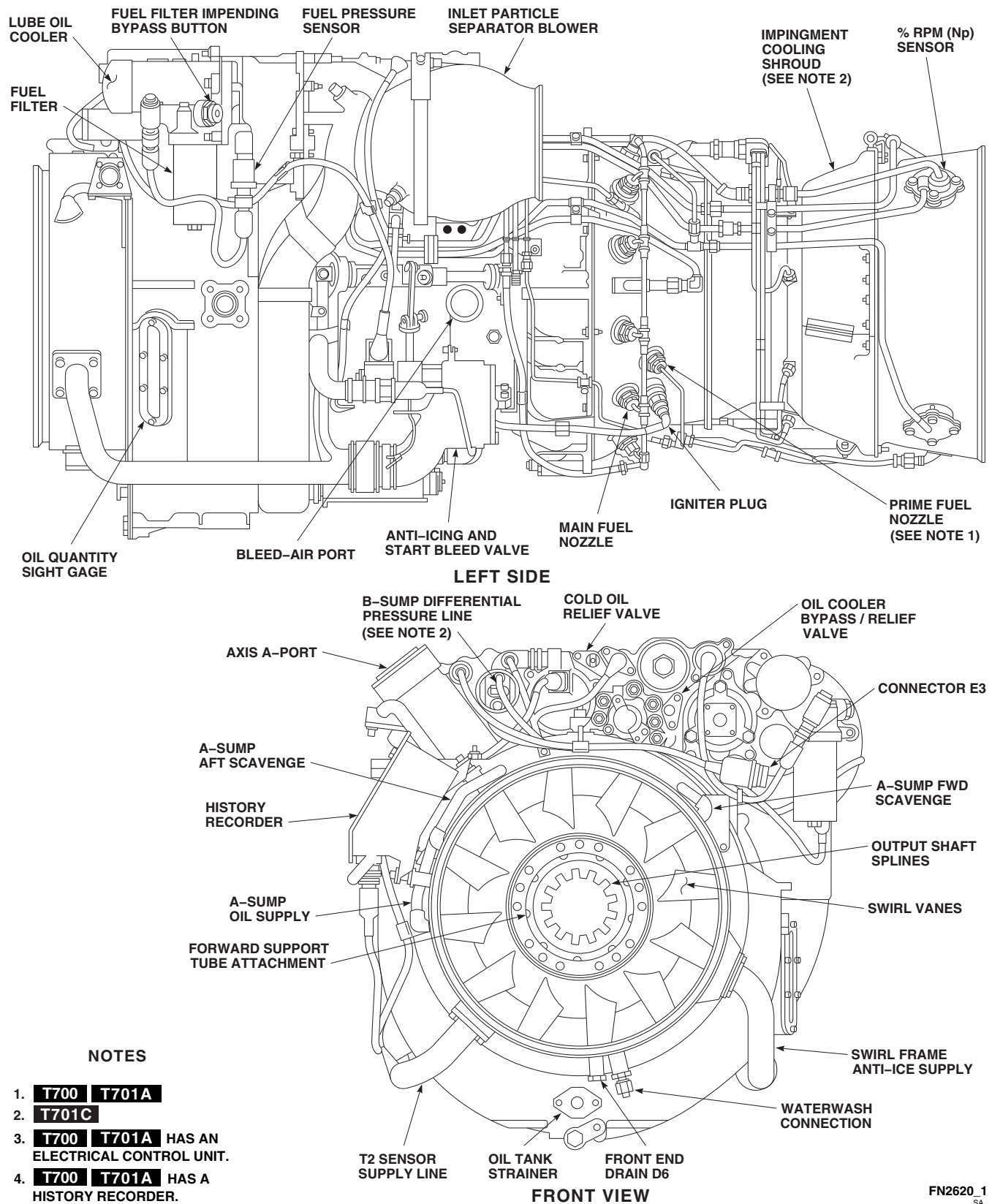


FIGURE 4-1-1-2. ENGINE (SHEET 1 OF 2)

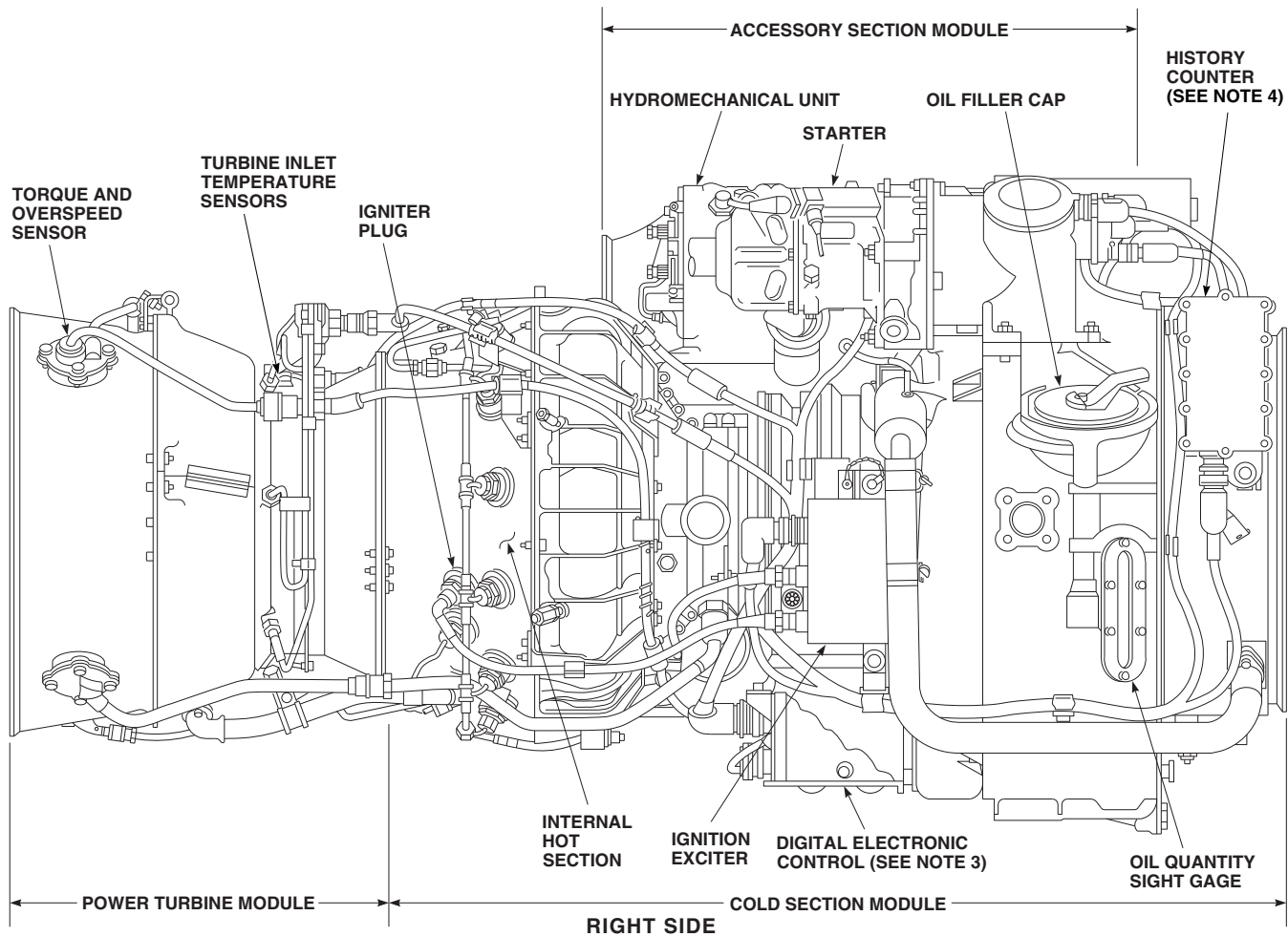
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FIGURE 4-1-1-2. ENGINE (SHEET 2 OF 2)

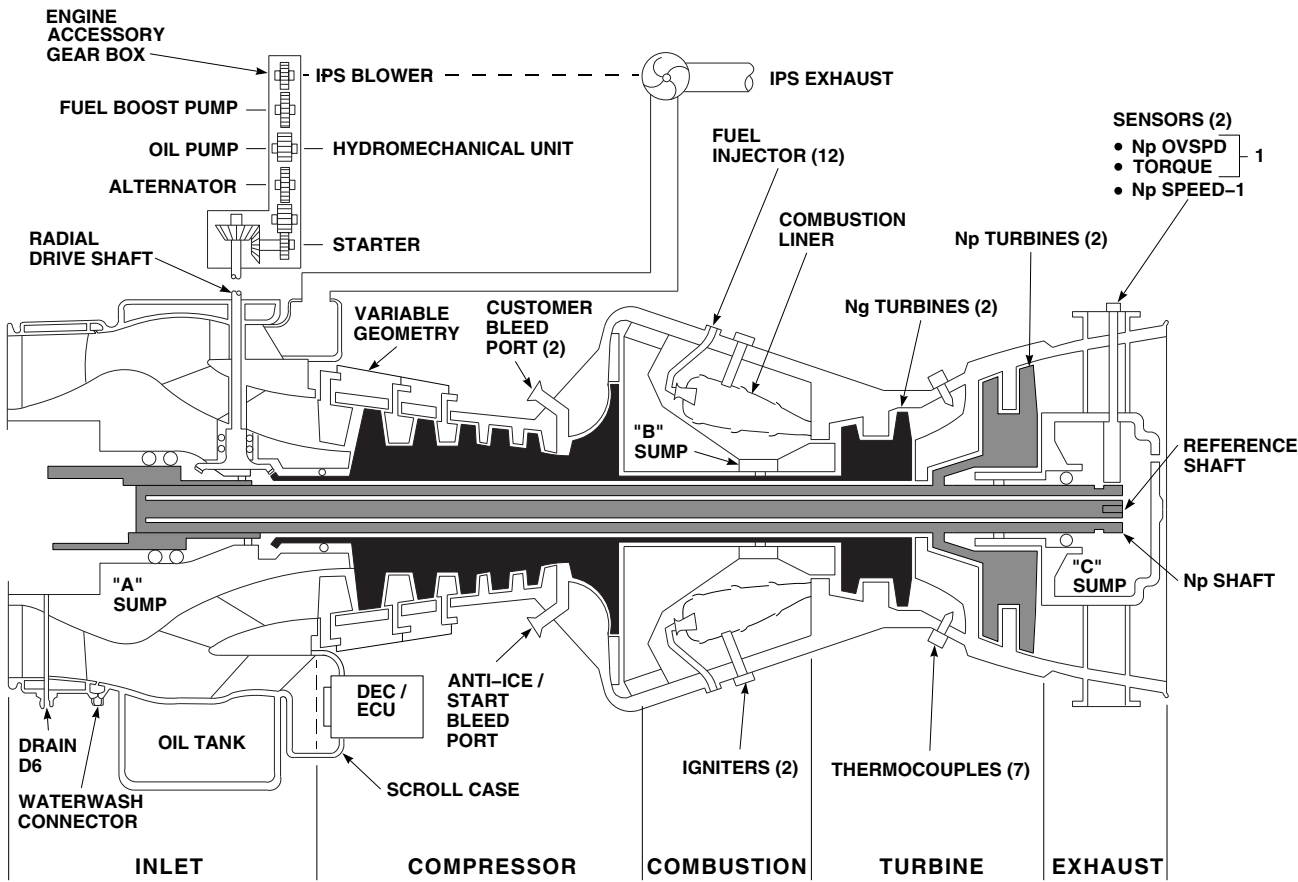


FIGURE 4-1-1-3. ENGINE OPERATION SCHEMATIC

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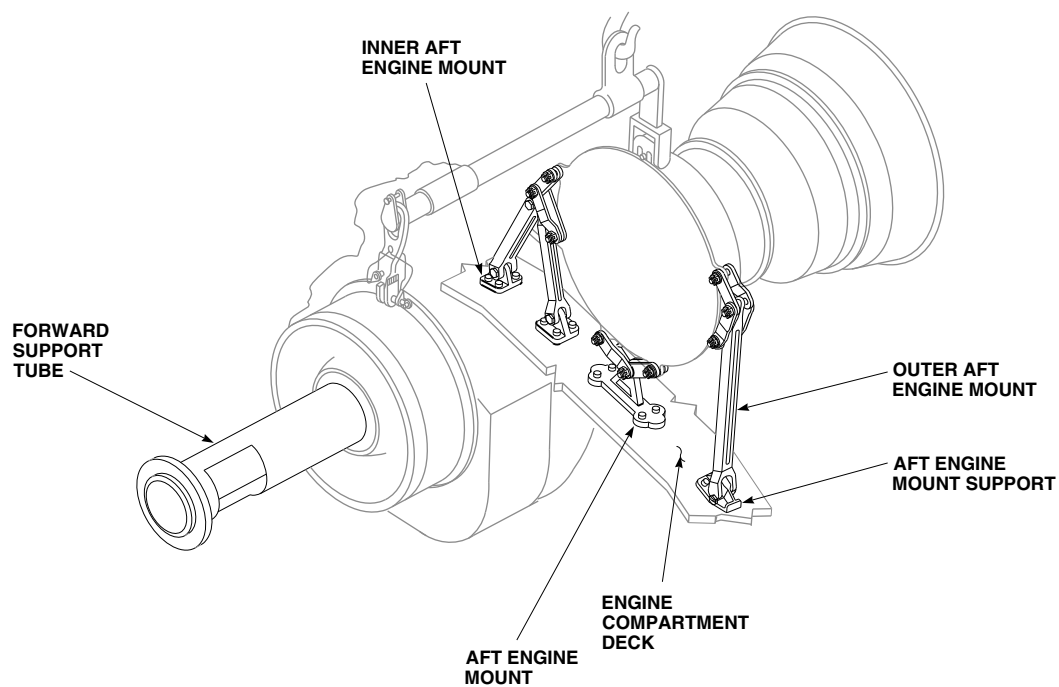
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FIGURE 4-1-1-4. ENGINE MOUNTS

4-1-2 ENGINE CONTROL SYSTEM DESCRIPTION AND DATA.

4-1-2.1 Load-Demand System.

The load-demand system supplies a collective pitch signal to the load-demand spindle on the engine hydromechanical fuel control through a push-pull control. The load-demand spindle automatically adjusts the engine gas generator (Ng) speed to a level about equal to the rotor load, thereby reducing transient droop (Figure 4-1-2-1).

T701C The engine provides additional transient droop improvement within the digital electronic control (DEC). A load demand signal from a potentiometer in the collective mixer and rotor speed signal from a magnetic pickup in the input module are both fed to the engine DEC where they are processed to provide better anticipation of load demand, thereby reducing transient rotor droop.

4-1-2.2 Power Control System.

The power control system for each engine operates through engine power control levers on the control quadrant. These levers permit starting and start-aborting. Each lever has four positions OFF, IDLE, FLY, and LOCKOUT. The power control lever is connected through a push-pull control to the power-available spindle on the engine fuel control. The spindle can either mechanically stopcock fuel or set permissible Ng speeds for operating between IDLE and FLY. Movement of the power control lever to LOCKOUT mechanically locks out the electrical control unit (ECU) or DEC input into the hydromechanical unit (HMU) and gives manual control over Ng speed, using the power control lever as a throttle.

4-1-2.3 Speed Control System.

The engine speed control system operates through a motor-driven potentiometer system and the ECU or DEC. When the ENG RPM switch on the collective stick is moved to INCR or DECR, an electrical signal goes through a potentiometer to the ECU or DEC on the engine. This signal, in turn, electrically adjusts the HMU on each engine increasing or decreasing engine power turbine speed Np (% RPM 1 or 2) and thereby increasing

main rotor speed (Nr) (% RPM RTR) to between 96% and 100%.

T701C A Nr sensor is added to the left accessory module. The Nr sensor senses main rotor RPM. This signal is sent to both engines' DEC's for engine speed control.■

4-1-2.4 Engine Controls Quadrant.

The engine controls quadrant, centered on the upper console, permits either the pilot or copilot to select engine speed, stopcock fuel, start engine, abort start, and control engine fire extinguisher. Four normally-open switches mounted in the quadrant actuate as levers are moved or the start button is pressed. Two lamps are in each fire extinguishing T-handle. The ENG POWER CONT lever positions are marked NUMBER 1 ENGINE and NUMBER 2 ENGINE, and identify the OFF, IDLE, FLY, and LOCKOUT positions. They are connected mechanically to each engine's HMU and are used to govern engine speeds. The HMU starts to open whenever the power control level is advanced more than 2° from OFF and increases proportionately with engine speed to FLY. If rotor brake is installed, a rotor brake interlock mechanism, installed in the controls quadrant, prevents the engine power control lever from being advanced beyond IDLE when the rotor brake is applied. A manual override lever is provided to override the rotor brake interlock when required. The engine start button is in the power control lever handle. The starter and starter override switches are actuated when the start button is pressed. The abort switch, mounted on the power control lever, is opened when the lever is pulled straight down. Each power control lever also includes a secondary stop, which prevents the inadvertent stop-cock of the engine when retarding the levers. The fire extinguishing arming switch is actuated when the fire extinguishing T-handle is pulled to the armed position. Lamps in the fire extinguishing T-handle go on when a fire is detected in the No. 1 or No. 2 engines. The fuel selector levers marked NO 1 ENG FUEL SYS and NO 2 ENG FUEL SYS allow pilot or copilot to select OFF, DIR (direct), or XFD (crossfeed) position for engine fuel supply. T-handles on the outboard side of either fuel selec-

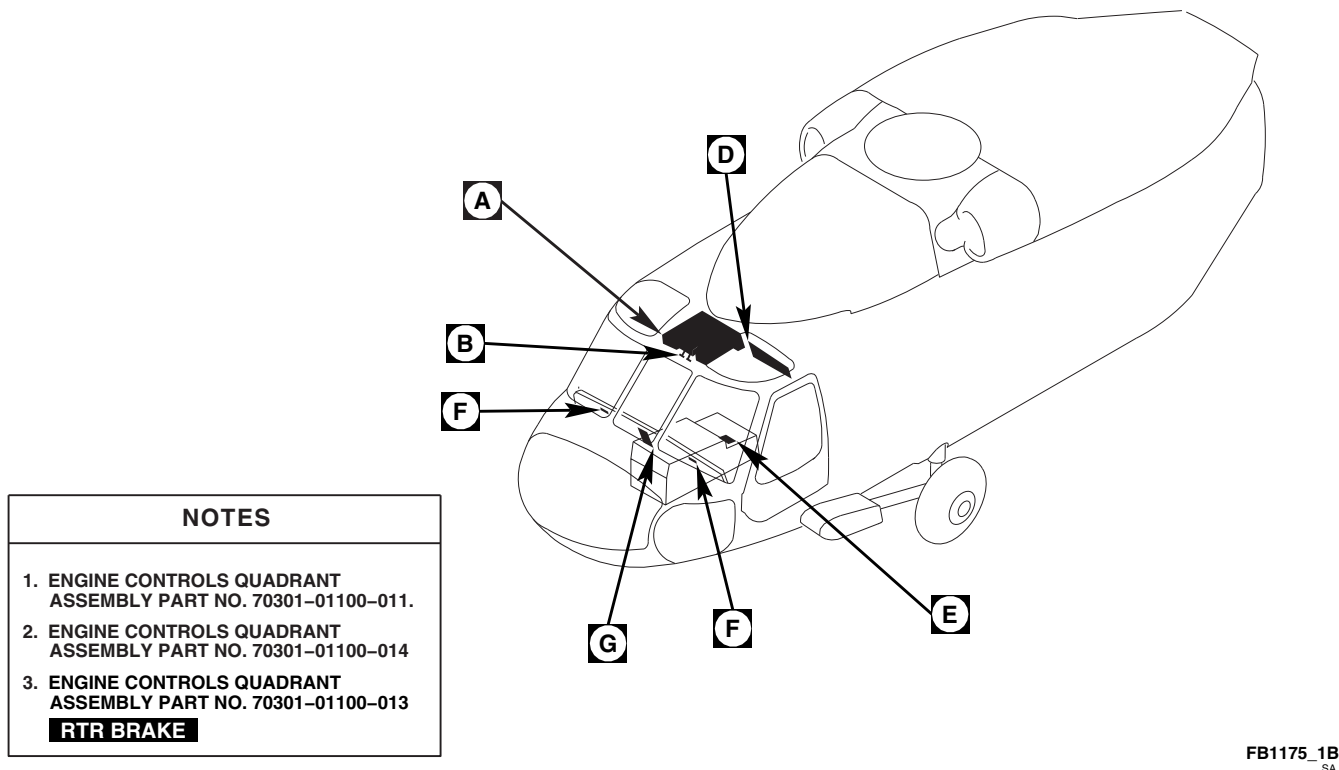
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FIGURE 4-1-2-1. ENGINE CONTROLS QUADRANT LOCATION DIAGRAM (SHEET 1 OF 5)

tor lever are used to direct the flow of the fire extinguishing agent to either engine compartment. Ball and push-pull cables connect both the power control levers and the fuel selector levers to their components.

The left, right, and center information panels on the quadrant are illuminated by 400 Hz power from the CONSOLE LT UPPER control on the upper console. For a further description of lighting, refer to console lighting (PARA 9-1-8).

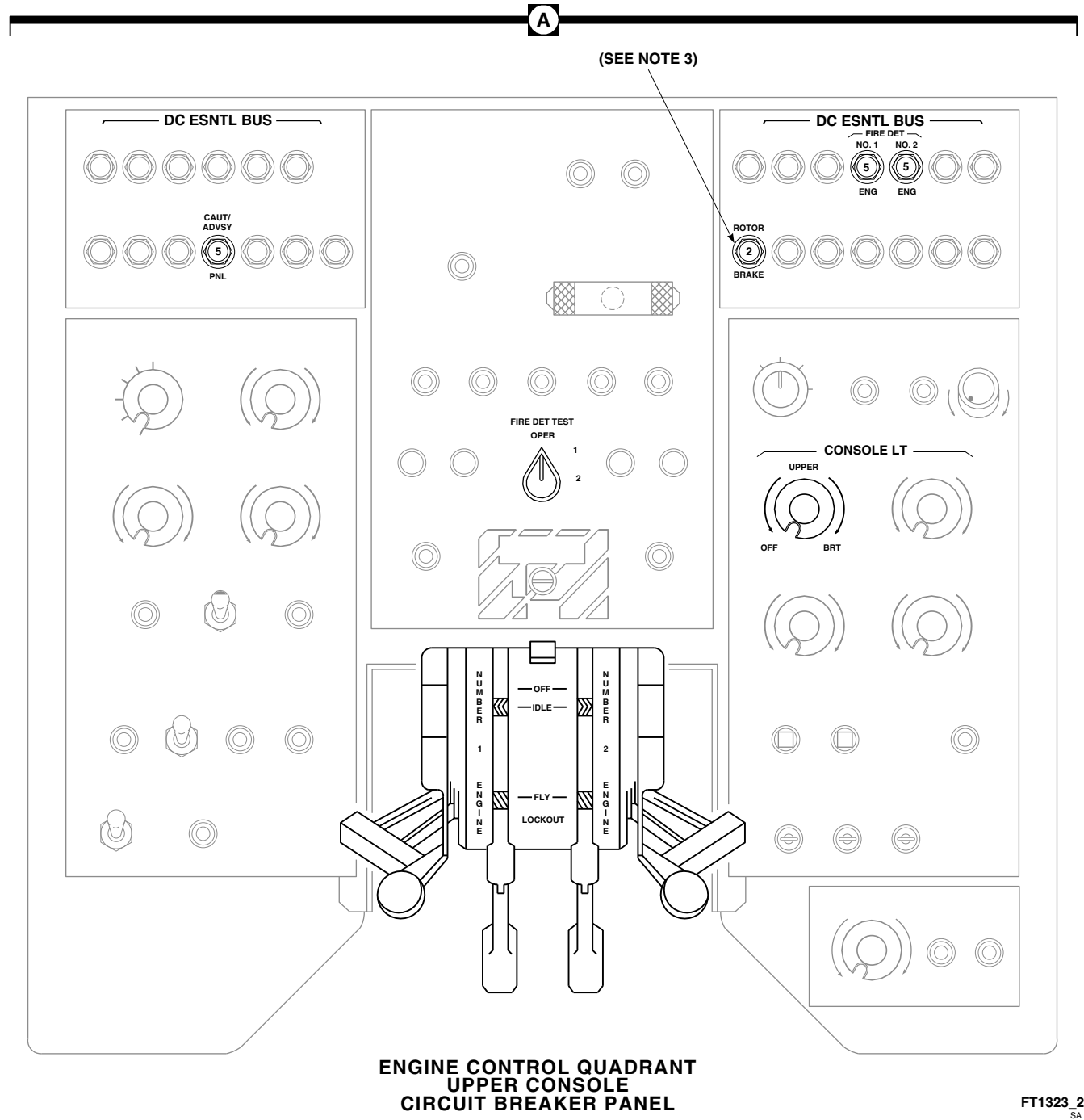


FIGURE 4-1-2-1. ENGINE CONTROLS QUADRANT LOCATION DIAGRAM (SHEET 2 OF 5)

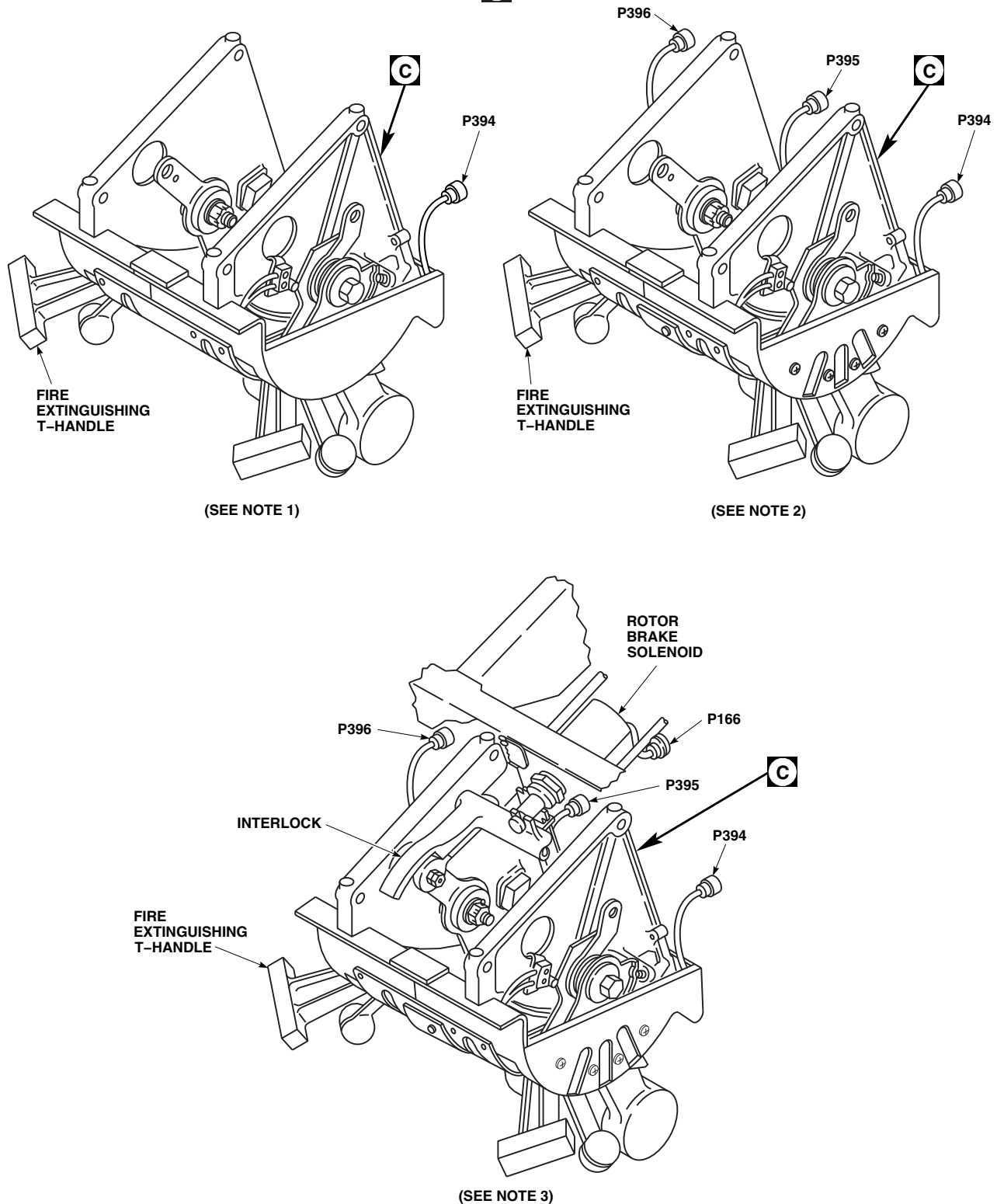
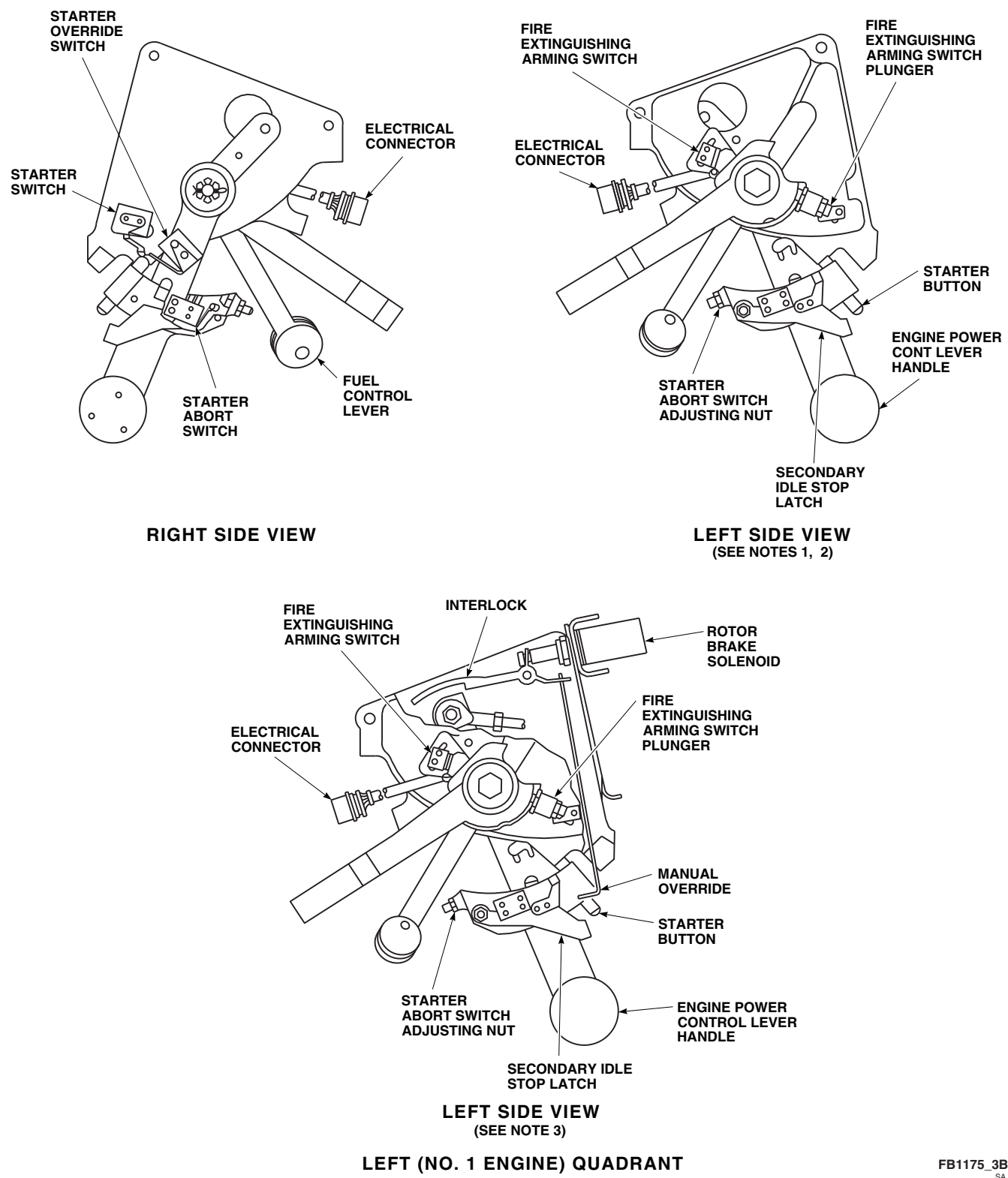
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FIGURE 4-1-2-1. ENGINE CONTROLS QUADRANT LOCATION DIAGRAM (SHEET 3 OF 5)

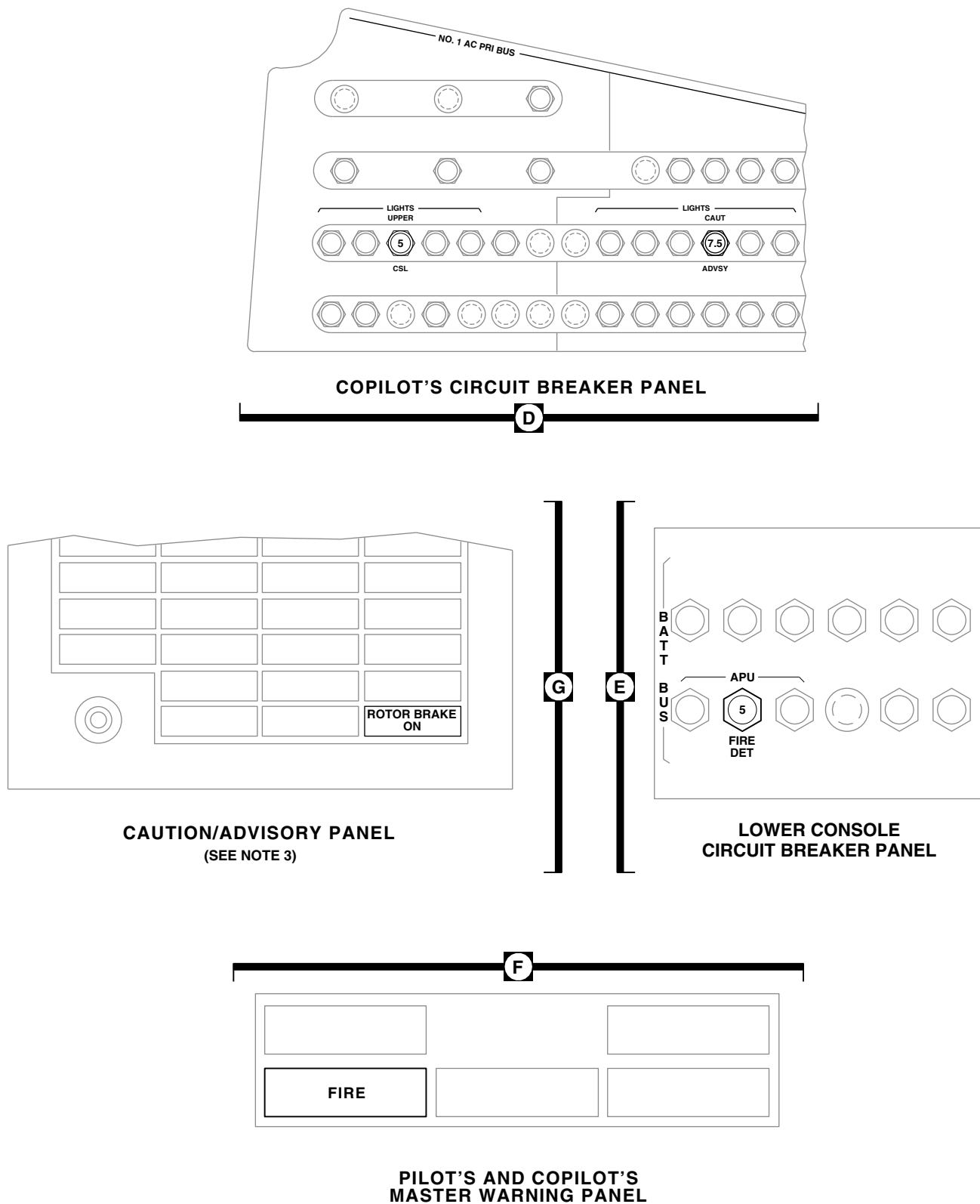
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FIGURE 4-1-2-1. ENGINE CONTROLS QUADRANT LOCATION DIAGRAM (SHEET 4 OF 5)



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FIGURE 4-1-2-1. ENGINE CONTROLS QUADRANT LOCATION DIAGRAM (SHEET 5 OF 5)

4-1-3 ENGINE START AND IGNITION SYSTEM DESCRIPTION AND DATA.

4-1-3.1 Engine Start and Ignition System.

The engine start and ignition system contains an electrically controlled pneumatic start system and an electrical capacitive-discharge ignition system. The pneumatics uses compressed air ducted to air turbine starters for engine starting (Figure 4-1-3-1 and Figure 4-1-3-2). Compressed air is obtained either from: (1) the auxiliary power unit (APU), (2) engine crossbleed, or (3) an external air supply. The APU has a check valve to control APU bleed-air flow to the starter of either engine. The engine crossbleed has a bleed-air manifold and a combination crossbleed shutoff and check valve for each engine to permit starting the opposite engine. The external air source supplies air through a combination external connector and check valve to either engine. Electrical power for the No. 1 engine start system is obtained from the dc essential bus circuit breaker panel on the upper console. Electrical power for the No. 2 engine start system is obtained from the No. 2 dc primary bus through the NO. 2 ENG START CONTR circuit breaker on the pilot's circuit breaker panel. An ENGINE IGNITION switch, marked ON and OFF, is at the center of the main instrument panel. When it is ON, and when an engine start button is pressed, the capacitor discharge engine ignition system operates. The ignition system is a noncontinuous ac powered, low-voltage system. Ignition is automatically shut off when gas generator (Ng) speed reaches 52% to 65%. The system consists of an ignition exciter, two igniter plugs, ignition leads, switches, and relays. Electrical power for the ignition system is obtained from the engine-mounted alternator.

4-1-3.2 Engine Starting.

There are three ways to start the engine:

APU. The APU is started, and the AIR SOURCE HEAT/START switch on the upper console is placed to APU. When the start button is pressed, the start control valve is opened and its relay is energized. Activation of the start control valve relay also opens the engine prime shutoff valve, closes the APU start bypass valve, and lights the ENGINE STARTER capsule. Opening the start control

valve releases compressed air from the APU to the engine start motor. The engine fuel system is primed when the prime shutoff valve is opened. As engine speed increases, the speed sensor pickup produces a signal that activates the starter speed switch, holding the engine start relay in the energized position. When the engine reaches 52% to 65% Ng, the starter speed switch deactivates, closing the engine start control valve, which then closes the prime shutoff valve and turns off the ENGINE STARTER capsule.

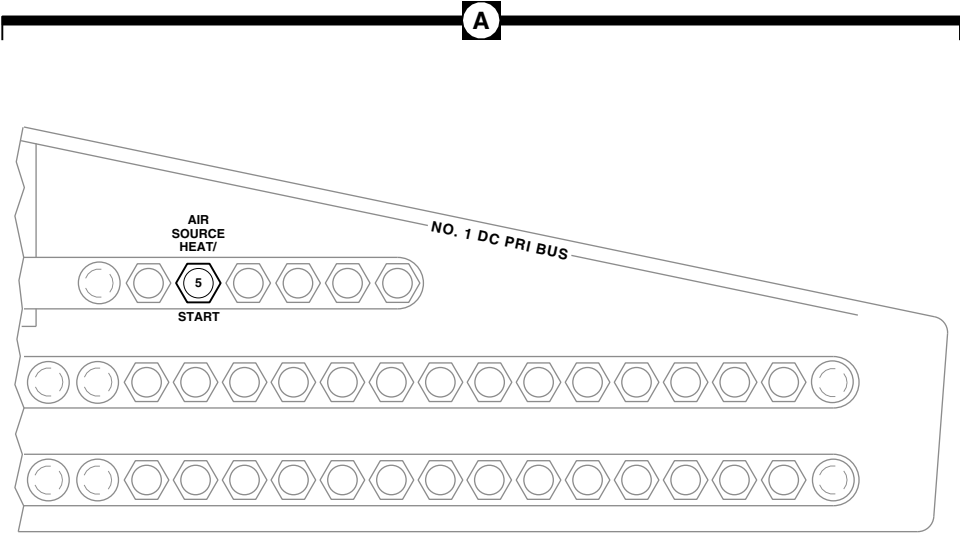
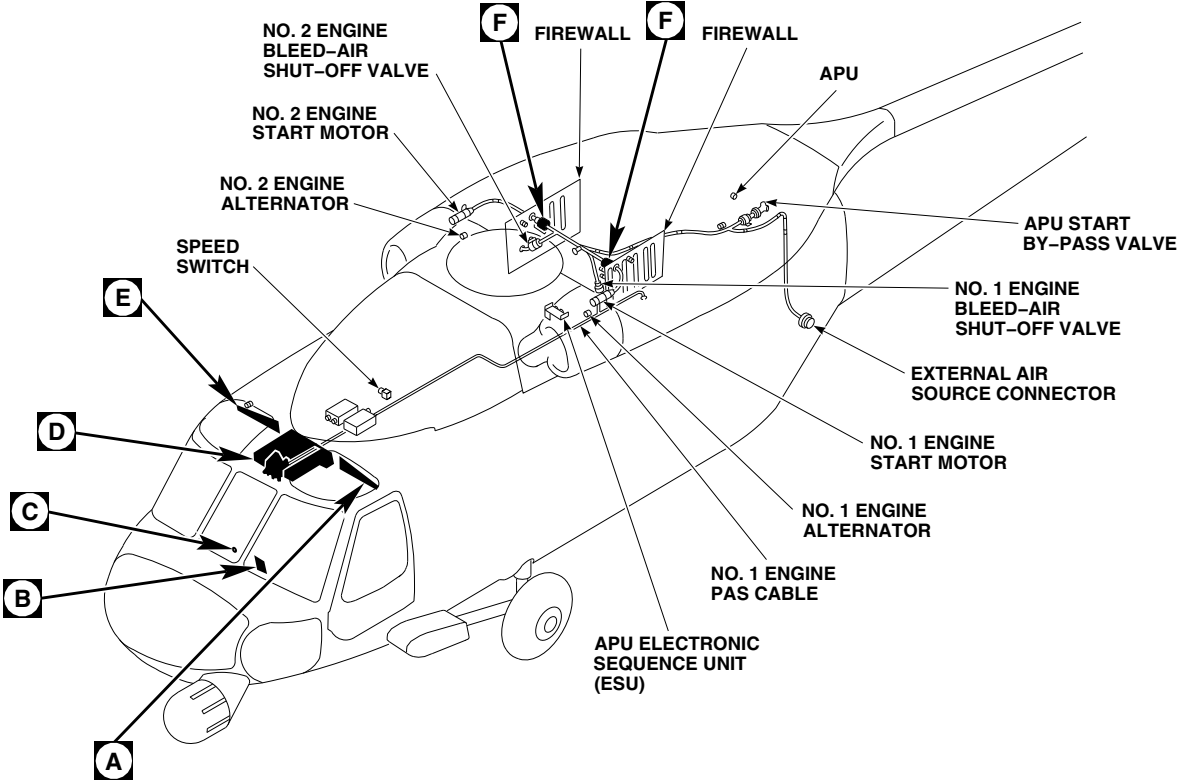
ENGINE CROSSBLEED. (ONE ENGINE OPERATING). When the AIR SOURCE HEAT/START switch on the upper console is placed to ENGINE, the engine crossbleed valve of the engine operating opens. The crossbleed valve of the engine not operating stays closed to prevent bleed-air from entering and turning the engine compressor in the opposite direction of start. With the ENG POWER CONT lever of the stopped engine at OFF, and the start button on that engine pressed, the start control valve of the engine being started opens. This releases between 27 and 31 psig bleed-air to the engine starter of the stopped engine. Activation of the start control valve opens the engine prime shutoff valve, priming the engine. The ENGINE STARTER capsule on the caution/advisory panel also goes on. As the engine accelerates to between 52% and 65% Ng, the start control valve closes, cutting off bleed-air to the starter. At this point ignition stops, the bleed-air shutoff valve opens, and the ENGINE STARTER capsule goes off. When the start control valve closes, the prime boost pump motor is turned off, the PRIME BOOST PUMP ON capsule on the caution/advisory panel goes off, and the engine prime shutoff valve closes. Placing the AIR SOURCE HEAT/START switch OFF closes both bleed-air shutoff valves.

EXTERNAL AIR SUPPLY. With the AIR SOURCE HEAT/START switch on the upper console OFF, and an external air supply connected to the helicopter, the engine start and

ignition system operates the same as when using the APU to start the engines.

Pressing the engine start button will also automatically turn on the prime boost pump motor, light the PRIME BOOST PUMP ON capsule on the caution/advisory panel, and open the prime shutoff valve on that engine.

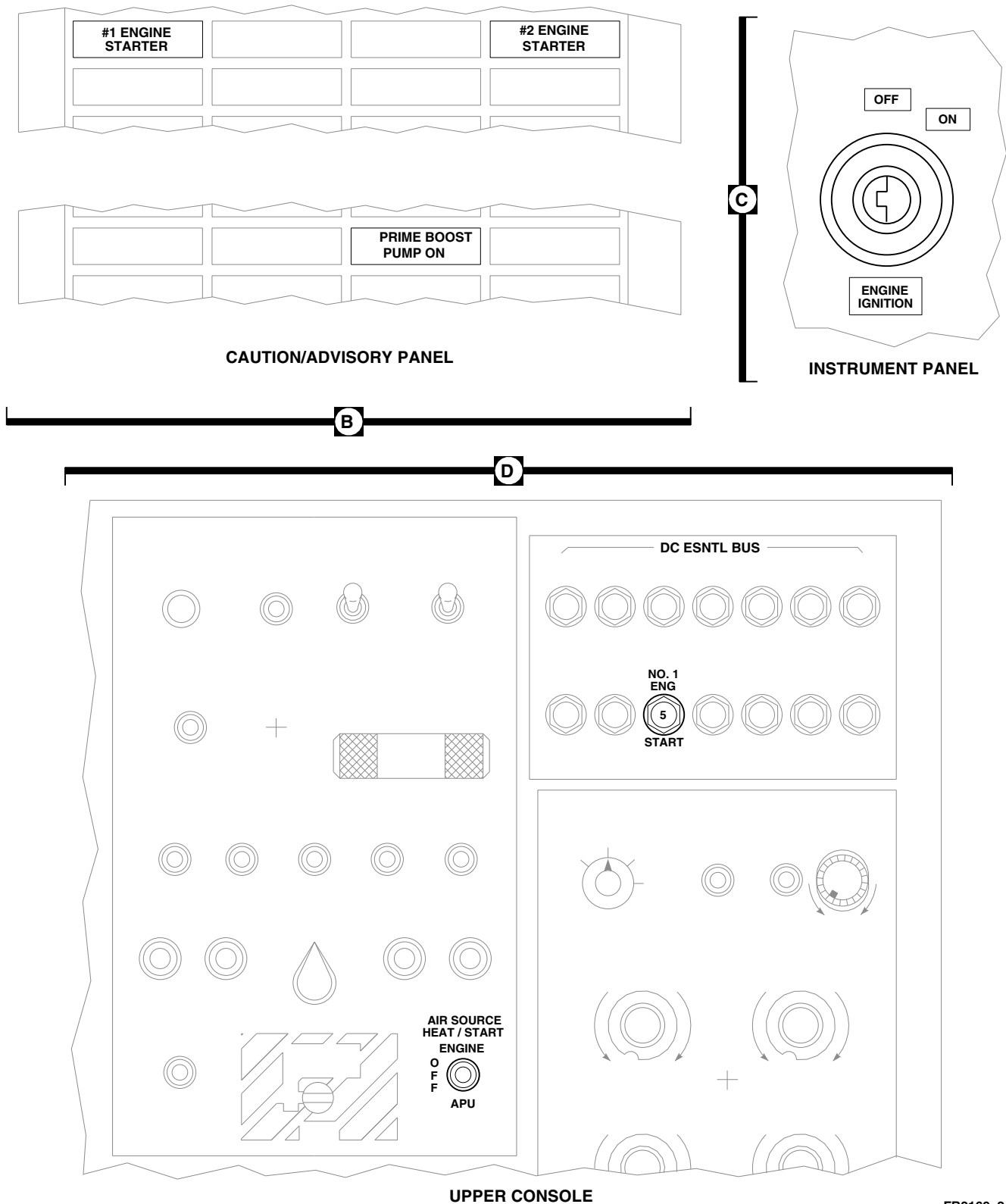
When the engine starter speed switch stops the ignition sequence, it also closes the prime shutoff valve, turns off the prime boost pump motor, and the PRIME BOOST PUMP ON capsule. Automatic fuel prime to engine will happen no matter what position the FUEL PUMP switch is in.



COPILLOT'S CIRCUIT BREAKER PANEL

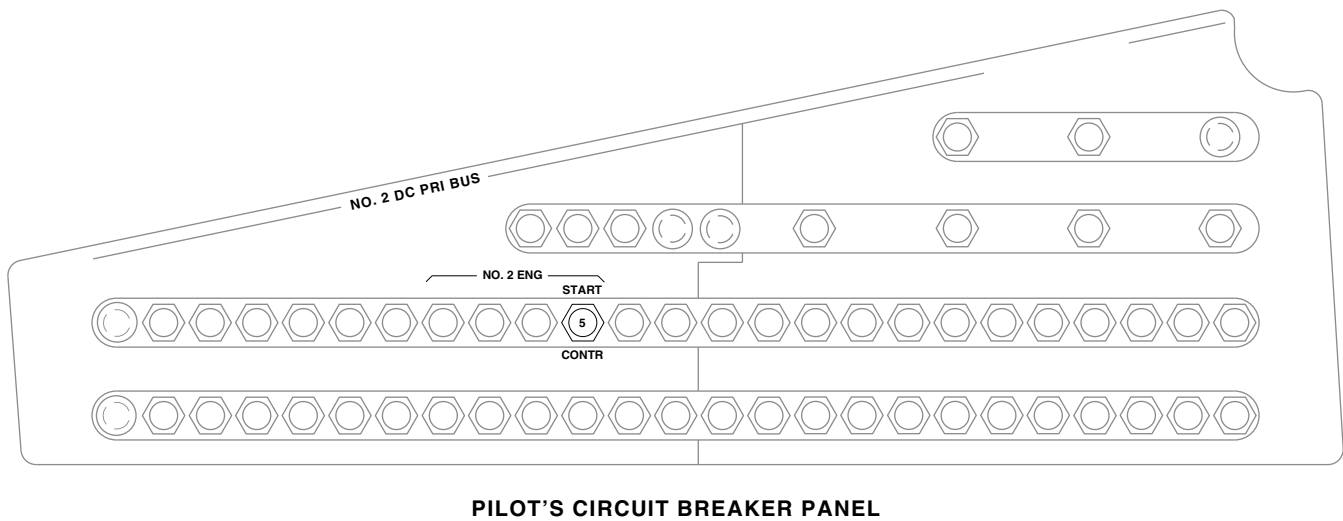
FIGURE 4-1-3-1. ENGINE START AND IGNITION SYSTEM LOCATION DIAGRAM (SHEET 1 OF 3)

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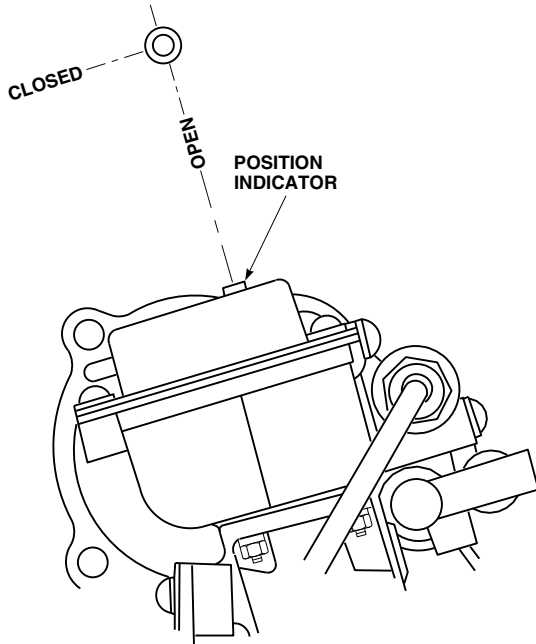
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**FIGURE 4-1-3-1. ENGINE START AND IGNITION SYSTEM LOCATION DIAGRAM
(SHEET 2 OF 3)**



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NO. 1 AND NO. 2 ENGINE
 START CONTROL VALVE

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FIGURE 4-1-3-1. ENGINE START AND IGNITION SYSTEM LOCATION DIAGRAM
 (SHEET 3 OF 3)

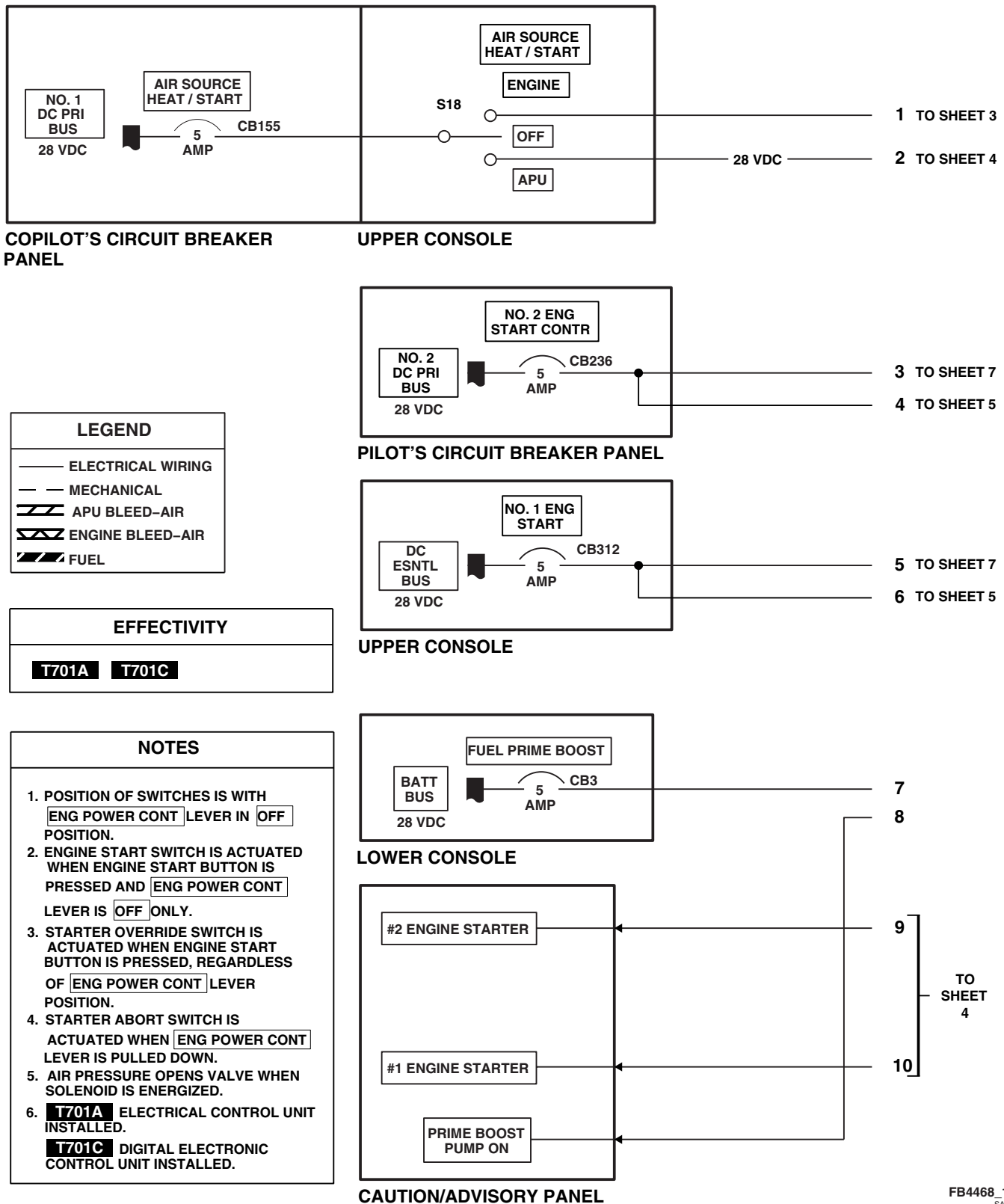


FIGURE 4-1-3-2. ENGINE START AND IGNITION SYSTEM BLOCK DIAGRAM
(SHEET 1 OF 8)

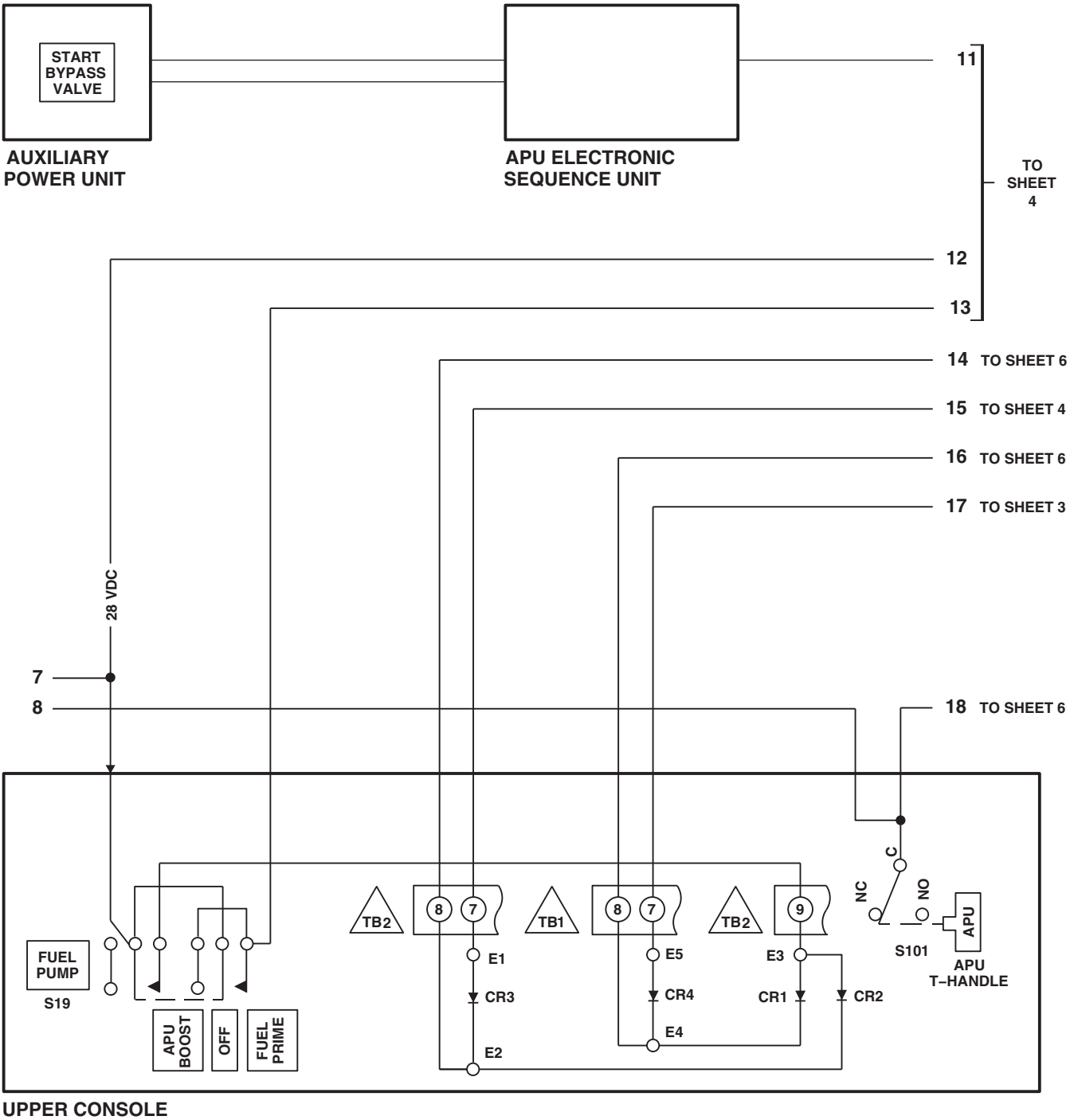
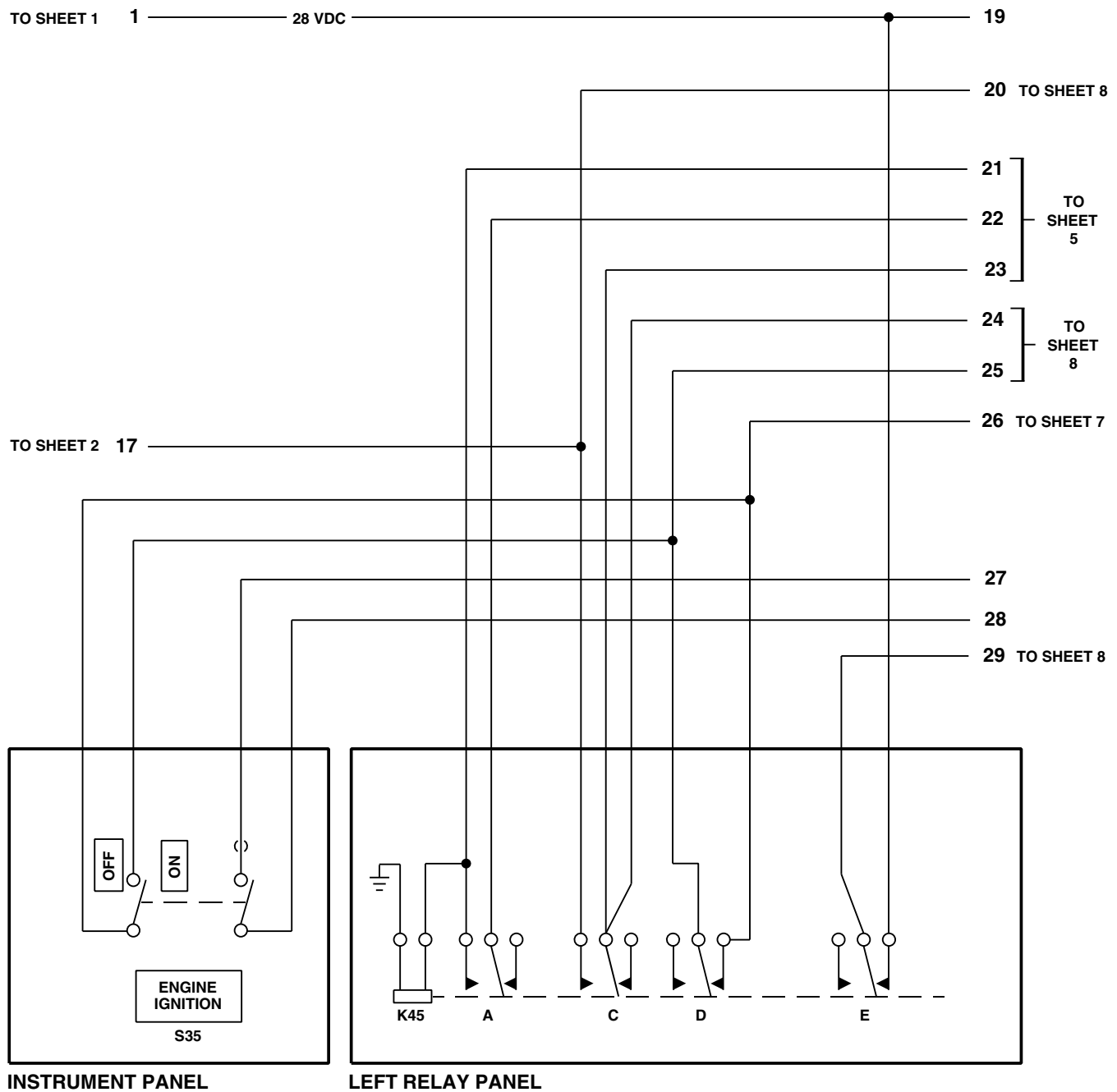


FIGURE 4-1-3-2. ENGINE START AND IGNITION SYSTEM BLOCK DIAGRAM (SHEET 2 OF 8)

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**FIGURE 4-1-3-2. ENGINE START AND IGNITION SYSTEM BLOCK DIAGRAM
(SHEET 3 OF 8)**

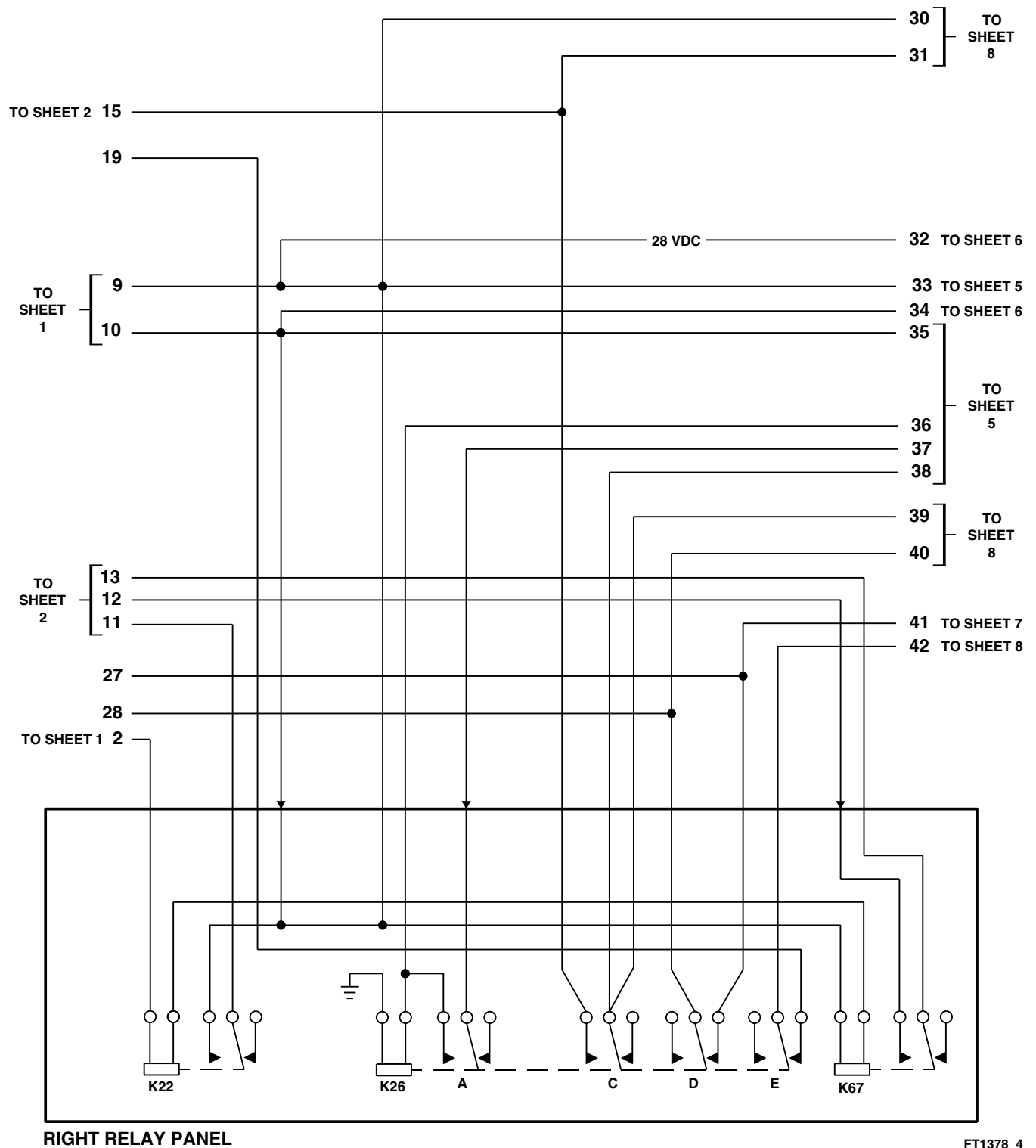
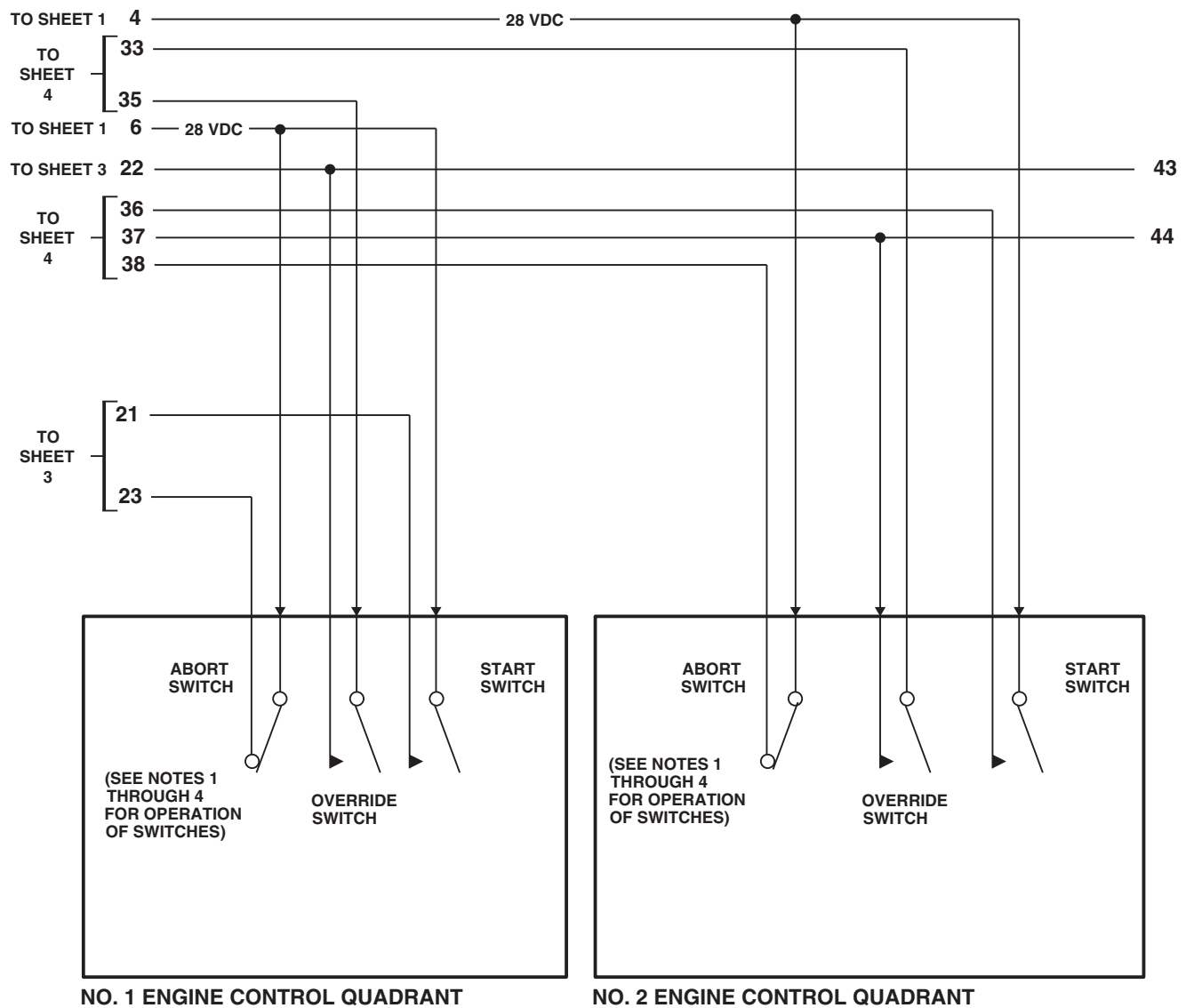


FIGURE 4-1-3-2. ENGINE START AND IGNITION SYSTEM BLOCK DIAGRAM
(SHEET 4 OF 8)

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FIGURE 4-1-3-2. ENGINE START AND IGNITION SYSTEM BLOCK DIAGRAM
(SHEET 5 OF 8)

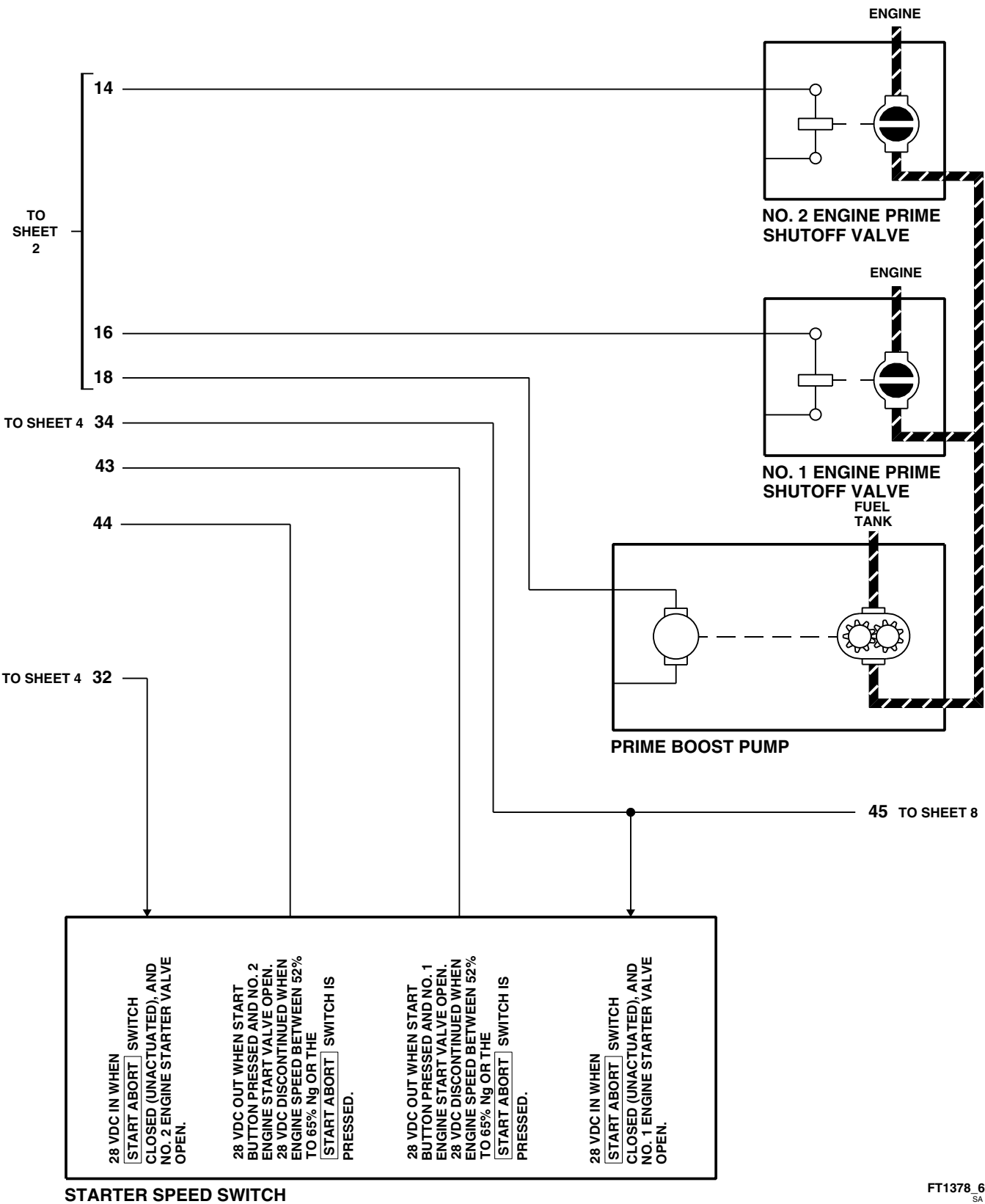
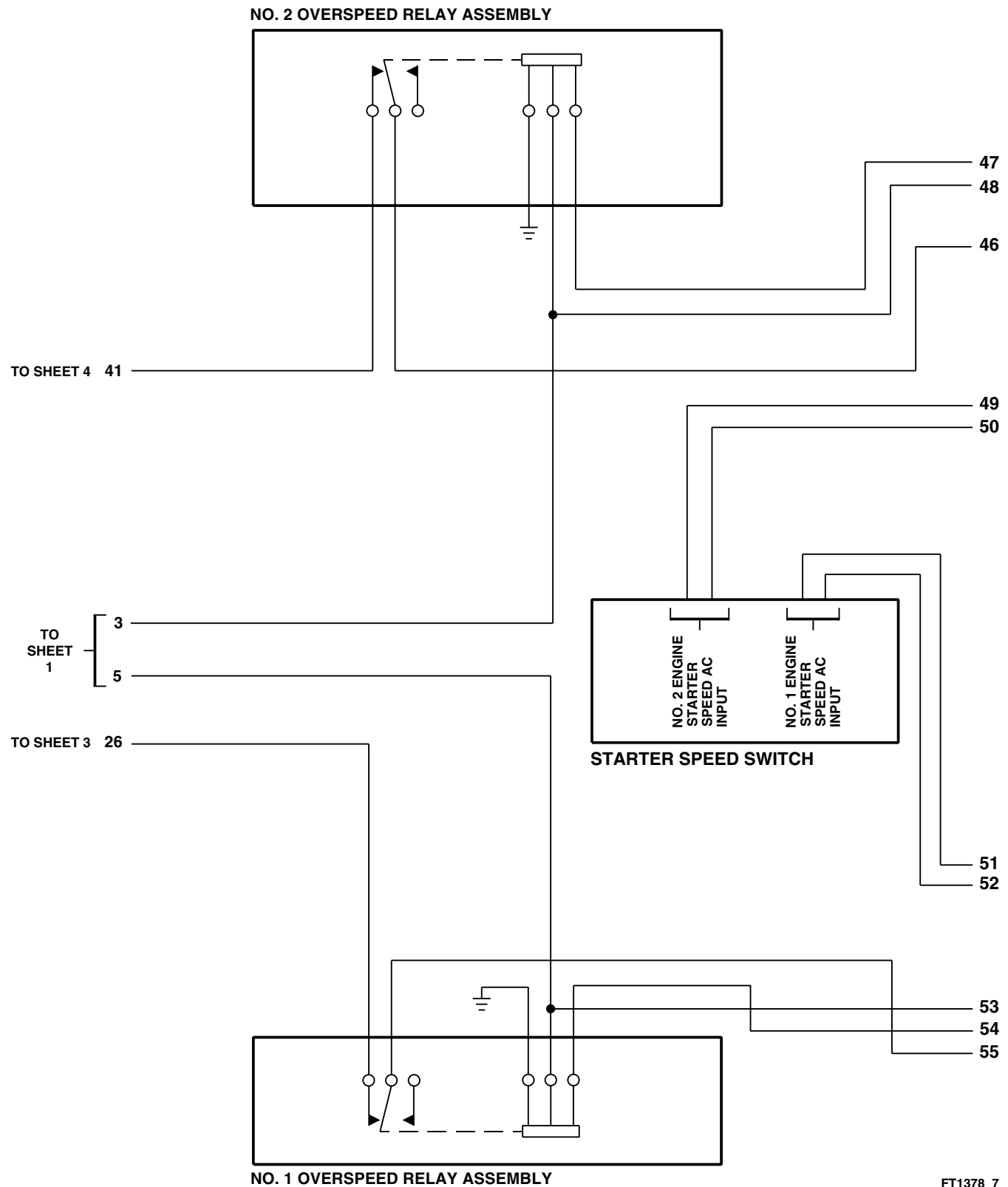


FIGURE 4-1-3-2. ENGINE START AND IGNITION SYSTEM BLOCK DIAGRAM
(SHEET 6 OF 8)

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**FIGURE 4-1-3-2. ENGINE START AND IGNITION SYSTEM BLOCK DIAGRAM
(SHEET 7 OF 8)**

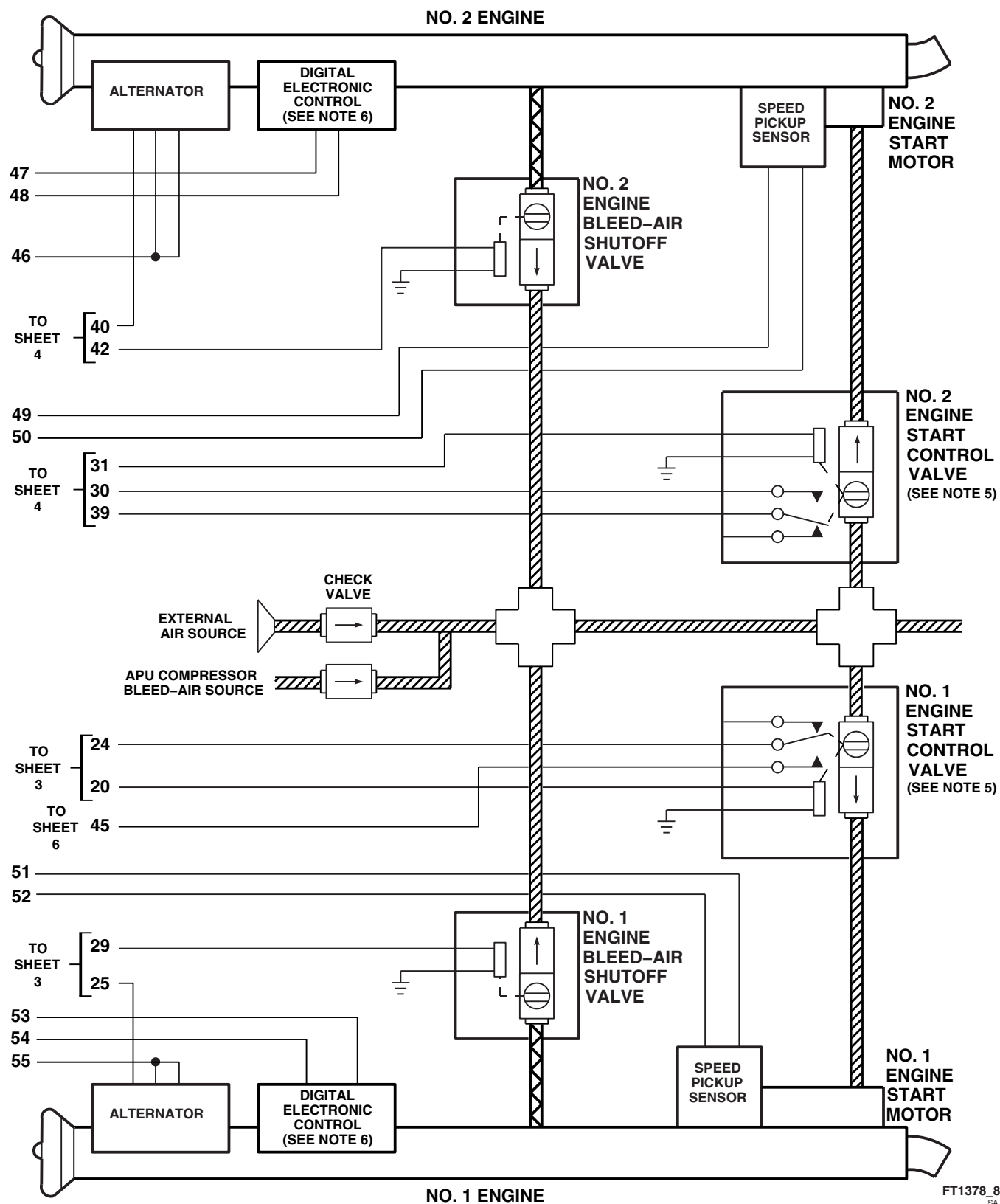


FIGURE 4-1-3-2. ENGINE START AND IGNITION SYSTEM BLOCK DIAGRAM
(SHEET 8 OF 8)

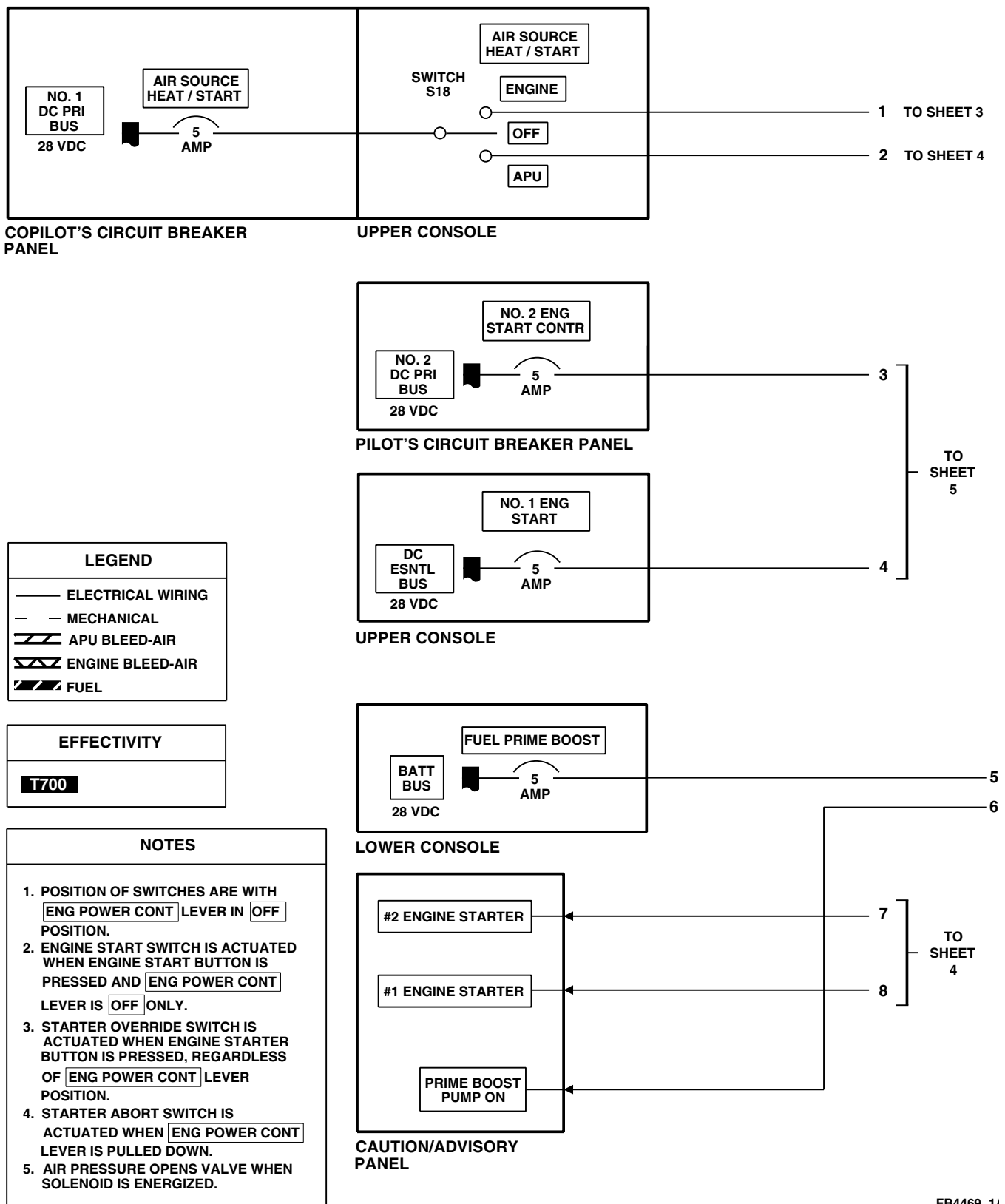
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FIGURE 4-1-3-3. ENGINE START AND IGNITION SYSTEM BLOCK DIAGRAM
(SHEET 1 OF 7)

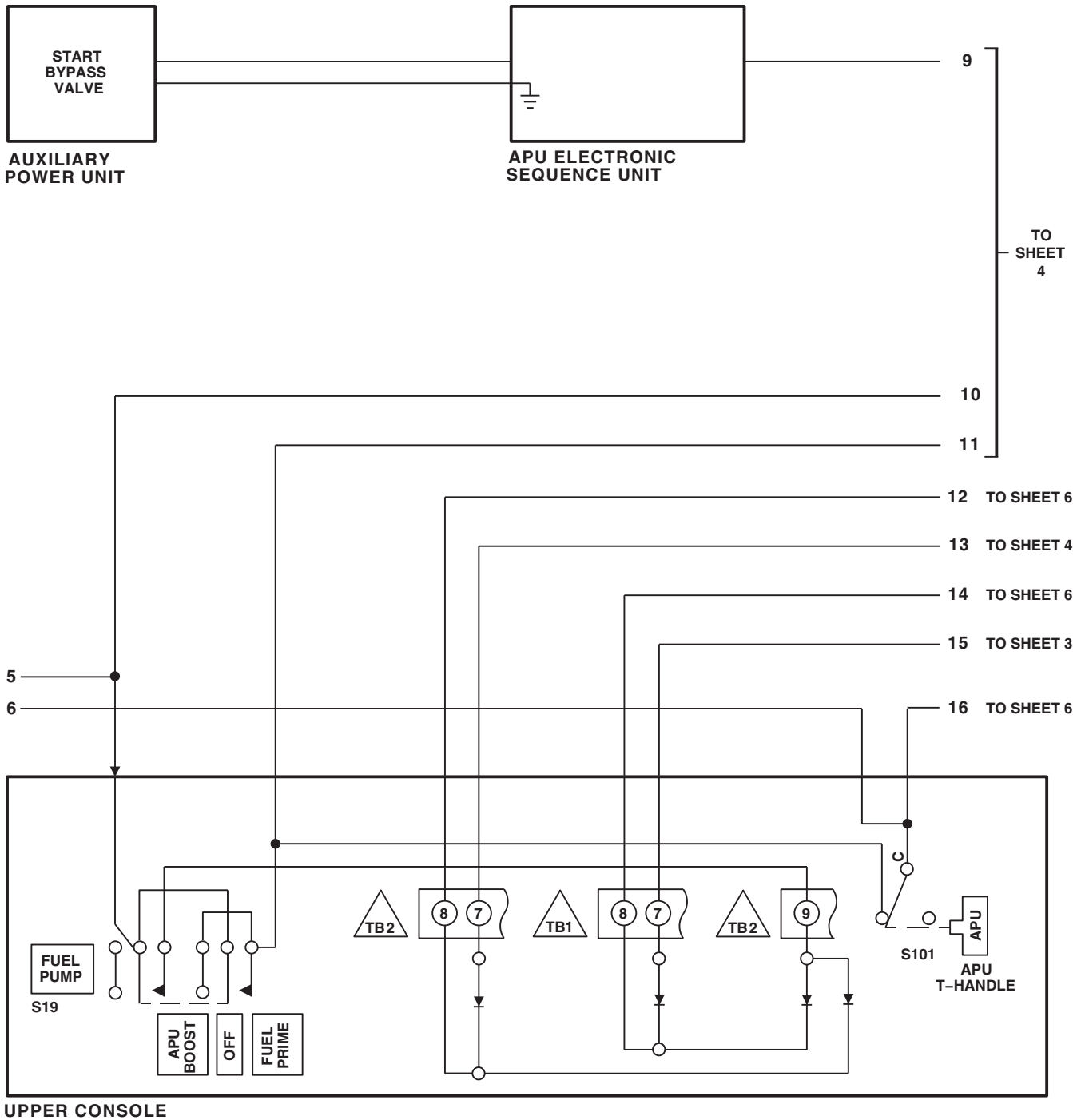
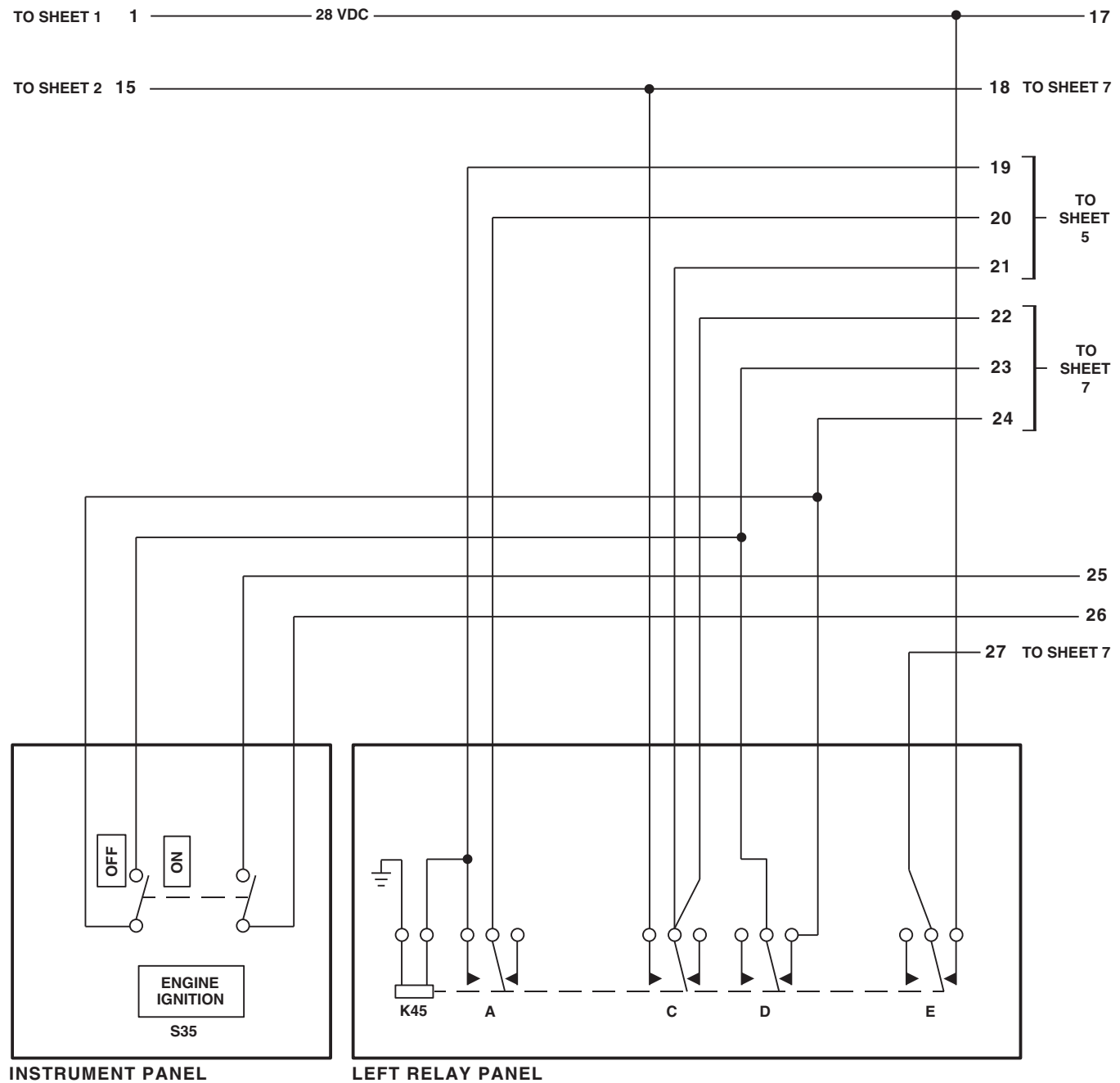


FIGURE 4-1-3-3. ENGINE START AND IGNITION SYSTEM BLOCK DIAGRAM
 (SHEET 2 OF 7)

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**FIGURE 4-1-3-3. ENGINE START AND IGNITION SYSTEM BLOCK DIAGRAM
(SHEET 3 OF 7)**

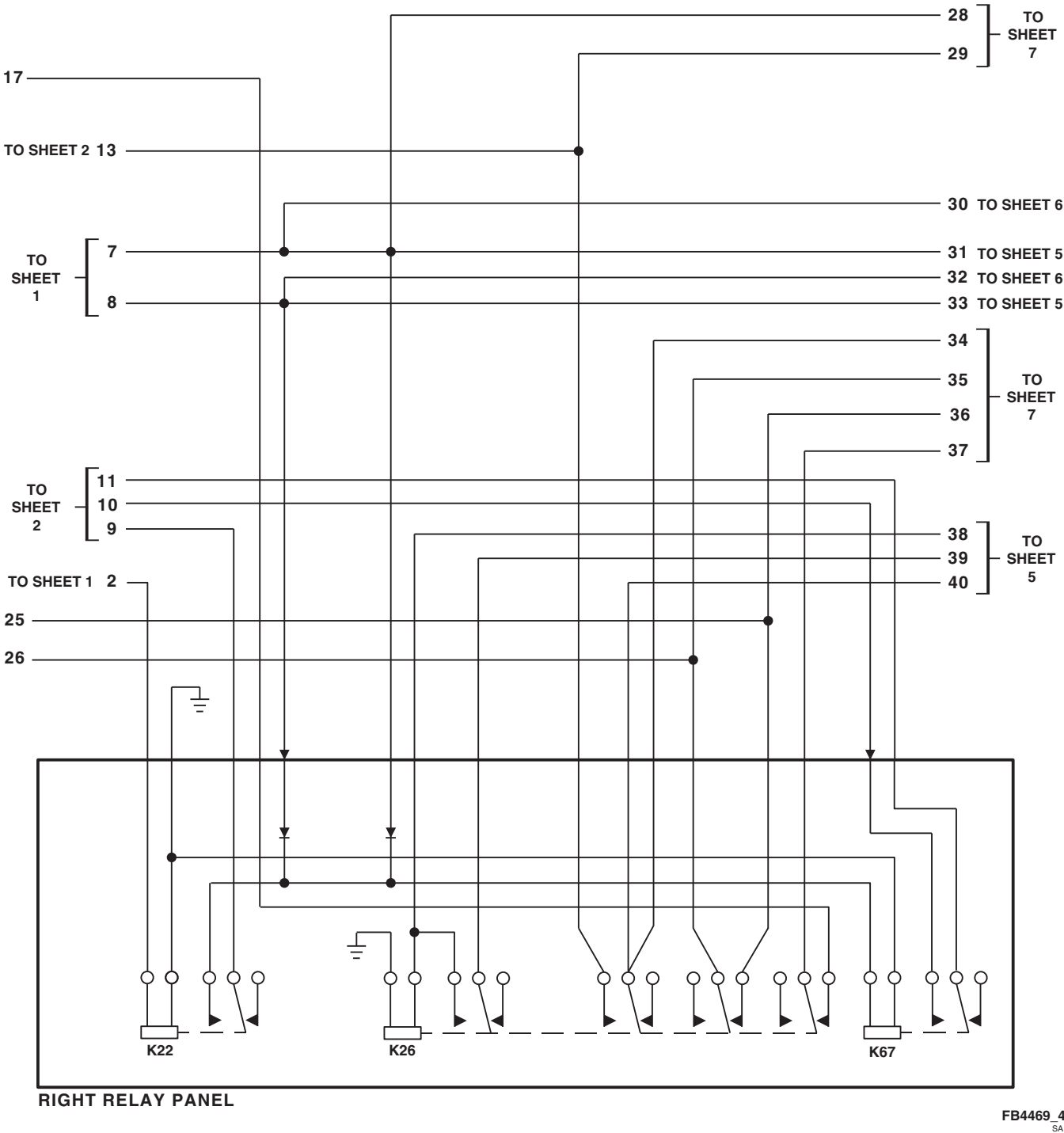


FIGURE 4-1-3.3. ENGINE START AND IGNITION SYSTEM BLOCK DIAGRAM
 (SHEET 4 OF 7)

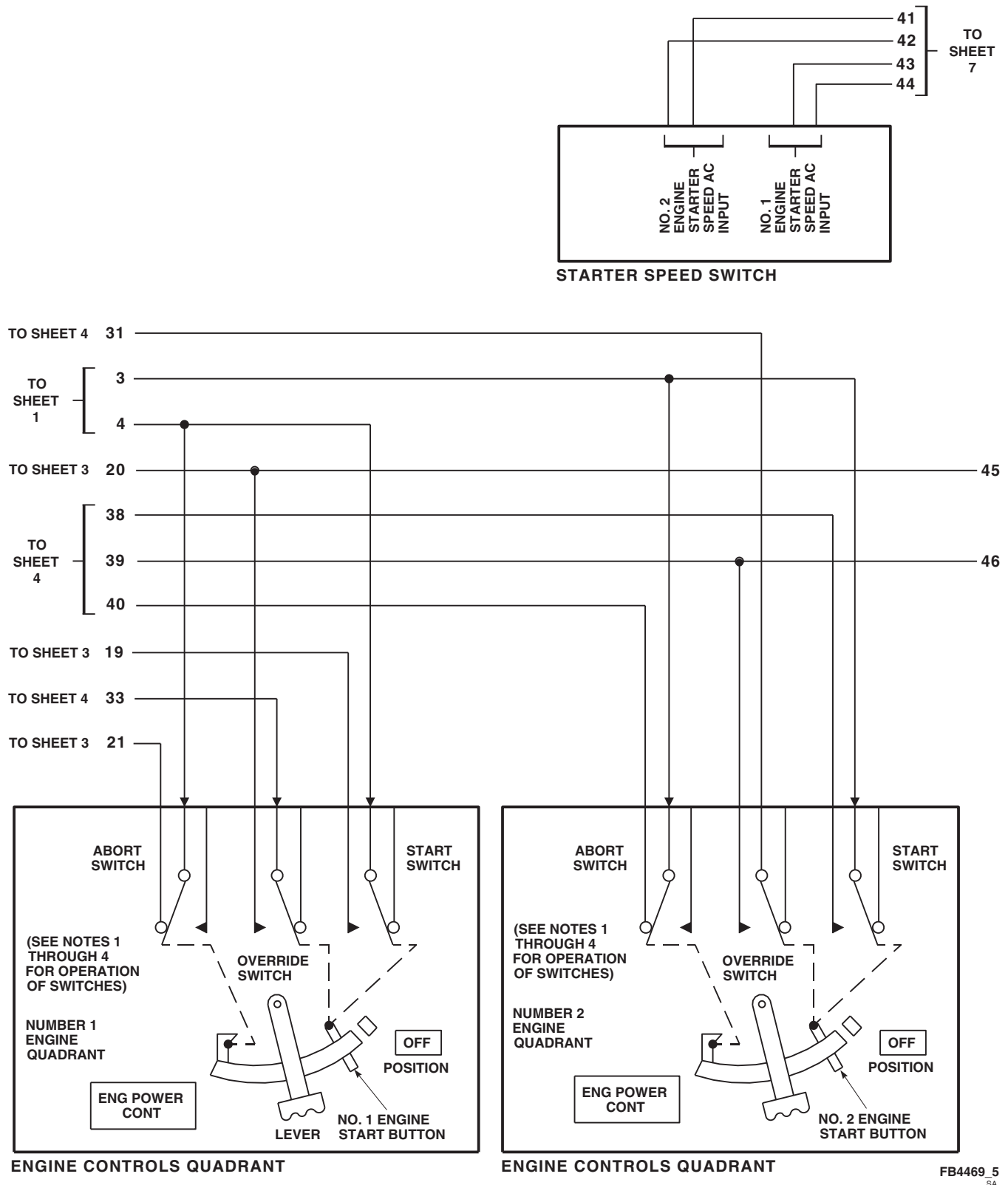


FIGURE 4-1-3-3. ENGINE START AND IGNITION SYSTEM BLOCK DIAGRAM
(SHEET 5 OF 7)

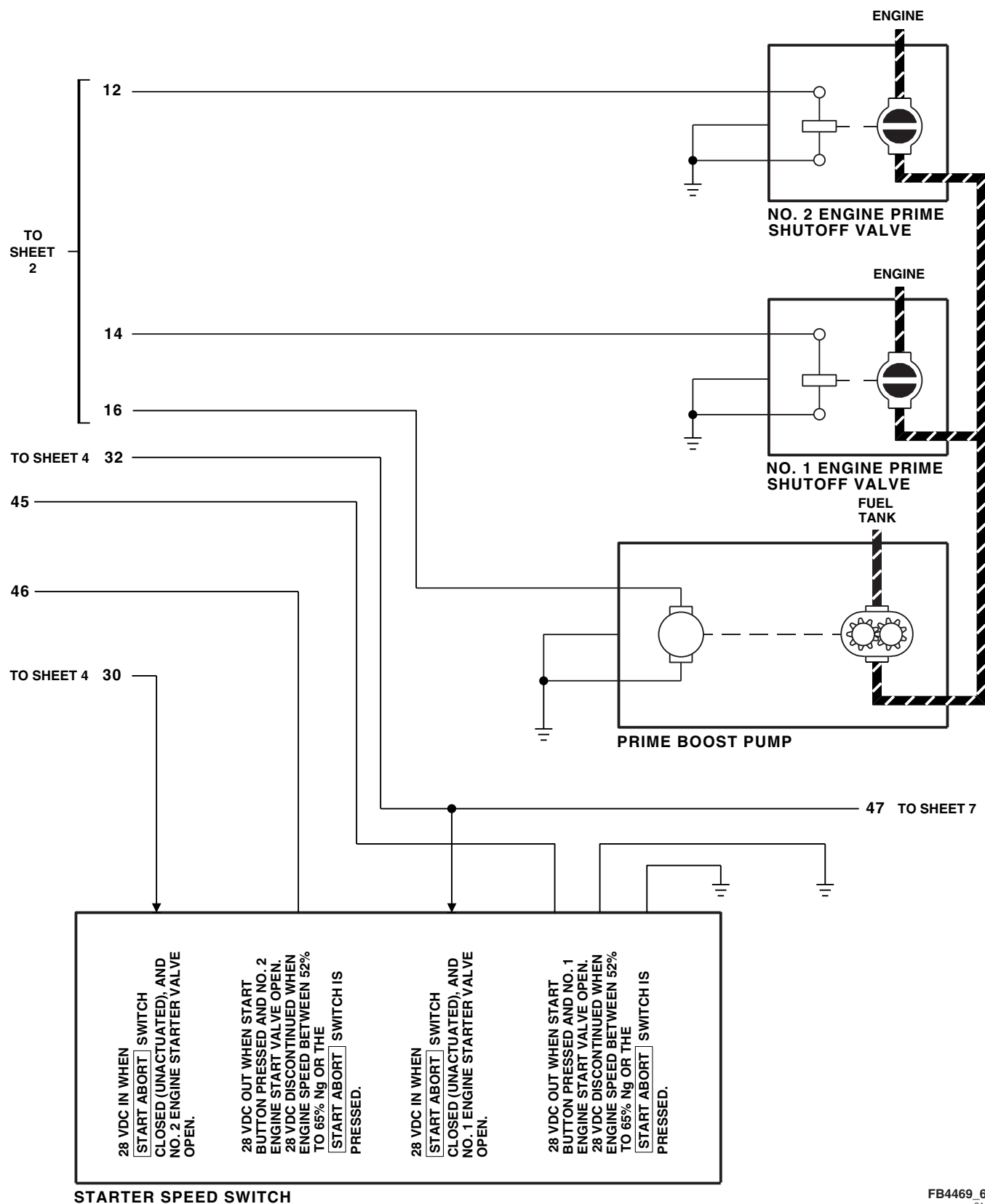


FIGURE 4-1-3-3. ENGINE START AND IGNITION SYSTEM BLOCK DIAGRAM (SHEET 6 OF 7)

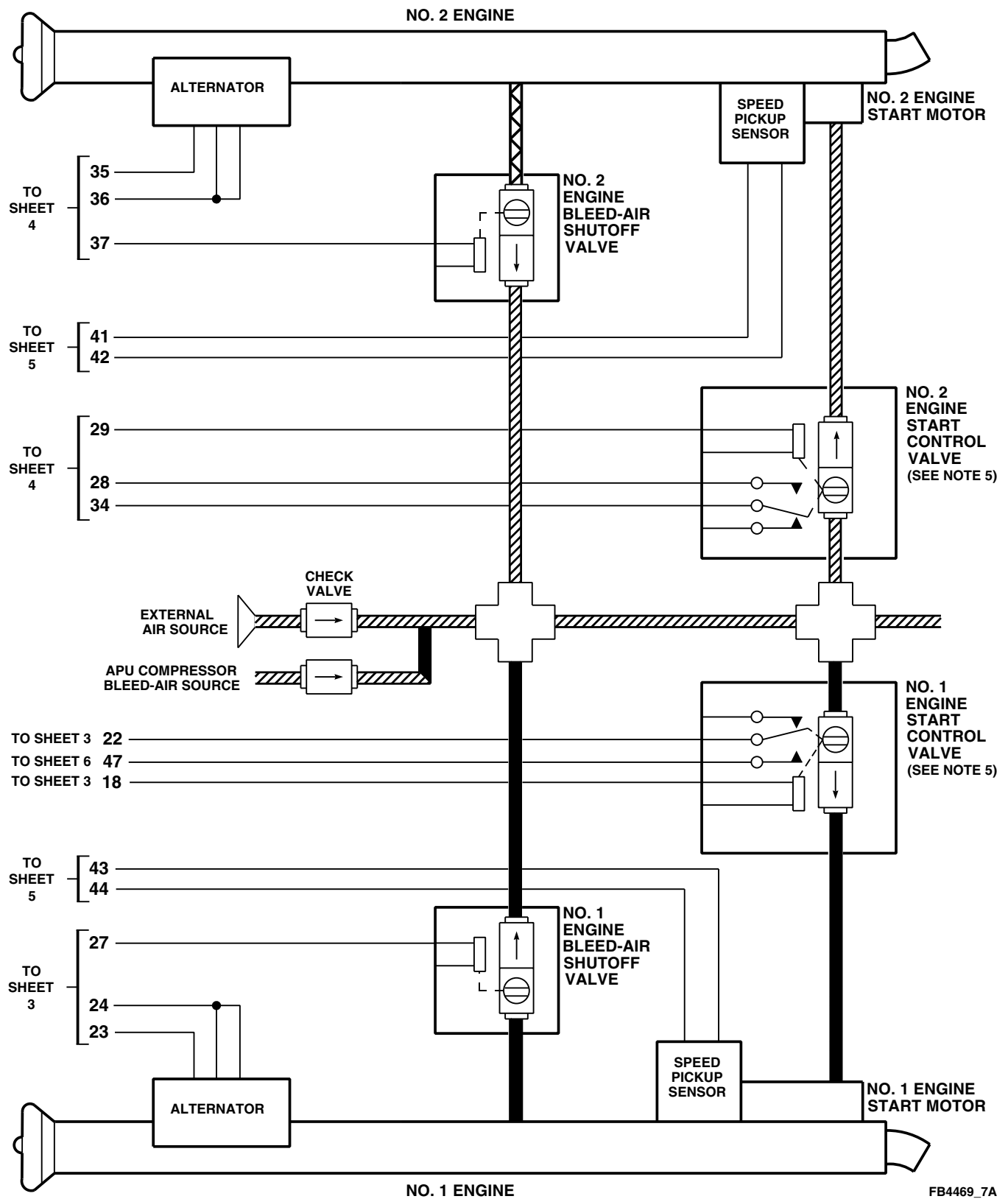


FIGURE 4-1-3-3. ENGINE START AND IGNITION SYSTEM BLOCK DIAGRAM
(SHEET 7 OF 7)

4-1-4 ENGINE SPEED GOVERNING SYSTEMS DESCRIPTION AND DATA.

4-1-4.1 Engine Speed Trim System.

The engine speed trim system consists of two speed trim switches, a pair of potentiometers, and a dc electric motor. The engine speed trim switches on each collective stick grip are used to increase or decrease engine power turbine speed (Np) (% RPM 1 and 2), thereby increasing main rotor speed (Nr) (% RPM RTR) to between 96% and 100%. The switches are spring loaded to the center position and marked ENG RPM, with positions INCR and DECR. Electrical power for the system is obtained from the No. 2 dc primary bus through the SPEED TRIM circuit breaker on the pilot's circuit breaker panel. The speed trim switches supply power to the dc motor through a gear to two potentiometers (one for each engine). These transmit a signal to each engine's electrical control unit (ECU) or digital electronic control (DEC) that automatically increases or decreases each engine's Np equally and simultaneously. The pilot can override copilot authority any time (Figure 4-1-4-1).

4-1-4.2 Engine Overspeed Protection System.

The engine overspeed protection system prevents destructive overspeed of the power turbine. The system is controlled by the engine-mounted ECU or DEC. Two identical electrical pickups mounted on the turbine section of the engine sense power turbine speed and torque. One pickup senses basic power turbine speed control and cockpit speed indication. The other pickup senses power turbine torque and provides a speed signal to the Np overspeed limiter in the ECU or DEC. When the Np overspeed limiter senses an overspeed signal, the overspeed solenoid in the sequence valve is activated, thereby cutting back on fuel flow. **T701A** **T701C** The system is set to operate at 120% Np (% RPM 1 or 2) and result in a fuel flow shutoff, by the overspeed protection relay, causing engine flameout. When % RPM is reduced, fuel flow is returned to the engine and engine ignition will remain on for an additional five seconds to allow for engine restart. The overspeed protection system is also used as a hot start prevention system. The DEC senses the hot start condition and sends a signal to the engine overspeed protection relay set-

ting that system in operation. **T700** The system is set to operate at 106% Np (% RPM 1 or 2) and will result in an initial reduction and eventual cycling of both Ng and Np until the cause of the overspeed is removed.

The engine overspeed test buttons on the upper console marked TEST A and TEST B, under the headings NO. 1 ENG OVSP and NO. 2 ENG OVSP, permit checkout of overspeed circuits to verify correct operation. The overspeed checkout system permits circuits A and B to be checked, thereby avoiding in-flight overspeed signals.

T701C The overspeed test buttons, when pressed individually, also suppresses/redisplays DEC diagnostic codes on the pilot display unit, and overrides the automatic hot start prevention system.

The engines are at 100% Np. When test switch B is actuated, and if test circuit A is working correctly, no change in either Np or Ng will result. Similarly with test circuit B. **T701C** When both switches A and B are closed simultaneously, an overspeed condition is sensed by the DEC and a signal is sent to the engine overspeed relays causing a shutoff of fuel flow to the engine. When the switches are released, the engine will restart and climb back to 100% Np (Figure 4-1-4-4 and Figure 4-1-4-5).

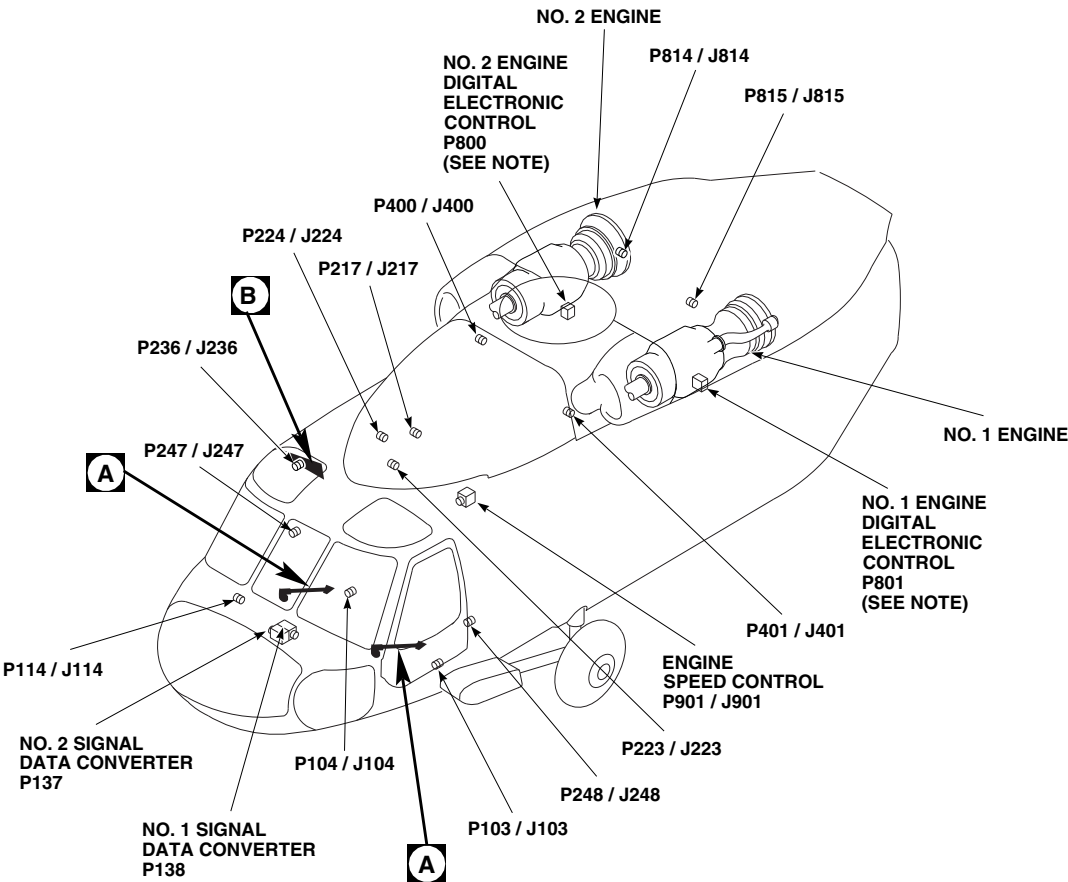
T701A When both switches A and B are closed simultaneously, the overspeed reference signal is reduced from the normal 119.6% Np to 96% Np. Since the engine has been set at 100% Np, the result is an overspeed signal to the overspeed and drain valve and activation of the engine ignition system, such that the engine will flame out and relight during the overspeed test. Power for the engine overspeed protection system is normally supplied by the engine alternator. Redundant power for the No. 1 engine overspeed protection system is supplied by the No. 1 primary ac bus through the NO. 1 ENG OVSP circuit breaker on the copilot's circuit breaker panel. Redundant power for No. 2 engine overspeed protection system is supplied by the No. 2 primary ac bus through the NO. 2 ENG OVSP circuit breaker on the pilot's circuit breaker panel (Figure 4-1-4-6 and Figure 4-1-4-7).

T700 When both switches A and B are closed

simultaneously, the overspeed reference signal is reduced from the normal 106% Np to 100% Np. Since the engine has been set at 100% Np, the result is an overspeed signal to the overspeed solenoid in the sequence valve. The engine will then experience Np and gas generator speed (Ng) cycling with 2% Np and 1% Ng variations. Power for the engine overspeed protection system is normally supplied by the engine alternator. Redundant power for the No. 1 engine overspeed protection system is supplied by the No. 1 primary ac bus through the NO. 1 ENG OVSP circuit breaker on the copilot's circuit breaker panel.

Redundant power for No. 2 engine overspeed protection system is supplied by the No. 2 primary ac bus through the NO. 2 ENG OVSP circuit breaker on the pilot's circuit breaker panel (Figure 4-1-4-6 and Figure 4-1-4-7).◀■

T701C ▶ A collective stick position sensor is added, located on the mixer assembly. The collective stick sensor converts collective stick position to a corresponding electrical signal, which is sent to the engine's digital electronic control units for engine speed trim adjustments (positive droop control).◀■

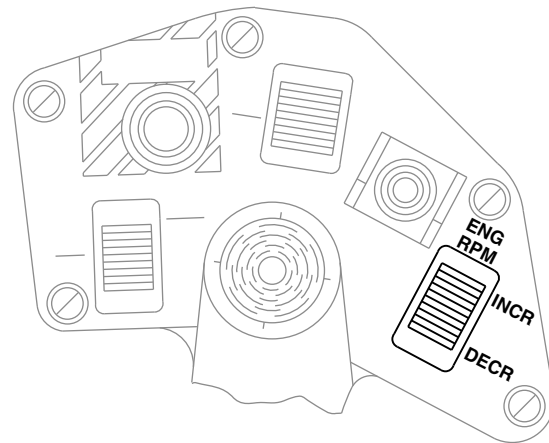


NOTE	
T700	T701A ELECTRICAL CONTROL UNIT INSTALLED.
T701C	DIGITAL ELECTRONIC CONTROL UNIT INSTALLED.

TERMINAL BOARD/DISCONNECT PLUG/RECEPTACLE	LOCATION/CONNECTION POINT
P103 / J103	COCKPIT TUB BL 32 LH, STA 247
P104 / J104	COCKPIT TUB BL 15 RH, STA 247
P114 / J114	COCKPIT BL 8 RH, STA 200
P137	NO. 2 SIGNAL DATA CONVERTER
P138	NO. 1 SIGNAL DATA CONVERTER
P217 / J217	CABIN CEILING BL 0, STA 284
P223 / J223	JUNCTION BOX, MAIN ROTOR PYLON DECK
P224 / J224	JUNCTION BOX, MAIN ROTOR PYLON DECK
P226 / J226	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 23 RH
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 23 RH

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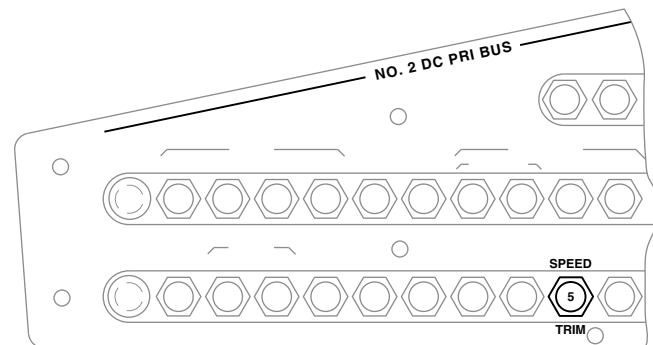
FIGURE 4-1-4-1. ENGINE SPEED GOVERNING (SPEED TRIM) SYSTEMS LOCATION DIAGRAM (SHEET 1 OF 2)



PILOT'S AND COPILOT'S
COLLECTIVE STICK



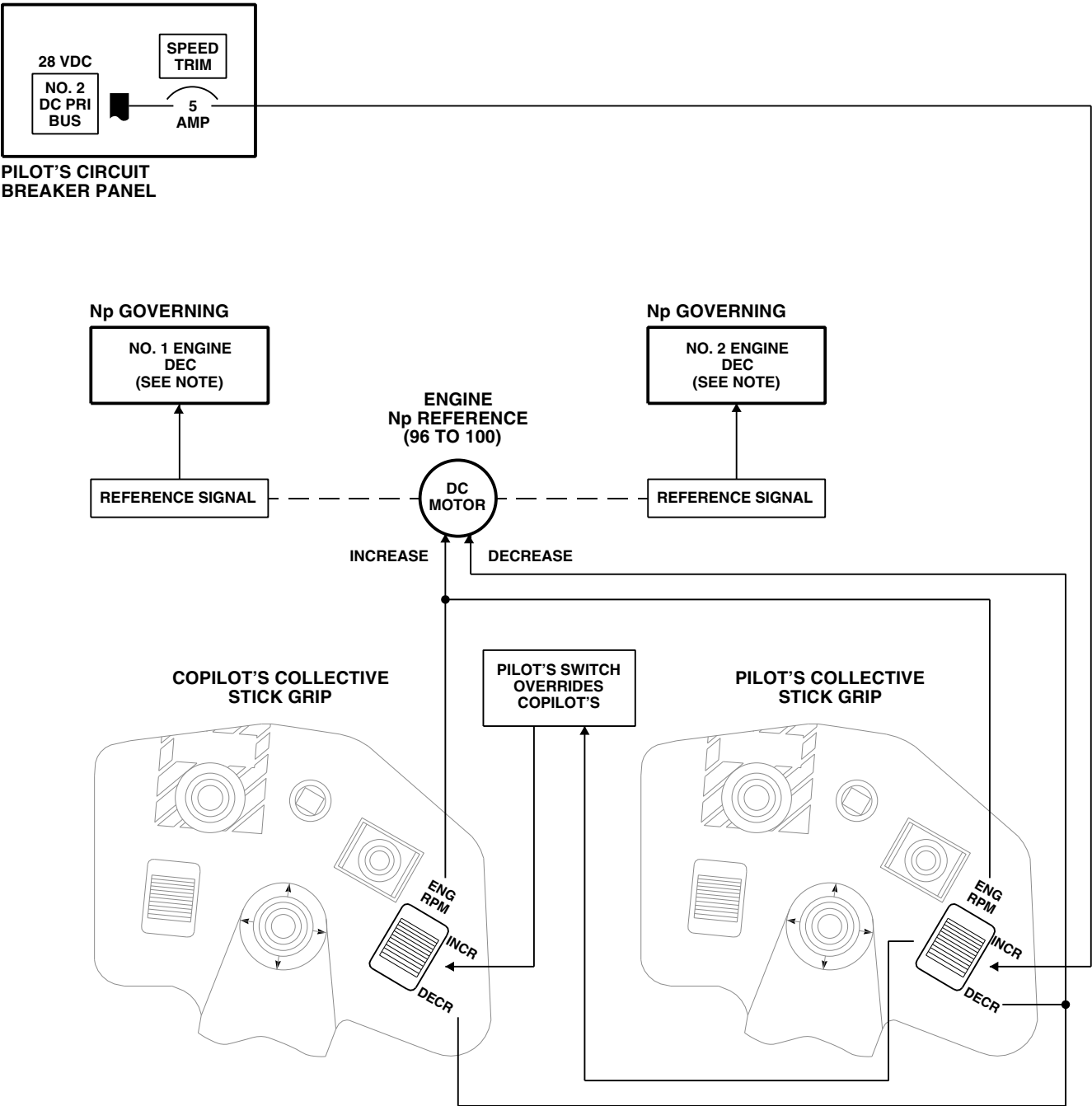
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P247 / J247	CABIN BL 40 RH, STA 248
P248 / J248	CABIN BL 40 LH, STA 248
P400 / J400	MAIN ROTOR PYLON DECK BL 24 RH, STA 318
P401 / J401	MAIN ROTOR PYLON DECK BL 24 LH, STA 318
P800	NO. 2 ENGINE DIGITAL ELECTRONIC CONTROL
P801	NO. 1 ENGINE DIGITAL ELECTRONIC CONTROL
P814 / J814	FIREWALL DISCONNECT BL 14 LH, STA 373
P815 / J815	FIREWALL DISCONNECT BL 14 LH, STA 373
P901 / J901	ENGINE SPEED CONTROL



PILOT'S CIRCUIT BREAKER PANEL

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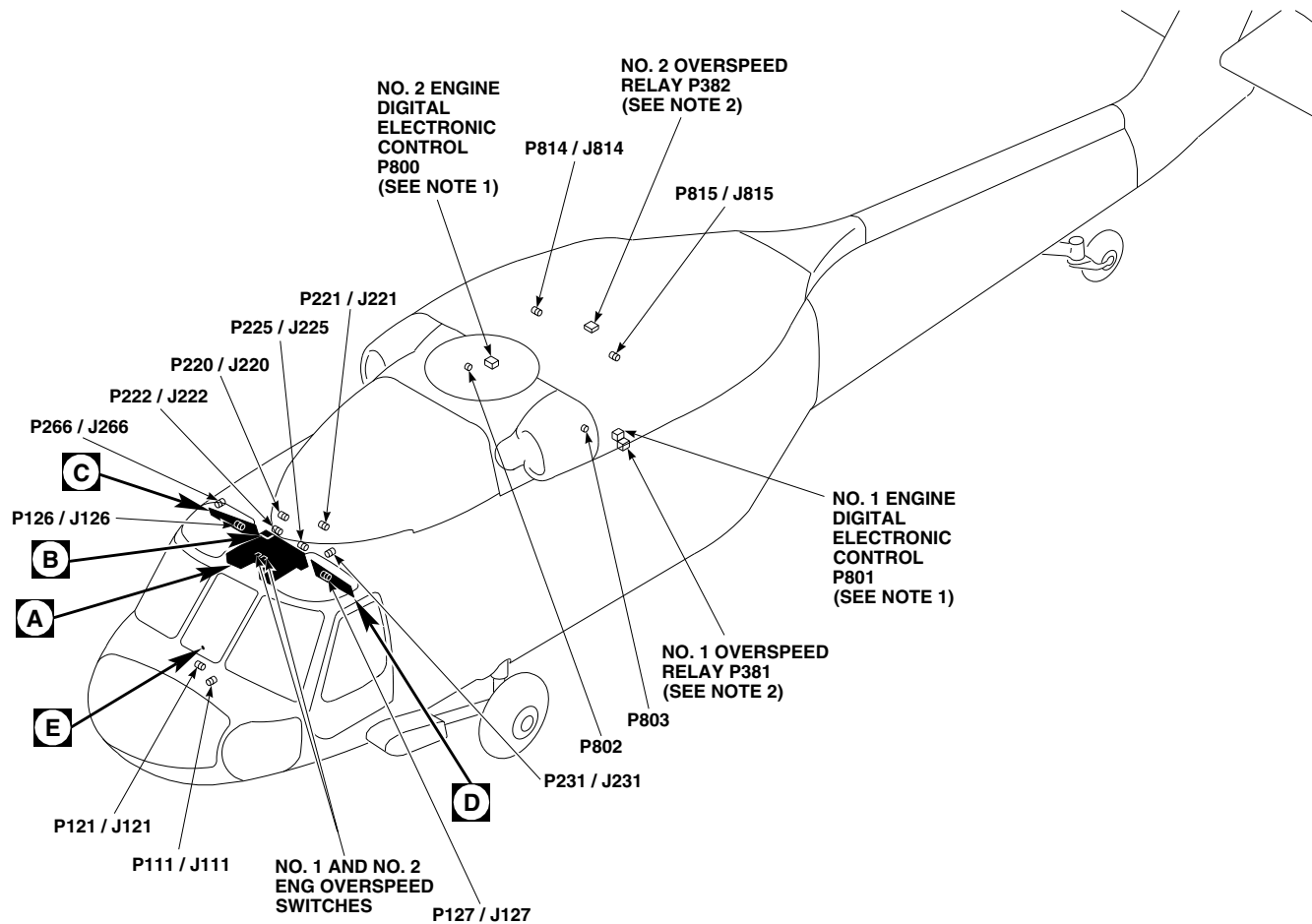
FIGURE 4-1-4-1. ENGINE SPEED GOVERNING (SPEED TRIM) SYSTEMS LOCATION
DIAGRAM (SHEET 2 OF 2)



NOTE	
T701C	HAS DIGITAL ELECTRONIC CONTROL (DEC).
T700 T701A	HAS AN ELECTRICAL CONTROL UNIT.

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FIGURE 4-1-4-2. ENGINE SPEED GOVERNING (SPEED TRIM) SYSTEMS BLOCK DIAGRAM



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P111 / J111	COCKPIT BL 7.5, STA 197
P121 / J121	COCKPIT BL 10 RH, STA 197
P126 / J126	COCKPIT BL 28 RH, STA 243
P127 / J127	COCKPIT BL 28 LH, STA 243
P220 / J220	JUNCTION BOX, MAIN ROTOR PYLON DECK
P221 / J221	JUNCTION BOX, MAIN ROTOR PYLON DECK
P222 / J222	JUNCTION BOX, MAIN ROTOR PYLON DECK

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P225 / J225	JUNCTION BOX, MAIN ROTOR PYLON DECK
P231 / J231	CABIN CEILING BL 8 RH, STA 247
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25.0 RH
P381 / J381	NO. 1 OVERSPEED RELAY (SEE NOTE 2)
P382 / J382	NO. 2 OVERSPEED RELAY (SEE NOTE 2)
P800	NO. 2 ENGINE DIGITAL ELECTRONIC CONTROL
P801	NO. 1 ENGINE DIGITAL ELECTRONIC CONTROL
P802	NO. 2 ENGINE INTERFACE DISCONNECT
P803	NO. 1 ENGINE INTERFACE DISCONNECT
P814 / J814	FIREWALL DISCONNECT BL 14 RH, STA 373
P815 / J815	FIREWALL DISCONNECT BL 14 LH, STA 373

NOTES

1. **T700** **T701A** ELECTRICAL
CONTROL UNIT INSTALLED.
T701C DIGITAL ELECTRONIC
CONTROL UNIT INSTALLED.
2. **T701A** **T701C**

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FIGURE 4-1-4-3. ENGINE SPEED GOVERNING (OVERSPEED PROTECTION) SYSTEMS LOCATION DIAGRAM (SHEET 1 OF 4)

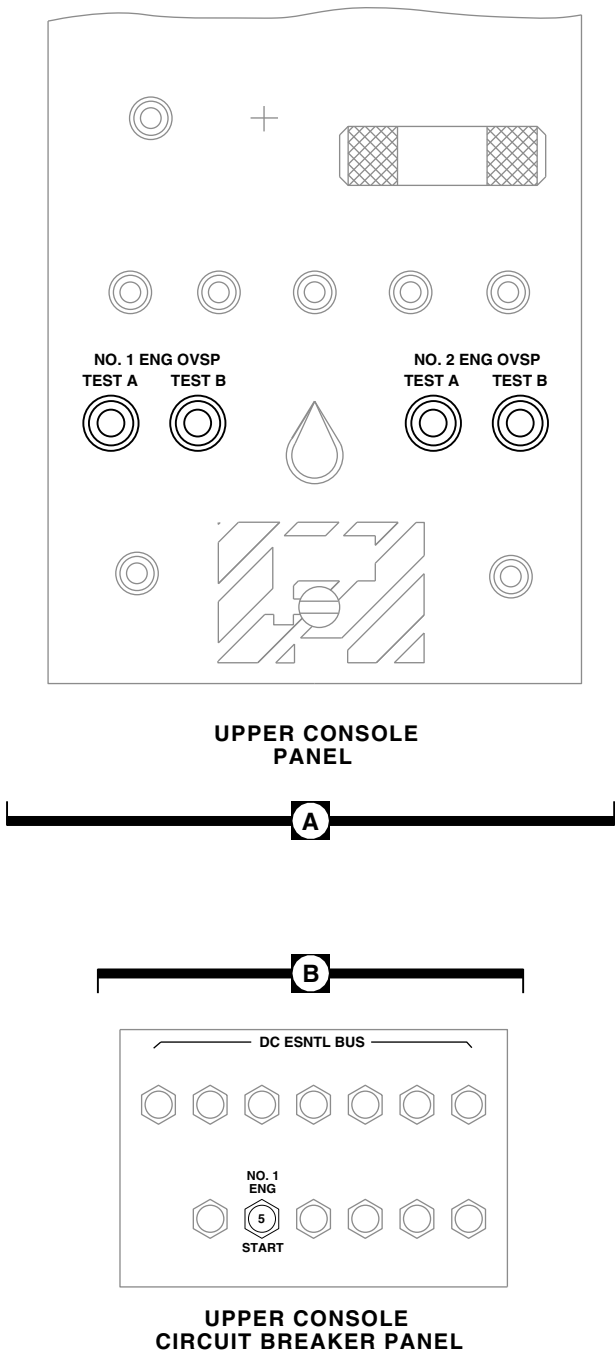


FIGURE 4-1-4-3. ENGINE SPEED GOVERNING (OVERSPEED PROTECTION) SYSTEMS LOCATION DIAGRAM (SHEET 2 OF 4)

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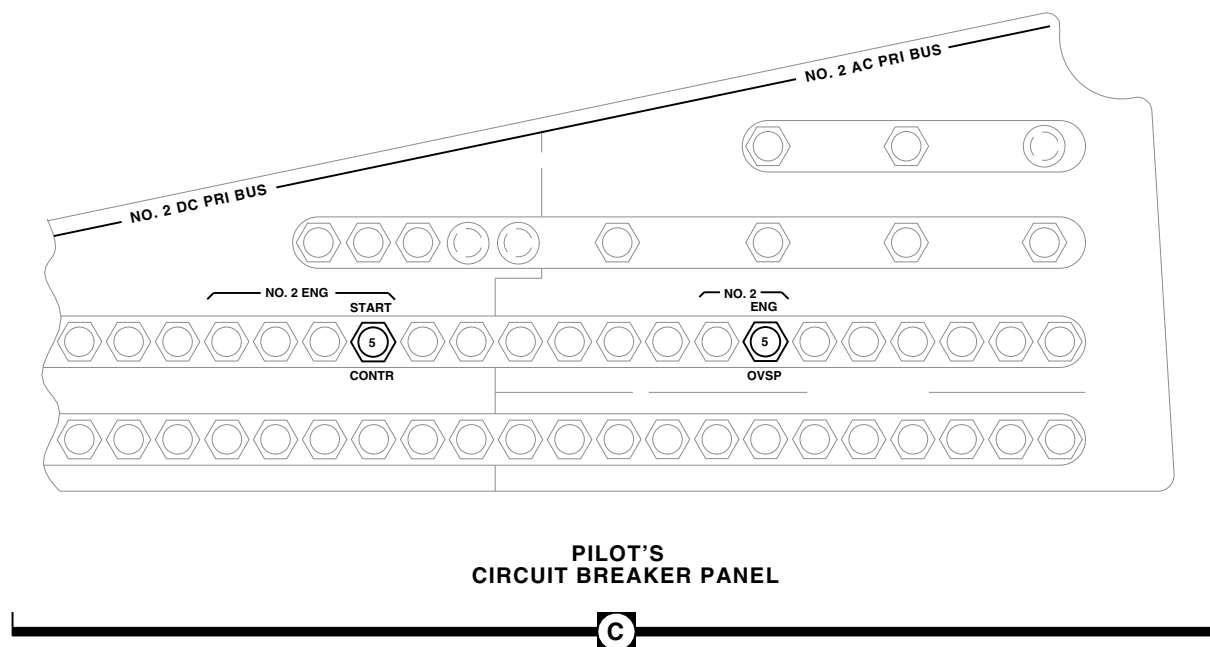
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FIGURE 4-1-4-3. ENGINE SPEED GOVERNING (OVERSPEED PROTECTION) SYSTEMS LOCATION DIAGRAM (SHEET 3 OF 4)

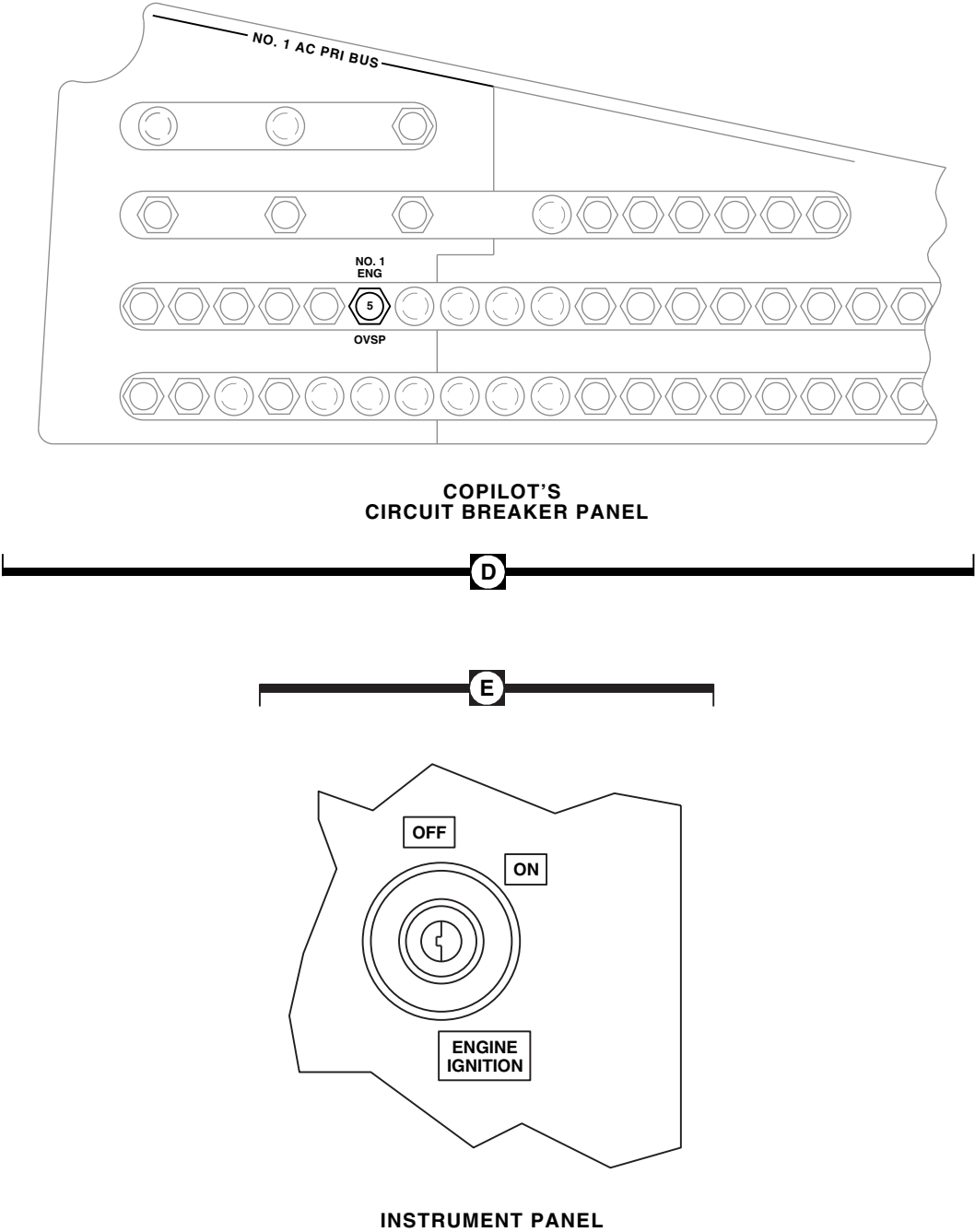


FIGURE 4-1-4-3. ENGINE SPEED GOVERNING (OVERSPEED PROTECTION) SYSTEMS LOCATION DIAGRAM (SHEET 4 OF 4)

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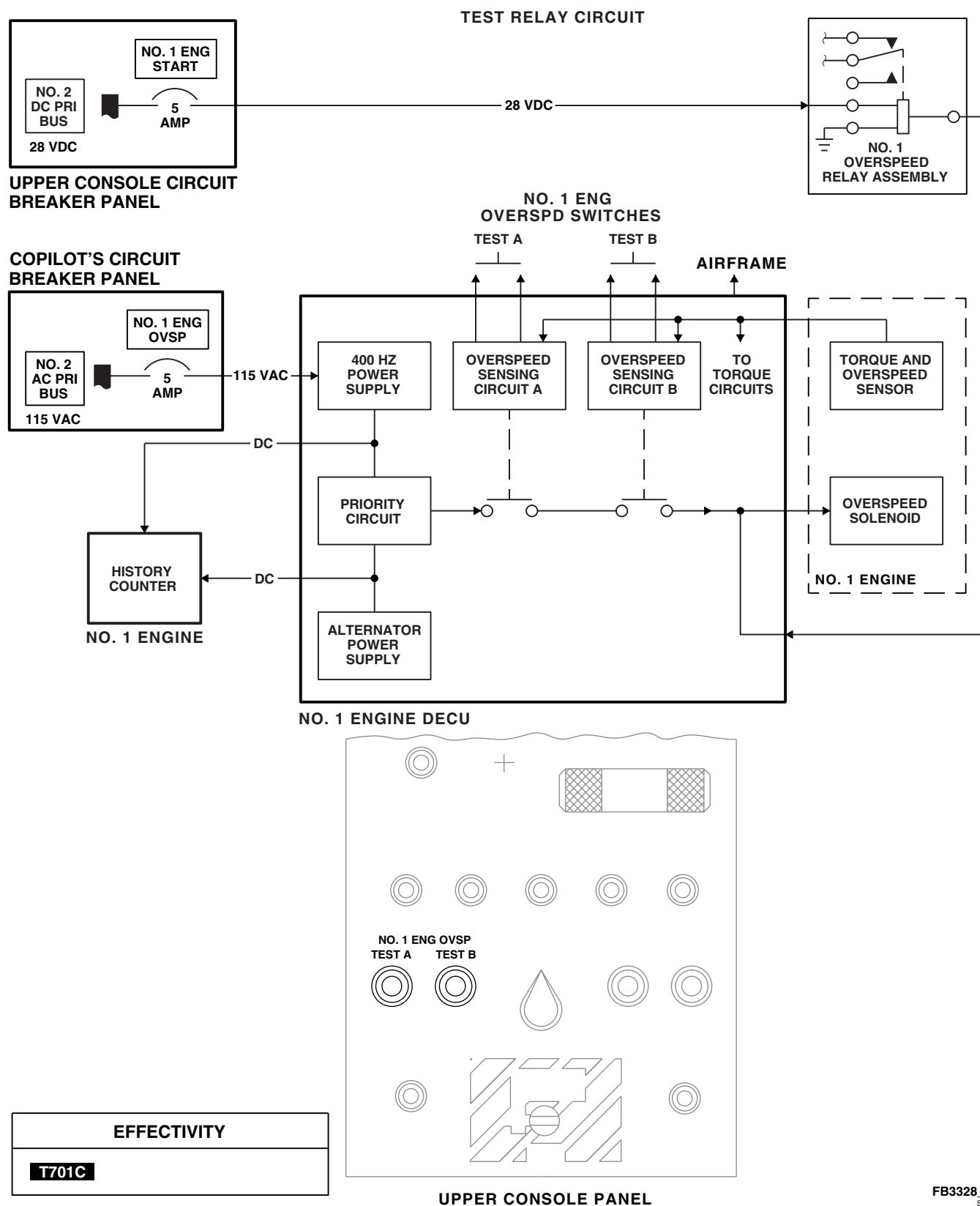


FIGURE 4-1-4-4. ENGINE NO. 1 SPEED GOVERNING SYSTEMS BLOCK DIAGRAM

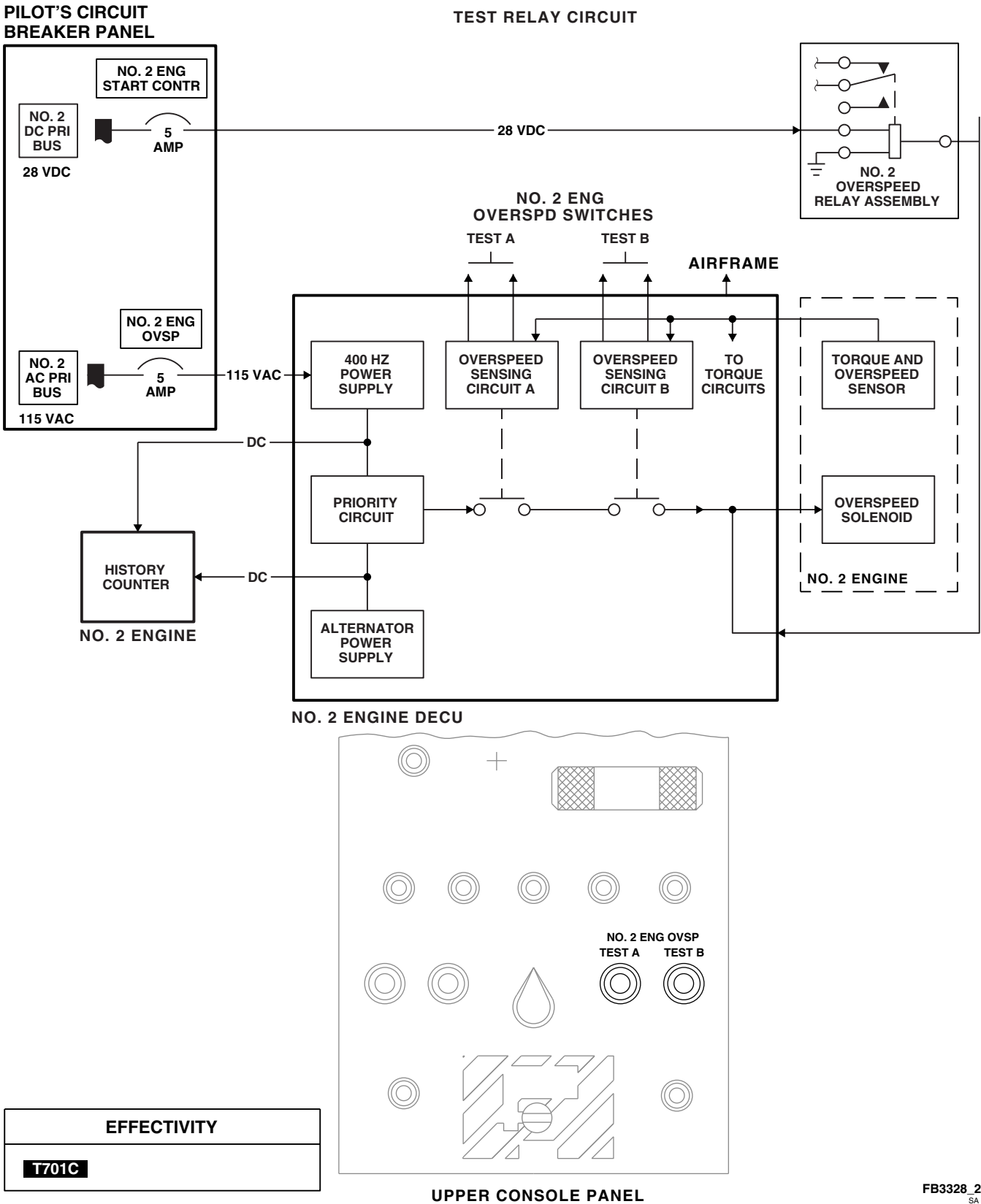


FIGURE 4-1-4-5. ENGINE NO. 2 SPEED GOVERNING SYSTEMS BLOCK DIAGRAM

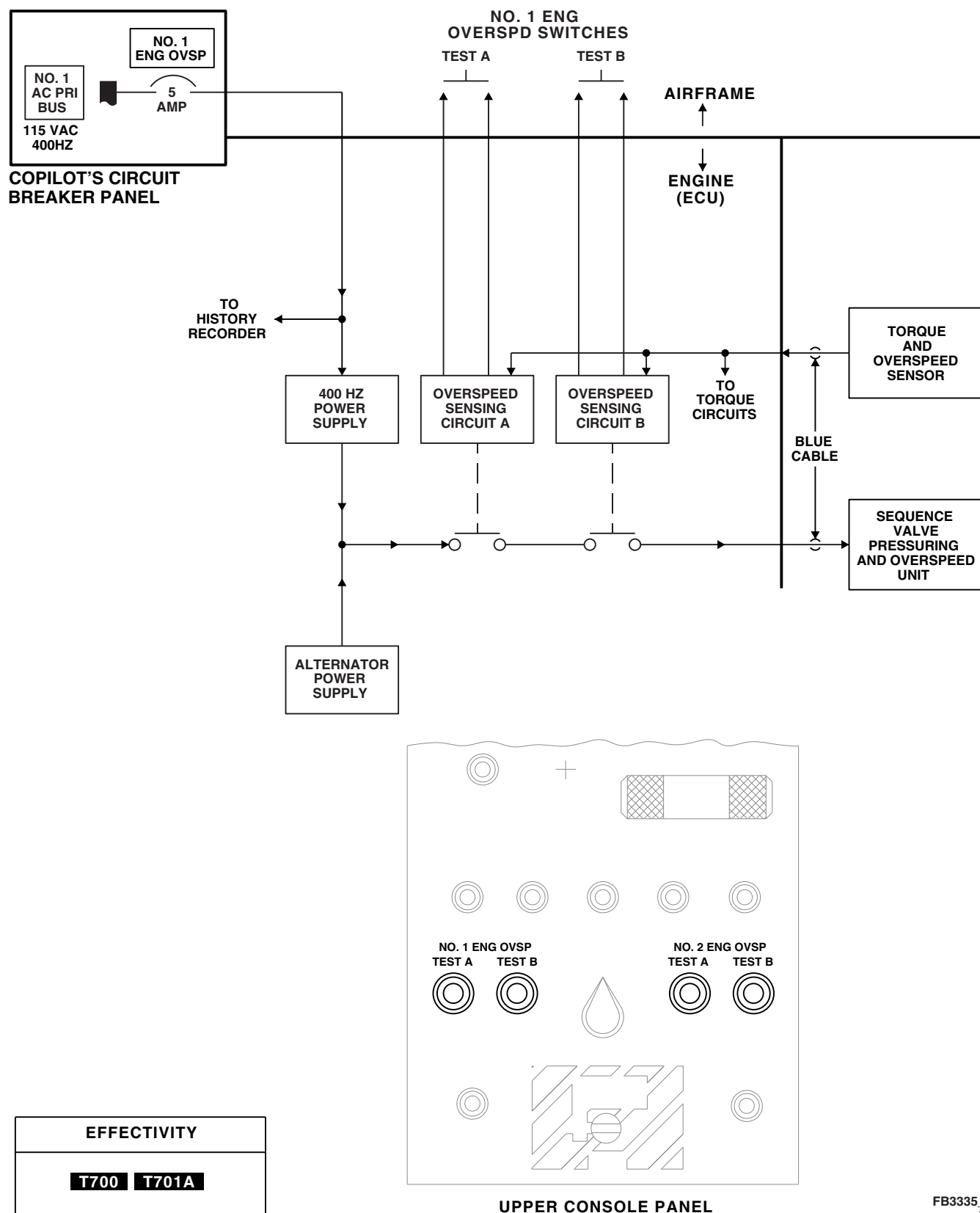


FIGURE 4-1-4-6. ENGINE NO. 1 SPEED GOVERNING SYSTEMS BLOCK DIAGRAM

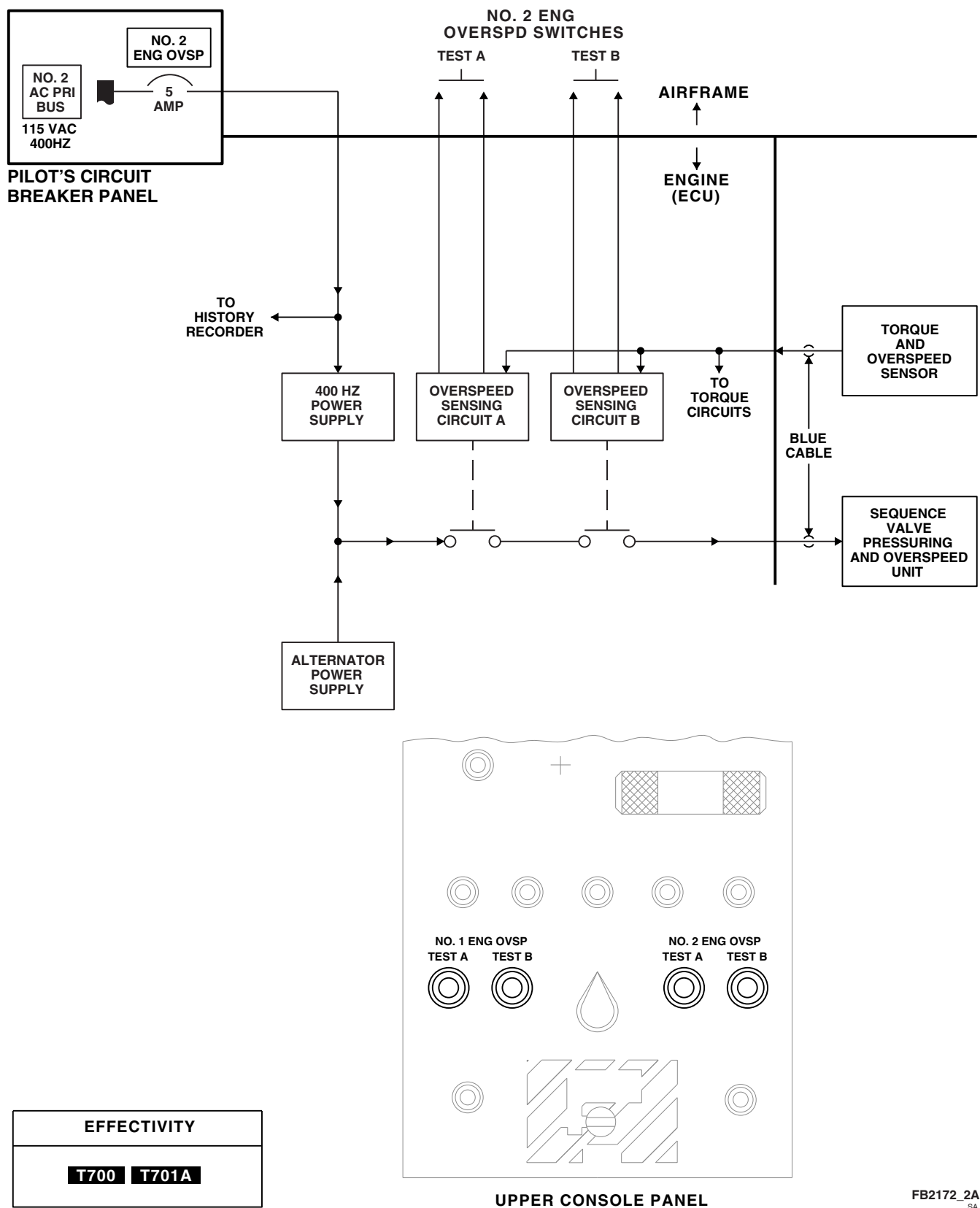


FIGURE 4-1-4-7. ENGINE NO. 2 SPEED GOVERNING SYSTEMS BLOCK DIAGRAM

4-1-5 ENGINE INDICATING SYSTEMS DESCRIPTION AND DATA.

4-1-5.1 Engine Indicating Systems.

The engine indicating systems consist of the power turbine gas temperature (TGT), oil temperature, oil pressure, engine torque, power turbine/rotor speed, and gas generator (Ng) speed indicating systems (Figure 4-1-5-3). Power for the No. 1 engine indicating systems is supplied by the No. 1 dc primary bus and No. 1 ac primary bus through the NO. 1 DC INST and NO. 1 AC INST circuit breakers. Power for the No. 2 engine indicating systems is supplied by the No. 2 dc primary bus and No. 2 ac primary bus through the NO. 2 DC INST and NO. 2 AC INST circuit breakers. All engine instruments are part of the instrument display system. For a complete description of this system, refer to PARA 8-1-3.

4-1-5.1.1 Power Turbine Gas Temperature Indicating System.

The TGT indicating system consists of a vertical scale indicator, seven thermocouples, and a harness. The harness is mounted on the power turbine module. The thermocouples are connected in parallel to provide an electrical output of the temperature sensed by the individual thermocouples. One vertical scale indicator and one digital indicator for each engine is on the instrument panel central display unit under the heading marked TGT TEMP. The scale has a range of 0° to 950°C. **T701C** The normal range is 0° to 810°C. **T700** **T701A** The normal range is 0° to 775°C.

4-1-5.1.2 Oil Temperature Indicating System.

The engine oil temperature indicating system consists of a temperature bulb in each engine and vertical scale indicators on the instrument panel. The ENG OIL TEMP indicators, one for each engine, are on the instrument panel central display unit and have a range of -50° to 180°C. Highest normal oil temperature operating limit is 135°C.

4-1-5.1.3 Oil Pressure Indicating System.

The engine oil pressure indicating system consists of a pressure switch in each engine and vertical

scale indicators on the instrument panel central display unit, one for each engine, under the marked heading ENG OIL PRESS. **T701C** The scale has a range of 12 - 170 psi. The normal range is 26 - 100 psi. **T700** **T701A** The scale has a range of 10 - 130 psi. The normal range is 40 - 100 psi.

4-1-5.1.4 Engine Torque Indicating System.

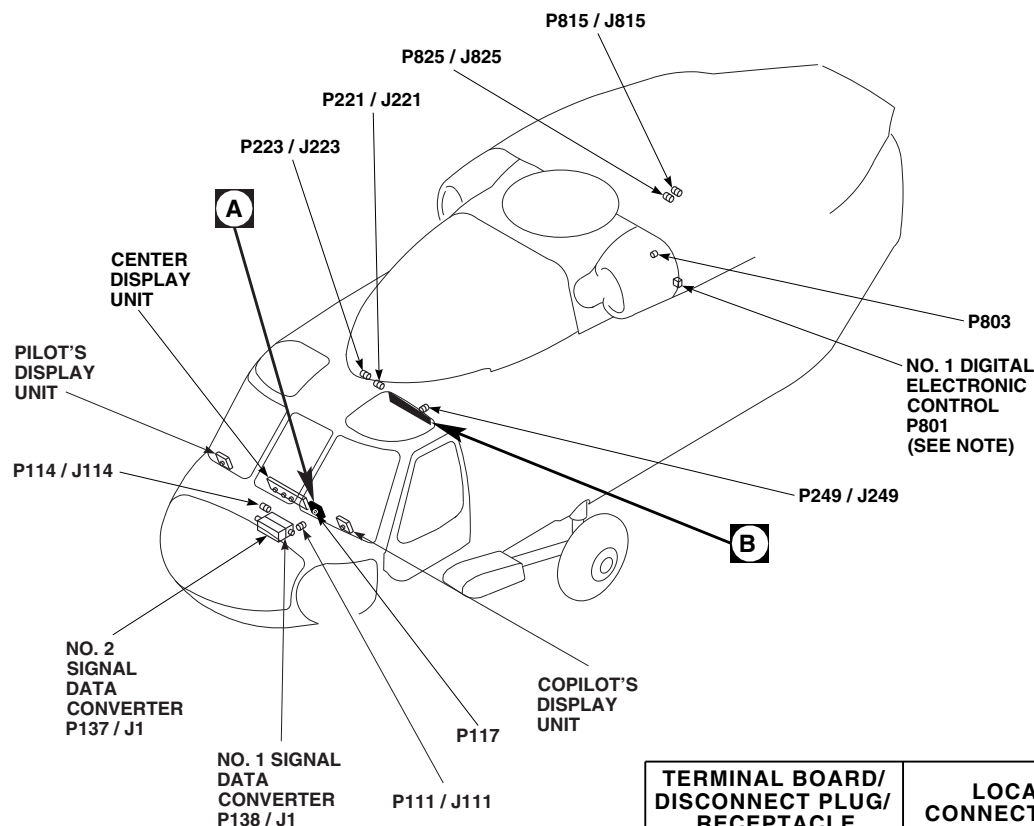
The engine torque indicating system consists of vertical scale indicators on the pilot's and copilot's display units and a torque sensor on each power turbine drive shaft. The torque indicating system shows the amount of power the engine is supplying to the main transmission by measuring the twist of the shaft. Each indicator is marked %TRQ. The scale has a range of 0% to 150% torque. Normal range is 0% to 100%.

4-1-5.1.5 Power Turbine/Rotor Speed Indicating System.

Power turbine and rotor speed are indicated for each engine on a single, vertical scale instrument with two scales. This is mounted on the pilot's and copilot's central display unit. Power turbine speed is indicated in percentage Np, and rotor speed in percent Nr. The power turbine speed system consists of electrical sensors on each engine drive shaft, and vertical scale indicators marked %RPM 1 and 2. The power turbine speed indicator scale is 0% to 130% RPM. Normal range is 96% to 101%. The rotor speed indicating system consists of a speed sense pickup on the rear of the right accessory input module, and a vertical scale indicator marked R. The rotor speed indicator scale is 0% - 130% RPM. Normal range is 96% - 101%.

4-1-5.1.6 Gas Generator Speed Indicating System.

The gas generator speed indicating system consists of an electrical signal from the alternator on the engine gear box and a vertical scale indicator marked Ng SPEED on the central display unit. The scale has a range of 0% - 110%. Normal range is 0% - 102%.



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P114 / J114	COCKPIT BL 8.0, STA 200
P117	CAUTION / ADVISORY PANEL
P137 / J1	NO. 2 SIGNAL DATA CONVERTER
P138 / J1	NO. 1 SIGNAL DATA CONVERTER
P221 / J221	JUNCTION BOX, MAIN ROTOR PYLON DECK
P223 / J223	JUNCTION BOX, MAIN ROTOR PYLON DECK
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23.0 LH
P801	NO. 1 ENGINE DIGITAL ELECTRONIC CONTROL (SEE NOTE)
P803	NO. 1 ENGINE INTERFACE DISCONNECT
P815 / J815	FIREWALL DISCONNECT BL 14 LH, STA 373
P825 / J825	FIREWALL DISCONNECT BL 14 LH, STA 358.25

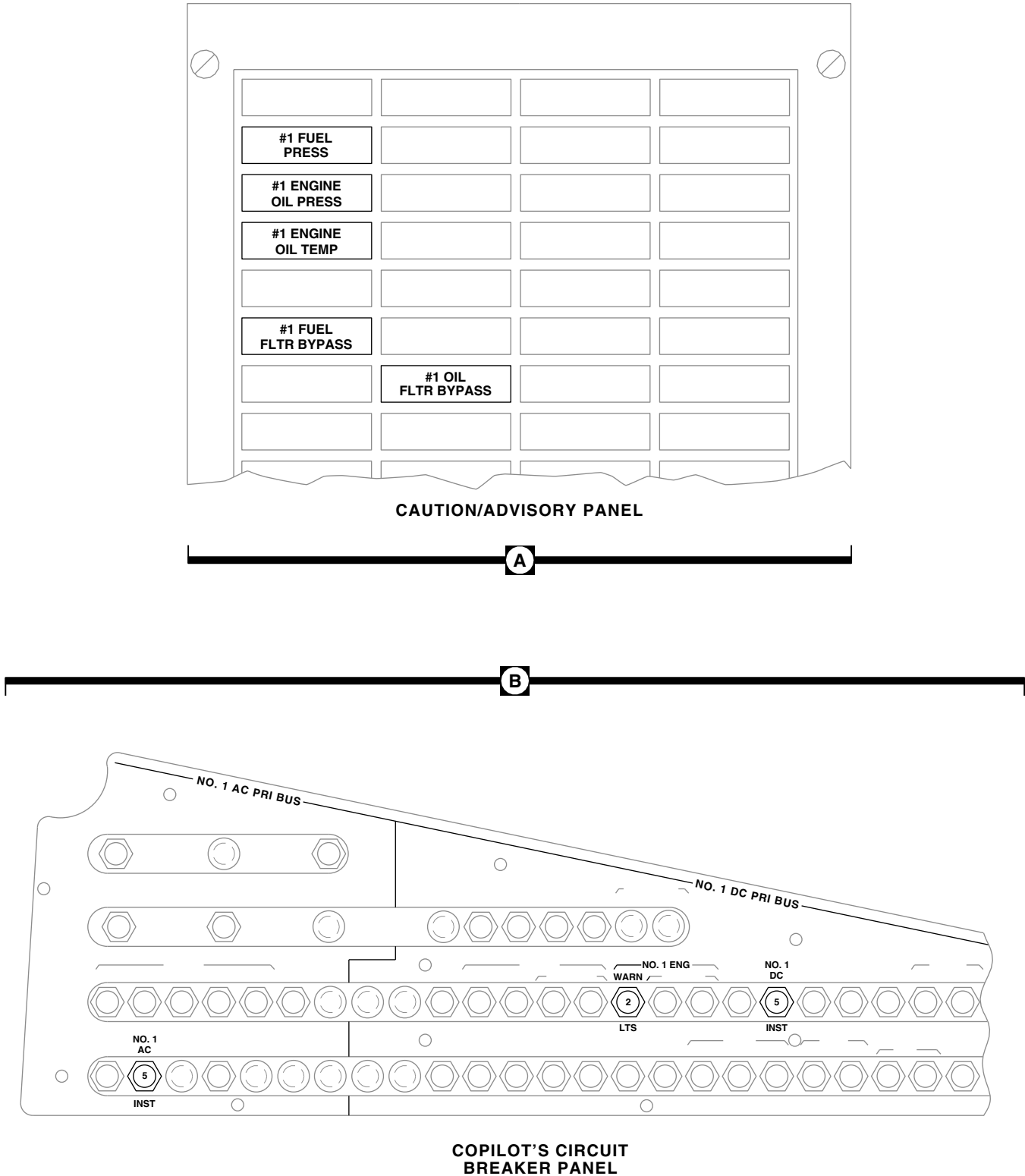
NOTE

T700 T701A ELECTRICAL CONTROL UNIT INSTALLED.

T701C DIGITAL ELECTRONIC CONTROL UNIT INSTALLED.

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FIGURE 4-1-5-1. NO. 1 ENGINE INDICATING SYSTEM LOCATION DIAGRAM
(SHEET 1 OF 2)



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**FIGURE 4-1-5-1. NO. 1 ENGINE INDICATING SYSTEM LOCATION DIAGRAM
(SHEET 2 OF 2)**

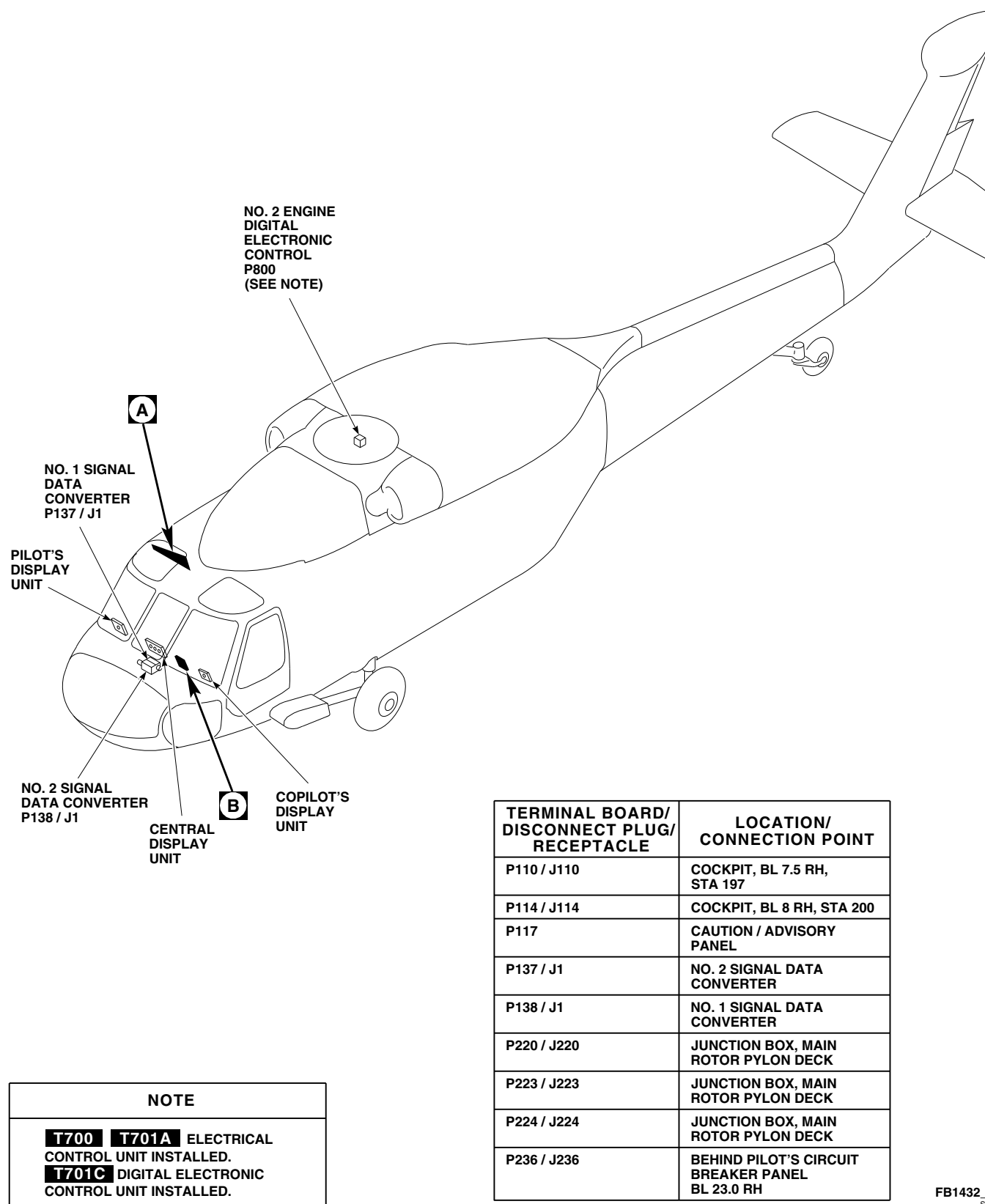
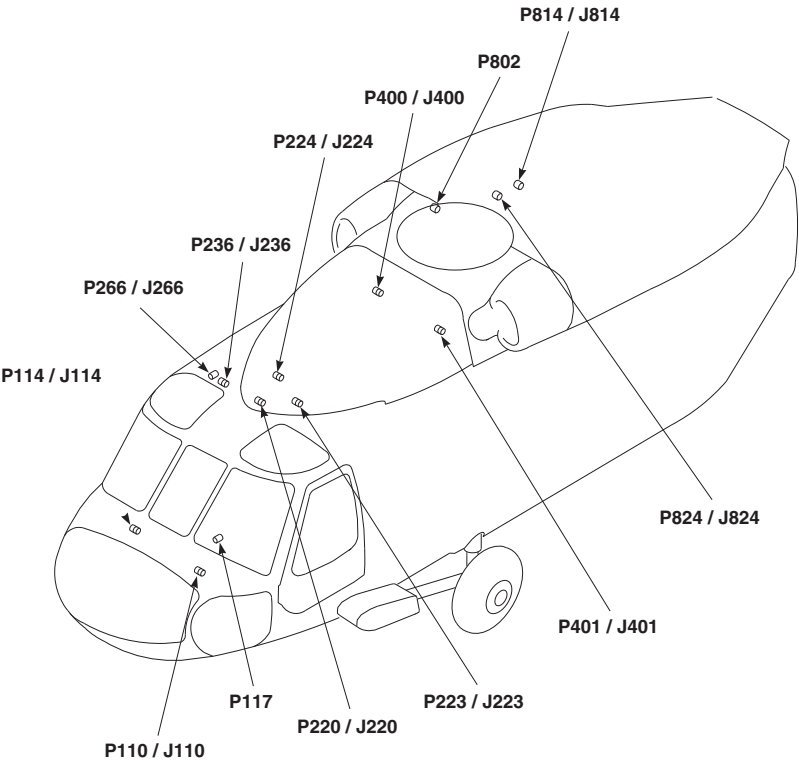


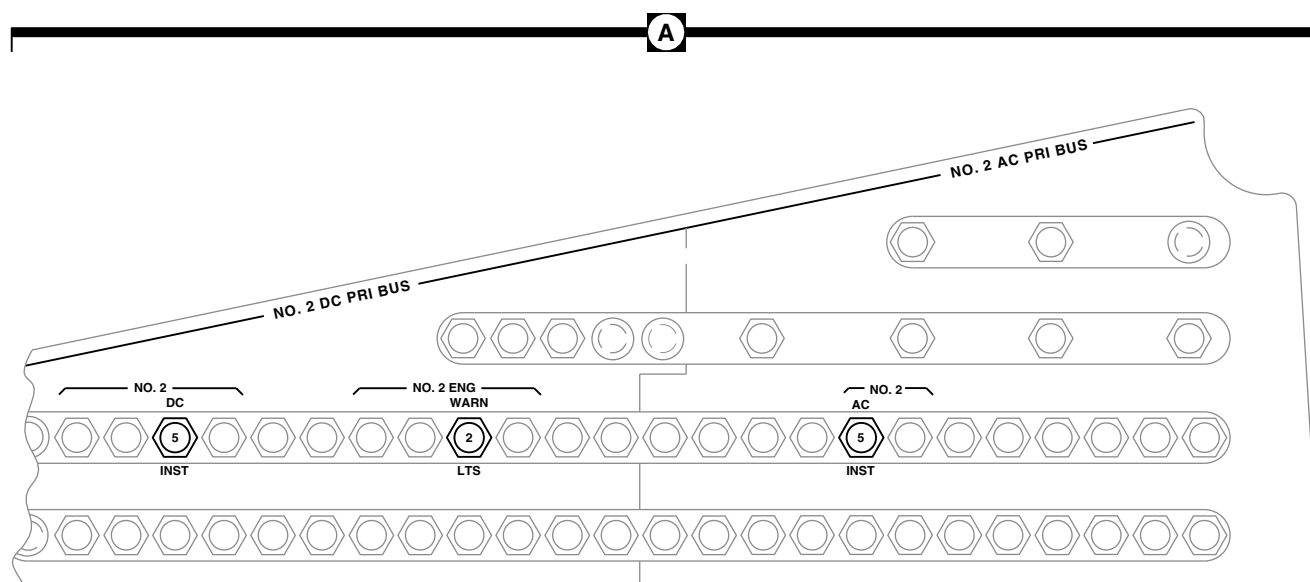
FIGURE 4-1-5-2. NO. 2 ENGINE INDICATING SYSTEM LOCATION DIAGRAM (SHEET 1 OF 4)



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25.0 RH
P400 / J400	MAIN ROTOR PYLON DECK BL 24 RH, STA 318
P401 / J401	MAIN ROTOR PYLON DECK BL 24 RH, STA 318
P800	NO. 2 ENGINE DIGITAL ELECTRONIC CONTROL (SEE NOTE)
P802	NO. 2 ENGINE INTERFACE DISCONNECT
P814 / J814	FIREWALL DISCONNECT BL 14 RH, STA 373
P824 / J824	FIREWALL DISCONNECT BL 14 RH, STA 356.25

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FIGURE 4-1-5-2. NO. 2 ENGINE INDICATING SYSTEM LOCATION DIAGRAM
 (SHEET 2 OF 4)



PILOT'S CIRCUIT BREAKER PANEL

FB1432_3
SAFIGURE 4-1-5-2. NO. 2 ENGINE INDICATING SYSTEM LOCATION DIAGRAM
(SHEET 3 OF 4)

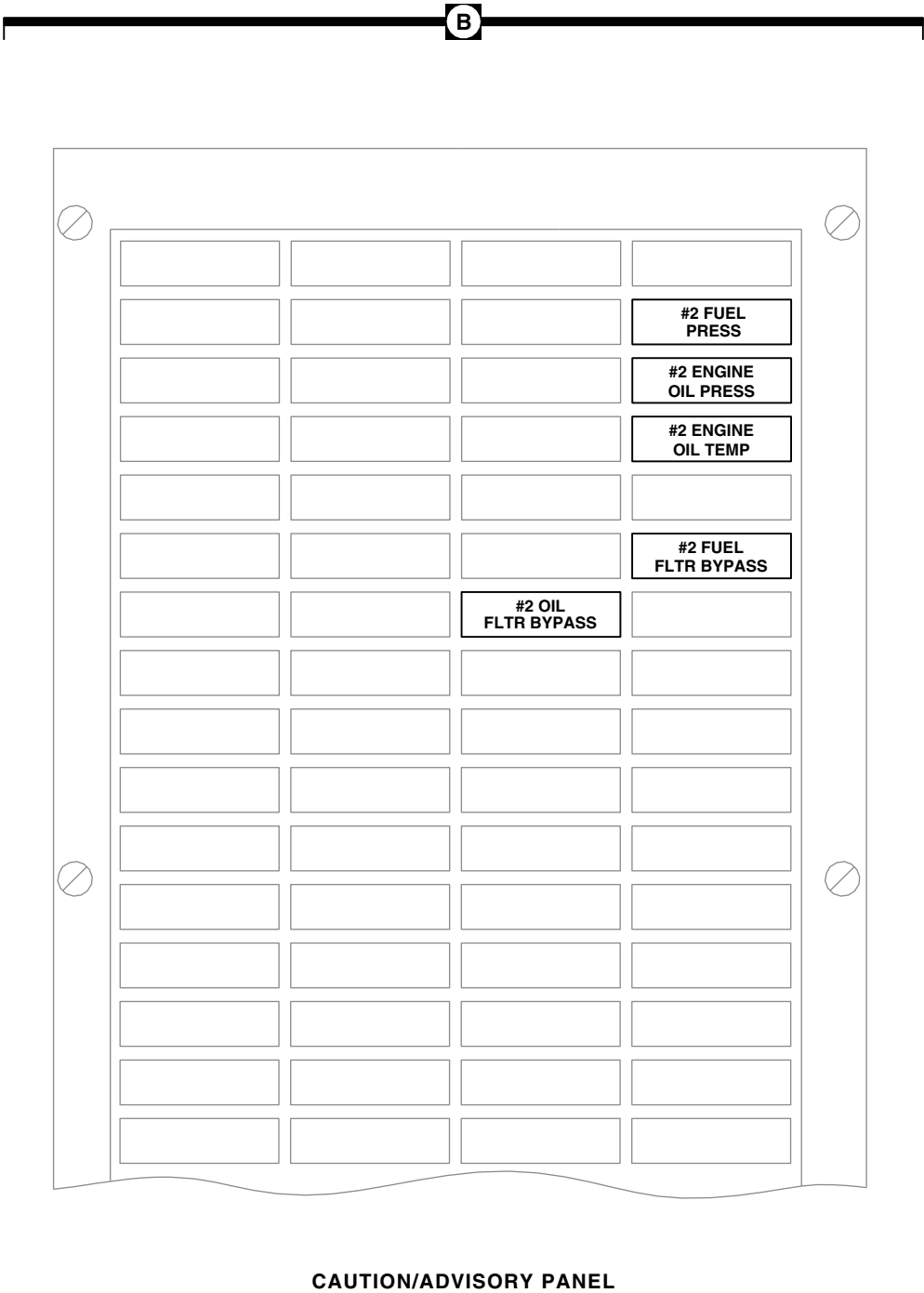
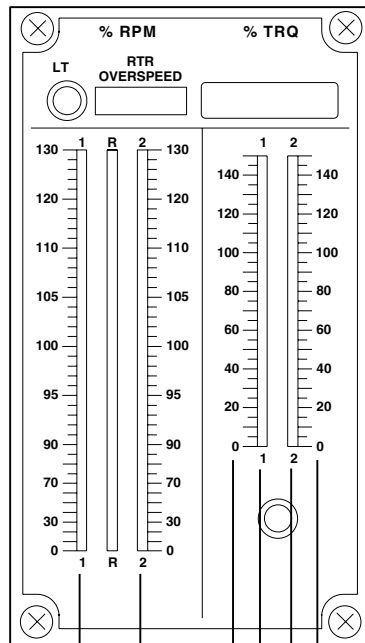


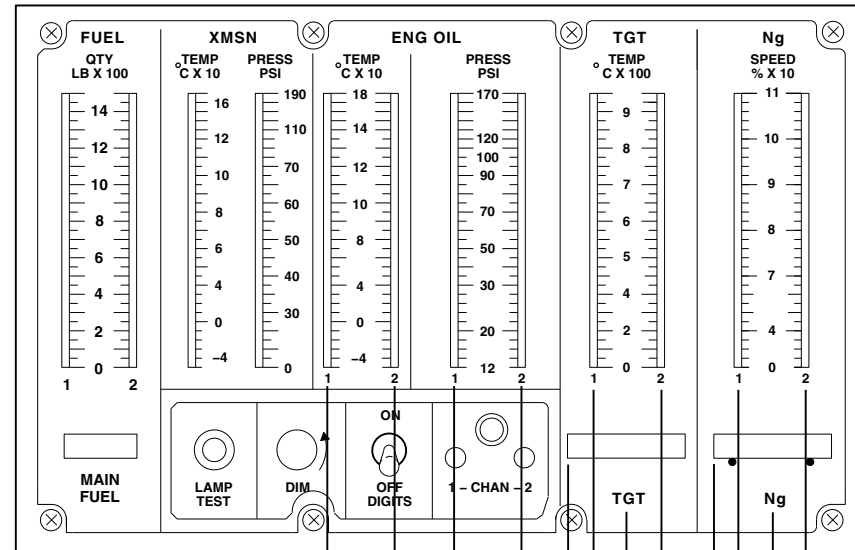
FIGURE 4-1-5-2. NO. 2 ENGINE INDICATING SYSTEM LOCATION DIAGRAM
 (SHEET 4 OF 4)

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COPILLOT'S DISPLAY UNIT



CENTRAL DISPLAY UNIT

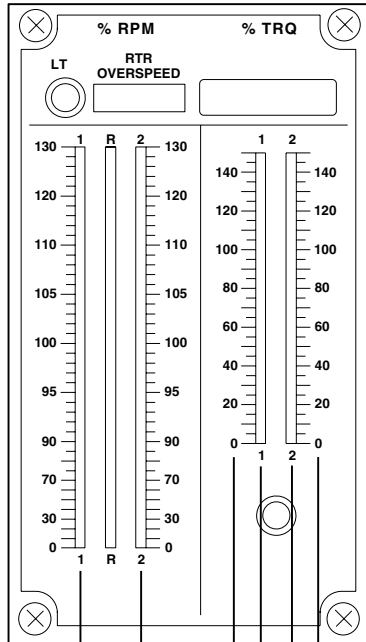


(SEE DETAIL A)

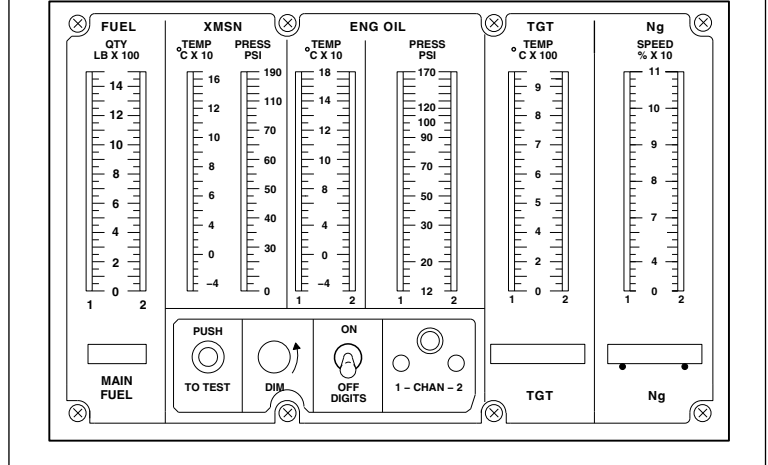
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FIGURE 4-1-5-3. ENGINE INDICATING SYSTEMS BLOCK DIAGRAM (SHEET 1 OF 2)

PILOT'S DISPLAY UNIT



CENTRAL DISPLAY UNIT



DETAIL A
(SEE NOTE 2)

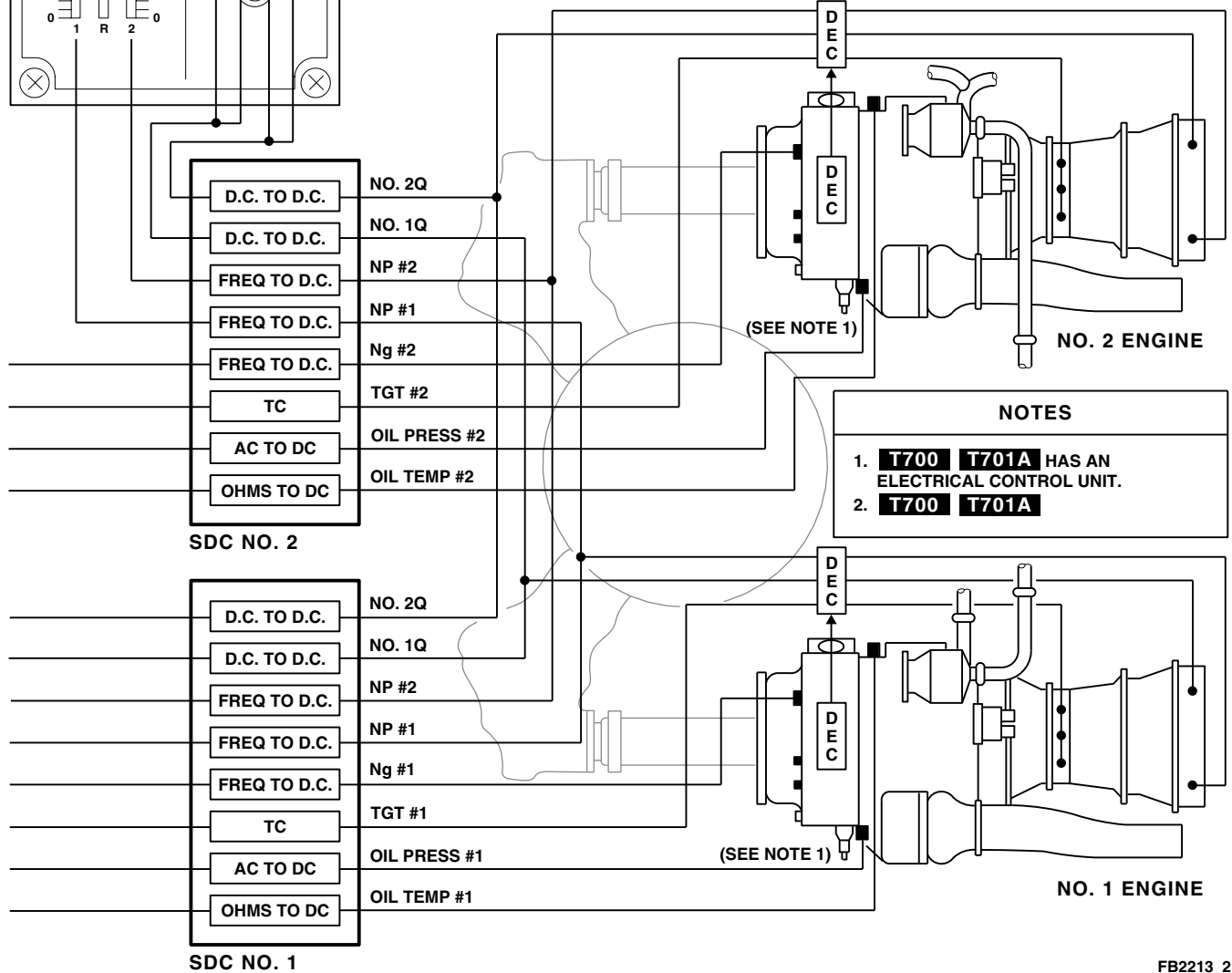


FIGURE 4-1-5-3. ENGINE INDICATING SYSTEMS BLOCK DIAGRAM (SHEET 2 OF 2)

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4-1-6 ENGINE WARNING SYSTEMS DESCRIPTION AND DATA.

4-1-6.1 Engine Warning Systems.

The engine warning systems consist of the fuel pressure, fuel filter bypass, oil pressure, oil temperature, oil filter bypass, chip detection, and starter warning system (Figures 4-1-6-1, 4-1-6-2, and 4-1-6-3).

4-1-6.2 Fuel Pressure Warning System.

The engine fuel pressure warning system for each engine consists of a pressure switch that turns on the caution lights at a decreasing pressure of between 8 and 9 psi. The pressure switch is connected to the fuel line between the engine-driven fuel boost pump and the hydromechanical unit (HMU) high-pressure pump. This visually indicates a possible malfunction in the engine-driven fuel boost pump or an air leak in the fuel system. Normal discharge pressure of fuel boost pump is not less than 30 psi at full power. The No. 1 and No. 2 engine fuel pressure caution capsules, marked #1 FUEL PRESS and #2 FUEL PRESS, on the caution/advisory panel, light when fuel pressure drops below 8 to 9 psi. Power for the No. 1 engine fuel pressure warning system is supplied by the No. 1 primary dc bus through the NO. 1 ENG WARN LTS circuit breaker on the copilot's circuit breaker panel. Power for the No. 2 engine fuel pressure warning system is supplied by the No. 2 primary dc bus through the NO. 2 ENG WARN LTS circuit breaker on the pilot's circuit breaker panel.

4-1-6.3 Fuel Filter Bypass Warning System.

The engine fuel filter bypass warning system for each engine consists of an electrical switch, impending bypass popout button, and caution lights. The button is mounted on the fuel filter assembly at the front left side of the engine accessory gear box. It is displayed when fuel pressure at the filter reaches 8 - 10 psi. The electrical switch, an integral part of the fuel filter assembly, visually indicates in the cockpit that fuel is bypassing the fuel filter. The caution capsules, marked #1 FUEL FLTR BYPASS and #2 FUEL FLTR BYPASS on the caution/advisory panel, go on when the fuel filter bypass valve opens at not less than 8 psi. Power for the No. 1 engine fuel filter bypass warn-

ing system is supplied by the No. 1 primary dc bus through the NO. 1 ENG WARN LTS circuit breaker on the copilot's circuit breaker panel. Power for the No. 2 engine fuel filter bypass warning system is supplied by the No. 2 primary dc bus through the NO. 2 ENG WARN LTS circuit breaker on the pilot's circuit breaker panel.

4-1-6.4 Oil Pressure Warning System.

The engine oil pressure warning system for each engine consists of a pressure switch and caution lights. The switch, connected to the engine main frame, lets you know when oil pressure is low. The caution capsules, marked #1 ENGINE OIL PRESS and #2 ENGINE OIL PRESS on the caution/advisory panel, light when oil pressure is below 25 psi. Power for the No. 1 engine oil pressure warning system is supplied by the No. 1 primary dc bus through the NO. 1 DC INST circuit breaker on copilot's circuit breaker panel. Power for the No. 2 engine oil pressure warning system is supplied by the No. 2 primary dc bus through the NO. 2 DC INST circuit breaker on the pilot's circuit breaker panel.

4-1-6.5 Oil Temperature Warning System.

The engine oil temperature warning system for each engine consists of a temperature sensing bulb and caution capsules. The bulb connected to the engine mainframe indicates a high oil temperature condition exists. The caution capsules, marked #1 ENGINE OIL TEMP and #2 ENGINE OIL TEMP on the caution/advisory panel, light when engine oil temperature exceeds 150°C. Power for the No. 1 engine oil temperature warning system is supplied by the No. 1 primary dc bus through the NO. 1 DC INST circuit breaker on the copilot's circuit breaker panel. Power for the No. 2 engine oil temperature warning system is supplied by the No. 2 primary dc bus through the NO. 2 DC INST circuit breaker on the pilot's circuit breaker panel.

4-1-6.6 Oil Filter Bypass Warning System.

The engine oil filter bypass warning system for each engine consists of an electrical switch, impending bypass popout button, and caution

lights. The bypass button, on the oil filter assembly, is displayed when oil pressure buildup at the filter is between 44 to 60 psi. The oil filter is mounted on the top front center of the engine accessory gearbox. The electrical switch, mounted on the engine accessory gear box, turns on the caution capsules, marked #1 OIL FLTR BYPASS and #2 OIL FLTR BYPASS on the caution/advisory panel, when oil pressure at the filter builds up to 60 or 80 psi. This indicates that oil will soon bypass the filter. Power for the No. 1 engine oil filter bypass warning is supplied by the No. 1 primary dc bus through the NO. 1 ENG WARN LTS circuit breaker on the copilot's circuit breaker panel. Power for the No. 2 engine oil filter bypass warning is supplied by the No. 2 primary dc bus through the NO. 2 ENG WARN LTS circuit breaker on the pilot's circuit breaker panel.

4-1-6.7 Chip Detector Warning System.

The engine chip detector warning system consists of a magnetic chip detector and caution capsules. The chip detector, mounted on the engine accessory gear box, tells when there are metal chips in the engine oil system. The caution capsules, marked CHIP #1 ENGINE and CHIP #2 ENGINE on the caution/advisory panel, light when metal chips collect on the chip detector. Power for the No. 1 engine chip detector warning system is supplied by the No. 1 primary dc bus through the NO. 1 ENG WARN LTS circuit breaker on the copilot's circuit breaker panel. Power for the No. 2 engine chip detector warning system is supplied by the No. 2 primary dc bus through the NO. 2 ENG WARN LTS circuit breaker on the pilot's circuit breaker panel.

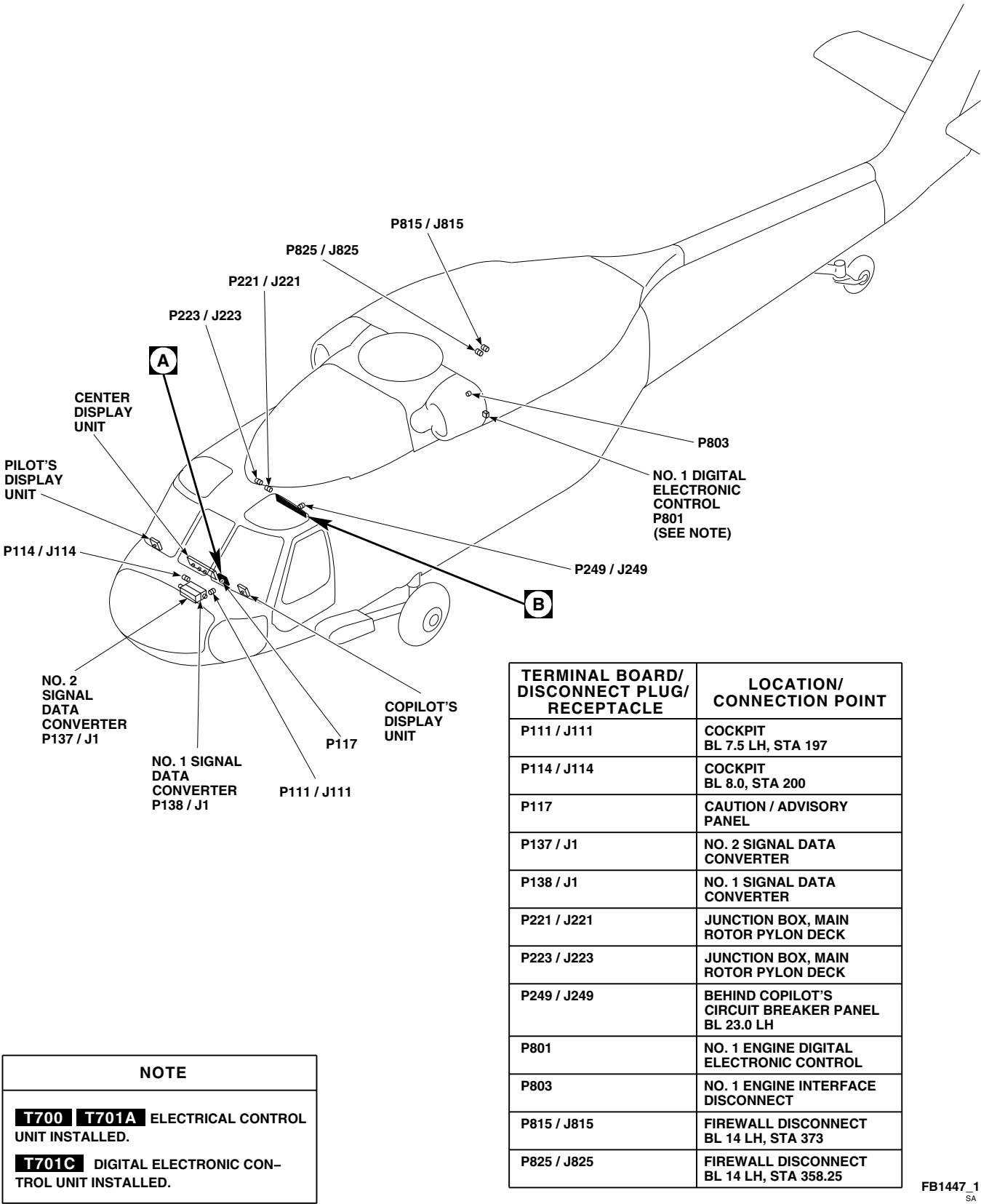


FIGURE 4-1-6-1. ENGINE WARNING SYSTEMS LOCATION DIAGRAM (SHEET 1 OF 9)

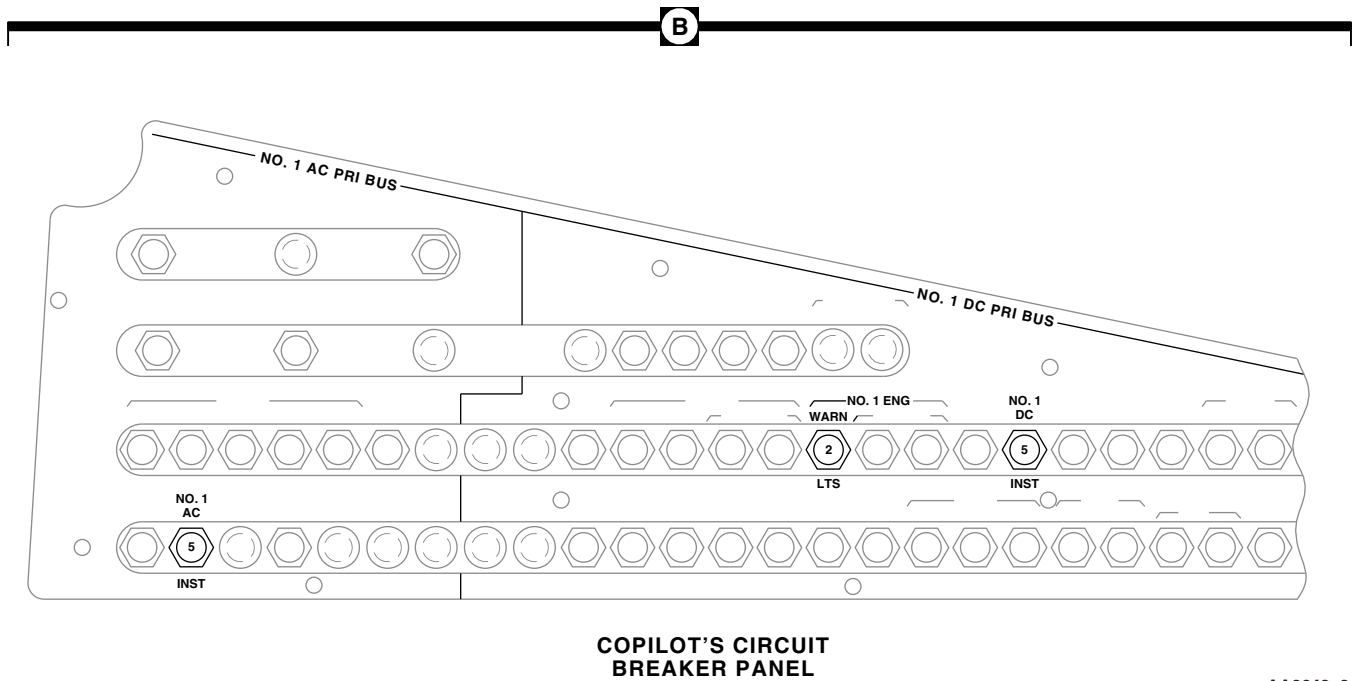
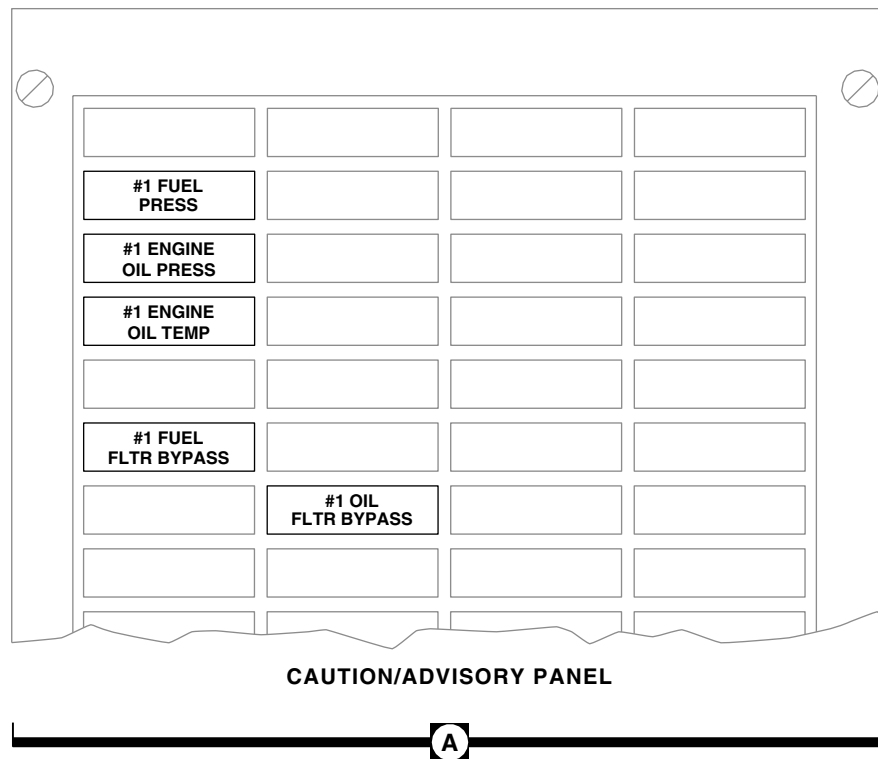


FIGURE 4-1-6-1. ENGINE WARNING SYSTEMS LOCATION DIAGRAM (SHEET 2 OF 9)

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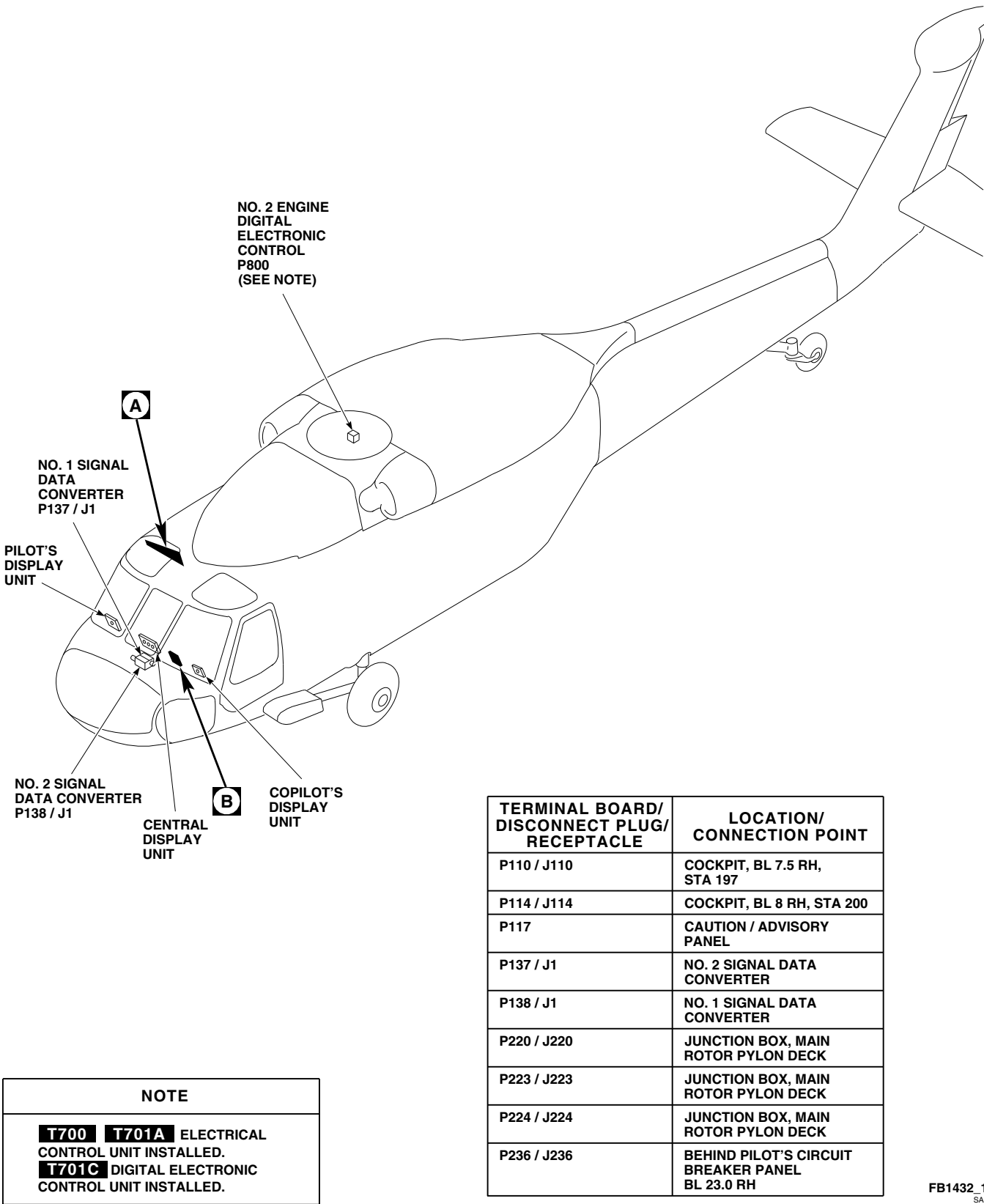
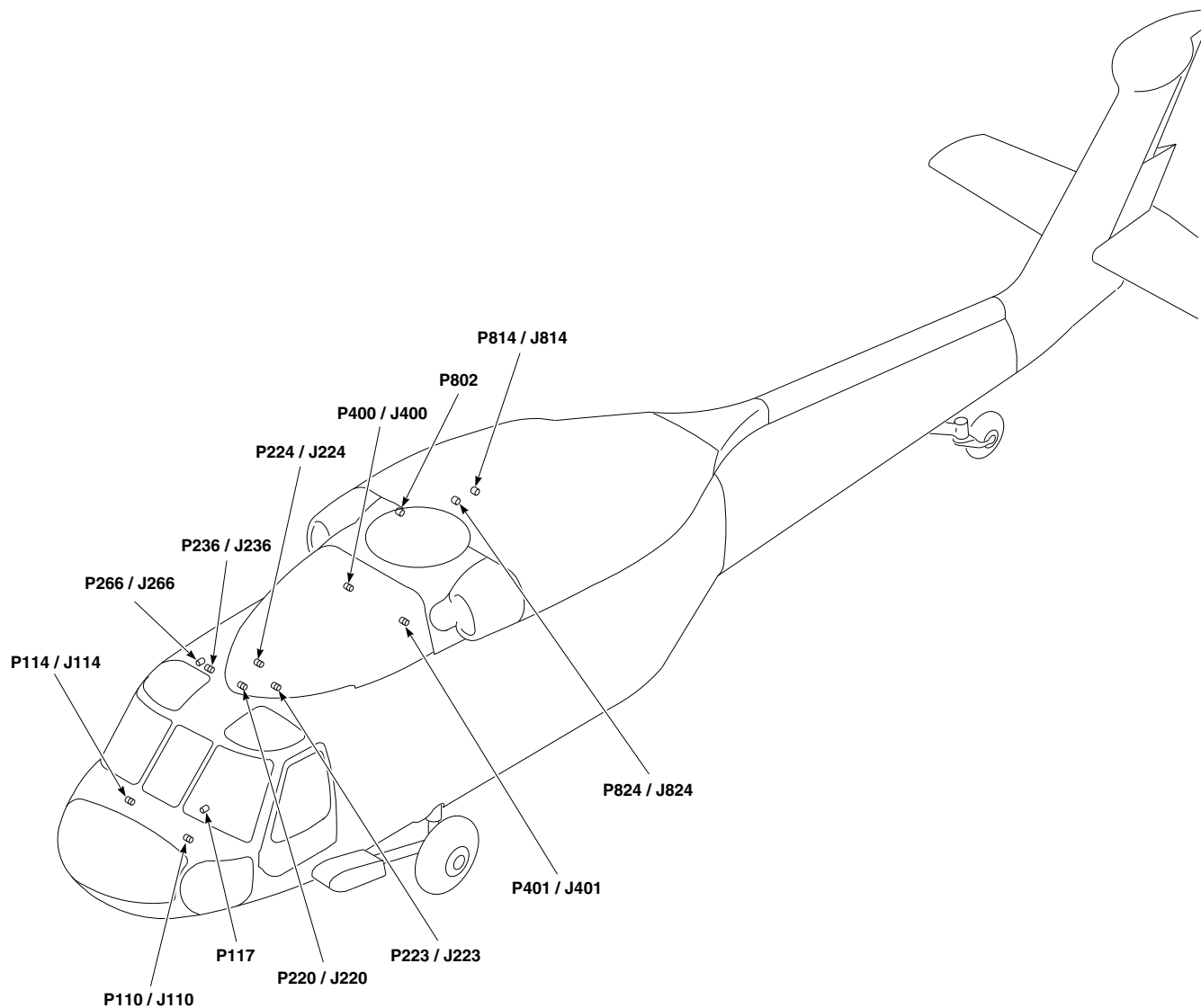


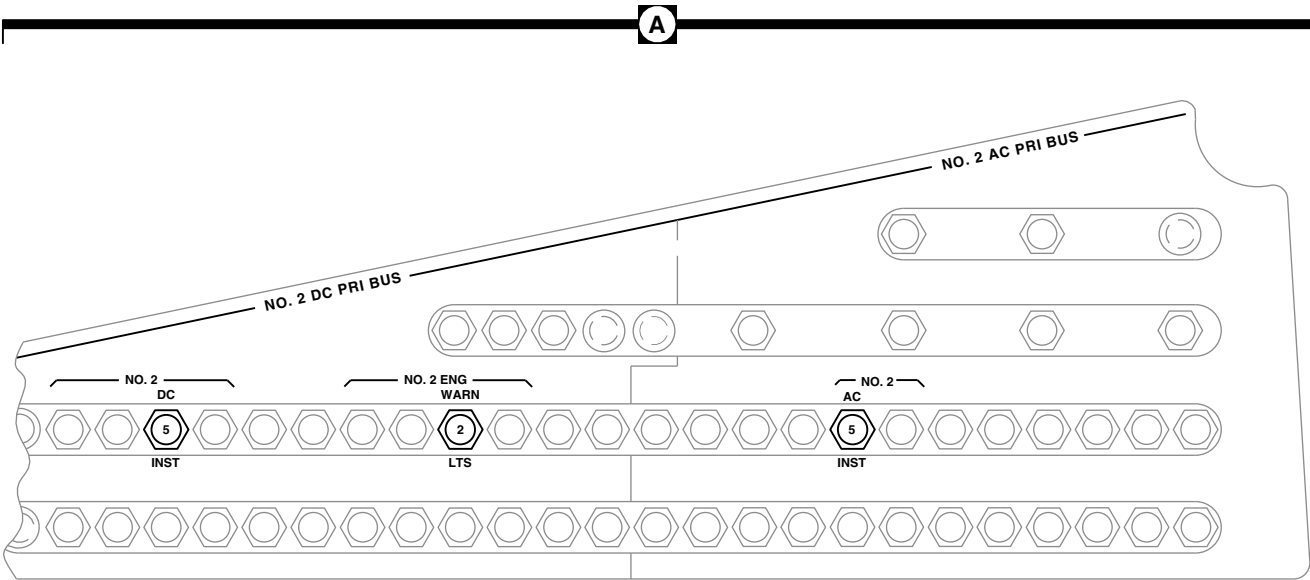
FIGURE 4-1-6-1. ENGINE WARNING SYSTEMS LOCATION DIAGRAM (SHEET 3 OF 9)



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25.0 RH
P400 / J400	MAIN ROTOR PYLON DECK BL 24 RH, STA 318
P401 / J401	MAIN ROTOR PYLON DECK BL 24 RH, STA 318
P800	NO. 2 ENGINE DIGITAL ELECTRONIC CONTROL
P802	NO. 2 ENGINE INTERFACE DISCONNECT
P814 / J814	FIREWALL DISCONNECT BL 14 RH, STA 373
P824 / J824	FIREWALL DISCONNECT BL 14 RH, STA 356.25

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FIGURE 4-1-6-1. ENGINE WARNING SYSTEMS LOCATION DIAGRAM (SHEET 4 OF 9)

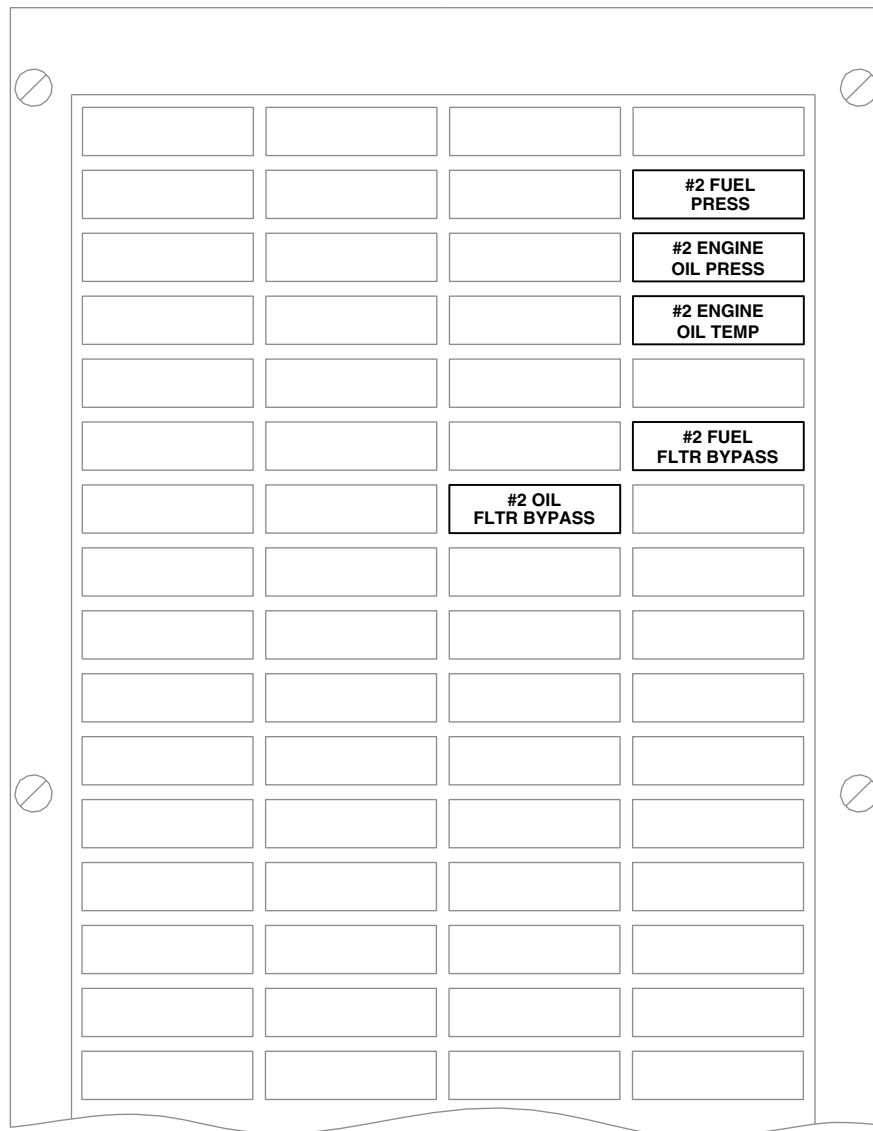


PILOT'S CIRCUIT BREAKER PANEL

FB1432_3
SA

FIGURE 4-1-6-1. ENGINE WARNING SYSTEMS LOCATION DIAGRAM (SHEET 5 OF 9)

B

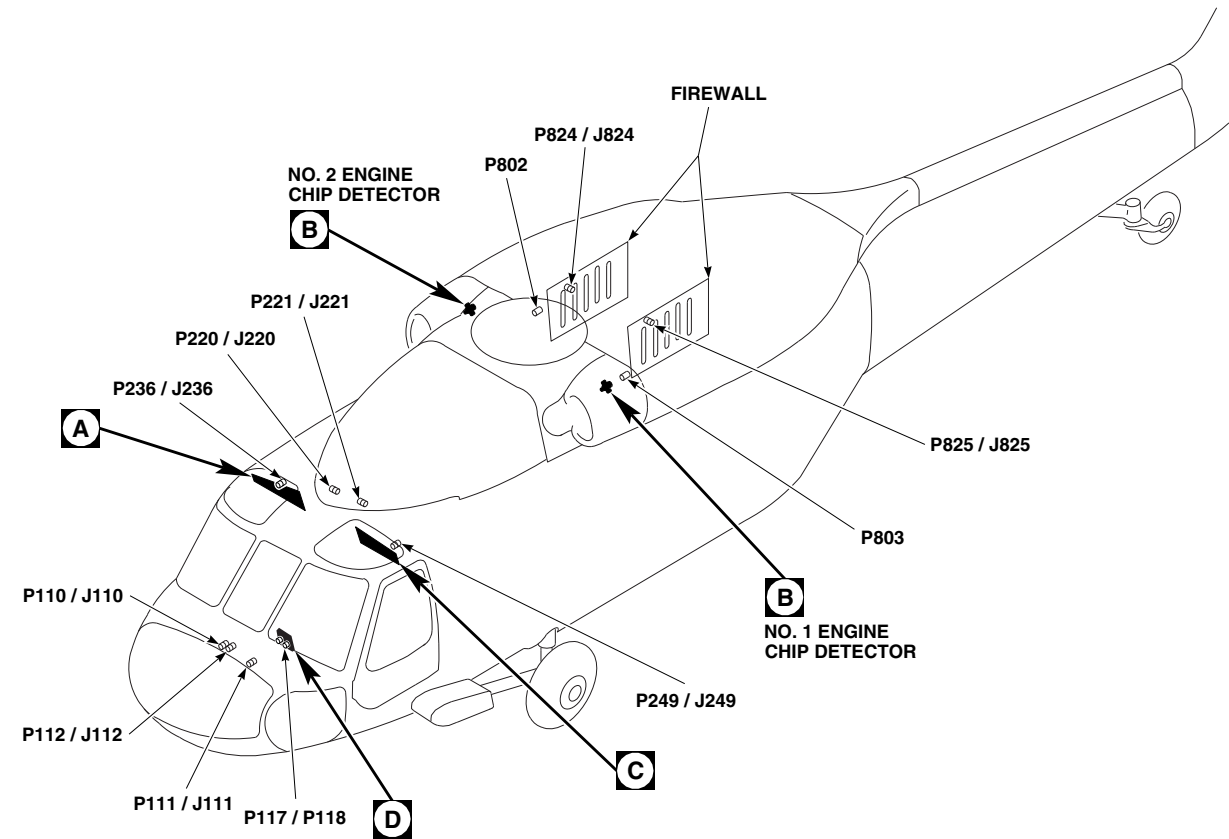


CAUTION/ADVISORY PANEL

AA2651_4
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FIGURE 4-1-6-1. ENGINE WARNING SYSTEMS LOCATION DIAGRAM (SHEET 6 OF 9)

4-1-6-8

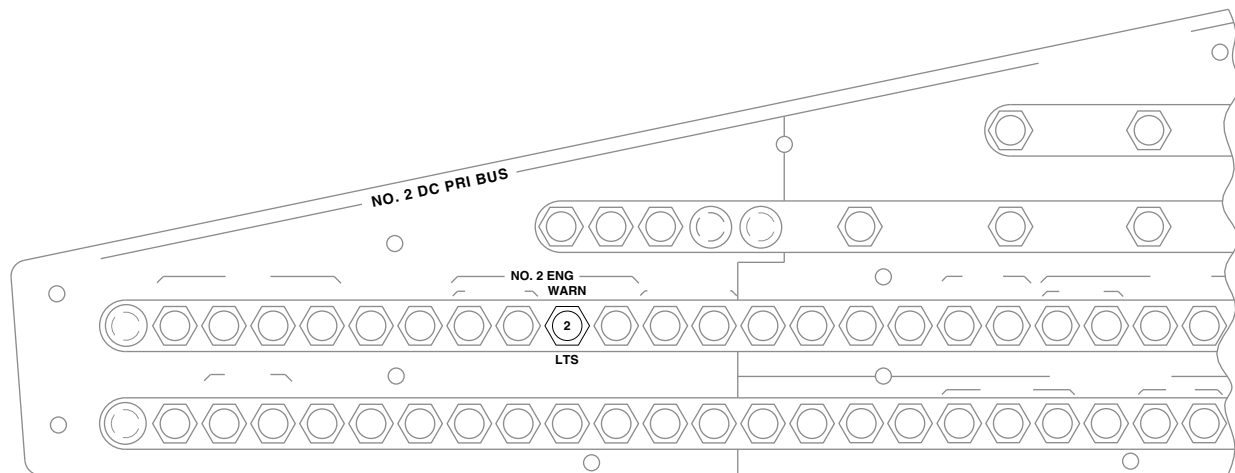


TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P110 / J110	COCKPIT, BL 7.5 RH, STA 197
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P112 / J112	COCKPIT, BL 6 RH, STA 197
P117	CAUTION / ADVISORY PANEL
P118	CAUTION / ADVISORY PANEL
P220 / J220	JUNCTION BOX, MAIN ROTOR PYLON DECK
P221 / J221	JUNCTION BOX, MAIN ROTOR PYLON DECK

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 23 RH
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 11 LH
P802	NO. 2 ENGINE INTERFACE DISCONNECT
P803	NO. 1 ENGINE INTERFACE DISCONNECT
P824 / J824	FIREWALL DISCONNECT BL 14 RH, STA 338.25
P825 / J825	FIREWALL DISCONNECT BL 14 LH, STA 358.25

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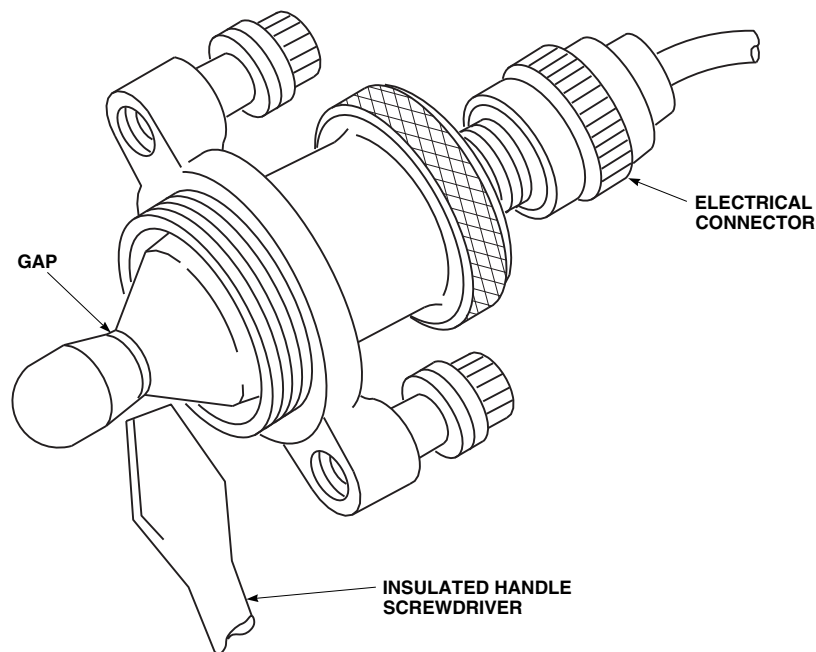
FIGURE 4-1-6-1. ENGINE WARNING SYSTEMS LOCATION DIAGRAM (SHEET 7 OF 9)



PILOT'S CIRCUIT BREAKER PANEL

A

B



ENGINE CHIP DETECTOR

AA2660_2
SA

FIGURE 4-1-6-1. ENGINE WARNING SYSTEMS LOCATION DIAGRAM (SHEET 8 OF 9)

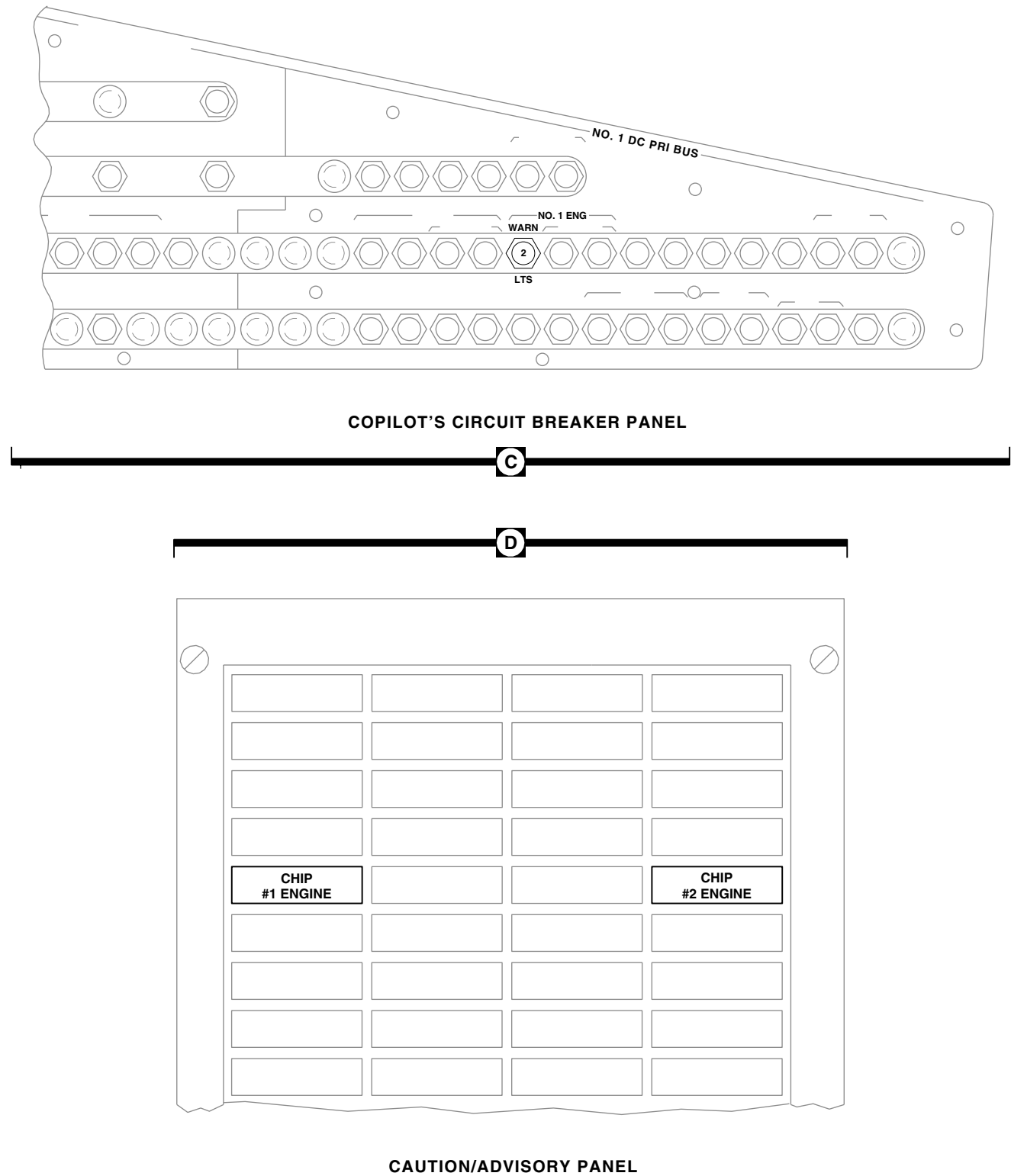


FIGURE 4-1-6-1. ENGINE WARNING SYSTEMS LOCATION DIAGRAM (SHEET 9 OF 9)

AA2660_3
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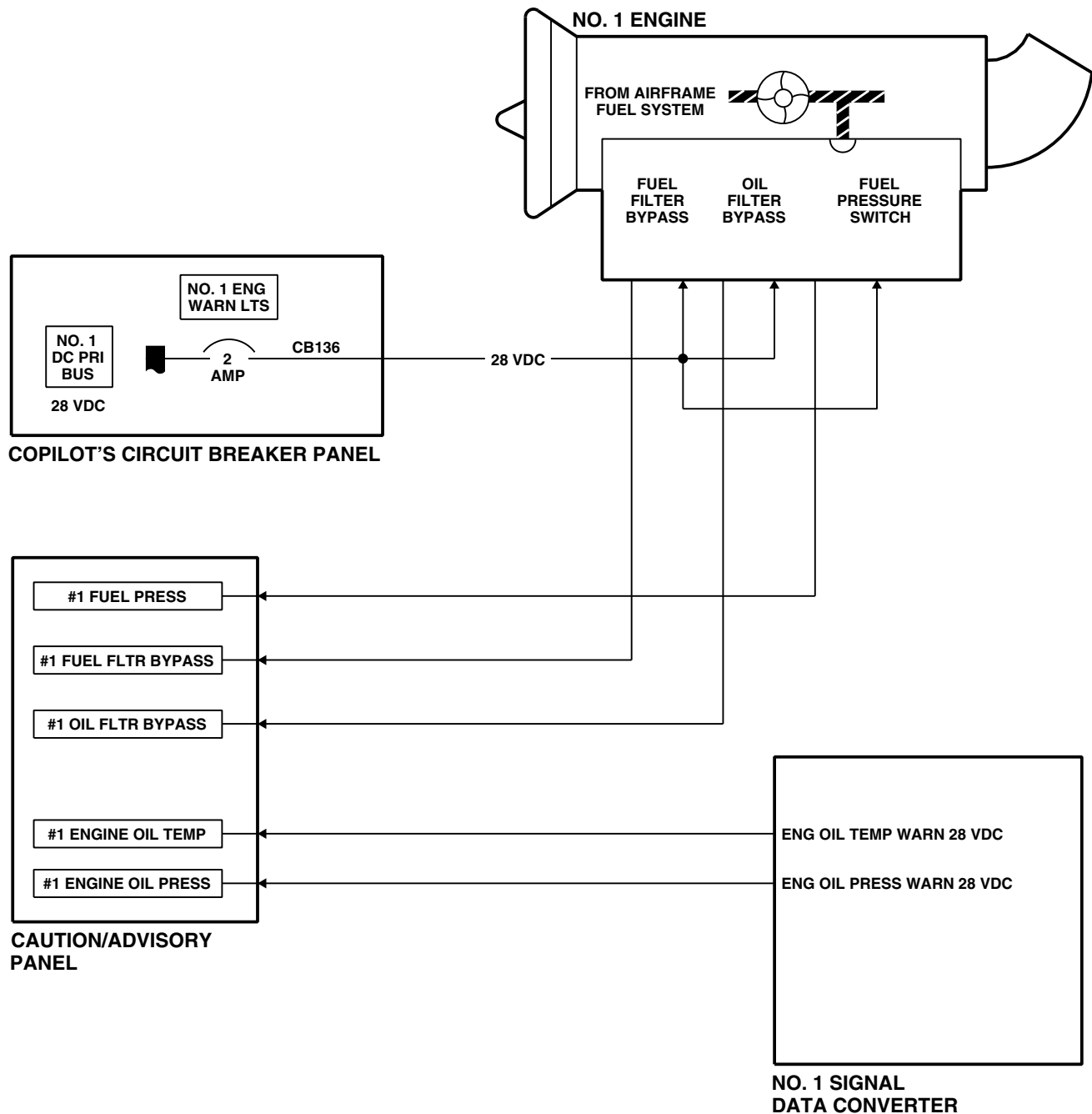
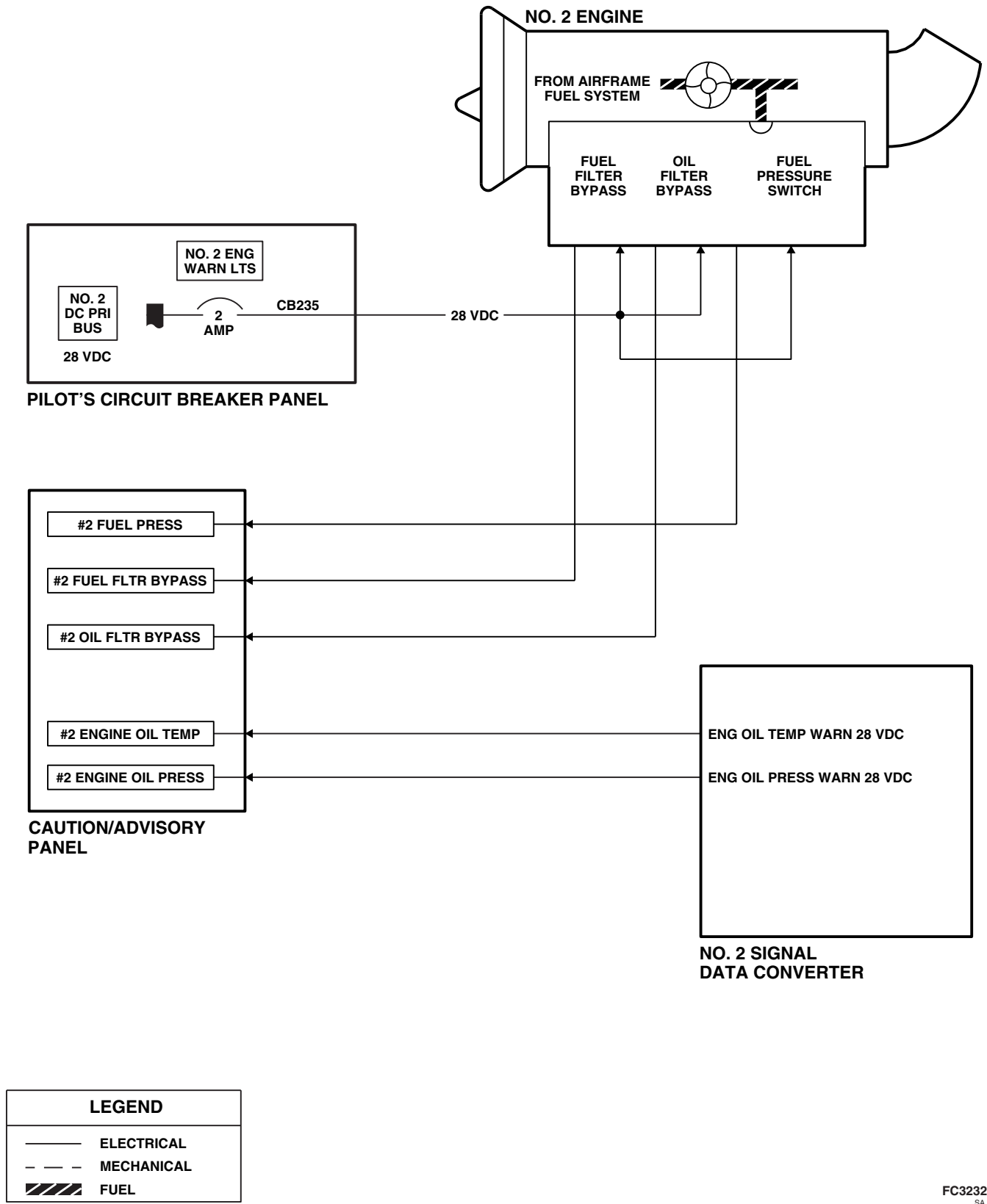


FIGURE 4-1-6-2. NO. 1 ENGINE WARNING SYSTEMS BLOCK DIAGRAM

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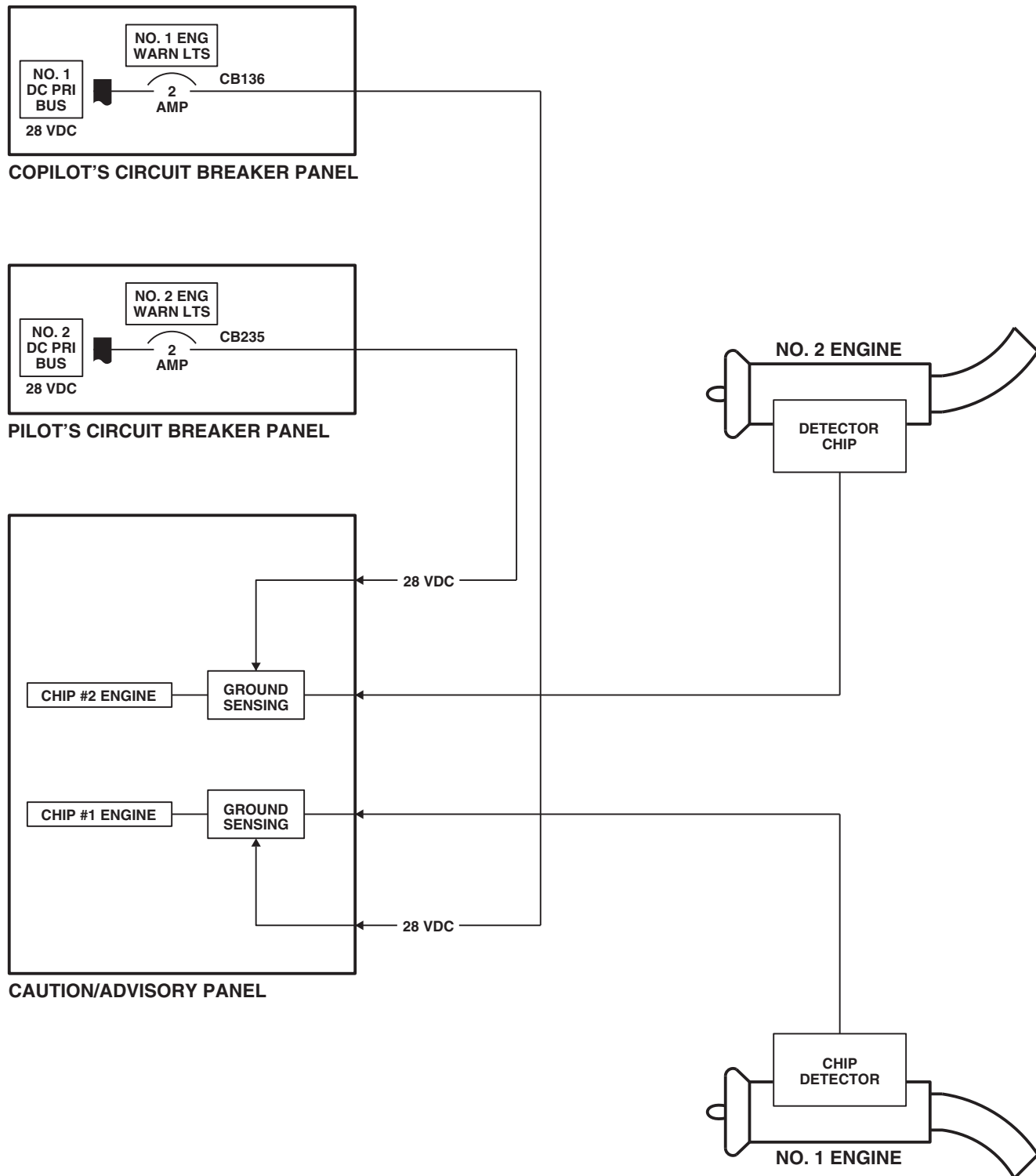
FC3227
SA

FIGURE 4-1-6-4. ENGINE CHIP DETECTOR WARNING SYSTEM BLOCK DIAGRAM

4-1-7 HOVER INFRARED SUPPRESSION SYSTEM (HIRSS) DESCRIPTION AND DATA.

4-1-7.1 Hover Infrared Suppression System (HIRSS).

The hover infrared suppressor system (HIRSS) provides improved helicopter survivability from heat seeking missiles during hover and cruise flight. The HIRSS kit contains a three-stage core and inner baffle which reduces exhaust gas radiation and prevents line-of-sight viewing of hot engine surfaces. The HIRSS channels hot exhaust gasses through its three-stage core and inner baffle by inducing the flow of cooling air from the engine bay and inlet scoop. The three-stage core and inner baffle cold surfaces are coated with low-reflectance material. Additional cooling is provided by ducting

hot exhaust gasses outboard and then forcing the exhaust gasses downward with downwash from the main rotor. An exhaust extender is attached to the HIRSS module to reduce fuselage surface temperature in the transition section and to allow operation of the aircraft with cargo doors open. Installation of each HIRSS module requires removal of standard engine exhaust modules and cargo door track fairings. HIRSS modules are installed on the basic airframe, with two additional mounts, and one angle. The rear fairings are installed using existing mounting points and hardware. While operating in a nonhostile environment, the inner baffle can be removed to enhance engine performance.

4-1-8 ENGINE ANTI-ICE SYSTEM DESCRIPTION AND DATA.

4-1-8.1 Engine Anti-Ice System.

The engine anti-ice system prevents ice buildup on the No. 1 and No. 2 engines and engine air inlets. When the system is on, hot bleed-air from the engine compressor fifth stage heats the swirl vanes, scroll vanes, and inlet guide vanes. The system consists of an engine anti-ice start bleed valve, an engine inlet anti-ice valve, and engine inlet temperature switch for each engine. The valves are controlled by switches on the upper console. The system may be operated automatically or manually (Figure 4-1-8-1). Power is supplied by the No. 1 dc primary bus and No. 2 dc primary bus on the copilot's and pilot's circuit breaker panels. The NO. 1 ENG ANTI-ICE WARN circuit breaker, located on the copilot's circuit breaker panel No. 1 dc primary bus, provides power to #1 ENG ANTI-ICE ON and #1 ENG INLET ANTI-ICE ON caution/advisory panel advisory capsules. The NO. 2 ENG ANTI-ICE WARN circuit breaker, located on the pilot's circuit breaker panel No. 2 dc primary bus, provides power to #2 ENG INLET ANTI-ICE ON and #2 ENG ANTI-ICE ON caution/advisory panel advisory capsules. The NO. 1 ENG ANTI-ICE circuit breaker, located on the copilot's circuit breaker panel No. 1 dc primary bus, and NO. 2 ENG ANTI-ICE circuit breaker located on the pilot's circuit breaker panel No. 2 dc primary bus, provide power to engine and engine inlet anti-ice control circuitry and valve solenoids. Caution/advisory panel #1 ENG ANTI-ICE ON and #1 ENG INLET ANTI-ICE ON capsules power is supplied by NO. 1 ENG ANTI-ICE WARN circuit breaker. When the No. 1 engine anti-ice valve is deenergized, a sensing switch in the valve closes, allowing 28 vdc to turn on #1 ENG ANTI-ICE ON advisory capsule. If No. 1 engine inlet temperature rises at or above 93°C (200°F), the inlet temperature switch in the inlet closes, and #1 ENG INLET ANTI-ICE ON advisory capsule goes on.

When the No. 1 engine start button is pressed, 28 vdc energizes K45. With K45 energized, 28 vdc from NO. 1 ENG ANTI-ICE circuit breaker is applied to the No. 1 engine inlet anti-ice valve. This keeps the valve closed, preventing air inlet anti-

icing operation during engine start. During No. 1 engine start, the engine start system will hold relay K45 energized until it is safe to operate No. 1 engine inlet anti-icing valve.

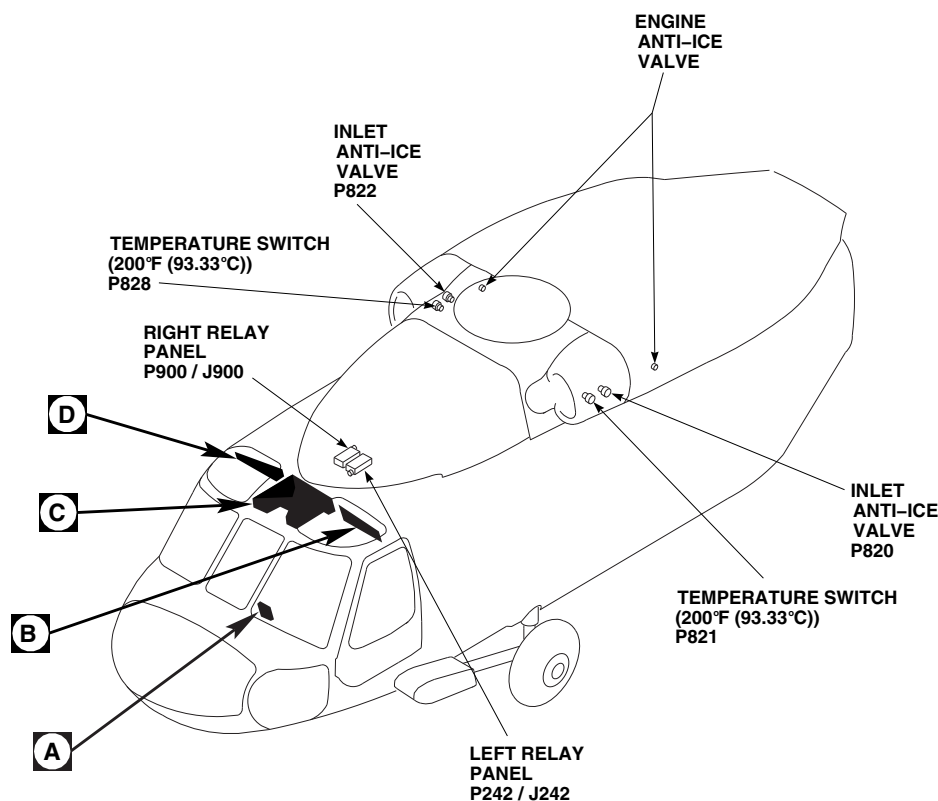
4-1-8.1.1 Engine Anti-Icing Valve.

The engine anti-icing valve is a solenoid-actuated, electromechanical, modulating bleed/start valve. Modulation (opening, closing) is controlled by an internal cam plate, which is connected by a push-rod to a crankshaft. The crankshaft, which is controlled by a servo in the engine hydromechanical unit (HMU), positions the cam plate to hold the valve open. If the ENG ANTI-ICE NO. 1 switch is OFF during engine start, the valve will be held open by the engine HMU and the engine anti-icing system will be on to provide startup heat to the engine until the engine power has increased to a predetermined level. In this case, power from the NO. 1 ENG START circuit breaker to the engine start system will close the valve as engine power is increased, shutting off the engine anti-ice system. If the ENG ANTI-ICE NO. 1 switch is ON, the valve will stay open. This allows constant bleed-air flow for engine anti-icing.

4-1-8.1.2 Engine Inlet Anti-Icing Valve.

The engine inlet anti-icing valve is a modulating solenoid-actuated, butterfly valve and actuator type. A reference regulator regulates inlet flow pressure to a constant value. An ambient-temperature biased servo provides airflow regulation as a function of ambient air temperature. With the ENG ANTI-ICE NO. 1 switch OFF, the solenoid is energized and the butterfly valve and actuator provide shutoff. With the ENG ANTI-ICE NO. 1 switch ON, airflow is regulated by actuator signals. The engine inlet anti-icing modulating valve is scheduled open based on outside air temperature (OAT). When the ENGINE ANTI-ICE switch is ON and the OAT is:

Less than 4°C (39°F), the valve will be open.
Between 4°C (39°F) and 13°C (55°F), the valve may or may not be open.
Above 13°C (55°F), the valve will be closed and the INLET ANTI-ICE ON advisory capsule will not go on.

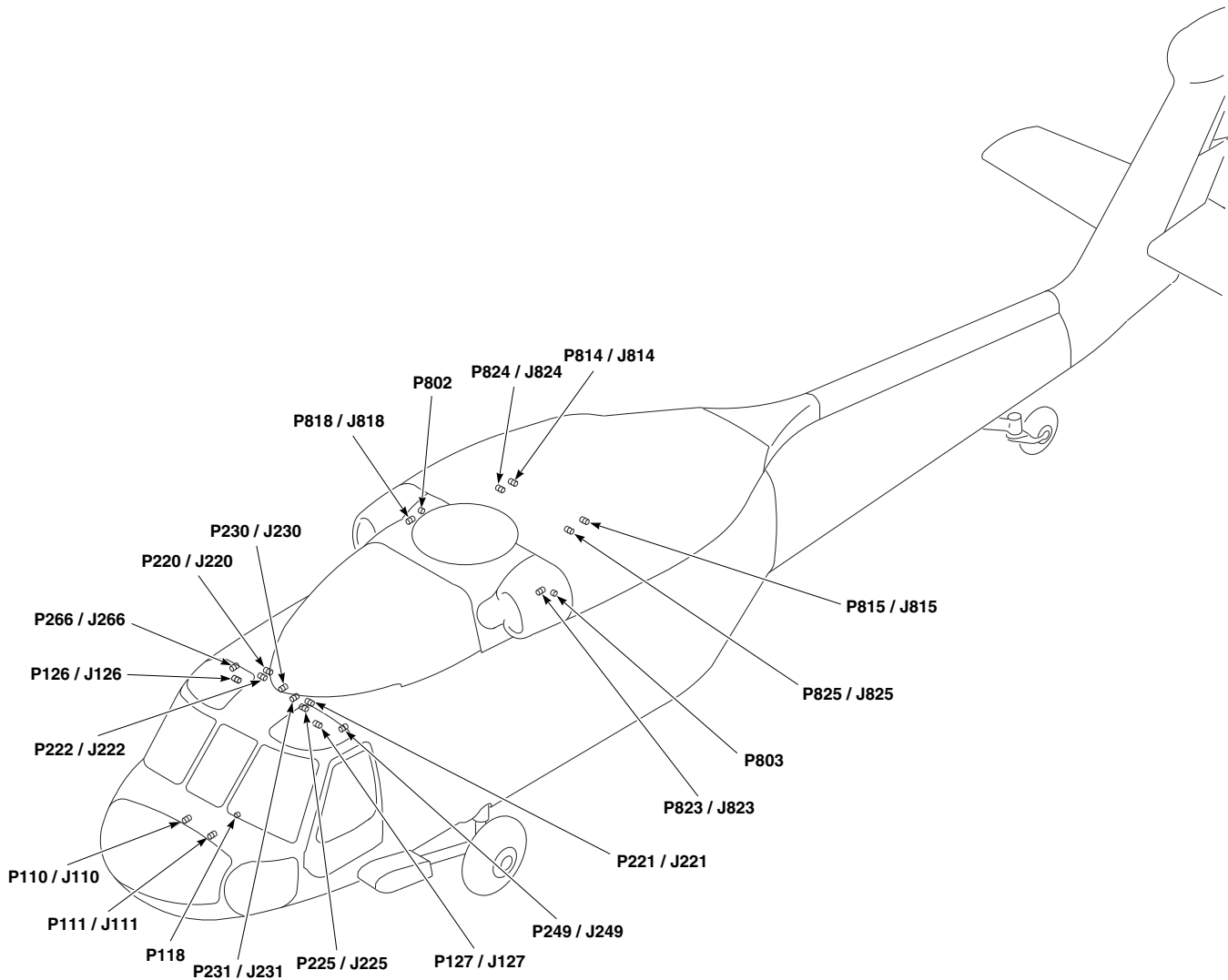


TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P110 / J110	COCKPIT, BL 7.5 RH, STA 197
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P118	CAUTION / ADVISORY PANEL
P126 / J126	COCKPIT CEILING BL 28 RH, STA 243
P127 / J127	COCKPIT CEILING BL 28 LH, STA 243
P220 / J220	JUNCTION BOX, MAIN ROTOR PYLON DECK
P221 / J221	JUNCTION BOX, MAIN ROTOR PYLON DECK

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P222 / J222	JUNCTION BOX, MAIN ROTOR PYLON DECK
P225 / J225	JUNCTION BOX, MAIN ROTOR PYLON DECK
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P231 / J231	CABIN CEILING, BL 9 LH, STA 247
P242 / J242	LH RELAY PANEL
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23.0 LH
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25.0 RH

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SA

FIGURE 4-1-8-1. ENGINE ANTI-ICE SYSTEM LOCATION DIAGRAM (SHEET 1 OF 5)

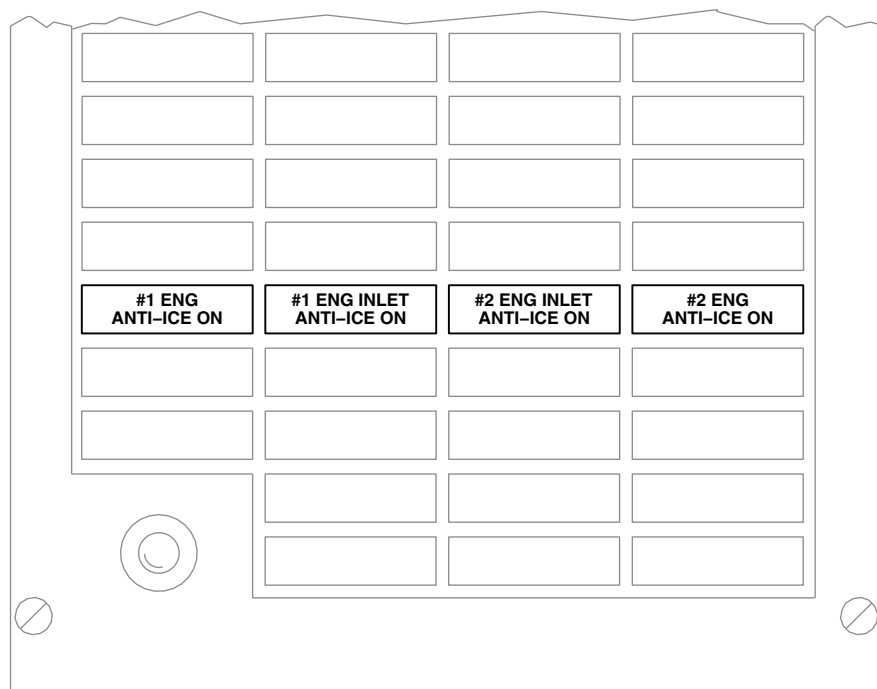


TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P802	NO. 2 ENGINE INTERFACE DISCONNECT
P803	NO. 1 ENGINE INTERFACE DISCONNECT
P814 / J814	FIREWALL DISCONNECT BL 14 RH, STA 373
P815 / J815	FIREWALL DISCONNECT BL 14 LH, STA 373
P818 / J818	NO. 2 ENGINE INLET
P820	NO. 1 ENGINE INLET ANTI-ICE VALVE

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P821	NO. 1 ENGINE INLET TEMPERATURE SWITCH
P822	NO. 2 ENGINE INLET ANTI-ICE VALVE
P823 / J823	NO. 1 ENGINE INLET
P824 / J824	FIREWALL DISCONNECT BL 14 RH, STA 358.25
P825 / J825	FIREWALL DISCONNECT BL 14 LH, STA 358.25
P828	NO. 2 ENGINE INLET TEMPERATURE SWITCH
P900 / J900	RH RELAY PANEL

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SA

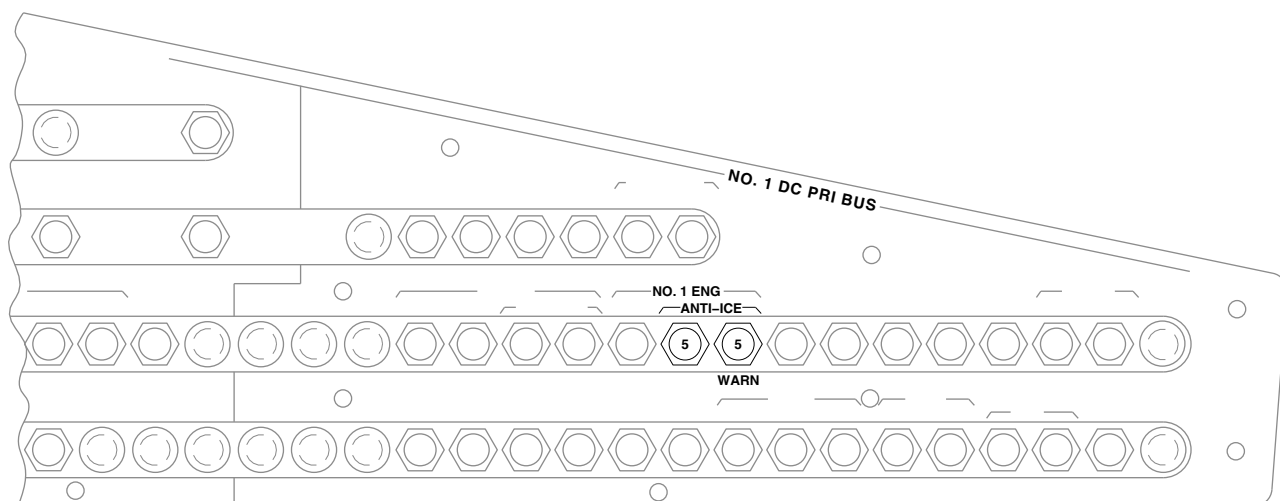
FIGURE 4-1-8-1. ENGINE ANTI-ICE SYSTEM LOCATION DIAGRAM (SHEET 2 OF 5)



CAUTION/ADVISORY PANEL

A

B

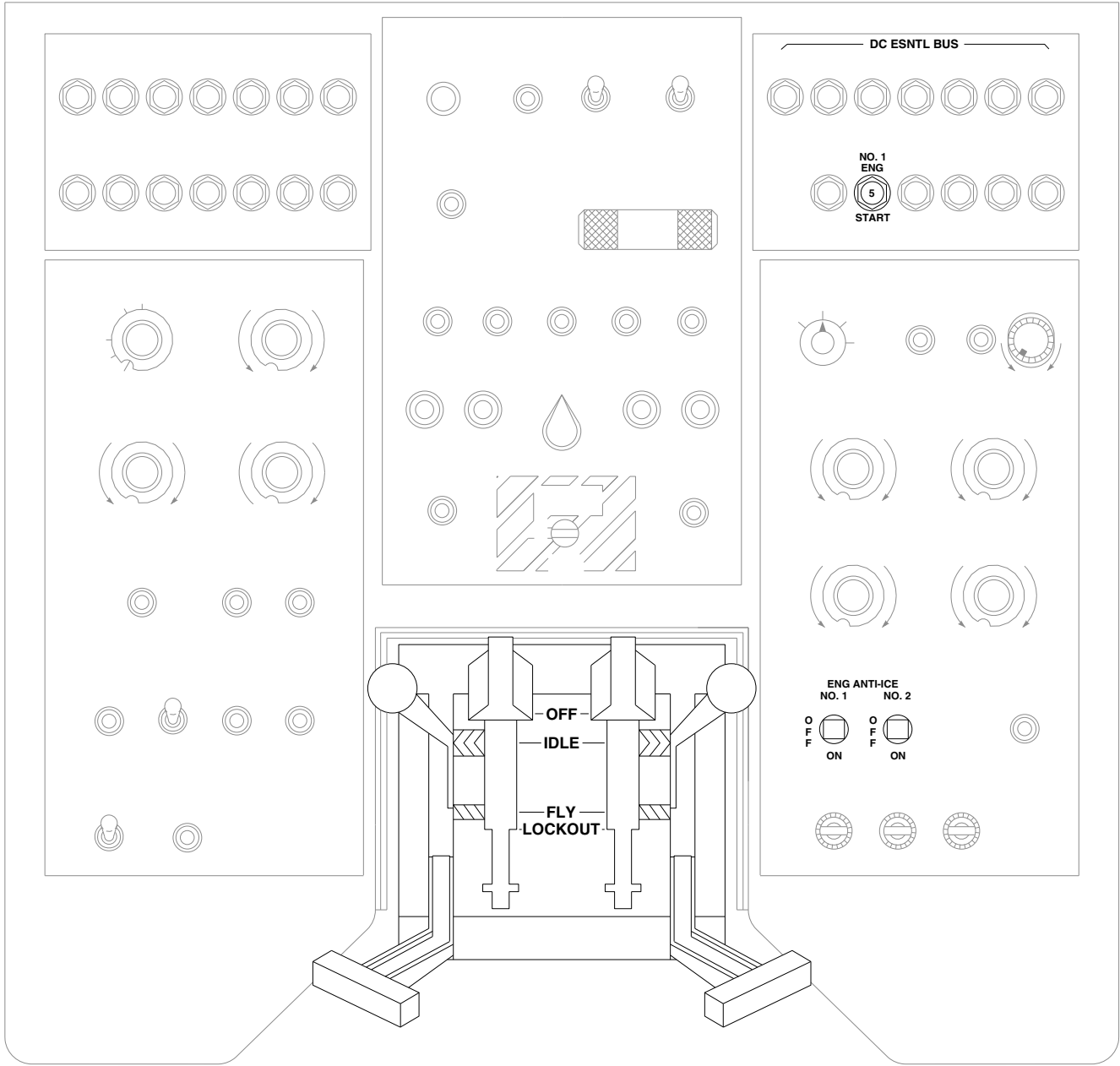


COPILOT'S CIRCUIT BREAKER PANEL

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SA

FIGURE 4-1-8-1. ENGINE ANTI-ICE SYSTEM LOCATION DIAGRAM (SHEET 3 OF 5)

C



UPPER CONSOLE PANEL

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SA

FIGURE 4-1-8-1. ENGINE ANTI-ICE SYSTEM LOCATION DIAGRAM (SHEET 4 OF 5)

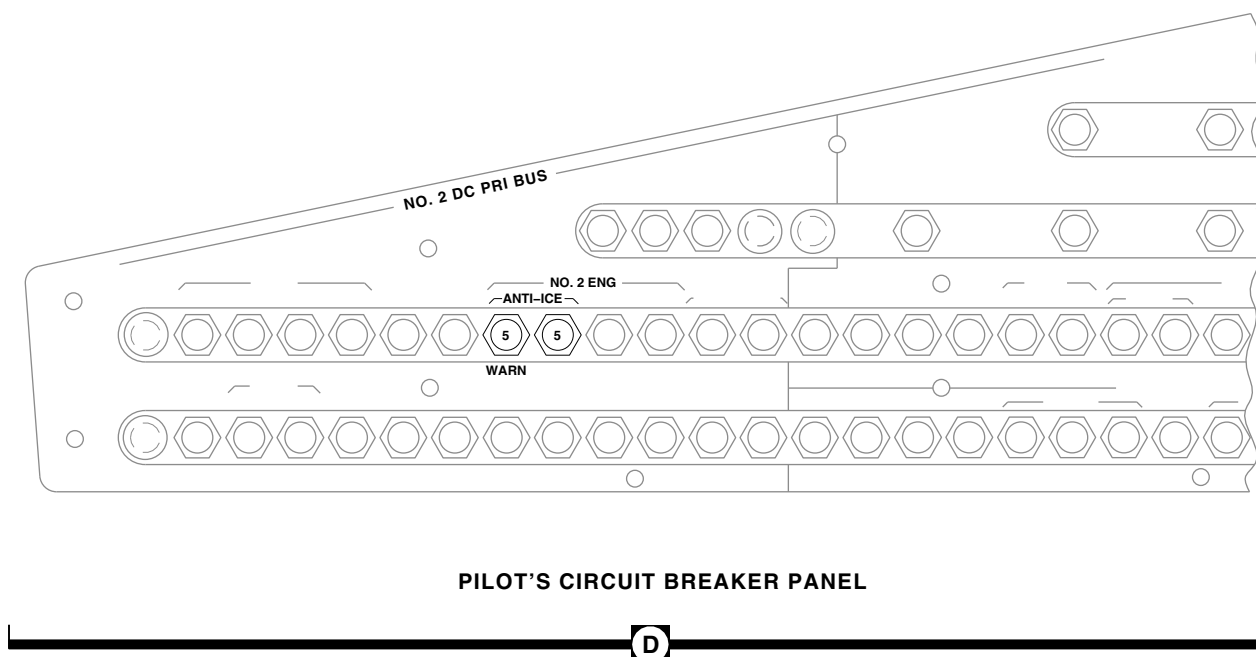
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FIGURE 4-1-8-1. ENGINE ANTI-ICE SYSTEM LOCATION DIAGRAM (SHEET 5 OF 5)

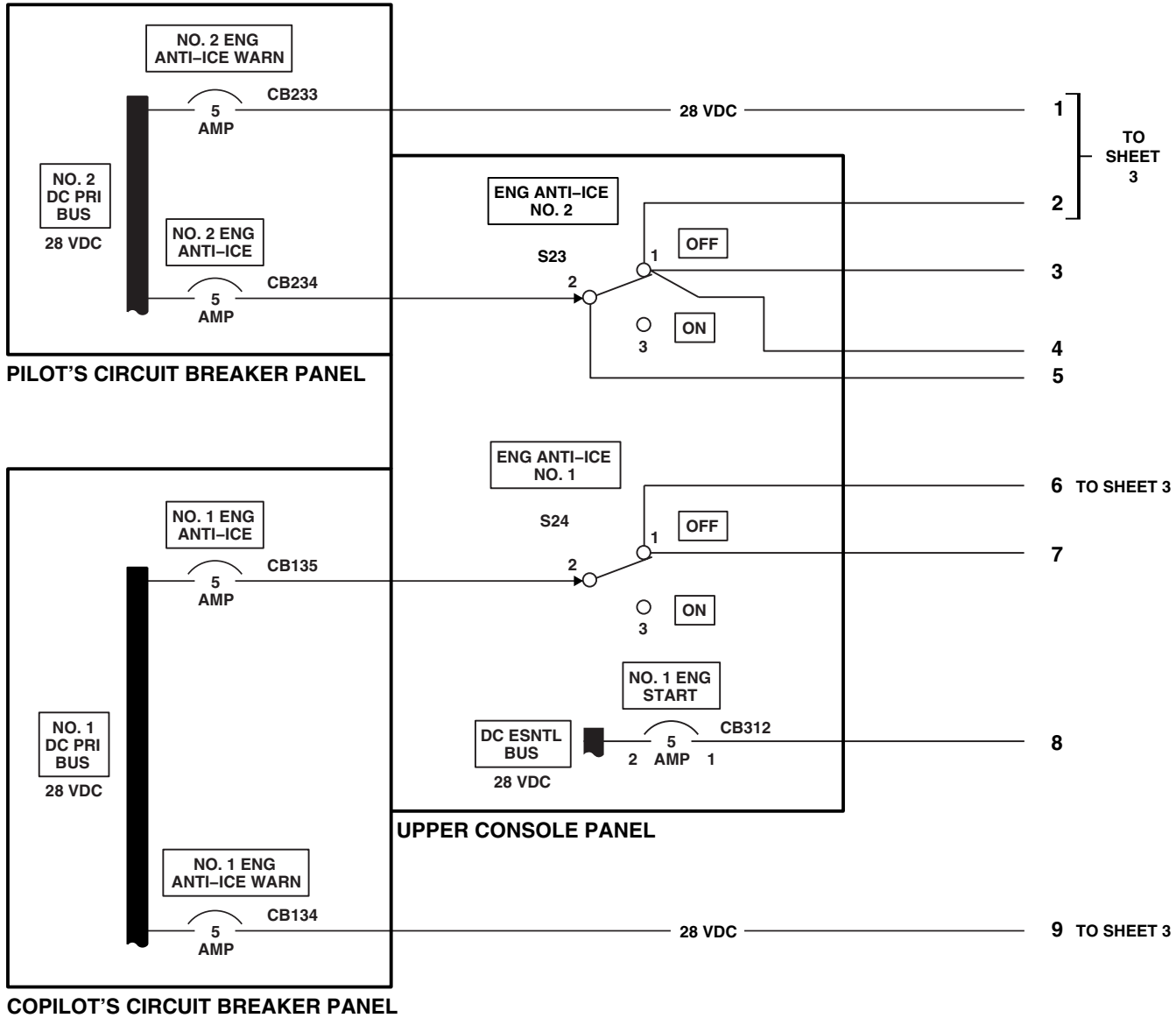


FIGURE 4-1-8-2. ENGINE ANTI-ICE SYSTEM BLOCK DIAGRAM (SHEET 1 OF 3)

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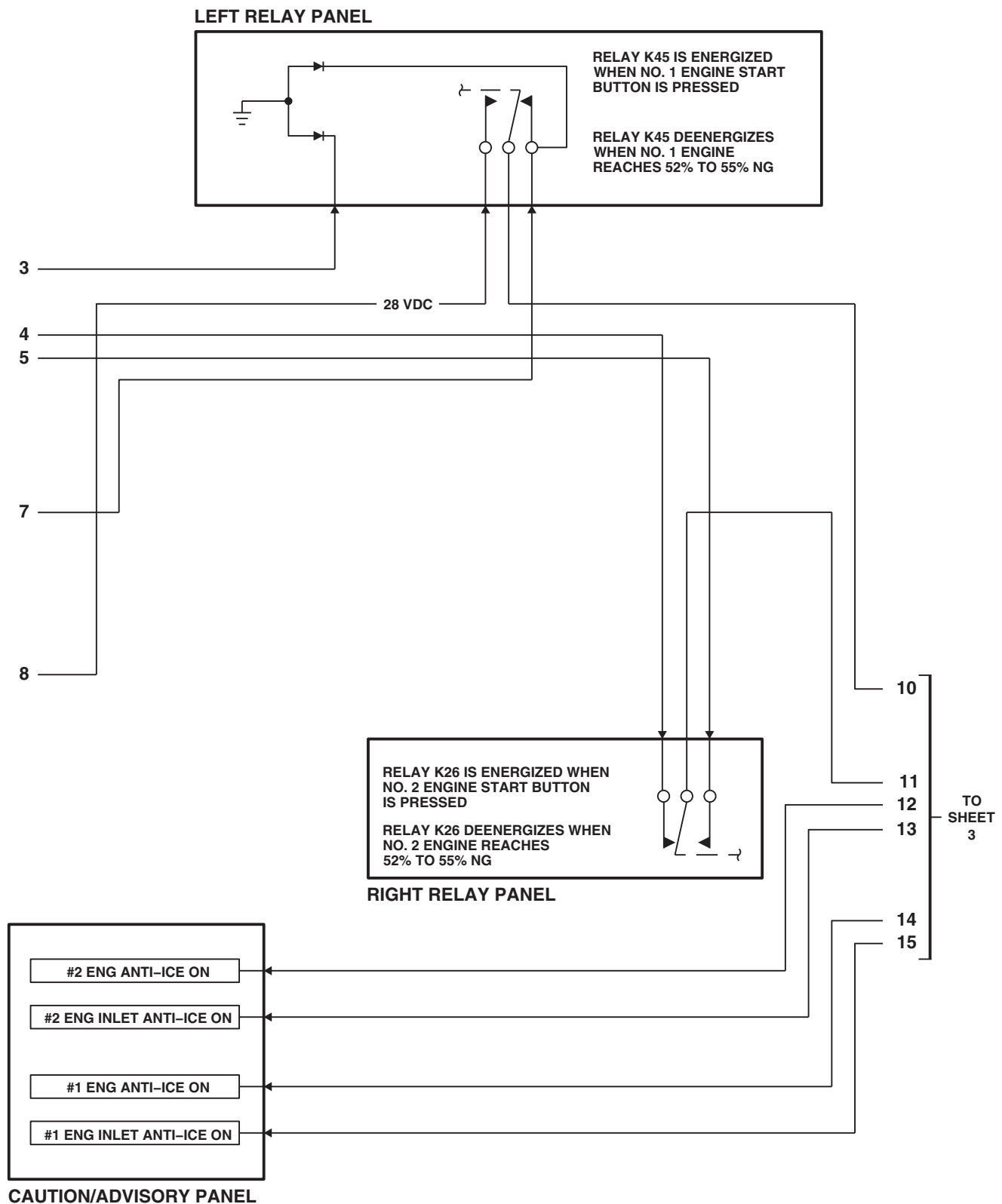
FC3226_2
SA

FIGURE 4-1-8-2. ENGINE ANTI-ICE SYSTEM BLOCK DIAGRAM (SHEET 2 OF 3)

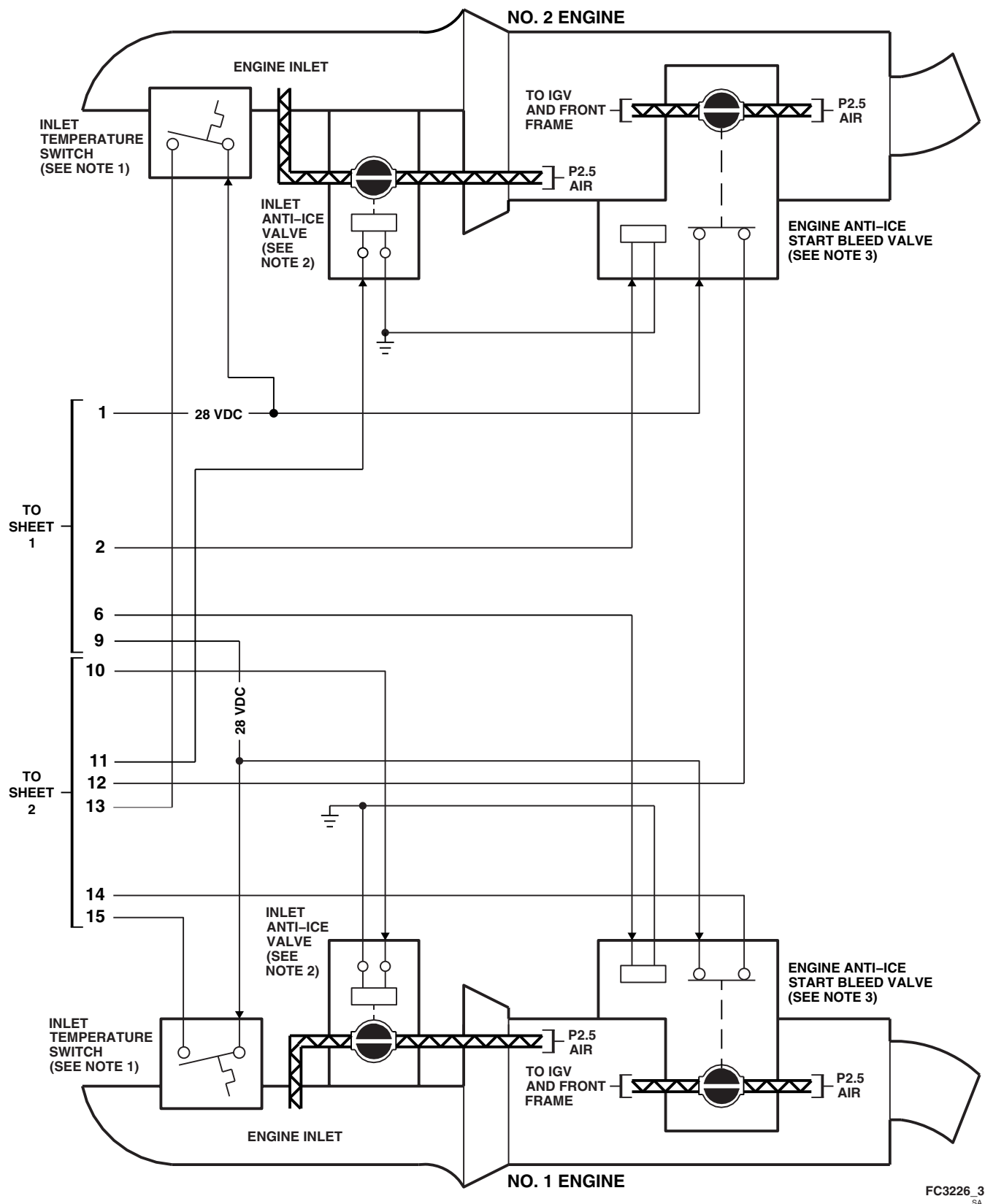


FIGURE 4-1-8-2. ENGINE ANTI-ICE SYSTEM BLOCK DIAGRAM (SHEET 3 OF 3)

SECTION 4-2

TROUBLESHOOTING

SECTION OVERVIEW

PARAGRAPH	TITLE
4-2-1	Engine and Engine Interface
4-2-2	No. 1 Engine Instruments and Warning Lights System
4-2-3	No. 2 Engine Instruments and Warning Lights System
4-2-4	Engine Overspeed Protection System
4-2-5	Engine Speed Trim System
4-2-6	Engine Chip Detector System
4-2-7	Engine Anti-Ice System
4-2-8	Engine Start and Ignition System
4-2-9	Engine Controls Quadrant (BENCH)

4-2-1/(4-2-2 Blank)

4-2-1 ENGINE AND ENGINE INTERFACE.

INITIAL SETUP

4-2-1.1 Introduction.

NOTE

As a general rule, if only one parameter of the engine is malfunctioning and all other indications are normal, the general cause is a malfunctioning airframe indicating system. Troubleshoot Instrument Display System (PARA 8-2-3).

- a. This operational/troubleshooting procedure provides fault isolation of engine malfunction

indication as reported by the pilot. An engine malfunction indication may be due to an airframe or an engine component malfunction. All troubleshooting tables here relate to airframe items first. If it has been determined that the airframe is not the cause, troubleshoot the engine (SEI 671 and TM 1-2840-248-23).

- b. The following table lists the self-test BITE indications for T700-GE-701C engines on the appropriate Torque Meter and references to troubleshooting procedures.

TABLE NO. 4-2-1-1. T701C		
BITE INDICATION	TROUBLE	TROUBLESHOOTING PROCEDURES
15%	DEC	SEI 671 and TM 1-2840-248-23
25%	Np Demand	SEI 671 and TM 1-2840-248-23
35%	Load Share	SEI 671 and TM 1-2840-248-23
45%	TGT	PARA 8-2-3/SEI 671 and TM 1-2840-248-23
55%	Alternator Power	SEI 671 and TM 1-2840-248-23
65%	Ng	PARA 8-2-3/SEI 671 and TM 1-2840-248-23
75%	Np	PARA 8-2-3/SEI 671 and TM 1-2840-248-23
85%	Torque and Overspeed	PARA 8-2-3/SEI 671 and TM 1-2840-248-23 (Torque), PARA 4-2-4/SEI 671 and TM 55- 2840-248-23 (Overspeed)
95%	Hot Start Prevention	SEI 671 and TM 1-2840-248-23
105%	Helicopter 400 Hz Power	PARA 9-2-1/SEI 671 and TM 1-2840-248-23 (Torque)
115%	Collective Channel	TABLE NO. 4-2-1-11
125%	Nr	TABLE NO. 4-2-1-10/SEI 671 and TM 1-2840-248-23 (Torque)

TABLE NO. 4-2-1-2. Engine Ng Low At Ground Idle Speed (Idle Speed Below Limits).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Troubleshoot Instrument Display System, % Ng 1 and 2 scales for proper operation (PARA 8-2-3).	
2. If Instrument Display System operates as specified, troubleshoot engine for low ground idle speed (SEI 671 and TM 1-2840-248-23). Go to 3.	
3. Procedure completed.	

TABLE NO. 4-2-1-3. Engine Ng High At Ground Idle Speed (Idle Speed Above Limits).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Troubleshoot Instrument Display System % Ng 1 and 2 scales for proper operation (PARA 8-2-3).	Step 1. If scales show proper operation, go to 2. Step 2. If scales do not show proper operation, repair/replace components as required (PARA 8-4-10). Go to 4.
2. Check power available Spindle System for proper rigging and operation (PARA 4-4-33).	Step 1. If rigging shows proper operation, go to 3. Step 2. If rigging does not show proper operation, rig power available spindle controls (PARA 4-4-33) or replace components as required (PARA 4-4-28). Go to 4.
3. Perform load demand spindle verification (TABLE NO. 4-2-1-14) to check for proper rigging and operation.	Step 1. If Load Demand Spindle System operates as specified, troubleshoot engine for high Ng during ground idle (SEI 671 and TM 1-2840-248-23). Go to 4. Step 2. If Load Demand Spindle System is not correct, repair/replace components as required (PARA 4-4-27). Go to 4.
4. Procedure completed.	

TABLE NO. 4-2-1-4. Engine Flames Out.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Move control quadrant ENG POWER CONT lever in cockpit to LOCKOUT. Go to 2. 2. Move control quadrant ENG FUEL SYS SELECTOR LEVER to detent. Go to 3. 3. Place FUEL PUMP switch to FUEL PRIME. Check that fuel is coming from overboard drain. Step 1. If fuel is coming from overboard drain, go to 4. Step 2. If fuel is not coming from overboard drain, troubleshoot Fuel Prime Boost System (PARA 10-2-3). Go to 5. 4. Visually check self-sealing breakaway main fuel valve on top of tank for a yellow stripe. Step 1. If yellow stripe is present, replace fuel valve (PARA 10-4-18). Go to 5. Step 2. If yellow stripe is not present, troubleshoot engine for flame out (SEI 671 and TM 1-2840-248-23). Go to 5. 5. Procedure completed.

TABLE NO. 4-2-1-5. Engine TGT Fluctuates At Ground Idle Speed.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Troubleshoot TGT 1 or 2 scale on instrument display unit (PARA 8-2-3). 2. If Instrument Display System operates as specified, troubleshoot engine for fluctuating TGT at ground idle speed (SEI 671 and TM 1-2840-248-23). Go to 3. 3. Procedure completed.

TABLE NO. 4-2-1-6. Engine TGT Exceeds Limiter Setting.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Troubleshoot TGT 1 or 2 scale on instrument display unit (PARA 8-2-3). 2. If Instrument Display System operates as specified, troubleshoot engine for exceeding TGT limiter (SEI 671 and TM 1-2840-248-23). Go to 3. 3. Procedure completed.

TABLE NO. 4-2-1-7. Engine Has Uncontrolled Acceleration (Ng).

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check ENG POWER CONT level for proper operation. - Step 1. If ENG POWER CONT level is normal, troubleshoot engine for uncontrolled acceleration (SEI 671 and TM 1-2840-248-23). Go to 2. - Step 2. If ENG POWER CONT level is not normal, repair/replace components in the PAS as required (PARA 4-4-28). Go to 2. 2. Procedure completed.

TABLE NO. 4-2-1-8. Engine Has Unstable Operation (Ng, Np, And Torque).

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check continuity between: - P137-N and P800-20 - P138-N and P801-20 - Step 1. If continuity is present, troubleshoot engine for unstable operation (SEI 671 and TM 1-2840-248-23). Go to 2. - Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 2. - P137-N and J814-30 - P138-N and J815-30

TABLE NO. 4-2-1-8. Engine Has Unstable Operation (Ng, Np, And Torque). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 2. 2. Procedure completed.

TABLE NO. 4-2-1-9. Torque Split Greater Than 5% Or Erratic.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check electrical connectors between DEC/ECU of both engines for proper connections.	Step 1. If connectors have proper connections, go to 2. Step 2. If connectors do not have proper connections, connect electrical connectors. Go to 10.
2. Check load demand cable spring pin (for each engine) for sheared condition.	Step 1. If sheared condition is present, go to 3. Step 2. If sheared condition is not present, go to 9.
3. Disconnect LDS cable from mixing unit lever. Push cable rod end in and out. Go to 4.	
4. Check that cable movement is smooth (no binding).	Step 1. If cable movement is smooth, go to 5. Step 2. If cable movement is not smooth, go to 6.
5. Replace collective bias spring pin (PARA 4-4-27). Go to 10.	
6. Disconnect load demand rotary input from HMU (PARA 4-4-18). Go to 7.	
7. Push cable rod end in and out. Go to 8.	
8. Check that cable movement is smooth (no binding).	

TABLE NO. 4-2-1-9. Torque Split Greater Than 5% Or Erratic. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If cable movement is smooth, replace spring pin in LDS cable (PARA 4-4-27). Go to 10. Step 2. If cable movement is not smooth, replace LDS cable (PARA 4-4-27). Go to 10. Step 3. If trouble remains, replace HMU (SEI 671 and TM 1-2840-248-23). Go to 10.
9. Check continuity between: P800-3 and P801-3 P800-22 and P801-22 P800-23 and P801-24 P800-24 and P801-23	Step 1. If continuity is present, troubleshoot engine for excessive torque split for erratic torque indication (SEI 671 and TM 1-2840-248-23). Go to 10. Step 2. If continuity is not present, repair/replace wiring between J814 and J815 as required (Appendix F). Go to 10. Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 10.
10. Procedure completed.	

TABLE NO. 4-2-1-10. BITE Code Indicates 125%.

T701C

TEST OR INSPECTION	CORRECTIVE ACTION
1. Do both engine BITE codes indicate 125%?	Step 1. If only No. 2 engine BITE code indicates 125%, go to 2. Step 2. If only No. 1 engine BITE code indicates 125%, go to 3. Step 3. If both engine BITE codes indicate 125%, replace main rotor speed sensor (PARA 6-4-28). Go to 4.
2. Check continuity between: P407-1 and P800-26 P407-2 and P800-25	

TABLE NO. 4-2-1-10. BITE Code Indicates 125%. (Cont)

T701C	TEST OR INSPECTION CORRECTIVE ACTION
	Step 1. If continuity is not present, repair/replace wiring (Appendix F) between: Go to 4. P407-1 and J814-4 P407-2 and J814-3 Step 2. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 4. 3. Check continuity between: P407-1 and P801-26 P407-2 and P801-25 Step 1. If continuity is not present, repair/replace wiring (Appendix F) between: Go to 4. P407-1 and J815-4 P407-2 and J815-3 Step 2. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 4. 4. Procedure completed.

TABLE NO. 4-2-1-11. BITE Code Indicates 115%, Or During Power Recovery, Np Does Not Increase To Match Nr.

T701C	TEST OR INSPECTION CORRECTIVE ACTION
	1. Do both engine BITE codes indicate 115%? Step 1. If only No. 2 engine BITE code indicates 115%, go to 2. Step 2. If only No. 1 engine BITE code indicates 115%, go to 4. Step 3. If both engine BITE codes indicate 115%, replace No. 3 collective stick position sensor (PARA 11-4-79). Go to 6. 2. Check continuity between: J351-4 and P800-29 J351-5 and P800-30 J351-6 and P800-31 Step 1. If continuity is present, go to 3.

TABLE NO. 4-2-1-11. BITE Code Indicates 115%, Or During Power Recovery, Np Does Not Increase To Match Nr. (Cont)

T701C	TEST OR INSPECTION CORRECTIVE ACTION
	Step 2. If continuity is not present, repair/replace wiring (Appendix F) between: Go to 6. J351-4 and J814-20 J351-5 and J814-22 J351-6 and J814-24 Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 6. 3. Is No. 3 collective stick position sensor adjusted properly? Step 1. If sensor is adjusted properly, troubleshoot No. 2 engine (SEI 671 and TM 1-2840-248-23). Go to 6. Step 2. If sensor is not adjusted properly, adjust sensor (PARA 11-4-79). Go to 6. Step 3. If sensor cannot be adjusted, replace sensor (PARA 11-4-79). Go to 6. 4. Check continuity between: J351-1 and P801-29 J351-2 and P801-30 J351-3 and P801-31 Step 1. If continuity is present, go to 5. Step 2. If continuity is not present, repair/replace wiring (Appendix F) between: Go to 6. J351-1 and J814-20 J351-2 and J814-22 J351-3 and J814-24 Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 6. 5. Is No. 3 collective stick position sensor adjusted properly? Step 1. If sensor is adjusted properly, troubleshoot No. 1 engine (SEI 671 and TM 1-2840-248-23). Go to 6. Step 2. If sensor is not adjusted properly, adjust sensor (PARA 11-4-79). Go to 6. Step 3. If sensor cannot be adjusted, replace sensor (PARA 11-4-79). Go to 6. 6. Procedure completed.

TABLE NO. 4-2-1-12. TGT Spikes On Either Engine.

TEST OR INSPECTION CORRECTIVE ACTION	
1. Check that maintenance has been performed in the following areas: No. 1 or No. 2 engines No. 1 or No. 2 SDCs Flight control harness to cockpit feed through connectors	
Step 1. If maintenance has been performed, go to 2. Step 2. If maintenance has not been performed, go to 3.	
2. Check that all bonded and grounded connections are secure and free of corrosion and that electrical harnesses and connectors have not been damaged and are free of moisture.	
Step 1. If connectors are as specified, go to 3. Step 2. If connections are not as specified, repair/replace wiring as required (Appendix F). Go to 18.	
3. Disconnect yellow harness connector W4P5 from alternator. Check for specified resistance between: J2-1 and J2-1 (Infinity) J2-2 and J2-2 (Infinity) J2-3 and J2-3 (0.6 - 1.0 Ohm) 2-4 and J2-4 (0.6 - 1.0 Ohm) J2-5 and J2-5 (1.2 - 2.0 Ohms)	
Step 1. If resistance is as specified, go to 4. Step 2. If resistance is not as specified, replace alternator (SEI 671 and TM 1-2840-248-23). Go to 18.	
4. Check circuit at S39 connector on ECU/ DEC (SEI 671 and TM 1-2840-248-23).	
Step 1. If circuit is good, go to 5. Step 2. If circuit is not good, repair/replace wiring as required (SEI 671 and TM 1-2840-248-23). Go to 18.	
5. Disconnect connectors P814 and P824. Run No. 2 engine (Appropriate Flight Manual). Is TGT still spiking?	
Step 1. If TGT still spikes, go to 6. Step 2. If TGT is not spiking, go to 11.	
6. Check all associated engine compartment harnesses and connectors, SDC harness and connectors, and interconnecting wiring for damage and moisture intrusion.	

TABLE NO. 4-2-1-12. TGT Spikes On Either Engine. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
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Step 1. If harness and connectors are good, go to 7.

Step 2. If harness and connectors are not good, repair/replace wiring as required (Appendix F). Go to 18.

7. Is related spiking indication also associated with torque?

Step 1. If spiking is as specified, go to 8.

Step 2. If spiking is not as specified, go to TABLE NO. 4-2-1-13.

8. Is indication of torque spiking associated with No. 2 engine?

Step 1. If indication of torque is as specified, go to 9.

Step 2. If indication of torque is not as specified, go to 10.

9. Check continuity between:

P800-10 and P814-15

P800-11 and P814-14

P800-20 and P814-30

Step 1. If continuity is present, go to TABLE NO. 4-2-1-13.

Step 2. If continuity is not present, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 18.

10. Check continuity between:

P801-10 and P815-15

P801-11 and P815-16

Step 1. If continuity is present, go to TABLE NO. 4-2-1-13.

Step 2. If continuity is not present, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 18.

11. Reconnect connectors P814 and P824. Disconnect connectors P815 and P825. Run No. 1 engine (Appropriate Flight Manual). Is TGT still spiking?

Step 1. If TGT still spikes, go to 12.

Step 2. If TGT does not spike, go to 13.

12. Check all associated engine compartment harnesses and connectors, SDC harness and connectors, and interconnecting wiring for damage and moisture intrusion.

TABLE NO. 4-2-1-12. TGT Spikes On Either Engine. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
Step 1. If harness and connectors are good, go to 14 .	Step 2. If harness and connectors are not good, repair/replace wiring as required (Appendix F). Go to 18 .
13. Troubleshoot Engine Speed Trim System (PARA 4-2-5). Is TGT still spiking?	
Step 1. If TGT still spikes, go to 14 .	Step 2. If TGT does not spike, System is operational. Reconnect connectors P815 and P825. Go to 18 .
14. Is related spiking indication also associated with torque?	
Step 1. If spiking is related with torque, go to 15 .	Step 2. If spiking is not related with torque, go to TABLE NO. 4-2-1-13 .
15. Is indication of torque spiking associated with No. 1 engine?	
Step 1. If spiking is associated with No. 1 engine, go to 16 .	Step 2. If spiking is not associated with No. 1 engine, go to 17 .
16. Check continuity between:	
P801-10 and P815-15	
P801-11 and P815-16	
P801-20 and P815-30	
Step 1. If continuity is present, go to TABLE NO. 4-2-1-13 .	Step 2. If continuity is not present, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 18 .
17. Check continuity between:	
P800-10 and P814-15	
P800-11 and P814-14	
P800-20 and P814-30	
Step 1. If continuity is present, go to TABLE NO. 4-2-1-13 .	Step 2. If continuity is not present, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 18 .
18. Procedure completed.	

TABLE NO. 4-2-1-13. TGT And/Or Torque Spiking Appear On PDUs Without Associated Change In Engine Power.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Troubleshoot Instrument Display System (PARA 8-2-3).	
	Step 1. If Instrument Display System is malfunctioning, repair as required. Go to 14 .
	Step 2. If Instrument Display System is not malfunctioning, go to 2 .
2. Check No. 1 and No. 2 engine harness continuity.	
	Step 1. If continuity is present, go to 3 .
	Step 2. If continuity is not present, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 14 .
3. Disconnect No. 2 engine history recorder/history counter. Run engines (Appropriate Flight Manual). Check that TGT and/or torque spiking indications do not appear on PDU's without change in engine power.	
	Step 1. If TGT and/or torque spiking appear on PDU, go to 4 .
	Step 2. If TGT and/or torque spiking does not appear on PDU, go to 5 .
4. Check connectors between history counter/history recorder and DEC/ECU for contamination.	
	Step 1. If contamination is present, clean connectors (SEI 671 and TM 1-2840-248-23). Go to 14 .
	Step 2. If contamination is not present, replace history counter/history recorder (SEI 671 and TM 1-2840-248-23). Go to 14 .
	Step 3. If trouble remains, replace yellow electrical cable (SEI 671 and TM 1-2840-248-23). Go to 14 .
5. Shutdown engines and disconnect No. 1 engine history counter/history recorder. Run engines (Appropriate Flight Manual). Has trouble been remedied?	
	Step 1. If trouble has been remedied, go to 6 .
	Step 2. If trouble has not been remedied, go to 7 .
6. Check connectors between history counter/history recorder and DECU/ECU for contamination.	

TABLE NO. 4-2-1-13. TGT And/Or Torque Spiking Appear On PDUs Without Associated Change In Engine Power. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
Step 1. If contamination is present, clean connectors (SEI 671 and TM 1-2840-248-23). Go to 14. Step 2. If contamination is not present, replace history counter/history recorder (SEI 671 and TM 1-2840-248-23). Go to 14. Step 3. If trouble remains, replace yellow electrical cable (SEI 671 and TM 1-2840-248-23). Go to 14.
7. Check yellow electrical cable for contamination.
Step 1. If contamination is present, clean connectors (SEI 671 and TM 1-2840-248-23). Go to 14. Step 2. If contamination is not present, go to 8.
8. Check continuity of yellow electrical cable.
Step 1. If continuity is present, go to 9. Step 2. If continuity is not present, replace cable (SEI 671 and TM 1-2840-248-23). Go to 14.
9. Check blue electrical cable for contamination.
Step 1. If contamination is present, clean connectors (SEI 671 and TM 1-2840-248-23). Go to 14. Step 2. If contamination is not present, go to 10.
10. Check continuity of blue electrical cable.
Step 1. If continuity is present, go to 11. Step 2. If continuity is not present, replace cable (SEI 671 and TM 1-2840-248-23). Go to 14.
11. Check green electrical cable for contamination.
Step 1. If contamination is present, clean connectors (SEI 671 and TM 1-2840-248-23). Go to 14. Step 2. If contamination is not present, go to 12.
12. Check continuity of green electrical cable.
Step 1. If continuity is present, go to 13.

TABLE NO. 4-2-1-13. TGT And/Or Torque Spiking Appear On PDUs Without Associated Change In Engine Power. (Cont)

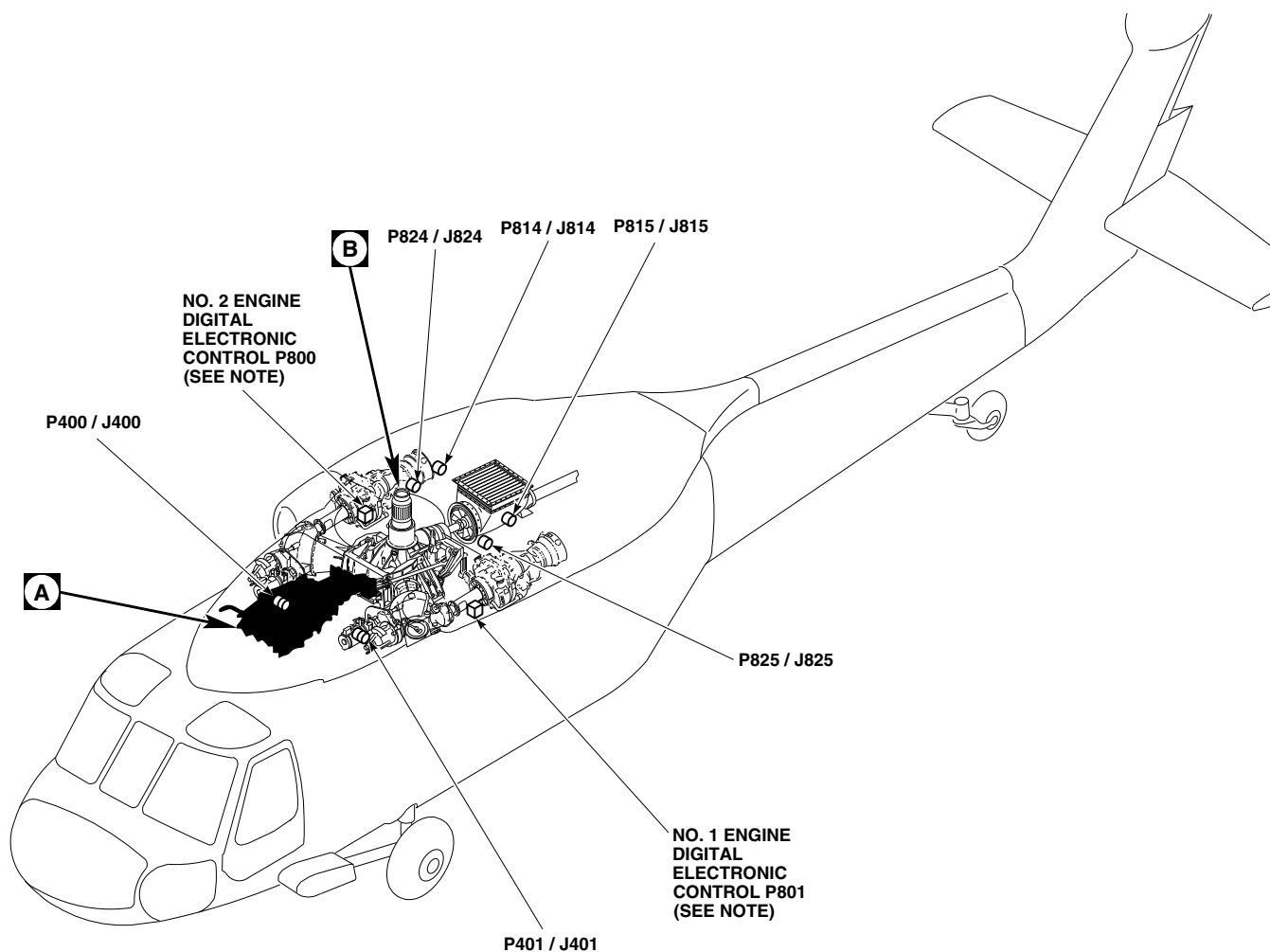
TEST OR INSPECTION CORRECTIVE ACTION
Step 2. If continuity is not present, replace cable (SEI 671 and TM 1-2840-248-23). Go to 14.
13. Troubleshoot Np Sensor System (SEI 671 and TM 1-2840-248-23).
Step 1. If Np Sensor System is malfunctioning, replace Np Sensor System (SEI 671 and TM 1-2840-248-23). Go to 14.
Step 2. If Np Sensor System is not malfunctioning, replace engine as required (SEI 671 and TM 1-2840-248-23). Go to 14.
14. Procedure completed.

TABLE NO. 4-2-1-14. Load Demand Spindle Verification.

TEST OR INSPECTION CORRECTIVE ACTION
1. Disconnect LDS cable from mixer unit lever assembly. Go to 2.
2. Measure LDS force (T1) (PARA 4-3-3).
Step 1. If force is greater than 45 pounds, go to 3.
Step 2. If force is less than or equal to 45 pounds, reconnect LDS cable to mixer unit lever assembly. Go to 10.
3. Disconnect LDS rotary input from HMU (PARA 4-4-18). Go to 4.
4. Measure LDS force (T2) (PARA 4-3-3).
Step 1. If force (T2) is greater than T1 minus 12 pounds, go to 5.
Step 2. If force (T2) is less or equal to T1 minus 12 pounds, go to 7.
5. Disconnect LDS cable from rotary input assembly (PARA 4-4-18). Go to 6.
6. Measure cable friction force (T3) (PARA 4-3-3).
Step 1. If force (T3) is greater than 3 pounds, replace LDS cable (PARA 4-4-27). Go to 10.

TABLE NO. 4-2-1-14. Load Demand Spindle Verification. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If force (T3) is less than or equal to 3 pounds, replace rotary input assembly (PARA 4-4-18). Go to 10.
7. Measure HMU spindle torque (Q) (SEI 671 and TM 1-2840-248-23).	Step 1. If spindle torque is greater than 15 inch-pounds, replace HMU (SEI 671 and TM 1-2840-248-23). Go to 10. Step 2. If spindle torque is less or equal to 15 inch-pounds, go to 8.
8. Disconnect LDS cable from rotary input assembly (PARA 4-4-18). Go to 9.	
9. Measure cable friction force (T3) (PARA 4-3-3).	Step 1. If force (T3) is greater than 3 pounds, replace LDS cable (PARA 4-4-27). Go to 10. Step 2. If force (T3) is less or equal to 3 pounds, replace rotary input assembly (PARA 4-4-18). Go to 10.
10. Procedure completed.	



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
J351	NO. 3 COLLECTIVE STICK POTENTIOMETER
P400 / J400	MAIN ROTOR PYLON DECK BL 24 RH, STA 318
P401 / J401	MAIN ROTOR PYLON DECK BL 24 LH, STA 318
P407	MAIN ROTOR SPEED SENSOR
P800	NO. 2 ENGINE DIGITAL ELECTRONIC CONTROL
P801	NO. 1 ENGINE DIGITAL ELECTRONIC CONTROL
P814 / J814	FIREWALL DISCONNECT BL 14 RH, STA 373
P815 / J815	FIREWALL DISCONNECT BL 14 LH, STA 373
P824 / J824	FIREWALL DISCONNECT BL 14 RH, STA 356.25
P825 / J825	FIREWALL DISCONNECT BL 14 LH, STA 358.25

NOTE

T700 T701A ELECTRICAL
CONTROL UNIT INSTALLED.
T701C DIGITAL ELECTRONIC
CONTROL UNIT INSTALLED.

FB1435
SA

FIGURE 4-2-1-1. ENGINE AND ENGINE INTERFACE LOCATION DIAGRAM (SHEET 1 OF 2)

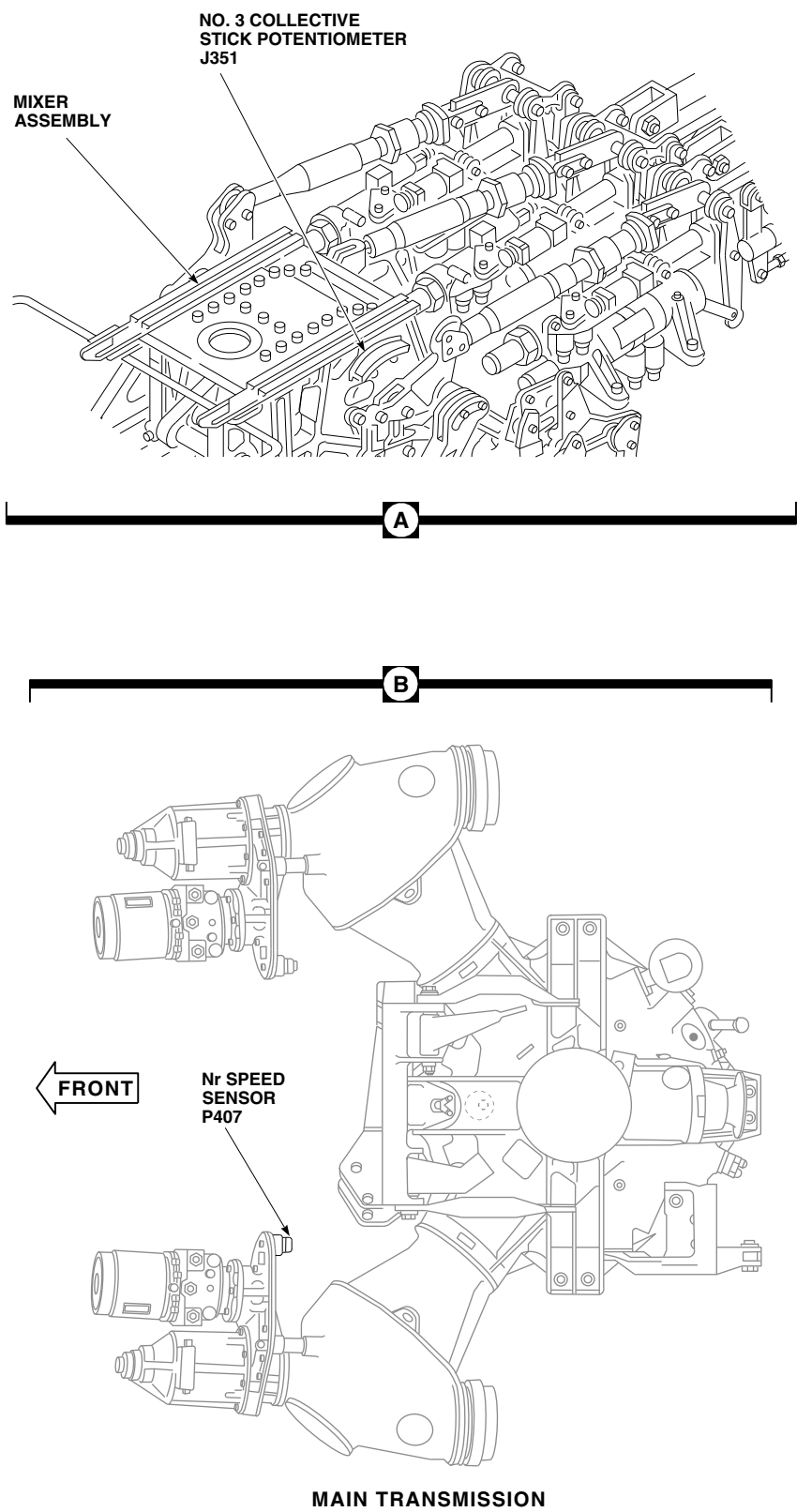


FIGURE 4-2-1-1. ENGINE AND ENGINE INTERFACE LOCATION DIAGRAM (SHEET 2 OF 2)

AA2641_2
SA

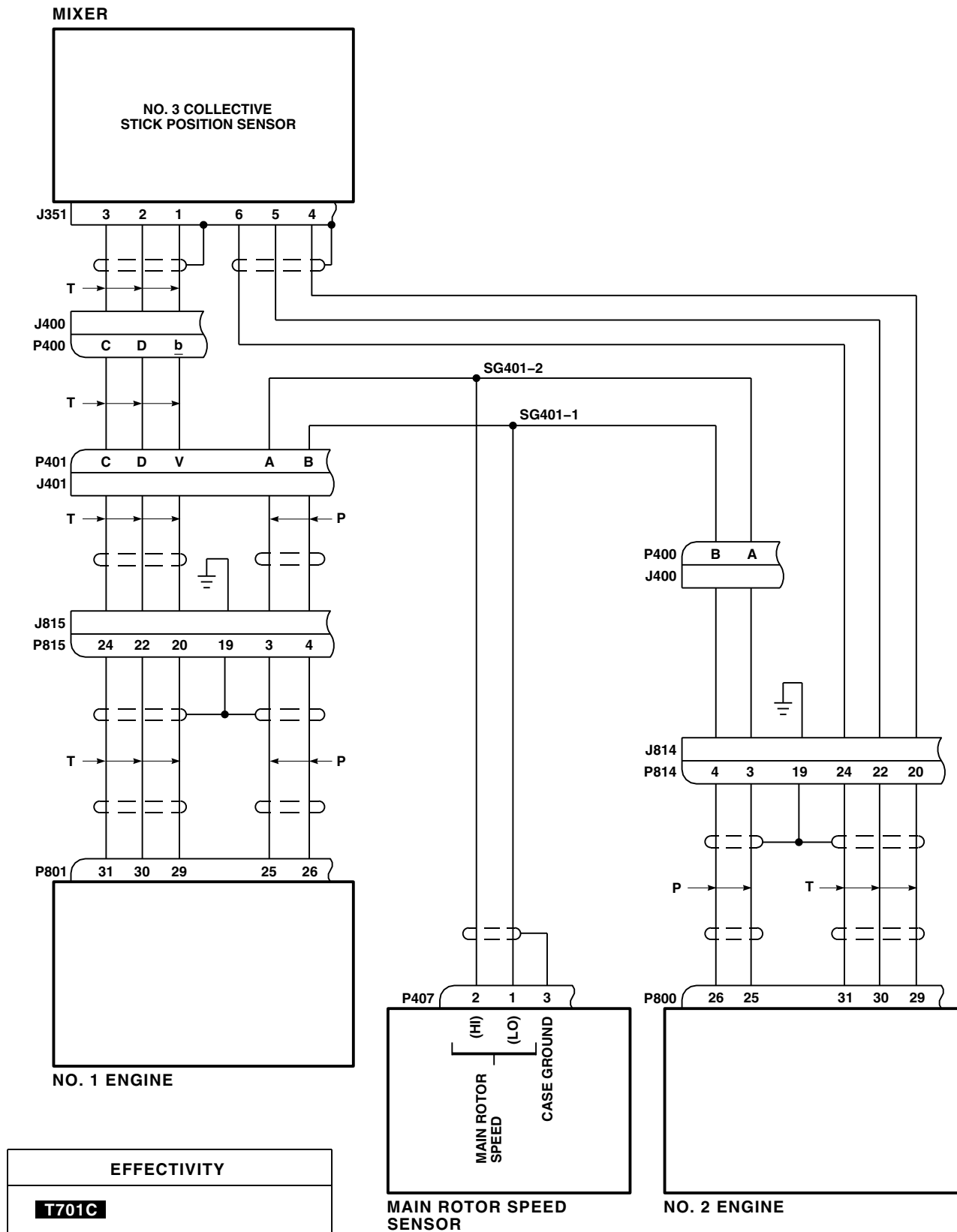


FIGURE 4-2-1-2. ENGINE AND ENGINE INTERFACE SCHEMATIC DIAGRAM

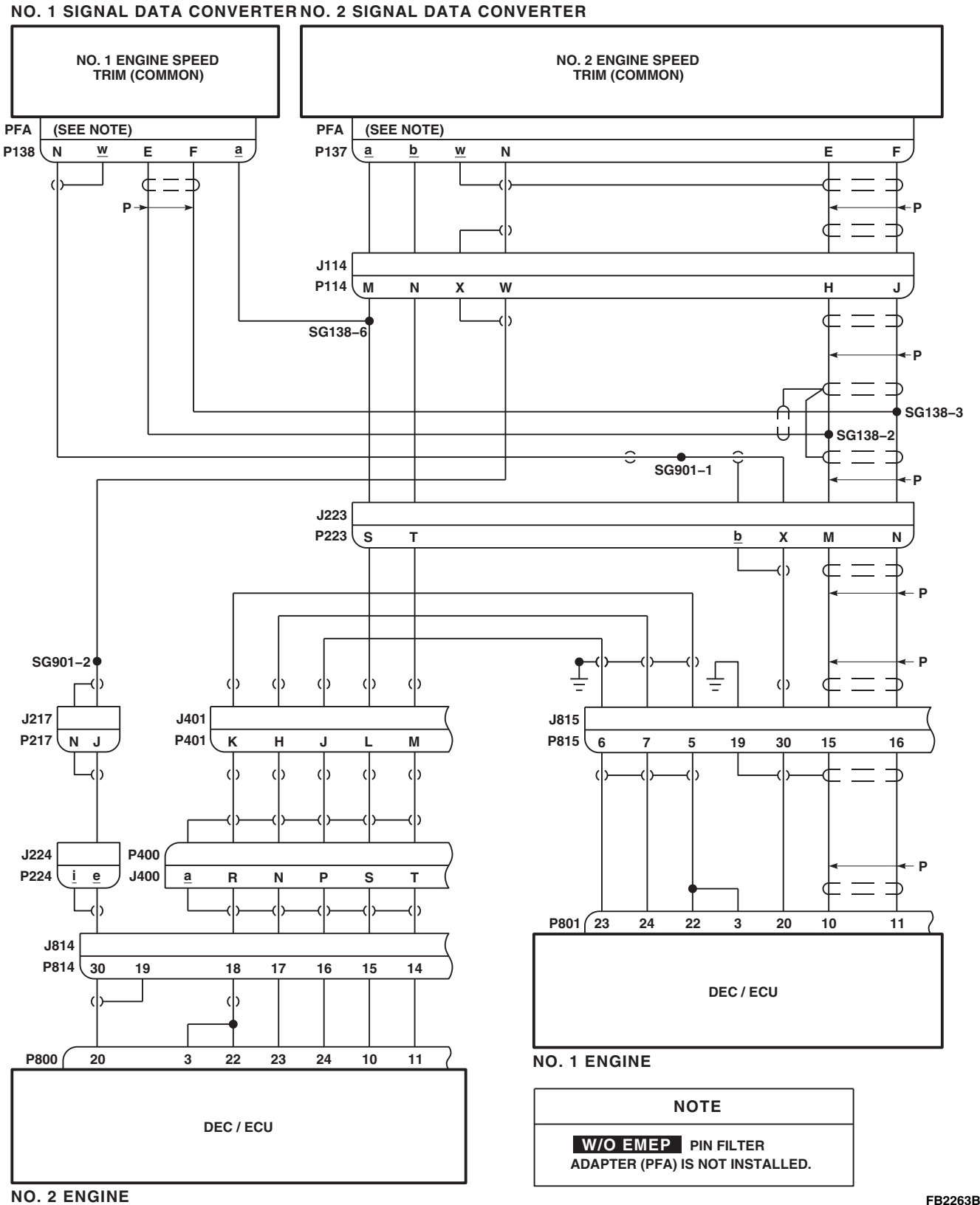


FIGURE 4-2-1-3. ENGINE AND ENGINE INTERFACE SCHEMATIC DIAGRAM

4-2-2 NO. 1 ENGINE INSTRUMENTS AND WARNING LIGHTS SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer Toolkit
- Engine Repairman’s Toolkit
- Insulated Jumper Wire

References

- Appropriate Flight Manual
- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational
- No. 1 Fuel Tank Serviced

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

4-2-2.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).

RESULT:

- #1 FUEL PRESS capsule on caution/advisory panel shall go on.
- #1 ENGINE OIL PRESS capsule on caution/advisory panel shall go on.
 - If result (1) is not as specified, go to TABLE NO. 4-2-2-1.
 - If result (2) is not as specified, go to TABLE NO. 4-2-2-2.
- Install an insulated jumper wire between P803-13 and P803-14.

RESULT: #1 OIL FLTR BYPASS capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 4-2-2-3.
- Remove jumper wire from P803.
 - Install an insulated jumper wire between P803-15 and P803-16.

RESULT: #1 FUEL FLTR BYPASS capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 4-2-2-4.
- Remove jumper wire from P803.
 - Connect P803 to No. 1 engine.
 - Turn off electrical power (PARA 1-6-16).
 - With qualified personnel at controls, run No. 1 engine up to idle speed (Appropriate Flight Manual).

RESULT: #1 FUEL PRESS capsule shall be off.

- If result is not as specified, go to TABLE NO. 4-2-2-5.

- j. With qualified personnel at controls, run No. 1 engine with main rotor turning at 100% Nr until engine instruments stabilize.

RESULT: Engine status shall be as follows:

CAUTION/ ADVISORY PANEL

CAPSULE STATUS

#1 FUEL PRESS

OFF

#1 ENGINE OIL
PRESS

OFF

#1 ENGINE OIL
TEMP

OFF

#1 FUEL FLTR
BYPASS

OFF

#1 OIL FLTR
BYPASS

OFF

CENTRAL DISPLAY UNIT

INDICATION

Ng SPEED 1
TGT TEMP 1

(Refer to Appropriate MTF for normal operating limits)

ENG OIL TEMP 1

ENG OIL PRESS 1

PILOT'S DISPLAY UNIT

INDICATION

% RPM 1
% TRQ 1

(Refer to Appropriate MTF for normal operating limits)

- If any of these caution/advisory panel capsules are on:

CAPSULE

GO TO

#1 ENGINE OIL
PRESS

TABLE NO. 4-2-2-6

#1 ENGINE OIL
TEMP

TABLE NO. 4-2-2-7

- If #1 FUEL PRESS capsule is on, troubleshoot engine for fuel low pressure caution light on during flight (SEI 671 and TM 1-2840-248-23).
- If #1 FUEL FLTR BYPASS or #1 OIL FLTR BYPASS capsule is on, troubleshoot engine (SEI 671 and TM 1-2840-248-23).
- If No. 1 engine oil temperature (#1 ENGINE OIL TEMP) has no indication, go to TABLE NO. 4-2-2-7.
- If No. 1 engine gas generator speed (% Ng 1) has no indication, go to TABLE NO. 4-2-2-8.
- If No. 1 engine turbine inlet temperature (TGT TEMP 1) has no indication, go to TABLE NO. 4-2-2-9.
- If No. 1 engine oil pressure (#1 ENGINE OIL PRESS):
 - (a) Is too high (10 psi above normal), go to TABLE NO. 4-2-2-10.
 - (b) Has no indication, go to TABLE NO. 4-2-2-11.
 - (c) Fluctuates greater than 5 psi, go to TABLE NO. 4-2-2-10.
 - (d) Is below minimum limits (45 psi), go to TABLE NO. 4-2-2-6.
- If No. 1 engine power turbine speed (% RPM 1) has no indication, go to TABLE NO. 4-2-2-12.
- If No. 1 engine has no torque indication (% TRQ 1), go to TABLE NO. 4-2-2-13.

- k. Shut down No. 1 engine (Appropriate Flight Manual).

TABLE NO. 4-2-2-1. #1 FUEL PRESS Capsule Does Not Go On.

TEST OR INSPECTION		CORRECTIVE ACTION	
		1. Check helicopter configuration.	
	Step 1. EMEP ➤	Go to 2.◀	
	Step 2. W/O EMEP ➤	Go to 4.◀	
EMEP	2. Remove and check pin filter adapter P117 (PARA 9-4-98).		
	Step 1. If pin filter adapter is good, go to	3.	
	Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98).	Go to 9.	
EMEP	3. Install pin filter adapter P117 (PARA 9-4-98). Go to 4.		
	4. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST.		
	Step 1. If #1 FUEL PRESS capsule goes on, go to	5.	
	Step 2. If #1 FUEL PRESS capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to	9.	
	5. Check for 28 VDC between P117-2 and ground.		
	Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to	9.	
	Step 2. If voltage is not as specified, go to	6.	
	6. Check for 28 VDC between P803-23 and ground.		
	Step 1. If voltage is as specified, go to	7.	
	Step 2. If voltage is not as specified, go to	8.	
	7. Check continuity between P117-2 and P803-24 .		
	Step 1. If continuity is present, replace fuel pressure switch (PARA 10-4-14). Go to	9.	
	Step 2. If continuity is not present, repair/replace wiring (Appendix F) between P117-2 and J825-10. Go to	9.	
	Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to	9.	

TABLE NO. 4-2-2-1. #1 FUEL PRESS Capsule Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
8. Check for 28 VDC between terminal 1 (NO. 1 ENG WARN LTS circuit breaker, NO. DC PRI BUS - copilot's circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and J825-9 (Appendix F). Go to 9. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 9. Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 9.
9. Procedure completed.	

TABLE NO. 4-2-2-2. #1 ENGINE OIL PRESS Capsule Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check helicopter configuration.	Step 1. EMEP Go to 2. W/O EMEP Go to 4.
2. Remove and check pin filter adapters P117 and P138 (PARA 9-4-98).	Step 1. If pin filter adapters are good, go to 3. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 7.
3. Install pin filter adapters (PARA 9-4-98). Go to 4.	
4. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST.	Step 1. If #1 ENGINE OIL PRESS capsule goes on, go to 5. Step 2. If #1 ENGINE OIL PRESS capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 7.
5. Check for 28 VDC between P117-3 and ground.	

TABLE NO. 4-2-2-2. #1 ENGINE OIL PRESS Capsule Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 7 . Step 2. If voltage is not as specified, go to 6 .
6. Check continuity between P117-3 and P138-X.	Step 1. If continuity is present, troubleshoot Instrument Display System (PARA 8-2-3). Go to 7 . Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 7 .
7. Procedure completed.	

TABLE NO. 4-2-2-3. #1 OIL FLTR BYPASS Capsule Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST.	Step 1. If #1 OIL FLTR BYPASS capsule goes on, go to 2 . Step 2. If #1 OIL FLTR BYPASS capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 9 .
2. Check helicopter configuration.	Step 1. EMEP Go to 3 . Step 2. W/O EMEP Go to 5 .
EMEP 3. Remove and check pin filter adapter P117 (PARA 9-4-98).	Step 1. If pin filter adapter is good, go to 4 . Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 9 .
EMEP 4. Install pin filter adapter P117 (PARA 9-4-98). Go to 5.	
5. Check for 28 VDC between P117-23 and ground.	

TABLE NO. 4-2-2-3. #1 OIL FLTR BYPASS Capsule Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 9. Step 2. If voltage is not as specified, go to 6. 6. Remove jumper wire from P803. Go to 7. 7. Check for continuity between P803-13 and P803-23. Step 1. If continuity is present, go to 8. Step 2. If continuity is not present, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 9. 8. Check for continuity between P803-14 and P117-23. Step 1. If continuity is present, troubleshoot Engine Oil Filter Bypass System (SEI 671 and TM 1-2840-248-23). Go to 9. Step 2. If continuity is not as specified, repair/replace wiring between P117-23 and J825-5 (Appendix F). Go to 9. Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 9. 9. Procedure completed.

TABLE NO. 4-2-2-4. #1 FUEL FLTR BYPASS Capsule Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check helicopter configuration. Step 1. EMEP Go to 2. W/O EMEP Go to 4. 2. Remove and check pin filter adapter P117 (PARA 9-4-98). Step 1. If pin filter adapter is good, go to 3. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 8.

TABLE NO. 4-2-2-4. #1 FUEL FLTR BYPASS Capsule Does Not Go On. (Cont)

	TEST OR INSPECTION CORRECTIVE ACTION	
EMEP	3. Install pin filter adapter P117 (PARA 9-4-98). Go to 4. 4. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Step 1. If #1 FUEL FLTR BYPASS capsule goes on, go to 5. Step 2. If #1 FUEL FLTR BYPASS capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 8. 5. Check for 28 VDC between P117-6 and ground. Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 8. Step 2. If voltage is not present, go to 6. 6. Remove jumper wire from P803. Check for 28 VDC between P803-15 and ground. Step 1. If voltage is as specified, go to 7. Step 2. If voltage is not as specified, repair/replace wiring between NO. 1 ENG WARN LTS circuit breaker terminal 1 and J825-9 (Appendix F). Go to 8. Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 8. 7. Check continuity between P117-6 and P803-16. Step 1. If continuity is present, troubleshoot Engine Fuel Filter Bypass System (SEI 671 and TM 1-2840-248-23). Go to 8. Step 2. If continuity is not present, repair/replace wiring between P117-6 and J825-6 (Appendix F). Go to 8. Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 8. 8. Procedure completed.	

TABLE NO. 4-2-2-5. #1 FUEL PRESS Capsule Remains On When Engine Is At Idle Speed.

TEST OR INSPECTION CORRECTIVE ACTION

1. With No. 1 engine operating at ground idle speed, place upper console FUEL PUMP switch to FUEL PRIME. Go to 2.

2. Check that #1 FUEL PRESS capsule goes off.

Step 1. If capsule goes off, go to **3**.

Step 2. If capsule does not go off, troubleshoot engine for #1 FUEL PRESS caution capsule on below flight idle speed (SEI 671 and TM 1-2840-248-23).
Go to **6**.

3. Shut down No. 1 engine (Appropriate Flight Manual). Go to 4.

4. Disconnect P803 from No. 1 engine fuel pressure switch. Go to 5.

5. Check that #1 FUEL PRESS capsule goes off.

Step 1. If capsule goes off, troubleshoot Airframe Fuel System for possible leaks between engine and fuel tanks, repair/replace components as required.
Go to **6**.

Step 2. If capsule does not go off, repair/replace wiring, as required, (Appendix F) between the following, then go to **6**:
J825-9 and P249-V
J825-10 and P117-2

Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23).
Go to **6**.

Step 4. If trouble remains, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **6**.

6. Procedure completed.

CAUTION

If engine ran for more than 1 minute while oil pressure was below minimum limits, and indicating system was operating properly, replace engine.

TABLE NO. 4-2-2-6. No. 1 Engine Oil Pressure (#1 ENGINE OIL PRESS Capsule Is On) Is Below Minimum Limits.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Visually check engine for external oil leaks.	Step 1. If leaks are found, repair/replace items causing leak, check oil tank level and service as necessary (SEI 671 and TM 1-2840-248-23). Go to 2. Step 2. If no leaks are found, go to TABLE NO. 4-2-2-10.
2. Procedure completed.	

TABLE NO. 4-2-2-7. No. 1 Engine Oil Temperature (#1 ENGINE OIL TEMP Capsule Is On) Is Too High Or Has No Indication.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check engine oil tank for proper level.	Step 1. If tank has proper oil level, go to 2. Step 2. If tank does not have proper oil level, troubleshoot Engine Oil System (SEI 671 and TM 1-2840-248- 23). Go to 18.
2. Troubleshoot Instrument Display System, ENG OIL TEMP 1 scale (PARA 8-2-3).	Go to 3.
3. If Instrument Display System is good, go to 4.	
4. Install an insulated jumper wire between P803-9 and P803-10.	Go to 5.
5. Check helicopter configuration.	Step 1. EMEP Go to 6.

TABLE NO. 4-2-2-7. No. 1 Engine Oil Temperature (#1 ENGINE OIL TEMP Capsule Is On) Is Too High Or Has No Indication. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. W/O EMEP ➔ Go to 8.◀
EMEP	6. Remove and check pin filter adapter P138 (PARA 9-4-98).
	Step 1. If pin filter adapter is good, go to 7.
	Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98).
	Go to 18.
EMEP	7. Install pin filter adapter (PARA 9-4-98). Go to 8.
	8. Check continuity between P138-g and P138-h.
	Step 1. If continuity is present, go to 9.
	Step 2. If continuity is not present, go to 15.
	9. Remove jumper wire from P803. Go to 10.
	10. Install No. 2 engine oil temperature sensor into No. 1 engine. Go to 11.
	11. Operate No. 1 engine (Appropriate Flight Manual). Go to 12.
	12. Check that No. 1 engine temperature is normal.
	Step 1. If No. 1 engine temperature is normal, go to 13.
	Step 2. If No. 1 engine temperature is not normal, troubleshoot Engine Oil System
	(SEI 671 and TM 1-2840-248-23). Go to 18.
	13. Reinstall No. 2 engine oil temperature sensor (PARA 4-4-2). Go to 14.
	14. Install a new oil temperature sensor in No. 1 engine (PARA 4-4-2). Go to 18.
	15. Remove jumper wire from P803. Go to 16.
	16. Repair/replace wiring, as required, between the following (Appendix F), then
	go to 17:
	P138-g and J815-8
	P138-h and J815-9
	17. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23).
	Go to 18.

TABLE NO. 4-2-2-7. No. 1 Engine Oil Temperature (#1 ENGINE OIL TEMP Capsule Is On) Is Too High Or Has No Indication. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
18.	Procedure completed.

TABLE NO. 4-2-2-8. No. 1 Engine Gas Generator Speed (% Ng 1) Has No Indication.

TEST OR INSPECTION	CORRECTIVE ACTION
1.	Troubleshoot engine Ng indicating system (SEI 671 and TM 1-2840-248-23). Go to 2.
2.	If indicating system is good, go to 3.
3.	Troubleshoot Instrument Display System, % Ng 1 scale (PARA 8-2-3). Step 1. If display system is good, repair/replace wiring, as required, between the following (Appendix F), then go to 4: P138-c and J815-10 P138-d and J815-11 Step 2. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 4.
4.	Procedure completed.

TABLE NO. 4-2-2-9. No. 1 Engine Turbine Inlet Temperature (TGT TEMP 1) Has No Indication.

TEST OR INSPECTION	CORRECTIVE ACTION
1.	Troubleshoot Instrument Display System, TGT TEMP 1 scale (PARA 8-2-3). Go to 2.
2.	If Instrument Display System is good, go to 3.
3.	Install an insulated jumper wire between P801-16 and P801-17. Go to 4.

TABLE NO. 4-2-2-9. No. 1 Engine Turbine Inlet Temperature (TGT TEMP 1) Has No Indication. (Cont)

**TEST OR INSPECTION
CORRECTIVE ACTION**

4. Check helicopter configuration.

Step 1. **EMEP** ➤ Go to 5.◀

Step 2. **W/O EMEP** ➤ Go to 7.◀

EMEP**5. Remove and check pin filter adapter P138 (PARA 9-4-98).**

Step 1. If pin filter adapter is good, go to 6.

Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98).
Go to 13.

EMEP**6. Install pin filter adapter (PARA 9-4-98). Go to 7.****7. Check continuity between P138-e and P138-f.**

Step 1. If continuity is present, go to 8.

Step 2. If continuity is not present, go to 10.

8. Remove jumper wire. Go to 9.**9. Troubleshoot engine for TGT not indicating (SEI 671 and TM 1-2840-248-23).
Go to 13.****10. Remove jumper wire. Go to 11.****11. Repair/replace wiring, as required, between the following (Appendix F), then
go to 12:**

P138-e and J815-21

P138-f and J815-23

**12. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23).
Go to 13.****13. Procedure completed.**

TABLE NO. 4-2-2-10. No. 1 Engine Oil Pressure (ENG OIL PRESS 1) Fluctuates Greater Than 5 PSI Or Is Too High (10 PSI Above Normal Operating Pressure).

	TEST OR INSPECTION CORRECTIVE ACTION	
	1. Check engine oil tank for proper level. Step 1. If engine oil is at proper level, go to 2. Step 2. If engine oil is not at proper level, troubleshoot Engine Oil System (SEI 671 and TM 1-2840-248-23). Go to 22.	
	2. Troubleshoot Instrument Display System, ENG OIL PRESS 1 scale (PARA 8-2-3). Go to 3.	
	3. If Instrument Display System is good, go to 4.	
	4. Install an insulated jumper wire between P803-6 and P803-7. Go to 5.	
	5. Check helicopter configuration. Step 1. EMEP Go to 6. Step 2. W/O EMEP Go to 8.	
EMEP	6. Remove and check pin filter adapter P138 (PARA 9-4-98). Step 1. If pin filter adapter is good, go to 7. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 22.	
EMEP	7. Install pin filter adapter (PARA 9-4-98). Go to 8.	
	8. Check continuity between P138-i and P138-j. Step 1. If continuity is present, go to 9. Step 2. If continuity is not present, go to 19.	
	9. Install insulated jumper wire between P803-5 and P803-7. Go to 10.	
	10. Check continuity between P138-j and P138-k. Step 1. If continuity is present, go to 11. Step 2. If continuity is not present, go to 17.	
	11. Remove jumper wire from P803. Go to 12.	

TABLE NO. 4-2-2-10. No. 1 Engine Oil Pressure (ENG OIL PRESS 1) Fluctuates Greater Than 5 PSI Or Is Too High (10 PSI Above Normal Operating Pressure). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	12. Remove No. 2 engine oil pressure sensor and install in No. 1 engine. Go to 13. 13. With qualified personnel at controls, motor No. 1 engine (Appropriate Flight Manual). Go to 14. 14. If a positive oil pressure indication appears during motoring of engine, do a normal start and check for normal oil pressure. Go to 15. 15. Check that oil pressure indication is within limits. Step 1. If oil pressure is within limits, go to 16. Step 2. If oil pressure is not within limits, troubleshoot Engine Oil System (SEI 671 and TM 1-2840-248-23). Go to 22. 16. No. 1 Engine Oil System is operational. Install a new oil pressure sensor in No. 2 engine (SEI 671, and TM 1-2840-248-23). Go to 22. 17. Remove jumper wire. Go to 18. 18. Repair/replace wiring between P138-k and J815-12 (Appendix F). Go to 22. Step 1. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 20. 19. Remove jumper wire. Go to 20. 20. Repair/replace wiring, as required, between the following (Appendix F), then go to 21: P138-i and J815-13 P138-j and J815-14 21. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 22. 22. Procedure completed.



If engine ran for more than 1 minute while oil pressure was below minimum limits, and indicating system was operating properly, replace engine.

TABLE NO. 4-2-2-11. No. 1 Engine Oil Pressure Has No Indication.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Make sure engine’s green harness is connected to sensor. Step 1. If harness is connected, go to TABLE NO. 4-2-2-10. Step 2. If harness is not connected, reconnect harness to sensor. Go to 2.
	2. Procedure completed.

TABLE NO. 4-2-2-12. No. 1 Engine Power Turbine Speed (% RPM 1) Has No Indication.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Troubleshoot Instrument Display System, % RPM 1 scale (PARA 8-2-3). Go to 2.
	2. If display system is good, go to 3.
	3. Install an insulated jumper wire between P801-8 and P801-9. Go to 4.
	4. Check helicopter configuration. Step 1. EMEP Go to 5. Step 2. W/O EMEP Go to 7.
EMEP	5. Remove and check pin filter adapter P138 (PARA 9-4-98). Step 1. If pin filter adapter is good, go to 6. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 13.
EMEP	6. Install pin filter adapter (PARA 9-4-98). Go to 7.

TABLE NO. 4-2-2-12. No. 1 Engine Power Turbine Speed (% RPM 1) Has No Indication. (Cont)

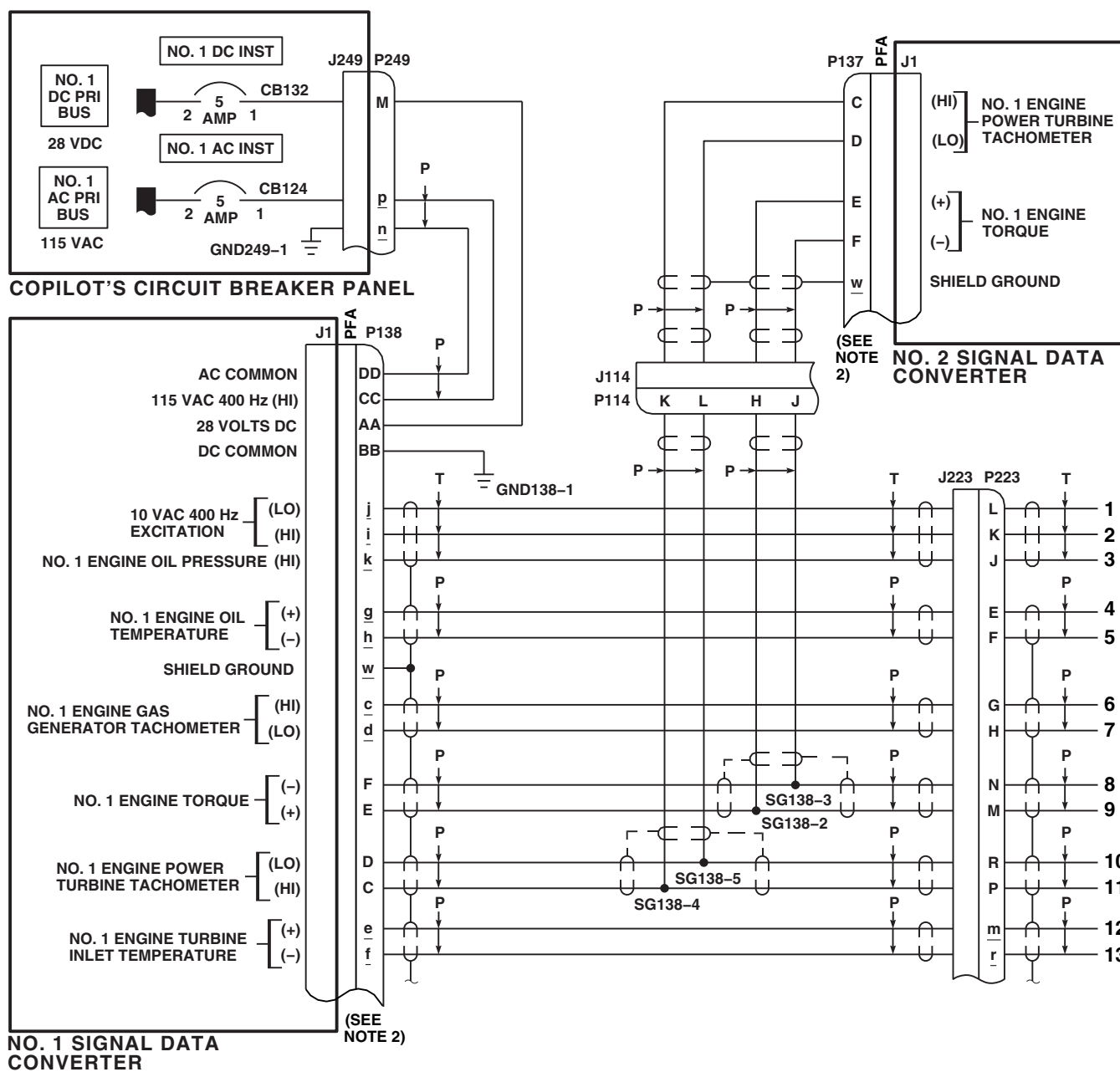
TEST OR INSPECTION	CORRECTIVE ACTION
7. Check continuity between P138-C and P138-D.	Step 1. If continuity is present, go to 8. Step 2. If continuity is not present, go to 10.
8. Remove jumper wire. Go to 9.	
9. Troubleshoot engine Np indicating system (SEI 671 and TM 1-2840-248-23). Go to 13.	
10. Remove jumper wire. Go to 11.	
11. Repair/replace wiring, as required, between the following (Appendix F), then go to 12:	P138-C and J815-17 P138-D and J815-18
12. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 13.	
13. Procedure completed.	

TABLE NO. 4-2-2-13. No. 1 Engine Torque (% TRQ 1) Has No Indication.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Troubleshoot Engine Overspeed Protection System (PARA 4-2-4). Go to 2.	
2. If Engine Overspeed Protection System is good, go to 3.	
3. Troubleshoot Instrument Display System, % TRQ 1 scale (PARA 8-2-3). Go to 4.	
4. If Instrument Display System is good, go to 5.	
5. Repair/replace wiring, as required, between the following (Appendix F), then go to 6:	

TABLE NO. 4-2-2-13. No. 1 Engine Torque (% TRQ 1) Has No Indication. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
P138-E and J815-15 P138-F and J815-16
6. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 7.
7. Procedure completed.



NOTES

1. **T700** **T701A** ELECTRICAL CONTROL UNIT INSTALLED.
T701C DIGITAL ELECTRONIC CONTROL UNIT INSTALLED.
2. **W/O EMEP** PIN FILTER ADAPTER (PFA) IS NOT INSTALLED.

FB1448_1A
SA

FIGURE 4-2-2-1. NO. 1 ENGINE INSTRUMENT SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 2)

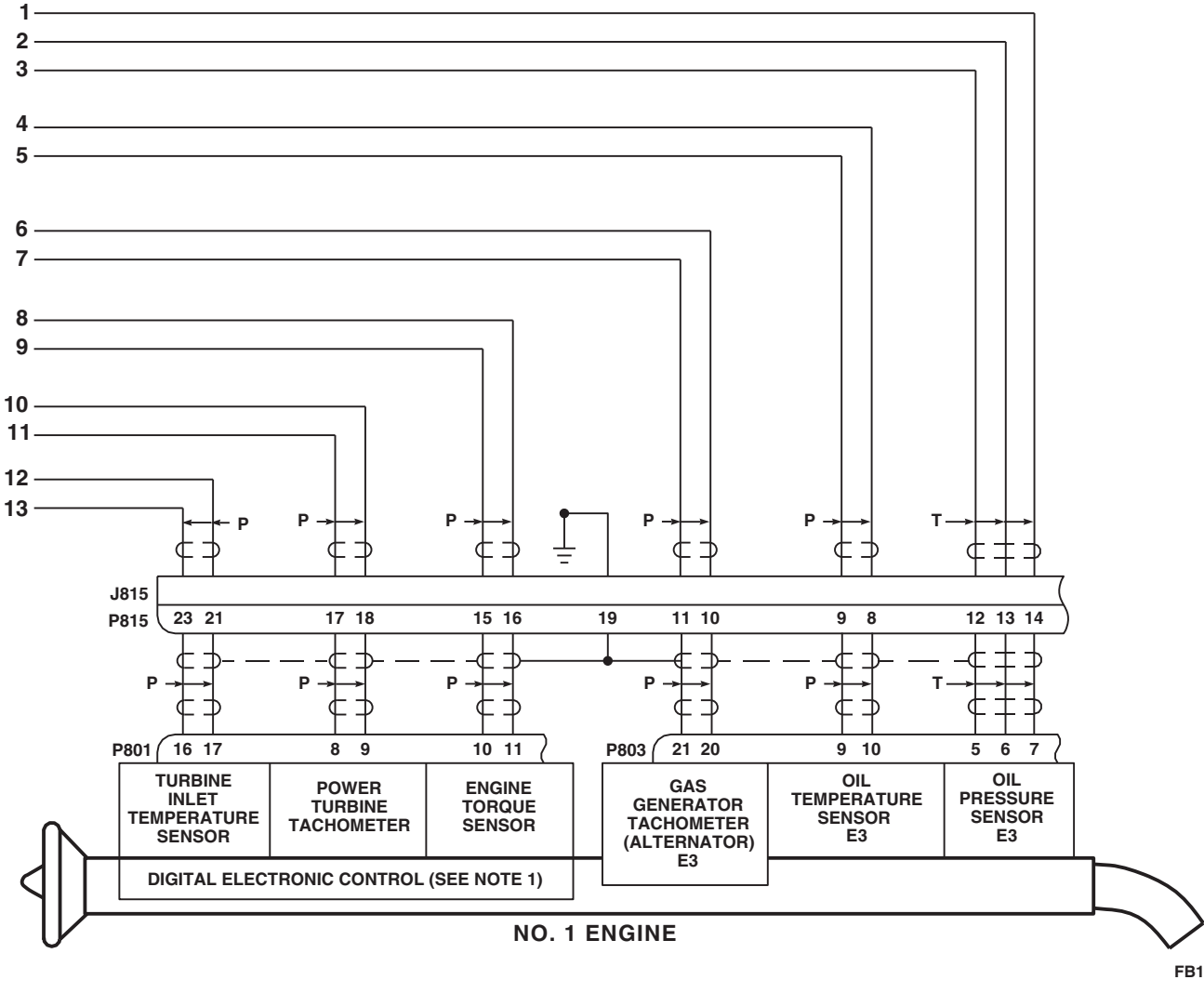


FIGURE 4-2-2-1. NO. 1 ENGINE INSTRUMENT SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 2)

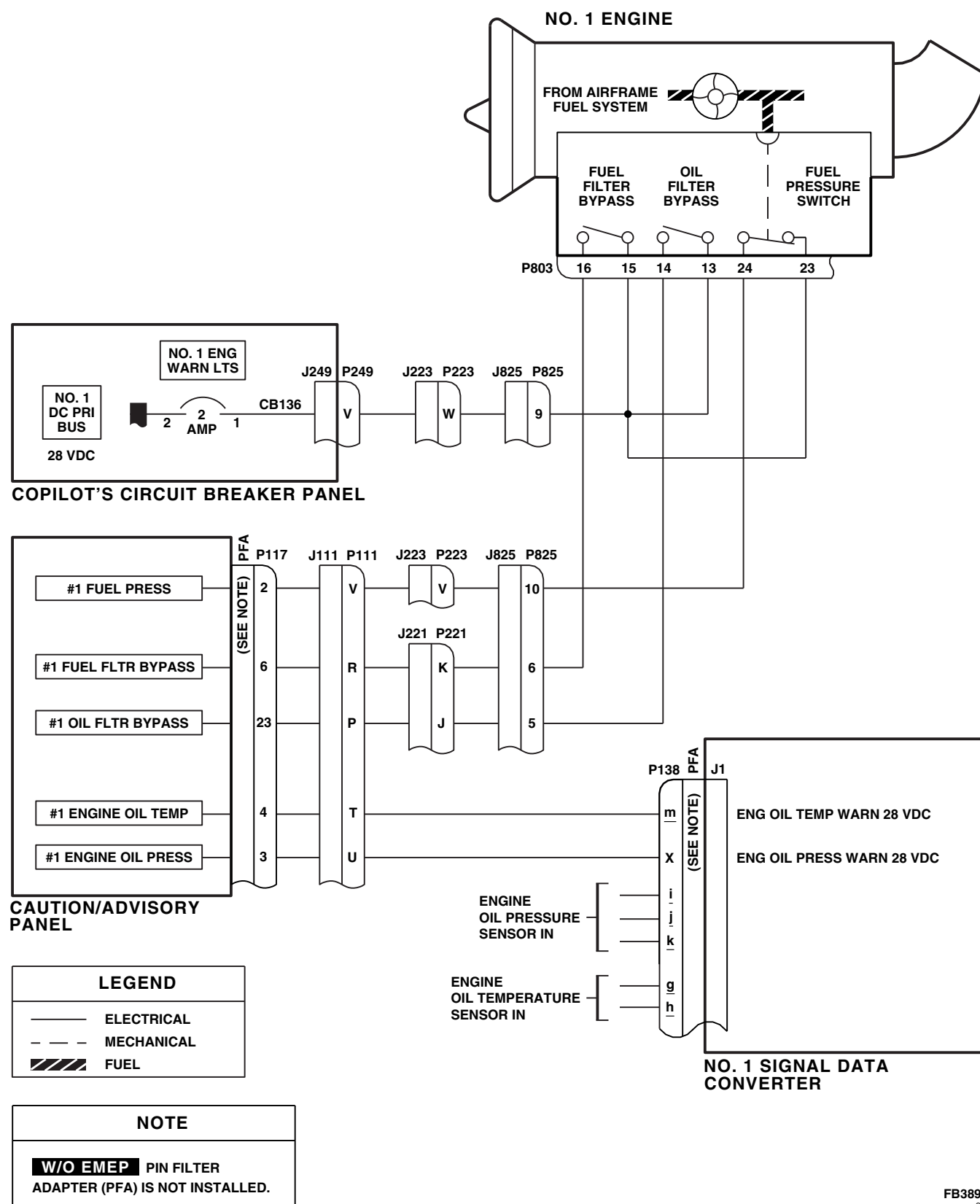
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FIGURE 4-2-2-2. NO. 1 ENGINE WARNING LIGHTS SCHEMATIC DIAGRAM

4-2-3 NO. 2 ENGINE INSTRUMENTS AND WARNING LIGHTS SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer Toolkit
- Engine Repairman’s Toolkit
- Insulated Jumper Wire

References

- Appropriate Flight Manual
- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available or APU Operational
- No. 2 Fuel Tank Serviced

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

4-2-3.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).

RESULT:

- #2 FUEL PRESS capsule on caution/advisory panel shall be on.
- #2 ENGINE OIL PRESS capsule on caution/advisory panel shall be on.
 - If result (1) is not as specified, go to TABLE NO. 4-2-3-1.
 - If result (2) is not as specified, go to TABLE NO. 4-2-3-2.
- Install an insulated jumper wire between P802-13 and P802-14.

RESULT: #2 OIL FLTR BYPASS capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 4-2-3-3.
- Remove jumper wire from P802.
 - Install an insulated jumper wire between P802-15 and P802-16.

RESULT: #2 FUEL FLTR BYPASS capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 4-2-3-4.
- Remove jumper wire from P802.
 - Connect P802 to No. 2 engine.
 - Turn off electrical power (PARA 1-6-16).
 - With qualified personnel at controls, run No. 2 engine up to idle speed (Appropriate Flight Manual).

RESULT: #2 FUEL PRESS capsule shall go off.

- If result is not as specified, go to TABLE NO. 4-2-3-5.

- j. With qualified personnel at controls, run No. 2 engine with main rotor turning at 100% Nr until engine instruments stabilize (Appropriate Flight Manual).

RESULT: Engine status shall be as follows:

CAUTION/ ADVISORY PANEL

CAPSULE STATUS

#2 FUEL PRESS

OFF

#2 ENGINE OIL
PRESS

OFF

#2 ENGINE OIL
TEMP

OFF

#2 FUEL FLTR
BYPASS

OFF

#2 OIL FLTR
BYPASS

OFF

CENTRAL DISPLAY UNIT

INDICATION

Ng SPEED 2
TGT TEMP 2

(Refer to Appropriate MTF for normal operating limits)

ENG OIL TEMP 2

ENG OIL PRESS 2

PILOT'S AND COPILOT'S DIS- PLAY UNIT

INDICATION

% RPM 2
% TRQ 2

(Refer to Appropriate MTF for normal operating limits)

- If any of these caution/advisory panel capsules are on:

CAPSULE

GO TO

#2 ENGINE OIL
PRESS

TABLE NO. 4-2-3-6

#2 ENGINE OIL
TEMP

TABLE NO. 4-2-3-7

- If #2 FUEL PRESS capsule is on, troubleshoot engine for fuel low pressure caution light on during flight (SEI 671 and TM 1-2840-248-23).
- If #2 FUEL FLTR BYPASS or #2 OIL FLTR BYPASS capsule is on, troubleshoot engine (SEI 671 and TM 1-2840-248-23).
- If No. 2 engine oil temperature (ENG OIL TEMP 2) has no indication, go to TABLE NO. 4-2-3-7.
- If No. 2 engine gas generator speed (% Ng 2) has no indication, go to TABLE NO. 4-2-3-8.
- If No. 2 engine turbine inlet temperature (TGT TEMP 2) has no indication, go to TABLE NO. 4-2-3-9.
- If No. 2 engine oil pressure (ENG OIL PRESS 2):
 - (a) Is too high (10 psi above normal), go to TABLE NO. 4-2-3-10.
 - (b) Has no indication, go to TABLE NO. 4-2-3-11.
 - (c) Fluctuates greater than 5 psi, go to TABLE NO. 4-2-3-10.
 - (d) Is below minimum limits (45 psi), go to TABLE NO. 4-2-3-6.
- If No. 2 engine power turbine speed (% RPM 2) has no indication, go to TABLE NO. 4-2-3-12.

- If No. 2 engine has no torque indication (% TRQ 2), go to TABLE NO. 4-2-3-13.
- k. Shut down No. 2 engine (Appropriate Flight Manual).

TABLE NO. 4-2-3-1. #2 FUEL PRESS Capsule Does Not Go On.

TEST OR INSPECTION	
CORRECTIVE ACTION	
1. Check helicopter configuration.	
Step 1. EMEP ➤ Go to 2.◀	
Step 2. W/O EMEP ➤ Go to 4.◀	
EMEP	
2. Remove and check pin filter adapter P117 (PARA 9-4-98).	
Step 1. If pin filter adapter is good, go to 3.	
Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 9.	
EMEP	
3. Install pin filter adapter P117 (PARA 9-4-98). Go to 4.	
4. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST.	
Step 1. If #2 FUEL PRESS capsule goes on, go to 5.	
Step 2. If #2 FUEL PRESS capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 9.	
5. Check for 28 VDC between P117-50 and ground.	
Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 9.	
Step 2. If voltage is not as specified, go to 6.	
6. Check for 28 VDC between P802-23 and ground.	
Step 1. If voltage is as specified, go to 7.	
Step 2. If voltage is not as specified, go to 8.	
7. Check continuity between P117-50 and J824-10.	
Step 1. If continuity is present, replace fuel pressure switch (PARA 10-4-14). Go to 9.	
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 9.	
Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 9.	

TABLE NO. 4-2-3-1. #2 FUEL PRESS Capsule Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
8. Check for 28 VDC between terminal 1 (NO. 2 ENG WARN LTS circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and J824-9 (Appendix F). Go to 9. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 9.
9. Procedure completed.	

TABLE NO. 4-2-3-2. #2 ENGINE OIL PRESS Capsule Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check helicopter configuration.	Step 1. EMEP Go to 2. Step 2. W/O EMEP Go to 4.
2. Remove and check pin filter adapter P117 (PARA 9-4-98).	Step 1. If pin filter adapter is good, go to 3. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 7.
3. Install pin filter adapter P117 (PARA 9-4-98). Go to 4.	
4. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST.	Step 1. If #2 ENGINE OIL PRESS capsule goes on, go to 5. Step 2. If #2 ENGINE OIL PRESS capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 7.
5. Check for 28 VDC between P117-51 and ground.	Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 7. Step 2. If voltage is not as specified, go to 6.

TABLE NO. 4-2-3-2. #2 ENGINE OIL PRESS Capsule Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
6. Check continuity between P117-51 and P137-X.	Step 1. If continuity is present, troubleshoot Instrument Display System (PARA 8-2-3). Go to 7. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.
7. Procedure completed.	

TABLE NO. 4-2-3-3. #2 OIL FLTR BYPASS Capsule Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check helicopter configuration.	Step 1. EMEP Go to 2. W/O EMEP Go to 4.
EMEP 2. Remove and check pin filter adapter P117 (PARA 9-4-98).	Step 1. If pin filter adapter is good, go to 3. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 9.
EMEP 3. Install pin filter adapter P117 (PARA 9-4-98). Go to 4.	
4. Check for 28 VDC between P117-39 and ground.	Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 9. Step 2. If voltage is not as specified, go to 5.
5. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST.	Step 1. If #2 OIL FLTR BYPASS capsule goes on, go to 6. Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 9.

TABLE NO. 4-2-3-3. #2 OIL FLTR BYPASS Capsule Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
6. Remove jumper wire from P802. Go to 7.	
7. Check for continuity between P802-13 and P802-23.	Step 1. If continuity is present, go to 8. Step 2. If continuity is not present, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 9.
8. Check for continuity between P117-39 and P802-14.	Step 1. If continuity is present, troubleshoot Engine Oil Filter Bypass System (SEI 671 and TM 1-2840-248-23). Go to 9. Step 2. If continuity is not present, repair/replace wiring between P117-39 and J824-5 (Appendix F). Go to 9. Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 9.
9. Procedure completed.	

TABLE NO. 4-2-3-4. #2 FUEL FLTR BYPASS Capsule Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check helicopter configuration.	Step 1. EMEP Go to 2. Step 2. W/O EMEP Go to 4.
2. Remove and check pin filter adapter P117 (PARA 9-4-98).	Step 1. If pin filter adapter is good, go to 3. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 8.
3. Install pin filter adapter P117 (PARA 9-4-98). Go to 4.	

TABLE NO. 4-2-3-4. #2 FUEL FLTR BYPASS Capsule Does Not Go On. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
4. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST.	
Step 1. If #2 FUEL FLTR BYPASS capsule goes on, go to 5 . Step 2. If #2 FUEL FLTR BYPASS capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 8 .	
5. Check for 28 VDC between P117-54 and ground.	
Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 8 . Step 2. If voltage is not as specified, go to 6 .	
6. Remove jumper wire from P802. Check for 28 VDC between P802-15 and ground.	
Step 1. If voltage is as specified, go to 7 . Step 2. If voltage is not as specified, repair/replace wiring between terminal 1 (NO. 2 ENG WARN LTS circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and P802-15 (Appendix F). Go to 8 .	
7. Check continuity between P117-54 and P802-16.	
Step 1. If continuity is present, troubleshoot Engine Fuel Filter Bypass System (SEI 671 and TM 1-2840-248-23). Go to 8 . Step 2. If continuity is not present, repair/replace wiring between P117-54 and J824-6 (Appendix F). Go to 8 . Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 8 .	
8. Procedure completed.	

TABLE NO. 4-2-3-5. #2 FUEL PRESS Capsule Stays On When Engine Is At Idle Speed.

TEST OR INSPECTION	CORRECTIVE ACTION
--------------------	-------------------

1. With No. 2 engine operating at ground idle speed, place upper console FUEL PUMP switch to FUEL PRIME. Go to 2.

2. Check that #2 FUEL PRESS capsule goes on.

Step 1. If capsule goes off, go to 3.

Step 2. If capsule does not go off, troubleshoot engine for #2 FUEL PRESS caution capsule at below flight idle speed (SEI 671 and TM 1-2840-248-23). Go to 6.

3. Shut down No. 2 engine (Appropriate Flight Manual). Go to 4.

4. Disconnect electrical connector P802 from No. 2 engine fuel pressure switch. Go to 5.

5. Check that #2 FUEL PRESS capsule goes off.

Step 1. If capsule goes off, troubleshoot Airframe Fuel System for possible leaks between engine and fuel tanks. Repair/replace components as required. Go to 6.

Step 2. If capsule does not go off, repair/replace wiring, as required, (Appendix F) between the following, then go to 6:

P117-50 and J824-10

P236-V and J824-9

Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 6.

Step 4. If trouble still remains, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 6.

6. Procedure completed.

CAUTION

If engine ran for more than 1 minute while oil pressure was below minimum limits, and indicating system was operating properly, replace engine.

TABLE NO. 4-2-3-6. No. 2 Engine Oil Pressure (#2 ENGINE OIL PRESS Capsule Is On) Is Below Minimum.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Visually check engine for external oil leaks.	Step 1. If leaks are present, repair/replace item causing leak, check oil tank level and service as necessary (SEI 671 and TM 1-2840-248-23). Go to 2. Step 2. If leaks are not present, go to TABLE NO. 4-2-3-10.
2. Procedure completed.	

TABLE NO. 4-2-3-7. No. 2 Engine Oil Temperature (#2 ENGINE OIL TEMP Capsule Is On) Is Too High Or Has No Indication.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check engine oil tank for proper level.	Step 1. If oil level is at proper level, go to 2. Step 2. If oil level is not at proper level, troubleshoot Engine Oil System (SEI 671 and TM 1-2840-248-23). Go to 16.
2. Troubleshoot Instrument Display System, ENG OIL TEMP 2 scale (PARA 8-2-3).	Go to 3.
3. If Instrument Display System is good, go to 4.	
4. Install an insulated jumper wire between P802-9 and P802-10. Go to 5.	
5. Check helicopter configuration.	Step 1. EMEP Go to 6.

TABLE NO. 4-2-3-7. No. 2 Engine Oil Temperature (#2 ENGINE OIL TEMP Capsule Is On) Is Too High Or Has No Indication. (Cont)

	TEST OR INSPECTION CORRECTIVE ACTION
	Step 2. W/O EMEP ➔ Go to 8.◀
EMEP	6. Remove and check pin filter adapter P137 (PARA 9-4-98).
	Step 1. If pin filter adapter is good, go to 7.
	Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 18.
EMEP	7. Install pin filter adapter P137 (PARA 9-4-98). Go to 8.
	8. Check continuity between P137-g and P137-h.
	Step 1. If continuity is present, go to 9.
	Step 2. If continuity is not present, go to 15.
	9. Remove jumper wire from P802. Go to 10.
	10. Install No. 1 engine oil temperature sensor into No. 2 engine. Go to 11.
	11. Operate No. 2 engine (Appropriate Flight Manual). Go to 12.
	12. Check that No. 2 engine temperature is normal.
	Step 1. If temperature is normal, go to 13.
	Step 2. If temperature is abnormal, troubleshoot Engine Oil System (SEI 671 and TM 1-2840-248-23). Go to 18.
	13. Install No. 1 engine oil temperature sensor (PARA 4-4-2). Go to 14.
	14. Replace faulty oil temperature sensor in No. 2 engine (PARA 4-4-2). Go to 18.
	15. Remove jumper wire from P802. Go to 16.
	16. Repair/replace wiring, as required, between the following (Appendix F), then go to 17: P137-g and J814-5 P137-h and J814-6
	17. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 18.

TABLE NO. 4-2-3-7. No. 2 Engine Oil Temperature (#2 ENGINE OIL TEMP Capsule Is On) Is Too High Or Has No Indication. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
18. Procedure completed.	

TABLE NO. 4-2-3-8. No. 2 Engine Gas Generator (% Ng 2) Has No Indication.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check helicopter configuration.	
	Step 1. EMEP Go to 2.
	Step 2. W/O EMEP Go to 4.
EMEP	2. Remove and check pin filter adapter P137 (PARA 9-4-98). Step 1. If pin filter adapter is good, go to 3. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 7.
EMEP	3. Install pin filter adapter P137 (PARA 9-4-98). Go to 4.
	4. Check continuity between P137- <u>c</u> and J814-7. Step 1. If continuity is present, go to 5. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 7. Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 7.
	5. Troubleshoot Engine Ng Indicating System (SEI 671 and TM 1-2840-248-23). Step 1. If indicating system is good, go to 6.
	6. Troubleshoot Instrument Display System, % Ng scale (PARA 8-2-3). Go to 7.
	7. Procedure completed.

TABLE NO. 4-2-3-9. No. 2 Engine Turbine Inlet Temperature (TGT TEMP 2) Has No Indication.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Troubleshoot Instrument Display System, TGT TEMP 2 scale (PARA 8-2-3). Go to 2. 2. If Instrument Display System is good, go to 3. 3. Install an insulated jumper wire between P800-16 and P800-17. Go to 4. 4. Check helicopter configuration. Step 1. EMEP Go to 5. Step 2. W/O EMEP Go to 7. 5. Remove and check pin filter adapter P137 (PARA 9-4-98). Step 1. If pin filter adapter is good, go to 6. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 13. 6. Install pin filter adapter P137 (PARA 9-4-98). Go to 7. 7. Check continuity between P137-e and P137-f. Step 1. If continuity is present, go to 8. Step 2. If continuity is not present, go to 10. 8. Remove jumper wire from P800. Go to 9. 9. Troubleshoot Engine Indicating System for TGT (SEI 671 and TM 1-2840-248-23). Go to 13. 10. Remove jumper wire from P800. Go to 11. 11. Repair/replace wiring, as required, between the following (Appendix F), then go to 12: P137-e and J814-21 P137-f and J814-23 12. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 13.

TABLE NO. 4-2-3-9. No. 2 Engine Turbine Inlet Temperature (TGT TEMP 2) Has No Indication. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	13. Procedure completed.

TABLE NO. 4-2-3-10. No. 2 Engine Oil Pressure (ENG OIL PRESS 2) Fluctuates Greater Than 5 PSI Or Is Too High (10 PSI Above Normal Operating Pressure).

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check engine oil tank for proper level. Step 1. If oil has proper level, go to 2. Step 2. If oil does not have proper level, troubleshoot Engine Oil System (SEI 671 and TM 1-2840-248-23). Go to 24.
	2. Troubleshoot Instrument Display System, ENG OIL PRESS 2 scale (PARA 8-2-3). Go to 3.
	3. If Instrument Display System is good, go to 4.
	4. Install an insulated jumper wire between P802-6 and P802-7. Go to 5.
	5. Check helicopter configuration. Step 1. EMEP Go to 6. Step 2. W/O EMEP Go to 8.
EMEP	6. Remove and check pin filter adapter P137 (PARA 9-4-98). Step 1. If pin filter adapter is good, go to 7. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 24.
EMEP	7. Install pin filter adapter P137 (PARA 9-4-98). Go to 8.
	8. Check continuity between P137-i and P137-j. Step 1. If continuity is present, go to 9. Step 2. If continuity is not present, go to 21.

TABLE NO. 4-2-3-10. No. 2 Engine Oil Pressure (ENG OIL PRESS 2) Fluctuates Greater Than 5 PSI Or Is Too High (10 PSI Above Normal Operating Pressure). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
9.	Install an insulated jumper wire between P802-5 and P802-7. Go to 10.
10.	Check continuity between P137-j and P137-k. Step 1. If continuity is present, go to 11. Step 2. If continuity is not present, go to 18.
11.	Remove jumper wire from P802. Go to 12.
12.	Remove No. 1 engine oil pressure sensor and install in No. 2 engine. Go to 13.
13.	With a qualified personnel at the controls, motor No. 2 engine (Appropriate Flight Manual). Go to 14.
14.	If a positive oil pressure indication appears during motoring of engine, do a normal start and check for normal oil pressure. Go to 15.
15.	Check pressure indication during motoring of engine. Go to 16.
16.	If a positive oil pressure indication appears during motoring of engine, do a normal start and go to 17.
17.	Check that oil indication is within limits. Step 1. If indication is within limits, No. 2 Engine Oil System is operational, install a new oil pressure sensor in No. 1 engine (PARA 4-4-2). Go to 24. Step 2. If indication is not within limits, troubleshoot Engine Oil System (SEI 671 and TM 1-2840-248-23). Go to 24.
18.	Remove jumper wire from P802. Go to 19.
19.	Repair/replace wiring between P137-k and J814-9 (Appendix F). Go to 20.
20.	If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 24.
21.	Remove jumper wire from connector P802. Go to 22.
22.	Repair/replace wiring, as required, between the following (Appendix F), then go to 23:

TABLE NO. 4-2-3-10. No. 2 Engine Oil Pressure (ENG OIL PRESS 2) Fluctuates Greater Than 5 PSI Or Is Too High (10 PSI Above Normal Operating Pressure). (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
P137-i and J814-10 P137-j and J814-11
23. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 24.
24. Procedure completed.



If engine ran for more than 1 minute while oil pressure was below minimum limits, and indicating system was operating properly, replace engine.

TABLE NO. 4-2-3-11. No. 2 Engine Oil Pressure Has No Indication.

TEST OR INSPECTION CORRECTIVE ACTION
1. Make sure engine's green harness is connected to sensor.
Step 1. If harness is connected, go to TABLE NO. 4-2-3-10 . Step 2. If harness is not connected, reconnect connector to sensor. Go to 2.
2. Procedure completed.

TABLE NO. 4-2-3-12. No. 2 Engine Power Turbine Speed (%RPM 2) Has No Indication.

TEST OR INSPECTION CORRECTIVE ACTION
1. Troubleshoot Instrument Display System, %RPM 2 scale (PARA 8-2-3). Go to 2.
2. If display system is good, go to 3.

TABLE NO. 4-2-3-12. No. 2 Engine Power Turbine Speed (%RPM 2) Has No Indication. (Cont)

**TEST OR INSPECTION
CORRECTIVE ACTION**

3. Install an insulated jumper wire between P800-8 and P800-9. Go to 4.

4. Check helicopter configuration.

Step 1. **EMEP** Go to 5. 

Step 2. **W/O EMEP** Go to 7. 

EMEP 5. Remove and check pin filter adapter P137 (PARA 9-4-98).

Step 1. If pin filter adapter is good, go to 6.

Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98).
Go to 13.

EMEP 6. Install pin filter adapter P137 (PARA 9-4-98). Go to 7.

7. Check continuity between P137-Y and P137-Z.

Step 1. If continuity is present, go to 8.

Step 2. If continuity is not present, go to 10.

8. Remove jumper wire from P800. Go to 9.

9. Troubleshoot engine Np Indicating System (SEI 671 and TM 1-2840-248-23).
Go to 13.

10. Remove jumper wire from P800. Go to 11.

11. Repair/replace wiring, as required, between the following (Appendix F), then
go to 12:

P137-Y and J814-12

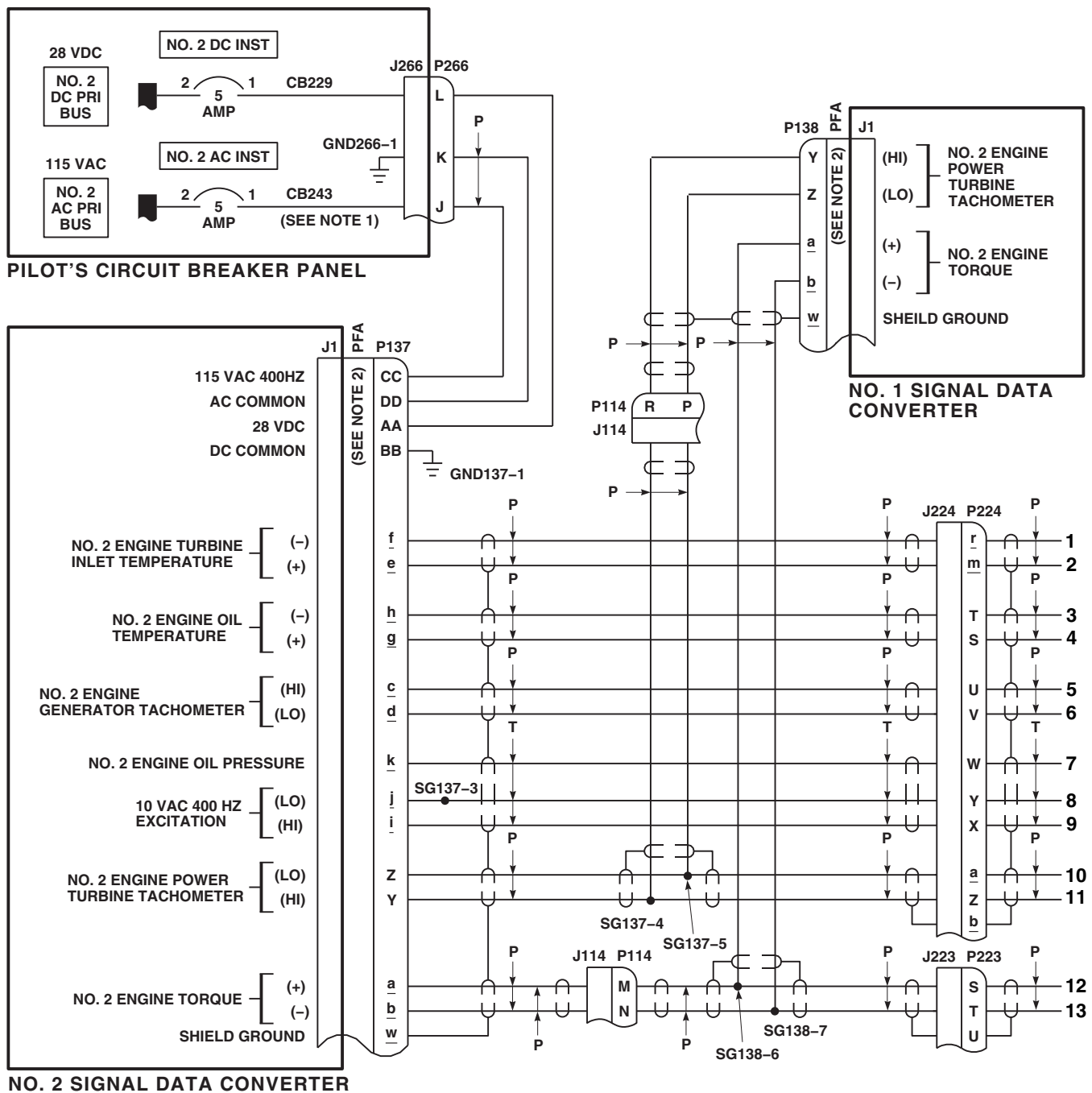
P137-Z and J814-13

12. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23).
Go to 13.

13. Procedure completed.

TABLE NO. 4-2-3-13. No. 2 Engine Torque (% TRQ 2) Has No Indication.

TEST OR INSPECTION CORRECTIVE ACTION
1. Troubleshoot Engine Overspeed Protection System (PARA 4-2-4). Go to 2.
2. If Engine Overspeed Protection System is good, go to 3.
3. Troubleshoot Instrument Display System, % TRQ 2 scale (PARA 8-2-3). Go to 4.
4. If Instrument Display System is good, go to 4.
5. Repair/replace wiring, as required, between the following (Appendix F), then go to 6: P137- <u>a</u> and J814-15 P137- <u>b</u> and J814-14
6. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 7.
7. Procedure completed.

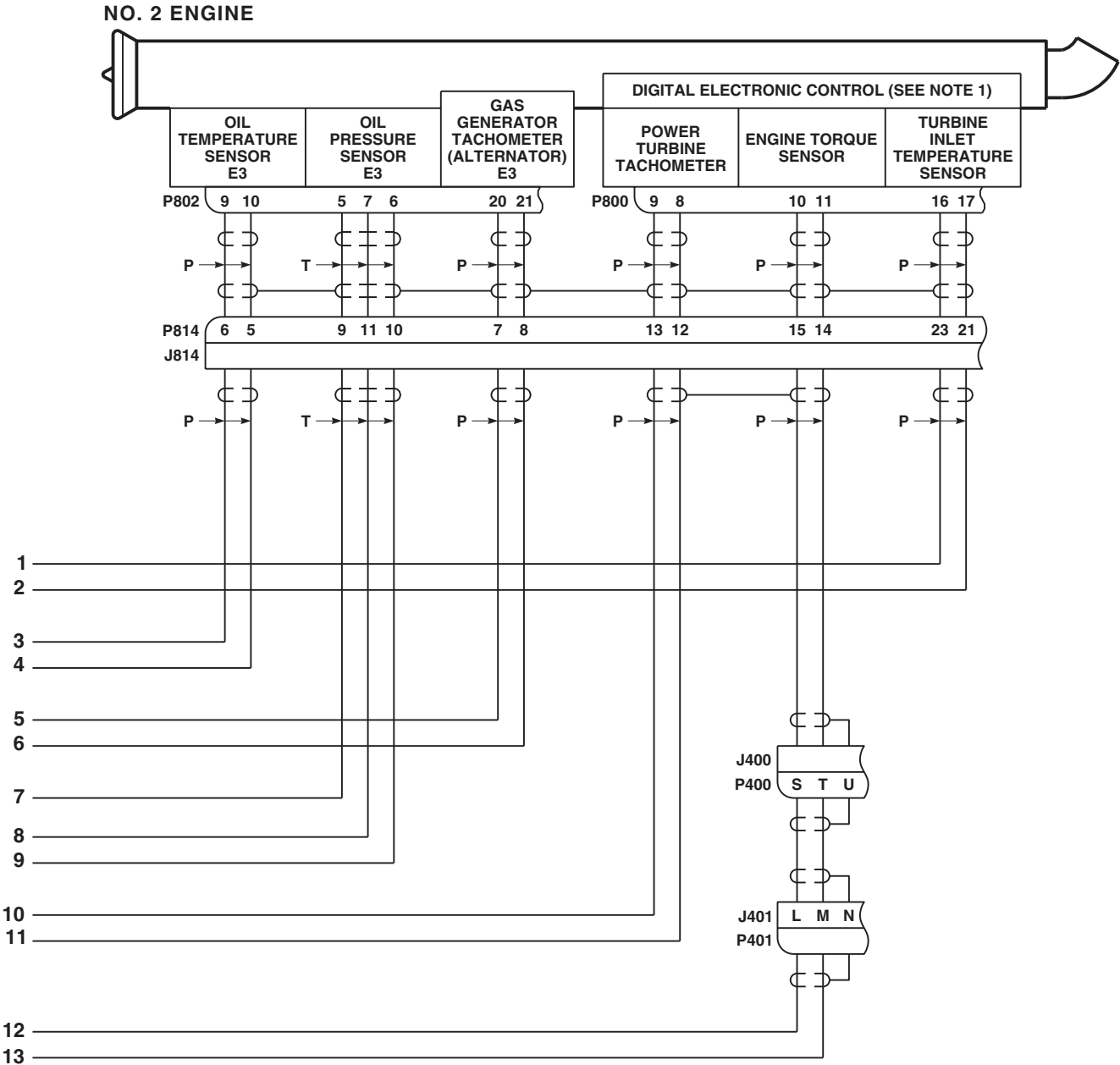


NOTES

1. **T700** **T701A** ELECTRICAL CONTROL UNIT INSTALLED.
T701C DIGITAL ELECTRONIC CONTROL UNIT INSTALLED.
2. **W/O EMEP** PIN FILTER ADAPTER (PFA) IS NOT INSTALLED.

FB1433_1A
SA

FIGURE 4-2-3-1. NO. 2 ENGINE INSTRUMENT SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 2)



FB1433_2B
SA

FIGURE 4-2-3-1. NO. 2 ENGINE INSTRUMENT SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 2)

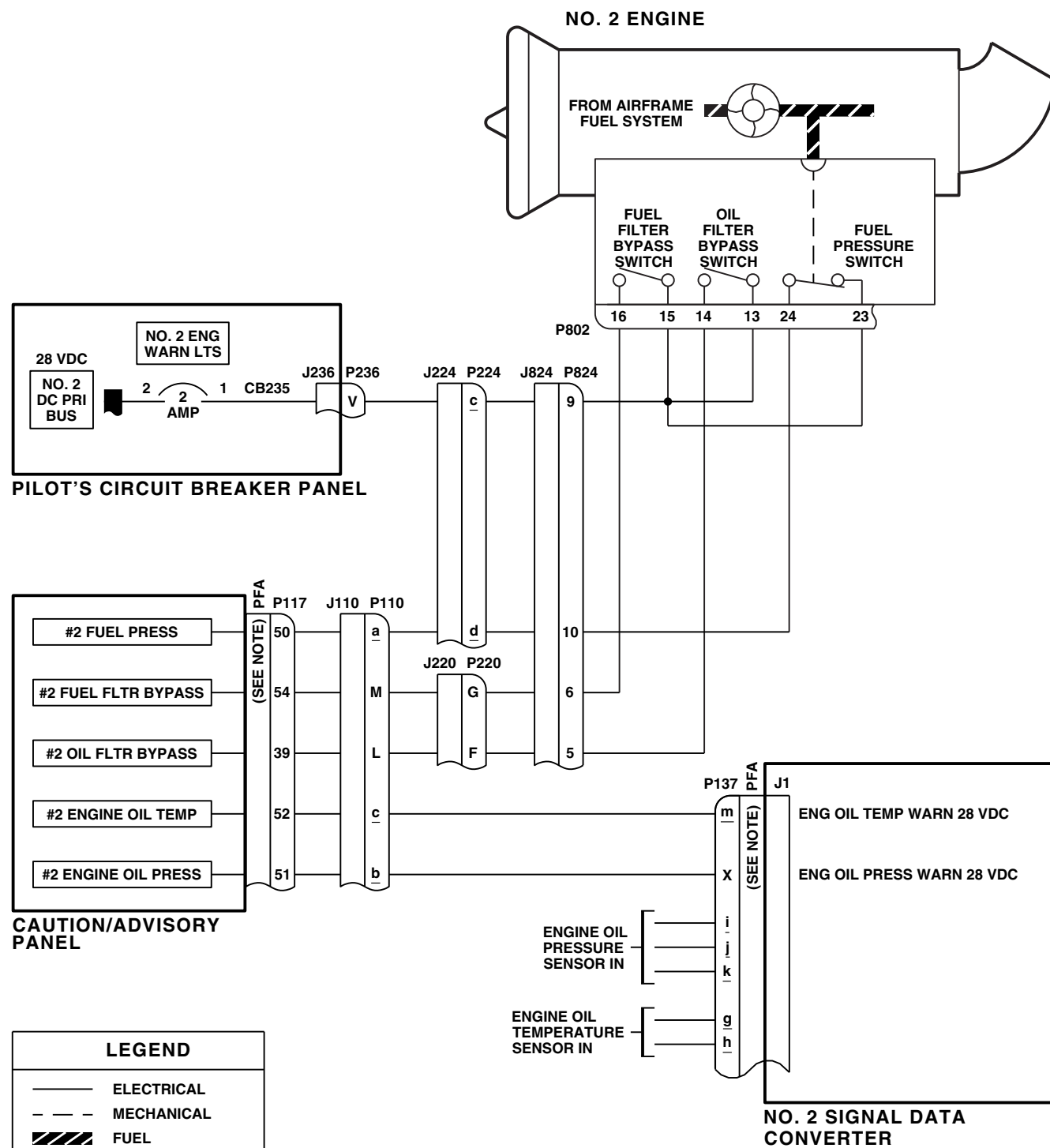
FB3896
SA

FIGURE 4-2-3-2. NO. 2 ENGINE WARNING LIGHTS SYSTEM SCHEMATIC DIAGRAM

4-2-4 ENGINE OVERSPEED PROTECTION SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer Toolkit

References

- Appropriate Flight Manual

Equipment Conditions

- Auxiliary Power Unit (APU) Operational
- No. 1 and No. 2 Engines Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

4-2-4.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- Start APU (Appropriate Flight Manual).
- With qualified personnel at controls, run No. 1 and No. 2 engines at 100% Nr (Appropriate Flight Manual).

RESULT: Engines shall operate normally.

- If either engine does not start, troubleshoot Engine Start And Ignition System (PARA 4-2-8).
- d. Press upper console NO. 1 ENG OVSP TEST A button.

RESULT: No. 1 engine gas generator speed (Ng SPEED 1) shall not decrease.

- If result is not as specified, go to TABLE NO. 4-2-4-1.
- e. Press upper console NO. 1 ENG OVSP TEST B button.

RESULT: No. 1 engine gas generator speed (Ng SPEED 1) shall not decrease.

- If result is not as specified, go to TABLE NO. 4-2-4-2.
- f. Simultaneously press upper console NO. 1 ENG OVSP TEST A and NO. 1 ENG OVSP TEST B buttons.

RESULT:

- (1) **S70-II** The engine shall have a reduction in Np, Ng, and torque.■
- (2) No. 1 gas generator speed (Ng) shall cycle.
- (3) When test buttons are released, the engine shall reignite and speed shall return to normal.

- If result (1) is not as specified, go to TABLE NO. 4-2-4-3.
- If results (2) and (3) are not as specified, troubleshoot engine (SEI 671 and TM 1-2840-248-23).
- g. Press upper console NO. 2 ENG OVSP TEST A button.

RESULT: No. 2 engine gas generator speed (Ng SPEED 2) shall not decrease.

- If result is not as specified, go to TABLE NO. 4-2-4-4.

h. Press upper console NO. 2 ENG OVSP TEST B button.

RESULT: No. 2 engine gas generator speed (Ng SPEED 2) shall not decrease.

- If result is not as specified, go to TABLE NO. 4-2-4-5.

i. Simultaneously press upper console NO. 2 ENG OVSP TEST A and NO. 2 ENG OVSP TEST B buttons.

RESULT:

- (1) **S70-II** The engine shall have a reduction in Np, Ng, and torque.■
- (2) No. 2 gas generator speed (Ng) and/or power turbine speed (Np) shall cycle.

(3) When test buttons are released, the engine shall reignite and speed shall return to normal.

- If result (1) is not as specified, go to TABLE NO. 4-2-4-6.

- If results (2) and (3) are not as specified, troubleshoot engine (SEI 671 and TM 1-2840-248-23).

j. Shut down engines (Appropriate Flight Manual).

RESULT: **S70-II** PDU shall not indicate BITE codes.■

- If No. 1 BITE shows 85%, go to TABLE NO. 4-2-4-7.

- If No. 2 BITE shows 85%, go to TABLE NO. 4-2-4-8.

k. Shut down APU (Appropriate Flight Manual).

TABLE NO. 4-2-4-1. No. 1 Engine Overspeed Cuts In When NO. 1 ENG OVSP TEST A Button Is Pressed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between P801-6 and P801-14.	Step 1. If continuity is present, go to 2. Step 2. If continuity is not present, troubleshoot Engine Overspeed Test System (SEI 671 and TM 1-2840-248-23). Go to 4.
2. Check continuity between terminal 1 (NO. 1 ENG OVSP TEST B switch) and P801-14.	Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, repair/replace wiring between terminal 1 and J815-26 (Appendix F). Go to 4. Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 4.

TABLE NO. 4-2-4-1. No. 1 Engine Overspeed Cuts In When NO. 1 ENG OVSP TEST A Button Is Pressed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
3. Check continuity between terminal 2 (NO. 1 ENG OVSP TEST B switch) and P801-6.	Step 1. If continuity is present, replace switch (PARA 9-4-5). Go to 4. Step 2. If continuity is not present, repair/replace wiring between terminal 2 and J815-27 (Appendix F). Go to 4. Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 4.
4. Procedure completed.	

TABLE NO. 4-2-4-2. No. 1 Engine Overspeed Cuts In When NO. 1 ENG OVSP TEST B Button Is Pressed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between P801-6 and P801-12.	Step 1. If continuity is present, go to 2. Step 2. If continuity is not present, troubleshoot Engine Overspeed Test System (SEI 671 and TM 1-2840-248-23). Go to 4.
2. Check continuity between terminal 1 (NO 1. ENG OVSP TEST A switch) and P801-12.	Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, repair/replace wiring between terminal 1 and J815-25 (Appendix F). Go to 4. Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 4.
3. Check continuity between terminal 2 (NO 1. ENG OVSP TEST A switch) and P801-6.	Step 1. If continuity is present, replace switch S17 (PARA 9-4-5). Go to 4. Step 2. If continuity is not present, repair/replace wiring between terminal 2 and J815-27 (Appendix F). Go to 4.

TABLE NO. 4-2-4-2. No. 1 Engine Overspeed Cuts In When NO. 1 ENG OVSP TEST B Button Is Pressed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 4. 4. Procedure completed.

TABLE NO. 4-2-4-3. No. 1 Engine Does Not Shut Down With Overspeed Condition.

S70-II	TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check for 28 VDC between P801-28 and gnd.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 8.
	2. Check for 28 VDC between P381-C and P381-B.	Step 1. If voltage is as specified, go to 3. Step 2. If voltage is not as specified, go to 5.
	3. Check for 28 VDC between P381-D and P381-B.	Step 1. If voltage is as specified, replace No. 1 overspeed relay assembly (PARA 9-4-97). Go to 9. Step 2. If voltage is not as specified, go to 4.
	4. Check for 28 VDC between terminal 1 (NO. 1 ENG START circuit breaker, DC ESNTL BUS - upper console circuit breaker panel) and gnd.	Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P381-D (Appendix F). Go to 9. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 9.
	5. Check for 28 VDC between P381-C and gnd.	Step 1. If voltage is as specified, repair/replace wiring between P381-B and gnd (Appendix F). Go to 9.

TABLE NO. 4-2-4-3. No. 1 Engine Does Not Shut Down With Overspeed Condition. (Cont)

S70-II	TEST OR INSPECTION CORRECTIVE ACTION
	Step 2. If voltage is not as specified, go to 6. 6. Check continuity between P381-C and J815-1. Step 1. If continuity is present, go to 7. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 9. 7. Check continuity between P801-27 and P815-1. Step 1. If continuity is present, troubleshoot engine (SEI 671 and TM 1-2840-248-23). Go to 9. Step 2. If continuity is not present, replace engine harness SEI 671 and TM 1-2840-248-23). Go to 9. 8. Check for 28 VDC between J815-2 and gnd. Step 1. If voltage is as specified, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 9. Step 2. If voltage is not as specified, repair/replace wiring between J815-2 and terminal 1 (NO. 1 ENG START circuit breaker, DC ESNTL BUS - upper console circuit breaker panel) (Appendix F). Go to 9. 9. Procedure completed.

TABLE NO. 4-2-4-4. No. 2 Engine Overspeed Cuts In When NO. 2 ENG OVSP TEST A Button Is Pressed.

TEST OR INSPECTION CORRECTIVE ACTION
1. Check continuity between P800-6 and P800-14. Step 1. If continuity is present, go to 2. Step 2. If continuity is not present, troubleshoot Engine Overspeed Test System (SEI 671 and TM 1-2840-248-2). Go to 4.

TABLE NO. 4-2-4-4. No. 2 Engine Overspeed Cuts In When NO. 2 ENG OVSP TEST A Button Is Pressed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
2. Check continuity between terminal 1 (NO. 2 ENG OVSP TEST B switch) and P800-14.	Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, repair/replace wiring between terminal 1 and J814-28 (Appendix F). Go to 4. Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 4.
3. Check continuity between terminal 2 (NO. 2 ENG OVSP TEST B switch) and P800-6.	Step 1. If continuity is present, replace switch S13 (PARA 9-4-5). Go to 4. Step 2. If continuity is not present, repair/replace wiring between terminal 2 and J814-29 (Appendix F). Go to 4. Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 4.
4. Procedure completed.	

TABLE NO. 4-2-4-5. No. 2 Engine Overspeed Cuts In When NO. 2 ENG OVSP TEST B Button Is Pressed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between P800-6 and P800-12.	Step 1. If continuity is present, go to 2. Step 2. If continuity is not present, troubleshoot Engine Overspeed Test System (SEI 671 and TM 1-2840-248-23). Go to 4.
2. Check continuity between terminal 1 (NO. 2 ENG OVSP TEST A switch) and P800-12.	Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, repair/replace wiring between terminal 1 and J814-27 (Appendix F). Go to 4.

TABLE NO. 4-2-4-5. No. 2 Engine Overspeed Cuts In When NO. 2 ENG OVSP TEST B Button Is Pressed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 4.
3.	Check continuity between terminal 2 (NO. 2 ENG OVSP TEST A switch) and P800-6. Step 1. If continuity is present, replace switch (PARA 9-4-5). Go to 4. Step 2. If continuity is not present, repair/replace wiring between terminal 2 and J814-29 (Appendix F). Go to 4. Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 4.
4.	Procedure completed.

TABLE NO. 4-2-4-6. No. 2 Engine Does Not Shut Down With Overspeed Condition.

S70-II	TEST OR INSPECTION	CORRECTIVE ACTION
	1.	Check for 28 VDC between P800-28 and gnd. Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 8.
	2.	Check for 28 VDC between P382-C and P382-B. Step 1. If voltage is as specified, go to 3. Step 2. If voltage is not as specified, go to 5.
	3.	Check for 28 VDC between P382-D and P382-B. Step 1. If voltage is as specified, replace No. 2 overspeed relay assembly (PARA 9-4-97). Go to 9. Step 2. If voltage is not as specified, go to 4.

TABLE NO. 4-2-4-6. No. 2 Engine Does Not Shut Down With Overspeed Condition. (Cont)

S70-II	TEST OR INSPECTION CORRECTIVE ACTION
I	4. Check for 28 VDC between terminal 1 (NO. 2 ENG START CONTR circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and gnd. Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P382-D (Appendix F). Go to 9. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 9.
I	5. Check for 28 VDC between P382-C and gnd. Step 1. If voltage is as specified, repair/replace wiring between P382-B and gnd (Appendix F). Go to 9. Step 2. If voltage is not as specified, go to 6.
I	6. Check continuity between J814-1 and P382-C. Step 1. If continuity is present, go to 7. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 9.
I	7. Check continuity between P800-27 and P814-1. Step 1. If continuity is present, troubleshoot engine (SEI 671 and TM 1-2840-248-23). Go to 9. Step 2. If continuity is not present, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 9. 8. Check for 28 VDC between J814-2 and gnd. Step 1. If voltage is as specified, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 9. Step 2. If voltage is not as specified, repair/replace wiring between J814-2 and terminal 1 (NO. 2 ENG START CONTR circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) (Appendix F). Go to 9.
	9. Procedure completed.

TABLE NO. 4-2-4-7. No. 1 Engine BITE Code Indicates 85%.

S70-II	TEST OR INSPECTION CORRECTIVE ACTION
	1. Perform TABLE NO. 4-2-4-3. Go to 2. 2. Check continuity between J815-35 and P381-F. Step 1. If continuity is present, go to 3. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 5. 3. Check continuity between: P803-17 and P815-35 P803-19 and P815-35 Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 5. 4. Check continuity between P381-G and terminal 2 (engine ignition switch). Step 1. If continuity is present, replace No. 1 overspeed relay assembly (PARA 9-4-97). Go to 5. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 5. 5. Procedure completed.

TABLE NO. 4-2-4-8. No. 2 Engine BITE Code Indicates 85%.

S70-II	TEST OR INSPECTION CORRECTIVE ACTION
	1. Perform TABLE NO. 4-2-4-6. Go to 2. 2. Check continuity between J814-35 and P382-F. Step 1. If voltage is as specified, go to 3. Step 2. If voltage is not as specified, repair/replace wiring (Appendix F). Go to 5. 3. Check continuity between: P802-17 and P814-35 P802-19 and P814-35

TABLE NO. 4-2-4-8. No. 2 Engine BITE Code Indicates 85%. (Cont)

S70-II**TEST OR INSPECTION
CORRECTIVE ACTION**

Step 1. If continuity is present, go to **4**.

Step 2. If continuity is not present, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to **5**.

4. Check continuity between P382-G and terminal 5 (engine ignition switch).

Step 1. If continuity is present, replace No. 2 overspeed relay assembly (PARA 9-4-97). Go to **5**.

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **5**.

5. Procedure completed.

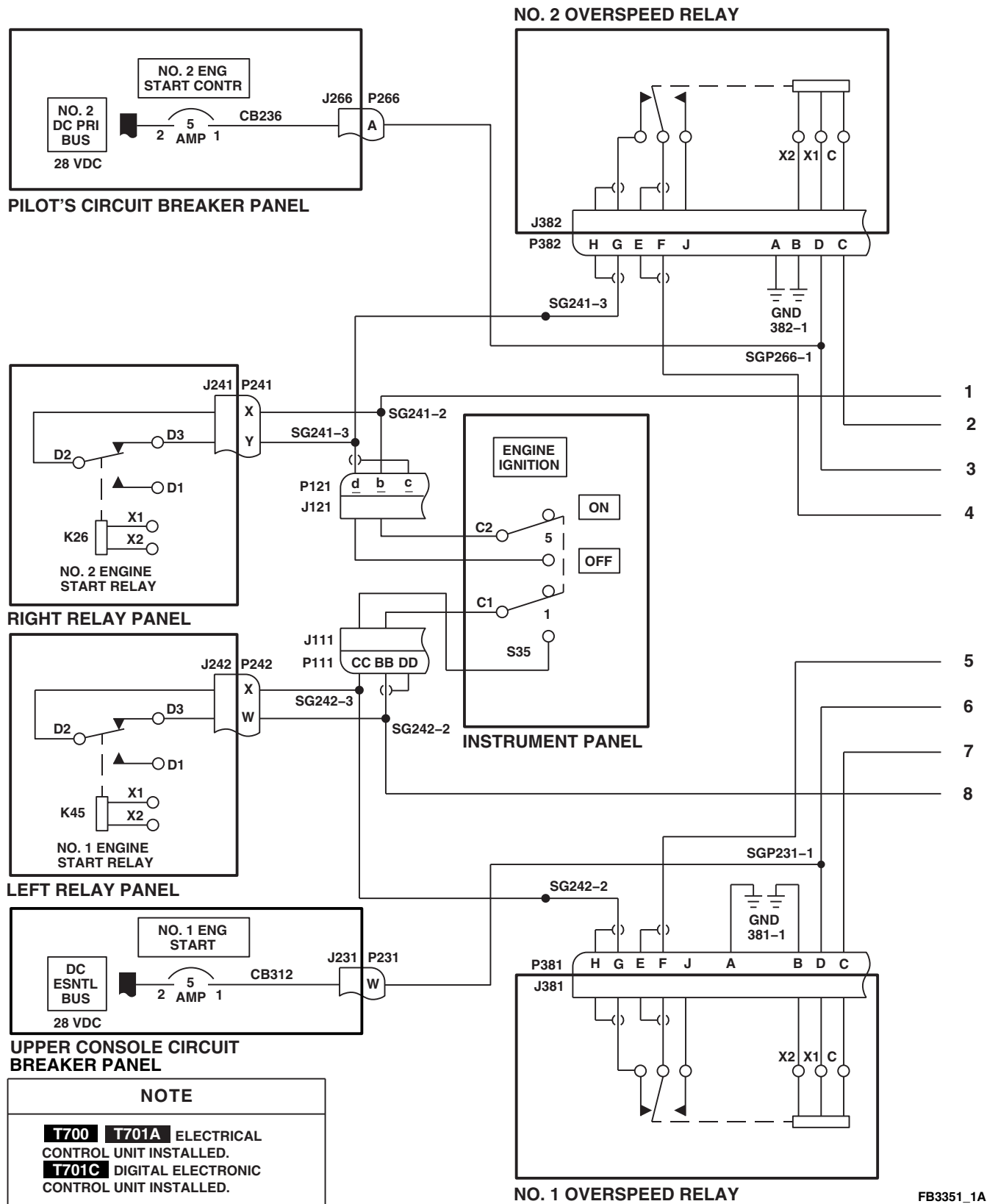


FIGURE 4-2-4-1. ENGINE OVERSPEED PROTECTION SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 3)

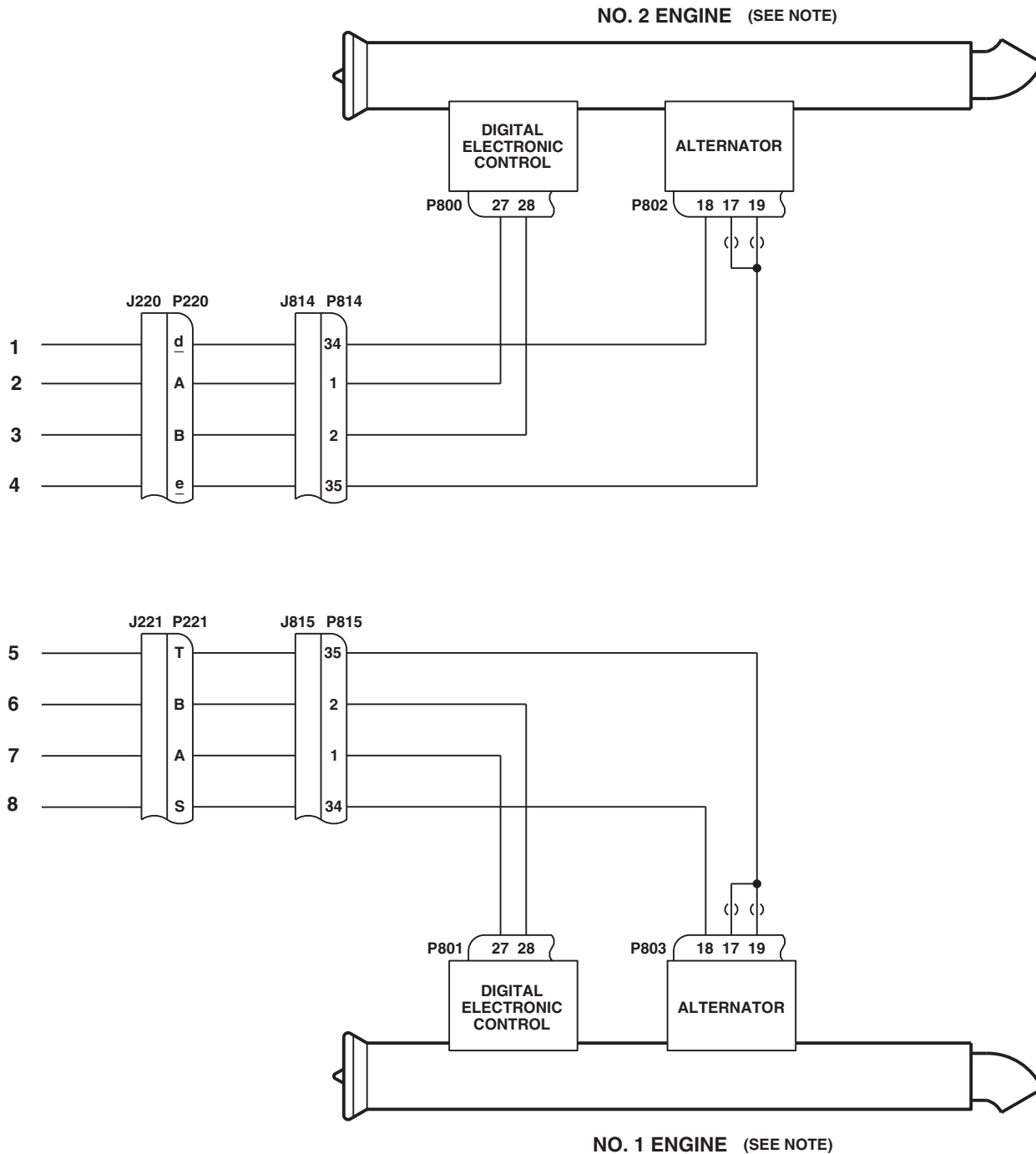
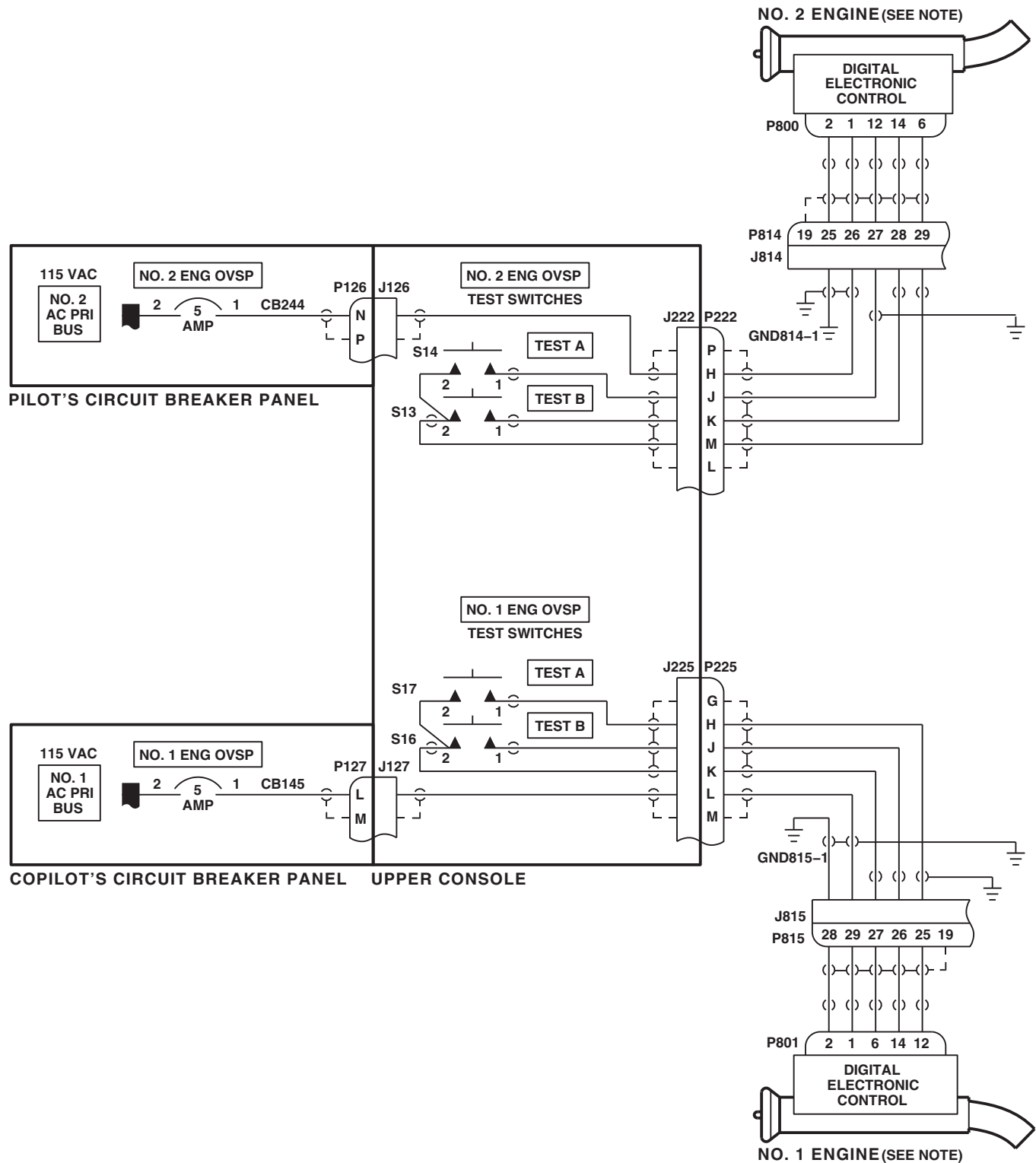
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FIGURE 4-2-4-1. ENGINE OVERSPEED PROTECTION SYSTEM SCHEMATIC DIAGRAM
(SHEET 2 OF 3)

4-2-4-12 Change 8



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FIGURE 4-2-4-1. ENGINE OVERSPEED PROTECTION SYSTEM SCHEMATIC DIAGRAM
(SHEET 3 OF 3)

4-2-5 ENGINE SPEED TRIM SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer Toolkit
- Engine Repairer’s Toolkit

References

- Appropriate Flight Manual

Equipment Conditions

- Auxiliary Power Unit (APU) Operational
- No. 1 and No. 2 Engines Operational
- No. 1 and No. 2 Fuel Tanks Serviced

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

NOTE

If one engine will not decrease below 100% turbine speed (Np), an associated torque split may be noticed.

4-2-5.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- Start APU (Appropriate Flight Manual).
- With qualified personnel at controls, start No. 1 and No. 2 engines. Operate main rotor at 100% rotor speed (Nr) until instruments stabilize (Appropriate Flight Manual).
- Push and hold pilot’s (or copilot’s) collective stick ENG RPM switch to DECR.

RESULT:

- % RPM 1 (Np) indication on both pilot display unit (PDUs) shall decrease to 96% (bottom of green range).
- % RPM 2 (Np) indication on both PDUs shall decrease to 96% (bottom of green range).

- No. 1 or No. 2 engine Np shall not go below 96% Np.
- If results (1) and (2) are not as specified, go to TABLE NO. 4-2-5-1.
- If result (1) is not as specified, go to TABLE NO. 4-2-5-2.
- If result (2) is not as specified, go to TABLE NO. 4-2-5-3.
- If result (3) is not as specified, replace engine speed control box (PARA 4-4-20).
- Push and hold pilot’s (or copilot’s) collective stick ENG RPM switch to INCR.

RESULT:

- % RPM 1 (Np) indication on both PDUs shall increase to 100% (top of green range).
- % RPM 2 (Np) indication on both PDUs shall increase to 100% (top of green range).

- If result (1) is not as specified, go to TABLE NO. 4-2-5-2.
- If result (2) is not as specified, go to TABLE NO. 4-2-5-3.
- f. Shut down both engines (Appropriate Flight Manual).
- g. Shut down APU (Appropriate Flight Manual).

TABLE NO. 4-2-5-1. No. 1 And No. 2 % Np Cannot Be Decreased Below 100%.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Place and hold copilot's collective stick ENG RPM switch to DECR. Go to 2.	
2. Check for 28 VDC between P901-J and ground.	
	Step 1. If voltage is as specified, go to 3. Step 2. If voltage is not as specified, go to 5.
3. Release copilot's collective stick ENG RPM switch. Go to 4.	
4. Check continuity between P901-L and ground.	
	Step 1. If continuity is present, replace engine speed control box (PARA 4-4-20). Go to 14. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 14.
5. Place and hold pilot's collective stick ENG RPM switch to DECR. Go to 6..	
6. Check for 28 VDC between P901-J and ground.	
	Step 1. If voltage is as specified, go to 7. Step 2. If voltage is not as specified, go to 11.
7. Release pilot's collective stick ENG RPM switch. Go to 8.	
8. Check for 28 VDC between J103-k and ground.	
	Step 1. If voltage is as specified, go to 9. Step 2. If voltage is not as specified, go to 10.
9. Check continuity between:	
	J103-h and J103-j J103-i and P901-J

TABLE NO. 4-2-5-1. No. 1 And No. 2 % Np Cannot Be Decreased Below 100%. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
Step 1. If continuity is present, troubleshoot copilot's collective stick (PARA 11-2-2). Go to 14 .	Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 14 .
10. Check continuity between J104-g and J103-k.	
Step 1. If continuity is present, troubleshoot pilot's collective stick (PARA 11-2-2). Go to 14 .	
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 14 .	
11. Check for 28 VDC between J104-k and ground.	
Step 1. If voltage is as specified, go to 12 .	
Step 2. If voltage is not as specified, go to 13 .	
12. Check continuity between: J104-h and J104-j J104-i and P901-J	
Step 1. If continuity is present, troubleshoot pilot's collective stick (PARA 11-2-2). Go to 14 .	
Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 14 .	
13. Check continuity between terminal 1 (SPEED TRIM circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and J104-k.	
Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to 14 .	
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 14 .	
14. Procedure completed.	

TABLE NO. 4-2-5-2. No. 1 % Np Cannot Be Increased Or Decreased.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check that power turbine (% Np) can be increased or decreased.	Step 1. If power turbine speed cannot be increased, go to 2. Step 2. If power turbine speed cannot be decreased, go to 5.
2. Push pilot's or copilot's ENG RPM switch to INCR until engine speed control motor stops. Go to 3.	
3. Check resistance between:	P801-18 and P801-20 (2300 ± 100 ohms (Ω)) P801-19 and P801-20 (250 ± 10 Ω) P801-20 and P801-21 (133 ± 9 Ω) Step 1. If resistance is as specified, troubleshoot engine (Np does not respond to Np demand trim) (Sikorsky Engineering Instructions (SEI) 671 and TM 55-2840-248-23). Go to 8. Step 2. If resistance is not as specified, go to 4.
4. Check for resistance of 133 ± 9 Ω between J901-A and J901-B.	Step 1. If resistance is as specified, repair/replace wiring, as required, between the following (Appendix F), then go to 8: J815-30 and P901-D J815-31 and P901-C J815-32 and P901-B J815-33 and P901-A Step 2. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 8. Step 3. If resistance is not as specified, replace engine speed control box (PARA 4-4-20). Go to 8.
5. Push pilot's or copilot's ENG RPM switch to DECR until engine speed control motor stops. Go to 6.	

TABLE NO. 4-2-5-2. No. 1 % Np Cannot Be Increased Or Decreased. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
6. Check resistance between:	P801-18 and P801-20 ($2300 \pm 100 \Omega$) P801-19 and P801-20 ($250 \pm 10 \Omega$) P801-20 and P801-21 ($880 \pm 20 \Omega$) Step 1. If resistance is as specified, troubleshoot engine (Np does not respond to Np demand trim) (SEI 671 and TM 55-2840-248-23). Go to 8. Step 2. If resistance is not as specified, go to 7.
7. Check for resistance of $2300 \pm 115 \Omega$ between J901-C and J901-D.	Step 1. If resistance is as specified, repair/replace wiring, as required, between the following (Appendix F), then go to 8: J815-30 and P901-D J815-31 and P901-C J815-32 and P901-B J815-33 and P901-A Step 2. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 8. Step 3. If resistance is not as specified, replace engine speed control box (PARA 4-4-20). Go to 8.
8. Procedure completed.	

TABLE NO. 4-2-5-3. No. 2 % Np Cannot Be Increased Or Decreased.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check that power turbine speed (% Np) can be increased or decreased.	Step 1. If power turbine speed cannot be increased, go to 2. Step 2. If power turbine speed cannot be decreased, go to 5.
2. Push pilot's or copilot's ENG RPM switch to INCR until engine speed control motor stops. Go to 3.	

TABLE NO. 4-2-5-3. No. 2 % Np Cannot Be Increased Or Decreased. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
3. Check resistance between:	P800-18 and P800-20 ($2300 \pm 100 \Omega$) P800-19 and P800-20 ($250 \pm 10 \Omega$) P800-20 and P800-21 ($133 \pm 9 \Omega$) Step 1. If resistance is as specified, troubleshoot engine (Np does not respond to Np demand trim) (SEI 671 and TM 55-2840-248-23). Go to 8. Step 2. If resistance is not as specified, go to 4.
4. Check for resistance of $133 \pm 9 \Omega$ between J901-E and J901-F.	Step 1. If resistance is as specified, repair/replace wiring, as required, between the following (Appendix F), then go to 8: J814-30 and P901-H J814-31 and P901-G J814-32 and P901-F J814-33 and P901-E Step 2. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 8. Step 3. If resistance is not as specified, replace engine speed control box (PARA 4-4-20). Go to 8.
5. Push pilot's or copilot's ENG RPM switch to DECR until engine speed control motor stops. Go to 6.	
6. Check resistance between:	P800-18 and P800-20 ($2300 \pm 100 \Omega$) P800-19 and P800-20 ($250 \pm 10 \Omega$) P800-20 and P800-21 ($880 \pm 20 \Omega$) Step 1. If resistance is as specified, troubleshoot engine (Np does not respond to Np demand trim) (SEI 671 and TM 55-2840-248-23). Go to 8. Step 2. If resistance is not as specified, go to 7.
7. Check for resistance of $2300 \pm 115 \Omega$ between J901-G and J901-H.	Step 1. If resistance is as specified, repair/replace wiring, as required, between the following (Appendix F), then go to 8: J814-30 and P901-H J814-31 and P901-G J814-32 and P901-F J814-33 and P901-E

TABLE NO. 4-2-5-3. No. 2 % Np Cannot Be Increased Or Decreased. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 8.
	Step 3. If resistance is not as specified, replace engine speed control box (PARA 4-4-20). Go to 8.
8. Procedure completed.	

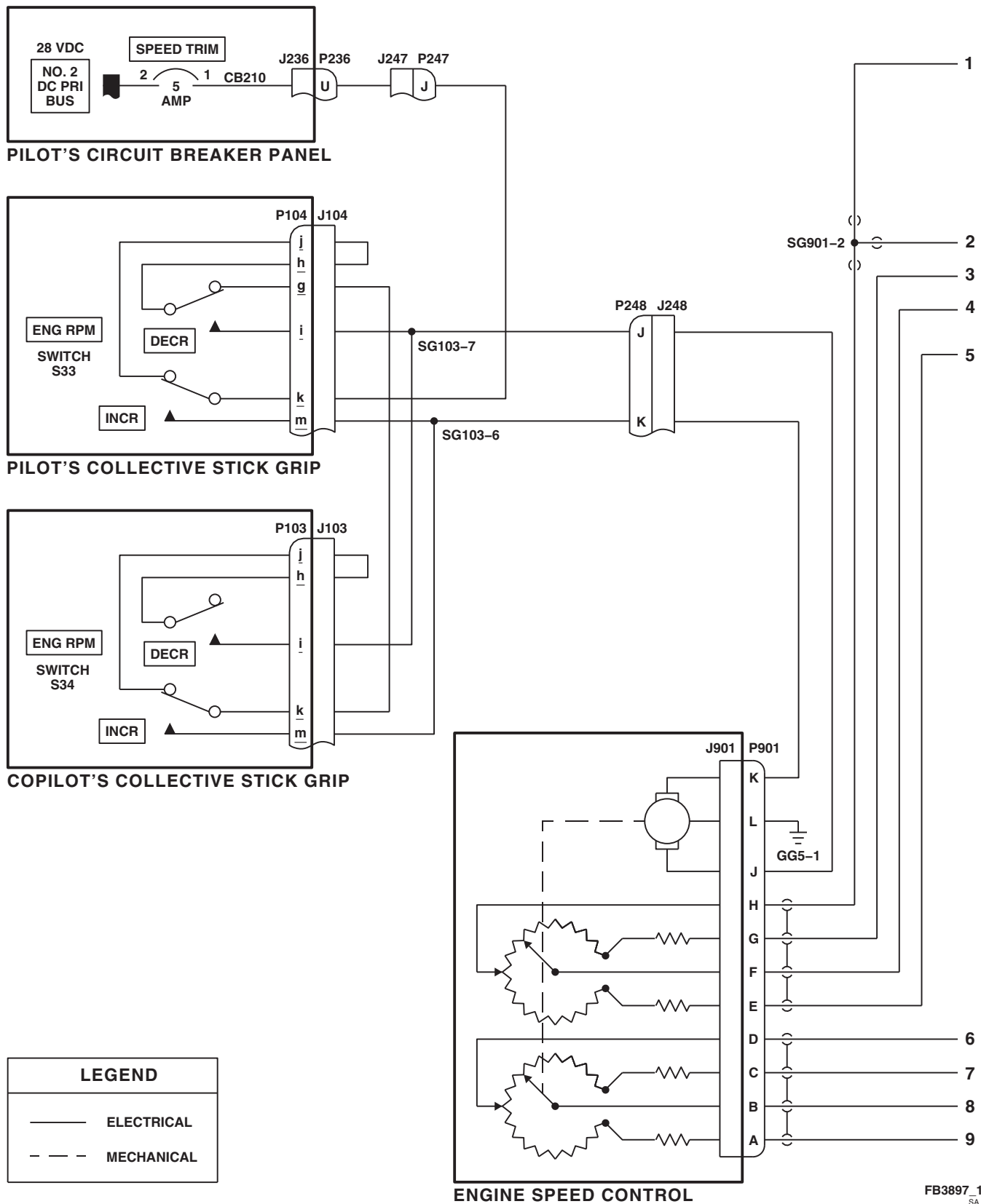
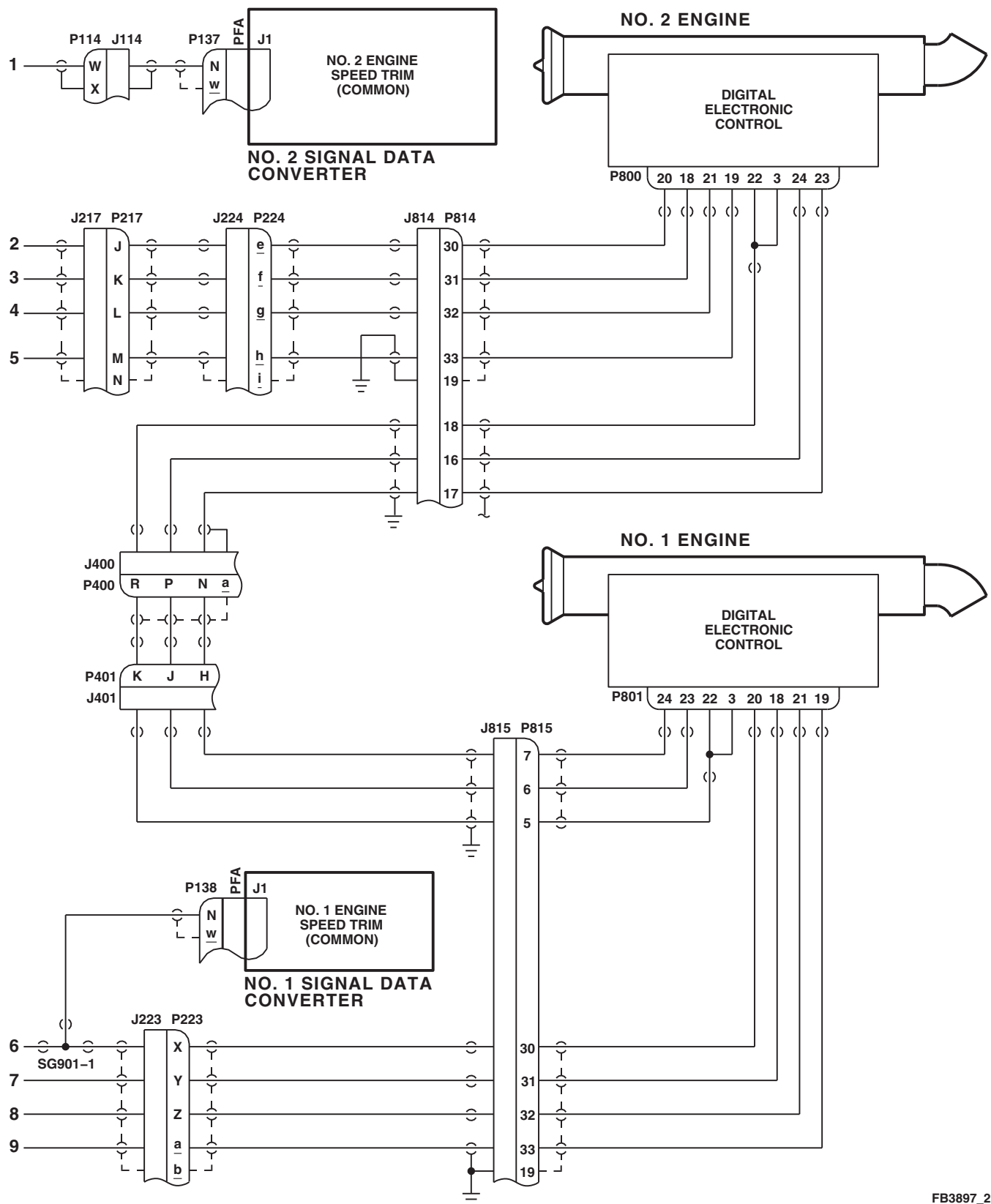


FIGURE 4-2-5-1. ENGINE SPEED TRIM SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 2)

4-2-5-8 Change 7



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FIGURE 4-2-5-1. ENGINE SPEED TRIM SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 2)

4-2-6 ENGINE CHIP DETECTOR SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer Toolkit

References

- PARA 1-6-16
- SEI 671

- TM 1-2840-248-23

Equipment Conditions

- External Electrical Power Available

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

Special Instructions

The Engine Chip Detector System operational/troubleshooting procedure is subdivided as follows:

- (1) Engine No. 1
- (2) Engine No. 2

4-2-6.1 Engine No. 1.

- Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- Remove No. 1 engine chip detector (SEI 671 and TM 1-2840-248-23).
- Turn on electrical power (PARA 1-6-16).

RESULT: CHIP #1 ENGINE capsule, on caution/advisory panel, shall be off.

- If result is not as specified, disconnect P803.
 - If CHIP #1 ENGINE capsule goes off, replace No. 1 engine chip detector (SEI 671 and TM 55-2840-248-23).
 - If CHIP #1 ENGINE capsule does not go off, disconnect P117.

- EMEP** Remove and check pin filter adapter. Replace if necessary (PARA 9-4-98).

- EMEP** Install pin filter adapter (PARA 9-4-98).

- If CHIP #1 ENGINE capsule goes off, repair/replace wiring between P117-5 and J825-7 (Appendix F).

- If CHIP #1 ENGINE capsule remains on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

NOTE

Before performing step, make sure that connectors P117 and P803 are connected.

- Position clean screwdriver with an insulated handle on chip detector.

RESULT: CHIP #1 ENGINE capsule, on caution/advisory panel, shall go on.

- If result is not as specified, go to TABLE NO. 4-2-6-1.

- e. Turn off electrical power (PARA 1-6-16).
- f. Install No. 1 engine chip detector (SEI 671 and TM 1-2840-248-23).

4-2-6.2 Engine No. 2.

- a. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- b. Remove No. 2 engine chip detector (SEI 671 and TM 1-2840-248-23).
- c. Turn on electrical power (PARA 1-6-16).

RESULT: CHIP #2 ENGINE capsule, on caution/advisory panel, shall be off.

- If result is not as specified, disconnect P802.
 - (a) If CHIP #2 ENGINE capsule goes off, replace No. 2 engine chip detector (SEI 671 and TM 55-2840-248-23).
 - (b) If CHIP #2 ENGINE capsule does not go off, disconnect P117.
- 1 **EMEP** Remove and check pin filter adapter. Replace if necessary (PARA 9-4-98).■

2 **EMEP** Install pin filter adapter (PARA 9-4-98).■

3 If CHIP #2 ENGINE capsule goes off, repair/replace wiring between P117-53 and J824-7 (Appendix F).

4 If CHIP #2 ENGINE capsule remains on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

NOTE

Before performing step, make sure that connectors P117, J824, and J825 are connected.

- d. Position clean screwdriver with an insulated handle on chip detector.

RESULT: CHIP #2 ENGINE capsule, on caution/advisory panel, shall go on.

- If result is not as specified, go to TABLE NO. 4-2-6-2.
- e. Turn off electrical power (PARA 1-6-16).
- f. Install No. 2 engine chip detector (SEI 671 and TM 1-2840-248-23).

TABLE NO. 4-2-6-1. CHIP #1 ENGINE Capsule Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2.	
2. Check that CHIP #1 ENGINE capsule goes on.	
Step 1. If capsule goes on, go to 3.	
Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 20.	
3. Install an insulated jumper wire between P803-11 and P803-12. Go to 4.	

TABLE NO. 4-2-6-1. CHIP #1 ENGINE Capsule Does Not Go On. (Cont)

	TEST OR INSPECTION CORRECTIVE ACTION	
	4. Check that CHIP #1 ENGINE capsule goes on.	
	Step 1. If capsule goes on, go to 5.	
	Step 2. If capsule does not go on, go to 7.	
	5. Remove jumper wire from P803. Go to 6.	
	6. Troubleshoot engine chip detector (SEI 671 and TM 1-2840-248-23). Go to 20.	
	7. Remove jumper wire from P803. Go to 8.	
	8. Check helicopter configuration.	
	Step 1. EMEP Go to 9.◀	
	Step 2. W/O EMEP Go to 11.◀	
EMEP	9. Remove and check pin filter adapters P117 and P118 (PARA 9-4-98).	
	Step 1. If pin filter adapters are good, go to 10.	
	Step 2. If pin filter adapters are not good, replace pin filter adapters (PARA 9-4-98). Go to 20.	
EMEP	10. Install pin filter adapters (PARA 9-4-98). Go to 11.	
	11. Check for 28 VDC between P118-g and ground.	
	Step 1. If voltage is as specified, go to 12.	
	Step 2. If voltage is not as specified, go to 19.	
	12. Install an insulated jumper wire between J117-5 and ground. Go to 13.	
	13. Check that CHIP #1 ENGINE caution capsule goes on.	
	Step 1. If capsule goes on, go to 14.	
	Step 2. If capsule does not go on, go to 16.	
	14. Remove jumper wire from J117. Go to 15.	

TABLE NO. 4-2-6-1. CHIP #1 ENGINE Capsule Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
15. Repair/replace wiring between P117-5 and J825-7 (Appendix F). Go to 20.	Step 1. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 20.
16. Check continuity between P803-12 and ground.	Step 1. If continuity is present, go to 17. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 20.
17. Remove jumper wire from J117. Go to 18.	
18. Troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 20.	
19. Check for 28 VDC between terminal 1 (NO. ENG WARN LTS circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and ground.	Step 1. If voltage is as specified, repair/replace wiring between terminal 1 (NO. ENG WARN LTS circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and P118-g (Appendix F). Go to 20. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 20.
20. Procedure completed.	

TABLE NO. 4-2-6-2. CHIP #2 ENGINE Capsule Does Not Go On.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2.	
2. Check that CHIP #2 ENGINE capsule goes on.	Step 1. If capsule goes on, go to 3. Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 20.
3. Install an insulated jumper wire between P802-11 and P802-12. Go to 4.	

TABLE NO. 4-2-6-2. CHIP #2 ENGINE Capsule Does Not Go On. (Cont)

	TEST OR INSPECTION CORRECTIVE ACTION
	4. Check that CHIP #2 ENGINE capsule goes on.
	Step 1. If capsule goes on, go to 5. Step 2. If capsule does not go on, go to 7.
	5. Remove jumper wire from P802. Go to 6.
	6. Troubleshoot engine chip detector (SEI 671 and TM 1-2840-248-23). Go to 20.
	7. Remove jumper wire from P802. Go to 8.
	8. Check helicopter configuration.
	Step 1. EMEP Go to 9.  Step 2. W/O EMEP Go to 11. 
EMEP	9. Remove and check pin filter adapters P117 and P118 (PARA 9-4-98).
	Step 1. If pin filter adapters are good, go to 10. Step 2. If pin filter adapters are not good, replace pin filter adapters (PARA 9-4-98). Go to 20.
EMEP	10. Install pin filter adapters (PARA 9-4-98). Go to 11.
	11. Check for 28 VDC between P118-h and ground.
	Step 1. If voltage is as specified, go to 12. Step 2. If voltage is not as specified, go to 19.
	12. Install insulated jumper wire between J117-53 and ground. Go to 13.
	13. Check that CHIP #2 ENGINE capsule goes on.
	Step 1. If capsule goes on, go to 14. Step 2. If capsule does not go on, go to 16.
	14. Remove jumper wire from J117-53. Go to 15.

TABLE NO. 4-2-6-2. CHIP #2 ENGINE Capsule Does Not Go On. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
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15. Repair/replace wiring between P117-53 and J824-7 (Appendix F). Go to 20.

Step 1. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23).
Go to **20**.

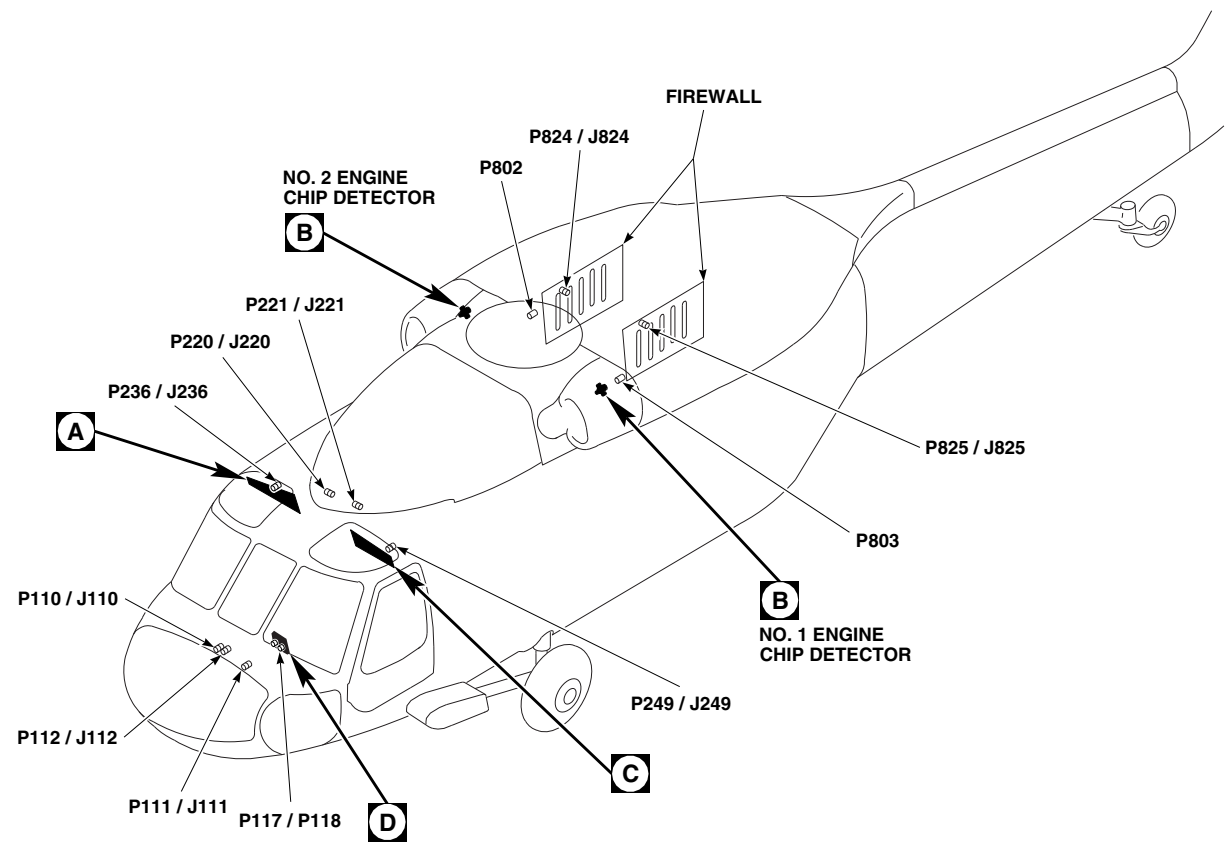
16. Check continuity between P802-12 and ground.

Step 1. If continuity is present, go to **17**.
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **20**.

17. Remove jumper wire from J117-53. Go to 18.**18. Troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 20.****19. Check for 28 VDC between terminal 1 (NO. 2 ENG WARN LTS circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and ground.**

Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P118-h (Appendix F). Go to **20**.
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **20**.

20. Procedure completed.

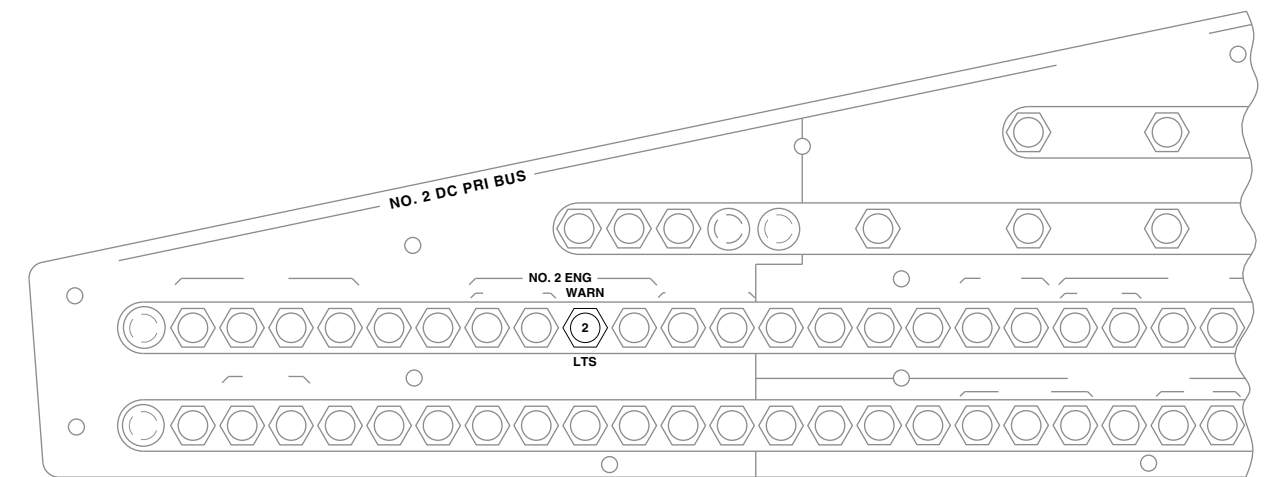


TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P110 / J110	COCKPIT, BL 7.5 RH, STA 197
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P112 / J112	COCKPIT, BL 6 RH, STA 197
P117	CAUTION / ADVISORY PANEL
P118	CAUTION / ADVISORY PANEL
P220 / J220	JUNCTION BOX, MAIN ROTOR PYLON DECK
P221 / J221	JUNCTION BOX, MAIN ROTOR PYLON DECK

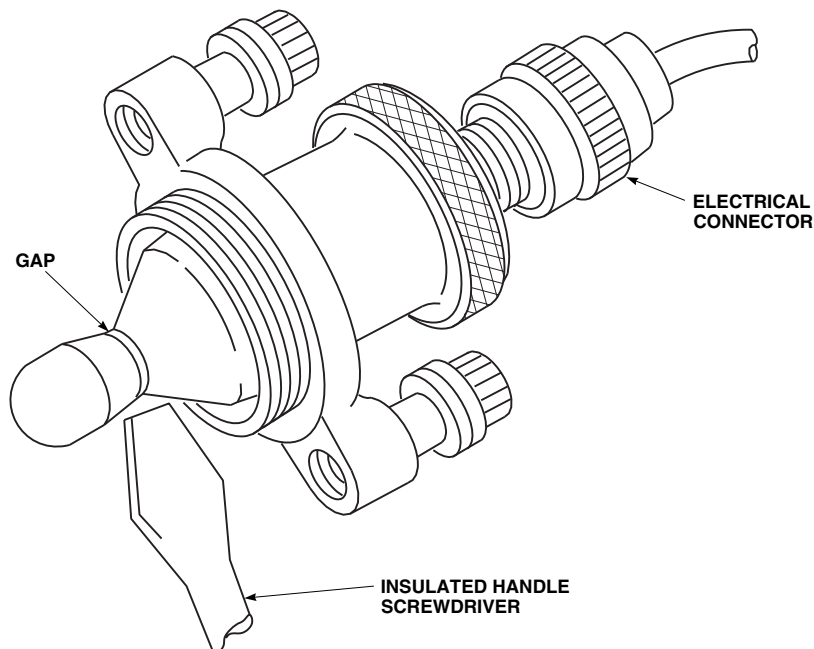
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 23 RH
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 11 LH
P802	NO. 2 ENGINE INTERFACE DISCONNECT
P803	NO. 1 ENGINE INTERFACE DISCONNECT
P824 / J824	FIREWALL DISCONNECT BL 14 RH, STA 338.25
P825 / J825	FIREWALL DISCONNECT BL 14 LH, STA 358.25

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FIGURE 4-2-6-1. ENGINE CHIP DETECTOR SYSTEM LOCATION DIAGRAM (SHEET 1 OF 3)



PILOT'S CIRCUIT BREAKER PANEL



ENGINE CHIP DETECTOR

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FIGURE 4-2-6-1. ENGINE CHIP DETECTOR SYSTEM LOCATION DIAGRAM (SHEET 2 OF 3)

4-2-6-8 Change 7

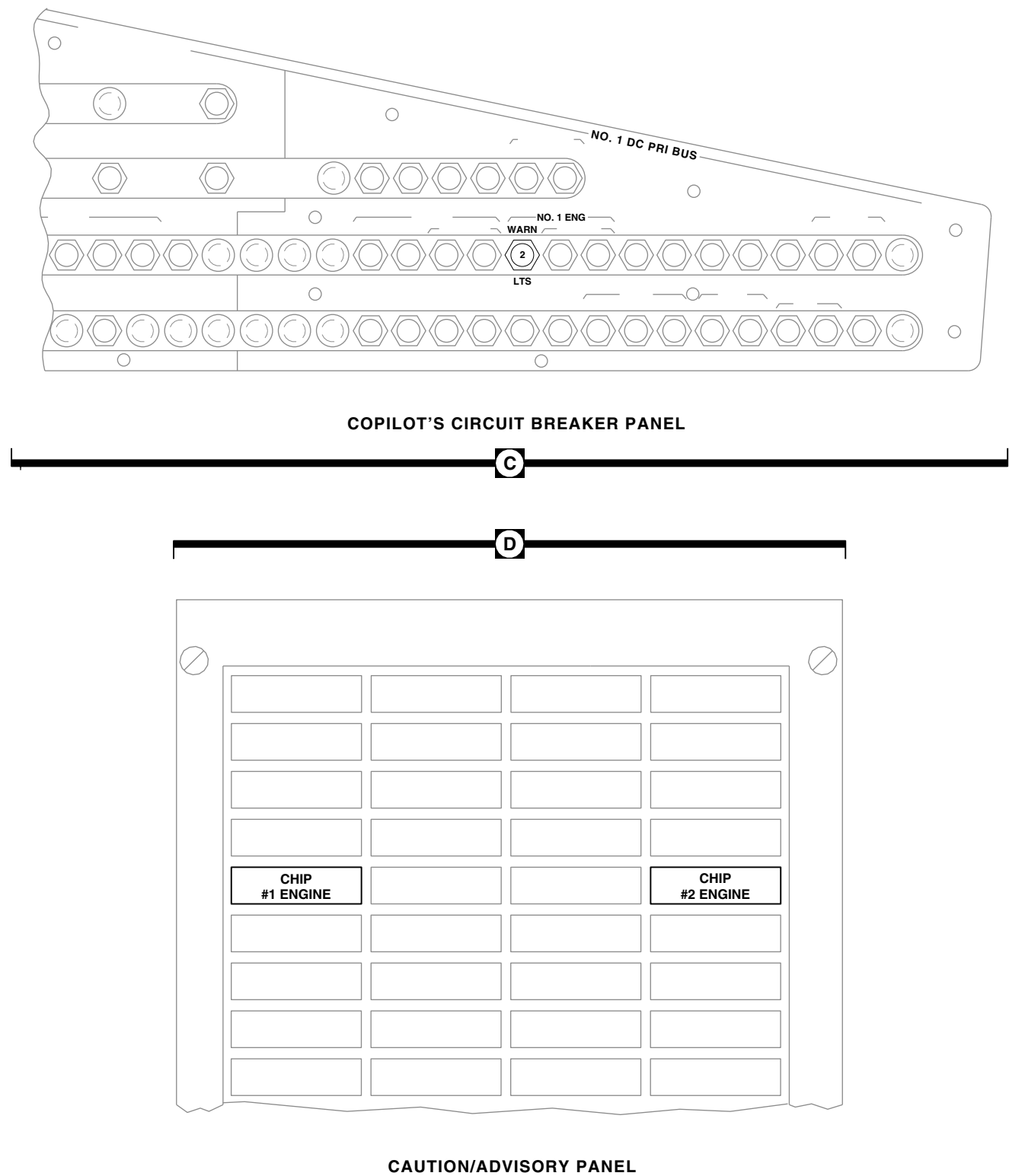


FIGURE 4-2-6-1. ENGINE CHIP DETECTOR SYSTEM LOCATION DIAGRAM (SHEET 3 OF 3)

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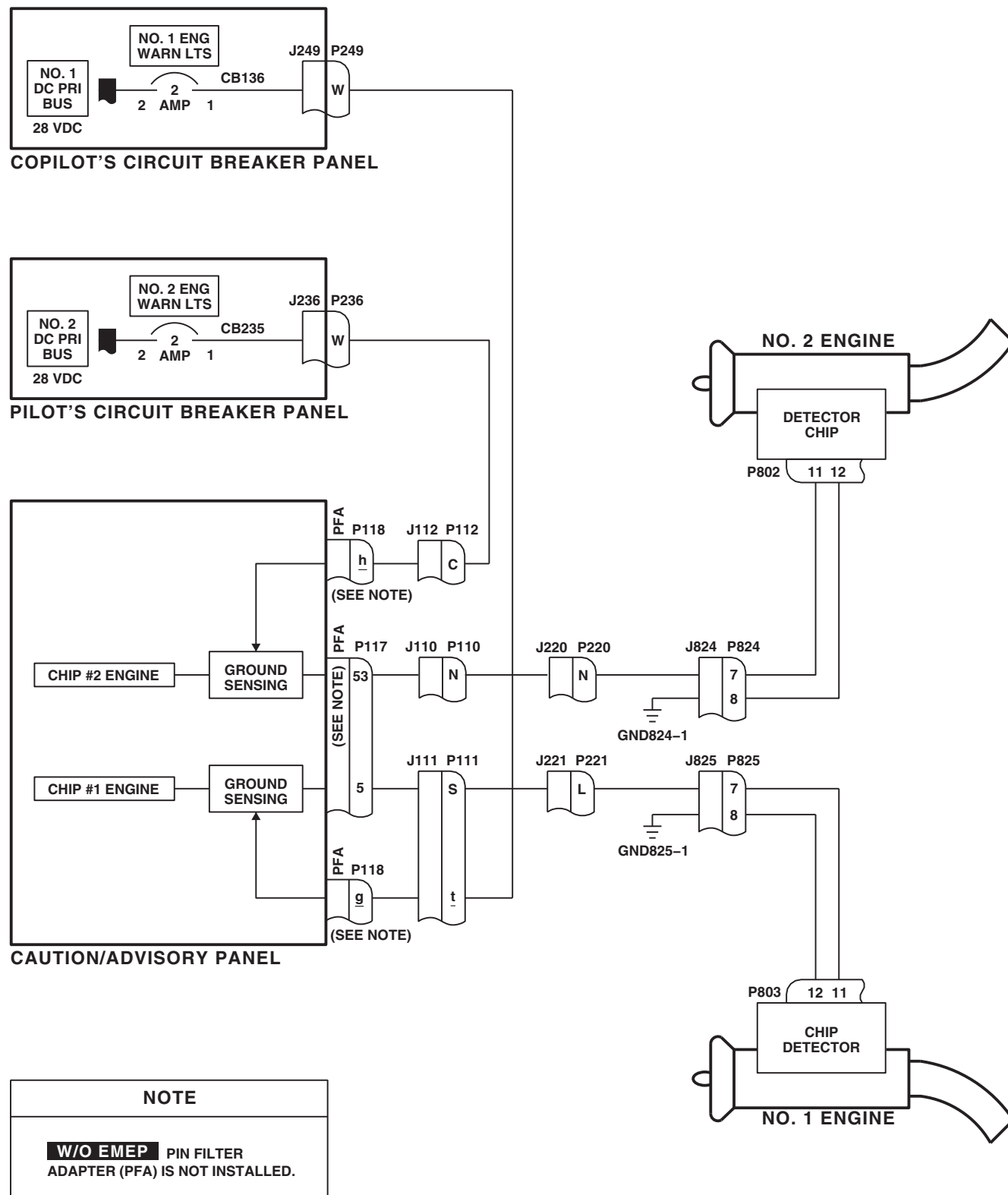
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FIGURE 4-2-6-2. ENGINE CHIP DETECTOR SYSTEM SCHEMATIC DIAGRAM

4-2-7 ENGINE ANTI-ICE SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer Toolkit
- Insulated Jumper Wire

References

- Appropriate Flight Manual

Equipment Conditions

- Auxiliary Power Unit (APU) Operational
- No. 1 and No. 2 Engines Operational
- No. 1 and No. 2 Fuel Tanks Serviced

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

Special Instructions

The Engine Anti-Ice System operational/troubleshooting procedure is subdivided as follows:

- (1) No. 1 Engine
- (2) No. 2 Engine

4-2-7.1 No. 1 Engine.

- a. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- b. Make sure upper console ENG ANTI-ICE NO. 1 switch is placed OFF.
- c. Start APU (Appropriate Flight Manual).
- d. With qualified personnel at controls, start No. 1 engine and operate up to ground idle speed (Appropriate Flight Manual).

RESULT:

- (1) #1 ENG ANTI-ICE ON capsule shall be on.
- (2) #1 ENG INLET ANTI-ICE ON capsule shall be off.
- If result (1) is not as specified, go to TABLE NO. 4-2-7-1.

- If result (2) is not as specified, replace No. 1 engine inlet anti-ice valve (PARA 12-4-22).
- e. Advance No. 1 engine controls quadrant ENG POWER CONT lever to FLY.
- f. Pull collective stick up as required to get Ng SPEED 1 above 90%.

RESULT: #1 ENG ANTI-ICE ON capsule shall go off when Ng SPEED 1 is above 90%.

- If result is not as specified, replace No. 1 engine anti-ice start bleed valve (SEI 671 and TM 1-2840-248-23).
- g. Place upper console ENG ANTI-ICE NO. 1 switch to ON.

RESULT:

- (1) #1 ENG ANTI-ICE ON capsule shall go on immediately.

- (2) No. 1 engine TGT shall increase.
- (3) No. 1 engine inlet anti-ice valve shall open, allowing engine bleed-air to heat up engine inlet.

NOTE

When operating at temperatures above 39.2° - 55.4°F (4° - 13°C), #1 ENG INLET ANTI-ICE ON capsule will not go on even when ENG ANTI-ICE NO. 1 switch is ON.

- (4) When engine inlet temperature is above 200°F (93.3°C), temperature switch shall close and #1 ENG INLET ANTI-ICE ON capsule shall go on.
- If results are not as specified, go to TABLE NO. 4-2-7-2.
- h. Place upper console ENG ANTI-ICE NO. 1 switch to OFF.

RESULT:

- (1) #1 ENG ANTI-ICE ON capsule shall go off immediately.
- (2) No. 1 engine TGT shall decrease.
- (3) #1 ENG INLET ANTI-ICE ON capsule shall go off when engine inlet is cooled down below 200°F (93.3°C).
- If result (1) is not as specified, go to TABLE NO. 4-2-7-3.
- If result (3) is not as specified, go to TABLE NO. 4-2-7-4.
- i. Lower copilot's collective stick.

RESULT: #1 ENG ANTI-ICE ON capsule shall go on.

- If result is not as specified, go to TABLE NO. 4-2-7-1.

- j. Shut down No. 1 engine (Appropriate Flight Manual).
- k. Shut down APU (Appropriate Flight Manual).

4-2-7.2 No. 2 Engine.

- a. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- b. Make sure upper console ENG ANTI-ICE NO. 2 switch is placed OFF.
- c. Start APU (Appropriate Flight Manual).
- d. With qualified personnel at controls, start No. 2 engine and operate up to ground idle speed (Appropriate Flight Manual).

RESULT:

- (1) #2 ENG ANTI-ICE ON capsule shall be on.
- (2) #2 ENG INLET ANTI-ICE ON capsule shall be off.
- If result (1) is not as specified, go to TABLE NO. 4-2-7-5.
- If result (2) is not as specified, replace No. 2 engine inlet anti-ice valve (PARA 12-4-22).
- e. Advance No. 2 engine controls quadrant ENG POWER CONT lever to FLY.
- f. Pull collective stick up as required to get Ng SPEED 2 above 90%.

RESULT: #2 ENG ANTI-ICE ON capsule shall go off when Ng SPEED 2 is above 90%.

- If result is not as specified, replace No. 2 engine anti-ice start bleed valve (SEI 671 and TM 1-2840-248-23).
- g. Place upper console ENG ANTI-ICE NO. 2 switch to ON.

RESULT:

- (1) #2 ENG ANTI-ICE ON capsule shall go on immediately.
- (2) No. 2 engine TGT shall increase.
- (3) No. 2 engine inlet anti-ice valve shall open, allowing engine bleed-air to heat up engine inlet.

NOTE

When operating at temperatures above 39.2° to 55.4°F (4° - 13°C), #2 ENG INLET ANTI-ICE ON capsule will not go on even when ENG ANTI-ICE NO. 2 switch is ON.

- (4) When engine inlet temperature is above 200°F (93.3°C), temperature switch shall close and #2 ENG INLET ANTI-ICE ON capsule shall go on.
- If results are not as specified, go to TABLE NO. 4-2-7-6.
- h. Place upper console ENG ANTI-ICE NO. 2 switch to OFF.

RESULT:

- (1) #2 ENG ANTI-ICE ON capsule shall go off immediately.
- (2) No. 2 engine TGT shall decrease.
- (3) #2 ENG INLET ANTI-ICE ON advisory capsule shall go off when engine inlet is cooled down below 200°F (93.3°C).

- If result (1) is not as specified, go to TABLE NO. 4-2-7-7.
- If result (3) is not as specified, go to TABLE NO. 4-2-7-8.

i. Lower pilot's collective stick.

RESULT: #2 ENG ANTI-ICE ON capsule shall go on.

- If result is not as specified, go to TABLE NO. 4-2-7-5.
- j. Shut down No. 2 engine (Appropriate Flight Manual).
- k. Shut down APU (Appropriate Flight Manual).

TABLE NO. 4-2-7-1. #1 ENG ANTI-ICE ON Capsule Does Not Go On When Engine Is At Ground Idle Speed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Place and hold BRT/DIM-TEST switch (caution/advisory panel) to TEST.	Go to 2.
2. Check if #1 ENGINE ANTI-ICE ON capsule goes on.	Step 1. If capsule goes on, go to 3. Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 16.
3. Install an insulated jumper wire between P803-3 and P803-4.	Go to 4.

TABLE NO. 4-2-7-1. #1 ENG ANTI-ICE ON Capsule Does Not Go On When Engine Is At Ground Idle Speed. (Cont)

	TEST OR INSPECTION CORRECTIVE ACTION
	4. Check if #1 ENG ANTI-ICE ON capsule goes on. Step 1. If capsule goes on, go to 5. Step 2. If capsule does not go on, go to 7.
	5. Remove jumper wire from P803. Go to 6.
	6. Troubleshoot No. 1 engine anti-ice start bleed valve (SEI 671 and TM 1-2840-248-23). Go to 16.
	7. Check helicopter configuration. Step 1. EMEP Go to 8. Step 2. W/O EMEP Go to 10.
EMEP	8. Remove and check pin filter adapter P118 (PARA 9-4-98). Step 1. If pin filter adapter is good, go to 9. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 16.
EMEP	9. Install pin filter adapter (PARA 9-4-98). Go to 10.
I	10. Check for 28 VDC between P118-A and gnd. Step 1. If voltage is as specified, go to 11. Step 2. If voltage is not as specified, go to 13.
	11. Remove jumper wire from P803. Go to 12.
	12. Troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 16.
	13. Remove jumper wire from P803. Go to 14.
I	14. Check for 28 VDC between P803-3 and gnd. Step 1. If voltage is as specified, repair/replace wiring between J825-12 and P118-A (Appendix F). Go to 16. Step 2. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 15.

TABLE NO. 4-2-7-1. #1 ENG ANTI-ICE ON Capsule Does Not Go On When Engine Is At Ground Idle Speed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 3. If voltage is not as specified, go to 15 .
15. Check for 28 VDC between terminal 1 (NO. 1 ENG ANTI-ICE WARN circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and gnd.	
	Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and J815-36 (Appendix F). Go to 16 .
	Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 16 .
16. Procedure completed.	

TABLE NO. 4-2-7-2. #1 ENG INLET ANTI-ICE ON Capsule Does Not Go On When Anti-Ice System Is Used.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Visually check rubber hoses, clamps, and tubing for possible damage between engine inlet and engine anti-ice valve.	
	Step 1. If damage is present, repair/replace clamps, tubing, or rubber hoses as required (PARA 4-4-4). Go to 17 .
	Step 2. If damage is not present, go to 2 .
2. Place and hold BRT/DIM-TEST switch (caution/advisory panel) to TEST. Go to 3.	
3. Check if #1 ENG INLET ANTI-ICE ON capsule goes on.	
	Step 1. If capsule goes on, go to 4 .
	Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 17 .
4. Install an insulated jumper wire between P821-A and P821-B. Go to 5.	

TABLE NO. 4-2-7-2. #1 ENG INLET ANTI-ICE ON Capsule Does Not Go On When Anti-Ice System Is Used. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
5. Check if #1 ENG INLET ANTI-ICE ON capsule goes on.	
Step 1. If capsule goes on, go to 6 .	
Step 2. If capsule does not go on, go to 8 .	
6. Remove jumper wire from P821. Go to 7.	
7. Replace No. 1 engine inlet temperature switch (PARA 12-4-23). Go to 17.	
8. Check helicopter configuration.	
Step 1. EMEP Go to 9 .	
Step 2. W/O EMEP Go to 11 .	
EMEP	9. Remove and check pin filter adapter P118 (PARA 9-4-98).
Step 1. If pin filter adapter is good, go to 10 .	
Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 17 .	
EMEP	10. Install pin filter adapter (PARA 9-4-98). Go to 11.
I	11. Check for 28 VDC between P118-D and gnd.
Step 1. If voltage is as specified, go to 12 .	
Step 2. If voltage is not as specified, go to 14 .	
12. Remove jumper wire from P821. Go to 13.	
13. Troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 17.	
14. Remove jumper wire from P821. Go to 15.	
I	15. Check for 28 VDC between P821-A and gnd.
Step 1. If voltage is as specified, repair/replace wiring between P118-D and J825-11 (Appendix F). Go to 17 .	
Step 2. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 17 .	
Step 3. If voltage is not as specified, go to 16 .	

TABLE NO. 4-2-7-2. #1 ENG INLET ANTI-ICE ON Capsule Does Not Go On When Anti-Ice System Is Used. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
16. Check for 28 VDC between terminal 1 (NO. 1 ENG ANTI-ICE WARN circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and gnd.
Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P821-A (Appendix F). Go to 17 .
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 17 .
17. Procedure completed.

TABLE NO. 4-2-7-3. #1 ENG ANTI-ICE ON Capsule Does Not Go Off Immediately.

TEST OR INSPECTION CORRECTIVE ACTION
1. With ENG ANTI-ICE ON NO. 1 switch placed to OFF, check for 28 VDC between P803-1 and P803-2.
Step 1. If voltage is as specified, troubleshoot engine anti-ice/start bleed valve (SEI 671 and TM 1-2840-248-23). Go to 3 .
Step 2. If voltage is not as specified, go to 2 .
2. Check for 28 VDC between P803-1 and gnd.
Step 1. If voltage is as specified, repair/replace gnd wire on P803-2 (Appendix F). Go to 3 .
Step 2. If voltage is not as specified, repair/replace wiring between P225-N and J815-39 (Appendix F). Go to 3 .
Step 3. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 3 .
3. Procedure completed.

TABLE NO. 4-2-7-4. #1 ENG INLET ANTI-ICE ON Capsule Does Not Go Off When Engine Inlet Is Cooled Down Below 93.3°C (200°F).

TEST OR INSPECTION	CORRECTIVE ACTION
1. With ENG ANTI-ICE NO. 1 switch placed OFF, check for 28 VDC between P820-2 and P820-5.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.
2. With No. 1 engine inlet cooled down, check continuity between J821-A and J821-B of inlet temperature switch.	Step 1. If continuity is present, replace No. 1 engine inlet temperature switch (PARA 12-4-23). Go to 13. Step 2. If continuity is not present, replace No. 1 engine inlet anti-ice valve (PARA 12-4-22). Go to 13.
3. Check for 28 VDC between P820-2 and gnd.	Step 1. If voltage is as specified, repair/replace wiring between P820-5 and gnd (Appendix F). Go to 13. Step 2. If voltage is not as specified, go to 4.
4. Install an insulated jumper wire between P242-R and P242-S. Go to 5.	
5. Check for 28 VDC between P820-2 and gnd.	Step 1. If voltage is as specified, go to 6. Step 2. If voltage is not as specified, go to 8.
6. Remove jumper wire from P242. Go to 7.	
7. Replace left relay panel (PARA 9-4-2). Go to 13.	
8. Remove jumper wire from P242. Go to 9.	
9. Check for 28 VDC between P242-R and gnd.	Step 1. If voltage is as specified, repair/replace wiring between P242-S and J815-37 (Appendix F). Go to 13. Step 2. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 13. Step 3. If voltage is not as specified, go to 10.

1. With ENG ANTI-ICE NO. 1 switch placed OFF, check for 28 VDC between P820-2 and P820-5.

Step 1. If voltage is as specified, go to 2.
 Step 2. If voltage is not as specified, go to 3.

2. With No. 1 engine inlet cooled down, check continuity between J821-A and J821-B of inlet temperature switch.

Step 1. If continuity is present, replace No. 1 engine inlet temperature switch (PARA 12-4-23). Go to 13.
 Step 2. If continuity is not present, replace No. 1 engine inlet anti-ice valve (PARA 12-4-22). Go to 13.

3. Check for 28 VDC between P820-2 and gnd.

Step 1. If voltage is as specified, repair/replace wiring between P820-5 and gnd (Appendix F). Go to 13.
 Step 2. If voltage is not as specified, go to 4.

4. Install an insulated jumper wire between P242-R and P242-S. Go to 5.

5. Check for 28 VDC between P820-2 and gnd.

Step 1. If voltage is as specified, go to 6.
 Step 2. If voltage is not as specified, go to 8.

6. Remove jumper wire from P242. Go to 7.

7. Replace left relay panel (PARA 9-4-2). Go to 13.

8. Remove jumper wire from P242. Go to 9.

9. Check for 28 VDC between P242-R and gnd.

Step 1. If voltage is as specified, repair/replace wiring between P242-S and J815-37 (Appendix F). Go to 13.
 Step 2. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 13.
 Step 3. If voltage is not as specified, go to 10.

TABLE NO. 4-2-7-4. #1 ENG INLET ANTI-ICE ON Capsule Does Not Go Off When Engine Inlet Is Cooled Down Below 93.3°C (200°F). (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
10. Check for 28 VDC between terminal 1 (ENG ANTI-ICE NO. 1 switch) and gnd. Step 1. If voltage is as specified, repair/replace wiring between P242-R and terminal 1 (Appendix F). Go to 13. Step 2. If voltage is not as specified, go to 11. 11. Check for 28 VDC between terminal 1 (NO. 1 ENG ANTI-ICE circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and gnd. Step 1. If voltage is as specified, go to 12. Step 2. If voltage is not as specified, repair/replace circuit breaker (PARA 9-4-1). Go to 13. 12. Check continuity between terminal 1 (NO. 1 ENG ANTI-ICE circuit breaker) and terminal 2 (ENG ANTI-ICE NO. 1 switch). Step 1. If continuity is present, replace switch (PARA 9-4-5). Go to 13. Step 2. If continuity is not present, repair/replace wiring to switch (Appendix F). Go to 13. 13. Procedure completed.

TABLE NO. 4-2-7-5. #2 ENG ANTI-ICE ON Capsule Does Not Go On When Engine Is At Ground Idle Speed.

TEST OR INSPECTION CORRECTIVE ACTION
1. Place and hold BRT/DIM-TEST switch (caution/advisory panel) to TEST. Go to 2. 2. Check if #2 ENG ANTI-ICE ON capsule goes on. Step 1. If capsule goes on, go to 3. Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 16.

TABLE NO. 4-2-7-5. #2 ENG ANTI-ICE ON Capsule Does Not Go On When Engine Is At Ground Idle Speed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
3. Install an insulated jumper wire between P802-3 and P802-4. Go to 4.	
4. Check if #2 ENG ANTI-ICE ON capsule goes on.	Step 1. If capsule goes on, go to 5. Step 2. If capsule does not go on, go to 7.
5. Remove jumper wire from P802. Go to 6.	
6. Troubleshoot No. 2 engine anti-ice start bleed valve (SEI 671 and TM 1-2840-248-23). Go to 16.	
7. Check helicopter configuration.	Step 1. EMEP Go to 8. Step 2. W/O EMEP Go to 10.
EMEP 8. Remove and check pin filter adapter P118 (PARA 9-4-98).	Step 1. If pin filter adapter is good, go to 9. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 16.
EMEP 9. Install pin filter adapter (PARA 9-4-98). Go to 10.	
I 10. Check for 28 VDC between P118-P and gnd.	Step 1. If voltage is as specified, go to 11. Step 2. If voltage is not as specified, go to 13.
11. Remove jumper wire from P802. Go to 12.	
12. Troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 16.	
13. Remove jumper wire from P802. Go to 14.	

TABLE NO. 4-2-7-5. #2 ENG ANTI-ICE ON Capsule Does Not Go On When Engine Is At Ground Idle Speed. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
14. Check for 28 VDC between P802-3 and gnd.
Step 1. If voltage is as specified, repair/replace wiring between P118-P and J824-11 (Appendix F). Go to 16. Step 2. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 16. Step 3. If voltage is not as specified, go to 15.
15. Check for 28 VDC between terminal 1 (NO. 2 ENG ANTI-ICE WARN circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and gnd.
Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and J814-39 (Appendix F). Go to 16. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 16.
16. Procedure completed.

TABLE NO. 4-2-7-6. #2 ENG INLET ANTI-ICE ON Capsule Does Not Go On When Anti-Ice System Is Used.

TEST OR INSPECTION CORRECTIVE ACTION
1. Visually check rubber hoses, clamps, and tubing for possible damage between engine inlet and engine anti-ice valve.
Step 1. If damage is present, repair/replace clamps, tubing, or rubber hoses as required (PARA 4-4-4). Go to 17. Step 2. If damage is not present, go to 2.
2. Place and hold BRT/DIM-TEST switch (caution/advisory panel) to TEST. Go to 3.
3. Check if #2 ENG INLET ANTI-ICE ON capsule goes on.
Step 1. If capsule goes on, go to 4.

TABLE NO. 4-2-7-6. #2 ENG INLET ANTI-ICE ON Capsule Does Not Go On When Anti-Ice System Is Used. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 17. 4. Install an insulated jumper wire between P828-A and P828-B. Go to 5. 5. #2 ENG INLET ANTI-ICE ON capsule should go on. Step 1. If capsule goes on, go to 6. Step 2. If capsule does not go on, go to 8. 6. Remove jumper wire from P828. Go to 7. 7. Replace No. 2 engine inlet temperature switch (PARA 12-4-23). Go to 17. 8. Check helicopter configuration. Step 1. EMEP Go to 9. W/O EMEP Go to 11. EMEP 9. Remove and check pin filter adapter P118 (PARA 9-4-98). Step 1. If pin filter adapter is good, go to 10. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 17. EMEP 10. Install pin filter adapter (PARA 9-4-98). Go to 11. W/O EMEP 11. Check for 28 VDC between P118-J and gnd. Step 1. If voltage is as specified, go to 12. Step 2. If voltage is not as specified, go to 14. 12. Remove jumper wire from P828. Go to 13. 13. Troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 17. 14. Remove jumper wire from P828. Go to 15. W/O EMEP 15. Check for 28 VDC between P828-A and gnd.

TABLE NO. 4-2-7-6. #2 ENG INLET ANTI-ICE ON Capsule Does Not Go On When Anti-Ice System Is Used. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
Step 1. If voltage is as specified, repair/replace wiring between P118-J and J824-12 (Appendix F). Go to 17. Step 2. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 17. Step 3. If voltage is not as specified, go to 16. 16. Check for 28 VDC between terminal 1 (NO. 2 ENG ANTI-ICE WARN circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and gnd. Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P828-A (Appendix F). Go to 17. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 17. 17. Procedure completed.

TABLE NO. 4-2-7-7. #2 ENG ANTI-ICE ON Capsule Does Not Go Off Immediately.

TEST OR INSPECTION CORRECTIVE ACTION
1. With ENG ANTI-ICE NO. 2 switch placed to OFF, check for 28 VDC between P802-1 and P802-2. Step 1. If voltage is as specified, troubleshoot engine anti-ice start bleed valve (SEI 671 and TM 1-2840-248-23). Go to 3. Step 2. If voltage is not as specified, go to 2. 2. Check for 28 VDC between P802-1 and gnd. Step 1. If voltage is as specified, repair/replace gnd wire to P802-2 (Appendix F). Go to 3. Step 2. If voltage is not as specified, repair/replace wiring between P222-R and J814-36 (Appendix F). Go to 3. Step 3. If trouble remanis, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 3.

TABLE NO. 4-2-7-7. #2 ENG ANTI-ICE ON Capsule Does Not Go Off Immediately. (Cont)

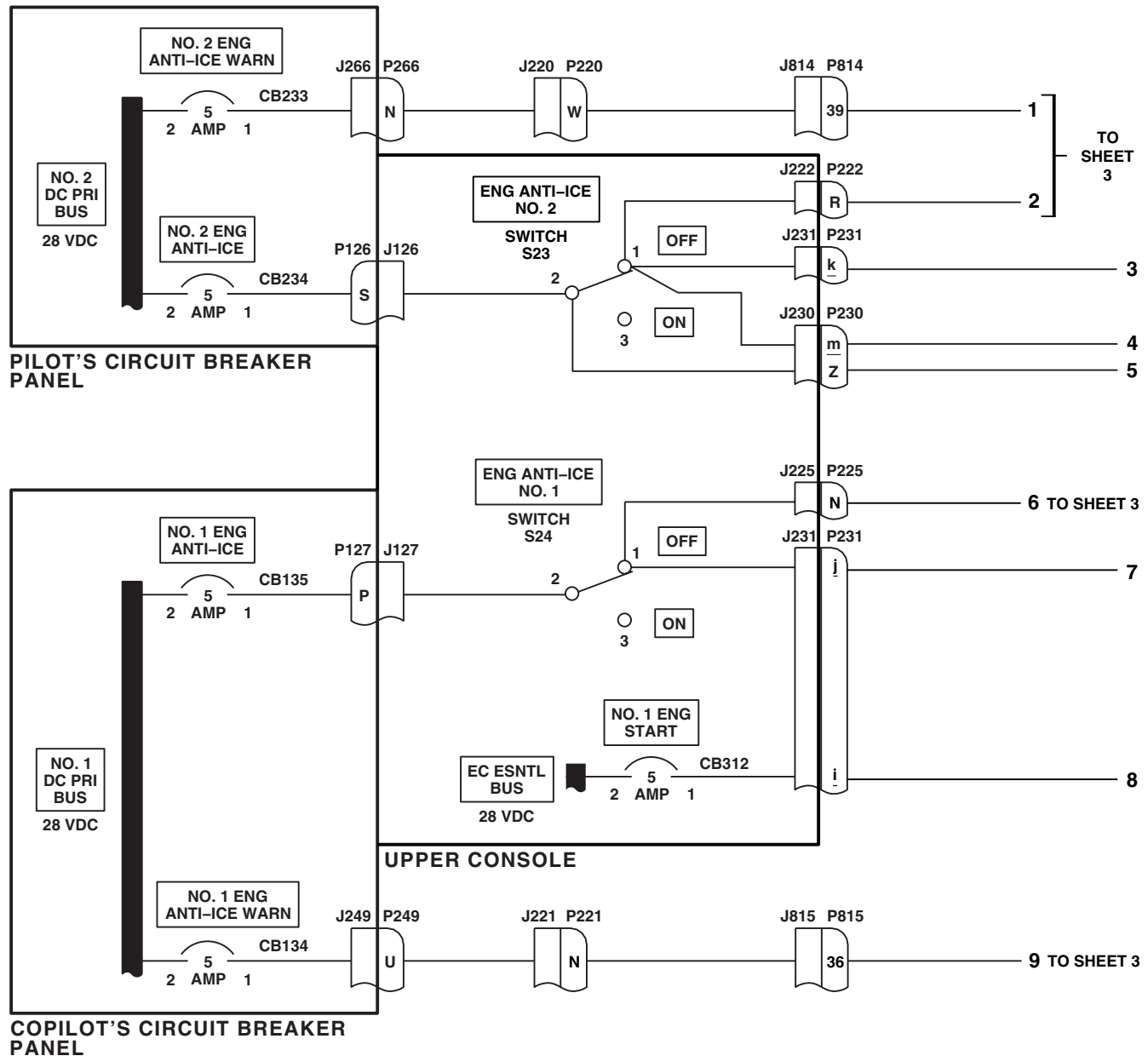
TEST OR INSPECTION	CORRECTIVE ACTION
3. Procedure completed.	

TABLE NO. 4-2-7-8. #2 ENG INLET ANTI-ICE ON Capsule Does Not Go Off When Engine Inlet Is Cooled Down Below 93.3°C (200°F).

TEST OR INSPECTION	CORRECTIVE ACTION
1. With ENG ANTI-ICE NO. 2 switch placed OFF, check for 28 VDC between P822-2 and P822-5.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.
2. With No. 2 engine inlet cooled down, check continuity between J828-A and J828-B of inlet temperature switch.	Step 1. If continuity is present, replace No. 2 engine inlet temperature switch (PARA 12-4-23). Go to 13. Step 2. If continuity is not present, replace No. 2 engine inlet anti-ice valve (PARA 12-4-22). Go to 13.
3. Check for 28 VDC between P822-2 and gnd.	Step 1. If voltage is as specified, repair/replace gnd wire to J814-37 (Appendix F). Go to 13. Step 2. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 13. Step 3. If voltage is not as specified, go to 4.
4. Install an insulated jumper wire between P900-K and P900-L. Go to 5.	
5. Check for 28 VDC between P822-2 and gnd.	Step 1. If voltage is as specified, go to 6. Step 2. If voltage is not as specified, go to 8.
6. Remove jumper wire from P900. Go to 7.	

TABLE NO. 4-2-7-8. #2 ENG INLET ANTI-ICE ON Capsule Does Not Go Off When Engine Inlet Is Cooled Down Below 93.3°C (200°F). (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
7. Troubleshoot right relay panel (PARA 9-2-19). Go to 13.	
8. Remove jumper wire from P900. Go to 9.	
9. Check for 28 VDC between P900-K and gnd.	
Step 1. If voltage is as specified, repair/replace wiring between P900-L and J814-38 (Appendix F). Go to 13 .	
Step 2. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 13 .	
Step 3. If voltage is not as specified, go to 10 .	
10. Check for 28 VDC between terminal 1 (ENG ANTI-ICE NO. 2 switch) and gnd.	
Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P900-K (Appendix F). Go to 13 .	
Step 2. If voltage is not as specified, go to 11 .	
11. Check for 28 VDC between terminal 1 (NO. 2 ENG ANTI-ICE circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and gnd.	
Step 1. If voltage is as specified, go to 12 .	
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 13 .	
12. Check continuity between terminal 1 (NO. 2 ENG ANTI-ICE circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and ENG ANTI-ICE NO. 2 switch S23 terminal 2.	
Step 1. If continuity is present, replace switch (PARA 9-4-5). Go to 13 .	
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 13 .	
13. Procedure completed.	



NOTES

1. SWITCH CLOSING WHEN ENGINE INLET TEMPERATURE IS ABOVE 200°F (93.3°C).
2. VALVE OPENS WHEN DEENERGIZED.
3. VALVE OPENS WHEN DEENERGIZED. THEN POSITION SWITCH WILL CLOSE. POSITION SWITCH IS ALSO CLOSED UNTIL ENGINE COMPRESSOR SPEED IS ABOVE 86% TO 87% NG.
4. **W/O EMEP** PIN FILTER ADAPTER (PFA) IS NOT INSTALLED.

LEGEND

- ELECTRICAL
- - - MECHANICAL
-  ENGINE BLEED-AIR

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SA

FIGURE 4-2-7-1. ENGINE ANTI-ICE SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 3)

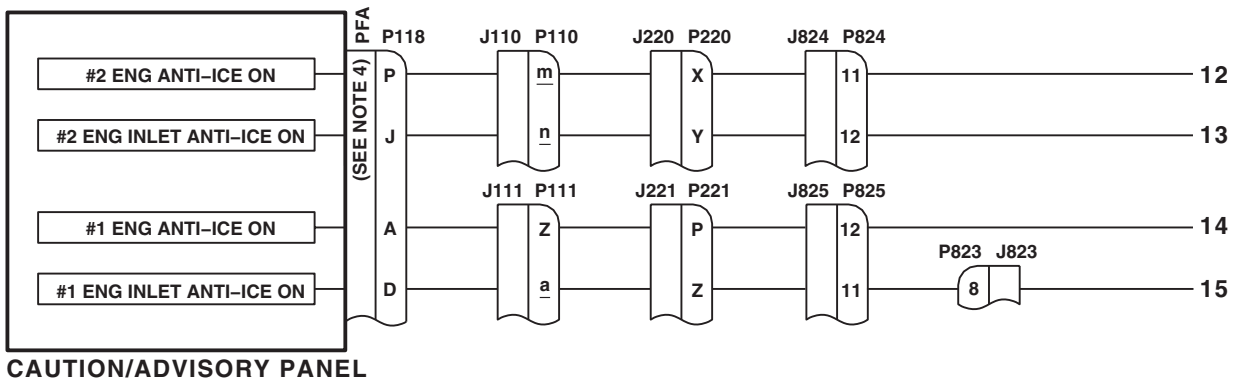
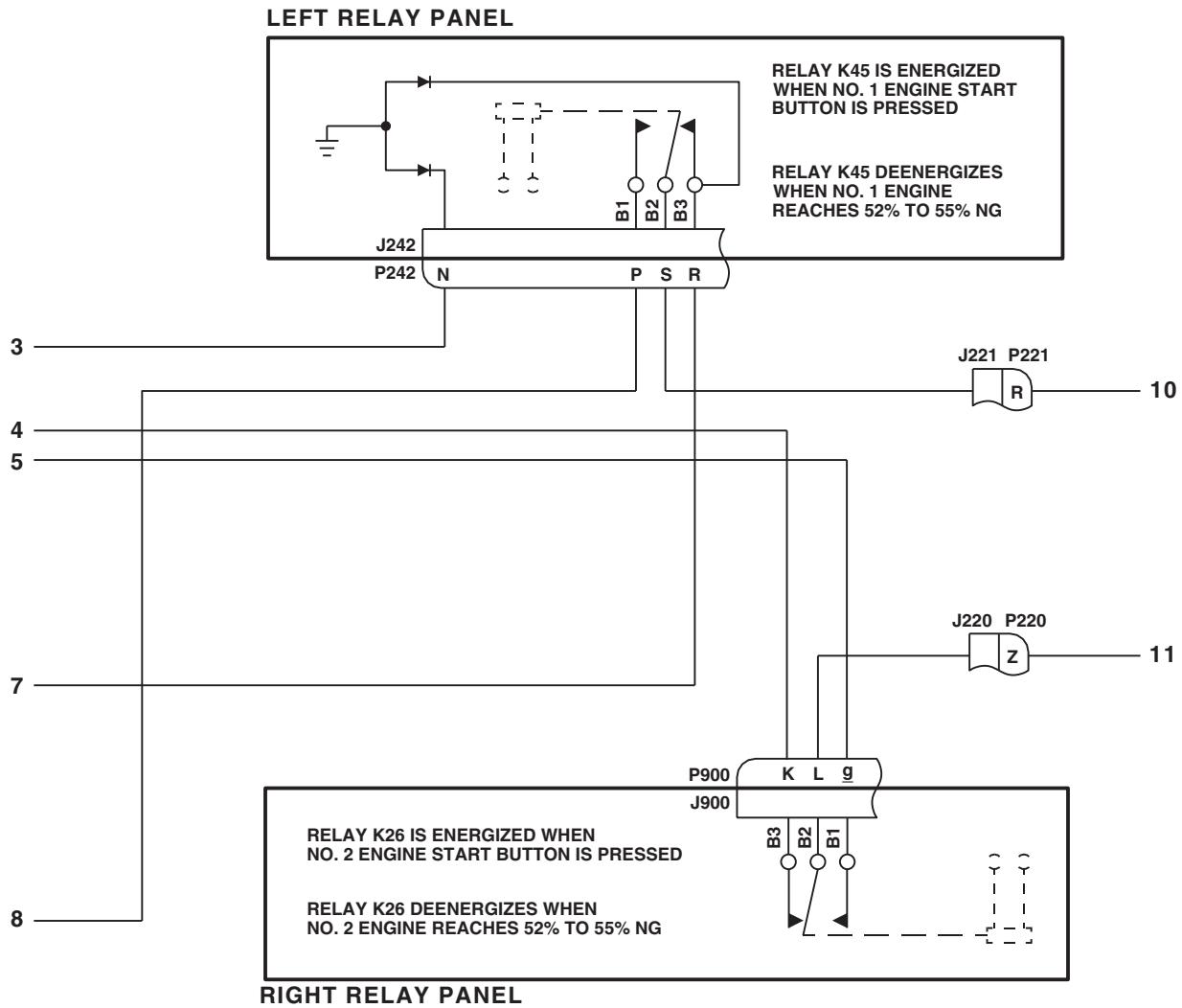


FIGURE 4-2-7-1. ENGINE ANTI-ICE SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 3)

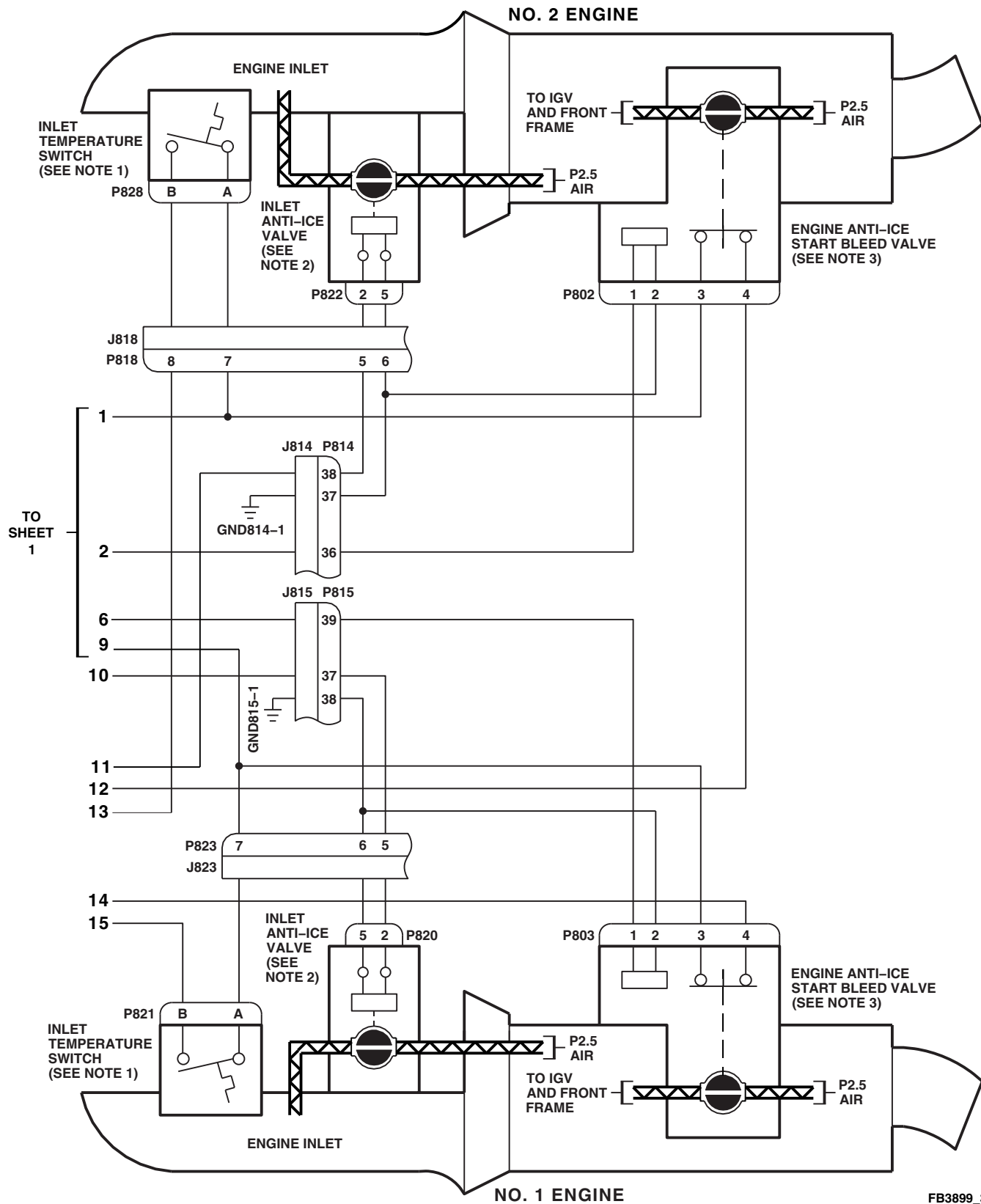
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SA

FIGURE 4-2-7-1. ENGINE ANTI-ICE SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 3)

4-2-8 ENGINE START AND IGNITION SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer Toolkit

References

- Appropriate Flight Manual
- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available and APU Operational
- No. 1 and No. 2 Fuel Tanks Serviced

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

Special Instructions

The Engine Start and Ignition System operational/troubleshooting procedure is subdivided as follows:

- (1) No. 1 Engine
- (2) No. 2 Engine

4-2-8.1 No. 1 Engine.

- a. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- b. Place upper console FUEL PUMP switch to OFF.
- c. Place upper console BATT switch to ON (PARA 1-6-16).

RESULT: PRIME BOOST PUMP ON capsule shall not come on.

- If result is not as specified, go to TABLE NO. 4-2-8-3.

- d. Start APU (Appropriate Flight Manual).

RESULT: #1 ENG OUT capsule on pilot’s and copilot’s master caution panel shall be on.

- If result is not as specified, troubleshoot Instrument Display System (PARA 8-2-3).

- e. With qualified personnel at controls, start No. 1 engine and operate up to IDLE speed (Appropriate Flight Manual). Refer to TABLE NO. 4-2-8-1 for trouble analysis.

CHART NO. 4-2-8-1. No. 1 Engine Starting.

FAULT INDICATION

No. 1 Engine Does Not Start (Compressor Not Rotating) (#1 ENGINE STARTER Capsule Is On)

Engine Starter Motors Engine, But Cannot Reach Proper Ng Speed To Start Engine

GO TO TABLE NO.

TABLE NO. 4-2-8-4

TABLE NO. 4-2-8-5

CHART NO. 4-2-8-1. No. 1 Engine Starting. (Cont)

FAULT INDICATION	GO TO TABLE NO.
No. 1 Engine Does Not Start (Compressor Not Rotating) (#1 ENGINE STARTER Capsule Is Off)	TABLE NO. 4-2-8-6
PRIME BOOST PUMP ON Capsule Is Not On When Start Button Is Pressed	TABLE NO. 4-2-8-7
No. 1 Engine Starter Stops (#1 ENGINE STARTER Capsule Goes Off) When Start Button Is Released	TABLE NO. 4-2-8-8
No. 1 Engine Starts, But #1 ENGINE STARTER Capsule Is Off (APU Is Not Stalling)	TABLE NO. 4-2-8-9
No. 1 Engine Motors, But Does Not Ignite And Accelerate To Idle Speed	TABLE NO. 4-2-8-10
No. 1 Engine Starter Motors Engine (#1 ENGINE STARTER Capsule Is On), But APU Stalls (Popping Sound Is Heard)	TABLE NO. 4-2-8-11
No. 1 Or No. 2 Engine Power Control Lever Is More Difficult To Move Than Normal	TABLE NO. 4-2-8-12
Engine No. 1 Or No. 2 Has A Hung Start (TGT Increases, But Hangs)	TABLE NO. 4-2-8-13
Engine No. 1 Or No. 2 TGT Abnormally High (TGT Reaches 810°C/1490°F Before Idle Speed Is Reached)	TABLE NO. 4-2-8-14
If ground idle speed is too high or low	PARA 4-2-1
#1 ENGINE STARTER Capsule Does Not Go Off (Engine Speed Above 55% Ng)	PARA 8-2-3/TABLE NO. 4-2-8-16
#1 Or #2 ENGINE STARTER Capsule Goes Off Before 52% Ng Engine Speed	TABLE NO. 4-2-8-15
PRIME BOOST PUMP ON Capsule Goes On When Power Is Applied To Helicopter	TABLE NO. 4-2-8-3

- f. Shut down No. 1 engine (Appropriate Flight Manual).

RESULT: #1 ENG OUT capsule shall go on before 55% Ng.

- If result is not as specified, troubleshoot Instrument Display System (PARA 8-2-3).

- g. With qualified personnel at controls and instrument panel ENGINE IGNITION switch at OFF, press and hold No. 1 engine start button until a cockpit Ng indication is noted (above 10% Ng), and then, pull engine controls quadrant ENG POWER CONT lever down (Appropriate Flight Manual).

RESULT: Engine starter shall motor engine. When engine POWER CONT lever is pulled down, engine shall stop motor-ing.

- If result is not as specified, go to TABLE NO. 4-2-8-17.

- h. Remove P314 from starter speed switch located in cabin ceiling.

- i. Place container under No. 1 engine overboard drain line.

- j. With qualified personnel at controls and APU operating, have pilot press and hold No. 1 engine start button. Advance engine controls quadrant ENG POWER CONT lever to between OFF and IDLE (Appropriate Flight Manual).

RESULT: No. 1 engine starter shall continue to motor No. 1 engine when ENG POWER CONT lever is away from OFF.

- If result is not as specified, go to TABLE NO. 4-2-8-18.

- k. Move engine controls quadrant ENG POWER CONT lever to LOCKOUT.

RESULT: Fuel from No. 1 engine overboard drain line shall drain into container.

- If result is not as specified, go to TABLE NO. 4-2-8-19.

- l. Connect P314 to starter speed switch.

- m. Shut down APU (Appropriate Flight Manual).

4-2-8.2 No. 2 Engine.

- a. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.

- b. Place upper console FUEL PUMP switch to OFF, if not already done.

- c. Place upper console BATT switch to ON, if not already done.

RESULT: PRIME BOOST PUMP ON capsule shall not come on.

- If result is not as specified, go to TABLE NO. 4-2-8-3.

- d. Start APU (Appropriate Flight Manual).

RESULT: #2 ENG OUT capsule on pilot's and copilot's master caution panel shall be on.

- If result is not as specified, troubleshoot Instrument Display System (PARA 8-2-3).

- e. With qualified personnel at controls, start No. 2 engine and operate engine up to IDLE speed (Appropriate Flight Manual). Refer to TABLE NO. 4-2-8-2 for trouble analysis.

CHART NO. 4-2-8-2. No. 2 Engine Starting.

FAULT INDICATION	GO TO TABLE NO.
No. 2 Engine Does Not Start (Compressor Not Rotating) (#2 ENGINE STARTER Capsule Is On)	TABLE NO. 4-2-8-20
Engine Starter Motors Engine, But Cannot Reach Proper Ng Speed To Start Engine	TABLE NO. 4-2-8-5
No. 2 Engine Does Not Start (Compressor Not Rotating) (#2 ENGINE STARTER Capsule Is Off)	TABLE NO. 4-2-8-21
PRIME BOOST PUMP ON Capsule Is Not On When Start Button Is Pressed	TABLE NO. 4-2-8-7
No. 2 Engine Starter Stops (#2 ENGINE STARTER Capsule Goes Off) When Start Button Is Released	TABLE NO. 4-2-8-22
No. 2 Engine Starts, But #2 ENGINE STARTER Capsule Is Off (APU Is Not Stalling)	TABLE NO. 4-2-8-23
No. 2 Engine Starter Motors Engine (#2 ENGINE STARTER Capsule Is On), But APU Stalls (Popping Sound Is Heard)	TABLE NO. 4-2-8-24
No. 2 Engine Motors, But Does Not Ignite And Accelerate To Idle Speed	TABLE NO. 4-2-8-25
No. 1 Or No. 2 Engine Power Control Lever Is More Difficult To Move Than Normal	TABLE NO. 4-2-8-12
Engine No. 1 Or No. 2 Has A Hung Start (TGT Increases, But Hangs)	TABLE NO. 4-2-8-13
Engine No. 1 Or No. 2 TGT Abnormally High (TGT Reaches 810°C/1490°F Before Idle Speed Is Reached)	TABLE NO. 4-2-8-14
If ground idle speed is too high or low	PARA 4-2-1
#2 ENGINE STARTER Capsule Does Not Go Off (Engine Speed Above 55% Ng)	PARA 8-2-3/TABLE NO. 4-2-8-26
#1 Or #2 ENGINE STARTER Capsule Goes Off Before 52% Ng Engine Speed	TABLE NO. 4-2-8-15
PRIME BOOST PUMP ON Capsule Goes On When Power Is Applied To Helicopter	TABLE NO. 4-2-8-3

- f. Shut down No. 2 engine (Appropriate Flight Manual).

RESULT: #2 ENG OUT capsule shall go on before 55% Ng.

- If result is not as specified, troubleshoot Instrument Display System (PARA 8-2-3).

- g. With qualified personnel at controls and instrument panel ENGINE IGNITION switch at OFF, press and hold No. 2 engine start button until a cockpit Ng indication is noted (above 10% Ng), then pull engine controls quadrant ENG POWER CONT lever down.

RESULT: Engine starter shall motor engine. When ENG POWER CONT lever is pulled down, engine shall stop motor-ing.

- If result is not as specified, go to TABLE NO. 4-2-8-17.
- h. Remove P314 from starter speed switch located in cabin ceiling.
- i. Place container under No. 2 engine overboard drain line.

- j. With qualified personnel at controls and APU operating, have pilot press and hold No. 2 engine start button. Advance engine control quadrant ENG POWER CONT lever to between OFF and IDLE (Appropriate Flight Manual).

RESULT: No. 2 engine starter shall continue to motor No. 2 engine when ENG POWER CONT lever is away from OFF.

- If result is not as specified, go to TABLE NO. 4-2-8-27.

- k. Move engine control quadrant ENG POWER CONT lever to LOCKOUT.

RESULT: Fuel from No. 2 engine overboard drain line shall drain into container.

- If result is not as specified, go to TABLE NO. 4-2-8-28.
- l. Connect P314 to starter speed switch.
- m. Shut down APU (Appropriate Flight Manual).

TABLE NO. 4-2-8-3. PRIME BOOST PUMP ON Capsule Goes On When Power Is Applied To Helicopter.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Is prime boost pump operating?	Step 1. If pump is operating, go to 2. Step 2. If pump is not operating, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 3.
2. Check continuity between J241-B and J241-C.	Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to 3. Step 2. If continuity is not present, replace switch S19 (PARA 9-4-5). Go to 3.

TABLE NO. 4-2-8-3. PRIME BOOST PUMP ON Capsule Goes On When Power Is Applied To Helicopter. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	3. Procedure completed.

TABLE NO. 4-2-8-4. No. 1 Engine Does Not Start (Compressor Not Rotating) (#1 ENGINE STARTER Capsule Is On).

TEST OR INSPECTION	CORRECTIVE ACTION
I	1. Troubleshoot APU System for faulty start bypass valve (PARA 15-2-1 or 15-2-2). Step 1. If valve is malfunctioning, replace valve (PARA 7-4-53). Go to 7. Step 2. If valve is operating properly, go to 2. 2. Check all clamps and rubber connectors on pneumatic lines between starter and APU check valve for proper connection or damage. Step 1. If connections are proper and no damage is present, go to 3. Step 2. If connections are not proper and/or damage is present, repair/replace clamps or connectors as required (PARA 7-4-4). Go to 7. 3. Disconnect starter from engine (PARA 7-4-1). Go to 4. NOTE Turning the gear train requires considerable effort to overcome resistance. 4. Check starter for sheared/seized shaft by grasping shaft spline end and rotating shaft clockwise, making sure turbine wheel also rotates. Step 1. If starter has sheared/seized shaft, replace starter (PARA 7-4-1). Go to 7. Step 2. If starter does not have sheared/seized shaft, go to 5. 5. Troubleshoot engine for seized condition (SEI 671 and TM 1-2840-248-23). Go to 6. 6. Install starter (PARA 7-4-1). Go to 7. 7. Procedure completed.

TABLE NO. 4-2-8-5. Engine Starter Motors Engine, But Cannot Reach Proper Ng Speed To Start Engine.

TEST OR INSPECTION	CORRECTIVE ACTION
1.	With helicopter on level ground, verify starter has not been over serviced by removing oil fill plug and letting excess oil drain from starter. Go to 2.
2.	Check that oil is level with oil fill port. Step 1. If oil is level with oil fill port, go to 3. Step 2. If oil is not level with oil fill port, replace starter (PARA 7-4-1). Go to 4.
3.	Check that engine starter can reach proper Ng. Step 1. If engine starter can reach proper Ng, Engine Start System is operational. Go to 4. Step 2. If engine starter can not reach proper Ng, replace starter (PARA 7-4-1). Go to 4.
4.	Procedure completed.

TABLE NO. 4-2-8-6. No. 1 Engine Does Not Start (Compressor Not Rotating) (#1 ENGINE STARTER Capsule Is Off).

TEST OR INSPECTION	CORRECTIVE ACTION
1.	With APU shut down and No. 1 engine start button pressed, check for 28 VDC between P440-1 and ground. Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.
2.	Check continuity between P440-2 and ground. Step 1. If continuity is present, replace No. 1 engine start control valve (PARA 7-4-2). Go to 16. Step 2. If continuity is not present, repair/replace wiring between P440-2 and ground wire (Appendix F). Go to 16.

TABLE NO. 4-2-8-6. No. 1 Engine Does Not Start (Compressor Not Rotating) (#1 ENGINE STARTER Capsule Is Off). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
3. Check continuity between J105-J and J105-K. Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, replace engine control quadrant assembly (PARA 4-4-21). Go to 16. 4. With APU shut down, press No. 1 engine start button and check continuity between J105-G and J105-H. Step 1. If continuity is present, go to 5. Step 2. If continuity is not present, go to 15. 5. Check for 28 VDC between P105-K and ground. Step 1. If voltage is as specified, go to 6. Step 2. If voltage is not as specified, go to 14. 6. Check for 28 VDC between P105-G and ground. Step 1. If voltage is as specified, go to 7. Step 2. If voltage is not as specified, repair/replace wiring between terminal 1 (NO. 1 ENG START circuit breaker, DC ESNTL BUS - upper console circuit breaker) and P105-G (Appendix F). Go to 16. 7. With No. 1 engine start button pressed, check for 28 VDC between P242-T and ground. Step 1. If voltage is as specified, go to 8. Step 2. If voltage is not as specified, repair/replace wiring between P105-H and P242-T (Appendix F). Go to 16. 8. Install an insulated jumper wire between P242-U and P242-Y. Go to 9. 9. Check for 28 VDC between P440-1 and ground. Step 1. If voltage is as specified, go to 10. Step 2. If voltage is not as specified, go to 12. 10. Remove jumper wire from P242. Go to 11.	

TABLE NO. 4-2-8-6. No. 1 Engine Does Not Start (Compressor Not Rotating) (#1 ENGINE STARTER Capsule Is Off). (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
11. Replace left relay panel (PARA 9-4-2). Go to 16.
12. Remove jumper wire from P242. Go to 13.
13. Repair/replace wiring, as required, between the following (Appendix F), then go to 16: P105-J and P242-U P242-Y and P440-1
14. Check for 28 VDC between terminal 1 (NO. 1 ENG START circuit breaker, DC ESNTL BUS - upper console circuit breaker panel) and ground. Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P105-K (Appendix F). Go to 16. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 16.
15. Check No. 1 engine starter switch for proper adjustment (PARA 4-4-23). Step 1. If switch has proper adjustment, replace switch (PARA 4-5-13). Go to 16. Step 2. If switch does not have proper adjustment, adjust switch (PARA 4-4-23). Go to 16.
16. Procedure completed.

TABLE NO. 4-2-8-7. PRIME BOOST PUMP ON Capsule Is Not On When Start Button Is Pressed.

TEST OR INSPECTION CORRECTIVE ACTION
1. Check for 28 VDC between P241-B and ground. Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 4.
2. Check for 28 VDC between P246-G and ground.

**TABLE NO. 4-2-8-7. PRIME BOOST PUMP ON Capsule Is Not On When Start Button Is Pressed.
(Cont)**

TEST OR INSPECTION CORRECTIVE ACTION	
Step 1. If voltage is as specified, troubleshoot Fuel Prime Boost System (PARA 10-2-3). Go to 5. Step 2. If voltage is not as specified, go to 3.	
3. Check continuity between P241-C and P246-G.	
Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to 5. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 5.	
4. Check for 28 VDC between P246-B and ground.	
Step 1. If voltage is as specified, repair/replace wiring between P241-B and P246- B (Appendix F). Go to 5.	
Step 2. If voltage is not as specified, troubleshoot Fuel Prime Boost System (PARA 10-2-3). Go to 5.	
5. Procedure completed.	

TABLE NO. 4-2-8-8. No. 1 Engine Starter Stops (#1 ENGINE STARTER Capsule Goes Off) When Start Button Is Released.

TEST OR INSPECTION CORRECTIVE ACTION	
1. Disconnect P314 from starter speed switch. Go to 2.	
2. Install an insulated jumper wire between P314-C and P314-D. Go to 3.	
3. Make sure NO. 1 ENG POWER CONT lever is OFF. Go to 4.	
4. With qualified personnel at controls and APU operating, press No. 1 engine start button, then release. Go to 5.	
5. Check that #1 ENGINE STARTER capsule goes on and engine continues to motor after start button is released.	
Step 1. If capsule goes on and engine continues to motor, go to 6.	
Step 2. If capsule does not go on and/or engine does not motor, go to 9.	

TABLE NO. 4-2-8-8. No. 1 Engine Starter Stops (#1 ENGINE STARTER Capsule Goes Off) When Start Button Is Released. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
6. Shut down APU (Appropriate Flight Manual). Go to 7.	
7. Remove jumper wire from P314. Go to 8.	
8. Replace starter speed switch (PARA 4-5-13). Go to 19.	
9. Remove jumper wire from P314. Do not install connector to starter speed switch. Go to 10.	
10. With qualified personnel at controls and APU operating, press No. 1 engine start button and hold. Go to 11.	
11. Advance NO. 1 ENG POWER CONT lever to a position between OFF and IDLE. Go to 12.	
12. Check that No. 1 engine continues to motor with ENG POWER CONT lever away from OFF.	
Step 1. If No. 1 engine continues to motor, go to 13. Step 2. If No. 1 engine does not continue to motor, go to 16.	
13. Shut down APU (Appropriate Flight Manual). Go to 14.	
14. Connect P314 to starter speed switch. Go to 15.	
15. Repair/replace wiring, as required, between the following (Appendix F), then go to 19:	
P105-T and P314-D P314-C and P440-5	
16. Shut down APU (Appropriate Flight Manual). Go to 17.	
17. Connect P314 to starter speed switch. Go to 18.	
18. Go to TABLE NO. 4-2-8-18. Go to 19.	
19. Procedure completed.	

TABLE NO. 4-2-8-9. No. 1 Engine Starts, But #1 ENGINE STARTER Capsule Is Off (APU Is Not Stalling).

**TEST OR INSPECTION
CORRECTIVE ACTION**

1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2.

2. Check that #1 ENGINE STARTER capsule goes on.

Step 1. If capsule goes on, go to 3.

Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 12.

3. Install an insulated jumper wire between P440-3 and P440-5. Go to 4.

4. Check helicopter configuration.

Step 1. **EMEP** Go to 5. 

Step 2. **W/O EMEP** Go to 7. 

5. Remove and check pin filter adapter P117 (PARA 9-4-98).

Step 1. If pin filter adapter is good, go to 6.

Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98).
Go to 12.

6. Install pin filter adapter (PARA 9-4-98). Go to 7.

7. Check for 28 VDC between P117-7 and ground.

Step 1. If voltage is as specified, go to 8.

Step 2. If voltage is not as specified, go to 10.

8. Remove jumper wire from P440. Go to 9.

9. Troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 12.

10. Remove jumper wire from P440. Go to 11.

11. Repair/replace wiring between P117-7 and P244-M (Appendix F). Go to 12.

12. Procedure completed.

TABLE NO. 4-2-8-10. No. 1 Engine Motors, But Does Not Ignite And Accelerate To Idle Speed.

TEST OR INSPECTION	CORRECTIVE ACTION
1.	Move ENG POWER CONT lever on control quadrant in cockpit to LOCKOUT and hold. Go to 2.
2.	Move ENG FUEL SYS selector lever on control quadrant to DIR detent. Go to 3.
3.	Place FUEL PUMP switch to FUEL PRIME. Go to 4.
4.	Check that fuel is coming from overboard drain. Step 1. If fuel is coming from overboard drain, go to 5. Step 2. If fuel is not coming from overboard drain, troubleshoot Fuel Prime Boost System (PARA 10-2-3). Go to 20.
5.	Troubleshoot engine for no ignite (SEI 671 and TM 55-284-248-23). Step 1. If engine is good, go to 6.
6.	Install an insulated jumper wire between P803-17 and P803-18. Go to 7.
7.	Check continuity between P111-BB and P111-CC. Step 1. If continuity is present, go to 8. Step 2. If continuity is not present, go to 18.
8.	Check continuity between P242-W and P242-X. Step 1. If continuity is present, go to 9. Step 2. If continuity is not present, go to 16.
9.	Remove jumper wire from P803. Go to 10.
10.	Check helicopter configuration. Step 1. S70-II , go to 11. Step 2. S70-I , go to 14.
S70-II	11. Check continuity between: P803-17 and P815-35 P803-19 and P815-35

TABLE NO. 4-2-8-10. No. 1 Engine Motors, But Does Not Ignite And Accelerate To Idle Speed. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

- Step 1. If continuity is present, go to **12**.
Step 2. If continuity is not present, replace engine harness (SEI 671 and TM 55-284-248-23). Go to **20**.

S70-II**12. Check continuity between P381-F and J815-35.**

- Step 1. If continuity is present, go to **13**.
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **20**.

S70-II**13. Check continuity between and P111-BB and P381-G.**

- Step 1. If continuity is present, go to **20**.
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **20**.

S70-I**14. On instrument panel, turn ENGINE IGNITION switch to ON. Go to 15.****S70-I****15. Check continuity between J111-BB and J111-CC.**

- Step 1. If continuity is present, replace engine ignition switch (PARA 9-4-96).
Go to **20**.
Step 2. If trouble remains, troubleshoot Engine Overspeed Protection System (PARA 4-2-4). Go to **19**.
Step 3. If continuity is not present, replace left relay panel (PARA 9-4-2). Go to **20**.

16. Remove jumper wire from P803. Go to 17.**17. Repair/replace wiring, as required, between the following (Appendix F), then go to 20:**

P111-BB and P242-W
P111-CC and P242-X

18. Remove jumper wire from P803. Go to 19.**19. Repair/replace wiring, as required, between the following (Appendix F), then go to 20:**

P111-BB and J815-35
P111-CC and J815-34

- Step 1. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23).
Go to **20**.

TABLE NO. 4-2-8-10. No. 1 Engine Motors, But Does Not Ignite And Accelerate To Idle Speed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
20. Procedure completed.	

TABLE NO. 4-2-8-11. No. 1 Engine Starter Motors Engine (#1 ENGINE STARTER Capsule Is On), But APU Stalls (Popping Sound Is Heard).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Start APU (Appropriate Flight Manual). Go to 2.	
2. Motor No. 2 engine (Appropriate Flight Manual). Go to 3.	
3. Check if APU stalls.	Step 1. If APU stalls, go to 4. Step 2. If APU does not stall, shut down APU (Appropriate Flight Manual). Troubleshoot right relay panel (PARA 9-2-19). Go to 17.
4. Troubleshoot APU System, start bypass valve circuitry (PARA 15-2-1 or 15-2-2).	
	Step 1. If APU System is good, go to 5.
5. Install an insulated jumper wire between P440-3 and P440-5. Go to 6.	
6. Check helicopter configuration.	Step 1. WINTER , go to 7. Step 2. W/O WINTER , go to 8.
WINTER 7. Check for 28 VDC between P244B-M and ground.	
	Step 1. If voltage is as specified, go to 9. Step 2. If voltage is not as specified, replace winterization kit harness (PARA 16-6-2). Go to 17. Step 3. If trouble remains, go to 8.
8. Check for 28 VDC between P244-M and ground.	

TABLE NO. 4-2-8-11. No. 1 Engine Starter Motors Engine (#1 ENGINE STARTER Capsule Is On), But APU Stalls (Popping Sound Is Heard). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
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- Step 1. If voltage is as specified, go to **9**.
 Step 2. If voltage is not as specified, go to **13**.

9. Remove jumper wire from P440. Go to 10.

10. Check helicopter configuration.

- Step 1. **WINTER** , go to **11**.
 Step 2. **W/O WINTER** , go to **12**.

WINTER **11. Replace winterization kit harness (PARA 16-6-2).**

- Step 1. If trouble remains, go to **12**.

12. Repair/replace wiring between terminal 1 (AIR SOURCE HEAT/START circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and P244-T (Appendix F). Go to 17.

13. Remove jumper wire from P440. Go to 14.

14. Check helicopter configuration.

- Step 1. **WINTER** , go to **15**.
 Step 2. **W/O WINTER** , go to **16**.

WINTER **15. Replace winterization kit harness (PARA 16-6-2).**

- Step 1. If trouble remains, go to **16**.

16. Repair/replace wiring between P117-7 and P244-M (Appendix F). Go to 17.

17. Procedure completed.

TABLE NO. 4-2-8-12. No. 1 Or No. 2 Engine Power Control Lever Is More Difficult To Move Than Normal.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Disconnect engine power available rotary input from hydromechanical control unit (HMU). Go to 2.	
2. Cycle engine power control lever (on engine control quadrant) back and forth. Go to 3.	
3. Check that movement of lever is smooth.	Step 1. If movement of lever is smooth, replace engine HMU (SEI 671 and TM 55-284-248-23). Go to 7. Step 2. If movement of lever is not smooth, go to 4.
4. Disconnect power available spindle (PAS) push/pull cable from rotary input. Go to 5.	
5. Cycle engine power control lever. Go to 6.	
6. Check that movement of lever is smooth.	Step 1. If movement is smooth, replace engine power available rotary input (PARA 4-4-19). Go to 7. Step 2. If movement is not smooth, replace fuselage power available push/pull cable (PARA 4-4-28). Go to 7.
7. Procedure completed.	

TABLE NO. 4-2-8-13. Engine No. 1 Or No. 2 Has A Hung Start (TGT Increases, But Hangs).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Troubleshoot Engine Anti-Ice System, engine inlet anti-ice valve (PARA 4-2-7).	Step 1. If system is good, go to 2. Step 2. If system is not good, repair/replace components as required. Go to 5.
2. Start engine (Appropriate Flight Manual). Go to 3.	

TABLE NO. 4-2-8-13. Engine No. 1 Or No. 2 Has A Hung Start (TGT Increases, But Hangs). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	3. Check that engine start is normal. - Step 1. If engine start is normal, system is operational. Go to 4. - Step 2. If engine start is abnormal, troubleshoot engine for hung start (SEI 671 and TM 55-284-248-23). Go to 4. 4. Shut down engine (Appropriate Flight Manual). Go to 5. 5. Procedure completed.

TABLE NO. 4-2-8-14. Engine No. 1 Or No. 2 TGT Abnormally High (TGT Reaches 810°C/1490°F Before Idle Speed Is Reached).

TEST OR INSPECTION	CORRECTIVE ACTION
	1. On upper console panel, place FUEL PUMP switch to FUEL PRIME. Go to 2. 2. Start engine (Appropriate Flight Manual). Go to 3. 3. Check that TGT is normal during start. - Step 1. If TGT is normal, system is operational. Go to 7. - Step 2. If TGT is abnormal, go to 4. 4. Replace engine starter (PARA 7-4-1). Go to 5. 5. Start engine (Appropriate Flight Manual). Go to 6. 6. Check that TGT is normal during start. - Step 1. If TGT is normal, system is operational. Go to 7. - Step 2. If TGT is abnormal, troubleshoot engine for abnormally high TGT during start (SEI 671 and TM 55-284-248-23). Go to 7. 7. Shut down engine (Appropriate Flight Manual). Go to 8. 8. Procedure completed.

TABLE NO. 4-2-8-15. #1 Or #2 ENGINE STARTER Capsule Goes Off Before 52% Ng Engine Speed.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Replace starter speed switch (PARA 7-4-10). Go to 2. 2. Start engine and operate up to idle speed (Appropriate Flight Manual). Go to 3. 3. Check that ENGINE STARTER capsule goes off between 52% and 55% Ng. - Step 1. If capsule goes off when specified, Engine Start System is operational. Go to 4. - Step 2. If capsule does not go off when specified, replace engine starter (PARA 7-4-1). Go to 4. 4. Shut down engine (Appropriate Flight Manual). Go to 5. 5. Procedure completed.

TABLE NO. 4-2-8-16. #1 ENGINE STARTER Capsule Does Not Go Off (Engine Speed Above 55% Ng).

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Visually check No. 1 engine start control valve position indicator. Go to 2. 2. Check that indicator shows CLOSED. - Step 1. If indicator shows CLOSED, go to 3. - Step 2. If indicator does not show CLOSED, replace start control valve (PARA 7-4-2). Go to 12. 3. Check continuity between J440-3 and J440-4. - Step 1. If continuity is present, go to 4. - Step 2. If continuity is not present, replace No. 1 engine start control valve (PARA 7-4-2). Go to 12. 4. Install an insulated jumper wire between P807-1 and P807-3. Go to 5.

**TABLE NO. 4-2-8-16. #1 ENGINE STARTER Capsule Does Not Go Off (Engine Speed Above 55% Ng).
(Cont)**

TEST OR INSPECTION CORRECTIVE ACTION
5. Check continuity between P314-E and P314-F.
Step 1. If continuity is present, go to 6. Step 2. If continuity is not present, go to 10.
6. Remove jumper wire. Go to 7.
7. Replace starter speed switch (PARA 7-4-10). Go to 8.
8. Operate No. 1 engine up to idle speed (Appropriate Flight Manual). Go to 9.
9. Check that #1 ENGINE STARTER capsule is off.
Step 1. If capsule is off, Engine Start System is operational. Shut down engine (Appropriate Flight Manual). Go to 12. Step 2. If capsule is not off, shut down engine (Appropriate Flight Manual). Replace No. 1 engine starter (PARA 7-4-1). Go to 12.
10. Remove jumper wire from connector P807. Go to 11.
11. Repair/replace wiring, as required, between the following (Appendix F), then go to 12: P314-E and J825-3 P314-F and J825-2
Step 1. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 12.
12. Procedure completed.

TABLE NO. 4-2-8-17. No. 1 Or No. 2 Engine Starter Does Not Shutdown When Power Control Lever Is Pulled Down (Speed Below 52% Ng).

TEST OR INSPECTION CORRECTIVE ACTION
1. Adjust starter abort switch (PARA 4-5-14). Go to 2.

TABLE NO. 4-2-8-17. No. 1 Or No. 2 Engine Starter Does Not Shutdown When Power Control Lever Is Pulled Down (Speed Below 52% Ng). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
2.	Pull lever down and check continuity between J105-J and J105-K. Step 1. If continuity is present, replace starter abort switch (PARA 4-5-14). Go to 3. Step 2. If continuity is not present, Engine Start System is operational. Go to 3.
3.	Procedure completed.

TABLE NO. 4-2-8-18. No. 1 Engine Starter Does Not Motor Engine (Compressor Is Not Rotating) When Using Engine Starter Override.

TEST OR INSPECTION	CORRECTIVE ACTION
1.	With No. 1 engine start button pressed, check continuity between J105-S and J105-T. Step 1. If continuity is present, go to 2. Step 2. If continuity is not present, replace engine control quadrant starter override switch (PARA 4-5-15). Go to 14.
2.	Install an insulated jumper wire between P105-K and P105-T. Go to 3.
3.	Check for 28 VDC between P242-V and ground. Step 1. If voltage is as specified, go to 4. Step 2. If voltage is not as specified, go to 13.
4.	Remove jumper wire from P105. Go to 5.
5.	Install insulated jumper wire between P105-K and P105-S. Go to 6.
6.	Check for 28 VDC between J221-Y and ground. Step 1. If voltage is as specified, go to 7. Step 2. If voltage is not as specified, go to 9.
7.	Remove jumper wire from P105. Go to 8.

TABLE NO. 4-2-8-18. No. 1 Engine Starter Does Not Motor Engine (Compressor Is Not Rotating) When Using Engine Starter Override. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
8. Replace left relay panel (PARA 9-4-2). Go to 15.	
9. Remove jumper wire from P105. Go to 10.	
10. Check helicopter configuration.	
Step 1. WINTER , go to 11.	
Step 2. W/O WINTER , go to 12.	
WINTER	
11. Replace winterization kit harness (PARA 16-6-2).	
Step 1. If trouble remains, go to 12.	
12. Repair/replace wiring between and P105-S and P244-M (Appendix F). Go to 15.	
13. Remove jumper wire from P105. Go to 14.	
14. Repair/replace wiring between P105-T and P242-V (Appendix F). Go to 15.	
15. Procedure completed.	

TABLE NO. 4-2-8-19. No Fuel Is Coming Out Of No. 1 Engine Fuel Drain.

TEST OR INSPECTION CORRECTIVE ACTION	
1. Visually check drain tube for obvious obstructions.	
Step 1. If obstructions are present, remove obstructions (PARA 10-4-29). Go to 3.	
Step 2. If obstructions are not present, go to 2.	
2. Check for 28 VDC between TB1-7 and ground.	
Step 1. If voltage is as specified, troubleshoot Fuel Prime Boost System (PARA 10-2-3). Go to 3.	
Step 2. If voltage is not as specified, repair/replace wiring between TB1 and ground (Appendix F). Go to 3.	

TABLE NO. 4-2-8-19. No Fuel Is Coming Out Of No. 1 Engine Fuel Drain. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	3. Procedure completed.

TABLE NO. 4-2-8-20. No. 2 Engine Does Not Start (Compressor Not Rotating) (#2 ENGINE STARTER Capsule Is On).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Troubleshoot APU System for faulty start bypass valve (PARA 15-2-1 or 15-2-2).	Step 1. If APU System start bypass valve is faulty, replace valve (PARA 7-4-53). Go to 7. Step 2. If APU System valve is good, go to 2.
2. Check all clamps and rubber connections on pneumatic lines between starter and APU check valve for proper connections or damage.	Step 1. If connections are good, go to 3. Step 2. If connections are not good, repair/replace clamps or connections as required. Go to 7.
3. Disconnect starter from engine (PARA 7-4-1). Go to 4.	
NOTE Turning the gear train requires considerable effort to overcome resistance.	
4. Check starter for sheared/seized shaft by grasping shaft spline end and rotating clockwise, making sure turbine wheel also rotates.	Step 1. If starter has sheared/seized shaft, replace starter (PARA 7-4-1). Go to 7. Step 2. If starter does not have sheared/seized shaft, go to 5.
5. Check engine for seized condition (SEI 671 and TM 55-284-248-23). Go to 6.	
6. Install starter (PARA 7-4-1). Go to 7.	
7. Procedure completed.	

TABLE NO. 4-2-8-21. No. 2 Engine Does Not Start (Compressor Not Rotating) (#2 ENGINE STARTER Capsule Is Off).

TEST OR INSPECTION	CORRECTIVE ACTION
1. With APU shut down and external power on, press No. 2 engine start button and check for 28 VDC between P450-1 and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.
2. Check continuity between P450-2 and ground.	Step 1. If continuity is present, replace No. 2 engine start control valve (PARA 7-4-2). Go to 10. Step 2. If continuity is not present, repair/replace wiring between P450-2 and ground (Appendix F). Go to 10.
3. Check continuity between J106-J and J106-K.	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, replace engine control quadrant assembly start abort switch (PARA 4-5-14). Go to 10.
4. Press No. 2 engine start button and check continuity between J106-G and J106-H.	Step 1. If continuity is present, go to 5. Step 2. If continuity is not present, go to 9.
5. Check for 28 VDC between P106-K and ground.	Step 1. If voltage is as specified, go to 6. Step 2. If voltage is not as specified, go to 7.
6. Check for 28 VDC between P106-G and ground.	Step 1. If voltage is as specified, go to 8. Step 2. If voltage is not as specified, repair/replace wiring between terminal 1 (NO. 2 ENG START CONTR circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and P106-G (Appendix F). Go to 10.
7. Check for 28 VDC between terminal 1 (NO. 2 ENG START CONTR circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and ground.	

1. With APU shut down and external power on, press No. 2 engine start button and check for 28 VDC between P450-1 and ground.

Step 1. If voltage is as specified, go to 2.

Step 2. If voltage is not as specified, go to 3.

2. Check continuity between P450-2 and ground.

Step 1. If continuity is present, replace No. 2 engine start control valve (PARA 7-4-2). Go to 10.

Step 2. If continuity is not present, repair/replace wiring between P450-2 and ground (Appendix F). Go to 10.

3. Check continuity between J106-J and J106-K.

Step 1. If continuity is present, go to 4.

Step 2. If continuity is not present, replace engine control quadrant assembly start abort switch (PARA 4-5-14). Go to 10.

4. Press No. 2 engine start button and check continuity between J106-G and J106-H.

Step 1. If continuity is present, go to 5.

Step 2. If continuity is not present, go to 9.

5. Check for 28 VDC between P106-K and ground.

Step 1. If voltage is as specified, go to 6.

Step 2. If voltage is not as specified, go to 7.

6. Check for 28 VDC between P106-G and ground.

Step 1. If voltage is as specified, go to 8.

Step 2. If voltage is not as specified, repair/replace wiring between terminal 1 (NO. 2 ENG START CONTR circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and P106-G (Appendix F). Go to 10.

7. Check for 28 VDC between terminal 1 (NO. 2 ENG START CONTR circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and ground.

TABLE NO. 4-2-8-21. No. 2 Engine Does Not Start (Compressor Not Rotating) (#2 ENGINE STARTER Capsule Is Off). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P106-K (Appendix F). Go to 10. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 10.
8. Check continuity between:	P241-a and P450-1 P241-b and P106-J P241-R and P106-H
	Step 1. If continuity is as specified, troubleshoot right relay panel (PARA 9-2-19). Go to 10. Step 2. If continuity is not as specified, repair/replace wiring (Appendix F). Go to 10.
9. Check No. 2 engine starter switch for proper adjustment (PARA 4-4-23).	
	Step 1. If switch is adjusted properly, replace starter switch (PARA 4-5-13). Go to 10. Step 2. If switch is not adjusted properly, adjust switch (PARA 4-4-23). Go to 10.
10. Procedure completed.	

TABLE NO. 4-2-8-22. No. 2 Engine Starter Stops (#2 ENGINE STARTER Capsule Goes Off) When Start Button Is Released.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Disconnect P314 from starter speed switch. Go to 2.	
2. Install an insulated jumper wire between P314-J and P314-K. Go to 3.	
3. Make sure No. 2 engine POWER CONT LEVER is OFF. Go to 4.	
4. With qualified personnel at control and APU operating, press No. 2 engine start button, then release. Go to 5.	

TABLE NO. 4-2-8-22. No. 2 Engine Starter Stops (#2 ENGINE STARTER Capsule Goes Off) When Start Button Is Released. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
5. Check that #2 ENGINE STARTER capsule goes on and engine continues to motor after start button is released.	Step 1. If capsule goes on and engine motors, go to 6. Step 2. If capsule does not go on and/or engine does not motor, go to 9.
6. Shut down APU (Appropriate Flight Manual). Go to 7.	
7. Remove jumper wire from P314. Go to 8.	
8. Replace starter speed switch (PARA 7-4-10). Go to 19.	
9. Remove jumper wire from P314. Do not install connector to speed switch. Go to 10.	
10. With qualified personnel at controls and APU operating, press No. 2 engine start button and hold. Go to 11.	
11. Advance NO. 2 ENG POWER CONT lever to a position between OFF and IDLE. Go to 12.	
12. Check that No. 2 engine continues to motor with ENG POWER CONT lever away from OFF.	Step 1. If No. 2 engine continues to motor, go to 13. Step 2. If No. 2 engine does not continue to motor, go to 16.
13. Shut down APU (Appropriate Flight Manual). Go to 14.	
14. Connect P314 to starter speed switch. Go to 15.	
15. Repair/replace wiring, as required, between the following (Appendix F), then go to 19:	P106-T and P314-K P314-J and P450-5
16. Shut down APU (Appropriate Flight Manual). Go to 17.	
17. Reconnect P314 to starter speed switch. Go to 18.	

5. Check that #2 ENGINE STARTER capsule goes on and engine continues to motor after start button is released.

Step 1. If capsule goes on and engine motors, go to 6.

Step 2. If capsule does not go on and/or engine does not motor, go to 9.

6. Shut down APU (Appropriate Flight Manual). Go to 7.

7. Remove jumper wire from P314. Go to 8.

8. Replace starter speed switch (PARA 7-4-10). Go to 19.

9. Remove jumper wire from P314. Do not install connector to speed switch. Go to 10.

10. With qualified personnel at controls and APU operating, press No. 2 engine start button and hold. Go to 11.

11. Advance NO. 2 ENG POWER CONT lever to a position between OFF and IDLE. Go to 12.

12. Check that No. 2 engine continues to motor with ENG POWER CONT lever away from OFF.

Step 1. If No. 2 engine continues to motor, go to 13.

Step 2. If No. 2 engine does not continue to motor, go to 16.

13. Shut down APU (Appropriate Flight Manual). Go to 14.

14. Connect P314 to starter speed switch. Go to 15.

15. Repair/replace wiring, as required, between the following (Appendix F), then go to 19:

P106-T and P314-K

P314-J and P450-5

16. Shut down APU (Appropriate Flight Manual). Go to 17.

17. Reconnect P314 to starter speed switch. Go to 18.

TABLE NO. 4-2-8-22. No. 2 Engine Starter Stops (#2 ENGINE STARTER Capsule Goes Off) When Start Button Is Released. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
18.	Go to TABLE NO. 4-2-8-15. Go to 19.
19.	Procedure completed.

TABLE NO. 4-2-8-23. No. 2 Engine Starts, But #2 ENGINE STARTER Capsule Is Off (APU Is Not Stalling).

TEST OR INSPECTION	CORRECTIVE ACTION
1.	Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2.
2.	Check that #2 ENGINE STARTER capsule goes on. Step 1. If capsule goes on, go to 3. Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 12.
3.	Install an insulated jumper wire between P450-3 and P450-5. Go to 4.
4.	Check helicopter configuration. Step 1. EMEP Go to 5. Step 2. W/O EMEP Go to 7.
EMEP	5. Remove and check pin filter adapter P117 (PARA 9-4-98). Step 1. If pin filter adapter is good, go to 6. Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to 12.
EMEP	6. Install pin filter adapter (PARA 9-4-98). Go to 7.
	7. Check for 28 VDC between P117-55 and ground. Step 1. If voltage is as specified, go to 8. Step 2. If voltage is not as specified, go to 10.

TABLE NO. 4-2-8-23. No. 2 Engine Starts, But #2 ENGINE STARTER Capsule Is Off (APU Is Not Stalling). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
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|--|--|
| 8. Remove jumper wire from P450. Go to 9. | |
| 9. Troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 12. | |
| 10. Remove jumper wire from P450. Go to 11. | |
| 11. Repair/replace wiring between P117-55 and P241-W (Appendix F). Go to 12. | |
| 12. Procedure completed. | |

TABLE NO. 4-2-8-24. No. 2 Engine Starter Motors Engine (#2 ENGINE STARTER Capsule Is On), But APU Stalls (Popping Sound Is Heard).

TEST OR INSPECTION	CORRECTIVE ACTION
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- | | |
|---|--|
| 1. Start APU (Appropriate Flight Manual). Go to 2. | |
| 2. Motor No. 1 engine (Appropriate Flight Manual). Go to 3. | |
| 3. Check if APU stalls. | |
| Step 1. If APU stalls, go to 4. | |
| Step 2. If APU does not stall, shut down APU (Appropriate Flight Manual). Replace right relay panel (PARA 9-4-2). Go to 16. | |
| 4. Troubleshoot APU System, start bypass valve circuitry (PARA 15-2-1 or 15-2-2). | |
| Step 1. If the APU start bypass valve and circuitry operates properly, go to 5. | |
| 5. Install an insulated jumper wire between P450-3 and P450-5. Go to 6. | |
| 6. Check for 28 VDC between P241-W and ground. | |
| Step 1. If voltage is as specified, go to 7. | |
| Step 2. If voltage is not as specified, go to 14. | |

TABLE NO. 4-2-8-24. No. 2 Engine Starter Motors Engine (#2 ENGINE STARTER Capsule Is On), But APU Stalls (Popping Sound Is Heard). (Cont)

	TEST OR INSPECTION CORRECTIVE ACTION	
	7. Remove jumper wire from P450. Go to 8.	
	8. Check helicopter configuration.	
	Step 1. WINTER , go to 9.	
	Step 2. W/O WINTER , go to 10.	
WINTER	9. With AIR SOURCE HEAT/START switch in APU position, check for 28 VDC between P244B-T and ground.	
	Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to 16.	
	Step 2. If voltage is not as specified, replace winterization kit harness (PARA 16-6-2). Go to 16.	
	Step 3. If trouble remains, go to 10.	
	10. With AIR SOURCE HEAT/START switch in APU position, check for 28 VDC between P244-T and ground.	
	Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to 16.	
	Step 2. If voltage is not as specified, go to 11.	
	11. Check continuity between AIR SOURCE HEAT/START switch terminal 1 and P244-T.	
	Step 1. If continuity is present, go to 12.	
	Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 16.	
	12. Check for 28 VDC between AIR SOURCE HEAT/START switch terminal 2 and ground.	
	Step 1. If voltage is as specified, replace switch S18 (PARA 9-4-5). Go to 16.	
	Step 2. If voltage is not as specified, go to 13.	
	13. Check for 28 VDC between terminal 1 (AIR SOURCE HEAT/START circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and ground.	
	Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and switch S18 (Appendix F). Go to 16.	
	Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 16.	

TABLE NO. 4-2-8-24. No. 2 Engine Starter Motors Engine (#2 ENGINE STARTER Capsule Is On), But APU Stalls (Popping Sound Is Heard). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
14. Remove jumper wire from P450. Go to 15.	
15. Repair/replace wiring (Appendix F). Go to 16.	
16. Procedure completed.	

TABLE NO. 4-2-8-25. No. 2 Engine Motors, But Does Not Ignite And Accelerate To Idle Speed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Move ENG POWER CONT lever on control quadrant in cockpit to LOCKOUT and hold. Go to 2.	
2. Move ENG FUEL SYS selector lever on control quadrant to DIR detent. Go to 3.	
3. Place FUEL PUMP switch to FUEL PRIME. Go to 4.	
4. Check that fuel is coming from overboard drain.	
	Step 1. If fuel is coming from overboard drain, go to 5.
	Step 2. If fuel is not coming from overboard drain, troubleshoot Fuel Prime Boost System (PARA 10-2-3). Go to 20.
5. Troubleshoot engine for no ignite (SEI 671 and TM 55-284-248-23).	
	Step 1. If engine is good, go to 6.
6. Install an insulated jumper wire between P802-17 and P802-18. Go to 7.	
7. Check continuity between P121- <u>b</u> and P121- <u>d</u> .	
	Step 1. If continuity is present, go to 8.
	Step 2. If continuity is not present, go to 18.
8. Check continuity between P241-X and P241-Y.	

TABLE NO. 4-2-8-25. No. 2 Engine Motors, But Does Not Ignite And Accelerate To Idle Speed. (Cont)

	TEST OR INSPECTION CORRECTIVE ACTION
	Step 1. If continuity is present, go to 9. Step 2. If continuity is not present, go to 16.
	9. Remove jumper wire from P802. Go to 10.
	10. Check helicopter configuration.
	Step 1. S70-II , go to 11. Step 2. S70-I , go to 14.
S70-II	11. Check continuity between: P802-17 and P814-35 P802-19 and P814-35 Step 1. If continuity is present, go to 12. Step 2. If continuity is not present, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 20.
S70-II	12. Check continuity between J814-35 and P382-F. Step 1. If continuity is present, go to 13. Step 2. If continuity is not present, repair/replace wiring (SEI 671 and TM 55-2840-248-23). Go to 20.
S70-II	13. Check continuity between P121-d and P382-G. Step 1. If continuity is present, troubleshoot Engine Overspeed Protection System (PARA 4-2-4). Go to 20. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 20.
S70-I	14. On instrument panel, turn ENGINE IGNITION switch to ON. Go to 15.
S70-I	15. Check continuity between J121-b and J121-d. Step 1. If continuity is present, replace engine ignition switch (PARA 9-4-96). Go to 20. Step 2. If continuity is not present, replace right relay panel (PARA 9-4-2). Go to 20.
	16. Remove jumper wire from P802. Go to 19.

TABLE NO. 4-2-8-25. No. 2 Engine Motors, But Does Not Ignite And Accelerate To Idle Speed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
17. Repair/replace wiring, as required, between the following (Appendix F), then go to 20: P121-b and P241-X P121-d and P241-Y	
18. Remove jumper wire from P802. Go to 19.	
19. Repair/replace wiring, as required, between the following (Appendix F), then go to 20: P121-b and J814-34 P121-d and J814-35	
Step 1. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 20.	
20. Procedure completed.	

TABLE NO. 4-2-8-26. #2 ENGINE STARTER Capsule Does Not Go Off (Engine Speed Above 55% Ng).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Visually check No. 2 engine start control valve position indicator. Go to 2.	
2. Check if indicator shows CLOSED.	
Step 1. If indicator shows CLOSED, go to 3.	
Step 2. If indicator does not show CLOSED, replace start control valve (PARA 7-4-2). Go to 13.	
3. Check continuity between J450-3 and J450-4.	
Step 1. If continuity is present, go to 4.	
Step 2. If continuity is not present, replace start control valve (PARA 7-4-2). Go to 12.	
4. Install an insulated jumper wire between P806-1 and P806-3. Go to 5.	

**TABLE NO. 4-2-8-26. #2 ENGINE STARTER Capsule Does Not Go Off (Engine Speed Above 55% Ng).
(Cont)**

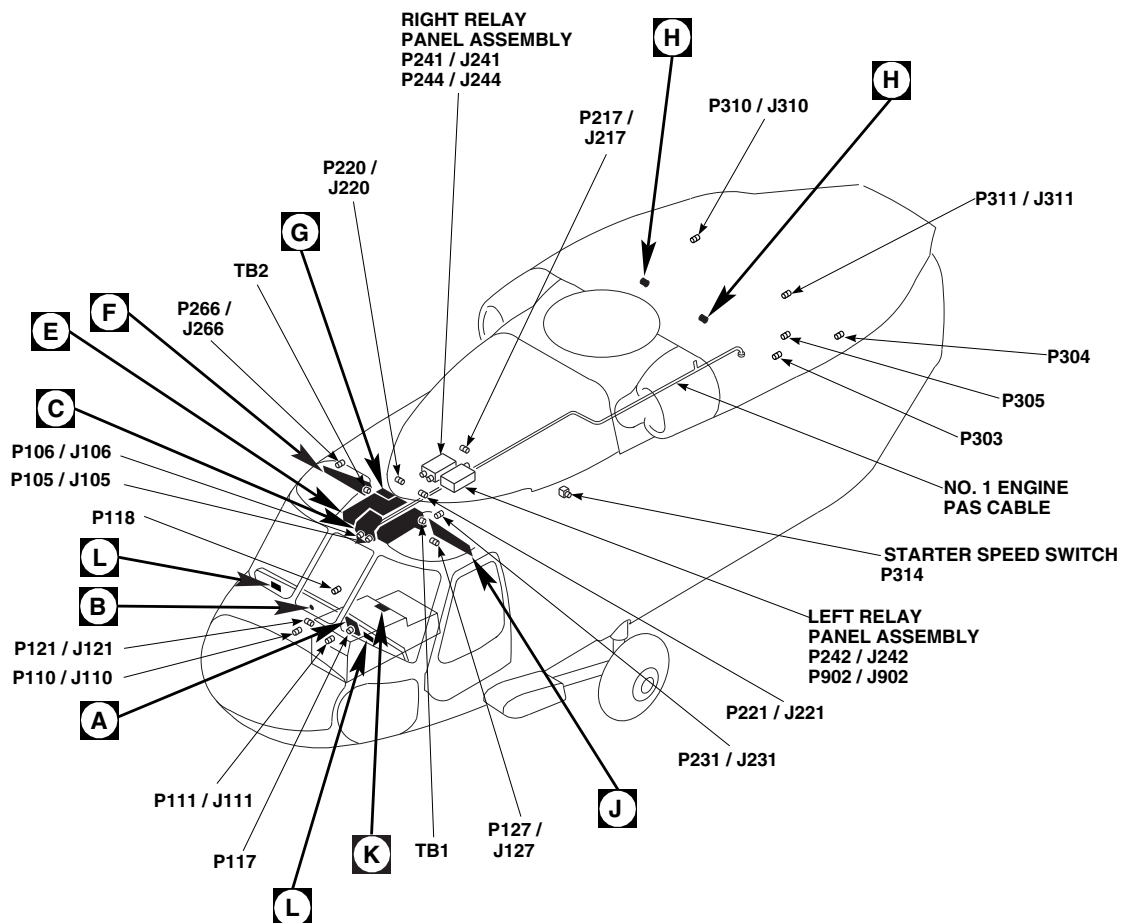
TEST OR INSPECTION CORRECTIVE ACTION
5. Check continuity between P314-L and P314-M.
Step 1. If continuity is present, go to 6. Step 2. If continuity is not present, go to 10.
6. Remove jumper wire from P806. Go to 7.
7. Replace starter speed switch (PARA 7-4-10). Go to 8.
8. Operate No. 2 engine up to idle speed (Appropriate Flight Manual). Go to 9.
9. Check if #2 ENGINE STARTER capsule goes off.
Step 1. If capsule is off, Engine Start System is operational. Shut down engine (Appropriate Flight Manual). Go to 13. Step 2. If capsule is not off, shut down engine (Appropriate Flight Manual). Replace No. 2 engine starter (PARA 7-4-1). Go to 12.
10. Shut down engine (Appropriate Flight Manual). Go to 11.
11. Remove jumper wire from P806. Go to 12.
12. Repair/replace wiring, as required, between (Appendix F), then go to 13: P314-L and J824-2 P314-M and J824-3
Step 1. If trouble remains, replace engine harness (SEI 671 and TM 1-2840-248-23). Go to 13.
13. Procedure completed.

TABLE NO. 4-2-8-27. No. 2 Engine Starter Does Not Motor Engine (Compressor Is Not Rotating) When Using Engine Starter Override.

TEST OR INSPECTION	CORRECTIVE ACTION
1. With No. 2 engine start button pressed, check continuity between J106-S and J106-T.	Step 1. If continuity is present, go to 2. Step 2. If continuity is not present, replace engine control quadrant starter override switch (PARA 4-5-15). Go to 13.
2. Install an insulated jumper wire between P106-K and P106-T. Go to 3.	
3. Check for 28 VDC between P241-U and ground.	Step 1. If voltage is as specified, go to 4. Step 2. If voltage is not as specified, go to 11.
4. Remove jumper wire from P106. Go to 5.	
5. Install an insulated jumper wire between P106-K and P106-S. Go to 6.	
6. Check for 28 VDC between J220-x and ground.	Step 1. If voltage is as specified, go to 7. Step 2. If voltage is not as specified, go to 9.
7. Remove jumper wire from P106. Go to 8.	
8. Replace right relay panel (PARA 9-4-2). Go to 13.	
9. Remove jumper wire from P106. Go to 10.	
10. Repair/replace wiring between P106-S and P220-x (Appendix F). Go to 13.	
11. Remove jumper wire from P106. Go to 12.	
12. Repair/replace wiring between and P106-T and P241-U (Appendix F). Go to 13.	
13. Procedure completed.	

TABLE NO. 4-2-8-28. No Fuel Coming Out Of No. 2 Fuel Drain.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Visually check drain tube for obvious obstructions.	Step 1. If obstructions are present, remove obstructions (PARA 10-4-29). Go to 3. Step 2. If obstructions are not present, go to 2.
2. Check for 28 VDC between TB2-7 and ground.	Step 1. If voltage is as specified, troubleshoot Fuel Prime Boost System (PARA 10-2-3). Go to 3. Step 2. If voltage is not as specified, repair/replace wiring (Appendix F). Go to 3.
3. Procedure completed.	



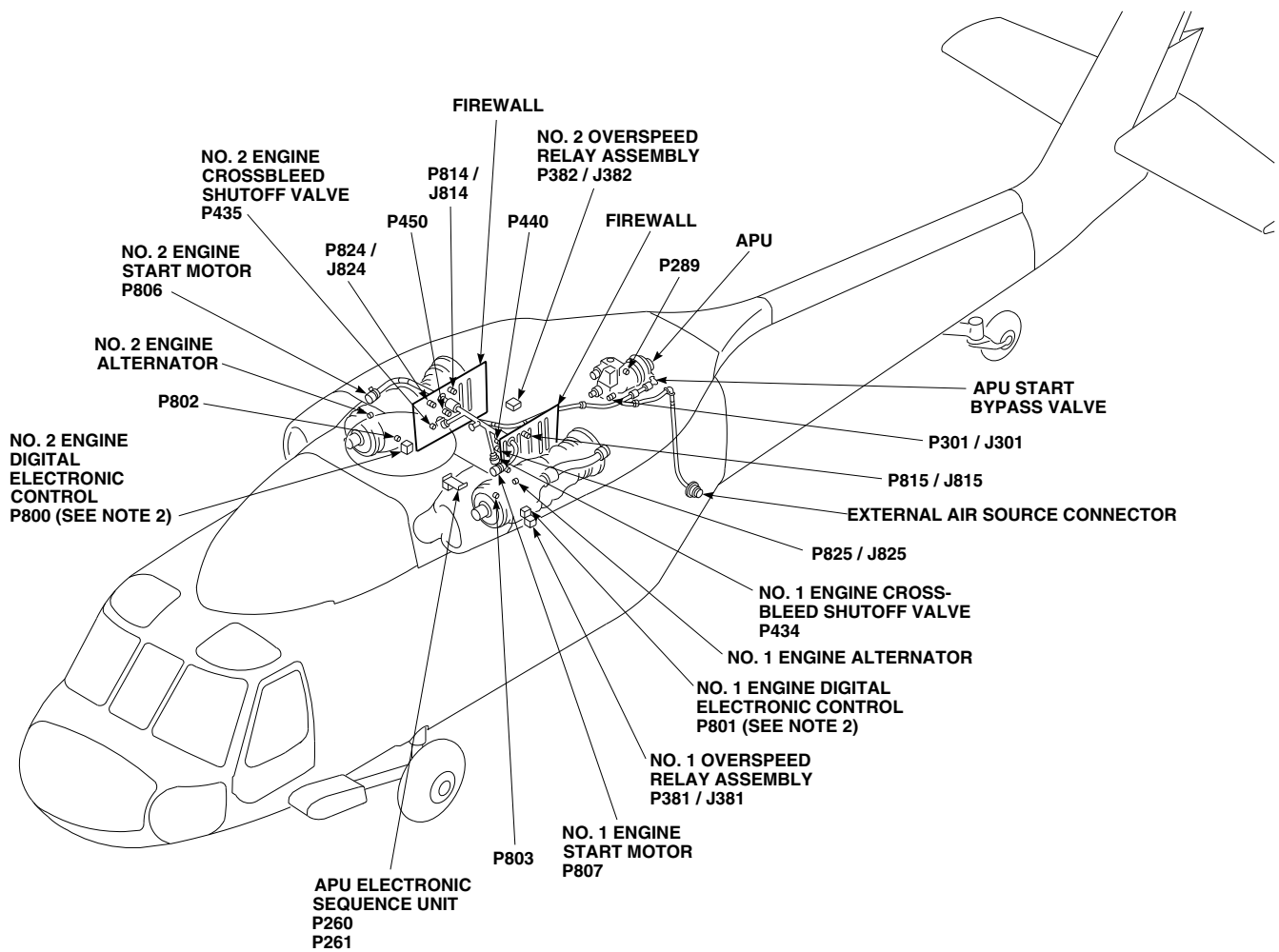
NOTES

1. **WINTER**
2. **T700 T701A** ELECTRICAL CONTROL UNIT INSTALLED.
T701C DIGITAL ELECTRONIC CONTROL UNIT INSTALLED.

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P105 / J105	NO. 1 ENGINE CONTROL QUADRANT
P106 / J106	NO. 2 ENGINE CONTROL QUADRANT
P107 / J107	
P110 / J110	COCKPIT BL 7.5 RH, STA 197
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P117	CAUTION / ADVISORY PANEL
P118	CAUTION / ADVISORY PANEL
P121 / J121	COCKPIT BL 10 RH, STA 197
P127 / J127	COCKPIT CEILING BL 28 LH, STA 243

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FIGURE 4-2-8-1. ENGINE START AND IGNITION SYSTEM LOCATION DIAGRAM
(SHEET 1 OF 6)

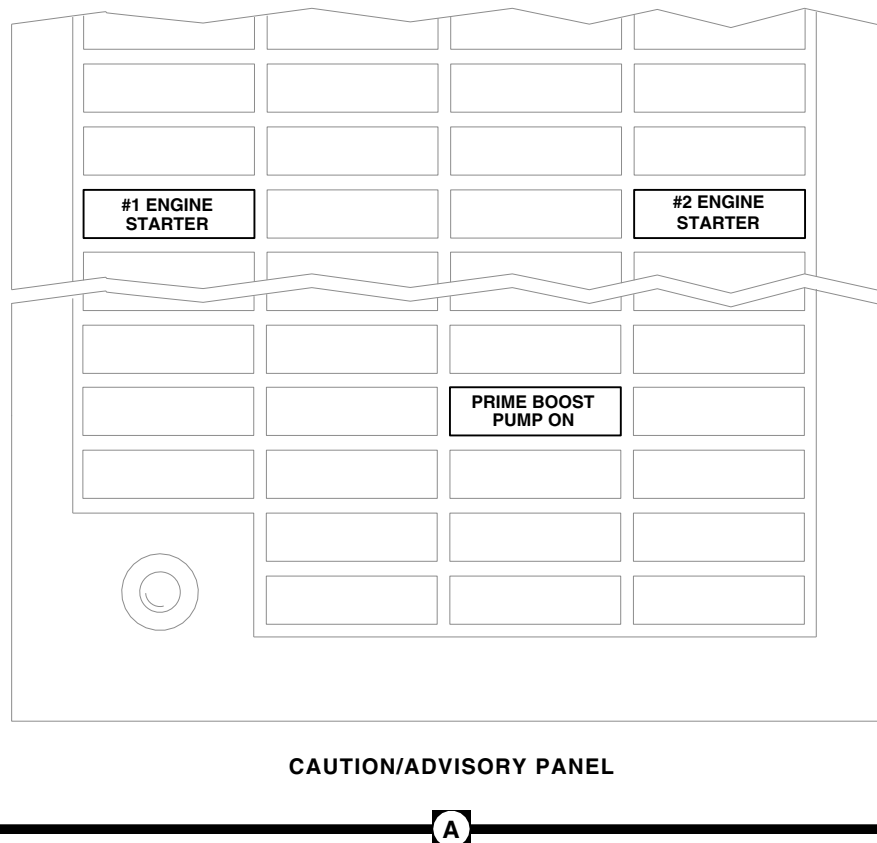


TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P217 / J217	CABIN CEILING BL 0, STA 284
P219 / J219	CABIN FLOOR BL 25 RH, STA 242
P220 / J220	JUNCTION BOX, MAIN ROTOR PYLON DECK
P221 / J221	JUNCTION BOX, MAIN ROTOR PYLON DECK
P231 / J231	CABIN CEILING BL 8 RH, STA 247
P241 / J241	RIGHT RELAY PANEL ASSEMBLY
P242 / J242	LEFT RELAY PANEL ASSEMBLY

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P244 / J244	RIGHT RELAY PANEL ASSEMBLY
P244 / J244A P244B / J244	WINTERIZATION KIT HARNESS ASSEMBLY (SEE NOTE 1)
P246 / J246	CABIN CEILING BL 5 RH, STA 247
P260 P261	APU ELECTRONIC SEQUENCE UNIT
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25 RH
P289	APU START BYPASS VALVE
P301 / J301	TRANSITION DECK STA 415
P303	PRIME BOOST PUMP
P304	NO. 1 ENGINE PRIME SHUTOFF VALVE

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SA

FIGURE 4-2-8-1. ENGINE START AND IGNITION SYSTEM LOCATION DIAGRAM (SHEET 2 OF 6)



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P305	NO. 1 ENGINE PRIME SHUTOFF VALVE
P310 / J310	CABIN CEILING, BL 17 RH, STA 387
P311 / J311	CABIN CEILING, BL 16.5 LH, STA 383
P314	STARTER SPEED SWITCH
P381 / J381	NO. 1 OVERSPEED RELAY ASSEMBLY
P382 / J382	NO. 2 OVERSPEED RELAY ASSEMBLY
P434	NO. 1 ENGINE CROSSBLEED SHUTOFF VALVE
P435	NO. 2 ENGINE CROSSBLEED SHUTOFF VALVE
P440	NO. 1 ENGINE START CONTROL VALVE
P450	NO. 2 ENGINE START CONTROL VALVE
P800	NO. 2 ENGINE DIGITAL ELECTRONIC CONTROL (SEE NOTE 2)

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P801	NO. 1 ENGINE DIGITAL ELECTRONIC CONTROL (SEE NOTE 2)
P802	NO. 2 ENGINE INTERFACE DISCONNECT
P803	NO. 1 ENGINE INTERFACE DISCONNECT
P806	NO. 2 ENGINE START MOTOR
P807	NO. 1 ENGINE START MOTOR
P814 / J814	FIREWALL DISCONNECT BL 14 RH, STA 373
P815 / J815	FIREWALL DISCONNECT BL 14 LH, STA 373
P824 / J824	FIREWALL DISCONNECT BL 14 RH, STA 358.25
P825 / J825	FIREWALL DISCONNECT BL 14 LH, STA 358.25
P902 / J902	LEFT RELAY PANEL ASSEMBLY
TB1	COCKPIT CEILING, BL 9 LH, STA 236
TB2	COCKPIT CEILING, BL 9 RH, STA 236

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FIGURE 4-2-8-1. ENGINE START AND IGNITION SYSTEM LOCATION DIAGRAM
(SHEET 3 OF 6)

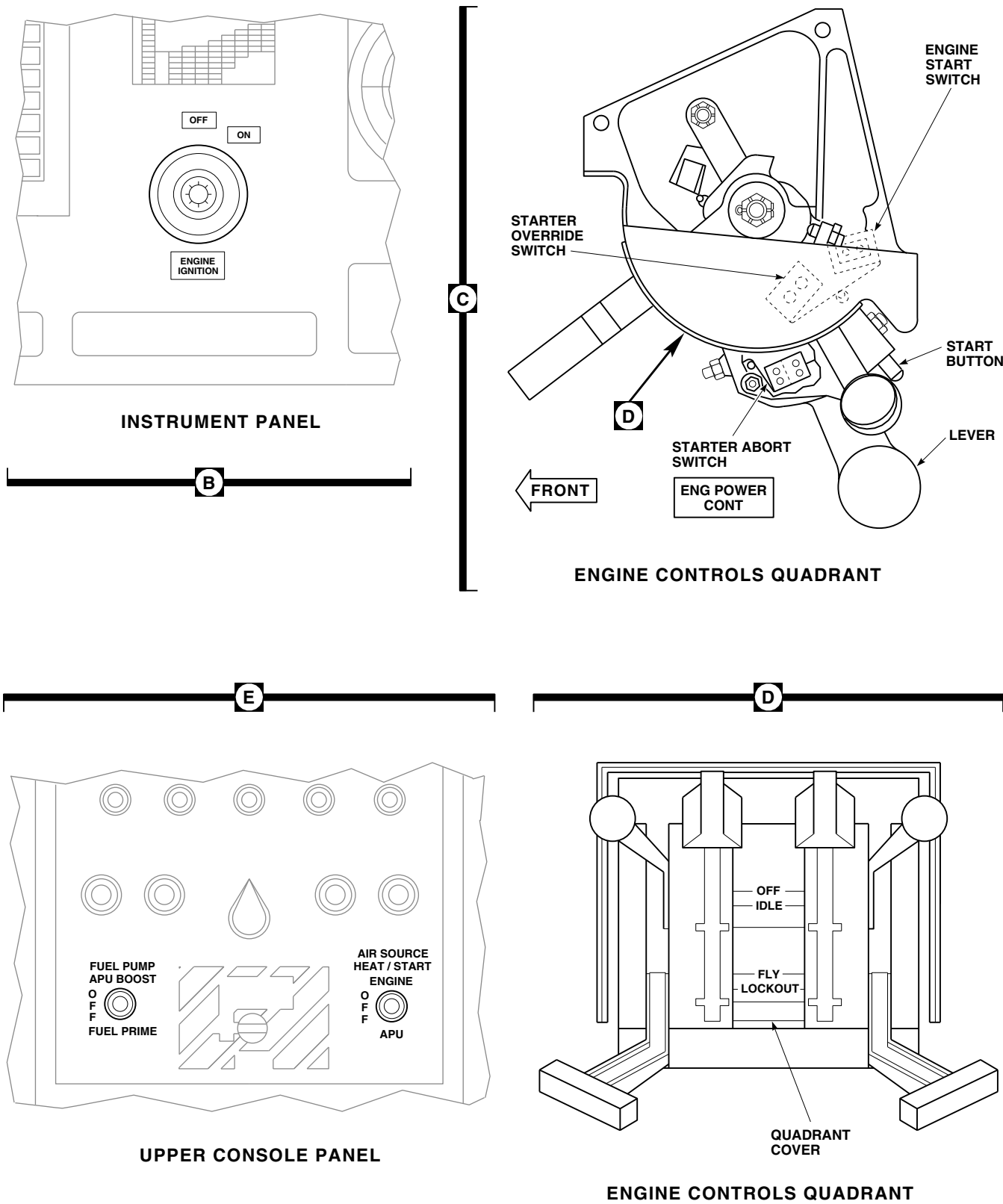
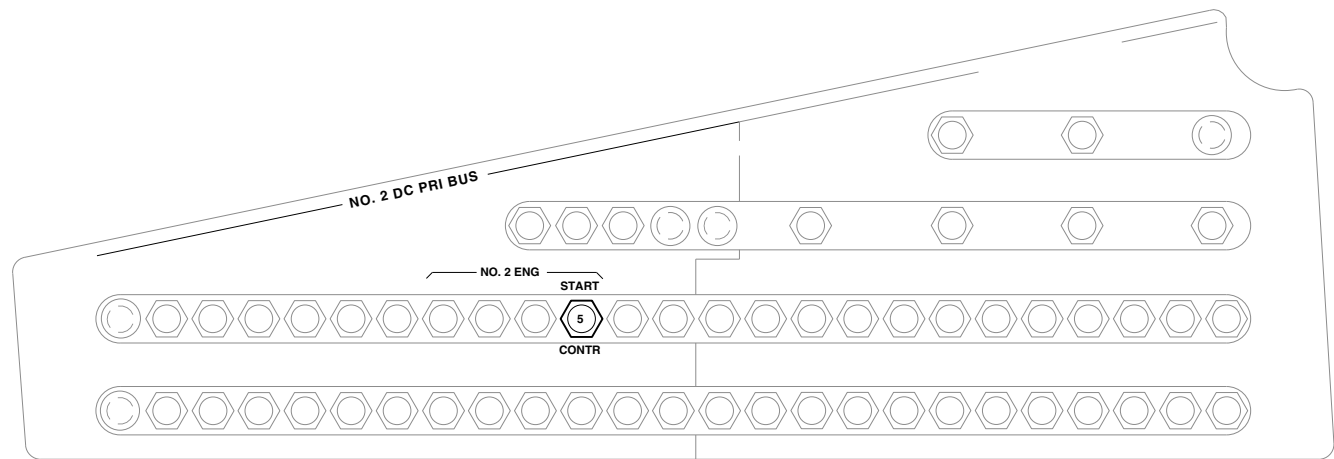


FIGURE 4-2-8-1. ENGINE START AND IGNITION SYSTEM LOCATION DIAGRAM
(SHEET 4 OF 6)

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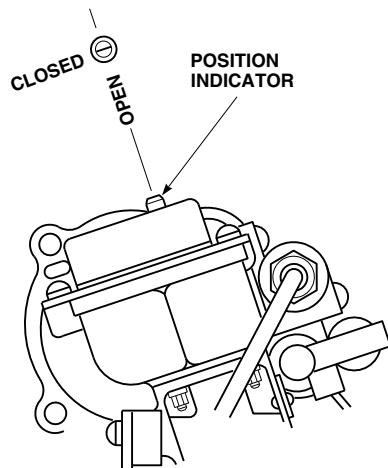


PILOT'S CIRCUIT BREAKER PANEL

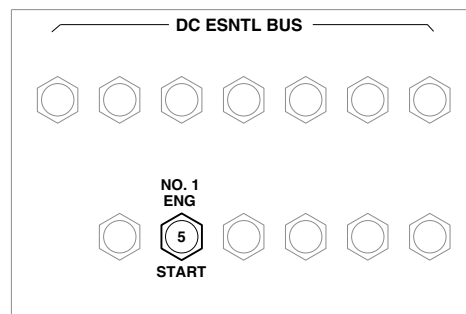
F

H

G



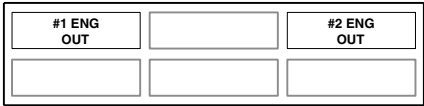
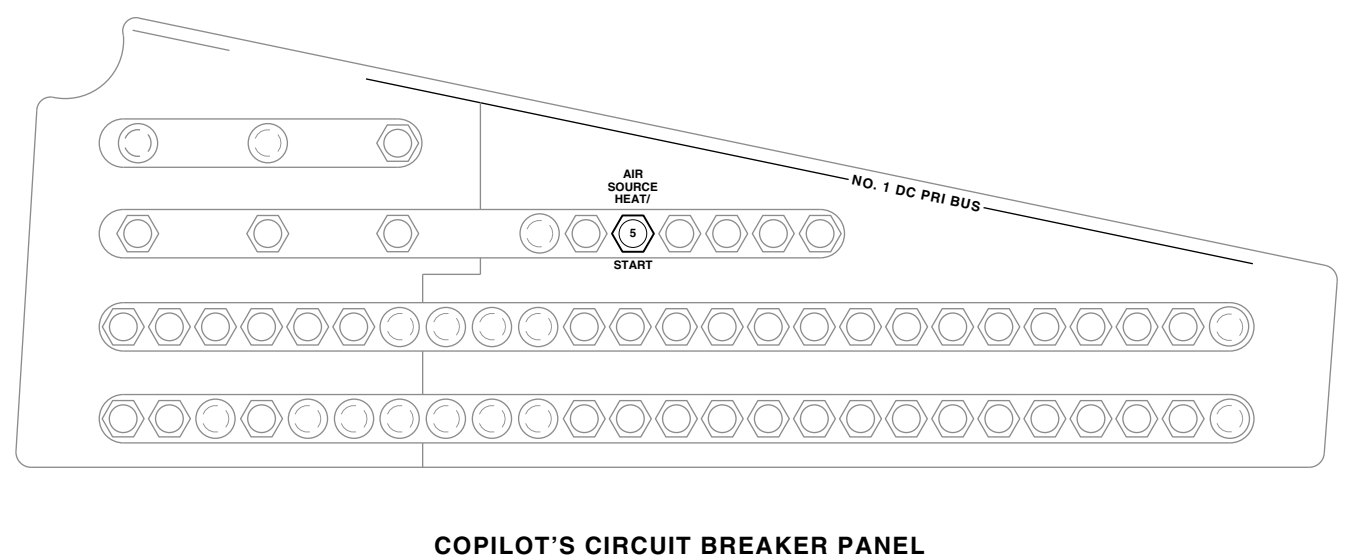
NO. 1 AND NO. 2 ENGINE
START CONTROL VALVE



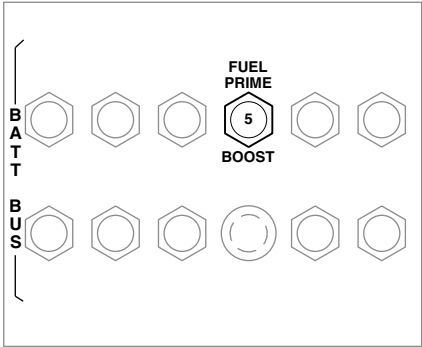
UPPER CONSOLE
CIRCUIT BREAKER PANEL

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FIGURE 4-2-8-1. ENGINE START AND IGNITION SYSTEM LOCATION DIAGRAM
(SHEET 5 OF 6)



PILOT'S AND COPILOT'S
MASTER WARNING PANEL



LOWER CONSOLE
CIRCUIT BREAKER PANEL

FT1321_6
SA

FIGURE 4-2-8-1. ENGINE START AND IGNITION SYSTEM LOCATION DIAGRAM
(SHEET 6 OF 6)

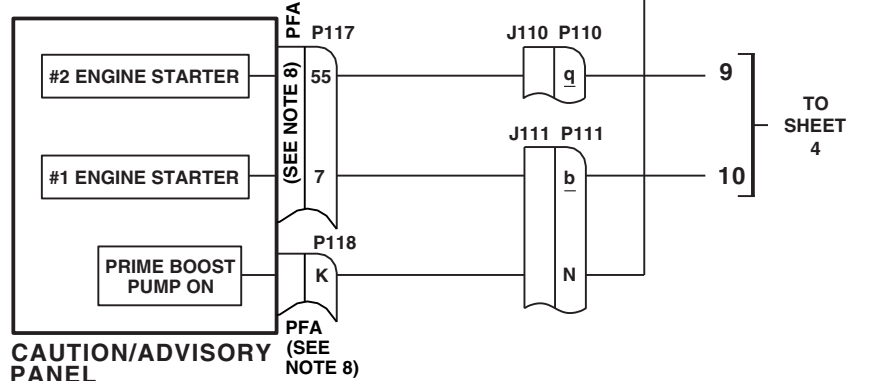
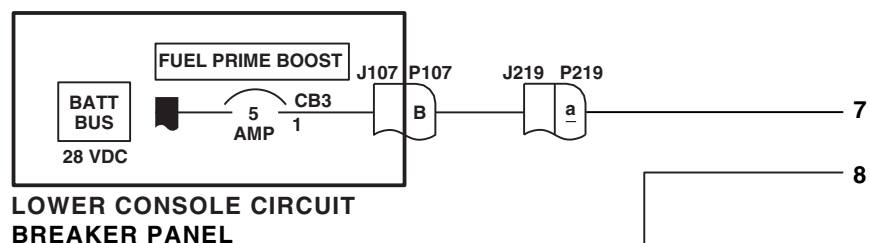
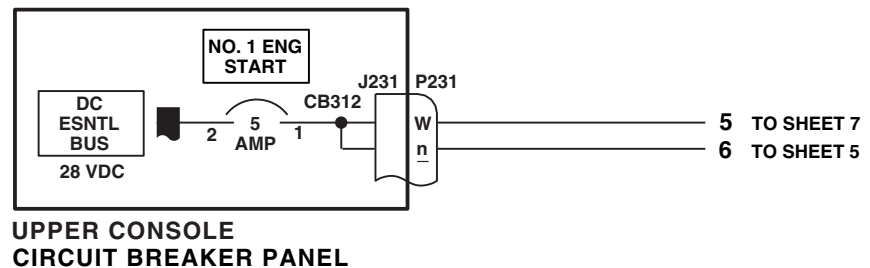
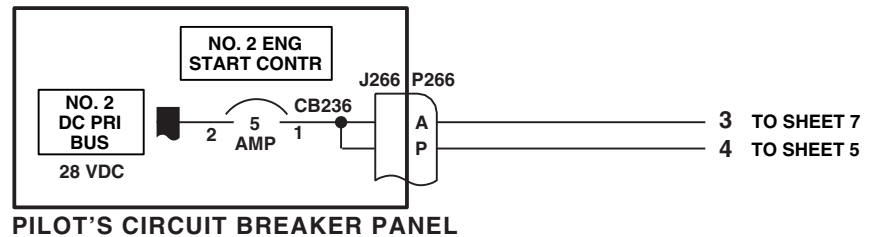
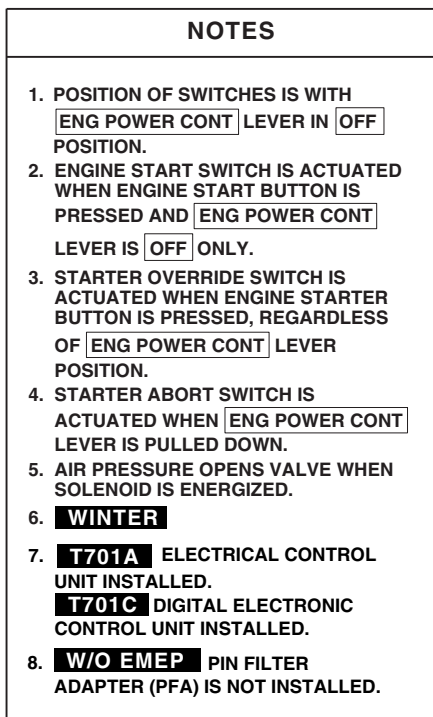
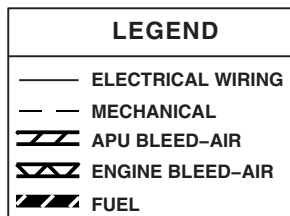
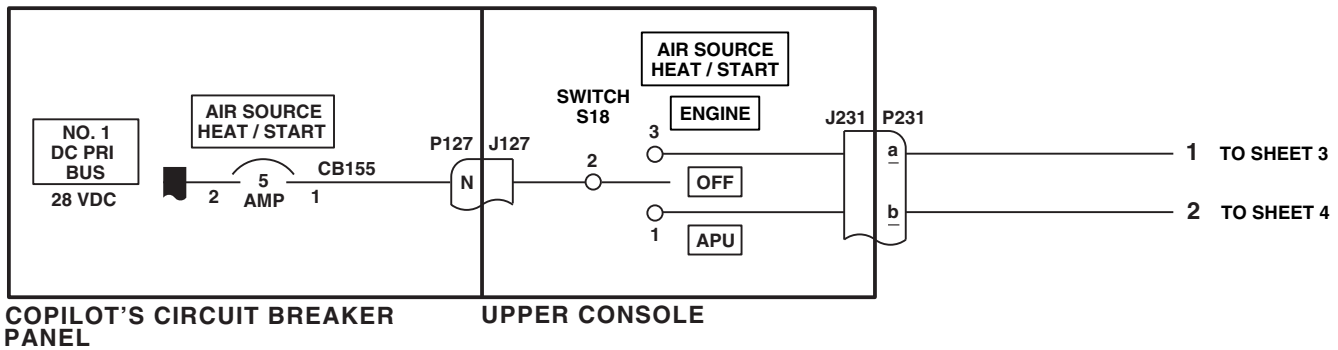
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FIGURE 4-2-8-2. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM
(SHEET 1 OF 10)

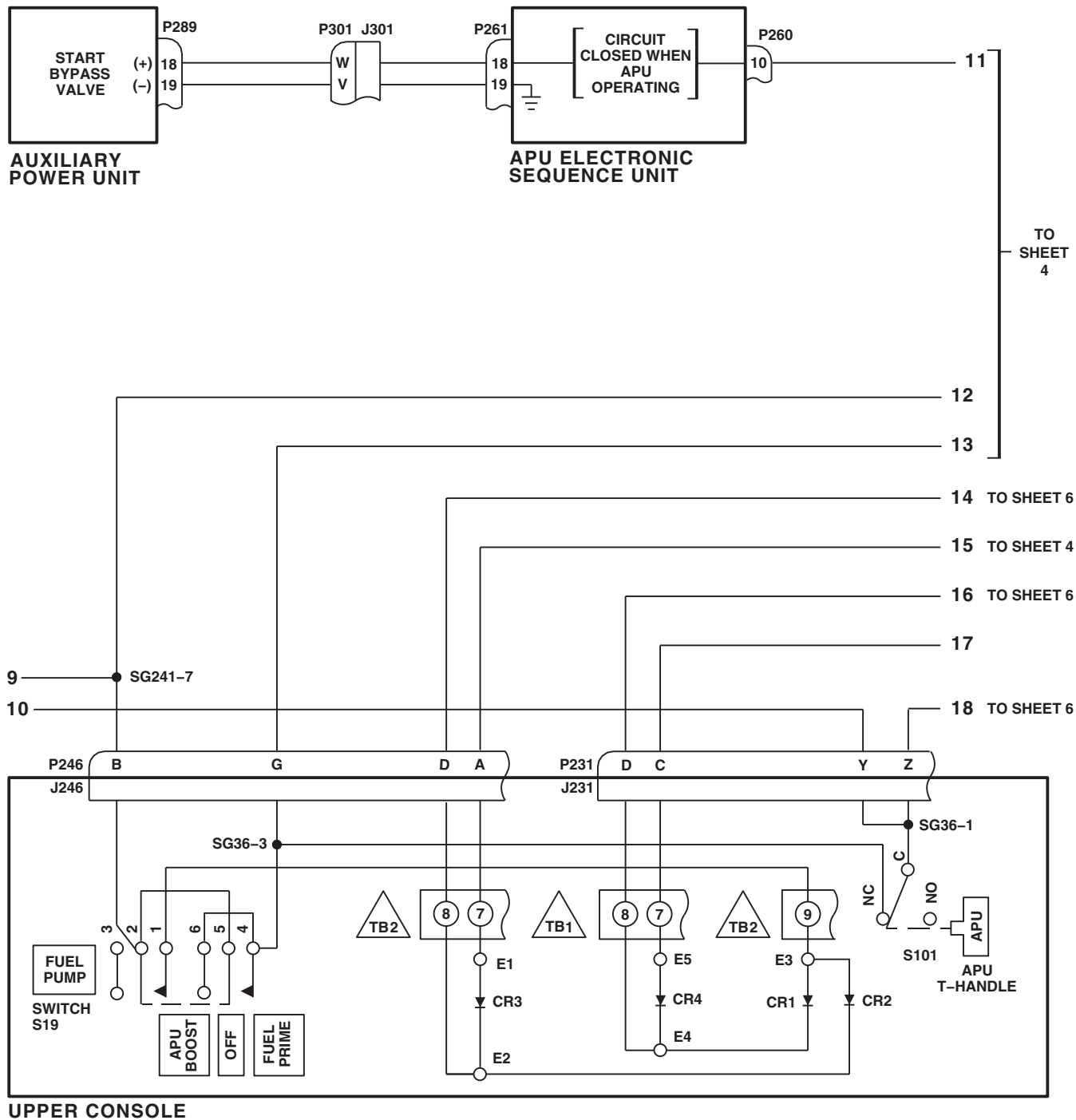


FIGURE 4-2-8-2. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 10)

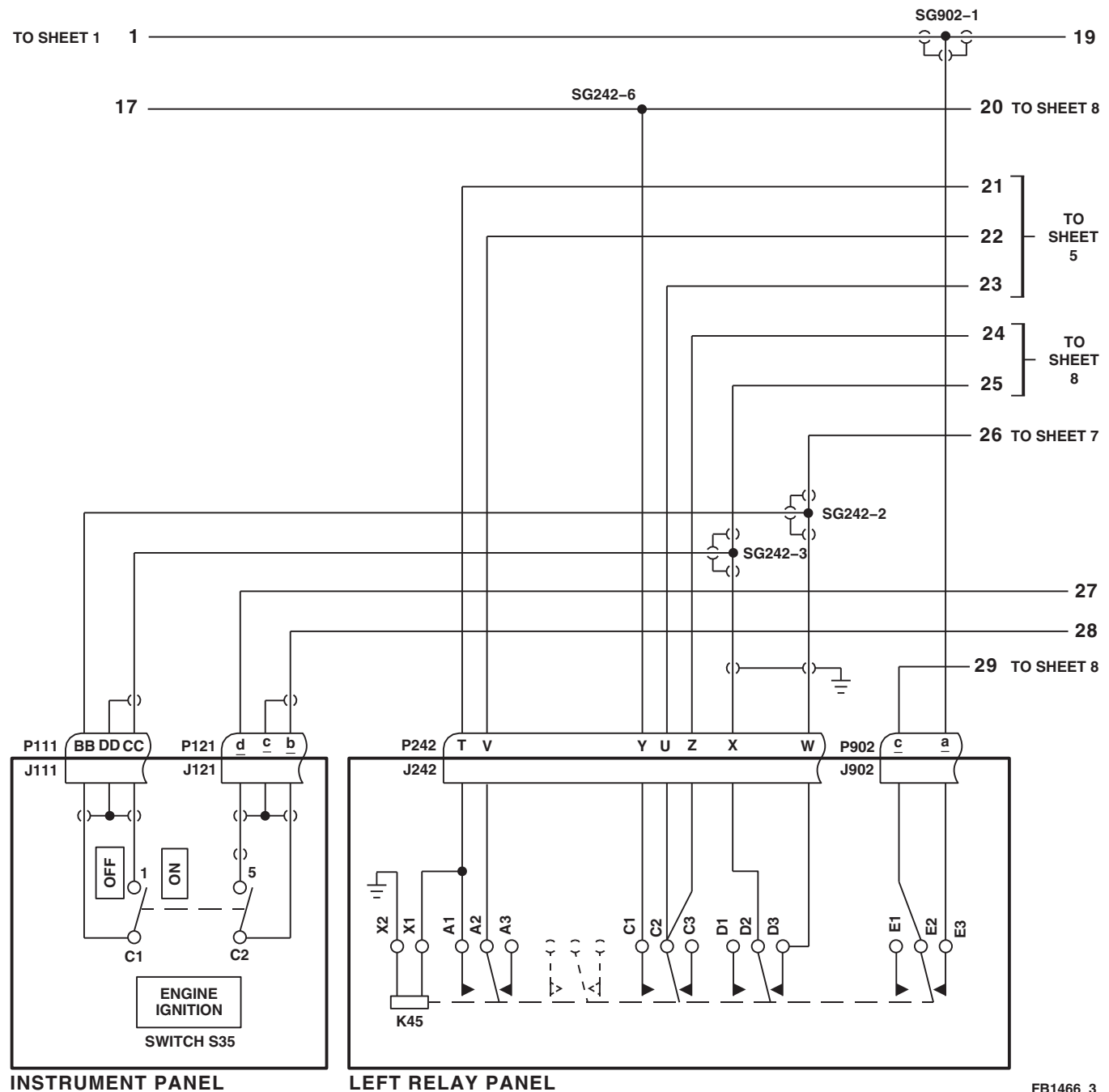


FIGURE 4-2-8-2. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM
(SHEET 3 OF 10)

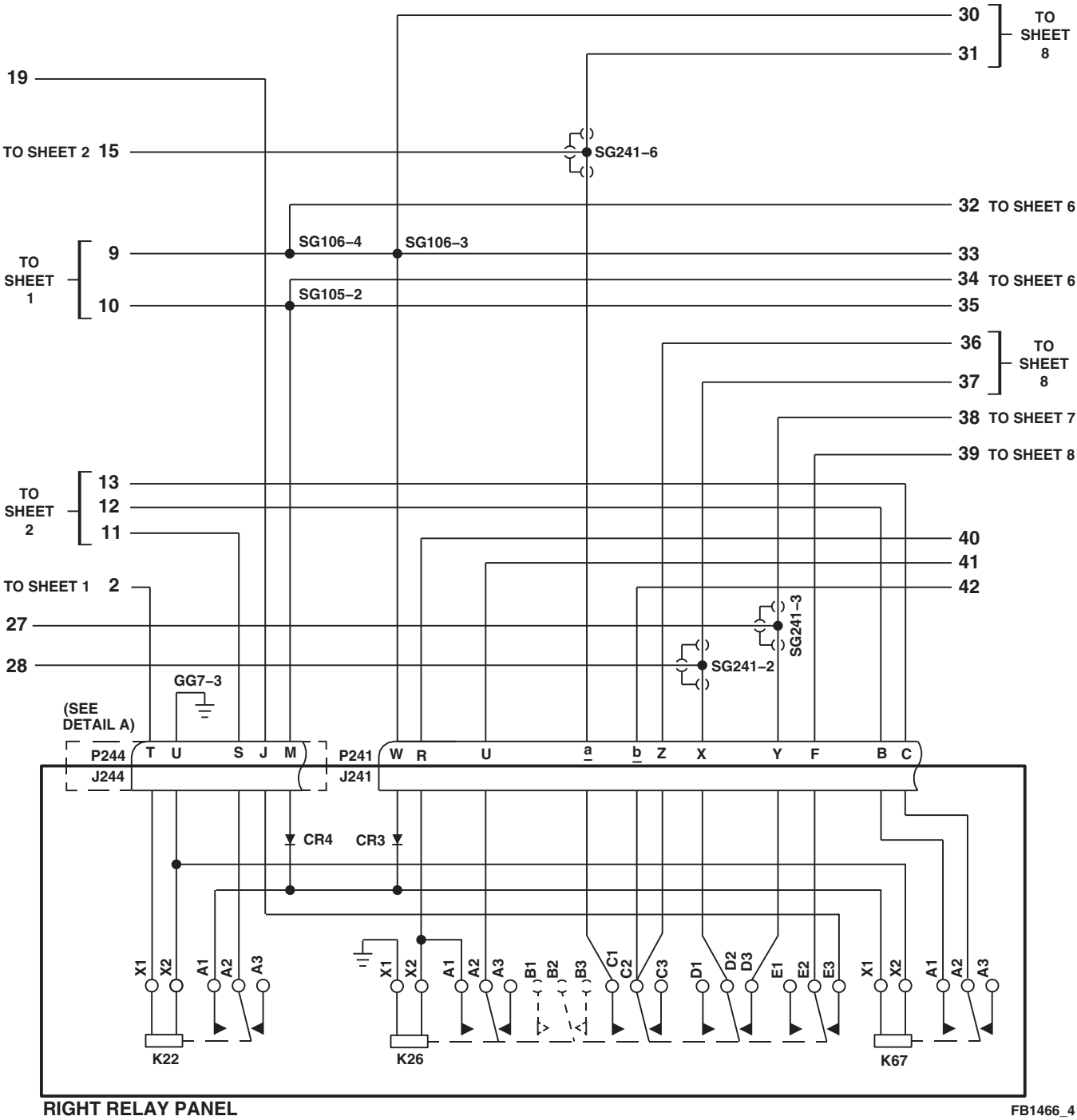


FIGURE 4-2-8-2. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 10)

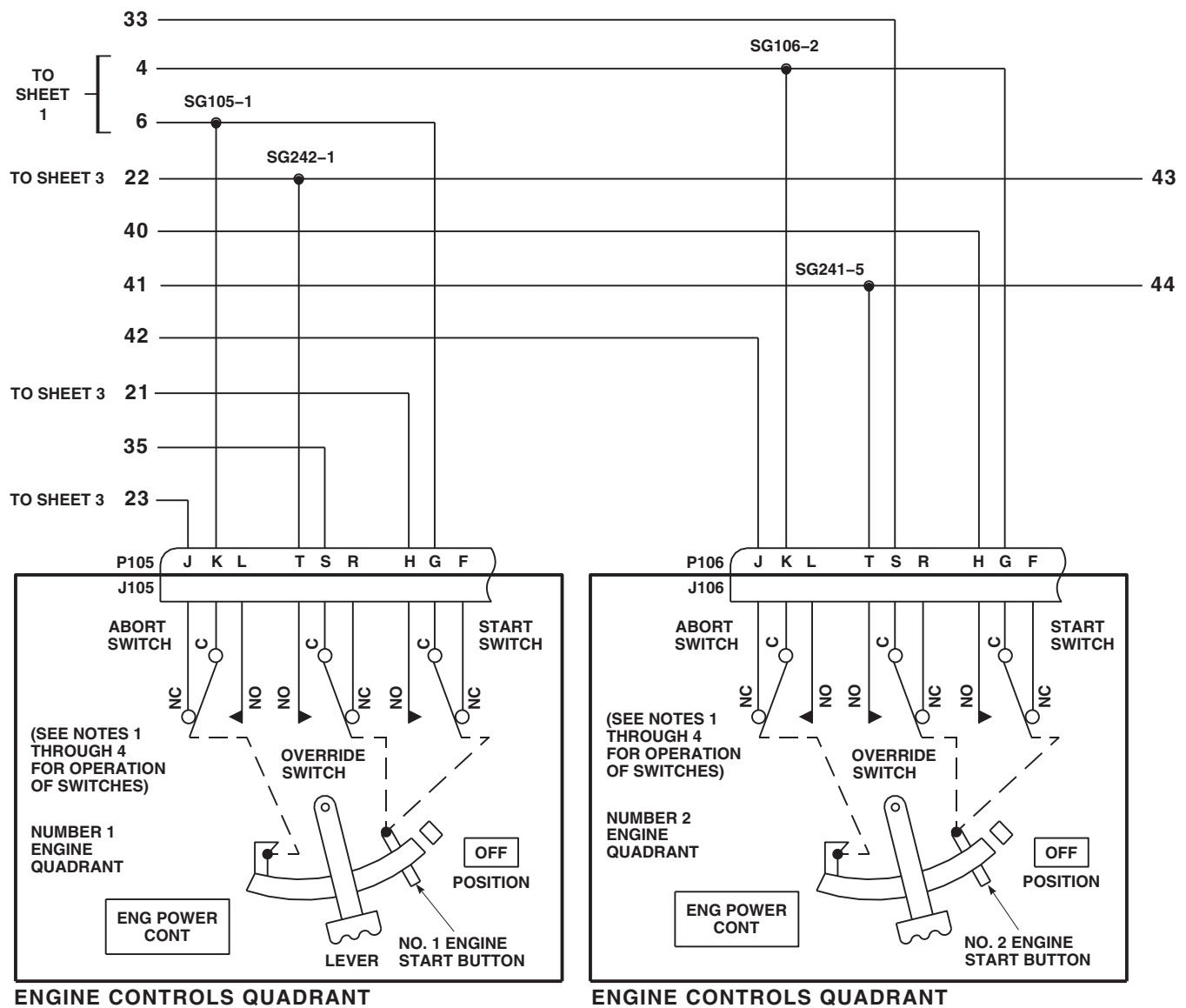
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FIGURE 4-2-8-2. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM
(SHEET 5 OF 10)

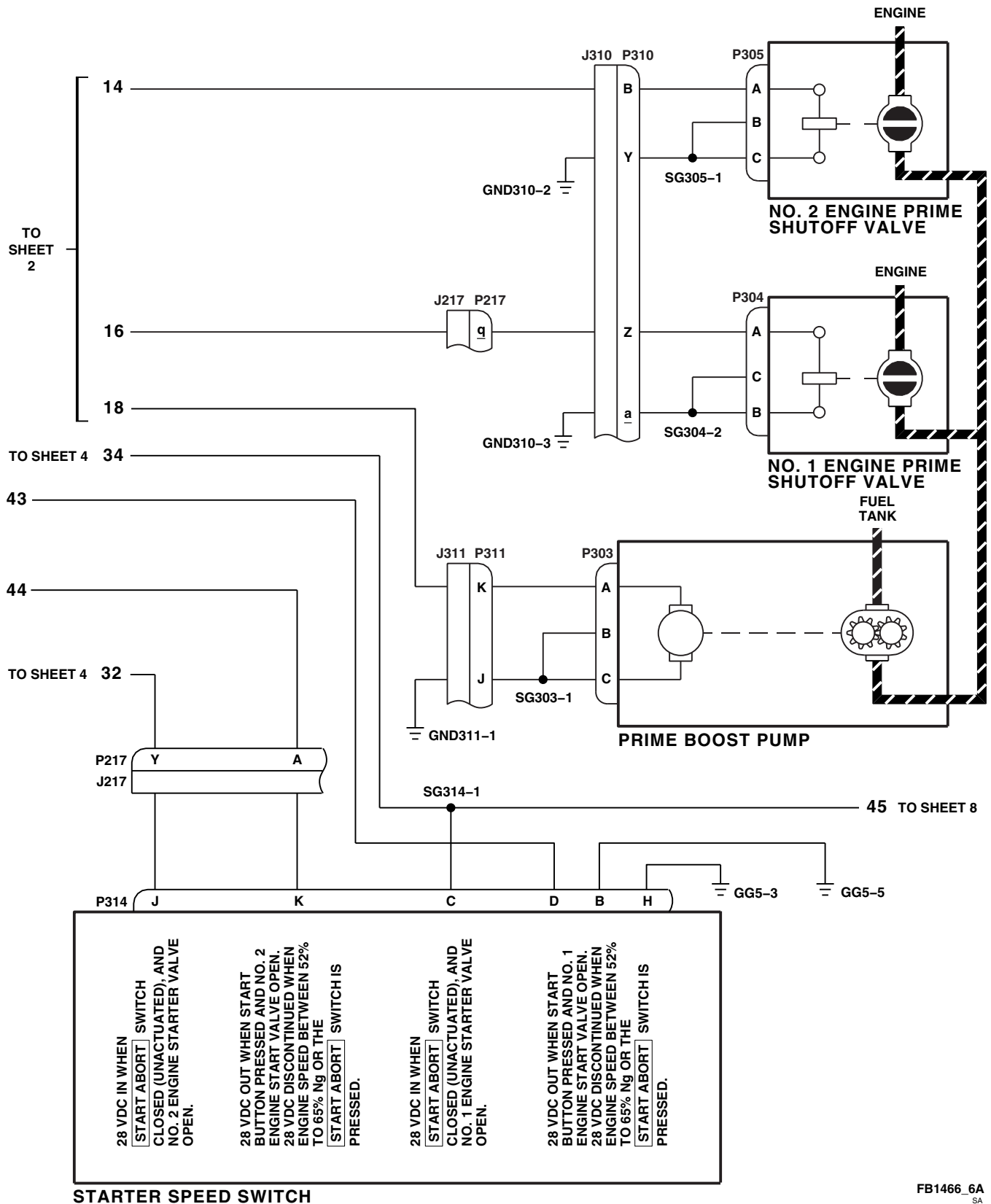


FIGURE 4-2-8-2. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM (SHEET 6 OF 10)

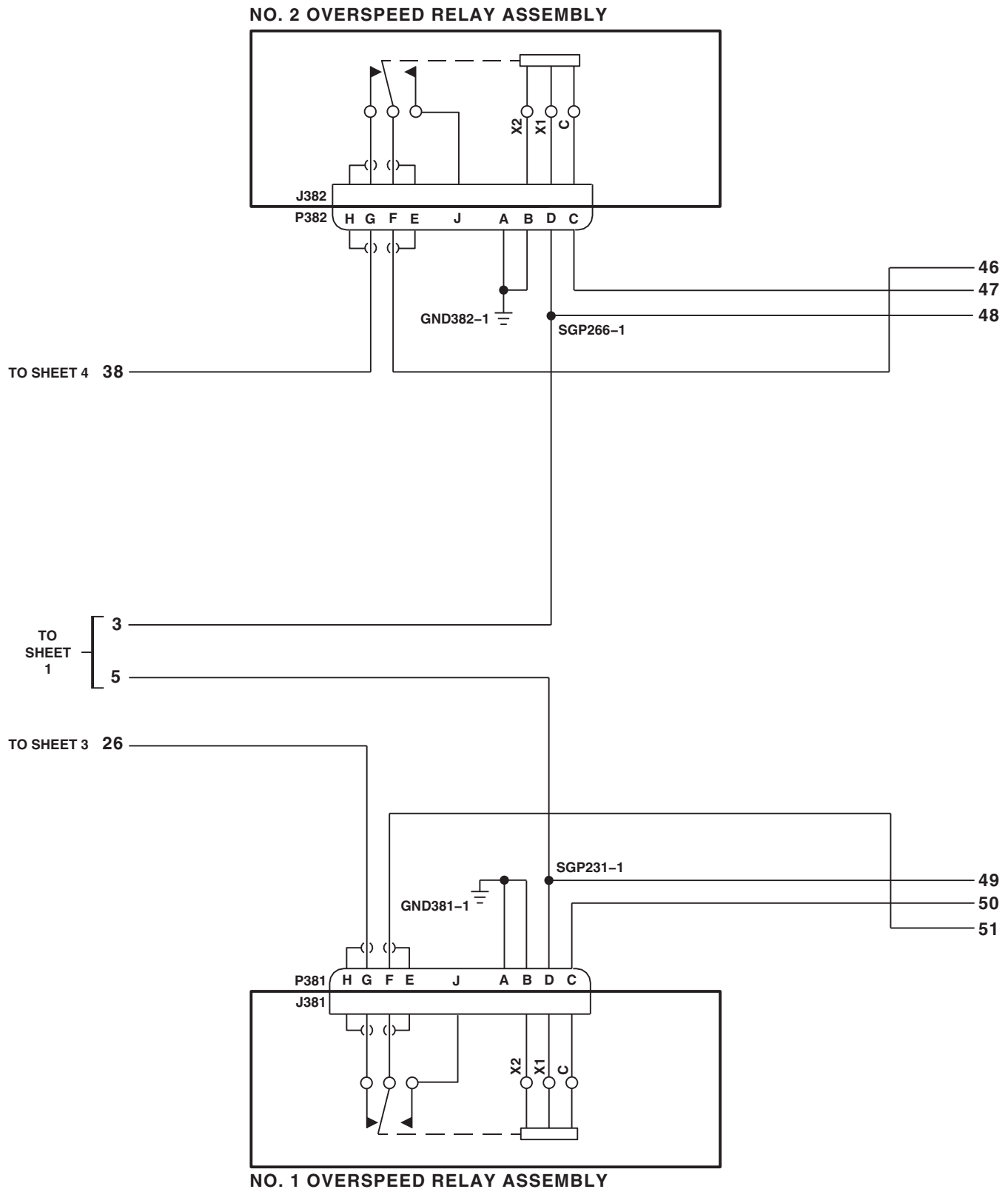
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FIGURE 4-2-8-2. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM
(SHEET 7 OF 10)

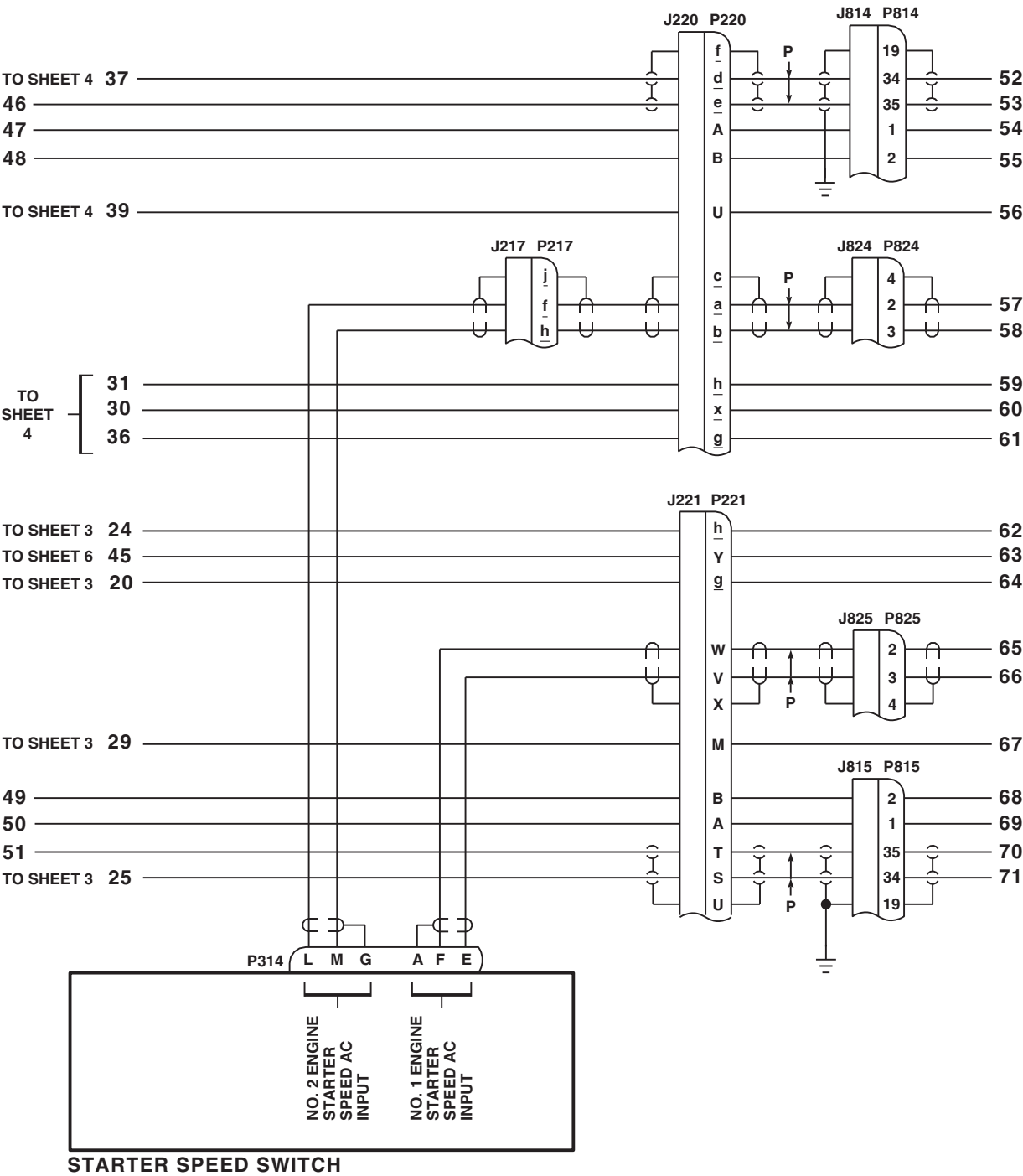


FIGURE 4-2-8-2. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM
 (SHEET 8 OF 10)

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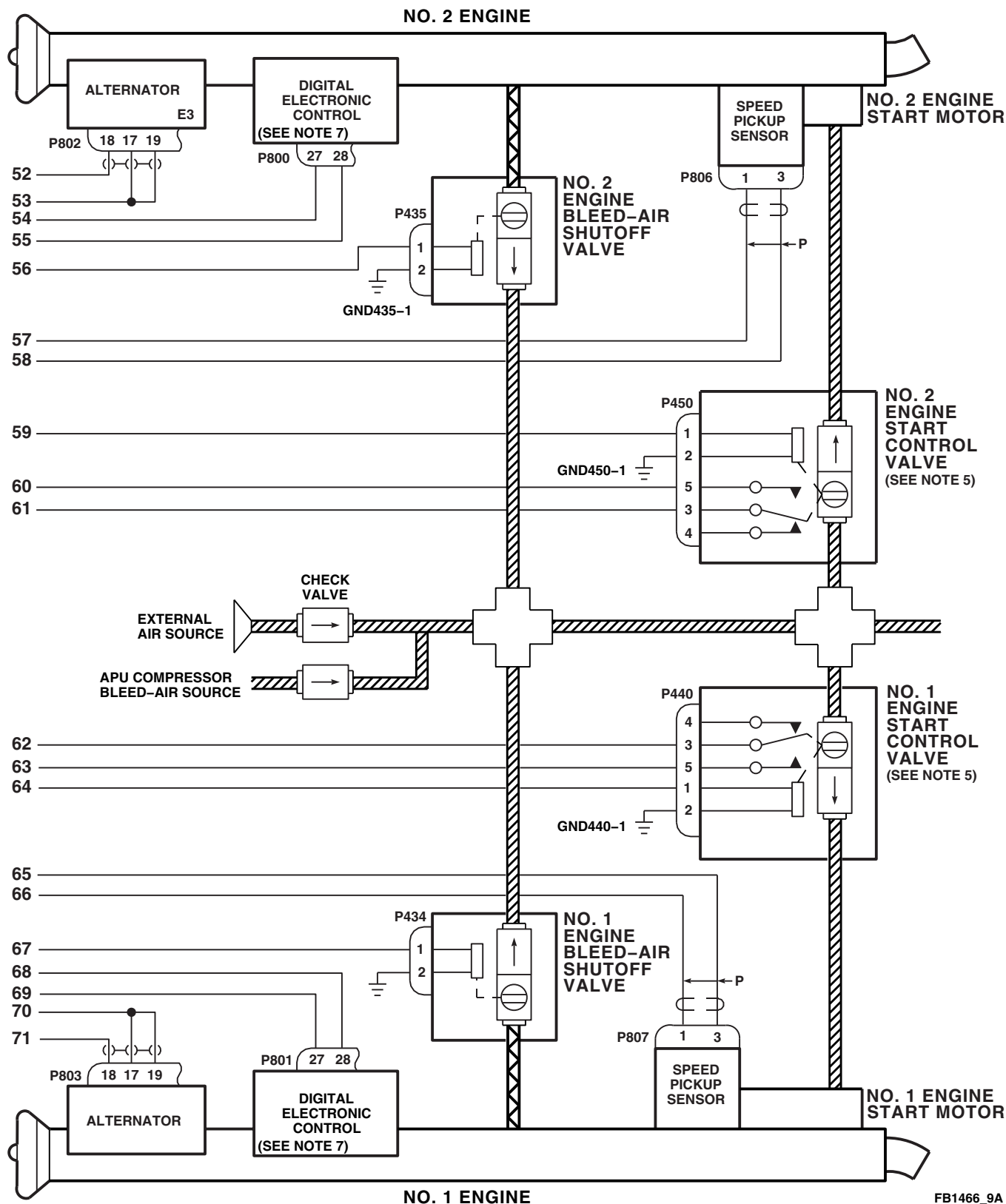
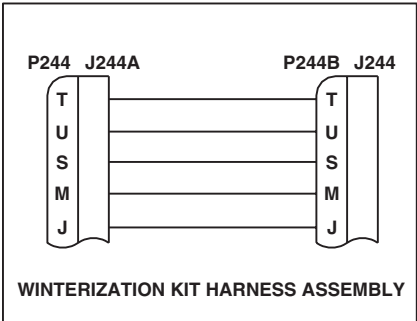
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FIGURE 4-2-8-2. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM
(SHEET 9 OF 10)



DETAIL A
 (SEE NOTE 6)

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 SA

FIGURE 4-2-8-2. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM
 (SHEET 10 OF 10)

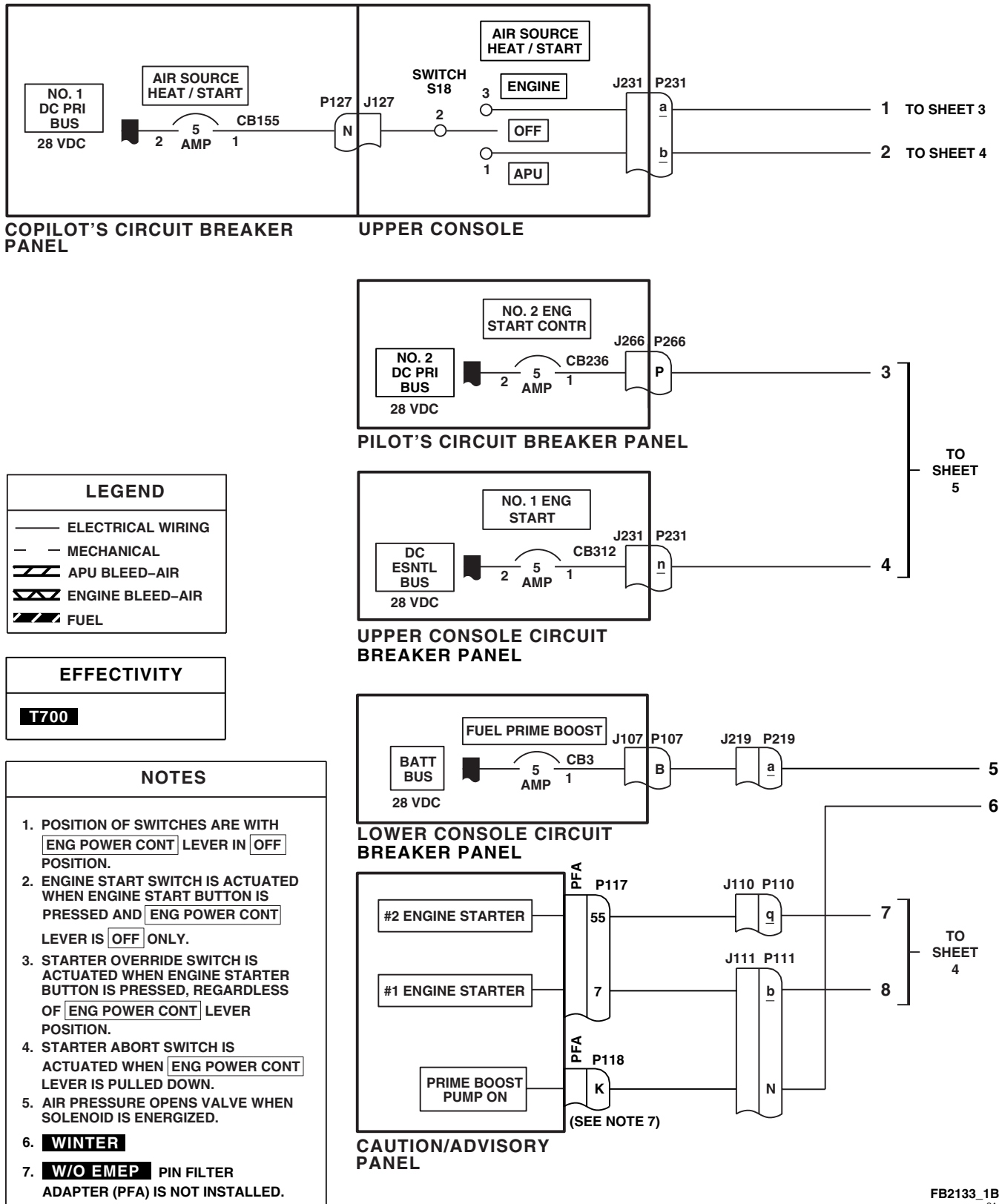


FIGURE 4-2-8-3. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 8)

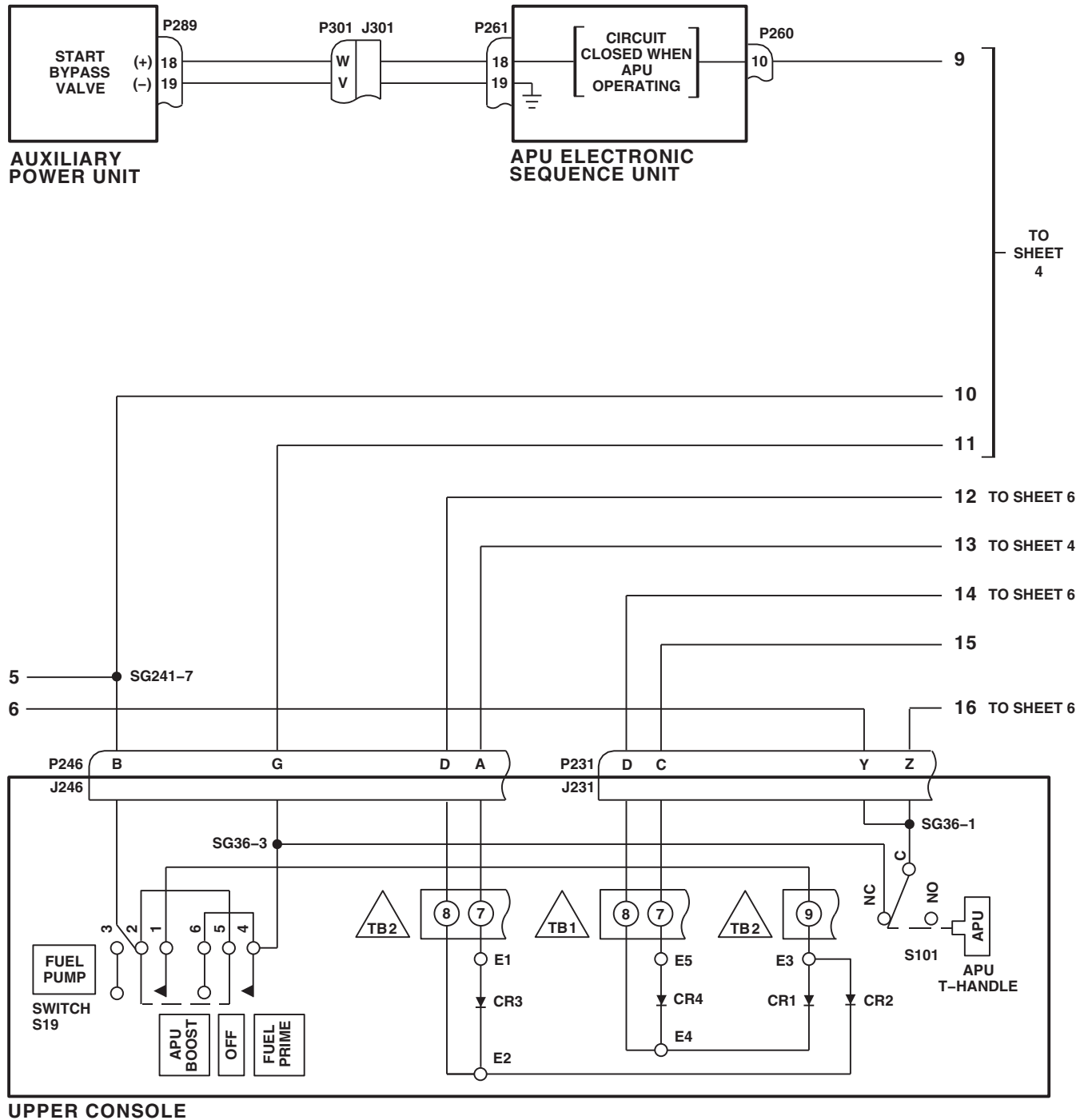


FIGURE 4-2-8-3. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 8)

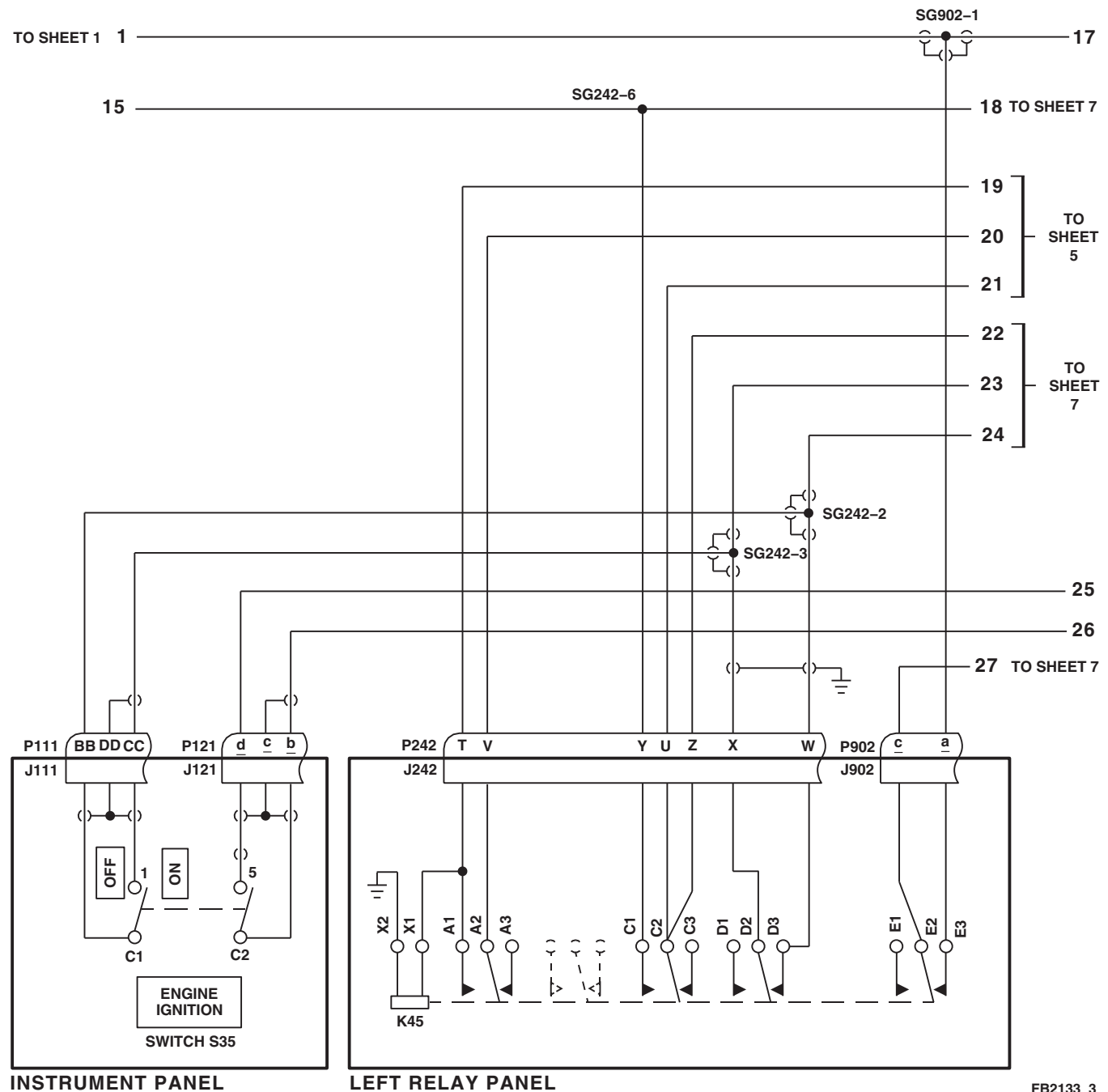
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FIGURE 4-2-8-3. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM
(SHEET 3 OF 8)

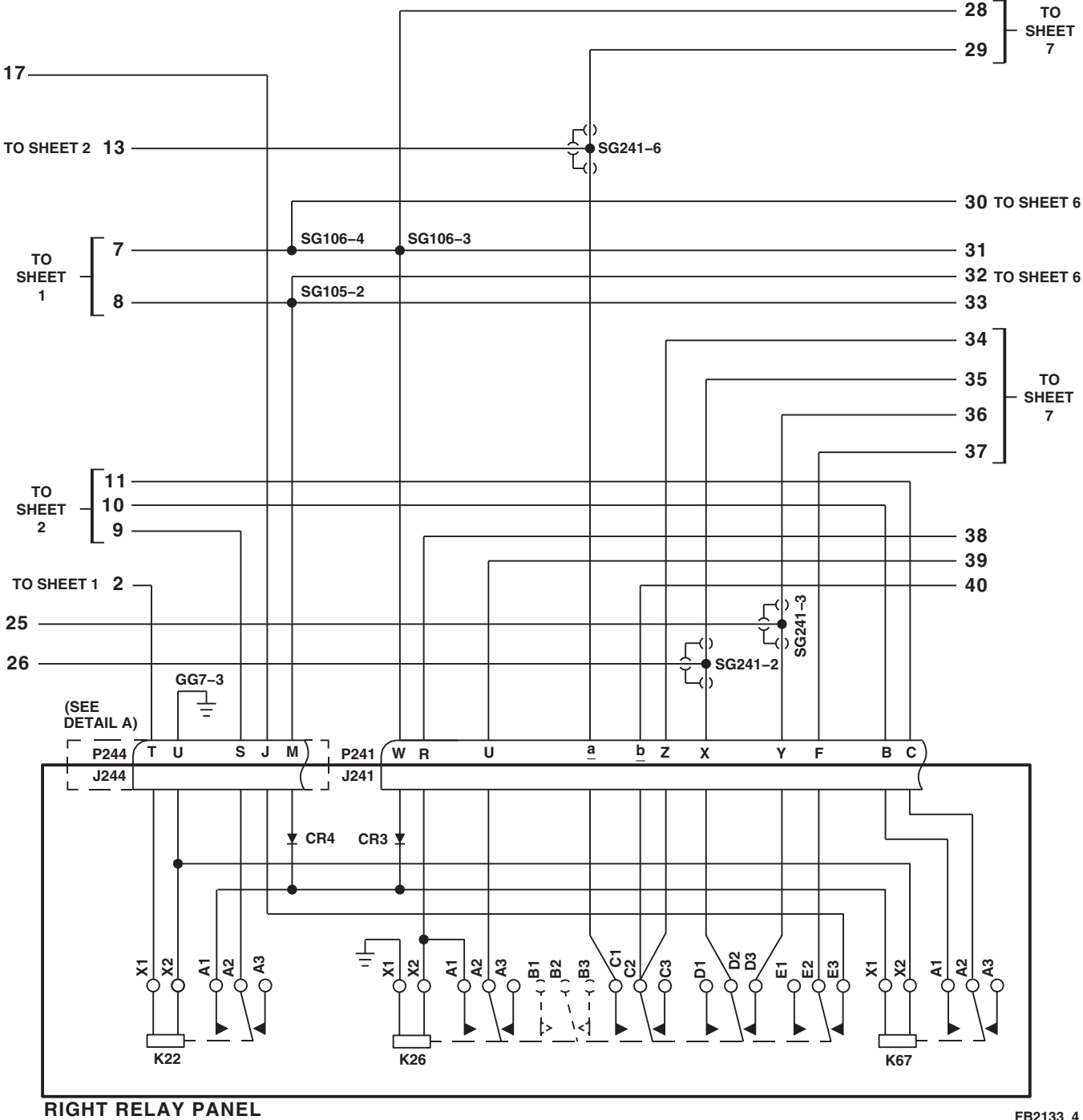


FIGURE 4-2-8-3. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 8)

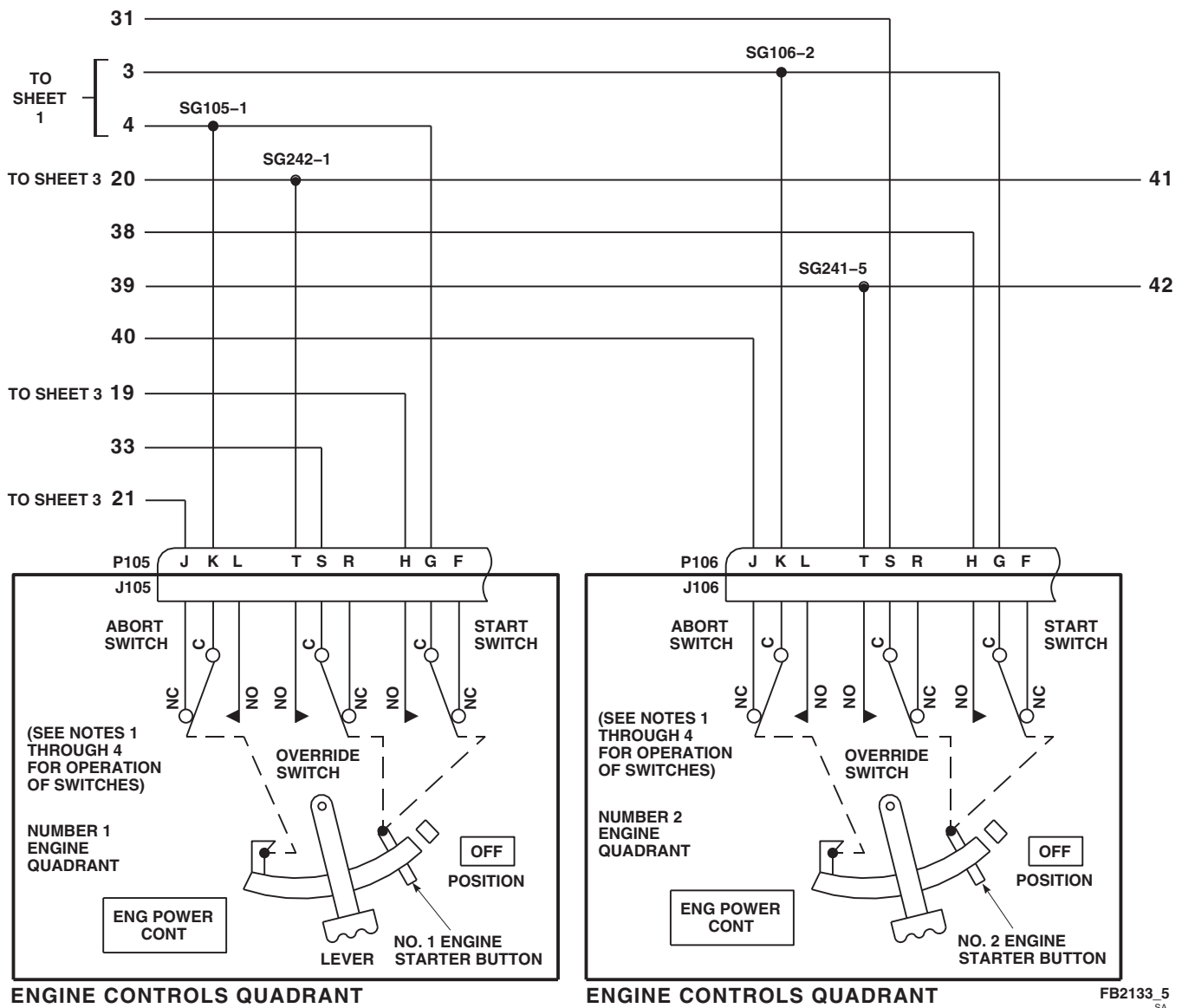
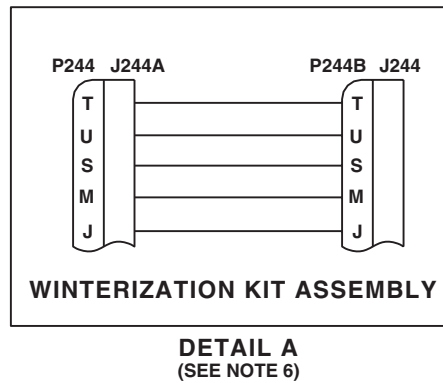


FIGURE 4-2-8-3. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM
(SHEET 5 OF 8)

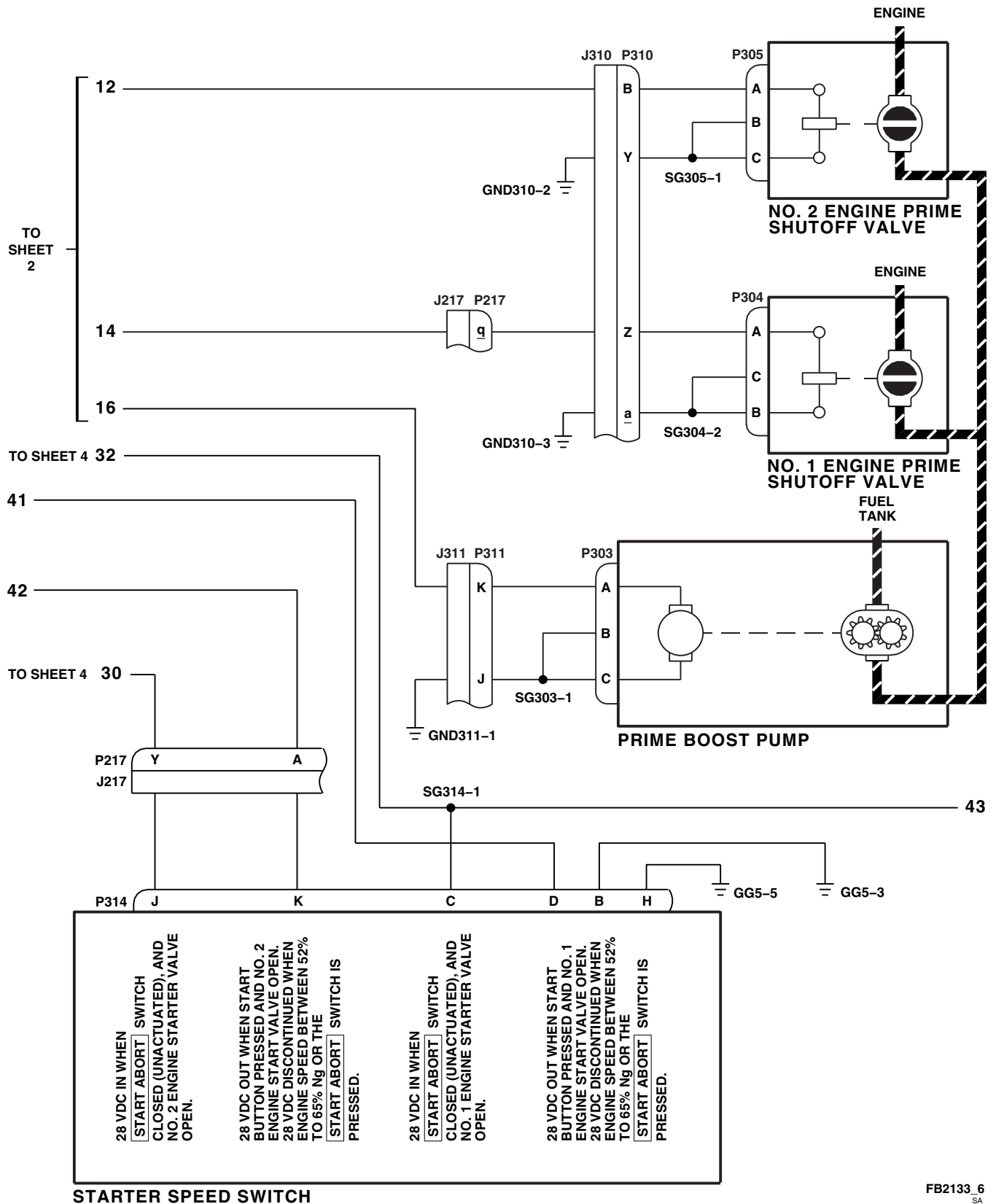


FIGURE 4-2-8-3. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM
(SHEET 6 OF 8)

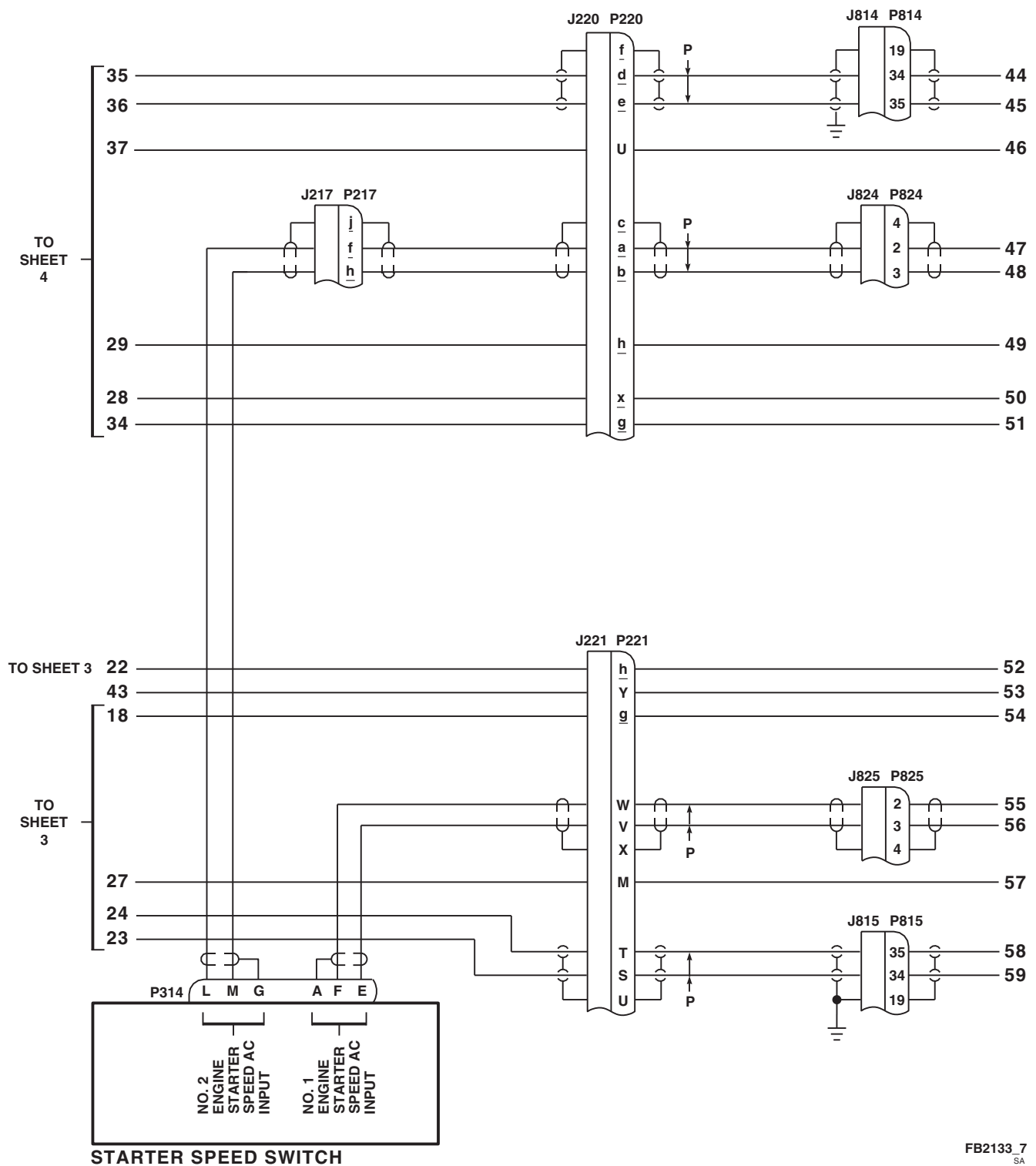
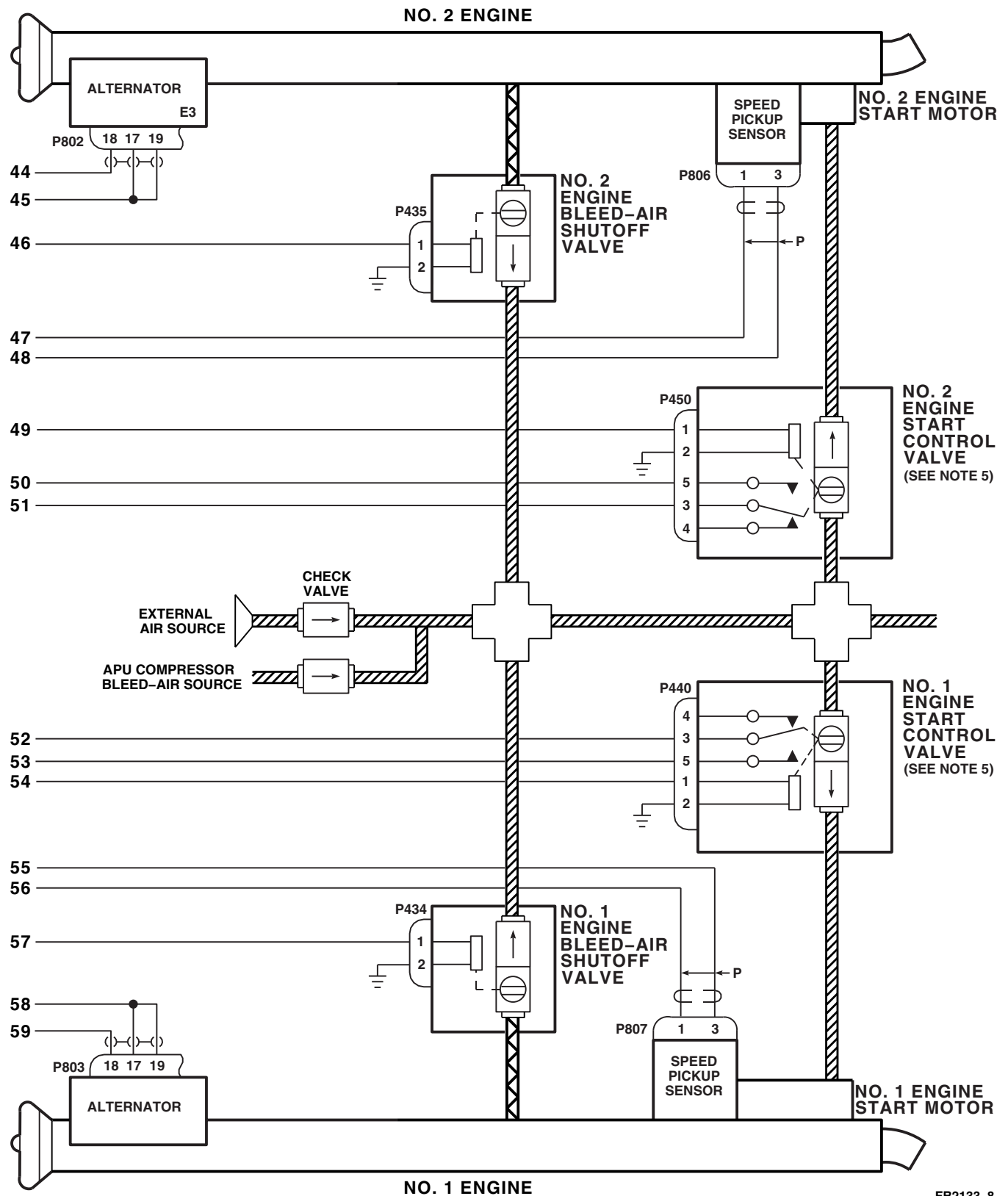


FIGURE 4-2-8-3. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM
(SHEET 7 OF 8)



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FIGURE 4-2-8-3. ENGINE START AND IGNITION SYSTEM SCHEMATIC DIAGRAM
(SHEET 8 OF 8)

4-2-9 ENGINE CONTROLS QUADRANT (BENCH).

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit

References

- PARA 1-6-16
- PARA 1-6-18

- PARA 4-4-21

Equipment Conditions

- External Electrical Power Available And/Or APU Operational

NOTE

- Operational/troubleshooting procedure of No. 1 or No. 2 engine controls quadrant is the same, except as noted.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

4-2-9.1 Operational/Troubleshooting Procedure.

- Remove engine controls quadrant (PARA 4-4-21).
- Inspect engine controls quadrant for broken or burnt wires, security of components, and general condition. Check quadrant levers for free movement. Check quadrant secondary idle stop latch for proper operation.

RESULT: Wires shall not be broken or burnt, components shall be installed and in good condition, and all levers shall have free movement. Secondary idle stop latch shall be in contact with secondary stop when engine POWER CONT lever is at IDLE position. If result is not as specified, replace starter switch (PARA 4-5-13).

- With No. 1 engine controls quadrant starter button not pressed, check continuity between J105-F and J105-G.

RESULT: Continuity shall be present.

- If result is not as specified, replace starter switch (PARA 4-5-13).

- With No. 1 engine controls quadrant starter button pressed, check continuity between J105-G and J105-H.

RESULT: Continuity shall be present.

- If result is not as specified, replace starter switch (PARA 4-5-13).

- With No. 1 engine controls quadrant starter button not pressed, check continuity between J105-R and J105-S.

RESULT: Continuity shall be present.

- If result is not as specified, replace starter override switch (PARA 4-5-15).

- With No. 1 engine controls quadrant starter button pressed, check continuity between J105-S and J105-T.

RESULT: Continuity shall be present.

- If result is not as specified, replace starter override switch (PARA 4-5-15).

- With No. 1 engine controls quadrant engine POWER CONT lever pushed up, check continuity between J105-J and J105-K.

RESULT: Continuity shall be present.

- If result is not as specified, replace starter abort switch (PARA 4-5-14).
- h. With No. 1 engine controls quadrant engine POWER CONT lever pulled down, check continuity between J105-K and J105-L.

RESULT: Continuity shall be present.

- If result is not as specified, replace starter abort switch (PARA 4-5-14).
- i. With No. 1 engine fire extinguishing T-handle in stowed position, check continuity between J105-P and J105-N.

RESULT: Continuity shall be present.

- If result is not as specified, replace fire extinguishing arming switch (PARA 4-5-16).
- j. With No. 1 engine fire extinguishing T-handle in armed position, check continuity between J105-N and J105-M.

RESULT: Continuity shall be present.

- If result is not as specified, replace fire extinguishing arming switch (PARA 4-5-16).
- k. Repeat operational/troubleshooting steps a. through j. for No. 2 engine controls quadrant using connector J106.

NOTE

Step l. is to be performed on engine controls quadrant part number 70301-01100-013 only.

- l. Check continuity between:

P394-1	and	P394-2
P395-1	and	P395-2
P396-1	and	P396-2

RESULT: Continuity shall be present.

- If result is not as specified, replace quadrant information panel (PARA 4-5-17).
- If trouble remains, replace engines controls quadrant (PARA 4-4-21).

NOTE

After installation of engine control quadrant, make sure that control levers are in the OFF or stowed position.

- m. Install engine controls quadrant (PARA 4-4-21).
- n. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- o. Turn on electrical power (PARA 1-6-16).

NOTE

The photosensitive material in the fire detectors is sensitive to red light. Sunlight filtered through smoke or haze, or at sunrise or sunset, may contain greater amounts of red light. This could trigger the fire detectors and cause a false fire indication. For this reason, fire detectors should be shielded from sunlight.

- p. Make sure that FIRE DET TEST switch, on upper console, is at OPER.

RESULT: No. 1 engine, No. 2 engine, and APU T-handle lights shall be off.

- If result is not as specified, troubleshoot Fire Detection System (PARA 12-2-1).
- q. Turn upper console FIRE DET TEST switch to position 1.

RESULT: No. 1 engine and No. 2 engine T-handle lights, and pilot's and copilot's master warning FIRE capsules shall be on.

- If result is not as specified, troubleshoot Fire Detection System (PARA 12-2-1).

- r. Turn upper console FIRE DET TEST switch to position 2.

RESULT: No. 1 engine and No. 2 engine T-handle lights, and pilot's and copilot's master warning FIRE capsules shall be on.

- If result is not as specified, troubleshoot Fire Detection System (PARA 12-2-1).
- s. Pull out FIRE DET NO. 1 ENG and FIRE DET NO. 2 ENG circuit breakers on upper console circuit breaker panel, and APU FIRE DET circuit breaker on lower console circuit breaker panel.
- t. Turn upper console FIRE DET TEST switch to OPER.
- u. Turn upper console CONSOLE LT UPPER OFF/BRT control to BRT.

RESULT: Engine controls quadrant panels shall go on as control is turned clockwise.

- If engine controls quadrant panels do not go on, check that panel lights on upper console are on.
 - (a) If upper console lights are not on, troubleshoot Upper and Lower Console Lights (PARA 9-2-5).
 - (b) If upper console lights are on, replace engine controls quadrant information panel (PARA 4-5-17).
 - (c) If trouble remains, replace engine controls quadrant (PARA 4-4-21).
- v. Pull out LIGHTS UPPER CSL circuit breaker on copilot's circuit breaker panel.

NOTE

RTR BRAKE Perform steps w. through ac.

- w. Engage rotor brake (PARA 1-6-18).

RESULT: ROTOR BRAKE ON capsule on caution/advisory panel shall go on.

- If result is not as specified, troubleshoot Rotor Brake System (PARA 6-2-3).
- x. Pull down and push engine control quadrant levers toward FLY.

RESULT: Levers shall not move past IDLE.

- If levers do not stop at IDLE, replace rotor brake solenoid (PARA 4-4-24).
- y. Place engine control quadrant levers to OFF.
- z. Disengage rotor brake and move handle into detent on master cylinder (PARA 1-6-18).
- aa. Pull down and push engine control quadrant levers toward FLY.

RESULT: Engine control quadrant lever shall move smoothly to FLY and ROTOR BRAKE ON capsule on caution/advisory panel shall be off.

- If engine control quadrant lever will not pass IDLE, replace rotor brake solenoid (PARA 4-4-24).
 - If ROTOR BRAKE ON capsule does not go off, troubleshoot Rotor Brake System (PARA 6-2-3).
- ab. Engage rotor brake (PARA 1-6-18).
- ac. Pull out ROTOR BRAKE and CAUT/ADVSY PNL circuit breakers on upper console circuit breaker panel, and LIGHTS CAUT/ADVSY circuit breaker on copilot's circuit breaker panel.
- ad. Turn off electrical power (PARA 1-6-16).

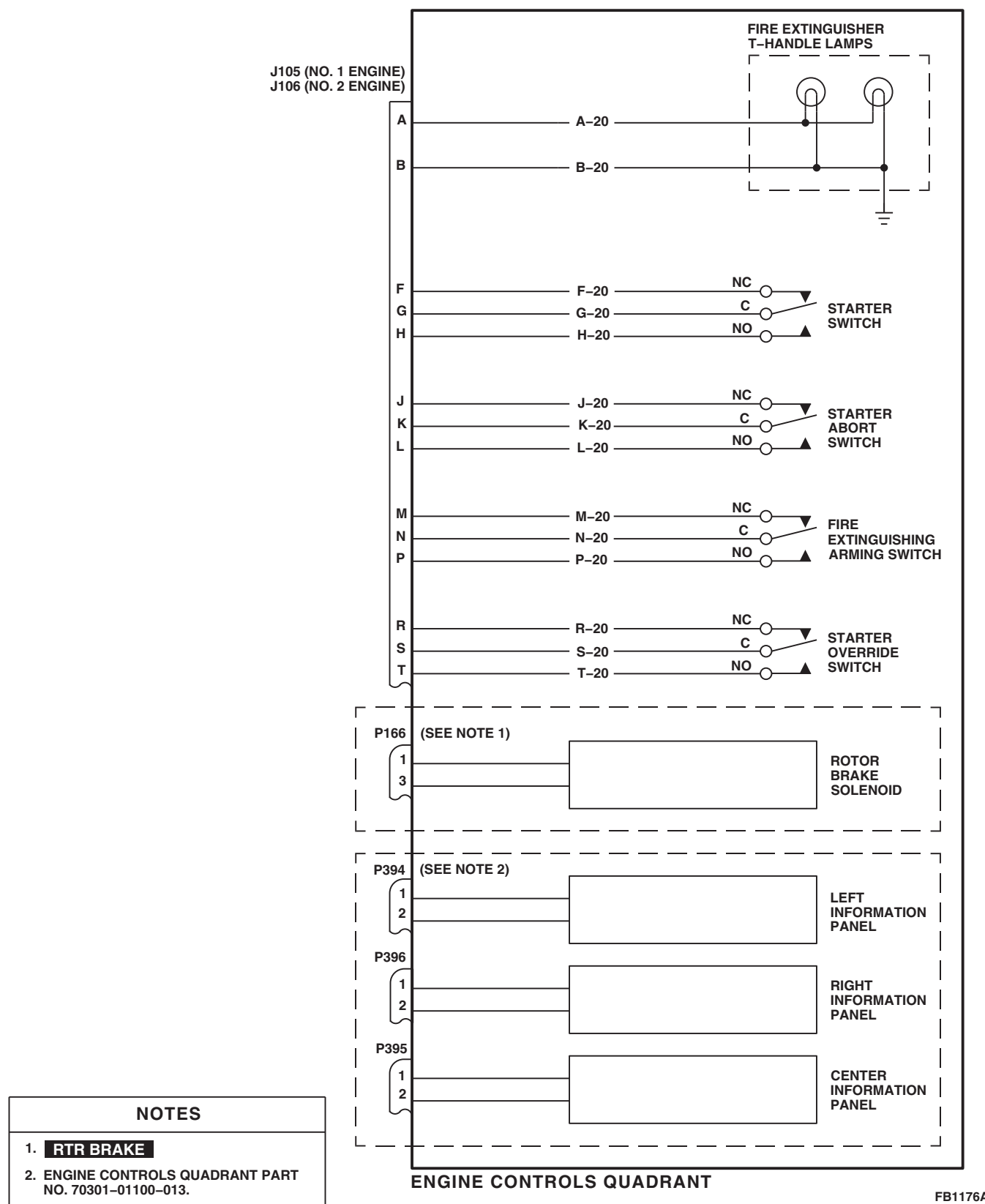


FIGURE 4-2-9-1. ENGINE CONTROL QUADRANT SCHEMATIC DIAGRAM

SECTION 4-3

ON AIRCRAFT INSPECTIONS

SECTION OVERVIEW

PARAGRAPH	TITLE
4-3-1	Engine Exhaust Module and Ejector Inspection
4-3-2	Engine Control Push/Pull Cable Inspection
4-3-3	Engine Load-Demand Push/Pull Cable Friction Check

4-3-1/(4-3-2 Blank)

4-3-1 ENGINE EXHAUST MODULE AND EJECTOR INSPECTION.

PARAGRAPH BREAKDOWN

	PAGE
4-3-1.1 Engine Exhaust Module and Ejector Inspection	4-3-1-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill, Rotary
- Drill Bit, 1/8-inch

- Fluorescent Penetrant Inspection Kit
- Plastic-Faced Hammer

References

- ASTM E1417
- PARA 4-5-6

4-3-1.1 Engine Exhaust Module and Ejector Inspection.

- a. Inspect exhaust module component, Table 4-3-1-1.

Table 4-3-1-1. Exhaust Module and Ejector Damage and Repair

COMPONENT	DAMAGE	REPAIR
Exhaust Fairings (Titanium)	Cracks, maximum four, each not more than 3-inches long, with distance between any two cracks no less than length of longer crack.	Weld per PARA 4-5-6.
Exhaust Ejector (Stainless Steel)	Dents, 2-inch maximum diameter or 0.250-inch maximum depth, except on welded seams, mating surfaces, and where cracks can occur.	If ejector has not been removed, no repair required. If ejector has been removed, use plastic-faced hammer with support to remove dents.
	Dents, more than 2-inches in diameter or 0.250-inch deep, on welded seams, mating surfaces, or capable of causing cracks.	Repair at appropriate sheet metal maintenance facility.

Table 4-3-1-1. Exhaust Module and Ejector Damage and Repair (Cont)

COMPONENT	DAMAGE	REPAIR
Exhaust Ejector (Stainless Steel)	Crack, single, 3-inches maximum length.	Stop-drill, using drill with 1/8-inch drill bit.
	Crack, single, more than 3-inches long.	Weld per PARA 4-5-6.
	Crack, continuous between two bolt-holes or nutplates.	Weld per PARA 4-5-6.
	Cracks, maximum four, each crack no more than 3-inches long and distance between two cracks more than 3-inches.	Weld per PARA 4-5-6.
	Attaching Angle (Stainless Steel)	Cracks, maximum two, each crack from 6 - 12-inches long maximum, with 12-inches minimum distance between cracks.
		Weld per PARA 4-5-6.
	Cracks, maximum four, each crack 6-inches long maximum, with distance between any two cracks equal to length of longer crack.	Weld per PARA 4-5-6.

- b. Repair exhaust module, Table 4-3-1-1 and PARA 4-5-6.

CAUTION

Do not repair cracks in ejector or ejector attaching angle with patches. Rivets can expand and cause more cracking.

- c. Visually inspect exhaust ejector for dents and cracks. Using fluorescent penetrant inspection

kit, determine length of crack by fluorescent penetrant inspection (ASTM E1417). Repair limits and methods are given in Table 4-3-1-1. If limit is exceeded, replace component.

- d. Visually inspect attaching angle for cracks. Using fluorescent penetrant inspection kit, inspect to determine length of crack (ASTM E1417). Repair limits and methods are given in Table 4-3-1-1. If limit is exceeded, replace component.

4-3-2 ENGINE CONTROL PUSH/PULL CABLE INSPECTION.

PARAGRAPH BREAKDOWN

		PAGE
4-3-2.1	Engine Control Push/Pull Cable Inspection	4-3-2-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 4-4-27
- PARA 4-4-29

4-3-2.1 Engine Control Push/Pull Cable Inspection.

NOTE

Visual inspection of cable spiral casing separation applies to engine load-demand, engine and fuselage power-available, and fuel selector valve push/pull cables.

- Visually inspect push/pull cable for separation of spiral casing from end fitting. **SEPARATION GREATER THAN 0.50-INCH NOT ALLOWED.** Separation up to 0.50-inch is allowable, but cable must be repaired as soon as practicable, but no later than next 10 hour/14 day inspection.
- Inspect rod end on engine load-demand cable for surface corrosion. **NONE ALLOWED.** If

corrosion is found, repair rod end (PARA 4-4-27).

- Inspect fuselage power-available push/pull cable as follows:
 - Inspect rod end for radial and axial play. **NONE ALLOWED.** If play is found, replace rod end (PARA 4-4-29).
 - Check stopnut self-locking feature by installing stopnut on adapter. **STOPNUT MUST NOT BE ABLE TO ROTATE BY HAND.** If stopnut can be rotated by hand, replace stopnut.
 - Inspect cable assembly for missing retaining ring or plunger. **NONE ALLOWED.** If either retaining ring or plunger are missing, replace cable (PARA 4-4-29).
- Check for clearance between cables and surrounding structure and components.

4-3-3 ENGINE LOAD-DEMAND PUSH/PULL CABLE FRICTION CHECK.

PARAGRAPH BREAKDOWN

		PAGE
4-3-3.1	Engine Load-Demand Push/Pull Cable Friction Check	4-3-3-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Powerplant Repairer’s Toolkit
- Spring Scale
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Tiedown Strap, Item 333, Appendix D

References

- Appendix D
- PARA 4-4-18
- PARA 4-4-27
- SEI 671
- TM 1-2840-248-23

4-3-3.1 Engine Load-Demand Push/Pull Cable Friction Check.

NOTE

The following procedures apply to both left and right side load-demand push/pull cables.

- Disconnect rod end of collective bias tube (part of engine load-demand push/pull cable) from collective bias lever on mixer (PARA 4-4-27).
- Install tiedown strap, Item 333, Appendix D, through rod end for spring scale attachment point.
- Attach spring scale to tiedown strap installed on rod end. Measure cable friction by pulling cable and measuring force at breakaway. Proceed as follows:

- If friction force is less than 45 pounds, proceed to step f.
- If friction force is greater than 45 pounds, disconnect load-demand system rotary input assembly from hydromechanical unit (HMU) (PARA 4-4-27).

- Attach spring scale to tiedown strap installed on rod end. Measure cable friction by pulling cable and measuring force at breakaway. Proceed as follows:
 - If friction force is less than 33 pounds, perform the following:
 - Perform torque check on engine HMU spindle. **TORQUE SHALL BE LESS THAN 15 INCH-POUNDS.**

- (b) If torque is greater than 15 inch-pounds, replace HMU (SEI 671 and TM 1-2840-248-23).
- (2) If friction force is greater than 33 pounds, perform the following:
 - (a) Remove load-demand rotary input (PARA 4-4-18).
 - (b) Attach spring scale to tiedown strap installed on rod end. Measure cable friction by pulling cable and measuring force at breakaway.
 - (c) If friction force is less than 3 pounds, replace load-demand rotary input (PARA 4-4-18).
 - (d) If friction force is greater than 3 pounds, replace engine load-demand push/pull cable (PARA 4-4-27).
 - e. Repeat step c. and step d. if load-demand push/pull cable, rotary input, or HMU was replaced during friction check.
 - f. Connect rod end of collective bias tube (part of engine load-demand push/pull cable) to collective bias lever on mixer (PARA 4-4-27).

SECTION 4-4

MAINTENANCE PROCEDURES

SECTION OVERVIEW

PARAGRAPH	TITLE
4-4-1	Engine
4-4-2	Demountable Power Package
4-4-3	Engine Line Replaceable Units (LRUs)
4-4-4	Engine Air Inlet
4-4-5	Aft Engine Mounts
4-4-6	Forward Support Tube/Engine Output Shaft
4-4-7	Engine Exhaust Module
4-4-8	Engine Exhaust Module Ejector
4-4-9	HIRSS
4-4-10	HIRSS Rear Fairings and Panels
4-4-11	HIRSS Suppressor Support Mounts
4-4-12	HIRSS Nacelle Fairing Support Mounts
4-4-13	HIRSS Nacelle Fairing
4-4-14	HIRSS Suppressor Core and Baffle
4-4-15	HIRSS Extender
4-4-16	HIRSS Bulkhead Repair
4-4-17	HIRSS Emissive Coating Repair
4-4-18	Load-Demand Rotary Input
4-4-19	Power-Available Rotary Input
4-4-20	Engine Speed Control Potentiometer

PARAGRAPH	TITLE
4-4-21	Engine Control Quadrant
4-4-22	Engine Control Quadrant Canopy Plunger
4-4-23	Engine Starter Switch Adjustment
4-4-24	Rotor Brake Solenoid
4-4-25	Engine Starter Abort Switch Adjustment
4-4-26	Engine Starter Override Switch Adjustment
4-4-27	Engine Load-Demand Push/Pull Cable
4-4-28	Engine Power-Available Push/Pull Cable
4-4-29	Fuselage Power-Available Push/Pull Cable
4-4-30	Fuel Selector Valve Push/Pull Cable
4-4-31	Fuel Selector Valve Control Box
4-4-32	Engine Load-Demand Push/Pull Cable Supports
4-4-33	Engine Power-Available System Rigging and Adjustment
4-4-34	Engine Mount Restraint Rings

4-4-1 ENGINE.

PARAGRAPH BREAKDOWN

	PAGE
4-4-1.1 Replace Engine	4-4-1-2

INITIAL SETUP

Tools

- Adapter, 21C7082G01
- Aircraft Mechanic’s Toolkit
- Cutting Wheel, High-Speed
- Feeler Gage
- File
- Hoist, Capacity to Support Engine
- Lifting Sling, 21C7081G02
- Maintenance Crane
- Maintenance Trailer
- Powerplant Repairer’s Toolkit
- Protection, Eye, Skin, and Breathing
- Rubber Mallet
- Spring Scale
- Straightedge, 3-inch
- Strapwrench
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Wood Blocks, 2 inch x 3 inch x 6 inch (5 cm x 8 cm x 15 cm)

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Abrasive Cloth, Item 9, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Hardware Kit, 70000-10100-043
- Hardware Kit, 70000-30100-043
- Hardware Kit, 70000-30100-044
- Lubricating Oil, Item 205, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 286, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 1-3-6
- PARA 1-6-16
- PARA 1-7-2
- PARA 1-7-20
- PARA 1-7-29
- PARA 1-7-37
- PARA 2-6-3
- PARA 2-6-5

- PARA 4-4-2
- PARA 4-4-4
- PARA 4-4-5
- PARA 4-4-6
- PARA 4-4-7
- PARA 6-3-8
- PARA 6-4-27
- PARA 18-4-1
- SEI 671
- TM 1-2840-248-23
- TM 1-70-VIB

4-4-1.1 Replace Engine.

NOTE

Replacement of No. 1 and No. 2 engine is the same, except as noted.

4-4-1.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Preserve engine before removal (SEI 671 and TM 1-2840-248-23).
- c. Remove engine air inlet (PARA 4-4-4).
- d. If cabin maintenance crane will be used, erect crane (PARA 18-4-1).
- e. Disconnect fuel hose from fuel boost pump swivel elbow. Install protective caps or plugs on hose and elbow (Figure 4-4-1-1, Sheet 2, Detail G).
- f. Disconnect engine combustion and engine manifold drain hoses from elbows on rear engine compartment deck (Figure 4-4-1-1, Sheet 2, Detail E). Install protective caps or plugs on hoses and elbows.
- g. Remove clamps securing common drain to engine compartment shield.
- h. On No. 1 engine only, remove clamp securing swirl frame drain to airframe (PARA 4-4-2).
- i. Disconnect electrical connectors, P815 and P825 (No. 1 engine), or P814 and P824 (No. 2 engine), from electrical receptacles on firewall (Figure 4-4-1-1, Sheet 1, Detail B).
- j. Install protective caps and plugs on electrical connectors.
- k. If installed, remove electrical harness clamp and bonding jumper from firewall.
- l. Disconnect power-available rotary input and load-demand rotary input from hydromechanical unit (Figure 4-4-1-1, Sheet 2, Detail D).
- m. Loosen clamps and remove starter bleed-air tube and coupling from starter and upper firewall tube (Figure 4-4-1-1, Sheet 1, Detail A). Keep starter bleed-air tube, coupling, and clamps with helicopter.
- n. Disconnect cross bleed-air tube as follows:
 - (1) **T700** **T701A** Loosen clamps and remove coupling securing cross bleed-air tube to lower firewall tube. Discard coupling. 
 - (2) **T701C** Loosen coupling securing cross bleed-air tube and sleeve to lower firewall tube. Slide sleeve and coupling onto cross bleed-air tube. 
- o. On No. 2 engine only, remove screw, washer, and spacer securing engine control cables to No. 2 engine inlet particle separator (IPS) blower duct (Figure 4-4-1-1, Sheet 1, Detail C).
- p. Disconnect couplings and remove No. 1 or No. 2 engine IPS blower ducts from IPS blowers and IPS collars in exhaust modules.
- q. Remove loop clamp, 70300-30200-101/-102, as follows (Figure 4-4-1-1, Sheet 2, Detail F):

4-4-1-2 Change 11

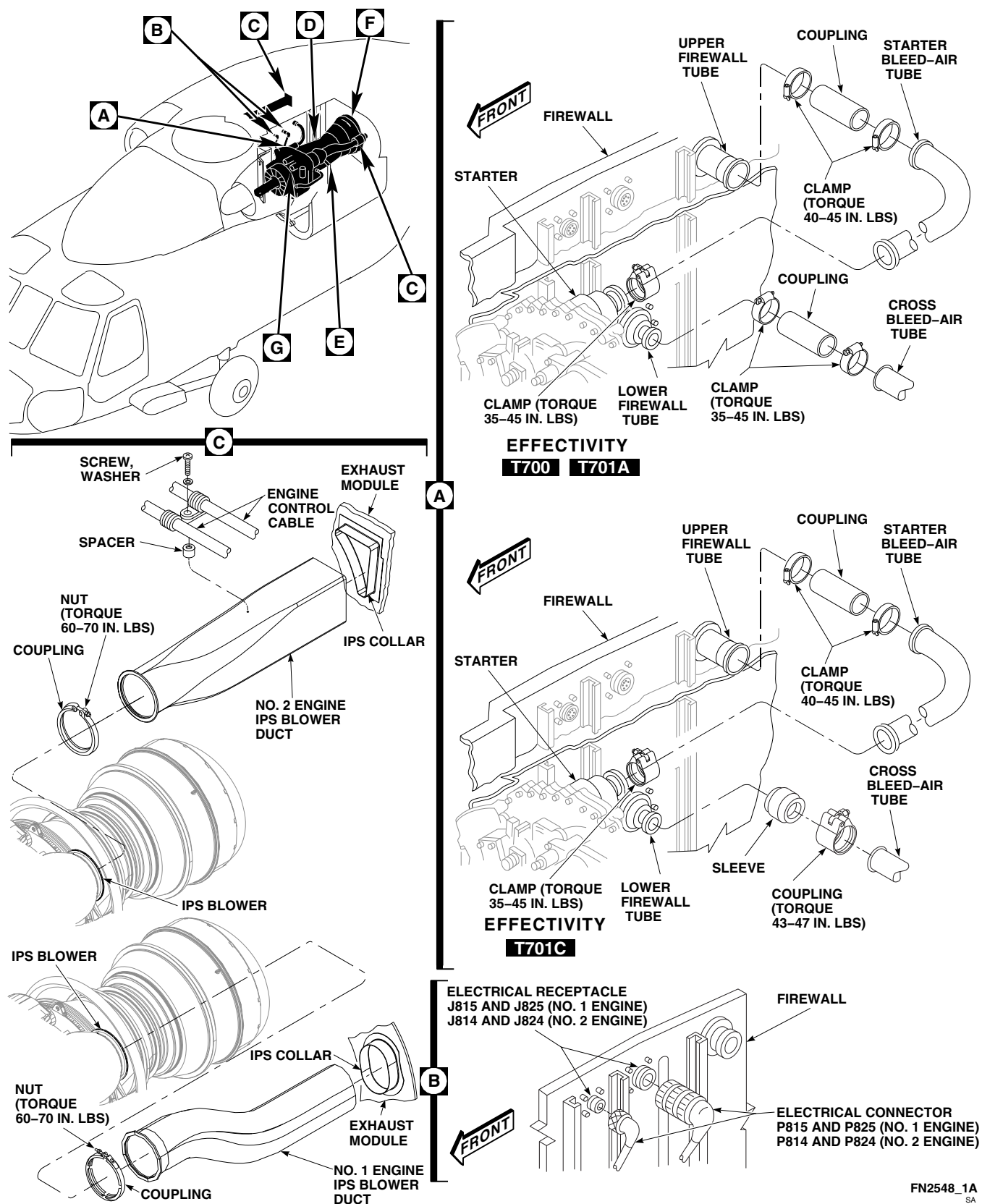


FIGURE 4-4-1-1. DISCONNECT/CONNECT ENGINE/AIRFRAME CONNECTIONS (SHEET 1 OF 2)

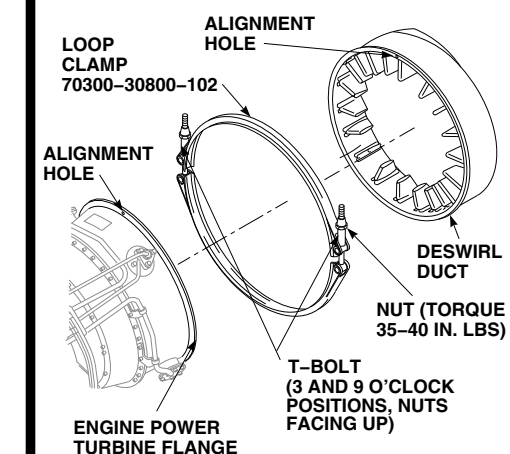
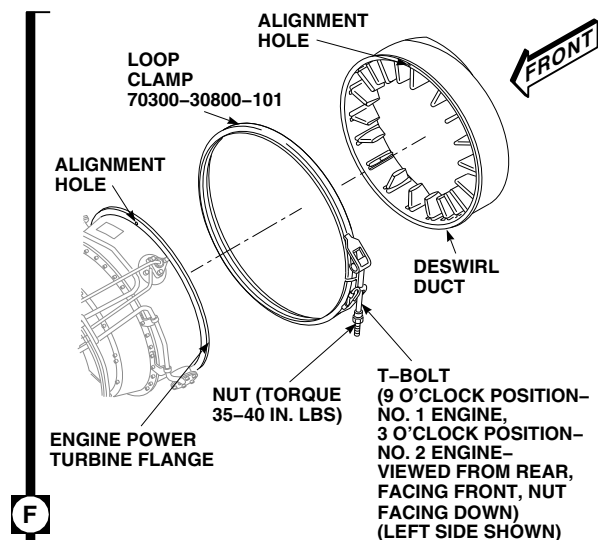
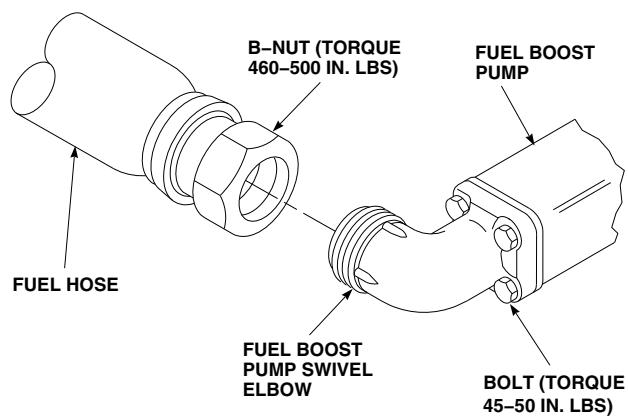
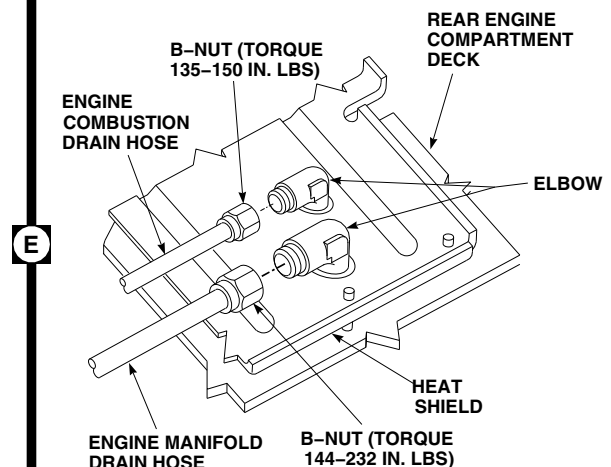
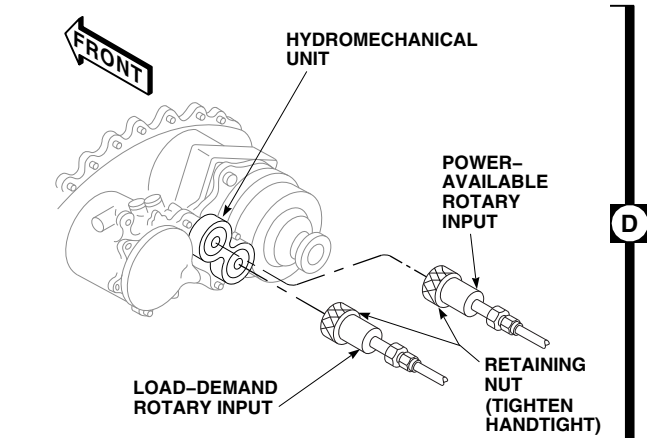
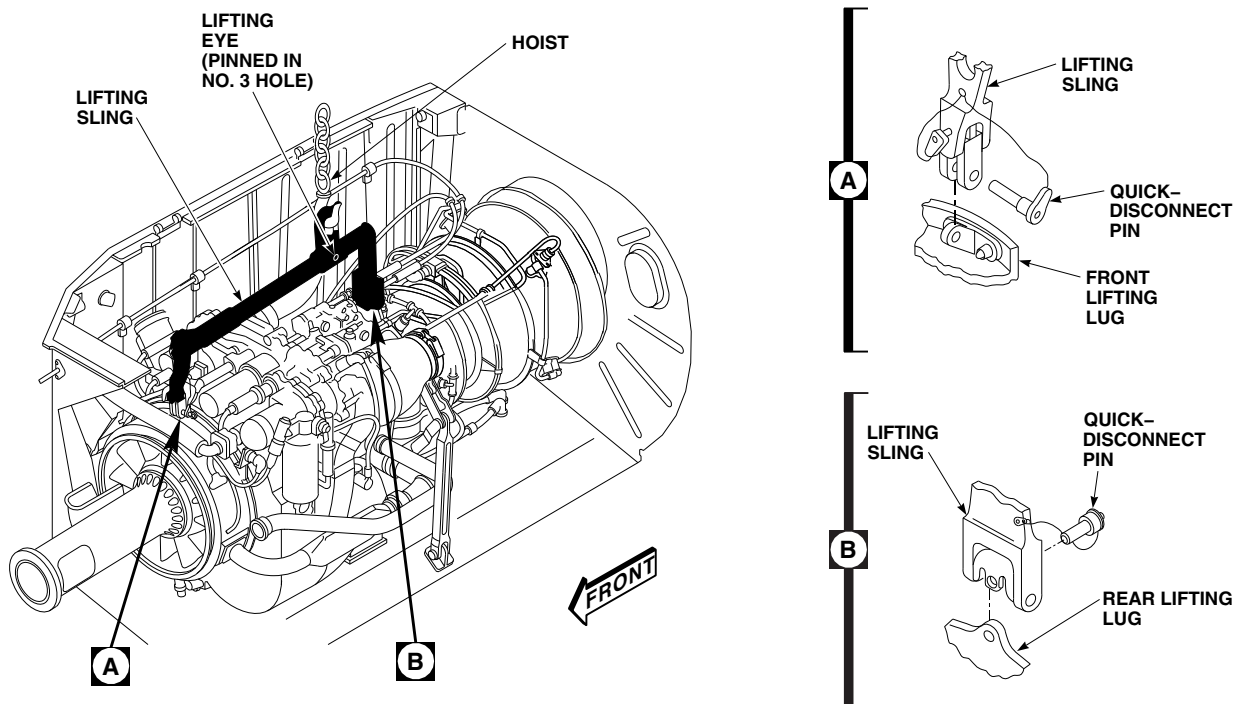
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FIGURE 4-4-1-1. DISCONNECT/CONNECT ENGINE/AIRFRAME CONNECTIONS
(SHEET 2 OF 2)



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FIGURE 4-4-1-2. ATTACH LIFTING SLING

- (1) **W/O HIRSS** Unclamp loop clamp from deswirl duct and engine power turbine flange. Place deswirl duct and loop clamp inside of exhaust fairing. ◀
 - (2) **HIRSS** Unclamp loop clamp from deswirl duct and engine power turbine flange. Position loop clamp on deswirl duct. Leave deswirl duct and loop clamp installed in HIRSS module. ◀
- r. Secure lifting sling on engine front and rear lifting lugs with quick-disconnect pins (Figure 4-4-1-2, Detail A and Detail B).
 - s. Pin lifting eye in No. 3 hole on lifting sling (Figure 4-4-1-2).

WARNING

- Injury to personnel and damage to equipment will result if caution is not

taken when hoisting engine. Keep area clear when hoisting engine.

- To prevent injury to personnel or damage to equipment, do not use maintenance crane or hoist as a source of excessive force to free engine from helicopter.
 - Hoist cable may break, causing injury to personnel or damage to equipment if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.
- t. Attach lifting sling to maintenance crane or hoist. Raise hoist until weight of engine is supported by sling.
 - u. Disconnect forward support mount as follows:
 - (1) Using strapwrench, hold engine output shaft from turning (Figure 4-4-1-3, Sheet 2, Detail B).

CAUTION

Excessive vibration will result if two or more nuts connecting engine output shaft to disc pack coupling are loosened or removed at any time. Engine output shaft, disc pack coupling, and connecting hardware is a balanced assembly. Only one nut connecting engine output shaft to disc pack coupling may be loosened or removed at any one time. Loose or removed nuts must be torqued to their proper value before another nut is loosened or removed.

NOTE

- Helicopters with input module input flange, 70351-08206-102, have self-locking nuts on front side of flange to secure disc pack coupling to flange.
 - Helicopters with input module input flange, 70351-08206-041, have nutplates on front side of flange to secure disc pack coupling to flange.
- (2) On helicopters with engine output shafts, 70361-08004-042/-043, remove bolts, countersunk washers, and nuts (if installed) securing disc pack coupling to input module input flange (Figure 4-4-1-3, Sheet 2, Detail B, and Sheet 4, Detail F).
 - (3) On helicopters with engine output shafts, 70361-08014-041, remove bolts securing diaphragm coupling to engine output shaft (Figure 4-4-1-3, Sheet 2, Detail B, and Sheet 4, Detail G).
 - (4) Remove bolts and washers securing forward support tube to elastomeric gimbal (Figure 4-4-1-3, Sheet 2, Detail B). Keep bolts and washers with elastomeric gimbal for installation. Replace any broken bolts on gimbal.
 - (5) If input module input flange, 70351-08206-041, is installed, check input

module input flange nutplate rivets for proper installation, looseness, and fretting on heads. **RIVETS MUST BE FLUSH WITH FLANGE SURFACE. NO LOOSENESS OR FRETTING ON RIVET HEADS ALLOWED.** If rivets are not flush with flange surface, are loose, or fretting is found on heads, replace input module input flange (PARA 6-4-27).

- (6) Perform engine output shaft inspection - after disconnect (PARA 1-7-37).

- v. Disconnect support mounts as follows:

WARNING

Bushings installed in rear engine mount links are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use eye and skin protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.

- (1) Remove bolts, countersunk washers, washers, and nuts securing inner and outer mount struts to inner and outer mount links (Figure 4-4-1-3, Sheet 1, Detail A). Keep attaching hardware and shim washers with mount strut.
- (2) Remove bolt, countersunk washer, washers, and nut securing center mount links to mount support. Keep attaching hardware and shim washers with mount support.

CAUTION

On helicopters with engine output shaft, 70361-08014-041, damage to shaft and diaphragm coupling will result if shaft is not completely separated from coupling.

- w. On helicopters with engine output shaft, 70361-08014-041, move engine slightly rear to

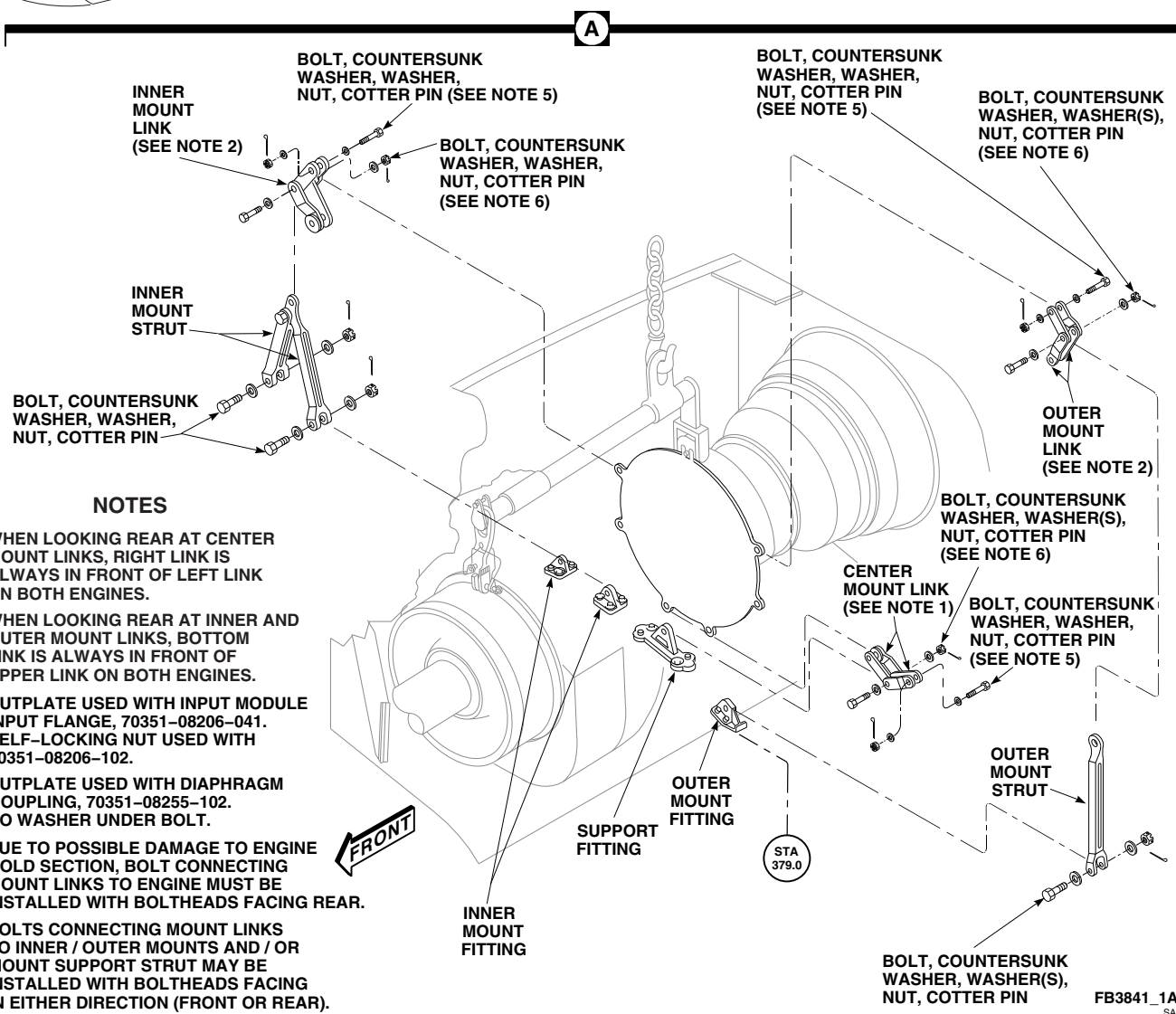
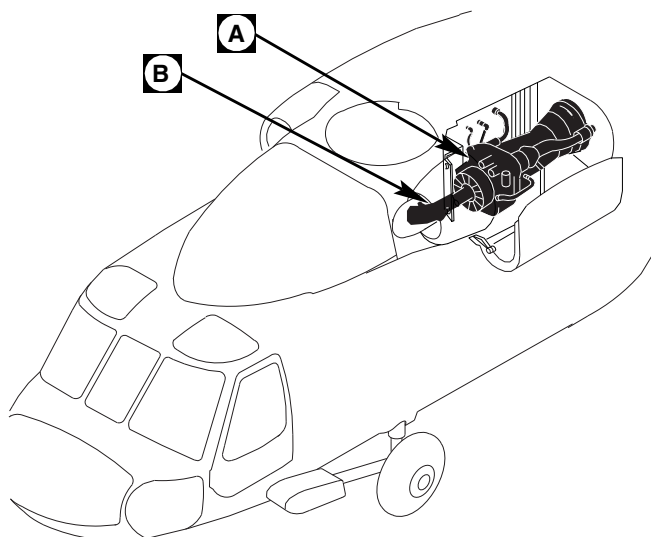
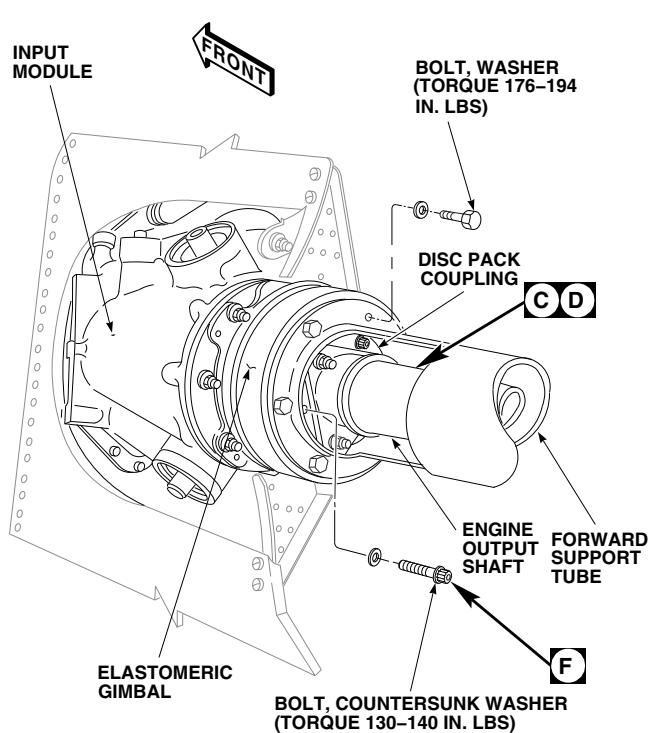


FIGURE 4-4-1-3. DISCONNECT/CONNECT ENGINE MOUNTS (SHEET 1 OF 4)

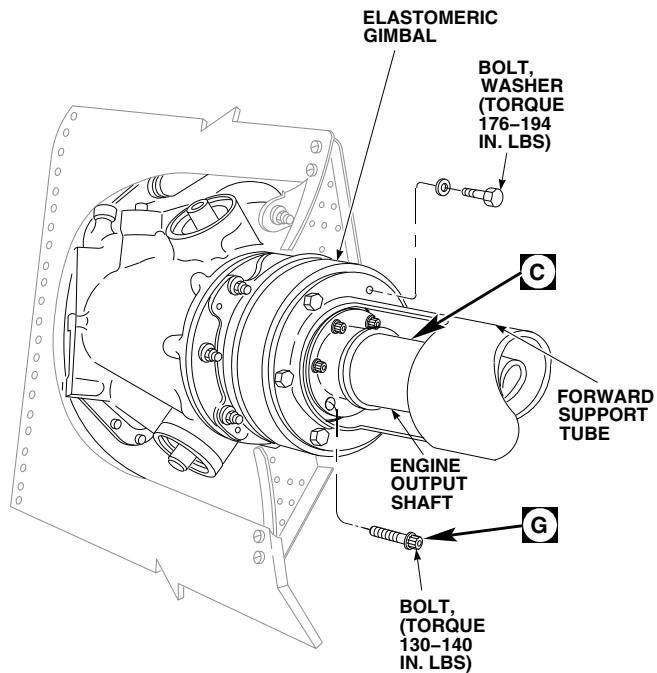


EFFECTIVITY

ON HELICOPTERS WITH ENGINE OUTPUT SHAFT, 70361-08004-042 OR 70361-08004-043.

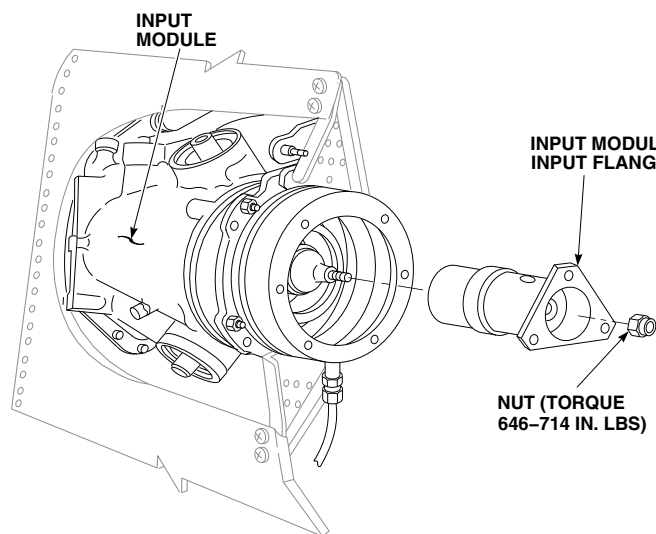
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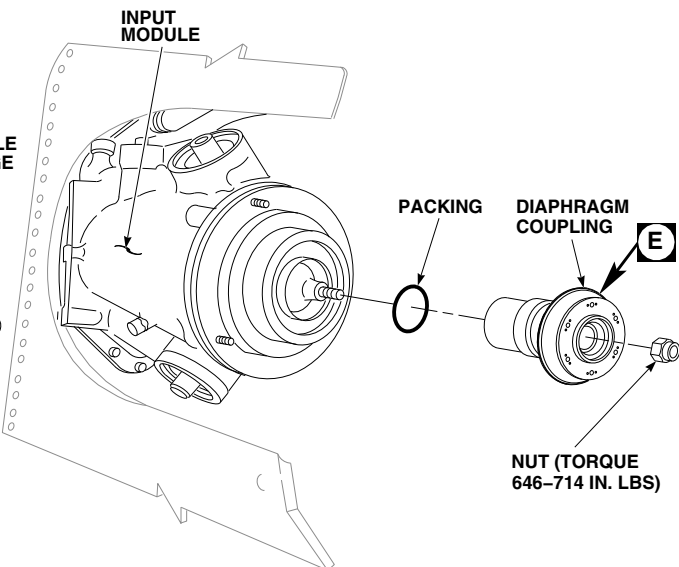
EFFECTIVITY

ON HELICOPTERS WITH ENGINE OUTPUT SHAFT, 70361-08014-041.



EFFECTIVITY

ON HELICOPTERS WITH ENGINE OUTPUT SHAFT, 70361-08004-042 OR 70361-08004-043.



EFFECTIVITY

ON HELICOPTERS WITH ENGINE OUTPUT SHAFT, 70361-08014-041.

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FIGURE 4-4-1-3. DISCONNECT/CONNECT ENGINE MOUNTS (SHEET 2 OF 4)

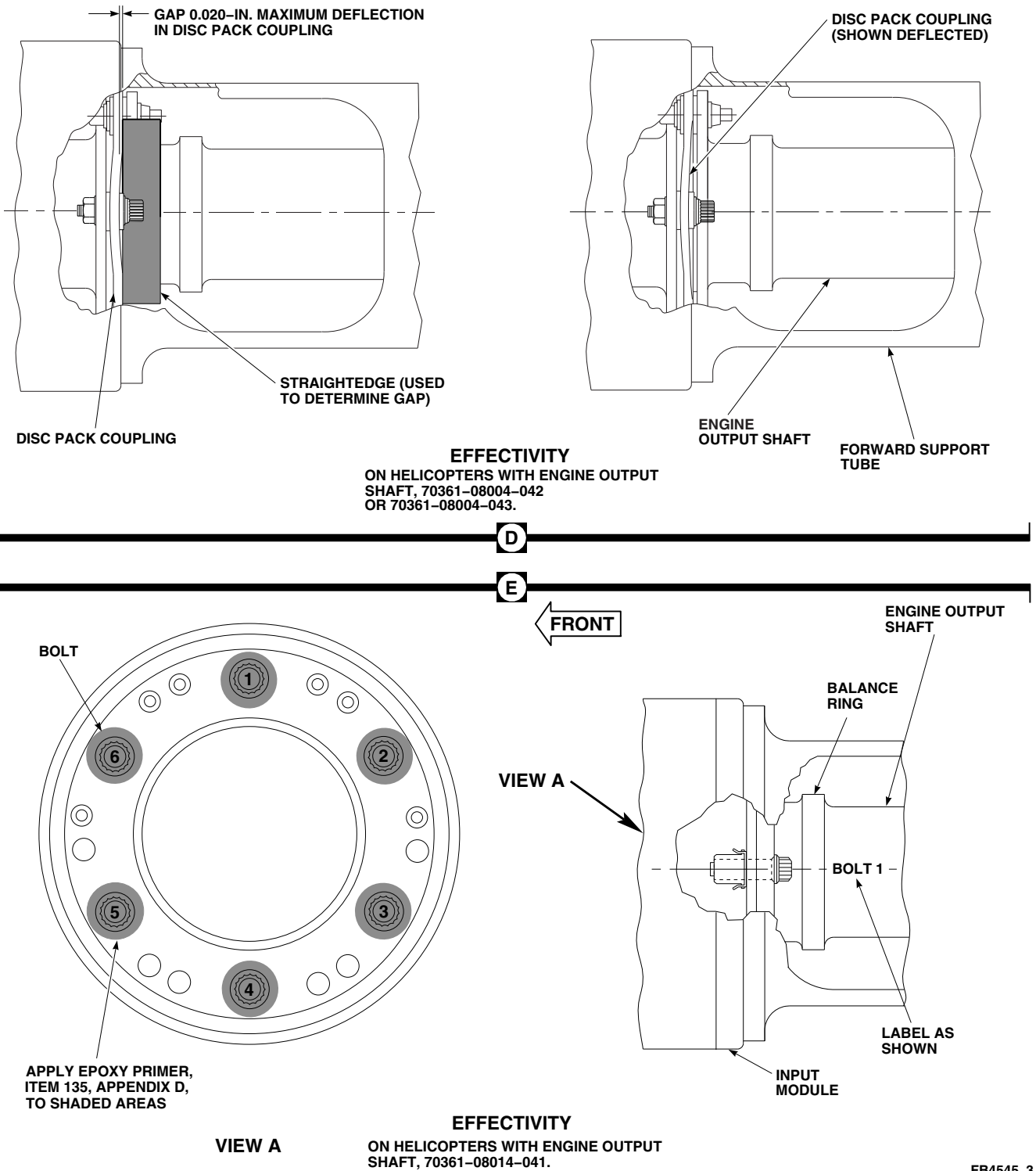


FIGURE 4-4-1-3. DISCONNECT/CONNECT ENGINE MOUNTS (SHEET 3 OF 4)

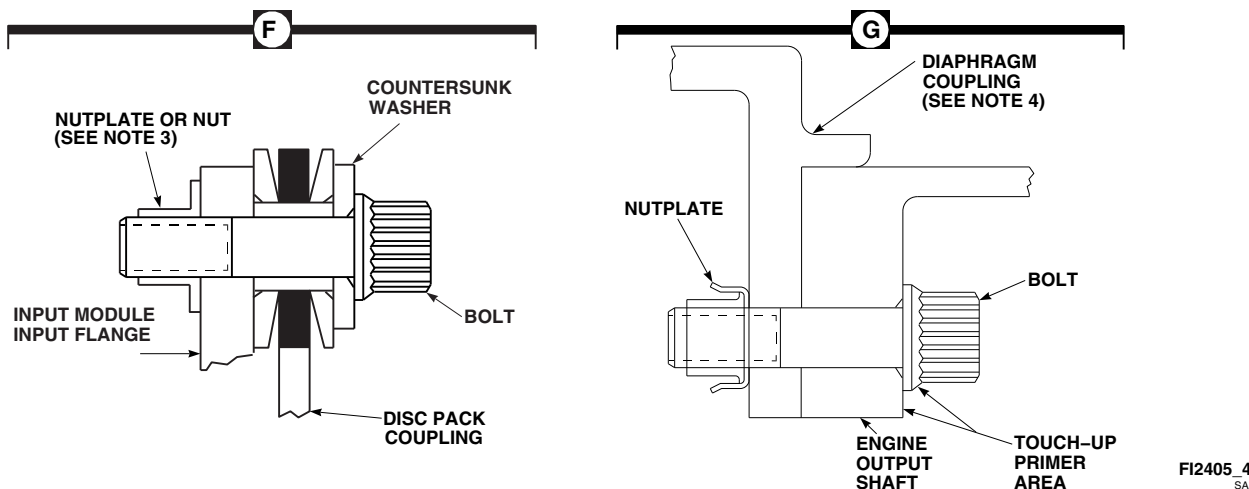


FIGURE 4-4-1-3. DISCONNECT/CONNECT ENGINE MOUNTS (SHEET 4 OF 4)

separate engine output shaft from input flange/coupling.

WARNING

- Injury to personnel and damage to equipment will result if caution is not taken when hoisting engine. Keep area clear when hoisting engine.
- To prevent injury to personnel or damage to equipment, do not use maintenance crane or hoist as a source of excessive force to free engine from helicopter.

CAUTION

Damage to engine and helicopter will result if engine is allowed to swing while removing from helicopter. Carefully move engine when removing from helicopter.

- x. Raise engine high enough to clear engine cowl-ing.
- y. Remove bolts, countersunk washers, washers, and nuts securing center mount links to flange under engine combustor section (Figure 4-4-1-

3, Sheet 1, Detail A). Keep links and attaching parts with demountable power package.

- z. Swing engine outward and lower it onto adapter (Figure 4-4-1-4).
- aa. Secure engine to adapter (SEI 671 and TM 1-2840-248-23).
- ab. Remove lifting sling from engine.

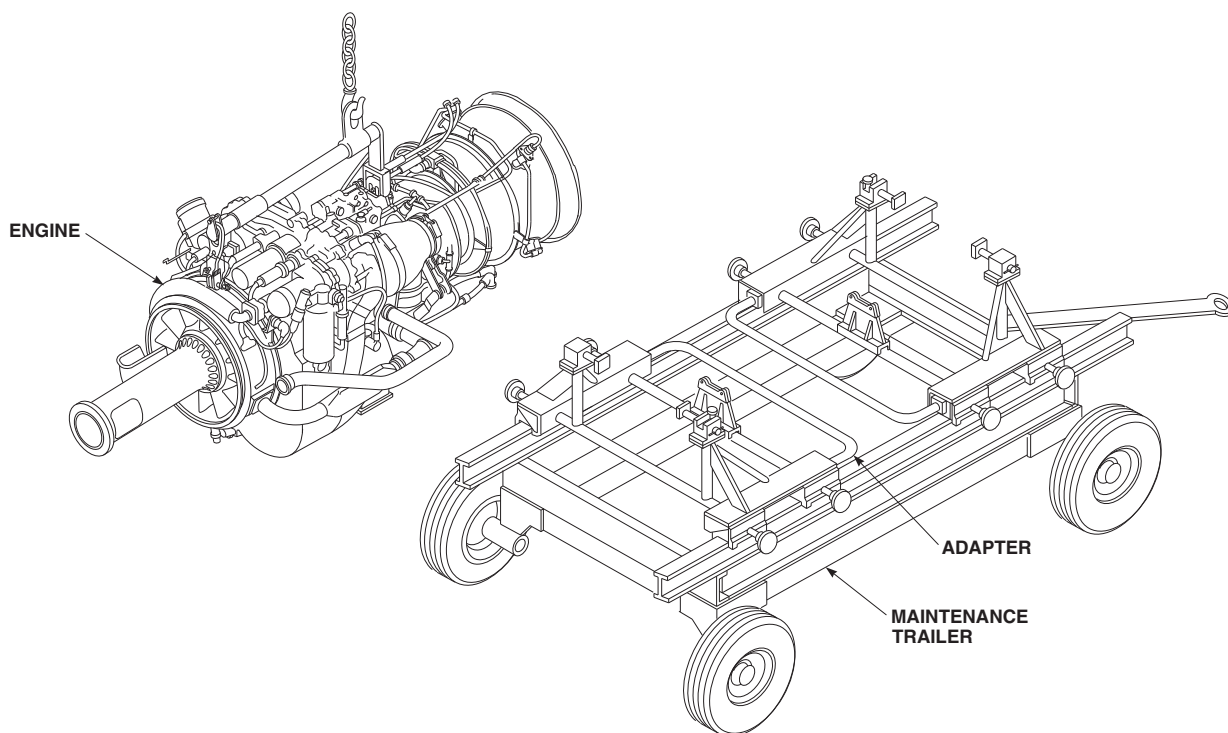
NOTE

If quick engine change is available, do not tear down demountable power package.

- ac. If engine is being replaced, tear down demountable power package and install components on new engine (PARA 4-4-2).

4-4-1.1.2 Inspect.

- a. Inspect input pinion axial play (PARA 6-3-8).
- b. On helicopters with engine output shaft, 70361-08004-042/-043, inspect engine output shaft and disc pack coupling (PARA 4-4-6).
- c. On helicopters with engine output shaft, 70361-08014-041, inspect engine output shaft (PARA 4-4-6).
- d. Inspect input module input flange (PARA 6-4-27).



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FIGURE 4-4-1-4. REPLACE ENGINE

e. Inspect elastomeric gimbal (PARA 6-3-8).

HIGH METAL, IS ACCEPTABLE.
If beyond limits, contour as needed.

f. Inspect deswirl duct as follows:

- (1) Check deswirl duct for cracks. **NONE ALLOWED.** If cracked, replace deswirl duct.
- (2) Check for nicks, scratches, and gouges on the following parts, and if necessary, repair:
 - (a) Clamp Flange: **ANY NUMBER, 1/32-INCH DEEP WITH NO HIGH METAL, IS ACCEPTABLE.** If beyond limits, contour as needed.
 - (b) Vanes and Doublers: **ANY NUMBER, 0.005-INCH DEEP WITH NO HIGH METAL OR SHARP CORNERS, ARE ACCEPTABLE.** If beyond limits, contour as needed.
 - (c) All Other Areas: **ANY NUMBER, 0.015-INCH DEEP WITH NO**

- (3) Check cowling for distortion. **COWLING MUST BE VISUALLY CIRCULAR AND CONCENTRIC WITH CASE.** If beyond limits, cold work.
- (4) Check vanes for distortion. **VANES MUST BE VISUALLY STRAIGHT AND PARALLEL WITH FLOW PATH.** If beyond limits, replace deswirl duct.
- (5) Check deswirl duct for missing vanes. **A TOTAL OF SIX VANES MAY BE MISSING OR REMOVED PROVIDED THAT NO TWO ADJACENT VANES ARE MISSING.** If beyond limits, replace deswirl duct.
- (6) Inspect for cracks in vanes, specifically in leading edge area of vanes. **CRACKS SHALL NOT EXCEED 1-INCH.** If cracks less than 1-inch are found, inspect deswirl duct every 30 hours. If cracks of

1-inch or more are found, remove vane as follows:

- (a) Remove deswirl duct from engine.

NOTE

Make sure removal of vane will not cause deswirl duct to be beyond limits.

- (b) Using high-speed cutting wheel, cut along base of vane.
 - (c) Grind smooth any rough edges.
- g. Inspect deswirl duct loop clamp for cracks near T-bolt attachment point(s). **NONE ALLOWED.** If cracked, replace deswirl duct loop clamp.

4-4-1.1.3 Install.

- a. Install loop clamp, 70300-30800-101/-102, as follows (Figure 4-4-1-1, Sheet 2, Detail F):
 - (1) **W/O HIRSS** Position deswirl duct and loop clamp inside exhaust fairing before installing engine in compartment.■
 - (2) **HIRSS** Install deswirl duct and loop clamp on HIRSS module by sliding duct inside finger seals of HIRSS core.■
- b. On No. 2 engine only, slide engine inlet particle separator (IPS) blower duct into exhaust fairing opening.
- c. Secure lifting sling on engine front and rear lifting lugs with quick-disconnect pins (Figure 4-4-1-2, Detail A and Detail B).
- d. Pin lifting eye in No. 3 hole on lifting sling (Figure 4-4-1-2).
- e. Attach lifting sling to hoist. Raise hoist until weight of engine is supported by sling.

CAUTION

Damage to engine and helicopter will result if engine is allowed to swing

while installing on helicopter. Carefully move engine when installing on helicopter.

- f. Remove bolts and washers securing engine to adapter (SEI 671 and TM 1-2840-248-23).

NOTE

- Figure 4-4-1-3, Sheet 1, Detail A depicts a general pattern for direction of bolt-heads, mount links, and mount struts. Boltheads for mount links and upper and lower bolts for mount struts may face in either direction, front or rear.
 - Mount links should have open fork ends (widest end) attached to mount support or mount strut.
 - For inner and outer mount links, upper mount link should have offset leg of fork facing rear and lower link should have offset leg of fork facing front.
 - When looking rear at inner and outer mount links, bottom link is always in front of upper link on both engines.
 - When looking rear at center mount links, right link is always in front of left link on both engines.
 - Forks should be in front of adjoining upper mount link fork with mount or strut between these forks.
 - Make sure countersunk washers are installed under bolthead with countersunk side facing bolthead.
 - Install washers, as required, under nut to obtain correct cotter pin position.
- g. Secure center mount links to flange under engine combustion section using bolts, countersunk washers, washers, and nuts. **TIGHTEN EACH NUT FINGERTIGHT, ADVANCE NUT TO NEXT CASTELLATION.** Install cotter pin.

CAUTION

- Disc pack coupling will be damaged if not connected to correct elastomeric gimbal.
- Make sure to use the correct elastomeric gimbal with input module input flange/coupling being used. Elastomeric gimbal, 70361-08015-102, must be used with input flange/coupling, 70351-08255-102, and engine output shaft, 70361-08014-041. Elastomeric gimbal, 70361-08001-101, 70361-08001-102 or 70361-08001-103, must be used with input flange, 70351-08206-102, and engine output shaft, 70361-08004-042, or 70361-08004-043, and disc pack, 70361-08002-041.

NOTE

- On helicopters with input module input flange, 70351-08206-102 or 70351-08260-041, hardware used to secure disc pack coupling to input flange is a balanced set and should not be replaced individually. If any individual pieces of hardware needs to be replaced, replace all hardware with balanced hardware kit, 70000-30100-043.
 - On helicopters with input module input flange/coupling, 70351-08255-102, hardware used to secure engine output shaft to input module input flange/coupling is a balanced set and should not be replaced individually. If any individual pieces of hardware needs to be replaced, replace all hardware with balanced hardware kit, 70000-30100-044.
- h. On helicopters with engine output shaft, 70361-08004-042/-043, inspect engine output shaft attachment bolts as follows:

CAUTION

Damage to bolts may result if bolts are not clean. Contaminates on engine output shaft bolts may lower fracture point. Prior to installation, make sure bolts and hardware are clean.

- (1) Check bolts and hardware for cleanness. If required, clean as necessary.
 - (2) Visually inspect bolts for signs of burrs or evidence of binding. **NONE ALLOWED.** If burrs or evidence of binding is found, replace all bolts and washers with correct hardware kit.
 - (3) Inspect bolts for binding in disc pack coupling by inserting bolts into disc pack only, and check for freedom of rotation. **BOLTS MUST ROTATE FREELY WHEN TURNED BY HAND.** If bolts do not rotate freely, replace all bolts and washers with correct hardware kit. If replacement bolts still do not rotate freely, replace engine output shaft (PARA 4-4-6) and input module input flange (PARA 6-4-27).
- i. On helicopters with engine output shaft, 70361-08014-041, inspect engine output shaft attachment bolts as follows:
- (1) Visually inspect bolts for signs of burrs or evidence of binding. **NONE ALLOWED.** If burrs or evidence of binding is found, replace all bolts with hardware kit, 70000-30100-044.
 - (2) Inspect bolts for binding in engine output shaft by inserting bolts into shaft only, and check for freedom of rotation. **BOLTS MUST ROTATE FREELY WHEN TURNED BY HAND.** If bolts do not rotate freely, replace all bolts and washers with correct hardware kit. If replacement bolts still do not rotate freely, replace engine output shaft (PARA 4-4-6) and input module input flange (PARA 6-4-27).

- j. Slide engine output shaft into engine as far as it will go.
- k. On helicopters with engine output shaft, 70361-08004-042/-043, perform torque check of nuts securing disc pack coupling to engine output shaft as follows:
 - (1) Use strapwrench on output shaft to prevent shaft from turning. Check disc pack coupling to engine output shaft nuts in tightening direction while holding bolts. **TORQUE SHALL NOT BE LESS THAN 90 INCH-POUNDS.** If any nut is less than 90 inch-pounds, replace engine output shaft (PARA 4-4-6).
 - (2) Reset torque wrench to 130 inch-pounds and torque check disc pack coupling to engine output shaft nuts in tightening direction while holding bolts. **TORQUE SHALL NOT BE LESS THAN 130 INCH-POUNDS.** If any nut is less than 130 inch-pounds, **RETORQUE NUT TO 140 INCH-POUNDS.**
- l. Perform torque check on input module input flange nut (Figure 4-4-1-3, Sheet 2, Detail C). **MINIMUM TORQUE IS 646 INCH-POUNDS. IF TORQUE IS LESS THAN 646 INCH-POUNDS, RETORQUE TO 646 - 714 INCH-POUNDS.**
- m. Remove bolts, washers, and nuts from input module input flange and retain for later use.
- n. Install bolts in input flange, 70351-08206-041. Make note if bolts can be fully threaded into nutplate by hand for later use.
- o. If input flange, 70351-08206-102, is installed, install bolts and nuts. If any one of the three bolts can be fully threaded by hand into nut, replace bolts, washers, and nuts as a matched set using hardware kit, 70000-30100-043.

CAUTION

On helicopters with engine output shaft, 70361-08004-042/-043, damage to input flange will result if disc pack coupling is not properly aligned with input module input flange. Bolts securing coupling to shaft may scrape against input flange. Make sure coupling-to-flange boltholes are properly aligned before placing engine in engine compartment.

- p. On helicopters with engine output shaft, 70361-08004-042/-043, rotate engine output shaft to align three mounting holes in shaft coupling with three holes in input flange.
- q. On helicopters with engine output shaft, 70361-08014-041, rotate engine output shaft to align six mounting holes in shaft with six holes in input module input flange/coupling.

WARNING

- Injury to personnel and damage to equipment will result if caution is not taken when hoisting engine. Keep area clear when hoisting engine.
- Hoist cable may break, causing injury to personnel or damage to equipment if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.
- r. Carefully raise and place engine in engine compartment, making sure forward support tube is flush on elastomeric gimbal.
- s. Connect support mounts as follows (Figure 4-4-1-3, Sheet 1, Detail A):

WARNING

Bushings installed in rear engine mount links are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use eye and skin protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.

CAUTION

- Damage to mount struts, mount links, and mounting bolt will result if force is used to install bolt. Bolt connecting mount struts to mount links is a close tolerance assembly. If bolt does not slide through strut and links, replace bolt.
- Damage to engine cold section may occur if bolts connecting mount links to engine are installed with boltheads facing front. Install boltheads connecting mount links to engine with boltheads facing rear. Bolts connecting mount links to inner/outer mounts and mount support strut may face in either direction.

NOTE

Make sure countersunk washer is installed under bolthead with countersunk side facing bolthead.

- (1) Secure inner and outer mount struts to inner and outer mount links with bolts, countersunk washers, washers, and nuts. **TORQUE NUTS TO 181 - 199 INCH-POUNDS.** Install cotter pins.

NOTE

- Make sure countersunk washer is installed under bolthead with countersunk side facing bolthead.

- Install washers, as required, under nut to obtain correct cotter pin position.
- (2) Secure center mount link to mount support with bolt, countersunk washer, washers, and nut. **TIGHTEN NUT UNTIL FIRMLY SEATED, THEN BACK OFF UNTIL WASHER UNDER NUT CAN BE TURNED BY HAND. RETIGHTEN NUT JUST ENOUGH SO WASHER CANNOT BE TURNED BY HAND. TIGHTEN NUT TO NEXT CASTELLATION.** Install cotter pin. Do not tighten nut more than one castellation.
 - (3) If required, additional engine mount instructions and torques are found in PARA 4-4-5.
- t. On helicopters with engine output shaft, 70361-08014-041, connect forward support engine mount as follows:

- (1) Position forward support tube on elastomeric gimbal (Figure 4-4-1-3, Sheet 2, Detail B).

CAUTION

Damage to gimbal and support tube will result if bolts are used to draw support tube to gimbal. Position support tube flush against gimbal before installing bolts.

- (2) Secure forward support tube to elastomeric gimbal with bolts and washers. **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.**
- (3) Connect engine output shaft to diaphragm coupling with bolts as follows:

CAUTION

Make sure correct hardware and stackup is used. Six bolts, MS9724-07, no washers allowed.

NOTE

- Bolts are selected by weight in sets of three and may be reused, provided there are no signs of fretting or other damage.
- If hardware kit is not available, individual bolts may be used, although pre-weighed kit, 70000-30100-044, is preferred. If individual bolts are used, weighing bolts is recommended. Difference in weight between three bolts should not exceed 0.025 grams.

- (a) Using ink marker, sequentially number boltholes 1 through 6 to align engine output shaft boltholes (Figure 4-4-1-3, Sheet 3, Detail E). Label boltholes rear of shaft balancing ring.
- (b) Secure engine output shaft to diaphragm coupling with six bolts, MS9724-07. Always use following torque sequence when torquing bolts: 1, 4, 2, 5, 3, 6. Use strap-wrench to prevent shaft from turning when torquing bolts.
- (c) Using torque sequence specified in step t.(3)(b), tighten each bolt in equal steps of two turns to evenly draw output shaft into position until shaft is flush against diaphragm coupling. Do not exceed 40 inch-pounds of torque when tightening bolts.
- (d) Using torque sequence specified in step t.(3)(b), **TORQUE BOLTS TO 90 INCH-POUNDS.**
- (e) Using torque sequence specified in step t.(3)(b), **TORQUE BOLTS TO 130 - 140 INCH-POUNDS. RETORQUE BOLTS TO 130 - 140 INCH POUNDS UNTIL NO MOTION IS FELT.**
- (f) Remove lifting sling from engine, remove maintenance crane and move

hoist out of way. Remove lifting sling from crane or hoist (PARA 18-4-1).

- (g) Make sure torque check of bolts connecting engine output shaft to input module input flange/coupling is due after 3 to 13 flight hours (PARA 1-7-2).
- u. On helicopters with engine output shaft, 70361-08004-042/-043, connect forward engine mount as follows:

- (1) Position forward support tube on elastomeric gimbal (Figure 4-4-1-3, Sheet 2, Detail B).

CAUTION

Damage to gimbal and support tube will result if bolts are used to draw support tube to gimbal. Position support tube flush against gimbal before installing bolts.

- (2) Secure forward support tube to elastomeric gimbal with bolts and washers. **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.**

WARNING
FSCAP

Verification of minimum torque of bolts is a critical characteristic.

NOTE

- Helicopters with input module input flange, 70351-08206-102, use self-locking nuts on forward side of flange to secure disc pack coupling to flange.
- Helicopters with input module input flange, 70351-08206-041, have nutplates

on forward side of flange to secure disc pack coupling to flange.

- Make sure countersunk washers are installed under bolthead with countersunk side facing bolthead.
- (3) Connect output shaft disc pack coupling to input module input flange with bolts, countersunk washers, and nuts (if required) as follows (Figure 4-4-1-3, Sheet 2, Detail B and Sheet 4, Detail F):
 - (a) Tighten each bolt in equal steps of two turns to evenly draw engine output shaft into position until coupling is flush against input module input flange. **DO NOT EXCEED 40 INCH-POUNDS OF TORQUE WHEN TIGHTENING BOLTS.**
 - (b) Place 3-inch straightedge across coupling face and measure for deflection (Figure 4-4-1-3, Sheet 3, Detail D).
 - (c) If deflection is more than 0.020-inch, lightly tap shaft against balance ring in forward direction using a 2 x 3 x 6 inch (5 x 8 x 15 cm) block of wood and rubber mallet, until coupling deflection is less than 0.020-inch.
 - (d) **IF GAP IS LESS THAN 0.020-INCH, TORQUE BOLTS TO 130 - 140 INCH-POUNDS.**
 - (e) If input module input flange 70351-08206-041 is used, and if bolts can be fully threaded into nutplate by hand, after torque has stabilized apply sealing compound, Item 286, Appendix D, to end of bolt on threads extending past nut. Make sure sealing compound is applied to thread near nut to allow it to seep into nut through capillary action.
 - (4) Remove lifting sling from engine and remove maintenance crane (PARA 18-4-

1) or move hoist out of way. Remove lifting sling from crane or hoist.

NOTE

Hardware used to secure disc pack coupling to input flange is a balanced set and cannot be replaced individually. If any individual piece of hardware is to be replaced, replace all hardware with the correct balanced hardware kit. If hardware kit is not available, hardware is to be selected in sets of three so that difference in weight between individual studs, bolts, washers, and nuts per set is not over 0.0250 gram.

- (5) If input flange, 70351-08206-102, is installed, install bolts and nuts. If any one of the three bolts can be fully threaded by hand into nut, replace bolts and nuts (PARA 6-4-27).
- (6) After torque has stabilized, apply sealing compound, Item 286, Appendix D, to end of bolt on threads extending past nut.
- (7) Make sure sealing compound is applied to threads near nut or nutplate to allow it to seep into nut or nutplate through capillary action.
- (8) Rotate engine output shaft to align mounting holes with input module input flange. This will prevent damage to input module input flange from disc pack coupling attaching bolts.

NOTE

- If input module input flange, 70351-08206-102, is installed, use hardware kit, 70000-10100-043, and three self-locking nuts, MS21043-4, making sure difference in weight between individual nuts is not over 0.0250 gram.
 - Hardware kit, 70000-30100-043, may be used with either input flange, 70351-08206-041 or 70351-08206-102.
- (9) Inspect engine output shaft attachment bolts as follows:

- (a) Visually inspect bolts for signs of burrs or evidence of binding. **NONE ALLOWED.** If burrs or evidence of binding is found, replace all bolts and washers with correct hardware kit.
- (b) Inspect bolts for binding in disc pack coupling by inserting bolts into disc pack only and checking for freedom of rotation. **BOLTS MUST ROTATE FREELY WHEN TURNED BY HAND.** If bolts do not rotate freely, replace all bolts and washers with correct hardware kit. If replacement bolts still do not rotate freely, replace engine output shaft assembly and input module input flange (PARA 4-4-6 and PARA 6-4-27).

NOTE

Discs which make up disc pack coupling are a balanced set and cannot be replaced individually. If any disc is damaged beyond limits, output shaft assembly must be replaced.

- (10) Inspect disc pack coupling for gaps between discs using feeler gage. Two or more discs between bolts on coupling flanges may separate or buckle. **NO SHARP BENDS OR CREASES IN DISC ALLOWED. MAXIMUM GAP BETWEEN DISCS MUST NOT EXCEED 0.030-INCH.** If disc pack coupling is damaged beyond limits, replace engine output shaft assembly.
- (11) Inspect disc pack coupling for dents and nicks. **DENTS AND NICKS LESS THAN 0.020-INCH ON THE EDGES OF INDIVIDUAL DISCS CAN BE BLENDED OUT.** Blend damage using abrasive cloth, Item 9, Appendix D, followed by abrasive cloth, Item 8, Appendix D. **REPLACE ENGINE OUTPUT SHAFT ASSEMBLY IF DAMAGE IS GREATER THAN 0.020-INCH.**
- (12) Perform torque check of nuts attaching disc pack coupling to engine output shaft as follows:

- (a) Use strap wrench on engine output shaft to prevent shaft from turning. Check disc pack coupling to engine output shaft nuts in tightening direction while holding bolts. **TORQUE SHALL NOT BE LESS THAN 90 INCH-POUNDS.** If any nut is less than 90 inch-pounds, replace engine output shaft.
- (b) Reset torque wrench to 130 inch-pounds and torque check disc pack coupling to engine output shaft nuts in tightening direction while holding bolts. **IF ANY NUT TURNS BELOW 130 INCH-POUNDS, RETORQUE NUT TO 140 INCH-POUNDS.**

WARNING

FSCAP

Verification of minimum torque of bolts is a critical characteristic.

NOTE

- Helicopters with input module input flange, 70351-08206-102, use self-locking nuts on front side of flange to secure disc pack coupling to flange.
- Helicopters with input module input flange, 70351-08206-041, have nutplates on forward side of flange to secure disc pack coupling to flange.
- (13) On helicopters with hardware kit, 70000-10100-043 or 70000-30100-043 installed, connect output shaft disc pack coupling to input module input flange with bolts, washers, and nuts (if required) as follows (Figure 4-4-1-3, Sheet 2, Detail B):
 - (a) Tighten each bolt in equal steps of two turns to evenly draw output shaft into position until coupling is flush against input module input

flange. **DO NOT EXCEED 40 INCH-POUNDS OF TORQUE WHEN TIGHTENING BOLTS.**

- (b) Place 3-inch straightedge across coupling face and measure for deflection (Figure 4-4-1-3, Sheet 2, Detail C).
 - (c) If deflection is more than 0.020-inch, lightly tap shaft against balance ring in forward direction using 2 x 3 x 6 inch (5 x 8 x 15 cm) block of wood and rubber mallet, until coupling deflection is less than 0.020-inch.
 - (d) **IF GAP IS LESS THAN 0.020-INCH, TORQUE BOLTS TO 130 - 140 INCH-POUNDS.**
 - (e) If input module input flange, 70351-08206-041 is used, and if bolts can be fully threaded into nutplate by hand, after torque has stabilized apply sealing compound, Item 286, Appendix D, to end of bolt on threads extending past nut. Make sure sealing compound is applied to thread near nut to allow it to seep into nut through capillary action.
- (14) Remove lifting sling from engine and move hoist out of way. Remove lifting sling from hoist.

NOTE

Minimum gap between pneumatic tubes coupling connection is 0.120-inch. If necessary, metal may be removed from tube end using file to achieve this gap. Do not remove any metal from bead on tubes.

- v. Connect cross bleed-air tube as follows (Figure 4-4-1-1, Sheet 1, Detail A):
 - (1) **T700 T701A** Connect cross bleed-air tube to lower firewall tube with new coupling. Place clamps next to beads on tubes. **TORQUE CLAMPS TO 35 - 45 INCH-POUNDS.**

- (2) **T701C** Connect cross bleed-air tube to lower firewall tube by sliding sleeve and coupling over cross bleed-air tube and lower firewall tube. **TORQUE COUPLING TO 43 - 47 INCH-POUNDS.**

- w. If new engine was installed, **TORQUE CLAMP SECURING CROSS BLEED-AIR TUBE TO ENGINE TO 35 - 45 INCH-POUNDS.**
- x. Secure starter bleed-air tube to starter with clamp. **TORQUE CLAMP TO 35 - 45 INCH-POUNDS.**

NOTE

Minimum gap between pneumatic tubes coupling connection is 0.120-inch. If necessary, metal may be removed from tube end using file to achieve this gap. Do not remove any metal from bead on tubes.

- y. Secure starter bleed-air tube to upper firewall tube with coupling and clamps. **TORQUE CLAMPS TO 40 - 45 INCH-POUNDS.**

CAUTION

Make sure rotary input alignment pin engages slot on hydromechanical unit (HMU). Slight axial/radial play is normal.

- z. Secure power-available rotary input and load-demand rotary input to HMU (Figure 4-4-1-1, Sheet 2, Detail D). **TIGHTEN RETAINING NUT HANDTIGHT UNTIL IT BOTTOMS IN SUPPORT BRACKET.**
- aa. Remove protective caps and plugs from electrical connectors and receptacles.
- ab. Inspect and treat electrical connectors (PARA 1-7-20).
- ac. Connect electrical connectors, P815 and P825 (No. 1 engine) or P814 and P824 (No. 2

engine), to electrical receptacles on firewall (Figure 4-4-1-1, Sheet 1, Detail B).

- ad. Remove caps and plugs from engine drain hoses and elbows on rear engine compartment deck.
- ae. Connect engine combustion drain hose to elbow on rear engine compartment deck (Figure 4-4-1-1, Sheet 2, Detail E). **TORQUE B-NUT TO 135 - 150 INCH-POUNDS.**
- af. Connect engine manifold drain hose to elbow on rear engine compartment deck. **TORQUE B-NUT TO 144 - 232 INCH-POUNDS.**
- ag. Install clamps securing common drain to engine compartment shield.
- ah. On No. 1 engine only, clamp engine swirl frame drain to airframe (PARA 4-4-2).

NOTE

Fuel hose for No. 1 engine should be positioned to avoid engine oil filler port and right front angled section of firewall. Additional contact between fuel hose and firewall is acceptable for both engines.

- ai. Remove protective cap and connect fuel hose to fuel boost pump swivel elbow (Figure 4-4-1-1, Sheet 2, Detail G).
- aj. Check for $\frac{3}{4}$ -inch clearance between fuel hose and fire extinguisher nozzle. If clearance is not correct, perform the following:
 - (1) Loosen bolts on fuel boost pump swivel elbow.
 - (2) Carefully move fuel boost pump elbow until at least $\frac{3}{4}$ -inch clearance is obtained between fuel hose and fire extinguisher nozzle.
 - (3) **TORQUE BOLTS ON BOOST PUMP SWIVEL ELBOW TO 45 - 50 INCH-POUNDS.**

- ak. **TORQUE B-NUT ON FUEL HOSE TO 460 - 500 INCH-POUNDS.**

- al. Inspect engine IPS blower duct for cracks. **NONE ALLOWED.** If necessary, repair IPS blower duct (PARA 2-6-3).
- am. Install engine IPS blower duct as follows (Figure 4-4-1-1, Sheet 1, Detail C):



Damage to engine cowling will result if coupling T-bolt securing engine IPS blower duct to IPS blower is installed incorrectly. Make sure coupling is positioned with T-bolt nut inboard of each engine.

NOTE

When installing new engine IPS blower duct, refer to PARA 4-4-7.

- (1) Position engine IPS blower duct between IPS collar on exhaust module and IPS blower. Secure duct to blower with coupling with T-bolt positioned inboard of engine.
- (2) **TORQUE COUPLING NUT TO 60 - 70 INCH-POUNDS.**
- (3) To seat coupling, tap lightly with rubber mallet and retorquing coupling nut. Continue tapping until two successive trials show no loss of torque.
- an. On No. 2 engine only, secure engine control cables to No. 2 engine IPS blower duct with screw, washer, and spacer.
- ao. Install deswirl duct to engine power turbine flange as follows (Figure 4-4-1-1, Sheet 2, Detail F):

NOTE

- Left and right deswirl ducts are identical.
 - Deswirl ducts have two alignment holes located on deswirl duct flange. When installing, one alignment hole shall be located top center while other alignment hole shall point inboard.
- (1) Make sure alignment holes in deswirl duct flange mate with alignment holes of engine power turbine flange.
 - (2) Visually inspect deswirl duct to make sure deswirl duct flange mates with engine power turbine flange through 360°. If complete mating of flanges is not achieved, perform the following:
 - (a) Remove deswirl duct.
 - (b) Using spring scale to measure load, apply no more than 15 pounds of load to remove warpage.
 - (c) If warpage cannot be removed using 15 pounds of load, replace deswirl duct.

NOTE

When using lubricating oil, Item 205, Appendix D, refer to General Electric SEI 671 for approved oils.

- (3) Using lubricating oil, Item 205, Appendix D, lubricate pressure surfaces of loop clamp and T-bolt(s).
- (4) Install loop clamp over lips of deswirl duct and engine power turbine flange.
- (5) Position loop clamp as follows:

NOTE

Loop clamp is not a rigging tool. Deswirl duct must be supported until clamp is torqued.

- (a) For loop clamp, 70300-30800-101 (one T-bolt), position loop clamp with T-bolt at 9 o'clock for No. 1 engine and 3 o'clock for No. 2 engine, with nut facing down.
- (b) For loop clamp, 70300-30800-102 (two T-bolts), position loop clamp with T-bolts at 3 o'clock and 9 o'clock positions, with nuts facing up.

NOTE

If torquing loop clamp, 70300-30800-102, alternate from nut to nut to maintain even loading on clamp.

- (6) **TORQUE LOOP CLAMP NUT(S) TO 35 - 40 INCH-POUNDS.**
- (7) To seat loop clamp, tap lightly with rubber mallet. Retorque loop clamp nut(s). Continue tapping until two successive trials show no loss in torque.
- (8) Loosen loop clamp nut(s) slightly. **RE-TORQUE LOOP CLAMP NUT(S) TO 35 - 40 INCH-POUNDS.**
- (9) Visually inspect latches to make sure no interference has occurred.
- (10) Visually inspect inside deswirl duct to make sure deswirl duct flange mates with engine power turbine flange through 360°.

- ap. Install engine air inlet (PARA 4-4-4).
- aq. Check engine oil level and service as required (PARA 1-3-6).
- ar. Close engine cowling.
- as. Perform engine output shaft balance (TM 1-70-VIB). Allow engine to cool down for 1 hour minimum.
- at. After engine output shaft balance, perform the following:
 - (1) Perform torque check of loop clamp nut(s). **TORQUE SHOULD BE 35 - 40**

INCH-POUNDS. If torque is below limit, perform the following:

- (a) Loosen loop clamp nut(s) slightly.
- (b) **RETORQUE LOOP CLAMP NUT(S) TO 35 - 40 INCH-POUNDS.**

- (2) Apply torque stripe to loop clamp nut(s) (PARA 2-6-5).

au. Perform torque stabilization as follows:

- (1) On helicopters with engine output shafts, 70361-08004-042/-043, perform engine output shaft torque check after initial ground run, after 1 to 3 flight hours, and after 25 to 30 flight hours as follows:
 - (a) Using strapwrench to keep engine output shaft from turning, check torque in tightening direction.
 - (b) **TORQUE MUST NOT BE LESS THAN 90 INCH-POUNDS.** If torque on any bolt or stud is below 90 inch-pounds, replace bolts, input module input flange, and engine output shaft. If using input flange, 70351-08206-102, also replace nuts.
 - (c) Using strapwrench to keep engine output shaft from turning, check torque in tightening direction. **TORQUE MUST NOT BE LESS THAN 130 INCH-POUNDS.** If torque on any bolt or stud is below 130 inch-pounds, **RETORQUE TO 130 - 140 INCH-POUNDS.** Repeat torque check after 1 to 3 flight hours.

- (2) On helicopters with engine output shaft, 70361-08014-041, perform engine output shaft torque check after initial ground run and again after 3 to 13 flight hours as follows:

- (a) Using strapwrench to keep engine output shaft from turning, check torque in tightening direction.
- (b) **TORQUE MUST NOT BE LESS THAN 90 INCH-POUNDS.** If torque on any bolt or stud is below 90 inch-pounds, replace diaphragm coupling, bolts, and engine output shaft.
- (c) Using strapwrench to keep engine output shaft from turning, check torque in tightening direction. **TORQUE MUST NOT BE LESS THAN 130 INCH-POUNDS.** If torque on any bolt or stud is below 130 inch-pounds, **RETORQUE TO 130 - 140 INCH-POUNDS.** Repeat torque check after 1 to 3 flight hours.
- (d) If torque does not stabilize after third consecutive torque check, replace all six bolts and diaphragm coupling.
- (e) Apply epoxy primer coating kit, Item 135, Appendix D, to flange in area around boltheads (Figure 4-4-1-3, Sheet 3, Detail E).

- av. Perform engine checkout procedure as part of maintenance operation check/maintenance test flight (SEI 671 and TM 1-2840-248-23).
- aw. Perform maintenance test flight (Appropriate MTF).

4-4-2 DEMOUNTABLE POWER PACKAGE.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
4-4-2.1	Tear Down Power Pack- age	4-4-2-1	4-4-2.3	Quick Engine Change (QEC)	4-4-2-11
4-4-2.2	Build Up Power Package	4-4-2-8			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Output Drive Shaft Puller, Locally-Made, Figure H-216, Appendix H
- Output Drive Shaft Puller, Locally-Made, Figure H-262, Appendix H
- Powerplant Repairer’s Toolkit
- Rubber Mallet
- Spring Scale
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Lubricating Oil, Item 205, Appendix D

- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs
- Spiral Wrap, Item 315, Appendix D
- Washer, NAS1149F0332P

References

- Appendix D
- Appendix H
- PARA 1-7-20
- PARA 2-6-5
- PARA 7-4-1
- SEI 671
- TM 1-2840-248-23

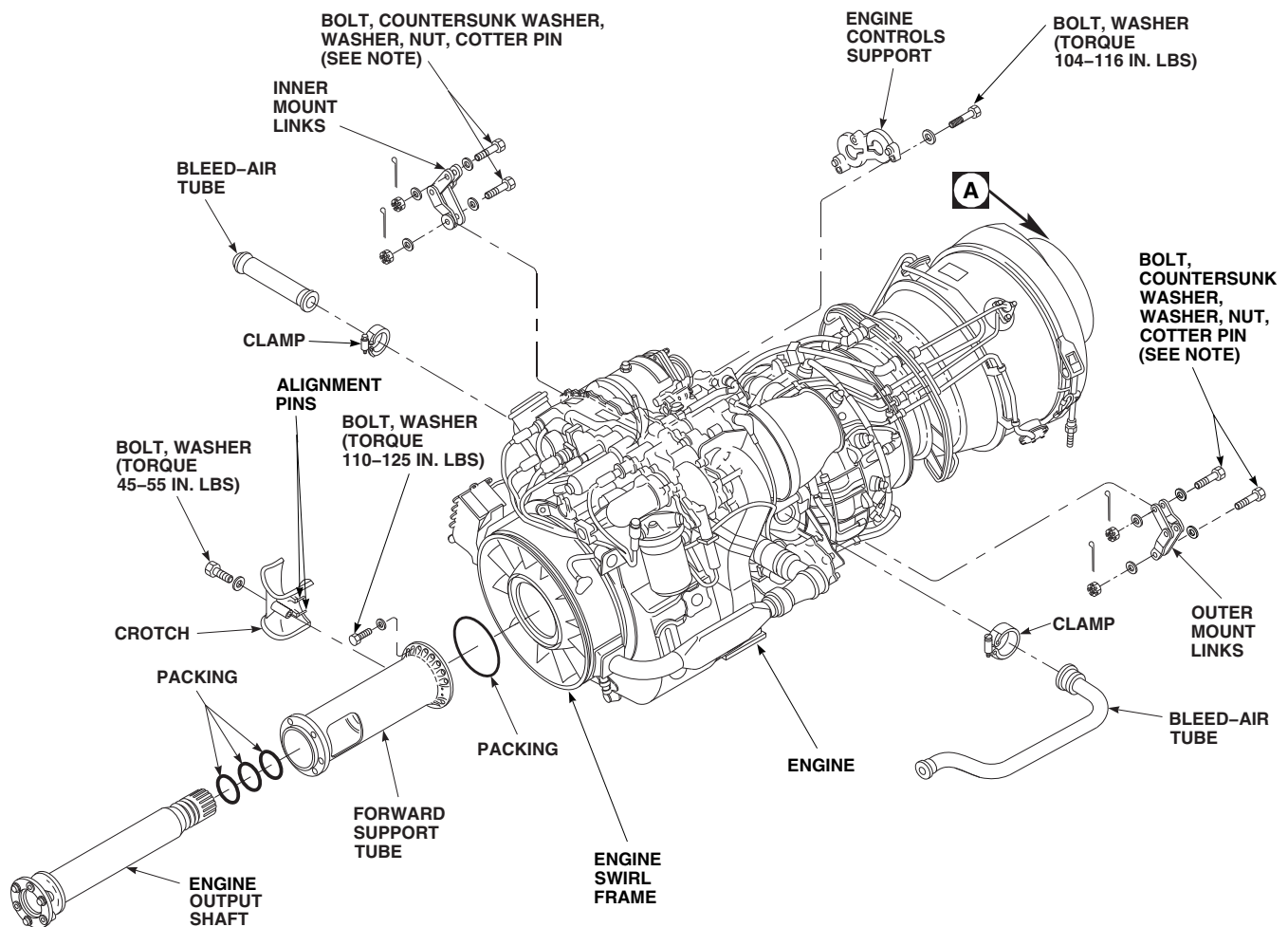
4-4-2.1 Tear Down Power Package.

NOTE

Teardown of No. 1 and No. 2 engine demountable power packages is the same, except as noted.

4-4-2.1.1 Remove Components.

- a. Remove engine starter (PARA 7-4-1).
- b. Remove electrical control unit or digital electronic control and hydromechanical control unit (HMU) (SEI 671 and TM 1-2840-248-23).
- c. If HMU is not to be installed on new engine, remove bolts and washers securing engine controls support to HMU (Figure 4-4-2-1, Sheet 1). Remove support.
- d. Loosen nut(s) on loop clamp, and remove deswirl duct from engine power turbine flange

**NOTE**

TIGHTEN EACH NUT FINGERTIGHT AND ADVANCE NUT TO NEXT CASTELLATION.

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FIGURE 4-4-2-1. REMOVE/INSTALL COMPONENTS (SHEET 1 OF 2)

(Figure 4-4-2-1, Sheet 2, Detail A and Detail B).

- e. **HIRSS** Loosen captive nuts and remove center body cone from engine exhaust.■
- f. **W/O HIRSS** Remove bolts, washers and cover plate from center tailpipe body. Remove nuts, washers, and center tailpipe body from engine exhaust.■

NOTE

Center mount links on flange under engine combustor section were removed when engine was removed.

- g. Remove bolts, countersunk washers, washers, and nuts securing inner and outer mount links to engine rear mounting flange (Figure 4-4-2-1, Sheet 1). Remove mount links.

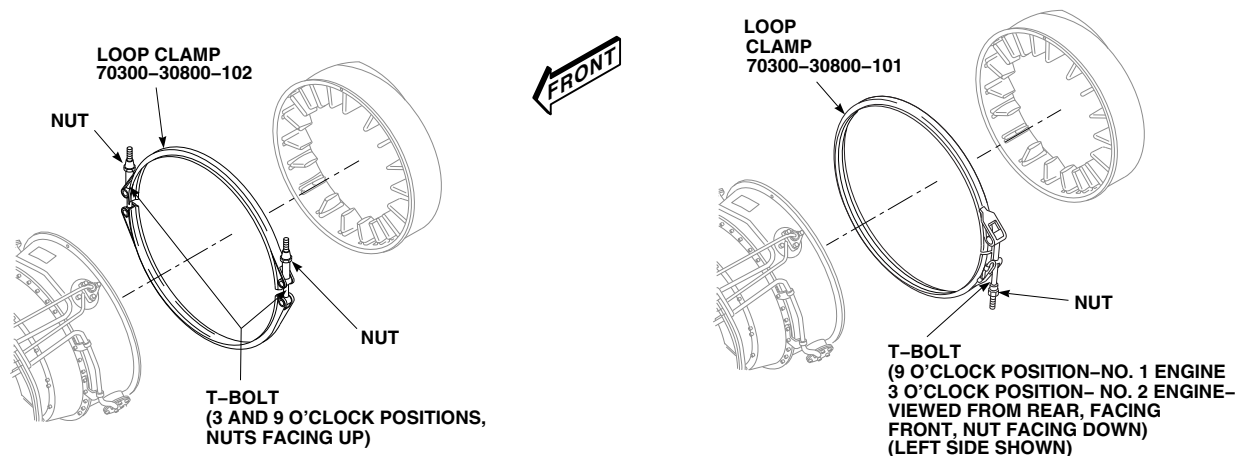
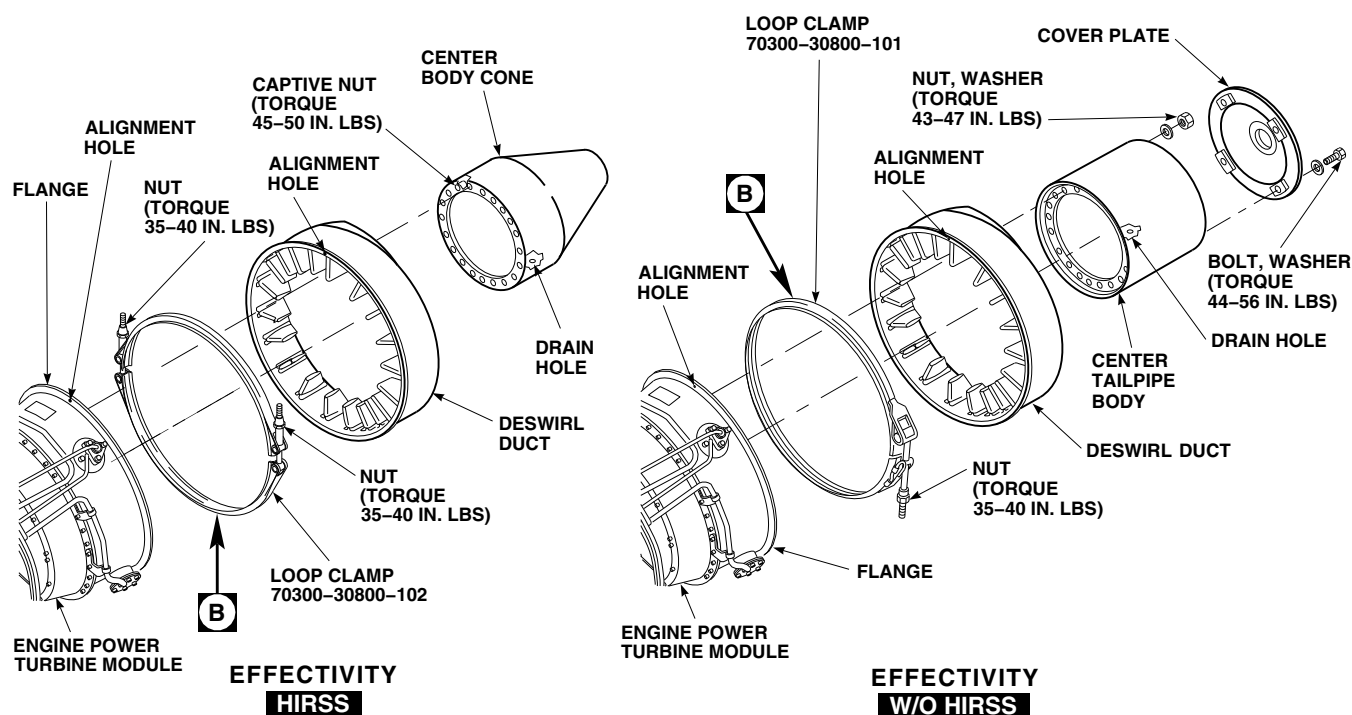
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FIGURE 4-4-2-1. REMOVE/INSTALL COMPONENTS (SHEET 2 OF 2)

Change 8

4-4-2-3

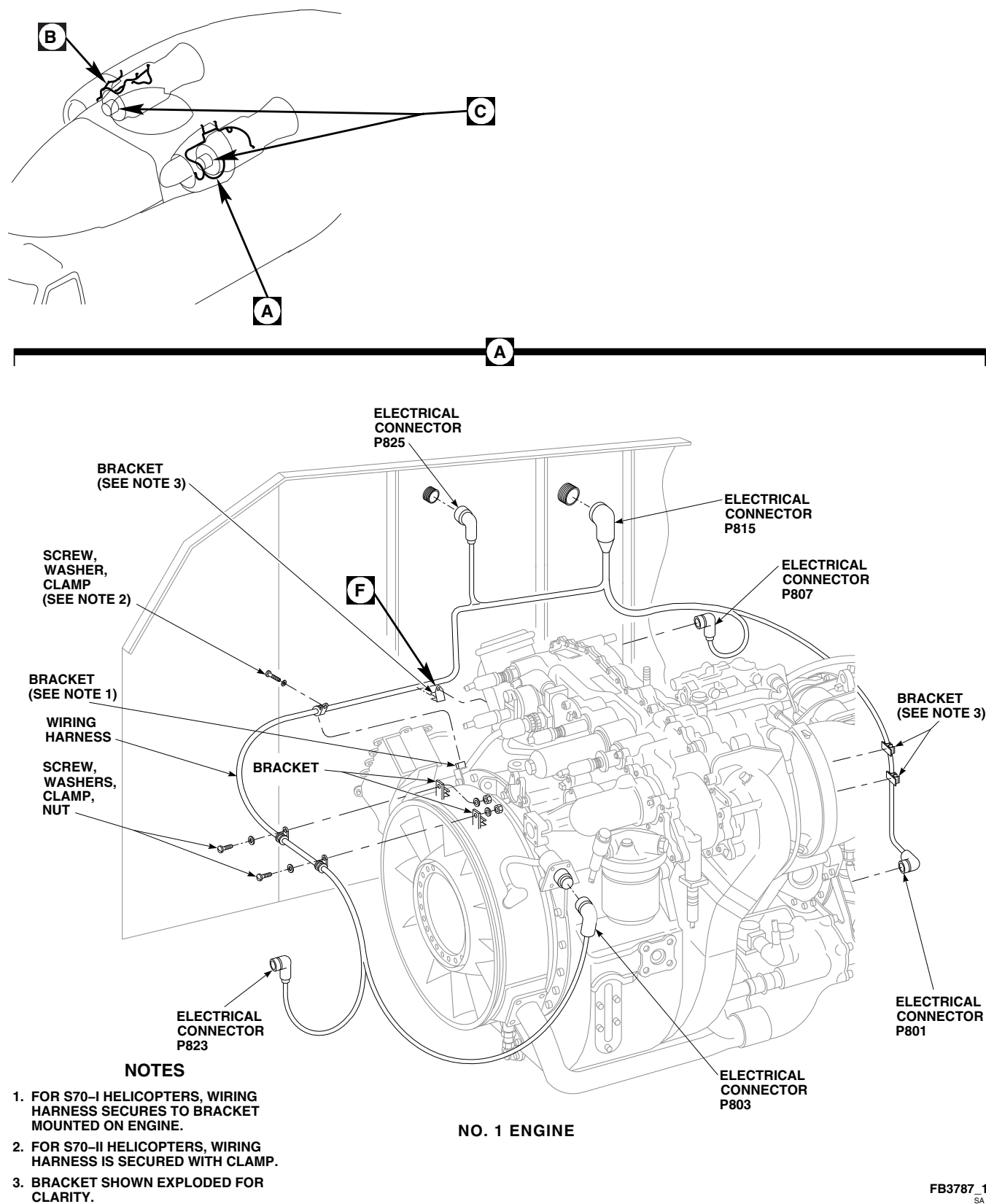


FIGURE 4-4-2.2. REMOVE/INSTALL WIRING HARNESS AND SENSING COMPONENTS (SHEET 1 OF 2)

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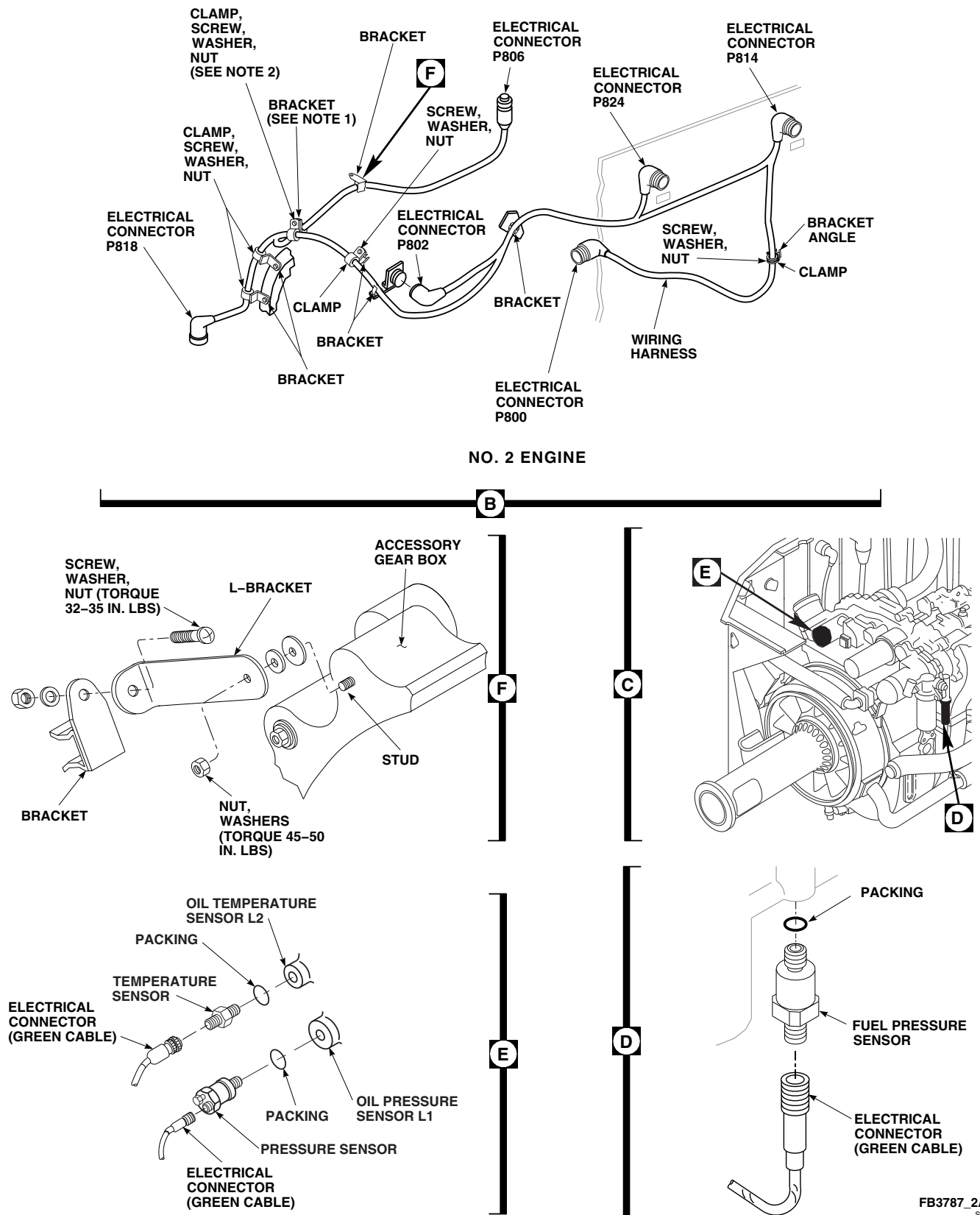


FIGURE 4-4-2-2. REMOVE/INSTALL WIRING HARNESS AND SENSING COMPONENTS (SHEET 2 OF 2)

- h. Remove bolt, washer, and crotch from forward support tube. Slide crotch from forward support tube.
- i. Pull engine output shaft from engine. If output drive shaft fails to release from engine, perform the following:
 - (1) Locally make output drive shaft puller (Figure H-216 or Figure H-262, Appendix H).
 - (2) Using output drive shaft puller, remove shaft from engine.
- j. Remove packings from engine output shaft. Discard packings.
- k. Remove bolts, washers, and forward support tube from engine swirl frame, and remove from engine.
- l. Remove packing from forward support tube. Discard packing.
- m. Remove clamps and bleed-air tubes from engine bleed ports.

4-4-2.1.2 Remove Wiring Harness and Sensing Components.

- a. Disconnect electrical connector, P803 (No. 1 engine) or P802 (No. 2 engine), from engine (Figure 4-4-2-2, Sheet 1, Detail A or Sheet 2, Detail B).
- b. Install protective caps and plugs on electrical connectors.
- c. Remove screws, washers, clamps, brackets, and nuts securing wiring harness to engine.
- d. Remove wiring harness from brackets and remove harness.
- e. Remove screw, washer, and nut securing bracket to L-bracket (Figure 4-4-2-2, Sheet 2, Detail F).

NOTE

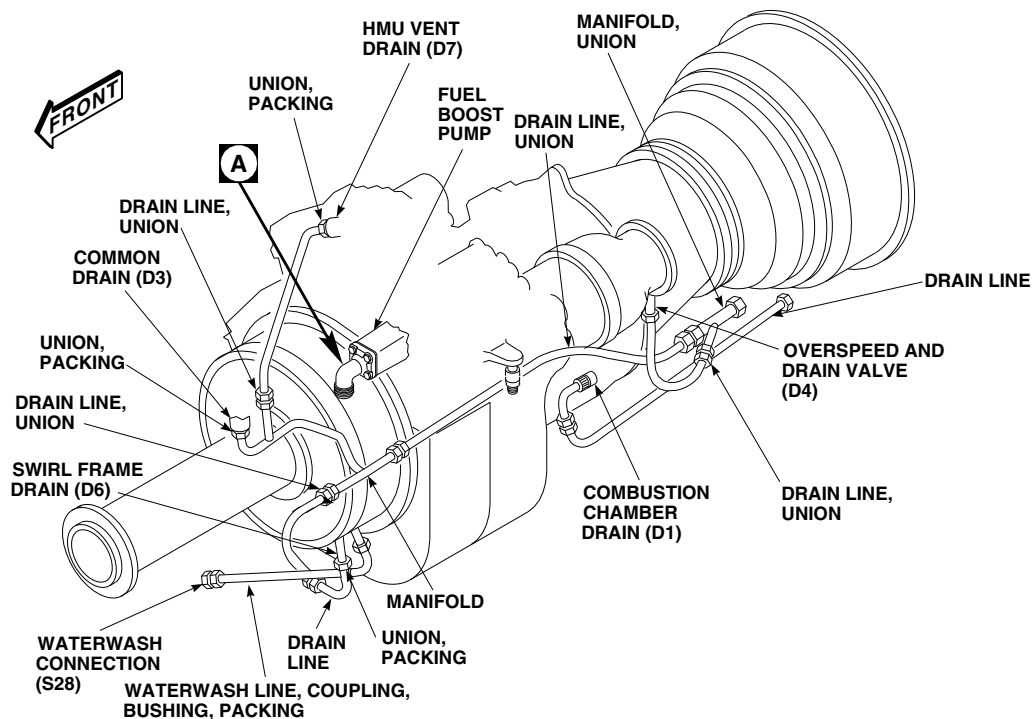
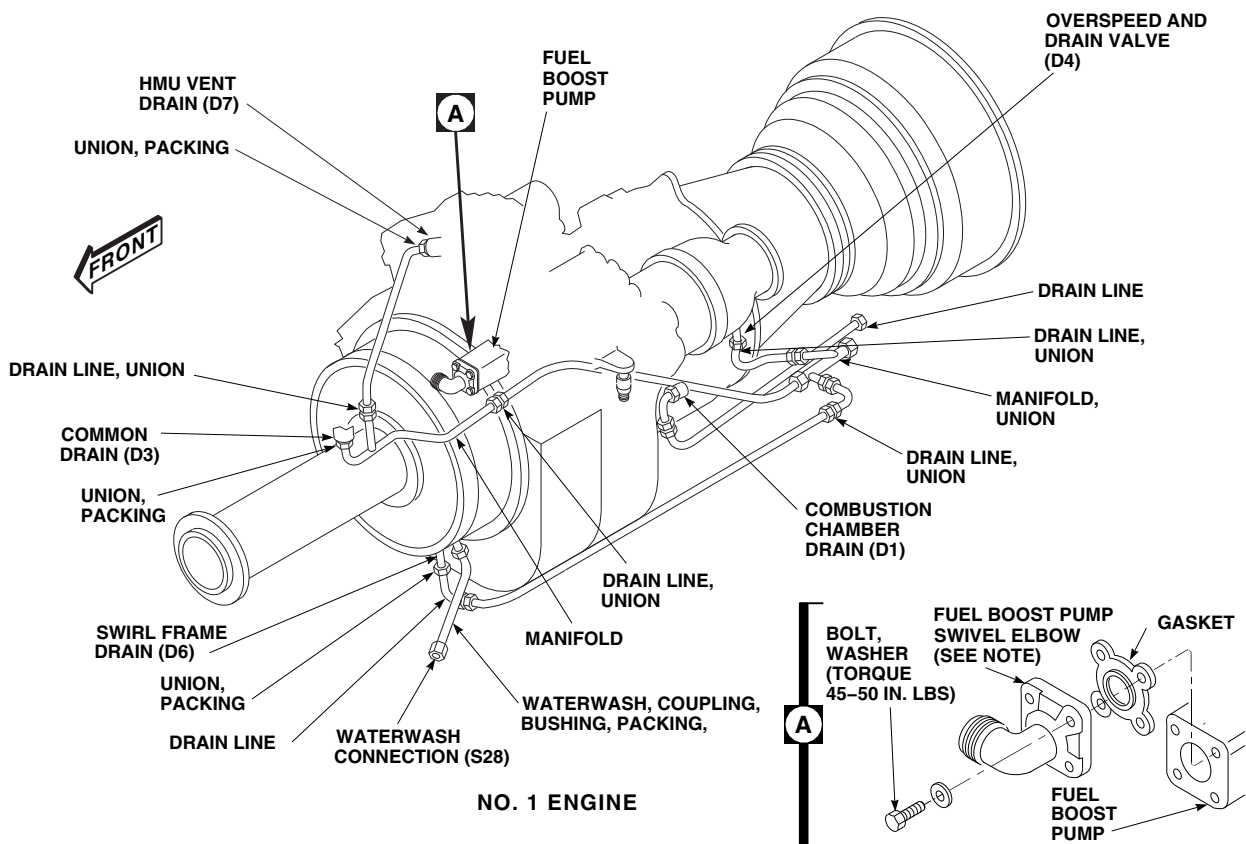
Inside washer is part of accessory gear box and is to remain on stud. Outside

washer will be removed and retained for use on replacement engine.

- f. Remove nut securing L-bracket and washers to stud on accessory gear box. Remove bracket and outside washer. Retain washer for use on replacement engine.
- g. Disconnect green cable electrical connectors from oil temperature, oil pressure, and fuel pressure sensors (Figure 4-4-2-2, Sheet 2, Detail D and Detail E).
- h. Install protective caps and plugs on electrical connectors.
- i. Remove oil temperature, oil pressure, and fuel pressure sensors from engine.

4-4-2.1.3 Remove Fuel and Drain Lines.

- a. Remove bolts, washers, swivel elbow, and gasket from fuel boost pump on engine (Figure 4-4-2-3, Detail A).
- b. Disconnect drain line from combustion chamber drain, D1 (Figure 4-4-2-3).
- c. Disconnect drain line from union at common drain, D3. Remove union and packing from common drain. Remove packing from union.
- d. Disconnect drain line at swirl frame drain, D6.
- e. Disconnect drain line from union at HMU vent drain, D7. Remove union and packing from HMU vent drain.
- f. Disconnect drain line from sequence drain valve/pressurizing and overspeed unit or overspeed and drain valve, D4. Remove manifold and drain lines from engine.
- g. Disconnect waterwash line hose from waterwash port (S28).
- h. Disassemble drain lines and manifolds and remove unions. Remove and discard packings.
- i. Install protective caps or plugs in engine drain lines and open ports.



NOTE

NO. 2 ENGINE

NO. 1 ENGINE SWIVEL ELBOW SHOWN.

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FIGURE 4-4-2-3. REMOVE/INSTALL FUEL AND DRAIN LINES

4-4-2.2 Build Up Power Package.

NOTE

Buildup of No. 1 and No. 2 engine demountable power packages is the same, except as noted.

4-4-2.2.1 Install Fuel and Drain Lines.

CAUTION

Damage to drain manifold will result if manifold chafes against engine main frame, swirl frame, or scroll case. If there is chafing, loosen nuts on lines, reposition lines, and retighten nuts. If chafing still exists, install clamps as required to maintain line separation.

NOTE

Line lengths and installation for No. 1 and No. 2 engines are different. Use correct figure.

- a. Prevent waterwash line from chafing work platform fitting by installing spiral wrap, Item 315, Appendix D, over entire length of hose and routing hose over anti-ice duct on No. 1 engine and over sensor tube on No. 2 engine.
- b. Connect waterwash line to waterwash port (S28) (Figure 4-4-2-3).
- c. Lubricate all packings with petrolatum, Item 234, Appendix D.

CAUTION

Damage to engine will result if antiseize compound enters fuel lines. Do not allow antiseize compound to enter fuel lines.

- d. Coat all male threads with antiseize compound, Item 74, Appendix D.

- e. Install and connect drain lines to engine.
- f. Lubricate gasket with petrolatum, Item 234, Appendix D. Install gasket and swivel elbow on fuel boost pump with bolts and washers. Position elbow for No. 1 or No. 2 engine installation. **TORQUE BOLTS TO 45 - 50 INCH-POUNDS** (Figure 4-4-2-3, Detail A).

4-4-2.2.2 Install Wiring Harness and Sensing Components.

- a. Remove protective caps and plugs from electrical connectors.
- b. Inspect and treat electrical connectors (PARA 1-7-20).
- c. Position wiring harness on engine to allow connection of electrical connector, P803 (No. 1 engine) or P802 (No. 2 engine) (Figure 4-4-2-2, Sheet 1, Detail A or Sheet 2, Detail B).
- d. Connect electrical connector, P803 (No. 1 engine) or P802 (No. 2 engine), to engine.
- e. Install bracket on accessory gear box as follows (Figure 4-4-2-2, Sheet 2, Detail F):
 - (1) Remove nut from stud on accessory gear box. Do not remove existing washer from stud.
 - (2) Install washer, retained from removal, and L-bracket over existing washer on accessory gear box stud. Secure L-bracket and washers with nut. **TORQUE NUT TO 45 - 50 INCH-POUNDS.**
 - (3) Install bracket to L-bracket and secure with screw, washer, and nut. **TORQUE NUT TO 32 - 35 INCH-POUNDS.**
 - (4) Install harness into bracket.
- f. Secure wiring harness to engine using screws, washers, clamps (where required), brackets (where required), and nuts (Figure 4-4-2-2, Sheet 1, Detail A or Sheet 2, Detail B).
- g. Lubricate packing with petrolatum, Item 234, Appendix D. Install packing on oil temperature sensor (Figure 4-4-2-2, Sheet 2, Detail E).

- h. Coat threads on oil temperature sensor with anti-seize compound, Item 74, Appendix D. Install oil temperature sensor in engine.
- i. Lubricate packing with petrolatum, Item 234, Appendix D. Install packing on oil pressure sensor.
- j. Coat threads on oil pressure sensor with anti-seize compound, Item 74, Appendix D. Install oil pressure sensor in engine.
- k. Lubricate packing with petrolatum, Item 234, Appendix D. Install packing on fuel pressure sensor (Figure 4-4-2-2, Sheet 2, Detail D).
- l. Coat threads on fuel pressure sensor with anti-seize compound, Item 74, Appendix D. Install fuel pressure sensor in engine.
- m. Remove protective caps and plugs from electrical connectors.
- n. Inspect and treat electrical connectors (PARA 1-7-20).
- o. Connect green cable electrical connectors to oil temperature, oil pressure, and fuel pressure sensors (Figure 4-4-2-2, Sheet 2, Detail D and Detail E).

4-4-2.2.3 Install Components.

WARNING

Injury to personnel and damage to equipment will result if bleed-air tube clamps interfere with engine variable geometry linkage. Make sure clamps are installed with clamp nuts and bolts straight up and down when connecting bleed-air tubes to engine bleed ports.

- a. Clamp bleed-air tubes to engine bleed ports (Figure 4-4-2-1, Sheet 1). Position clamps with nuts front and up. Do not tighten clamps on bleed-air tubes until installation in helicopter.
- b. Lubricate packings with petrolatum, Item 234, Appendix D. Install packings in grooves on

engine output shaft and on forward support tube.

- c. Carefully slide engine output shaft into front of engine. Place forward support tube with new packing against engine swirl frame so access window is outboard and shaft can be seen. Bolt forward support tube to swirl frame. **TORQUE BOLTS TO 110 - 125 INCH-POUNDS.**

NOTE

Crotch bolt should be aligned in lower hole of support tube for both No. 1 and No. 2 engines.

- d. Install crotch on forward support tube engaging alignment pins. Bolt crotch to forward support tube. Shim between crotch and support tube with washers, NAS1149F0332P, as required to mate crotch to engine swirl frame. **TORQUE BOLT TO 45 - 55 INCH-POUNDS.**

NOTE

- Center mount links on flange under engine combustor section will be installed when engine is installed.
- Upper mount link should have offset leg of fork facing rear and lower link should have offset leg of fork facing front.
- When looking rear, lower link is always in front of upper link on both engines.
- Make sure countersunk washers are installed under bolthead with countersunk side facing bolthead.
- Install washers, as required, under nut to obtain correct cotter pin position.
- e. Secure inner mount links to engine rear mounting flange with bolts, countersunk washers, washers, and nuts. **TIGHTEN EACH NUT FINGERTIGHT AND ADVANCE NUT TO NEXT CASTELLATION.** Install cotter pin.

NOTE

- Upper mount link should have offset leg of fork facing rear and lower link should have offset leg of fork facing front.
 - When looking rear, lower link is always in front of upper link on both engines.
 - Make sure countersunk washers are installed under bolthead with countersunk side facing bolthead.
 - Install washers, as required, under nut to obtain correct cotter pin position.
- f. Secure outer mount links to engine rear mounting flange with bolts, countersunk washers, washers, and nuts. **TIGHTEN EACH NUT FINGERTIGHT AND ADVANCE NUT TO NEXT CASTELLATION.** Install cotter pin.
- g. **HIRSS** Inspect center body cone as follows:■
- (1) Inspect center body cone for cracks. **NONE ALLOWED.** If cracked, replace center body cone.
 - (2) Inspect center body cone for dents. **NO MORE THAN FOUR DENTS OF 1/4-INCH DEPTH CONTAINING NO SHARP CORNERS ARE ALLOWED.** If possible, cold work any dents beyond limits back to normal contours and within limits.
 - (3) Inspect captive nuts for damage. **NONE ALLOWED.** If any captive nuts are damaged, replace center body cone.
- h. **HIRSS** Install center body cone as follows (Figure 4-4-2-1, Sheet 2, Detail A):■

NOTE

When using lubricating oil, Item 205, Appendix D, refer to General Electric SEI 671 for approved oils.

- (1) On engine power turbine module, coat threads of center body cone attachment studs with lubricating oil, Item 205, Appendix D.

- (2) Install center body cone, with drain hole on bottom, on flange of engine power turbine module with captive nuts. **TORQUE CAPTIVE NUTS TO 45 - 50 INCH-POUNDS.**

- i. **W/O HIRSS** Install center tailpipe body and cover plate as follows (Figure 4-4-2-1, Sheet 2, Detail A):■
- (1) Install center tailpipe body, with drain hole on bottom, on engine power turbine module, using washers and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
 - (2) Install cover plate to center tailpipe body with bolts and washers. **TORQUE BOLTS TO 44 - 56 INCH-POUNDS.**
- j. Install deswirl duct to engine power turbine flange as follows (Figure 4-4-2-1, Sheet 2, Detail A and Detail B):

NOTE

- Left and right deswirl ducts are identical.
 - Deswirl ducts have two alignment holes located on deswirl duct flange. When installing, one alignment hole shall be located top center while other alignment hole shall point inboard.
- (1) Make sure alignment holes in deswirl duct flange mate with alignment holes of engine power turbine flange.
 - (2) Visually inspect deswirl duct to make sure deswirl duct flange mates with engine power turbine flange through 360°. If complete mating of flanges is not achieved, perform the following:
 - (a) Remove deswirl duct.
 - (b) Using spring scale to measure load, apply no more than 15 pounds of load to remove warpage.

- (c) If warpage cannot be removed using 15 pounds of load, replace deswirl duct.

NOTE

When using lubricating oil, Item 205, Appendix D, refer to General Electric SEI 671 for approved oils.

- (3) Using lubricating oil, Item 205, Appendix D, lubricate pressure surfaces of loop clamp and T-bolt(s).
- (4) Install loop clamp over lips of deswirl duct and engine power turbine flange.
- (5) Position loop clamp as follows:

NOTE

Loop clamp is not a rigging tool. Deswirl duct must be supported until clamp is torqued.

- (a) For loop clamp, 70300-30800-101 (one T-bolt), position loop clamp with T-bolt at 9 o'clock for No. 1 engine and 3 o'clock for No. 2 engine, with nut facing down.
- (b) For loop clamp, 70300-30800-102 (two T-bolts), position loop clamp with T-bolts at 3 o'clock and 9 o'clock positions, with nuts facing up.

NOTE

If torquing loop clamp, 70300-30800-102, alternate from nut to nut to maintain even loading on clamp.

- (6) **TORQUE LOOP CLAMP NUT(S) TO 35 - 40 INCH-POUNDS** (Figure 4-4-2-1, Sheet 2, Detail A).

- (7) To seat loop clamp, tap lightly with rubber mallet. Retorque loop clamp nut(s). Continue tapping until two successive trials show no loss in torque.

- (8) Loosen loop clamp nut(s) slightly. **RE-TORQUE LOOP CLAMP NUT(S) TO 35 - 40 INCH-POUNDS.**

- (9) Visually inspect latches to make sure no interference has occurred.

- (10) Visually inspect inside deswirl duct to make sure deswirl duct flange mates with engine power turbine module through 360°.

- (11) Apply torque stripe to loop clamp nut(s) (PARA 2-6-5).

- k. Install electrical control unit or digital electronic control and hydromechanical control unit (HMU) (SEI 671 and TM 1-2840-248-23).

- l. If engine controls support is not already installed, install on HMU with bolts and washers (Figure 4-4-2-1, Sheet 1). **TORQUE BOLTS TO 104 - 116 INCH-POUNDS.**

- m. Install engine starter (PARA 7-4-1).

4-4-2.3 Quick Engine Change (QEC).

4-4-2.3.1 Tear Down QEC.

Remove engine starter from old engine (PARA 7-4-1).

4-4-2.3.2 Build Up QEC.

Install engine starter in new engine (PARA 7-4-1).

4-4-3 ENGINE LINE REPLACEABLE UNITS (LRUs).

PARAGRAPH BREAKDOWN

		PAGE
4-4-3.1	Replace Engine Line Replaceable Units (LRUs)	4-4-3-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Powerplant Repairer’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 4-3-3

- PARA 4-4-1
- PARA 4-4-2
- SEI 671
- TM 1-2840-248-23

4-4-3.1 Replace Engine Line Replacement Units (LRUs).

NOTE

The following procedures apply to both No. 1 and No. 2 engine LRUs, except as noted.

4-4-3.1.1 To Begin Removal.

- Turn off all electrical power (PARA 1-6-16).
- Open engine cowling.
- Remove and install engine LRUs (SEI 671 and TM 1-2840-248-23). For other items such as lines, tubes, hoses, IPS, access panels, and engine control units, use additional instructions following each LRUs listed.

4-4-3.1.2 Remove Accessory Section Module LRUs.

Inlet Particle Separator (IPS) Blower. After removing pneumatic tube from starter and firewall (No. 2 engine only) and IPS duct (PARA 4-4-1).

4-4-3.1.3 Remove Fuel System LRUs.

- Fuel Boost Pump. After disconnecting fuel hose from elbow, and removing elbow from pump (PARA 4-4-2).
- Hydromechanical Control Unit (HMU). After removing engine control power-available and load-demand rotary input assemblies and support from HMU.

4-4-3.1.4 Remove Electrical System LRUs.

- History Recorder/History Counter. On No. 1 engine firewall, remove access panel. Remove

history recorder/counter lower attachment bolts.

- b. Thermocouple Assembly. After removing IPS duct (PARA 4-4-1).

4-4-3.1.5 Remove Oil System LRUs.

Oil and Scavenge Pump. After removing left/right access panels in firewall. On No. 1 engine, disconnect main fuel line at elbow on fuel pump. Install protective caps on line and elbow.

4-4-3.1.6 Remove Air System LRUs.

Anti-Icing Bleed and Start Valve. On No. 2 engine, remove inlet particle separator blower and duct.

4-4-3.1.7 Install Accessory Section Module LRUs.

Inlet Particle Separator (IPS) Blower. Install IPS duct (PARA 4-4-1) and pneumatic tube to starter and firewall.

4-4-3.1.8 Install Fuel System LRUs.

- a. Fuel Boost Pump. Install elbow on pump and connect fuel hose to elbow (PARA 4-4-2).
- b. Hydromechanical Control Unit (HMU). Install support and engine control power-available and load-demand rotary input assemblies on HMU.

- c. Perform an engine load-demand push/pull cable friction check (PARA 4-3-3).

4-4-3.1.9 Install Electrical System LRUs.

- a. History Recorder/History Counter. Install lower attachment bolts through opening in No. 1 engine firewall and install access panel on firewall in oil cooler compartment.
- b. Thermocouple Assembly. Install IPS duct (PARA 4-4-1).

4-4-3.1.10 Install Oil System LRUs.

Oil and Scavenge Pump. Install left/right access panels in firewall. On No. 1 engine, uncap and connect main fuel line at elbow on fuel pump.

4-4-3.1.11 Install Air System LRUs.

Anti-Icing Bleed and Start Valve. On No. 2 engine, install IPS blower and duct.

4-4-3.1.12 To Complete Installation.

- a. Close engine cowling.
- b. Perform engine checkout procedure as part of maintenance operation check/maintenance test flight (SEI 671 and TM 1-2840-248-23).

4-4-4 ENGINE AIR INLET.

PARAGRAPH BREAKDOWN

	PAGE
4-4-4.1 Replace Engine Air Inlet	4-4-4-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill
- Drill Bit, 1⁄16-inch
- File
- Plastic Mallet
- Rotary File, 1⁄16-inch
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D

- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 12-4-5
- PARA 12-4-22
- PARA 14-4-9
- PARA 14-4-20

4-4-4.1 Replace Engine Air Inlet.

NOTE

Replacement of No. 1 and No. 2 engine air inlets is the same, except as noted.

4-4-4.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- If required, remove shroud assembly from No. 1 engine air inlet as follows (Figure 4-4-4-1, Sheet 1, Detail A):

- Remove screws and washers securing shroud assembly to engine air inlet and support assemblies.

NOTE

- Tag support assemblies as upper and lower prior to removal from engine air inlet.
- Remove screws and washers securing support assemblies to engine air inlet.

- c. If required, remove upper and lower shrouds from No. 2 engine air inlet by removing screws and washers securing shrouds to engine air inlet (Figure 4-4-4-1, Sheet 1, Detail B).
- d. Open engine cowling.
- e. Disconnect electrical connector, P823 (No. 1 engine) or P818 (No. 2 engine), from air inlet receptacle (Figure 4-4-4-1, Sheet 2, Detail C).
- f. Install protective caps and plugs on electrical connector and receptacle.
- g. Perform the following:
 - (1) **T700** **T701A** Loosen clamps and disconnect flexible hose coupling from engine air inlet and bleed-air tubes.■
 - (2) **T701C** Loosen coupling on engine air inlet tube and bleed-air tubes, and slide compression sleeve and coupling onto bleed-air tubes (Figure 4-4-4-1, Sheet 4, Detail K).■
- h. **T700** **T701A** Inspect flexible hose coupling for cracks, tears, hardening, or deterioration. **NONE ALLOWED.** If damaged, replace flexible hose coupling.■
- i. Remove bolt, washers, clamp, and spacer securing waterwash hose to engine air inlet (Figure 4-4-4-1, Sheet 2, Detail D).
- j. Disconnect fire extinguisher discharge tube from fitting on firewall (Figure 4-4-4-1, Sheet 2, Detail E).
- k. Remove screws and washers from clamps securing fire extinguisher discharge tube to engine air inlet (Figure 4-4-4-1, Sheet 2, Detail F).
- l. On No. 2 engine, if installed, perform the following:
 - (1) Disconnect ice detector electrical connector, P819 from J819, on firewall.
 - (2) Install protective caps and plugs on electrical connectors.
- m. Loosen both nuts on clamp equally. Disconnect coupling securing engine air inlet to engine swirl frame (Figure 4-4-4-1, Sheet 2, Detail C).
- n. On No. 2 engine air inlet, perform the following (Figure 4-4-4-1, Sheet 3, Detail G):
 - (1) Remove rescue hoist cowlings (PARA 14-4-9) and rescue hoist support fairing (PARA 14-4-20).
- o. If necessary, remove bolt, screws, washers, and inlet shroud fairing to allow removal of engine air inlet (Figure 4-4-4-1, Sheet 2, Detail C).
- p. Slide engine air inlet toward front and remove from helicopter.

WARNING

Injury to personnel and damage to equipment will result if rescue hoist is not lowered carefully. Rescue hoist weighs 105 pounds and must be firmly supported before removing outboard bolt and lowering hoist.

- (2) Remove outboard bolt, washers, and nut securing support arm to base support, and slowly allow rescue hoist to pivot downward to a stop.

CAUTION

Clearance between No. 2 engine air inlet and rescue hoist base support/support arm is small. Use care to avoid striking base support/support arm.

4-4-4.1.2 Disassemble.

- a. On No. 1 engine air inlet, remove screws, washers, and plate from anti-icing valve cover (Figure 4-4-4-2).
- b. Remove screws, washers, and anti-icing valve cover from engine air inlet.

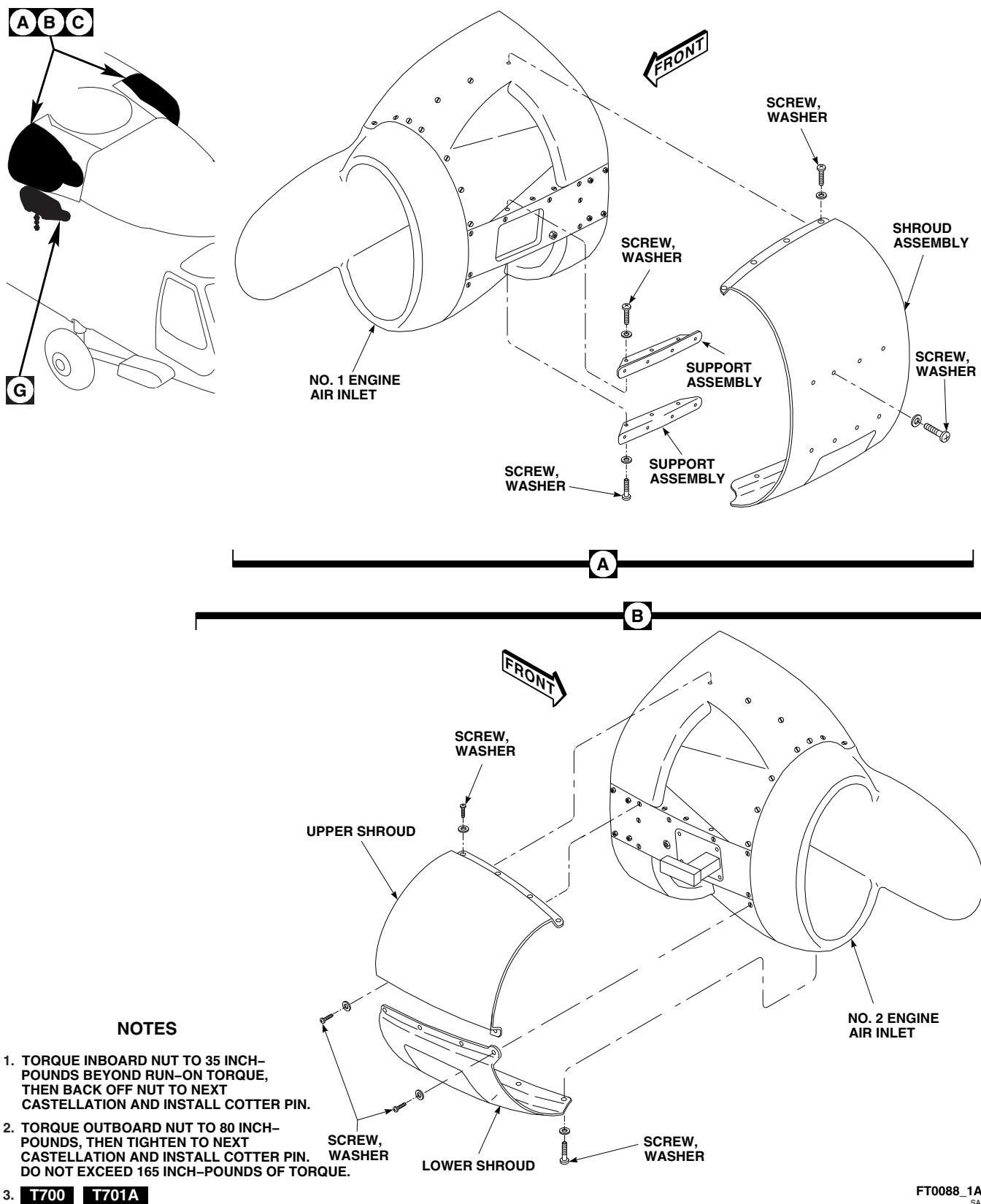


FIGURE 4-4-4-1. REPLACE ENGINE AIR INLET (SHEET 1 OF 4)

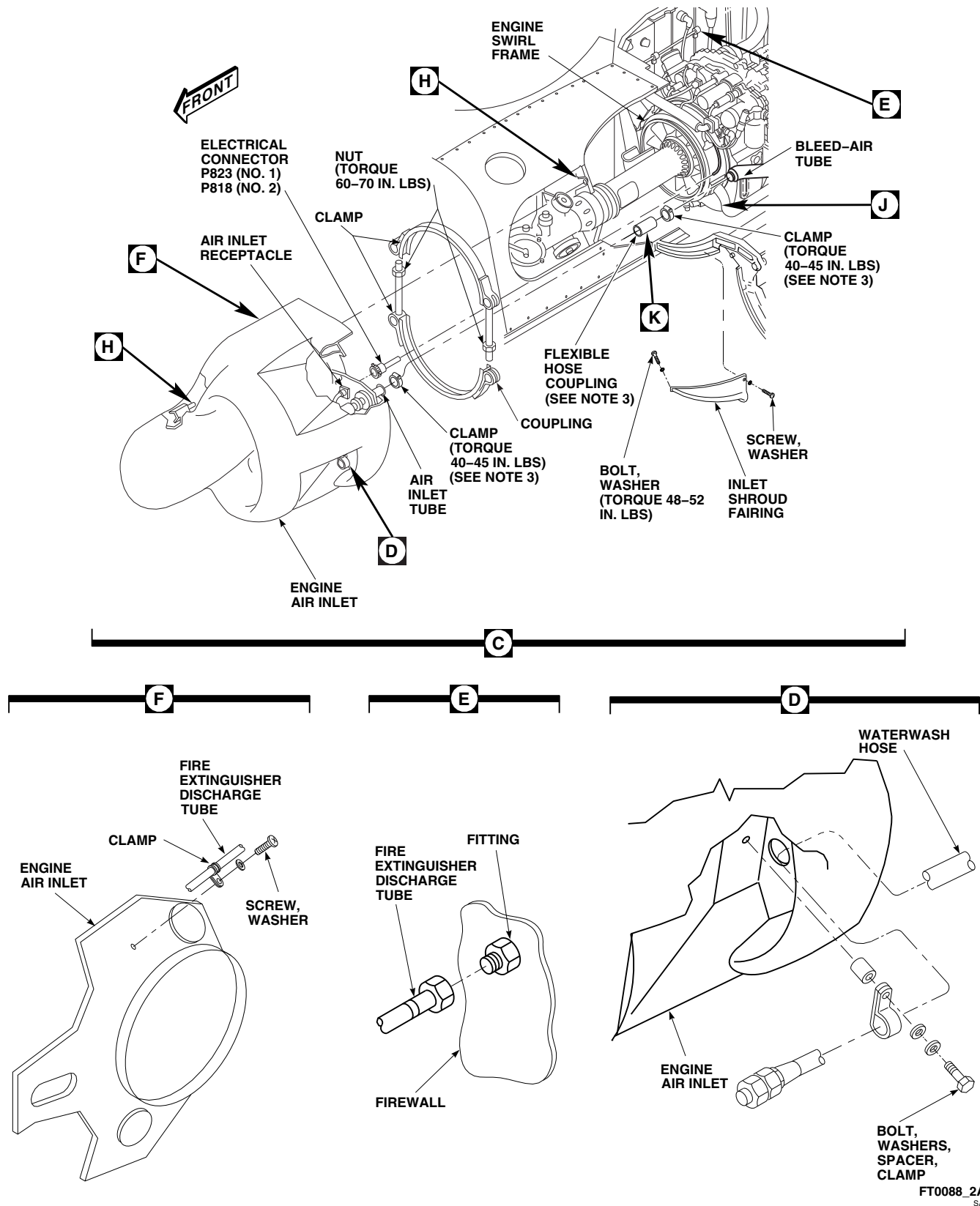


FIGURE 4-4-4-1. REPLACE ENGINE AIR INLET (SHEET 2 OF 4)

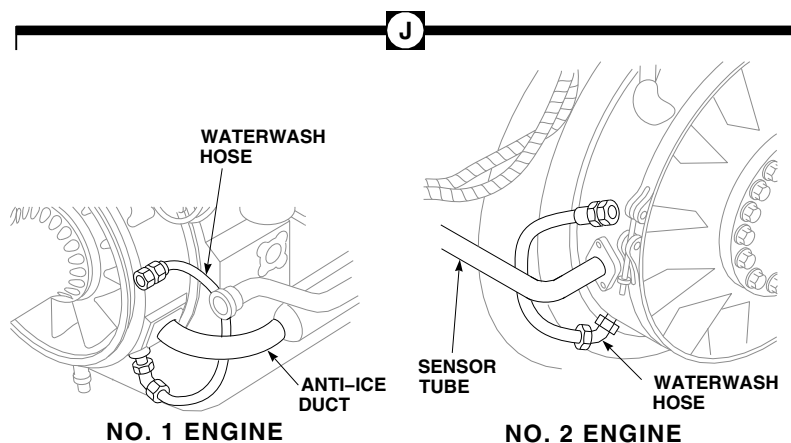
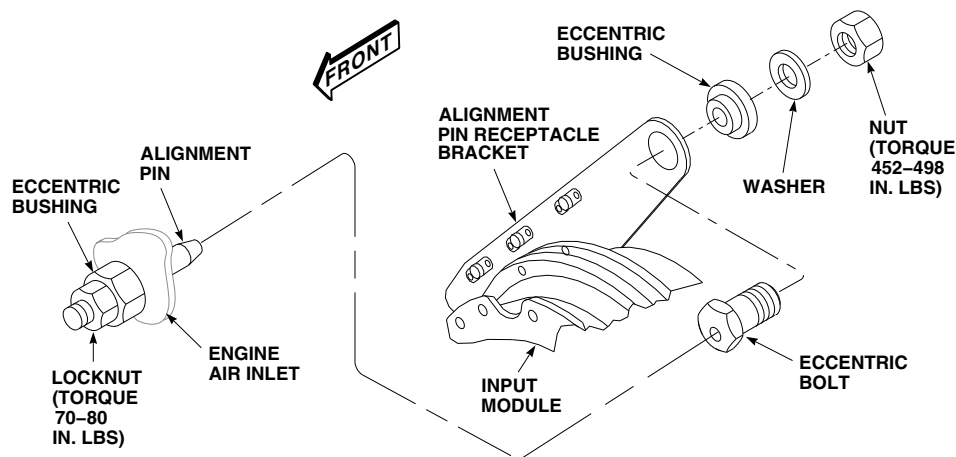
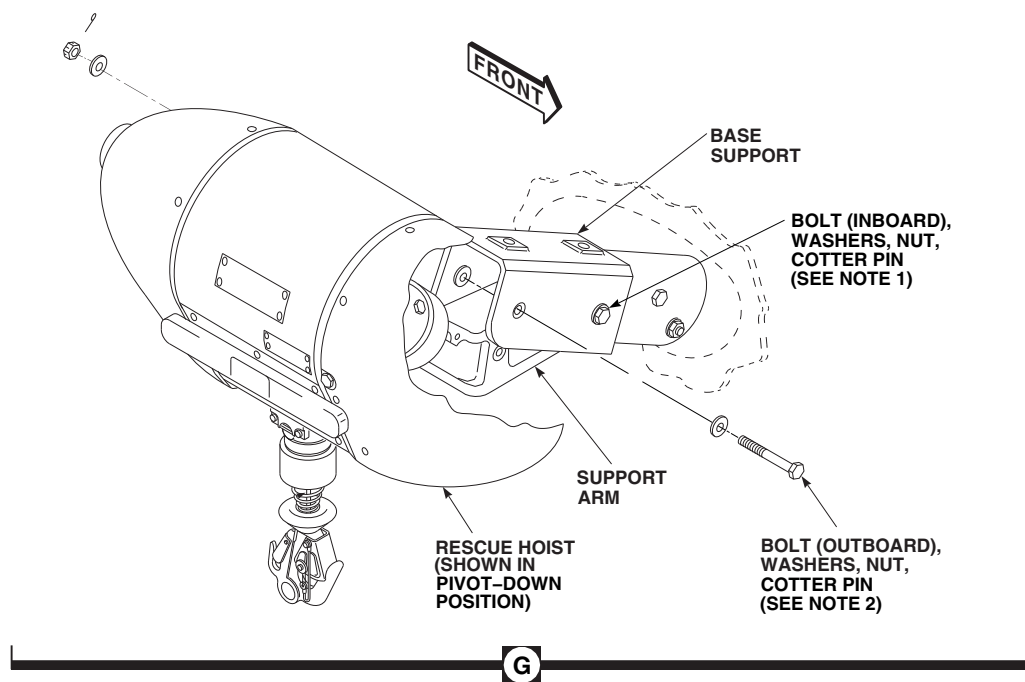


FIGURE 4-4-4-1. REPLACE ENGINE AIR INLET (SHEET 3 OF 4)

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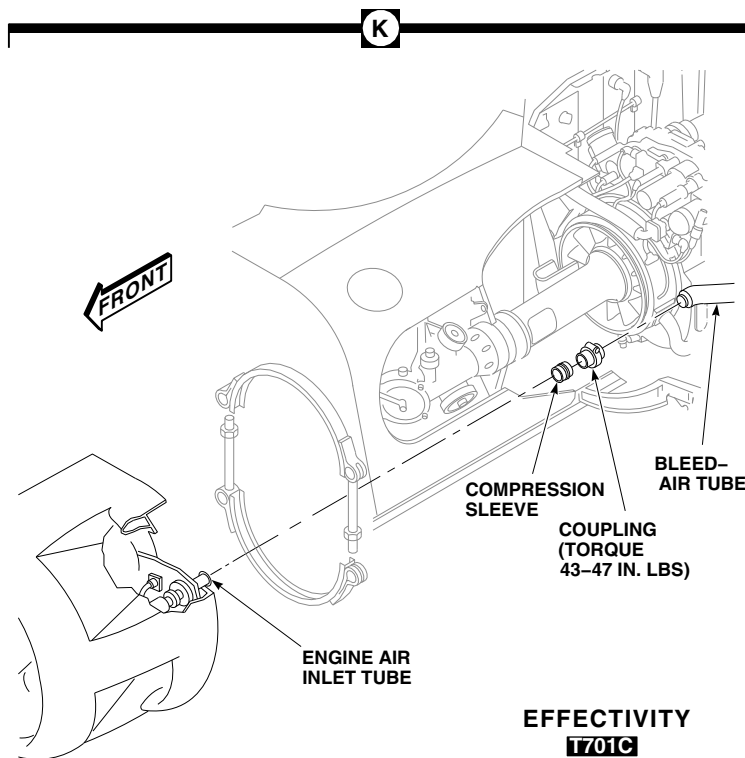


FIGURE 4-4-4-1. REPLACE ENGINE AIR INLET (SHEET 4 OF 4)

- c. Inspect grommet in anti-icing valve cover for damage or looseness. **NONE ALLOWED.** If damaged or loose, remove and discard grommet.
- d. If engine air inlet is to be replaced, remove engine air inlet anti-icing valve (PARA 12-4-22). Retain valve and attaching hardware for installation.

4-4-4.1.3 Inspect/Repair.

- a. Inspect bleed-air exhaust slots for cracks or sharp corners as follows (Figure 4-4-4-3, Detail A):
 - (1) Remove paint from ends of slots (PARA 2-6-5).

CAUTION

Damage to inner fiberglass wall will result if file or drill is inserted too deep. To prevent damage to inner fiberglass wall, do not insert file or drill more than 1/8-inch into slot.

- (2) Using rotary file or drill with 1/16-inch drill bit, round off ends of exhaust slots. Remove only enough material to eliminate sharp corners.
- (3) A group of no more than three adjacent slots may have one crack between a slot and the next adjacent slot. Multiple groups of three slots or less may have cracks, provided there are four slots between each group of slots having cracks. **CRACKS BEYOND LIMITS ARE NOT ALLOWED.** If limits are exceeded, replace engine air inlet.
- (4) **MORE THAN ONE CRACK FROM SLOT TO ADJACENT SLOT IS NOT ALLOWED.** If limit is exceeded, replace engine air inlet.
- (5) **CRACKS ORIGINATING FROM SEVERAL SLOTS CREATING BOXED AREA ARE NOT ALLOWED.** If limit is exceeded, replace engine air inlet.
- (6) Using drill with 1/16-inch drill bit, stop-drill any crack(s) within limits.

- (7) Touch up reworked area (PARA 2-6-5).
- b. Inspect engine air inlet for chafing next to alignment pin receptacle bracket. If chafing is present, perform the following:
 - (1) File corner of bracket. **MINIMUM THICKNESS BETWEEN EDGE OF BRACKET AND ALIGNMENT HOLE IS 0.20-INCH** (Figure 4-4-4-3, Detail B).
 - (2) Touch up reworked area (PARA 2-6-5).

4-4-4.1.4 Assemble.

- a. If new engine air inlet is being installed, install engine air inlet anti-icing valve (PARA 12-4-22).
- b. If grommet was discarded, install new grommet on anti-icing valve cover (Figure 4-4-4-2).
- c. Secure anti-icing valve cover on engine air inlet with screws and washers.
- d. On No. 1 engine air inlet, secure plate on anti-icing valve cover with screws and washers.

4-4-4.1.5 Install.

- a. Turn off all electrical power (PARA 1-6-16).



Clearance between No. 2 engine air inlet and rescue hoist base support/support arm is small. Use care to avoid striking base support/support arm.

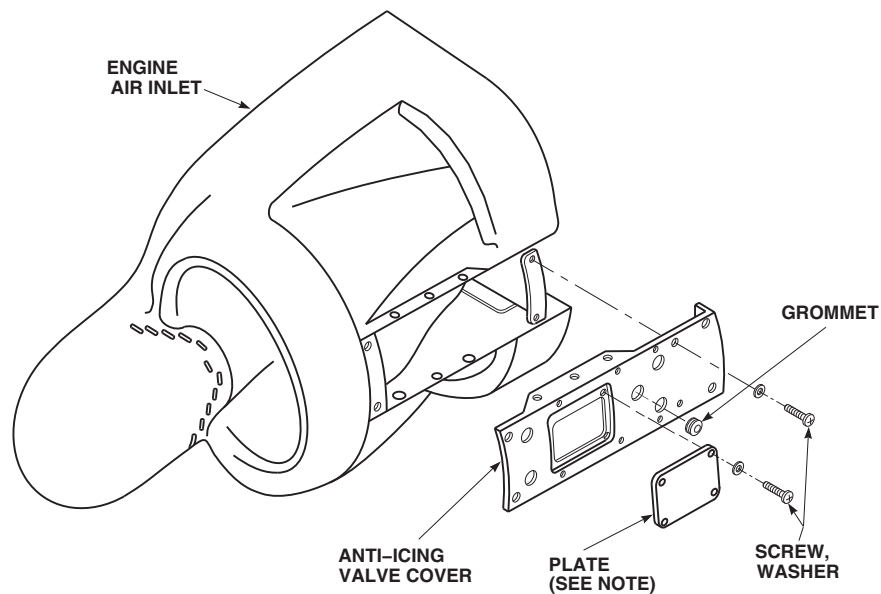
- b. Position engine air inlet on helicopter, making sure alignment pin is inserted in alignment pin receptacle bracket on input module (Figure 4-4-4-1, Sheet 3, Detail H).
- c. If required, adjust alignment pin on engine air inlet and receptacle bracket on input module as follows:
 - (1) On input module, loosen nut and center alignment pin receptacle bracket by turning eccentric bolt and eccentric bushing.

- (2) On engine air inlet, loosen locknut securing alignment pin.
- (3) Slide engine air inlet toward rear until it contacts front face of engine swirl frame (Figure 4-4-4-1, Sheet 2, Detail C).
- (4) On engine air inlet, adjust eccentric bushing on alignment pin until circumference of engine air inlet matches engine swirl frame (Figure 4-4-4-1, Sheet 3, Detail H). On input module, **TORQUE NUT SECURING ECCENTRIC BOLT TO 452 - 498 INCH-POUNDS.**
- (5) On engine air inlet, **TORQUE LOCKNUT SECURING ALIGNMENT PIN TO 70 - 80 INCH-POUNDS.**
- (6) If additional adjustment is required, remove engine air inlet and realign receptacle.

NOTE

Make sure flange on engine air inlet and swirl frame on engine are clean.

- d. Apply light coat of petrolatum, Item 234, Appendix D, to mating surfaces of engine air inlet and engine swirl frame.
- e. Slide engine air inlet back against engine swirl frame (Figure 4-4-4-1, Sheet 2, Detail C).
- f. Install coupling so inboard nut of clamp is facing up (outboard nut faces down).
- g. Tighten each nut a little at a time. While tightening, lightly tap all around clamp with plastic mallet to correctly seat clamp. **ONCE CLAMP IS CORRECTLY SEATED, TORQUE NUTS TO 60 - 70 INCH-POUNDS.** Continue tapping clamp until two successive trials show no loss of torque.
- h. If inlet shroud fairing was removed, perform the following:
 - (1) Apply corrosion preventive compound, Item 115, Appendix D, to fairing hardware.

**NOTE**

NO. 1 ENGINE AIR INLET ONLY.

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- (2) Secure inlet shroud fairing using screws, washers, and bolt. **TORQUE BOLT TO 48 - 52 INCH-POUNDS.**

- i. On No. 2 engine air inlet, perform the following (Figure 4-4-4-1, Sheet 3, Detail G):

WARNING

Injury to personnel and damage to equipment will result if rescue hoist is not lowered carefully. Rescue hoist weighs 105 pounds and must be firmly supported before installing outboard bolt.

- (1) Slowly pivot rescue hoist up and secure support arm to support base with outboard bolt, washers, and nut.
- (2) **TORQUE INBOARD NUT TO 35 INCH-POUNDS BEYOND RUN-ON TORQUE, THEN BACK OFF NUT TO NEXT CASTELLATION.** Install cotter pin.

CAUTION

Do not exceed 165 inch-pounds torque.

- (3) **TORQUE OUTBOARD NUT TO 80 INCH-POUNDS. THEN TIGHTEN NUT TO NEXT CASTELLATION.** Install cotter pin.
- (4) Install rescue hoist cowlings (PARA 14-4-9) and rescue hoist support fairing (PARA 14-4-20).
- j. Connect fire extinguisher discharge tube to fitting on firewall (Figure 4-4-4-1, Sheet 2, Detail E).
- k. Properly position and align fire extinguisher discharge tube (PARA 12-4-5).
- l. Install clamp, screw, and washer securing fire extinguisher discharge tube to engine air inlet (Figure 4-4-4-1, Sheet 2, Detail F).
- m. On No. 2 engine, perform the following:

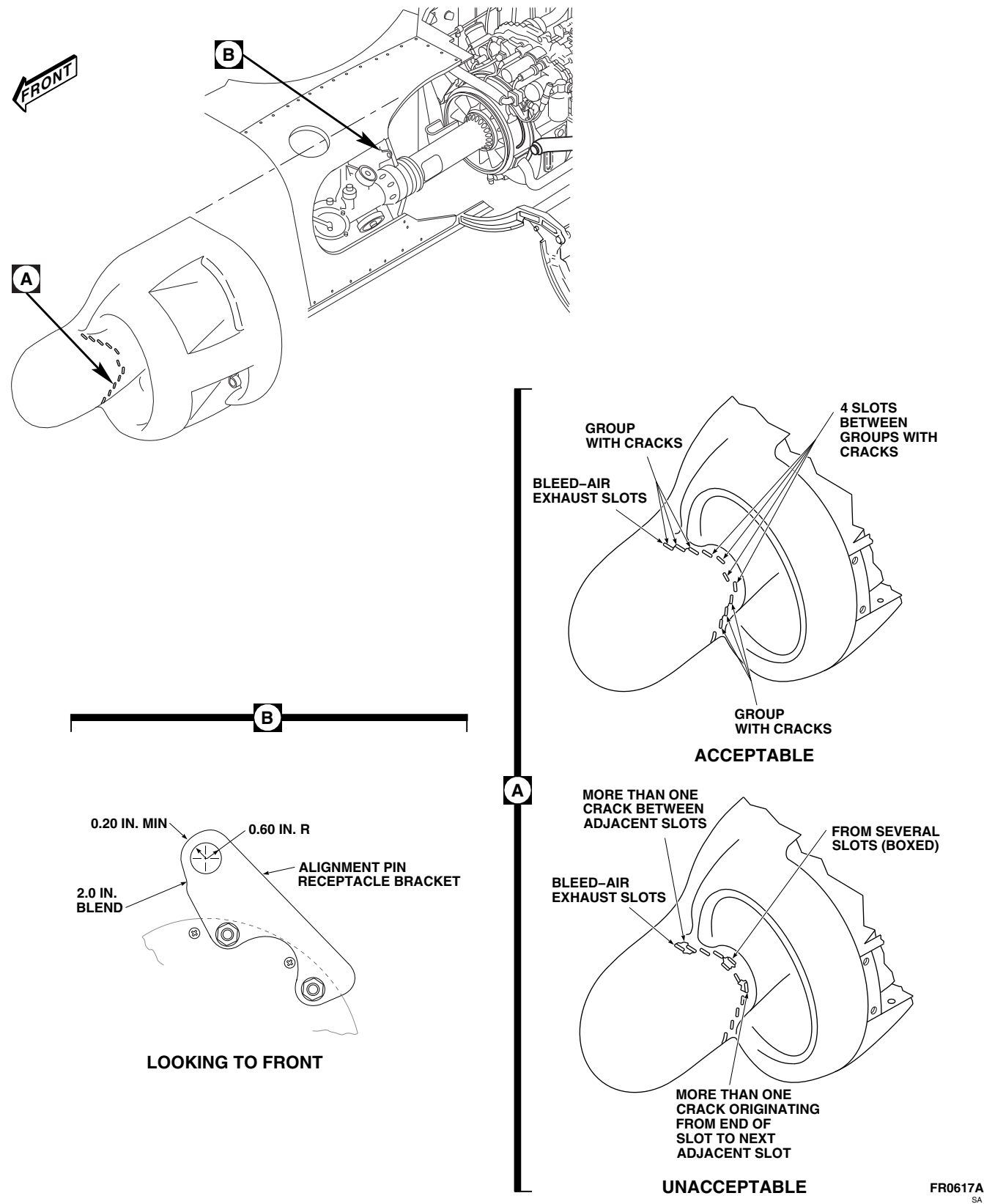


FIGURE 4-4-4-3. INSPECT/REPAIR ENGINE AIR INLET

- (1) Remove protective caps and plugs from electrical connectors.
- (2) Inspect and treat electrical connectors (PARA 1-7-20).
- (3) If installed, connect ice detector electrical connector, P819 to J819, on firewall.

CAUTION

Damage to waterwash hose will result if allowed to chafe against work platform. Position hose to avoid contact with work platform.

- n. Route waterwash hose over anti-ice duct (No. 1 engine) or over sensor tube (No. 2 engine) (Figure 4-4-4-1, Sheet 3, Detail J).
- o. Connect waterwash hose to engine air inlet using bolt, washers, clamp, and spacer (Figure 4-4-4-1, Sheet 2, Detail D).
- p. Remove protective caps and plugs from electrical connector and receptacle.
- q. Inspect and treat electrical connector and receptacle (PARA 1-7-20).
- r. Connect electrical connector, P823 (No. 1 engine) or P818 (No. 2 engine), to engine air inlet receptacle (Figure 4-4-4-1, Sheet 2, Detail C).
- s. Perform the following:
 - (1) **T700** **T701A** Connect flexible hose coupling to engine air inlet tube and bleed-air tubes with clamps. **TORQUE CLAMPS TO 40 - 45 INCH-POUNDS.**

NOTE

Minimum gap between pneumatic tubes coupling connection is 0.120-inch. If necessary, metal may be removed from tube end using file to achieve gap. Do not remove any metal from bead on tubes.

- (2) **T701C** Install compression sleeve and coupling over engine air inlet tube and bleed-air tubes (Figure 4-4-4-1, Sheet 4, Detail K). **TORQUE NUT TO 43 - 47 INCH-POUNDS.**
- t. If new engine was installed, **TORQUE CLAMP SECURING BLEED-AIR TUBE TO ENGINE TO 35 - 45 INCH-POUNDS.**
- u. Make sure area is clean and free of foreign material.
- v. Close and latch engine cowl.
- w. If removed, secure shroud assembly to No. 1 engine air inlet as follows (Figure 4-4-4-1, Sheet 1, Detail A):

NOTE

Make sure support assemblies are secured in the proper orientation (upper and lower) on engine air inlet.

- (1) Secure support assemblies to engine air inlet with screws and washers. Remove tags.
- (2) Secure shroud assembly to engine air inlet and support assemblies with screws and washers.
- x. If removed, secure upper and lower shrouds to No. 2 engine air inlet with screws and washers (Figure 4-4-4-1, Sheet 1, Detail B).

4-4-5 AFT ENGINE MOUNTS.

PARAGRAPH BREAKDOWN

		PAGE
4-4-5.1	Replace Aft Engine Mounts	4-4-5-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Dial Indicator
- Fluorescent Penetrant Inspection Kit
- Hoist, Capacity to Support Engine
- Lifting Sling
- Maintenance Crane
- Nonmetallic Scraper, Locally-Made
- Padding
- Protection, Skin and Breathing
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Vise

Materials/Parts

- Abrasive Cloth, Item 5, Appendix D
- Acetone, Item 25, Appendix D
- Adhesive, Item 55, Appendix D

- Corrosion Compound, Item 108, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Scouring Pad, Item 270, Appendix D
- Sealing Compound, Item 287, Appendix D
- Washer, NAS1149C0532R
- Washer, NAS1149C0563R
- Washer, NAS1149C0632R
- Washer, NAS1149C0663R
- Washer, NAS1149C0863R

References

- Appendix D
- ASTM E1417
- PARA 2-4-69
- PARA 4-4-1
- PARA 4-5-1
- PARA 4-5-2
- PARA 4-5-3
- PARA 18-4-1

4-4-5.1 Replace Aft Engine Mounts.**NOTE**

Replacement of No. 1 and No. 2 aft engine mounts is the same.

4-4-5.1.1 Remove.

- a. Open engine cowling.
- b. Loosen coupling and disconnect inlet particle separator (IPS) blower duct from IPS blower (PARA 4-4-1).
- c. If cabin maintenance crane will be used, erect crane (PARA 18-4-1).

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when hoisting engine. Keep area clear when hoisting engine.

- d. Install lifting sling on engine and connect maintenance crane or hoist to engine (PARA 4-4-1). Raise sling until weight of engine is supported by sling.

CAUTION

Do not attempt to remove or otherwise disturb engine mount-to-airframe shims.

- e. Using nonmetallic scraper, remove sealing compound from edges of fittings installed on engine compartment deck.

WARNING

Bushings installed in aft engine mount struts are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking da-

maged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.

- f. Remove bolts, countersunk washer, washer, and nuts securing outer mount strut to outer mount fitting and outer mount links to diffuser case. Remove outer mount strut and links from helicopter (Figure 4-4-5-1, Sheet 1, Detail A).
- g. Remove bolt, countersunk washer, washer, and nut securing outer mount links to outer mount strut. Remove links from strut.
- h. Remove bolts, countersunk washer, washer, and nuts securing inner mount struts to inner mount fittings and inner mount links to diffuser case. Remove inner mount strut and links from helicopter.
- i. Remove bolt, countersunk washer, washer, and nuts securing inner mount links to inner mount struts and bolt, countersunk washer, washer, and nut securing inner struts together. Remove links from struts.
- j. Remove bolt, countersunk washer, washer, and nut securing center mount links to support fitting.
- k. Remove bolts, countersunk washer, washer, and nuts securing center mount links to diffuser case. Remove links.
- l. Remove soundproofing panels as required (PARA 2-4-69).
- m. Remove bolts, countersunk washers, washers, and nuts securing inner mount fittings and support fitting to engine compartment deck. Remove fittings (Figure 4-4-5-1, Sheet 2, Detail B).
- n. Remove bolts, countersunk washers, washers, and nuts securing outer mount fitting and radius block to engine compartment deck and cabin door upper track (Figure 4-4-5-1, Sheet 2, Detail B and Detail C). Remove fitting.

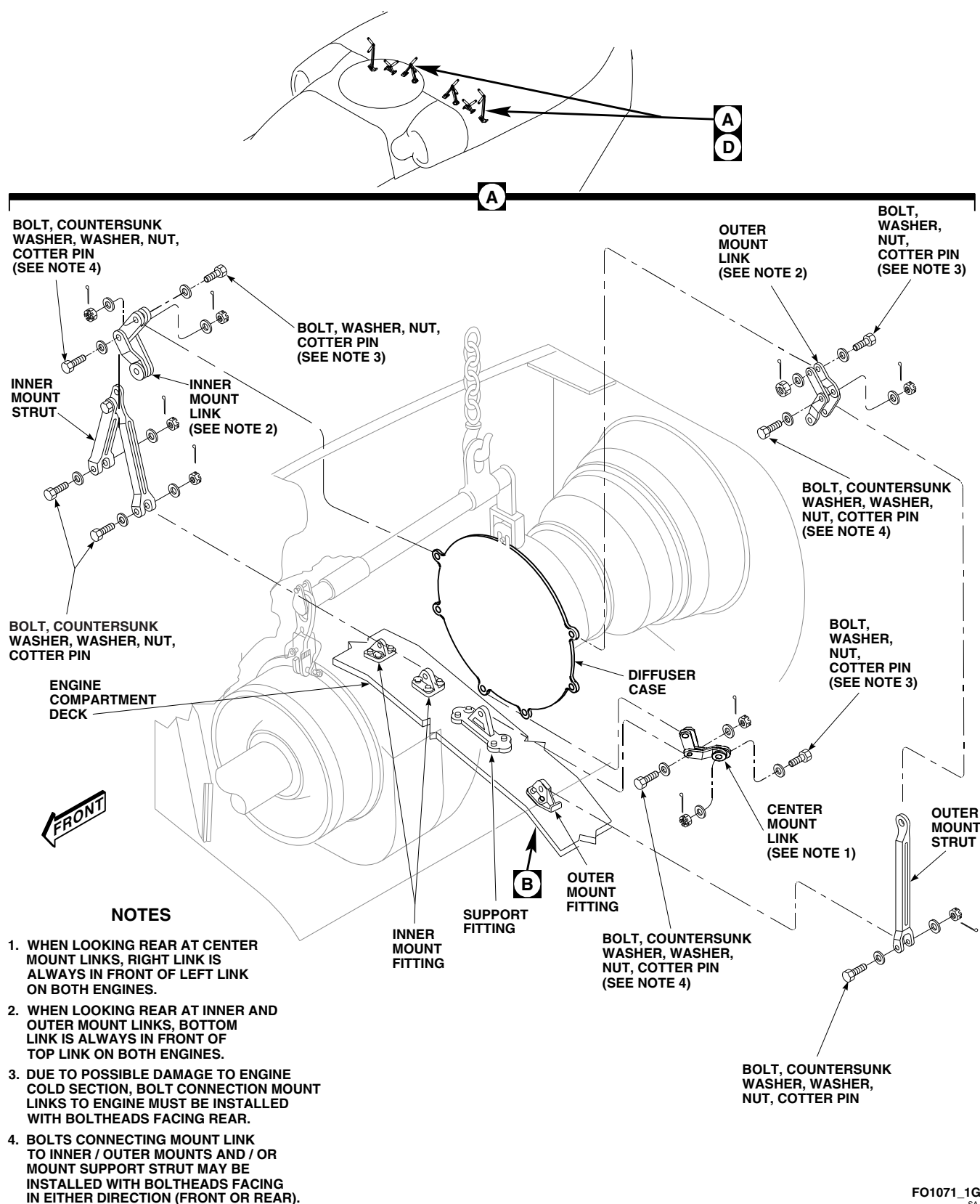


FIGURE 4-4-5-1. REPLACE AFT ENGINE MOUNTS (SHEET 1 OF 3)

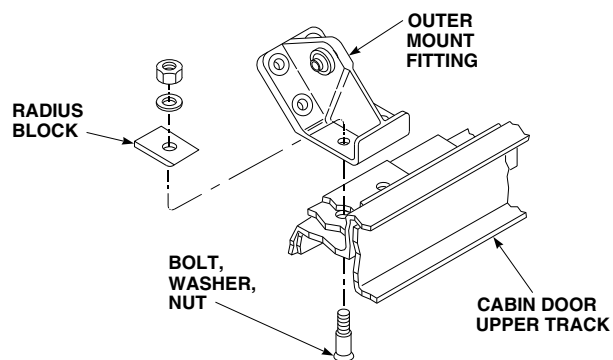
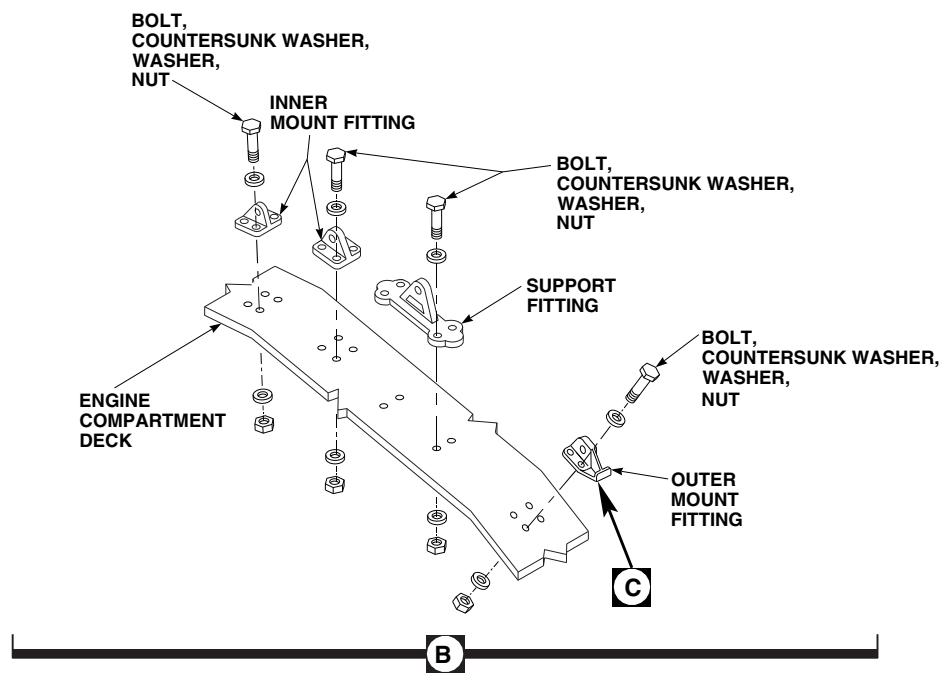
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FIGURE 4-4-5-1. REPLACE AFT ENGINE MOUNTS (SHEET 2 OF 3)

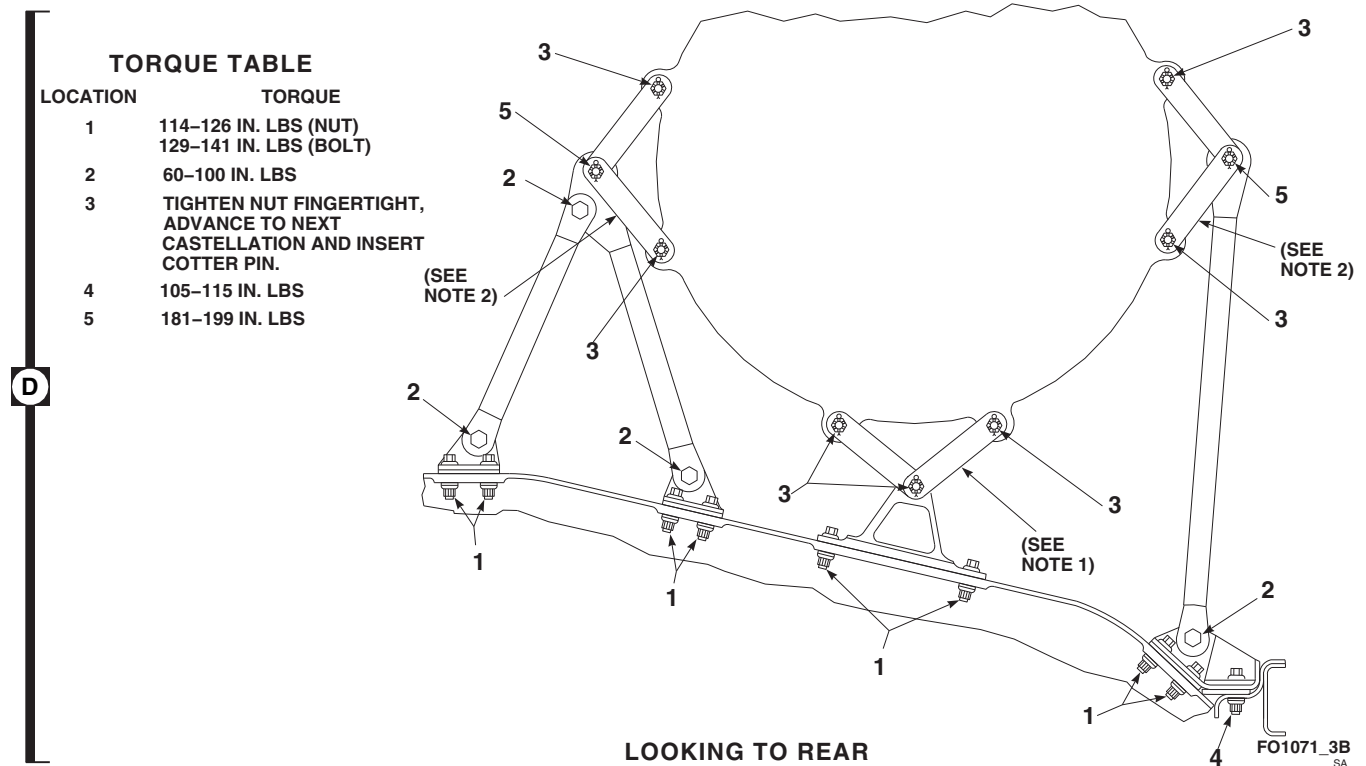


FIGURE 4-4-5-1. REPLACE AFT ENGINE MOUNTS (SHEET 3 OF 3)

4-4-5.1.2 Inspect/Repair.

WARNING

Bushings installed in aft engine mount struts are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.

a. Check bearing radial play as follows:

- (1) Install bolt and nut in bearing.
- (2) Clamp bolt in vise.
- (3) Clamp dial indicator to vise, place tip of indicator on fitting or strut.

- (4) **RADIAL PLAY MUST NOT EXCEED 0.005-INCH.** If limit is exceeded, replace bearing (PARA 4-5-2).

b. Check bearing axial play as follows:

- (1) Install bolt and nut in bearing.

CAUTION

Damage to strut or fitting will result if clamped in vise unprotected. Use padding to protect fitting or strut when clamping in vise.

- (2) Clamp fitting or strut in vise.
- (3) Clamp dial indicator on vise, place tip of indicator on bolt.
- (4) **AXIAL PLAY MUST NOT EXCEED 0.020-INCH.** If limit is exceeded, replace bearing (PARA 4-5-2).

- c. Check unbushed hole in inner mount strut for wear. **IF WEAR IS FOUND OR SUSPECTED, PERFORM DETAILED INSPECTION** (PARA 4-5-1).
- d. Inspect mount struts for corrosion, cracks, nicks, gouges, and scratches. If damage is found, **DETERMINE IF DAMAGE IS IN NICKEL-PLATING OR PARENT MATERIAL** as follows:

NOTE

NICKS, GOUGES, SCRATCHES, OR PITTING MUST NOT EXCEED 0.250-INCH LONG OR 0.032-INCH DEEP. If limits are exceeded, replace strut.

- (1) Using abrasive cloth, Item 5, Appendix D, remove nickel-plating in immediate suspect area. **DO NOT REMOVE PARENT MATERIAL.**
- (2) Determine if damage is in parent material. Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect suspect area for cracks. **NO CRACKS ALLOWED IN PARENT MATERIAL.** If cracked, replace strut.
- (3) Perform detailed inspection of remainder of strut (PARA 4-5-1).
- (4) Clean surfaces where plating was removed and/or is missing with corrosion compound, Item 108, Appendix D.
- (5) Using epoxy primer coating kit, Item 135, Appendix D, apply two coats of epoxy primer to surfaces where plating was removed and/or is missing.
- (6) Apply sealing compound, Item 287, Appendix D, to primed area.
- e. Inspect mount fittings for corrosion, cracks, nicks, gouges, and scratches. **IF DAMAGE IS FOUND OR SUSPECTED, PERFORM DETAILED INSPECTION** (PARA 4-5-2).
- f. Inspect mount links for corrosion, cracks, nicks, gouges, and scratches. **IF DAMAGE IS FOUND OR SUSPECTED, PERFORM DETAILED INSPECTION** (PARA 4-5-3).
- g. Inspect support fitting for corrosion, cracks, nicks, gouges, and scratches. **IF DAMAGE IS FOUND OR SUSPECTED, PERFORM DETAILED INSPECTION** (PARA 4-5-2).
- h. Inspect engine mount bolts for pitting. **NONE ALLOWED.** If pitting is found, replace engine mount bolts.
- i. Inspect engine mount bolts for corrosion. **NONE ALLOWED.** If corrosion is found, using scouring pad, Item 270, Appendix D, remove minor corrosion from engine mount bolt.
- j. Measure mount strut to mount fitting mounting bolts and inner mount strut to strut mount bolts at mating points on shank. **MINIMUM SHANK DIAMETER OF BOLTS, NAS6705DU14 AND NAS6705DU18, IS 0.3095-INCH.** If not within limit, replace bolts.
- k. Measure mount strut to mount link mounting bolts at mating points on shank. **MINIMUM SHANK DIAMETER OF BOLTS, NAS6706DU26, IS 0.3720-INCH.** If not within limit, replace bolts.
- l. Measure mount link to engine mount bolts at mating points on shank. **MINIMUM SHANK DIAMETER OF BOLTS, 74178-8-19, IS 0.4962-INCH.** If not within limit, replace bolts.
- m. Measure ID of inner and outer mount strut bushing and mount fitting bearing. **MAXIMUM INSIDE DIAMETER WHEN USING BOLTS, NAS6705DU14, IS 0.3145-INCH.** If ID exceeds limits, perform the following:
 - (1) Subtract mount bolt shank diameter from bushing or bearing inside diameter. **MAXIMUM GAP IS 0.005-INCH.**
 - (2) If limit is exceeded, replace mount strut bushing (PARA 4-5-1) or mount fitting bearing (PARA 4-5-2).
- n. Measure ID of inner and outer mount strut bearing and mount link bores which secure to

mount struts. **MAXIMUM INSIDE DIAMETER WHEN USING BOLTS, NAS6706DU26, IS 0.3770-INCH.** If ID exceeds limits, perform the following:

(1) Subtract mount bolt shank diameter from bearing or mount link bore inside diameter. **GAP MUST NOT EXCEED 0.005-INCH.**

(2) If limit is exceeded, replace mount strut bearing (PARA 4-5-1) and mount link.

o. Measure ID of mount link bores which secure to engine mounts. **INSIDE DIAMETER MUST NOT EXCEED 0.5012-INCH.** If ID is more than limits, perform the following:

(1) Subtract mount bolt shank diameter from mount link bore inside diameter. **GAP MUST NOT EXCEED 0.005-INCH.**

(2) If limit is exceeded, replace mount link.

4-4-5.1.3 Install.

- a. Using nonmetallic scraper and machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D, clean bottom and edges of mount fittings and area on engine compartment deck where mount fittings are installed. Do not allow solvent to dry on deck. Keep turning towel after each wipe.
- b. Wipe area dry with clean machinery towel, Item 211, Appendix D.
- c. Inspect mount fitting shims on engine compartment deck for damage and security. **NO DAMAGE OR LOOSENESS ALLOWED.** If shims are loose or damaged, perform the following:

NOTE

Up to three shims, totaling 0.270-inch may be installed at any one location. Count number of shims during removal so same amount is installed.

(1) Remove rivets or pins and collars securing shim(s) to top of engine compartment deck.

(2) Using nonmetallic scraper and machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D, clean deck area.

(3) Position shim(s) and install rivets or pins and collars.

NOTE

- Bolt length may be changed, if required, due to shim thickness.
- Make sure countersunk washers are installed under bolthead with countersunk side facing bolthead.
- d. Install inner mount fittings and support fitting to engine compartment deck with bolts, countersunk washers, washers, and nuts. **TORQUE NUTS TO 114 - 126 INCH-POUNDS OR BOLTS TO 129 - 141 INCH-POUNDS** (Figure 4-4-5-1, Sheet 2, Detail B and Sheet 3, Detail D).

NOTE

Make sure countersunk washers are installed under bolthead with countersunk side facing bolthead.

- e. Install outer mount fitting to engine compartment deck with bolts, countersunk washers, washers, and nuts. **TORQUE NUTS TO 114 - 126 INCH-POUNDS OR BOLTS TO 129 - 141 INCH-POUNDS.**

CAUTION

Damage to engine cold section may occur if bolts connecting mount links to engine are installed with boltheads facing front. Install boltheads connecting mount links to engine with boltheads facing rear. Bolts connecting mount links to inner/outer mounts and mount support may face in either direction.

- f. Install outer mount fitting and radius block to cabin door upper track with bolts, washers, and nuts. **TORQUE NUTS TO 105 - 115 INCH-**

POUNDS (Figure 4-4-5-1, Sheet 2, Detail C and Sheet 3, Detail D).

- g. Apply adhesive, Item 55, Appendix D, to all mating edges of engine mount shims, engine mount fittings, engine compartment deck, and all boltheads.

NOTE

- Shim nuts with washers, NAS1149C0863R, if required, to install cotter pin when connecting center mount links to mounting flange.
 - Make sure countersunk washers are installed under bolthead with countersunk side facing bolthead.
- h. Install center mount links on diffuser case and secure with bolts, countersunk washers washer, and nuts. **TIGHTEN EACH NUT FINGER-TIGHT, ADVANCE NUT TO NEXT CASTELLATION** (Figure 4-4-5-1, Sheet 1, Detail A and Sheet 3, Detail D). Install cotter pin.

CAUTION

Do not overtorque nut.

NOTE

- Shim nut with washers, NAS1149C0663R and NAS1149C0632R, if required, to install cotter pin when connecting center mount links to support fitting.
 - Make sure countersunk washers are installed under bolthead with countersunk side facing bolthead.
- i. Install center mount links on support fitting and secure with bolt, countersunk washer, washers, and nut. **TIGHTEN NUT UNTIL FIRMLY SEATED, THEN BACK OFF UNTIL WASH-**

ER UNDER NUT CAN BE TURNED BY HAND. RETIGHTEN NUT JUST ENOUGH SO WASHER CANNOT BE TURNED BY HAND. Proceed as follows:

NOTE

Do not tighten nut more than one castellation.

- (1) **IF NUT IS ALIGNED WITH HOLE IN BOLT, TIGHTEN NUT TO NEXT CASTELLATION.** Install cotter pin.
- (2) **IF NUT IS NOT ALIGNED WITH HOLE IN BOLT, CONTINUE TIGHTENING NUT UNTIL CASTELLATION IS ALIGNED WITH BOLT.** Install cotter pin.

WARNING

Bushings installed in aft engine mount struts are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.

NOTE

- Shim nut with washers, NAS1149C0563R and NAS1149C0532R, if required, to install cotter pin when connecting inner mount struts together.
- Make sure countersunk washers are installed under bolthead with countersunk side facing bolthead. Assemble inner mount struts with bolt, countersunk washer, washer(s), and nut. **TORQUE**

NUT TO 60 - 100 INCH-POUNDS. Install cotter pin.

NOTE

- Shim nut with washers, NAS1149C0663R or NAS1149C0632R, if required, to install cotter pin when connecting mount links to inner mount strut.
 - Make sure countersunk washers are installed under bolthead with countersunk side facing bolthead.
- j. Install inner mount links on inner mount strut with bolt, washers, and nut. **TORQUE NUT TO 181 - 199 INCH-POUNDS.** Install cotter pin.

NOTE

- Shim nuts with washers, NAS1149C0863R, if required, to install cotter pin when connecting inner mount links to engine.
 - Make sure countersunk washers are installed under bolthead with countersunk side facing bolthead.
- k. Install inner mount links on diffuser case with bolts, countersunk washers, washers, and nuts. **TIGHTEN EACH NUT FINGERTIGHT, ADVANCE NUT TO NEXT CASTELLATION.** Install cotter pin.

NOTE

- Shim nuts with washers, NAS1149C0563R or NAS1149C0532R, if required, to install cotter pin when connecting inner mount struts to mount fitting.
 - Make sure countersunk washers are installed under bolthead with countersunk side facing bolthead.
- l. Install inner mount struts to inner mount fittings with bolts, countersunk washers, washers,

and nuts. **TORQUE NUTS TO 60 - 100 INCH-POUNDS.** Install cotter pins.

WARNING

Bushings installed in aft engine mount struts are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.

NOTE

- Shim nut with washers, NAS1149C0663R or NAS1149C0632R, if required, to install cotter pin when connecting mount links to inner mount strut.
 - Make sure countersunk washers are installed under bolthead with countersunk side facing bolthead.
- m. Install outer mount links to outer mount strut with bolt, countersunk washer, washers, and nut. **TORQUE NUT TO 181 - 199 INCH-POUNDS.** Install cotter pin.

NOTE

- Shim nuts with washers, NAS1149C0863R, if required, to install cotter pin when connecting outer mount links to engine.
 - Make sure countersunk washers are installed under bolthead with countersunk side facing bolthead.
- n. Bolt outer mount links to engine with bolts, countersunk washers, washers, and nuts. **TIGHTEN EACH NUT FINGERTIGHT, ADVANCE NUT TO NEXT CASTELLATION.** Install cotter pin.

NOTE

- Shim nuts with washers, NAS1149C0563R or NAS1149C0532R, if required, to install cotter pin when connecting outer mount strut to mount fitting.
 - Make sure countersunk washers are installed under bolthead with countersunk side facing bolthead.
- o. Bolt outer mount strut to outer mount fitting with bolt, countersunk washer, washers, and nut. **TORQUE NUT TO 60 - 100 INCH-POUNDS.** Install cotter pin.
- p. Install soundproofing panels as required (PARA 2-4-69).

WARNING

Injury to personnel and damage to equipment will result if caution is not

taken when hoisting engine. Keep area clear when hoisting engine.

- q. Remove lifting sling from engine and remove maintenance crane or move hoist out of the way. Remove lifting sling from crane or hoist (PARA 18-4-1).
- r. If removed, install and connect inlet particle separator (IPS) duct to IPS blower with coupling. **TORQUE COUPLING NUT TO 60 - 70 INCH-POUNDS.**
- s. Make sure area is clean and free of foreign material.
- t. Close engine cowling.

4-4-6 FORWARD SUPPORT TUBE/ENGINE OUTPUT SHAFT.

PARAGRAPH BREAKDOWN

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4-4-6.1 Replace Forward Support Tube/Engine Output Shaft	4-4-6-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Borescope
- Drill
- Drill Bit, 13/64-inch
- Feeler Gage
- Fluorescent Penetrant Inspection Kit
- Hoist, Capacity to Support Engine
- Lifting Sling
- Nonmetallic Scraper, Locally-Made
- Output Drive Shaft Puller, Locally-Made, Figure H-216, Appendix H
- Output Drive Shaft Puller, Locally-Made, Figure H-262, Appendix H
- Powerplant Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Abrasive Cloth, Item 8, Appendix D

- Abrasive Cloth, Item 9, Appendix D
- Abrasive Cloth, Item 9A, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Adhesive, Item 34, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Hi-Lok Collar, HL87W-6
- Hi-Lok Fastener, HL64PB6-5
- Petrolatum, Item 234, Appendix D
- Rivet, MS90354-0604
- Sealing Compound, Item 292, Appendix D
- Tape, Insulation, Item 319A, Appendix D
- Washer, NAS1149F0332P

References

- Appendix D
- Appendix H
- Appropriate MTF
- ASTM E1417
- PARA 1-7-2

- PARA 2-6-5
- PARA 4-4-1
- PARA 4-4-4
- PARA 4-4-34
- PARA 4-5-5
- PARA 6-3-8
- SEI 671
- TM 1-2840-248-23
- TM 1-70-VIB

4-4-6.1 Replace Forward Support Tube/Engine Output Shaft.

NOTE

Replacement of No. 1 and No. 2 forward support tubes/engine output shafts is the same.

4-4-6.1.1 Remove.

- a. Open engine cowl.
- b. Disconnect engine/airframe connections (PARA 4-4-1).
- c. Remove engine air inlet (PARA 4-4-4).
- d. Attach lifting sling to engine (PARA 4-4-1).
- e. Disconnect engine mounts (PARA 4-4-1).

CAUTION

On helicopters with engine output shaft, 70361-08014-041, damage to engine output shaft and input flange/coupling will result if output shaft is not completely separated from input flange/coupling before removal.

- f. On helicopters with engine output shaft, 70361-08014-041, move engine slightly rear to separate output shaft from input flange/coupling.

WARNING

- Injury to personnel and damage to equipment will result if caution is not taken when hoisting engine. Keep area clear when hoisting engine.
- To prevent injury to personnel or damage to equipment, do not use maintenance crane or hoist as source of excessive force to free engine from helicopter.
- Hoist cable may break, causing injury to personnel or damage to equipment if hook is allowed to contact pulley when removing component from helicopter. Do not allow hook assembly to contact pulley.
- g. Raise engine high enough off engine compartment deck to clear input module of main transmission.
- h. Remove bolt and washer from crotch assembly and forward support tube. Slide crotch assembly from forward support tube (Figure 4-4-6-1, Sheet 1, Detail A).
- i. Pull engine output shaft from engine. If shaft fails to release from engine, locally make output drive shaft puller (Figure H-216 or Figure H-262, Appendix H).
- j. Remove packings from engine output shaft. Discard packings.

4-4-6-2 Change 11

- k. Remove bolts, washers, and forward support tube from engine swirl frame and remove from engine.
- l. On helicopters with forward support tube, 70361-10003-043, remove packing from forward support tube (Figure 4-4-6-1, Sheet 1, Detail B). Discard packing.
- m. Inspect input pinion for axial play (PARA 6-3-8).

4-4-6.1.2 Inspect Forward Support Tube.

NOTE

Some forward support tubes may have improper asbestos tape installed on restrainer rings instead of proper insulation tape. Asbestos tape can be identified by its white color, cloth-like texture, and 1/8-inch thickness.

- a. Check restrainer rings for asbestos tape. If asbestos tape is present, perform one of the following repairs:
 - (1) Remove asbestos tape and replace with insulation tape, Item 319A, Appendix D.
 - (2) Replace forward support tube.
 - (3) If replacement forward support tube or insulation tape is not available, remove asbestos tape from restrainer rings. Support tube may remain in service until replacement tube or insulation tape is available and maintenance is scheduled.
- b. Check restrainer rings for disbonding of insulation tape. If disbonding is found, perform the following:

NOTE

If access for use of feeler gage is inadequate, visual inspection is allowed.

- (1) Using feeler gage, measure depth and extent of separation between restrainer ring and insulation tape (Figure 4-4-6-1, Sheet 1, Detail C).

- (2) **MAXIMUM ALLOWABLE DISBOND OF INSULATION TAPE IS 20% OF TOTAL BOND SURFACE, BUT NOT MORE THAN 50% IN 45° SEGMENT.**
- (3) **NO DISBONDING ALLOWED AT SLOT IN RESTRAINER RING AND 1/2-INCH ADJACENT TO SLOT.**
- (4) If insulation tape is disbonded beyond limits, perform the following:
 - (a) Replace tape or forward support tube.
 - (b) If replacement forward support tube or insulation tape is not available, remove asbestos tape from restrainer rings. Support tube may remain in service until replacement tube or insulation tape is available and maintenance is scheduled.
- c. Check forward support tube for nicks, scratches, gouges, or damage. **NONE ALLOWED.** Repair or replace support tube, as required.

4-4-6.1.3 Inspect Engine Output Shaft.

- a. On helicopters with engine output shaft, 70361-08004-042/-043, inspect engine output shaft as follows:

NOTE

- When inspecting engine output shaft and disc pack coupling assembly, one attaching nut may be removed and replaced if corroded or to aid inspection and repair.
- Discs which make up disc pack coupling are a balanced set and cannot be replaced individually. If any disc is damaged beyond limits, engine output shaft assembly must be replaced.
- (1) Check engine output shaft for scratches, nicks, gouges, and corrosion (Table 4-4-6-1).
- (2) Check that repaired damage will not exceed limits given in Table 4-4-6-1. If

limits are exceeded, replace engine output shaft and disc pack assembly.

- (3) Check disc pack coupling for damage. If damage exceeds limits given in Table 4-4-6-1, replace engine output shaft and disc pack coupling.
- (4) Blend damage using abrasive cloth, Item 9, Appendix D, followed by abrasive cloth, Item 10, Appendix D.

NOTE

Discs which make up disc pack coupling are a balanced set and cannot be replaced individually. If any disc is damaged beyond limits, output shaft assembly must be replaced.

- (5) Inspect disc pack coupling for laminate cracks (Figure 4-4-6-1). **NONE ALLOWED.** If cracked, replace engine output shaft and disc pack coupling.
- (6) Inspect disc pack coupling for gaps between discs using feeler gage. Two or more discs between bolts on coupling flanges may separate or buckle. **NO SHARP BENDS OR CREASES IN DISC ALLOWED. MAXIMUM GAP BETWEEN DISCS MUST NOT EXCEED 0.030-INCH.** If disc pack coupling is damaged beyond limits, replace engine output shaft.
- b. On helicopters with engine output shaft, 70361-08014-041, inspect engine output shaft for scratches, nicks, gouges, and corrosion (Table 4-4-6-1).

NOTE

Perform disc coupling inspection through engine support tube inspection hole. If there is no inspection hole in engine support tube, perform inspection using borescope.

- c. On helicopters with engine output shaft, 70361-08014-041, check that repaired damage

on engine output shaft will not exceed limits given in Table 4-4-6-1. If repair exceeds limit, replace engine output shaft.

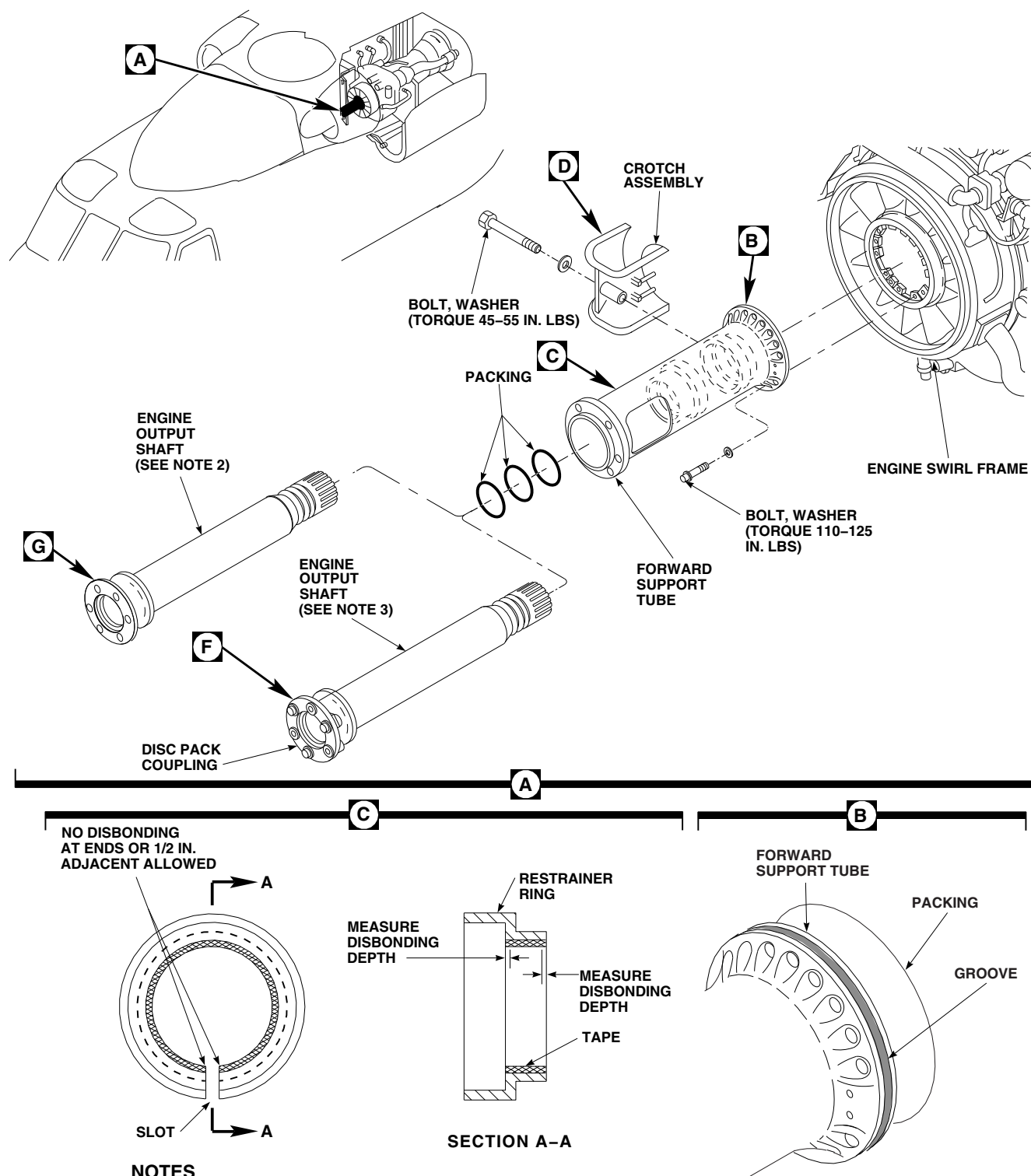
4-4-6.1.4 Inspect Crotch Assembly.

- a. Check crotch assembly for loose or missing pins. **NONE ALLOWED** (Figure 4-4-6-1, Sheet 2, Detail D). If loose or missing pins are found, replace as necessary (PARA 4-5-5).
- b. Check heat shield, firewall plate, backing strip, and segment ring for bends and distortion. **NONE ALLOWED** (Figure 4-4-6-1, Sheet 2, Detail D and Detail E). If bent or distorted, replace parts as required (PARA 4-5-5).
- c. Check crotch assembly for bends or distortion. **NONE ALLOWED.** If bent or distorted, replace crotch assembly (PARA 4-5-5).
- d. Check seal for cracks and tears. **NONE ALLOWED.** If cracked or torn, replace seal (PARA 4-5-5).

NOTE

Crotch assembly should be fluorescent penetrant inspected whenever repair to crotch assembly has been performed, when damage is suspected, or when another periodic or special inspection specifically requires fluorescent penetrant inspection of crotch assembly.

- e. Visually inspect crotch assembly. If cracks are detected or suspected, using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type 1, Method C, crotch assembly and segment ring for cracks (ASTM E1417). **NONE ALLOWED.** If cracked, replace crotch assembly or segment ring (PARA 4-5-5).
- f. Inspect crotch assembly, in areas other than crotch flange, for missing or damaged anodized finish. **NONE ALLOWED.** If anodized finish is missing or damaged, replace crotch assembly (PARA 4-5-5).
- g. Inspect crotch flange for scratches, gouges, wear grooves, and rub marks. If necessary, repair (PARA 4-5-5).



NOTES

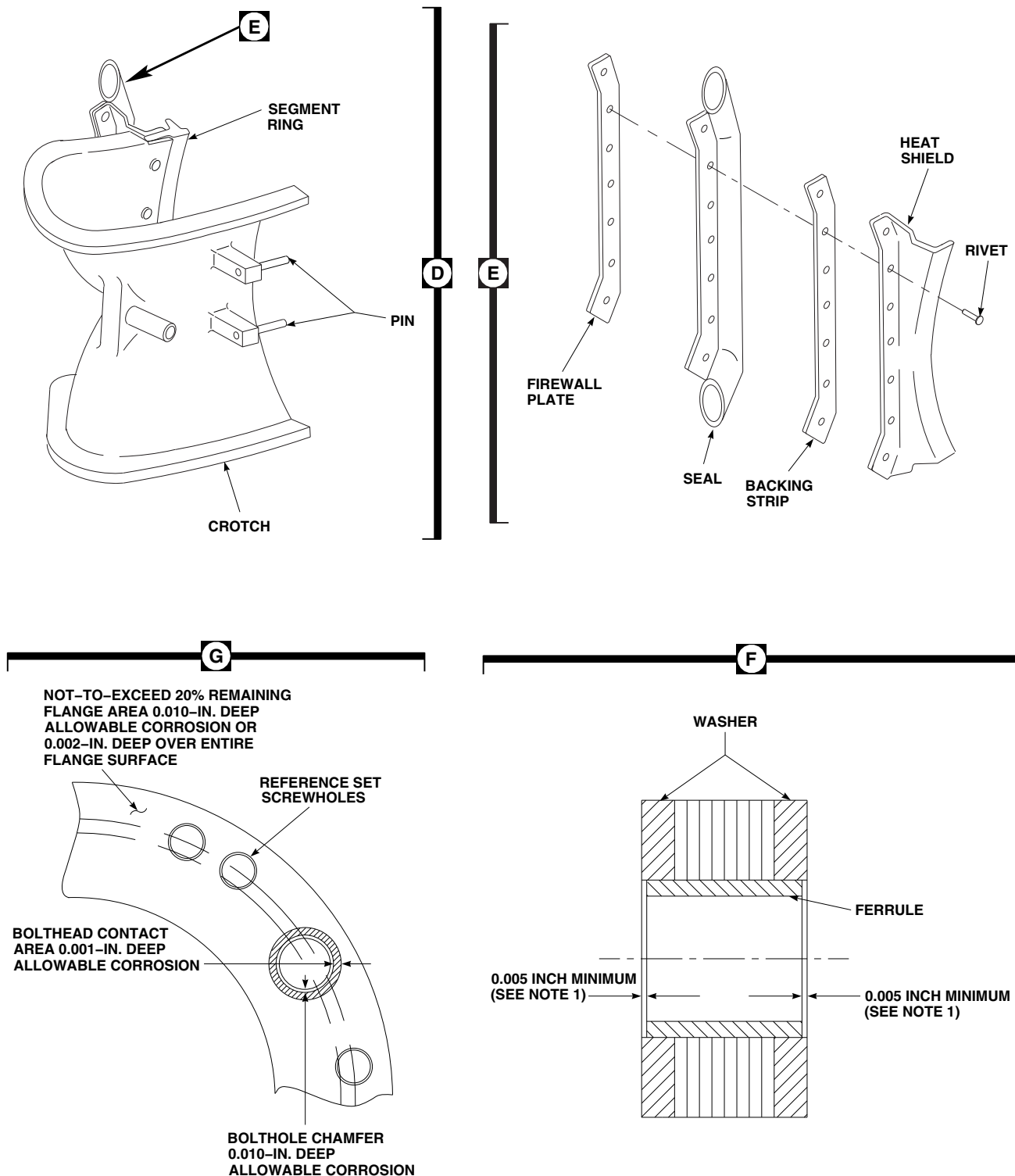
1. FERRULES MUST BE A MINIMUM OF 0.005 INCHES BELOW THE SURFACE OF THE WASHER.
2. 70361-08014-041.
3. 70361-08004-042, OR 70361-08004-043.

EFFECTIVITY

ON HELICOPTERS WITH FORWARD SUPPORT TUBE, 70361-10003-043.

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FIGURE 4-4-6-1. REPLACE FORWARD SUPPORT TUBE/ENGINE OUTPUT SHAFT (SHEET 1 OF 2)



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**FIGURE 4-4-6-1. REPLACE FORWARD SUPPORT TUBE/ENGINE OUTPUT SHAFT
(SHEET 2 OF 2)**

- h. Check crotch assembly for loose or missing rivets. **NONE ALLOWED.** If rivets are loose or missing, replace rivets (PARA 4-5-5).

NOTE

Make sure repair of nicks, scratches, and corrosion will not cause part to be beyond limits.

- i. Inspect heat shield, firewall plate, and backing strip for nicks and scratches. **DAMAGE MUST NOT BE MORE THAN 0.010-INCH DEEP OR MORE THAN 25% OF TOTAL AREA OF PART.** Blend out nicks and scratches using abrasive cloth, Item 8, Appendix D. If damage is greater than limits, replace damaged part(s) (PARA 4-5-5).
- j. Inspect firewall plate for corrosion. **CORROSION MUST NOT EXCEED 0.005-INCH DEEP OR MORE THAN 25% OF TOTAL AREA OF PART.** Blend out corrosion using abrasive cloth, Item 8, Appendix D. If corrosion exceeds limits, replace firewall plate (PARA 4-5-5).

4-4-6.1.5 Repair Engine Output Shaft.

CAUTION

On helicopters with engine output shaft, 70361-08004-042/-043, excessive vibration will result if two or more nuts connecting engine output shaft to disc pack coupling are loosened or removed at any time. Engine output shaft, disc pack coupling, and connecting hardware is a balanced assembly. Only one nut connecting engine output shaft to disc pack coupling may be loosened or removed at any one time. Loose or removed nuts must be torqued to their proper value before another nut is loosened or removed.

- a. On helicopters with engine output drive shaft, 70361-08004-042/-043, if required, remove one nut at a time to facilitate repairs to the engine shaft flange.

- b. Any scratches, through cadmium plating only, may be blended smooth using abrasive cloth, Item 8, Appendix D.
- c. Blend damaged area that is through cadmium plating into base metal, removing least amount of material, using abrasive cloth, Item 1A, Appendix D.
- d. Polish blended area to remove abrasive marks and match surface smoothness using abrasive cloth, Item 10, Appendix D.
- e. Clean reworked area with dry-cleaning solvent, Item 124, Appendix D.
- f. Using epoxy primer coating kit, Item 135, Appendix D, touch up paint on reworked area with epoxy primer (PARA 2-6-5).
- g. If removed, install nut on engine output shaft flange. **TORQUE BOLT TO 110 - 125 INCH-POUNDS** (Figure 4-4-6-1, Sheet 1).

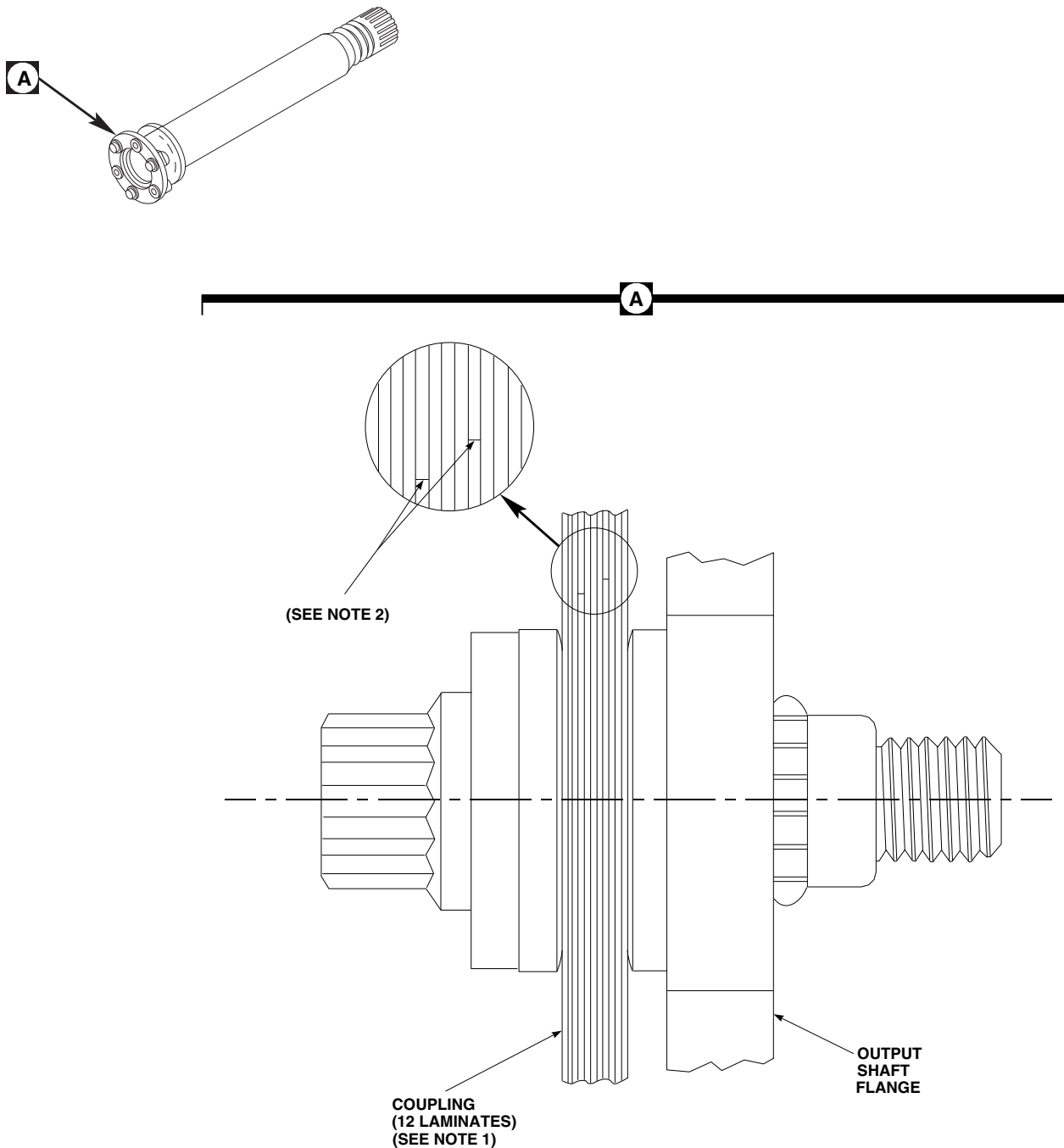
4-4-6.1.6 Repair Forward Support Tube.

- a. Repair forward support tube with scratches, dents, and gouges less than 0.015-inch deep x 0.030-inch wide x 1/2-inch long. If damage exceeds limits, replace support tube.

NOTE

Make sure polished-out damage will not be greater than allowable limits.

- b. Blend damaged area to at least 1/2-inch radius, removing least amount of material, using abrasive cloth, Item 1A, Appendix D.
- c. Polish blended area to remove abrasive marks and match surface smoothness using abrasive cloth, Item 10, Appendix D.
- d. Clean reworked area with dry-cleaning solvent, Item 124, Appendix D.
- e. Brush corrosion resistant coating, Item 118, Appendix D, on reworked area.



NOTES

1. COUPLING CRACKS MAY INITIATE IN OUTER LAMINATES, WHICH CAN BE EASILY SEEN ON THE SURFACE, OR INNER LAMINATES WHICH ONLY APPEAR ON THE OUTER SURFACE OF THE LAMINATE.
2. INNER LAMINATE CRACKS APPEAR AS THIN BLACK LINES ON THE OUTER SURFACE OF THE LAMINATE.

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FIGURE 4-4-6-2. INSPECT COUPLING

NOTE

Make sure polished-out damage will not be greater than allowable limits.

- f. Check support tube for loose or broken rivets securing restrainer rings. **NONE ALLOWED.** If necessary, replace loose or broken rivets as follows:



Damage to support tube will result if rivets are not replaced one at a time. Removing all rivets will cause loss of alignment. Replace rivets one at a time.

- (1) Remove loose or broken rivet.
- (2) Using nonmetallic scraper, remove sealing compound from area.
- (3) Measure hole diameter.
- (4) **IF HOLE DIAMETER IS LESS THAN 0.202-INCH, INSTALL RIVET, MS90354-0604, AS FOLLOWS:**
 - (a) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect hole for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace support tube.
 - (b) Apply corrosion resistant coating, Item 118, Appendix D, to hole.
 - (c) Using epoxy primer coating kit, Item 135, Appendix D, coat rivet with epoxy primer. Install rivet while primer is still wet.
- (5) **IF HOLE DIAMETER IS MORE THAN 0.202-INCH, INSTALL HI-LOK FASTENER, HL64PB6-5, AND COLLAR, HL87W-6, AS FOLLOWS:**
 - (a) Redrill hole using drill with 13/64-inch drill bit. Deburr hole.

- (b) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect hole for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace support tube.
 - (c) Apply corrosion resistant coating, Item 118, Appendix D, to hole.
 - (d) Using epoxy primer coating kit, Item 135, Appendix D, coat Hi-Lok fastener with epoxy primer. Install fastener and collar while primer is still wet.
-
- (6) After rivet or Hi-Lok fastener has been installed, check gap between restrainer ring and support tube. **MAXIMUM GAP ALLOWED BETWEEN RESTRAINER RING AND SUPPORT TUBE IS 0.020-INCH.**
 - (7) Touch up finish as required (PARA 2-6-5).
 - (8) Apply sealing compound, Item 292, Appendix D, around head of rivet or Hi-Lok fastener.
-
- g. Replace Rulon tape as follows:
 - (1) Remove one restrainer ring at a time (PARA 4-4-34). Note slot position for installation. When removing rivets, start with one adjacent to slot and work around to other side of slot.
 - (2) Remove tape and adhesive, and clean surface for bonding.
 - (3) Cut insulation tape, Item 319A, Appendix D, to 9.02-inches.
 - (4) Apply adhesive, Item 34, Appendix D, to restrainer ring, and position tape.
 - (5) Apply pressure to hold tape in position (1 to 15 psi).
 - (6) Allow to cure at 190 - 210°F (88 - 99°C) for 1 hour.

- (7) Inspect per PARA 4-4-6.1.2, step b.
- (8) Inspect holes in support tube and install restrainer ring per step f. above. Make sure to position slot in same location when installing rivets. When installing rivets, start with one adjacent to slot and work around to other side of slot.

4-4-6.1.7 Repair Forward Support Tube Flange.

- a. Repair forward support tube flange with scratches, dents or gouges less than 0.015-inch deep and 0.030-inch wide. There cannot be more than four damaged areas in 2-inch arc on flange. If damage exceeds limits, replace support tube.

NOTE

Make sure polished-out damage will not be greater than allowable limits.

- b. Blend damaged area to at least 0.060-inch radius, removing least amount of material, using abrasive cloth, Item 9A, Appendix D.
- c. Polish blended area to remove abrasive marks and match surface smoothness using abrasive cloth, Item 10, Appendix D.
- d. Clean reworked area with dry-cleaning solvent, Item 124, Appendix D.
- e. Brush corrosion resistant coating, Item 118, Appendix D, on reworked area.

4-4-6.1.8 Install.

NOTE

Packing is installed on forward support tube, 70361-10003-043, only.

- a. On helicopters with forward support tube, 70361-10003-043, lubricate packing with petrolatum, Item 234, Appendix D, and install in groove on forward support tube (Figure 4-4-6-1, Sheet 1, Detail B).
- b. Position forward support tube, with new packing (if installed), against engine swirl frame so

cutout is outboard and shaft can be seen. Install bolts and washers securing support tube to engine swirl frame. **TORQUE BOLTS TO 110 - 125 INCH-POUNDS** (Figure 4-4-6-1, Sheet 1, Detail A).

- c. Lubricate packings with petrolatum, Item 234, Appendix D. Install packings in grooves on output shaft.
- d. Carefully slide shaft into support tube and engine.
- e. If crotch assembly is new or has been repaired at another facility or maintenance level, perform inspection of crotch assembly (PARA 4-4-6.1.4) before installation.
- f. Install crotch assembly on support tube engaging alignment pins. Install bolt and washer securing crotch assembly to support tube. Shim between crotch assembly and support tube with washers, NAS1149F0332P, as required to mate crotch assembly to engine swirl frame. **TORQUE BOLT TO 45 - 55 INCH-POUNDS.**
- g. Perform torque check on nut of input module input flange. **TORQUE CHECK NUT IN TIGHTENING DIRECTION ONLY TO 646 - 714 INCH-POUNDS.**
- h. Inspect input pinion axial play (PARA 6-3-8).
- i. Make sure work area is clean and free of all tools and foreign material.
- j. Lower engine into engine compartment. Connect engine mounts (PARA 4-4-1).
- k. Connect engine/airframe connections (PARA 4-4-1).
- l. Perform engine output shaft vibration level check (TM 1-70-VIB).
- m. Install engine air inlet (PARA 4-4-4).
- n. Make sure area is clean and free of foreign material.
- o. Close engine cowling.

- p. Perform engine checkout procedure (SEI 671 and TM 1-2840-248-23) as part of maintenance test flight (Appropriate MTF).
- q. On helicopters with engine output shaft, 70361-08004-042/-043, maintain records as follows:
 - (1) Check helicopter logbook to make sure torque check of three bolts securing disc pack coupling to input module flange is shown due after initial ground run (PARA 1-7-2).
 - (2) Check helicopter logbook to make sure torque check of three bolts securing disc
- r. On helicopters with engine output shaft, 70361-08014-041, maintain records by checking helicopter logbook to make sure torque check of six bolts connecting engine output shaft to input module flange/coupling is due after 3 to 13 flight hours (PARA 1-7-2).
- s. Perform maintenance test flight (Appropriate MTF)

pack coupling to input module flange and three bolts securing output shaft to disc pack coupling is shown due after 1 to 3 hours (PARA 1-7-2).

Table 4-4-6-1. Inspect Engine Output Shaft

Damage	Depth (in.)	Width (in.)	Length (in.)	Number of Discs
Scratches on shaft	0.005	0.005		
Nicks, gouges on shaft	0.010	0.010		
Nicks, gouges on flange	0.015		0.125	
Corrosion on shaft	0.001			
(Refer to Figure 4-4-6-1, Sheet 2, Detail G)				
Corrosion on flange (Figure 4-4-6-1, Sheet 2, Detail G).				
-Bolthead contact area	0.001			
-Bolthole chamfer	0.010			
-Not-to-exceed 20% of flange surface or	0.020			
-Entire flange surface	0.002			
Disc pack coupling dents and nicks on outer edge of disc	0.020			No more than half

Table 4-4-6-1. Inspect Engine Output Shaft (Cont)

Damage	Depth (in.)	Width (in.)	Length (in.)	Number of Discs
Disc pack coupling lamination spread		0.030		No limit
Disc pack coupling dents and scratches	0.005			
Disc pack coupling cracks	None allowed (See Figure 4-4-6-2, Detail A)			
Inspect to make sure ferrules are seated below the surface of the washers	(See Figure 4-4-6-1, Sheet 2, Detail F)			
Disc pack coupling ferrule looseness (none allowed)				
Corrosion and fretting on surface of disc pack coupling washers	0.0005 all over			

4-4-7 ENGINE EXHAUST MODULE.

PARAGRAPH BREAKDOWN

		PAGE
4-4-7.1	Replace Engine Exhaust Module	4-4-7-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Plastic Hammer
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 1-6-16
- PARA 4-3-1
- PARA 4-4-1

4-4-7.1 Replace Engine Exhaust Module.

NOTE

Replacement of No. 1 and No. 2 engine exhaust modules is the same, except as noted.

4-4-7.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove screws, washers, and cover from upper rear cargo door track (Figure 4-4-7-1, Detail A).
- Open engine cowling.
- Remove screws, washers, and hinge cover from engine cowling rear hinge.
- Remove starter air tube on right side engine (PARA 4-4-1).
- On No. 2 engine inlet particle separator (IPS) blower duct only, remove screw, washer, and spacer securing engine control cables to duct (Figure 4-4-7-1, Detail B).

- Disconnect coupling and remove engine IPS blower duct from IPS blower and IPS collar (Figure 4-4-7-1, Detail A or Detail B). Remove duct.
- Remove screws and washers securing exhaust module to helicopter (Figure 4-4-7-1, Detail A). Remove module.
- If IPS collar is to be replaced, remove bolts and washers that secure IPS collar to exhaust module (Figure 4-4-7-1, Detail A or Detail B). Remove collar.
- Inspect and repair exhaust module component (PARA 4-3-1).

4-4-7.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- If inlet particle separator (IPS) collar is being replaced, install IPS blower duct collar on exhaust module with bolts and washers (Figure 4-4-7-1, Detail A or Detail B).

- c. Install replacement exhaust module on helicopter with screws and washers (Figure 4-4-7-1, Detail A).
- d. If installing new IPS blower duct, perform the following:
 - (1) Trial-fit blower duct in IPS collar (Figure 4-4-7-1, Detail A or Detail B).
 - (2) If duct will not fit in collar, or is distorted by collar, cut slits as wide as necessary in duct.
 - (3) Draw slits together, so that end of duct will slip fit in collar.
 - (4) Repair slit end.
- e. Install IPS blower duct in IPS collar. Secure blower duct to IPS blower with clamp. **TORQUE COUPLING NUT TO 60 - 70 INCH-POUNDS.** To seat clamp, tap lightly with plastic hammer. Retorque coupling nut. Continue tapping clamp until two successive trials show no loss in torque.
- f. On No. 2 engine only, secure engine control cables to top of IPS blower duct with screw, washer, and spacer (Figure 4-4-7-1, Detail B).
- g. Install starter air tube on right side engine (PARA 4-4-1).
- h. Install hinge cover over rear hinge of engine cowling with screws and washers (Figure 4-4-7-1, Detail A).
- i. Install cover over upper rear cargo door track with screws and washers.
- j. Make sure area is clean and free of foreign material.
- k. Close and latch engine cowling.

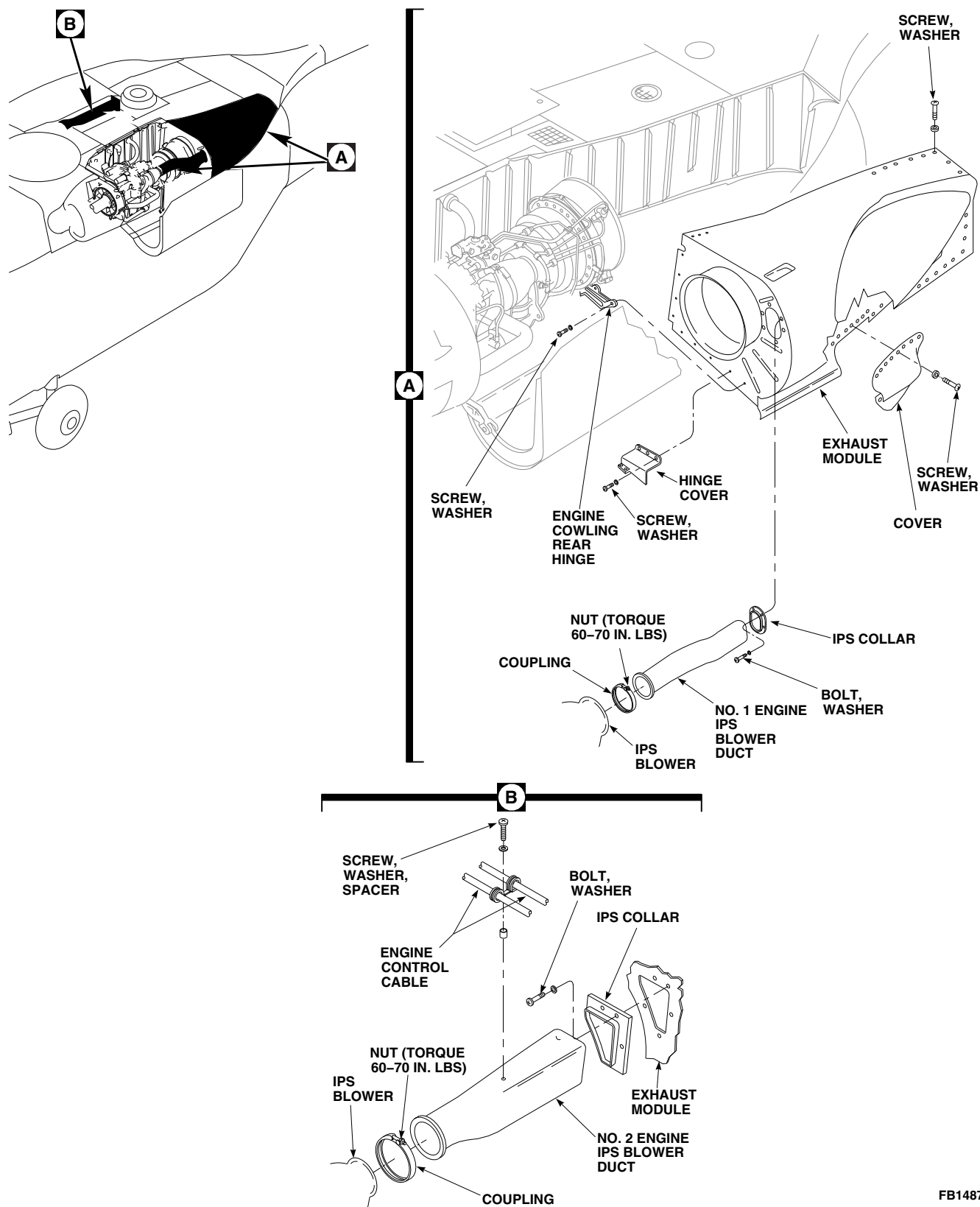


FIGURE 4-4-7-1. REPLACE ENGINE EXHAUST MODULE

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4-4-8 ENGINE EXHAUST MODULE EJECTOR.

PARAGRAPH BREAKDOWN

		PAGE
4-4-8.1	Replace Engine Exhaust Module Ejector	4-4-8-1

INITIAL SETUP

- PARA 4-4-7

Tools

- Aircraft Mechanic’s Toolkit

- PARA 4-5-6

References

- PARA 1-6-16
- PARA 4-3-1

4-4-8.1 Replace Engine Exhaust Module Ejector.

4-4-8.1.2 Inspect.

- Inspect and repair engine exhaust module ejector (PARA 4-3-1 and PARA 4-5-6).

NOTE

Replacement of left and right engine exhaust module ejectors is the same.

4-4-8.1.3 Install.

4-4-8.1.1 Remove.

- Remove engine exhaust module (PARA 4-4-7).
- Remove screws, washers, and nuts securing top edge of engine exhaust module ejector to upper fairing of exhaust module (Figure 4-4-8-1, Detail A).
- Remove screws and washers securing lower edge of engine exhaust module ejector to lower fairing of exhaust module.
- Remove screw and washer securing angle to bracket on inside of engine exhaust module.
- Remove engine exhaust module ejector.

- Turn off all electrical power (PARA 1-6-16).
- Place replacement exhaust ejector in exhaust module and slide ejector over braided seal on exhaust module firewall (Figure 4-4-8-1, Detail A).
- Install screw and washer securing angle to exhaust module bracket on inside of module.
- Install screws and washers securing lower edge of exhaust ejector to exhaust module.
- Install screws, washers, and nuts securing top edge of exhaust ejector to exhaust module.
- Install engine exhaust module (PARA 4-4-7).

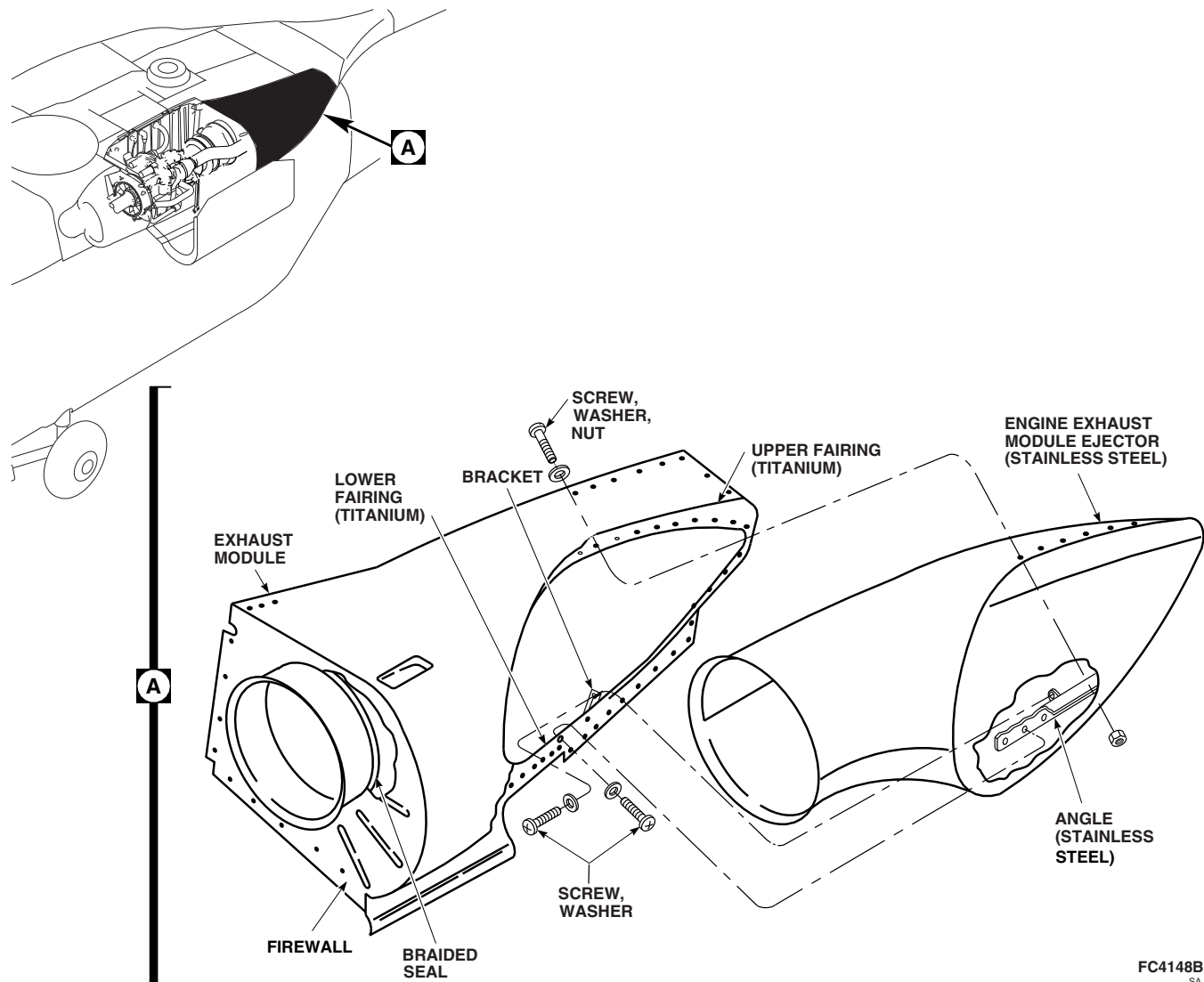
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FIGURE 4-4-8-1. REPLACE ENGINE EXHAUST MODULE EJECTOR

4-4-9 HIRSS.

PARAGRAPH BREAKDOWN

	PAGE
4-4-9.1 Replace HIRSS	4-4-9-2

INITIAL SETUP

Tools

- Adapter, 70700-20424-041
- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill
- Drill Bit, No. 40
- Drill Bit, ¼-inch
- Fluorescent Penetrant Inspection Kit
- Hoist Adapters, 70700-20468-041
- Hoist, Capacity to Support HIRSS
- Hoisting Sling
- Maintenance Crane
- Maintenance Trailer
- Plastic Mallet
- Rubber Mallet
- Spring Scale

- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lubricating Oil, Item 205, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appropriate MTF
- ASTM E1417
- PARA 2-6-5
- PARA 4-4-1
- PARA 4-4-2
- PARA 4-4-7
- PARA 4-4-10
- PARA 4-4-11
- PARA 4-4-12
- PARA 18-4-1
- SEI 671

4-4-9.1 Replace HIRSS.**NOTE**

Replacement of left and right HIRSS is the same, except where noted.

4-4-9.1.1 Remove.**NOTE**

Identify items removed with attaching hardware and keep together for installation.

- a. Position main rotor head to allow overhead clearance.
- b. Open engine cowling.
- c. Assemble cabin maintenance crane (PARA 18-4-1), hoist, or lifting device to lift HIRSS.
- d. On No. 2 engine only, remove screw, washer, and spacer securing engine control cables to No. 2 inlet particle separator (IPS) blower duct (Figure 4-4-9-1, Sheet 2, Detail D).
- e. Remove No. 2 engine starter air tube (PARA 4-4-1).
- f. Disconnect coupling and remove IPS blower duct from IPS blower and IPS collar in exhaust module.
- g. Remove HIRSS rear fairing and rear panel (PARA 4-4-10).
- h. Remove screws, washers, and splice strap from top of suppressor module bulkhead and oil cooler support beam (Figure 4-4-9-1, Sheet 1, Detail A).
- i. Install hoist adapters on module hoist points.

WARNING

To prevent injury to personnel or damage to equipment, before lifting suppressor module, make sure sling is installed properly.

- j. Secure hoisting sling legs between hoist adapters and ring.
- k. Secure ring to maintenance crane, hoist, or lifting device. Adjust leg length of sling as needed.

WARNING

- Injury to personnel and damage to equipment will result if caution is not taken when hoisting engine. Keep area clear when hoisting HIRSS module.
 - Hoist cable may break, causing injury to personnel or damage to equipment if hoist is used to free a sticking component.
 - Hoist cable may break, causing injury to personnel or damage to equipment if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.
- l. Raise hoist enough to take slack from sling.
 - m. Loosen loop clamp securing deswirl duct to engine power turbine flange and slide loop clamp over flanges onto deswirl duct (Figure 4-4-9-1, Sheet 3, Detail G).
 - n. Remove bolts and washers securing lower support angle to HIRSS bulkhead (Figure 4-4-9-1, Sheet 3, Detail F).
 - o. Remove fail-safe clevis bolt securing fail-safe clevis to fail-safe plate on HIRSS core (Figure 4-4-9-1, Sheet 2, Detail C).
 - p. Remove clevis bolt securing rear support link to rear suppressor support clevis on HIRSS core (Figure 4-4-9-1, Sheet 2, Detail B).
 - q. Remove bolts and washers securing suppressor module bulkhead to exhaust module angle and firewall support angle (Figure 4-4-9-1, Sheet 1, Detail A).

4-4-9-2 Change 9

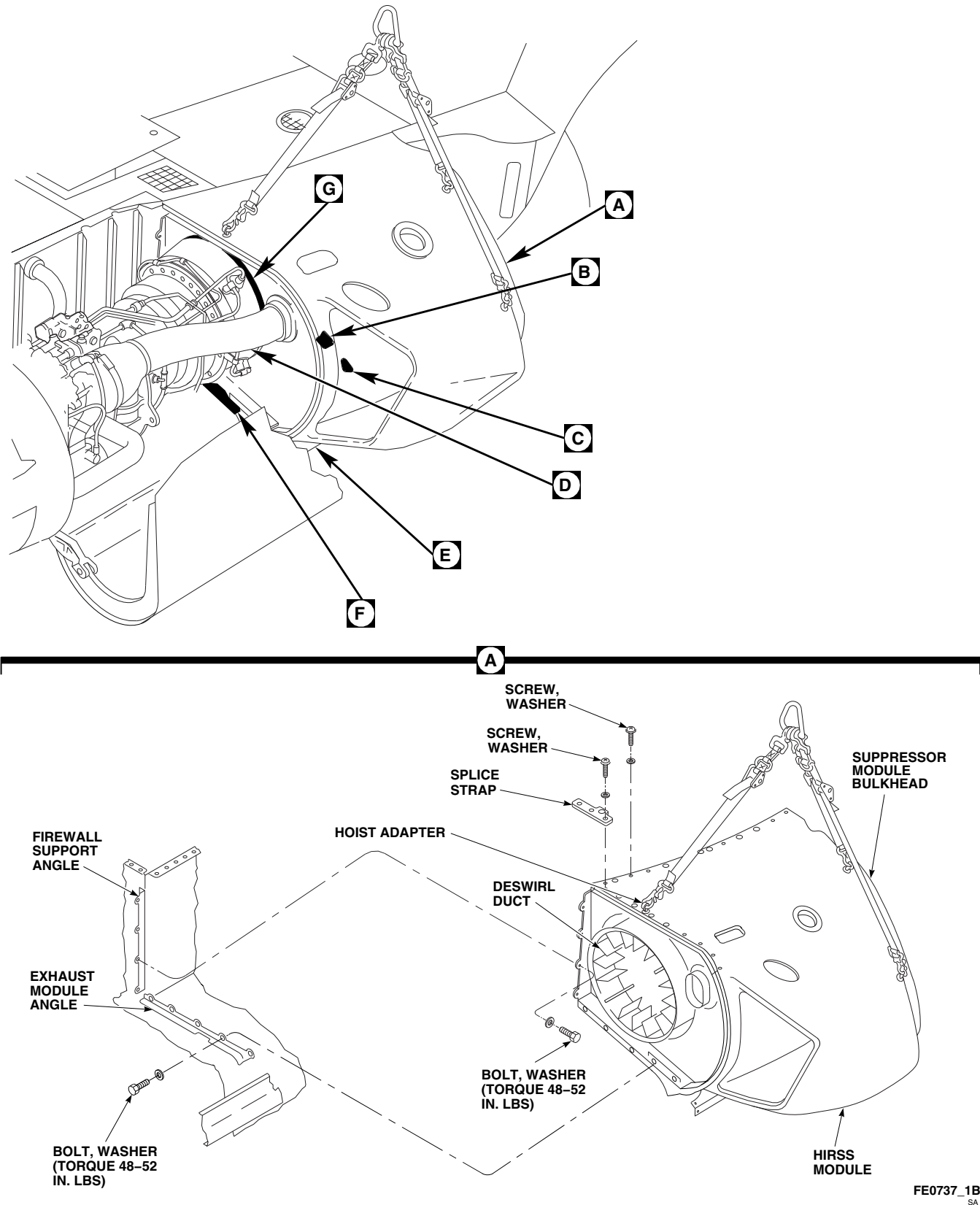


FIGURE 4-4-9-1. REPLACE HIRSS MODULE (SHEET 1 OF 3)

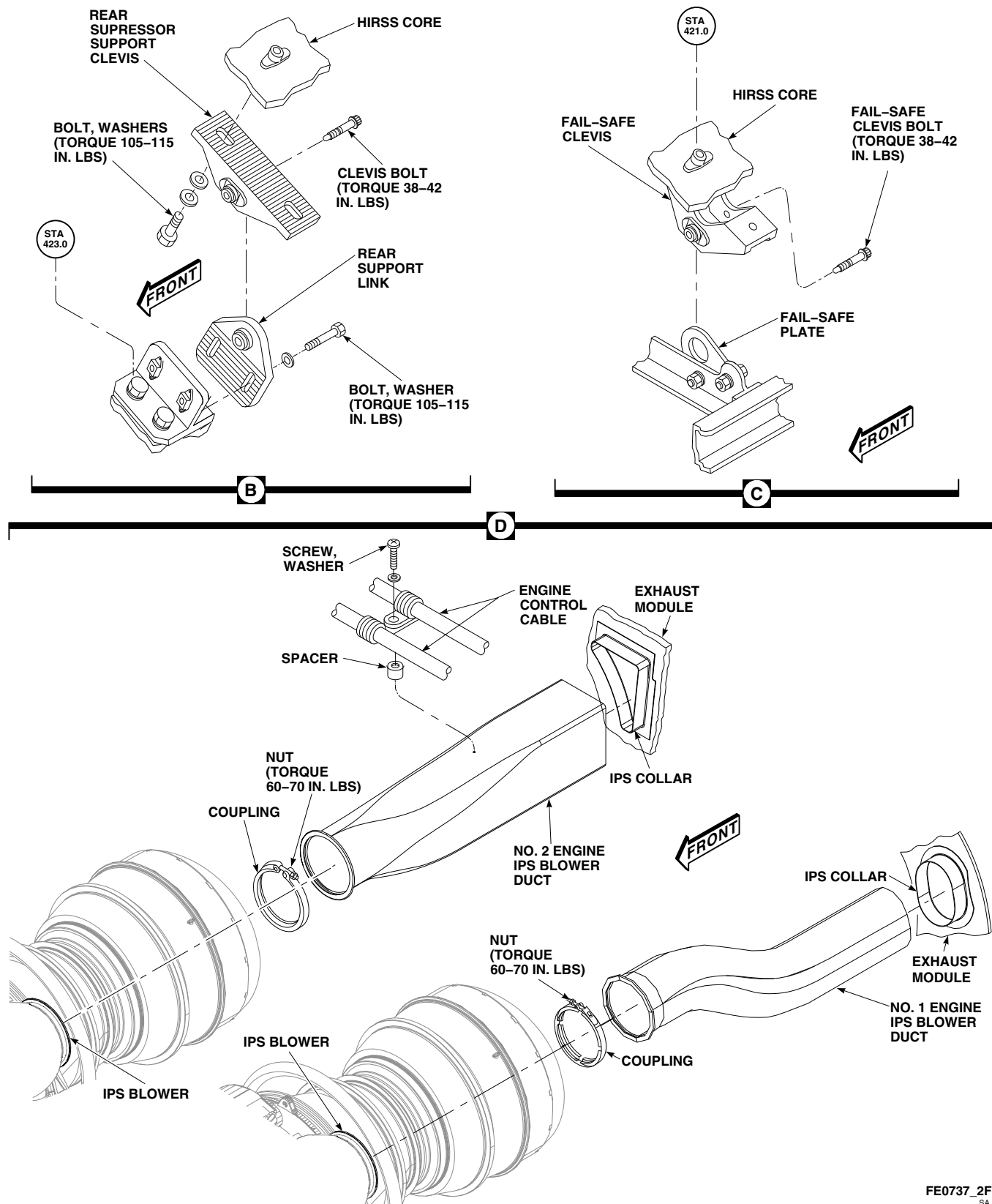
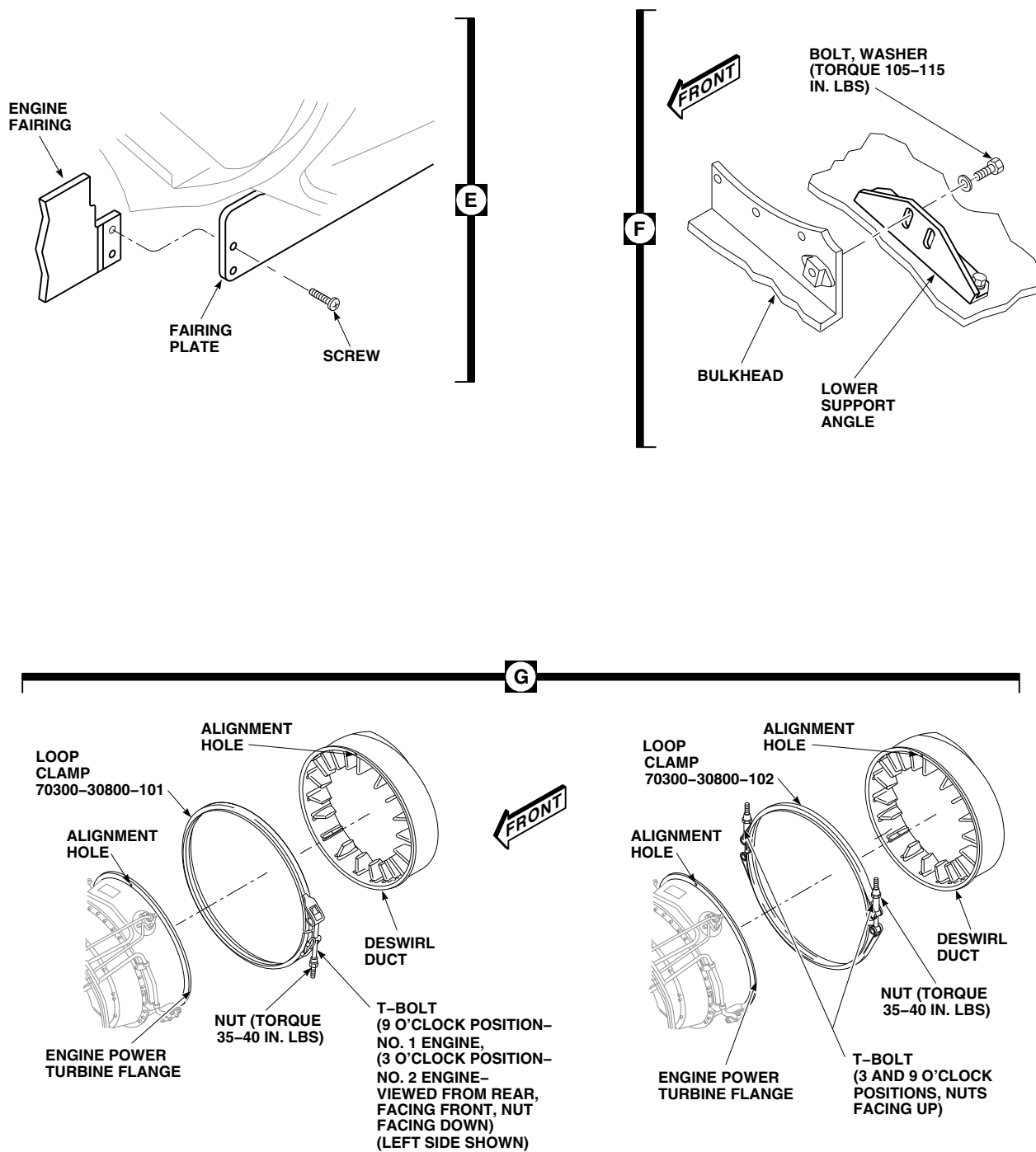
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FIGURE 4-4-9-1. REPLACE HIRSS MODULE (SHEET 2 OF 3)

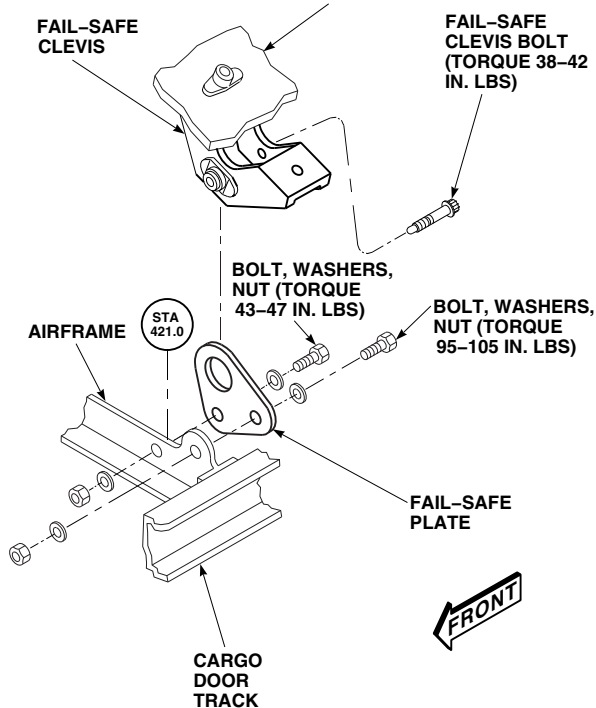
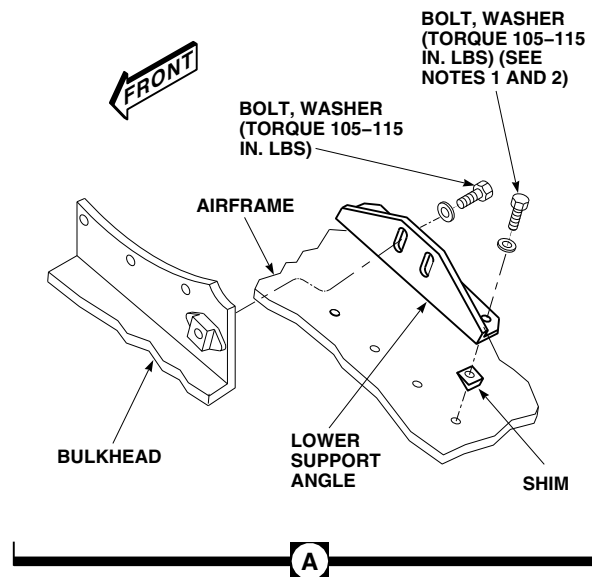
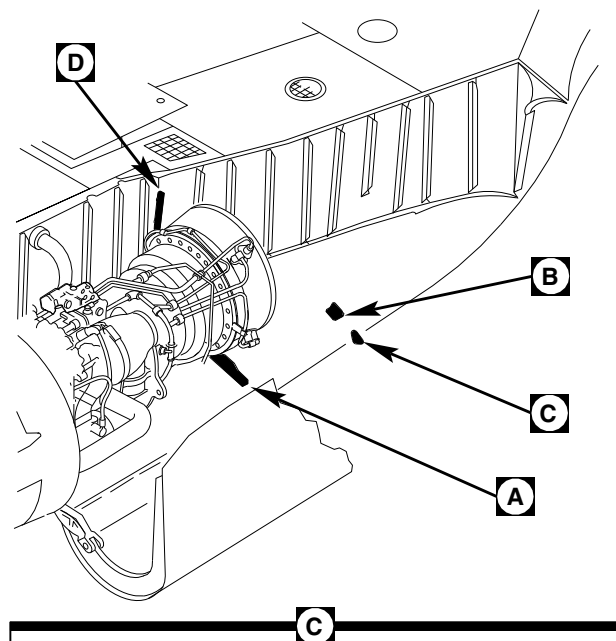
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Change 9



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FIGURE 4-4-9-1. REPLACE HIRSS MODULE (SHEET 3 OF 3)



NOTES

1. INSERT BOLTS AND WASHERS WHEN ANGLE IS REMOVED.
2. LONG BOLTS INSTALLED IN OUTBOARD AND INBOARD HOLES.
3. INSERT BOLTS AND WASHERS WHEN FITTING IS REMOVED.
4. DRILL 0.250 IN. DIA HOLES ON INSTALLATION, COORDINATE HOLES WITH HOLES AT FIREWALL.

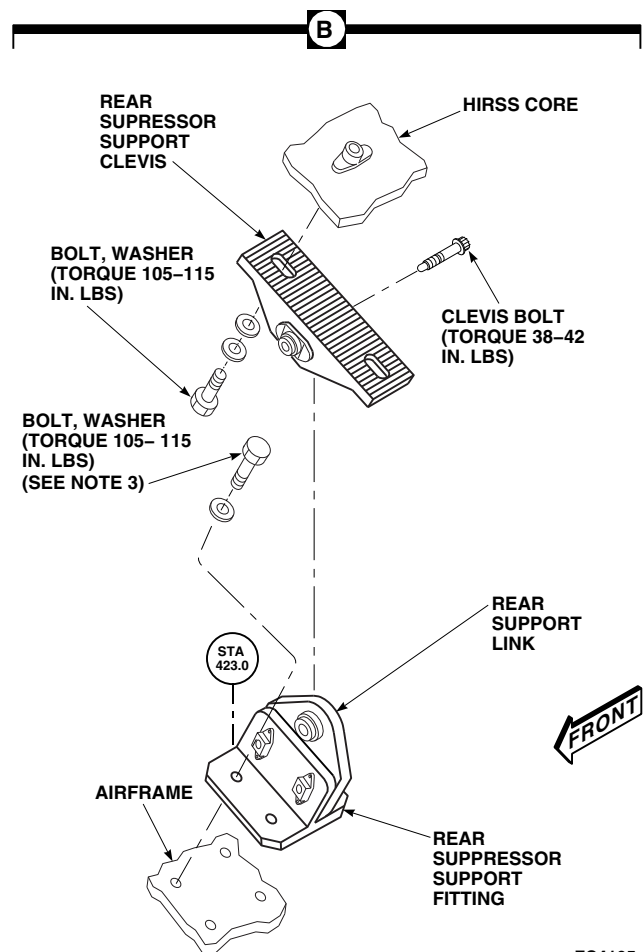
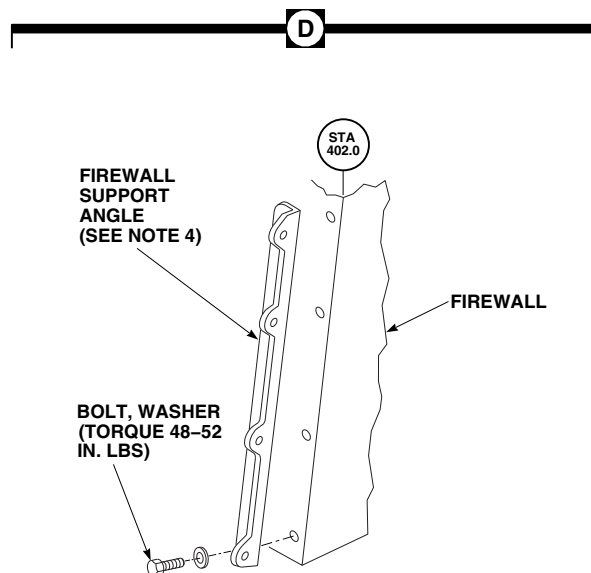


FIGURE 4-4-9-2. DISCONNECT HIRSS MOUNTS (SHEET 1 OF 2)

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FIGURE 4-4-9-2. DISCONNECT HIRSS MOUNTS (SHEET 2 OF 2)

- r. Remove bolts and washers securing rear panel to suppressor module.
- s. Remove screws securing fairing plate to engine fairing (Figure 4-4-9-1, Sheet 3, Detail E).
- t. Remove screws and washers securing nacelle fairing to firewall (Figure 4-4-9-1, Sheet 1, Detail A).
- w. Lower HIRSS module onto adapter, 70700-20424-041, and secure module with straps (Figure 4-4-9-3).
- x. Remove sling and hoist adapters.
- y. Inspect suppressor support mounts (PARA 4-4-9.1.2).
- z. If engine exhaust module is to be installed in place of HIRSS, perform the following:

NOTE

Deswirl duct remains installed on HIRSS module during HIRSS removal.

- u. Slide suppressor module and deswirl duct toward rear to clear center body cone and lift suppressor module off helicopter.
- v. Remove deswirl duct and loop clamp from suppressor module.

NOTE

Do not lay module on bulkhead lower flange.

- (1) Remove bolts, washers, and nuts securing fail-safe plate to cargo door track. Remove plate (Figure 4-4-9-2, Sheet 1, Detail C).
- (2) Remove bolts and washers securing rear suppressor support fitting. Remove fitting from airframe (Figure 4-4-9-2, Sheet 1, Detail B).
- (3) Apply sealing compound, Item 292, Appendix D, under boltheads. Install bolts and washers in rear suppressor support fitting nutplates.

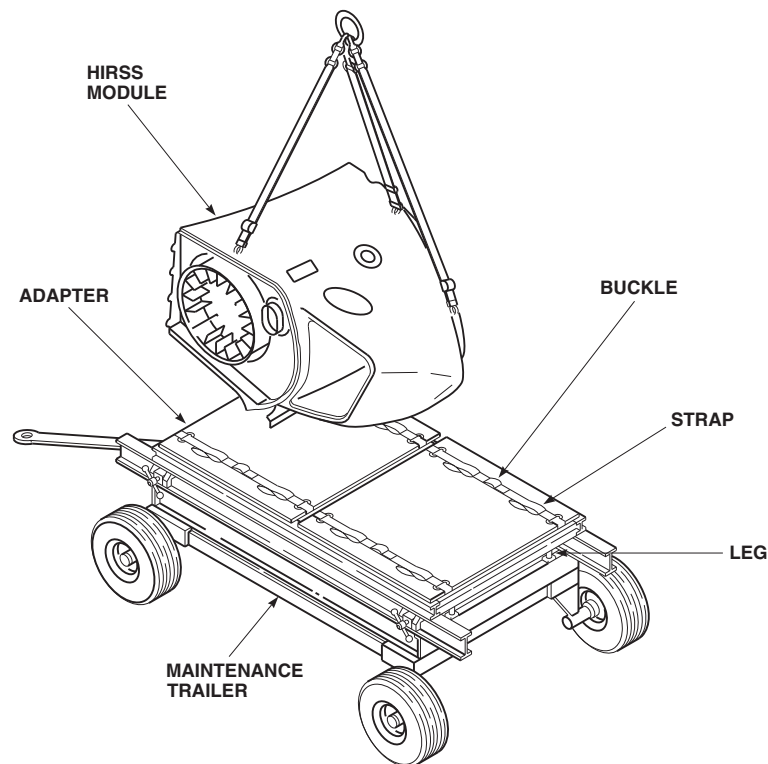
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FIGURE 4-4-9-3. HIRSS MODULE ADAPTER

(4) Remove bolts, washers, shims, and lower support angle from airframe (Figure 4-4-9-2, Sheet 1, Detail A).

(5) Apply sealing compound, Item 292, Appendix D, under boltheads. Install bolts and washers in lower support angle nut-plates.

(6) Stow HIRSS components and attaching hardware to protect from damage and to have ready for installation. Repair any damage before stowing.

(7) Remove center body cone (PARA 4-4-2).

(8) Install original center tailpipe body and cover plate (PARA 4-4-2).

(9) Install engine exhaust module (PARA 4-4-7).

aa. Remove maintenance crane, hoist, or lifting device (PARA 18-4-1).

ab. Make sure area is clean and free of foreign material.

ac. Close engine cowlings.

4-4-9.1.2 Inspect Suppressor and Nacelle Fairing Support Mounts.

a. Inspect suppressor support mount as follows:

(1) Visually check spherical bearing in rear support link for looseness or damage. **NONE ALLOWED.** If loose or damaged bearings are found, replace parts.

(2) Visually inspect rear suppressor support fitting, rear support link, and rear suppressor support clevis for flats or damage to adjustment ridges. **NONE ALLOWED.** If damaged ridges are found, repair ridges.

(3) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect fail-safe clevis, fail-safe plate, rear suppressor support clevis, rear suppressor support fitting, and link for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace parts (PARA 4-4-11).

- (4) Check fail-safe and rear suppressor support clevis bolts for wear. **DIAMETER OF CLEVIS BOLT SHANK IN CONTACT AREA MUST NOT BE LESS THAN 0.281-INCH.** If wear exceeds limit, replace bolt.
- b. Inspect nacelle fairing support mount as follows:
 - (1) Visually inspect spherical bearings in upper nacelle support fitting and lower nacelle support link for looseness or damage. **NONE ALLOWED.** If loose or damaged bearings are found, replace parts.
 - (2) Inspect upper support clevis for flats or damage to adjustment ridges. **NONE ALLOWED.** If damaged adjustment ridges are found, replace upper support clevis.
 - (3) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect lower nacelle support fitting and link for cracks (ASTM E1417). **NONE ALLOWED.** If cracked, replace lower nacelle support fitting and link (PARA 4-4-12).
 - (4) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect upper nacelle support fitting and clevis for cracks (ASTM E1417). **NONE ALLOWED.** If cracked, replace upper nacelle support fitting and clevis (PARA 4-4-12).
 - (5) Check upper and lower clevis bolts for wear. **DIAMETER OF CLEVIS BOLT SHANK IN CONTACT AREA MUST NOT BE LESS THAN 0.281-INCH.** If wear exceeds limit, replace bolt.

4-4-9.1.3 Install.

NOTE

Initial installation of HIRSS requires the following entry on helicopter basic weight and balance record: WEIGHT CHANGE +172.4, MOMENT ARM 420.1, MOMENT/1000 +72,431.

- a. If exhaust module is installed, perform the following:

- (1) Remove exhaust module (PARA 4-4-7).
- (2) Remove engine center tailpipe (PARA 4-4-2).

NOTE

If HIRSS has never been installed on helicopter, firewall support angle holes must be drilled.

- (3) Position firewall support angles on firewall at STA 402.0 (Figure 4-4-9-2, Sheet 2, Detail D). Mark four boltholes on firewall support angle.
- (4) Using drill with No. 40 drill bit, drill four holes in firewall support angle.
- (5) Using drill with 1/4-inch diameter drill bit, redrill holes. Deburr holes.
- (6) Secure firewall support angles to firewall at STA 402.0, with bolts and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**
- (7) Secure rear suppressor support fittings to airframe with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS** (Figure 4-4-9-2, Sheet 1, Detail B).
- (8) Secure fail-safe plates to airframe above cargo door track with bolts, washers, and nuts (Figure 4-4-9-2, Sheet 1, Detail C). **TORQUE NUT TO 95 - 105 INCH-POUNDS.**
- (9) Install bolt, washer, and nut. **TORQUE NUT TO 43 - 47 INCH-POUNDS.**
- (10) Install center body cone (PARA 4-4-2).

- b. Open engine cowling.
- c. Position main rotor head to allow overhead clearance.
- d. Position maintenance crane (PARA 18-4-1), hoist, or lifting device to lift HIRSS.

- e. Inspect deswirl duct (PARA 4-4-1).

NOTE

Deswirl duct must be temporarily installed on HIRSS module before module is installed on helicopter.

- f. Temporarily install deswirl duct to HIRSS module as follows:

NOTE

- Left and right deswirl ducts are identical.

- Deswirl ducts have two alignment holes located on deswirl duct flange. When installing, one alignment hole shall be located top-center while other alignment hole shall point inboard (Figure 4-4-9-1, Sheet 3, Detail G).

- (1) Temporarily install deswirl duct on HIRSS module by sliding duct inside finger seals on core.

- (2) Position loop clamp on deswirl duct beyond flange.

- g. Secure three hoist adapters to HIRSS module hoist points (Figure 4-4-9-1, Sheet 1, Detail A).

WARNING

To prevent injury to personnel or damage to equipment, before lifting suppressor module, make sure sling is installed properly.

- h. Secure hoisting legs between hoist adapters and sling apex. Adjust leg lengths to position module for installation.
- i. Secure ring to hoist or lifting device and raise sling enough to remove slack.

WARNING

- Injury to personnel and damage to equipment will result if caution is not

taken when hoisting engine. Keep area clear when hoisting HIRSS module.

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component.
- Hoist cable may break, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when hoisting components. Do not allow hook assembly to contact pulley.

CAUTION

Damage to center body cone and/or deswirl vanes will occur if HIRSS module is not carefully hoisted in place. Take care when hoisting and positioning HIRSS module on helicopter.

- j. Lift HIRSS module into position.
- k. Install screws and washers securing nacelle fairing to firewall. Do not tighten screws at this time.
- l. Secure rear suppressor support clevis to rear support link as follows (Figure 4-4-9-1, Sheet 2, Detail B):
- (1) Attempt to install clevis bolt securing support clevis to support link. If bolt is easily installed, tighten, but do not torque at this time.
 - (2) If bolt cannot be easily installed, perform the following:
 - (a) If required, loosen bolts securing rear support link to rear suppressor support fitting.
 - (b) If required, loosen bolts securing support clevis to HIRSS core.

NOTE

Rear support link should be centered between sides of suppressor support clevis after torque is applied.

- (c) Secure rear support link to rear suppressor support clevis on HIRSS core with clevis bolt. Do not torque bolt at this time.
- (d) Align rear support clevis and rear support link to obtain best possible alignment of clevis so no preload exists.

NOTE

Rear support link must seat with rear of suppressor support fitting when bolts are tightened.

- (e) Tighten, but do not torque, bolts securing support link to support fitting.
- (f) Tighten, but do not torque, bolts securing support clevis to HIRSS core.
- m. Secure fail-safe plate to fail-safe clevis on HIRSS core with fail-safe clevis bolt (Figure 4-4-9-1, Sheet 2, Detail C). Do not torque bolt at this time.
- n. Secure suppressor module bulkhead to exhaust module angle with bolts and washers (Figure 4-4-9-1, Sheet 1, Detail A). Do not torque bolts at this time.
- o. Secure suppressor module bulkhead to firewall support angle with bolts and washers. Do not torque bolts at this time.
- p. Secure lower support angle to HIRSS bulkhead with bolts and washers (Figure 4-4-9-1, Sheet 3, Detail F). Do not torque bolts at this time.
- q. Secure fairing plate to engine fairing with screws (Figure 4-4-9-1, Sheet 3, Detail E). Do not tighten screws at this time.

- r. Secure splice strap to top of suppressor module bulkhead and oil cooler support beam with screws and washers (Figure 4-4-9-1, Sheet 1, Detail A). Do not tighten screws at this time.
- s. Secure deswirl duct to engine power turbine flange as follows (Figure 4-4-9-1, Sheet 3, Detail G):

NOTE

- Left and right deswirl ducts are identical.
- Deswirl ducts have two alignment holes located on deswirl duct flange. When installing, one alignment hole shall be located top-center while other alignment hole shall point inboard when installed in HIRSS modules.
- (1) Make sure alignment holes in deswirl duct flange mate with alignment holes of engine power turbine flange.
- (2) Visually inspect deswirl duct to make sure deswirl duct flange mates with engine power turbine flange through 360°. If complete mating of flanges is not achieved, perform the following:
 - (a) Remove deswirl duct.
 - (b) Using spring scale to measure load, apply no more than 15 pounds of load to remove warpage.
 - (c) If warpage cannot be removed using 15 pounds of load, replace deswirl duct.

NOTE

When using lubricating oil, Item 205, Appendix D, refer to General Electric SEI 671 for approved oils.

- (3) Using lubricating oil, Item 205, Appendix D, lubricate pressure surfaces of loop clamp and T-bolt(s).
- (4) Secure loop clamp over lips of deswirl duct and engine power turbine flange.

- (5) Position loop clamp as follows:

NOTE

Loop clamp is not a rigging tool. Deswirl duct must be supported until clamp is torqued.

- (a) For loop clamp, 70300-30800-101 (one T-bolt), position loop clamp with T-bolt at 9 o'clock for No. 1 engine and 3 o'clock for No. 2 engine, with nut facing down.
- (b) If loop clamp, 70300-30800-102 (two T-bolts), position loop clamp with T-bolts at 3 o'clock and 9 o'clock positions, with nuts facing up.

NOTE

If torquing loop clamp, 70300-30800-102, alternate from nut to nut to maintain even loading on clamp.

- (6) **TORQUE LOOP CLAMP NUT(S) TO 35 - 40 INCH-POUNDS.**
- (7) To seat loop clamp, tap lightly with rubber mallet. Retorque loop clamp nut(s). Continue tapping until two successive trials show no loss in torque.
- (8) Loosen loop clamp nut(s) slightly. **RE-TORQUE LOOP CLAMP NUT(S) TO 35 - 40 INCH-POUNDS.**
- (9) Visually inspect latches to make sure no interference has occurred.
- (10) Visually inspect inside deswirl duct to make sure deswirl duct flange mates with engine power turbine flange through 360°.

- t. Torque hardware as follows:

- (1) Tighten all screws securing nacelle fairing to firewall.

NOTE

Rear support link must seat with rear of suppressor support fitting after bolts are torqued.

- (2) **TORQUE REAR SUPPRESSOR SUPPORT FITTING TO REAR SUPPORT LINK BOLTS TO 105 - 115 INCH-POUNDS** (Figure 4-4-9-1, Sheet 2, Detail B).

NOTE

Rear support link should be centered between sides of suppressor support clevis after torque is applied.

- (3) **TORQUE REAR SUPPRESSOR SUPPORT CLEVIS BOLT TO 38 - 42 INCH-POUNDS.**
- (4) **TORQUE REAR SUPPRESSOR SUPPORT CLEVIS TO HIRSS CORE BOLTS TO 105 - 115 INCH-POUNDS.**

NOTE

After torquing fail-safe clevis bolt, it is acceptable for a gap to exist between underside of bolthead and fail-safe clevis.

- (5) **TORQUE FAIL-SAFE CLEVIS BOLT TO 38 - 42 INCH-POUNDS** (Figure 4-4-9-1, Sheet 2, Detail C).
- (6) Tighten screws securing splice strap to oil cooler support beam and suppressor module bulkhead (Figure 4-4-9-1, Sheet 1, Detail A).
- (7) **TORQUE BOLTS SECURING SUPPRESSOR MODULE BULKHEAD TO FIREWALL SUPPORT ANGLE TO 48 - 52 INCH-POUNDS.**
- (8) **TORQUE BOLTS SECURING SUPPRESSOR MODULE BULKHEAD TO EXHAUST MODULE ANGLE TO 48 - 52 INCH-POUNDS.**
- (9) **TORQUE BOLTS SECURING LOWER SUPPORT ANGLE TO HIRSS BULKHEAD TO 105 - 115 INCH-POUNDS** (Figure 4-4-9-1, Sheet 3, Detail F).

- (10) Secure support to lower outboard side of bulkhead and upper cargo door track fairing with screws.
- (11) Tighten screws securing fairing plate to engine fairing (Figure 4-4-9-1, Sheet 3, Detail E).
- u. Install engine inlet particle separator (IPS) blower duct as follows (Figure 4-4-9-1, Sheet 2, Detail D):
 - (1) Position IPS blower duct between IPS collar on exhaust module and IPS blower. Secure duct to blower with coupling.
 - (2) **TORQUE COUPLING NUT TO 60 - 70 INCH-POUNDS.**
 - (3) To seat coupling, tap lightly with plastic mallet and retorque coupling nut. Continue tapping until two successive trials show no loss of torque.
- v. On No. 2 engine only, secure engine control cables to No. 2 engine IPS blower duct with screw, spacer, and washer.
- w. Install No. 2 engine starter air tube (PARA 4-4-1).
- x. Install HIRSS rear fairing and rear panel (PARA 4-4-10).
 - y. Remove hoisting sling from nacelle.
 - z. Remove hoist adapters from HIRSS module and install screws.
 - aa. Make sure area is clean and free of foreign material.
 - ab. Close engine cowling.
 - ac. Touch up paint on exposed fasteners (PARA 2-6-5).
 - ad. Run up engine for at least 10 minutes (Appropriate MTF). Allow engine to cool down for 1 hour minimum.
 - ae. After engine runup, perform the following:
 - (1) Perform torque check of loop clamp nut(s). **TORQUE SHOULD BE 35 - 40 INCH-POUNDS.** If torque is below limit, perform the following:
 - (a) Loosen loop clamp nut(s) slightly.
 - (b) **RETORQUE LOOP CLAMP NUT(S) TO 35 - 40 INCH-POUNDS.**
 - (2) Apply torque stripe to loop clamp nut(s) (PARA 2-6-5).

4-4-10 HIRSS REAR FAIRINGS AND PANELS.

PARAGRAPH BREAKDOWN

	PAGE		PAGE
4-4-10.1	Replace HIRSS Rear Fairings and Panels	4-4-10.1	
4-4-10.2	Repair HIRSS Rear Fairings and Panels	4-4-10-2	

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Bit, No. 52
- Drill, Rotary
- Rivet Gun

- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-6-3
- PARA 2-6-5
- PARA 4-4-13
- PARA 4-4-15

4-4-10.1 Replace HIRSS Rear Fairings and Panels.

NOTE

Replacement of left and right HIRSS rear fairings and panels is the same, except as noted.

- Remove screws and washers securing rear fairing and panel to firewall and nacelle. Remove rear fairing and panel (Figure 4-4-10-1, Detail A or Detail B).
- Disconnect panel from rear fairings by removing vertical row of bolts and washers.

4-4-10.1.1 Remove.

- Remove HIRSS extender (PARA 4-4-15).
- On right side only, remove bolts and washers securing suppressor module inlet particle separator blower duct to panel (PARA 4-4-13).

NOTE

If panel and fairing are to be removed as one piece, do not remove vertical row of bolts and washers securing panel to rear fairing.

4-4-10.1.2 Inspect.

- Check panels and rear fairings for cracks in fiberglass. **ANY NUMBER OF CRACKS UP TO 1-INCH LONG ARE ALLOWED.** Stop-drill cracks using No. 52 drill bit and rotary drill. Repair fairings (PARA 2-6-3).
- Check panels and rear fairings for loose, broken, or missing fasteners. **UP TO 15% OF FASTENERS MAY BE DAMAGED OR MISSING, WITH AT LEAST TWO GOOD FASTENERS BETWEEN EACH DAMAGED OR MISSING FASTENER.** If required, replace fasteners.

4-4-10.1.3 Install.

- a. Install panel to rear fairing with bolts and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS** (Figure 4-4-10-1, Detail A or Detail B).
- b. Make sure area is clean and free of foreign material.
- c. Install panel and rear fairing on firewall and nacelle with screws and washers.
- d. On right side only, secure rear panel to suppressor module inlet particle separator blower duct with bolts and washers (PARA 4-4-13).
- e. Install HIRSS extender (PARA 4-4-15).
- f. Touch up paint on exposed screws as required (PARA 2-6-5).

4-4-10.2 Repair HIRSS Rear Fairings and Panels.**4-4-10.2.1 Repair.**

- a. Remove panel from rear fairing (PARA 4-4-10.1).

NOTE

Rear panels, fairings, and outlet panels are constructed of fiberglass skin.

- b. Evaluate damage to fiberglass components (PARA 4-4-10.1 and PARA 2-6-3). If required, repair damaged areas of panels, fairings, or outlet panels (PARA 2-6-3).
- c. Evaluate limit of missing, broken, or loose nutplates (PARA 4-4-10.1). If required, replace nutplates as required (Figure 4-4-10-2, Detail A or Detail B).
- d. Replace stiffener on outlet panel as required (Figure 4-4-10-2, Detail C).
- e. Install rear panel and rear fairing (PARA 4-4-10.1).
- f. Touch up paint in repaired areas as required (PARA 2-6-5).

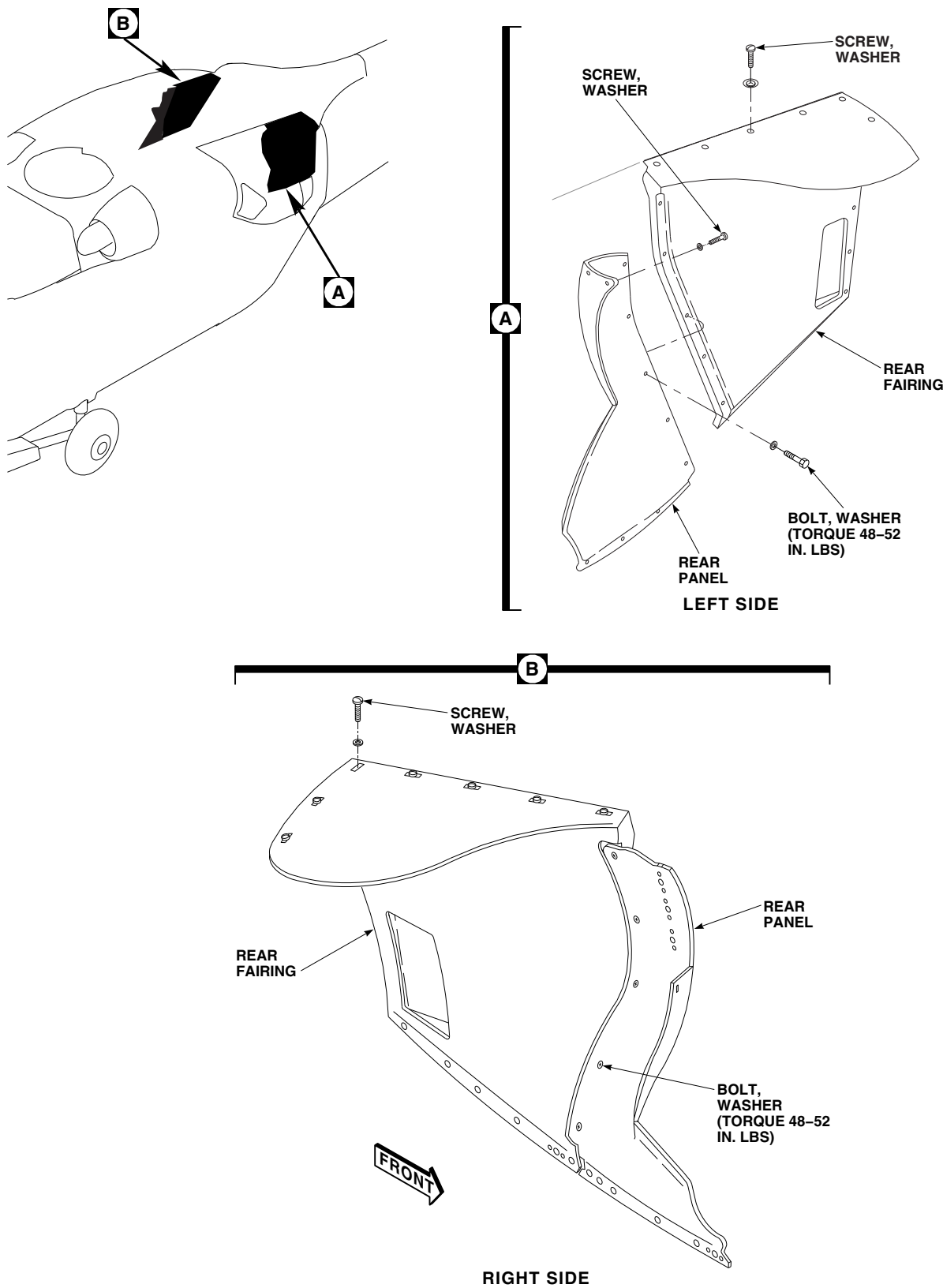
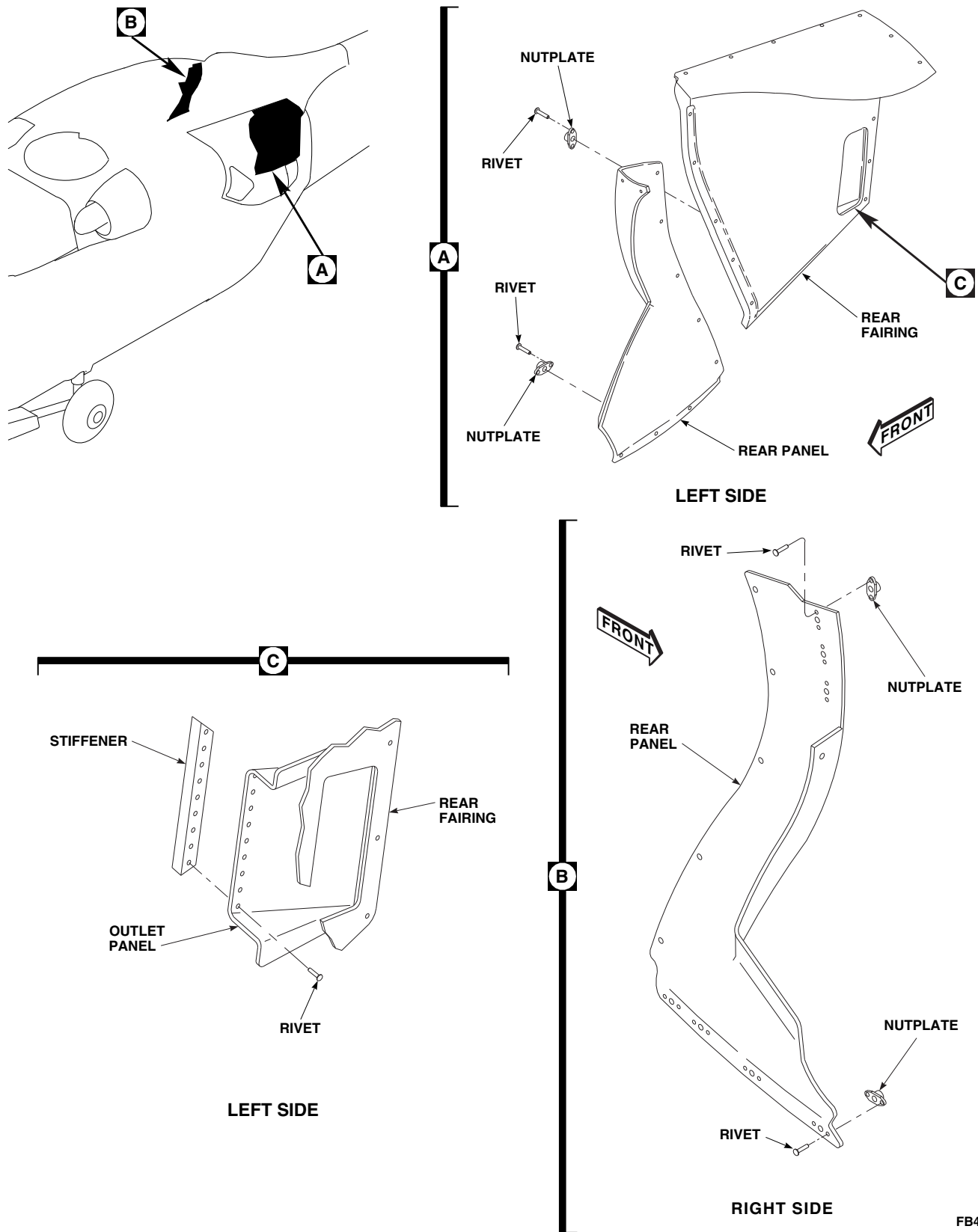


FIGURE 4-4-10-1. REPLACE HIRSS REAR FAIRINGS AND PANELS

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FIGURE 4-4-10-2. REPAIR HIRSS REAR FAIRINGS AND PANELS

4-4-11 HIRSS SUPPRESSOR SUPPORT MOUNTS.

PARAGRAPH BREAKDOWN

	PAGE
4-4-11.1 Replace HIRSS Suppres- sor Support Mounts	4-4-11-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 4-4-9

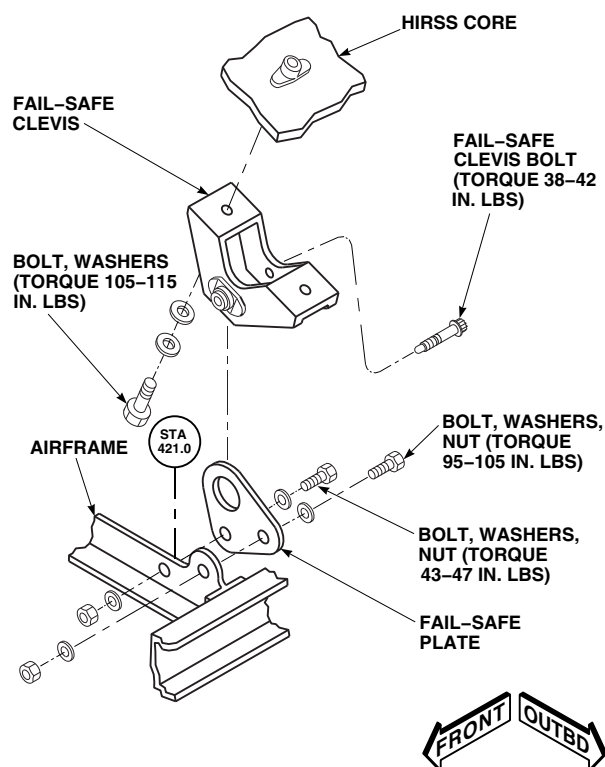
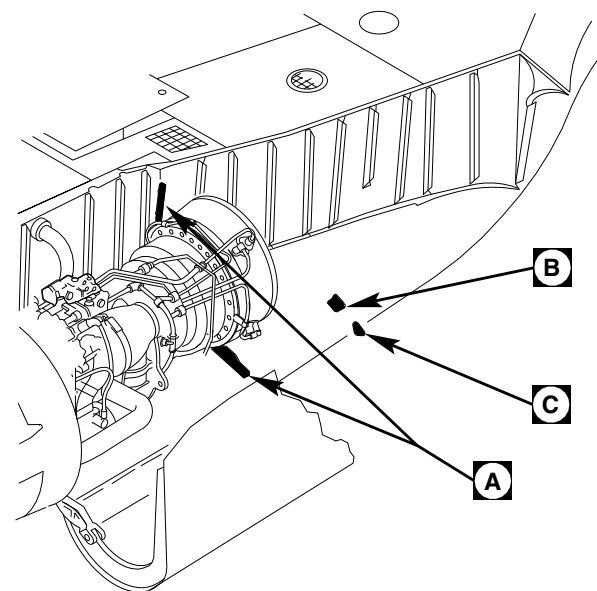
4-4-11.1 Replace HIRSS Suppressor Support Mounts.

NOTE

It is not necessary to remove HIRSS module if support mounts are replaced one at a time.

- If necessary, remove HIRSS module (PARA 4-4-9).
- Replace lower support angle as follows (Figure 4-4-11-1, Detail A):
 - If HIRSS is installed, remove bolts and washers securing lower support angle to bulkhead.
 - Remove bolts and washers securing support angle and shim to airframe. Remove support angle.
 - Apply sealing compound, Item 292, Appendix D, under boltheads and around edges of lower support angle.

- Position lower support angle and shim on airframe and install bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**
- If HIRSS is installed, install bolts and washers securing support angle to bulkhead. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**
- Replace firewall support angle as follows:
 - If HIRSS is installed, remove bolts and washers securing HIRSS to support angle.
 - Remove bolts and washers securing support angle to firewall. Remove support angle.
 - Position support angle on firewall and install bolts and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**
 - If HIRSS is installed, install bolts and washers securing support angle to HIRSS. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**



NOTES

1. INSERT BOLTS AND WASHERS WHEN ANGLE IS REMOVED.
2. LONG BOLTS INSTALLED IN OUTBOARD AND INBOARD HOLES.
3. INSERT BOLTS AND WASHERS WHEN FITTING IS REMOVED.

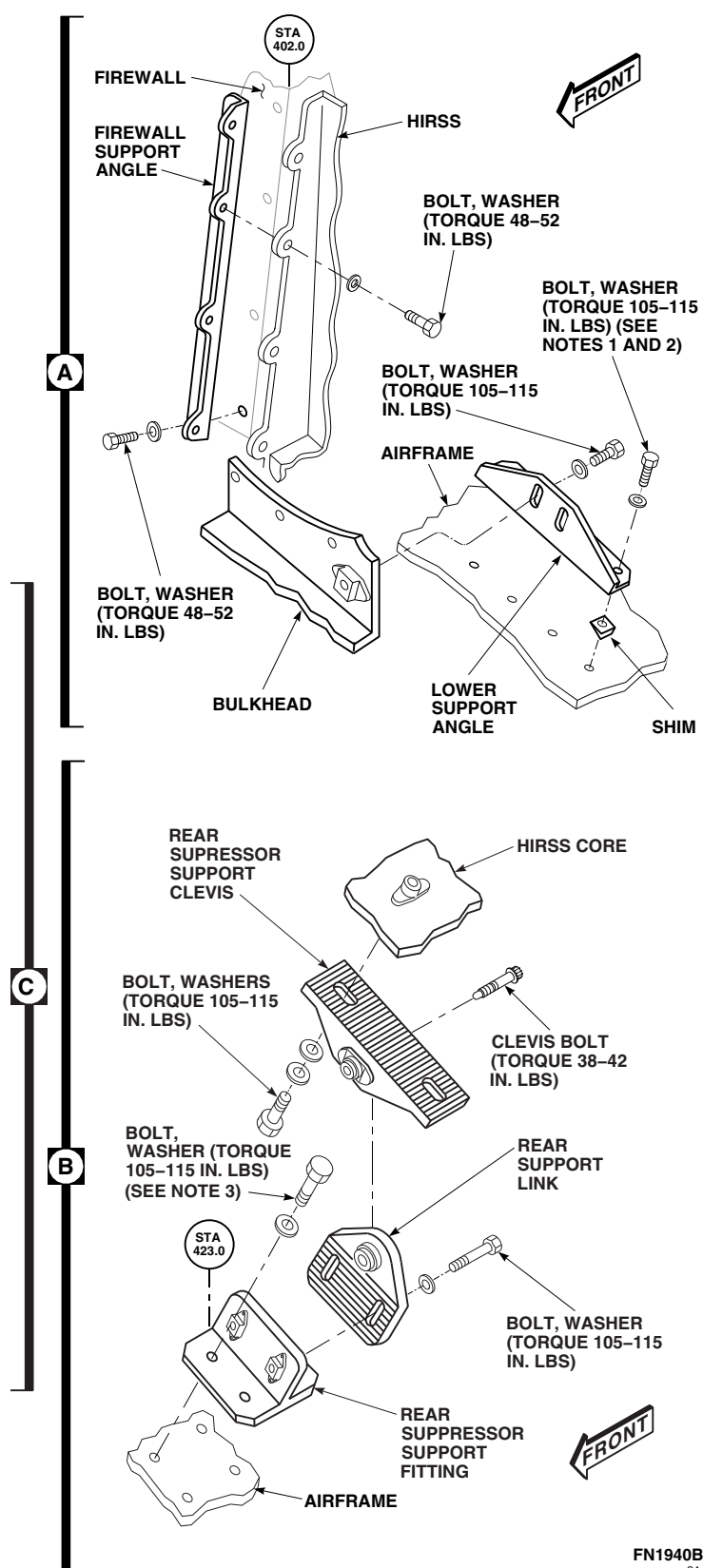
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FIGURE 4-4-11-1. REPLACE HIRSS SUPPRESSOR SUPPORT MOUNTS

- d. Replace rear suppressor support clevis, rear support link, and rear suppressor support fitting as follows (Figure 4-4-11-1, Detail B):

- (1) If HIRSS is installed, remove clevis bolt securing support clevis to support link.
- (2) Remove bolts and washers securing support clevis to HIRSS core. Remove clevis.
- (3) Remove bolts and washers securing support link to rear support fitting. Remove link.
- (4) Remove bolts and washers securing support fitting to airframe. Remove fitting.
- (5) Apply sealing compound, Item 292, Appendix D, under boltheads and around edges of support fitting.
- (6) Position support fitting on airframe and secure with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**
- (7) Position support clevis on HIRSS core and loosely secure with bolts and washers. Do not torque at this time.
- (8) Position support link on support fitting and loosely secure with bolts and washers. Do not torque at this time.

NOTE

If HIRSS is not installed, the following steps are not required. Final adjustment of support link and clevis, and torquing of bolts will be performed during HIRSS installation.

- (9) If HIRSS is installed, perform the following:
 - (a) Install clevis bolt securing support clevis to support link. **TORQUE BOLT TO 38 - 42 INCH-POUNDS.**
 - (b) Align support clevis and support link to obtain best possible alignment of clevis so that no preload exists.

- (c) **TORQUE BOLTS SECURING SUPPORT CLEVIS TO 105 - 115 INCH-POUNDS.**

NOTE

Rear support link must seat with the rear of suppressor support fitting after the bolts are torqued.

- (d) **TORQUE BOLTS SECURING SUPPORT LINK TO 105 - 115 INCH-POUNDS.**

- e. Replace fail-safe clevis and fail-safe plate as follows (Figure 4-4-11-1, Detail C):

- (1) If HIRSS is installed, remove fail-safe clevis bolt.
- (2) Remove bolts and washers securing fail-safe clevis to HIRSS core. Remove clevis.
- (3) Remove bolts, washers, and nuts securing fail-safe plate to airframe. Remove plate.
- (4) Install fail-safe plate on airframe and secure with bolts, washers, and nuts. **TORQUE OUTBOARD NUT TO 95 - 105 INCH-POUNDS. TORQUE INBOARD NUT TO 43 - 47 INCH-POUNDS.**
- (5) Install fail-safe clevis on HIRSS core and secure with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

NOTE

After torquing fail-safe clevis bolt, it is acceptable for a gap to exist between the underside of bolthead and the fail-safe clevis.

- (6) If HIRSS is installed, install fail-safe clevis bolt. **TORQUE BOLT TO 38 - 42 INCH-POUNDS.**
- f. If necessary, install HIRSS module (PARA 4-4-9).

4-4-12 HIRSS NACELLE FAIRING SUPPORT MOUNTS.

PARAGRAPH BREAKDOWN

	PAGE
4-4-12.1 Replace HIRSS Nacelle Fairing Support Mounts	4-4-12-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 4-4-13

4-4-12.1 Replace HIRSS Nacelle Fairing Support Mounts.

NOTE

Replacement of left and right, upper and lower, HIRSS nacelle fairing support mounts is the same.

4-4-12.1.1 Remove.

NOTE

It is not necessary to remove HIRSS nacelle fairing if mounts are replaced one at a time.

- If necessary, remove HIRSS nacelle fairing (PARA 4-4-13).
- Remove lower nacelle support link and lower nacelle support fitting as follows (Figure 4-4-12-1, Detail B):
 - If HIRSS nacelle fairing is installed, remove clevis bolt, washer, and nut securing lower nacelle support link to lower nacelle support fitting.

- Remove bolts and washers securing lower nacelle support fitting and shim(s) to lower nacelle fairing. Remove support fitting and shim(s).
 - Remove bolts, washers, and nuts securing lower nacelle support link to HIRSS core. Remove support link.
- c. Remove upper nacelle support link and upper nacelle support fitting as follows (Figure 4-4-12-1, Detail A):
- If HIRSS nacelle fairing is installed, remove clevis bolt securing upper nacelle support fitting to upper nacelle support clevis.
 - Remove bolts, washers, and nuts securing upper nacelle support fitting and shim(s) to upper nacelle fairing. Remove support fitting and shim(s).
 - Remove bolts, washers, and nuts securing upper nacelle support clevis to HIRSS core. Remove support clevis.

4-4-12.1.2 Install.

- a. Install lower nacelle support link and lower nacelle support fitting as follows (Figure 4-4-12-1, Detail B):

- (1) Secure lower nacelle support link to HIRSS core with bolts, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
- (2) Position lower nacelle support fitting on lower nacelle fairing. Install shim(s), as required, under support fitting to align support fitting with lower nacelle support link.
- (3) Secure lower nacelle support fitting to lower nacelle fairing with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

NOTE

Gap under bolthead is acceptable.

- (4) If HIRSS nacelle fairing is installed, install clevis bolt, washer, and nut securing lower nacelle support link to lower nacelle support fitting. **TORQUE BOLT TO 38 - 42 INCH-POUNDS.**

- b. Install upper nacelle support fitting and upper support clevis as follows (Figure 4-4-12-1, Detail A):

- (1) Secure upper nacelle support clevis to HIRSS core with bolts, washers, and nuts.

TORQUE NUTS TO 95 - 105 INCH-POUNDS.**NOTE**

- Maximum of three shims may be used under upper nacelle support fitting.
- Two different shims are available, a solid shim and a laminated shim. Laminated shim may appear to be two shims but should be counted as one. One shim is approximately 0.180-inch thick.

- (2) Position upper nacelle support fitting on upper nacelle fairing. Install shim(s), as required, under support fitting to align support fitting with support clevis.

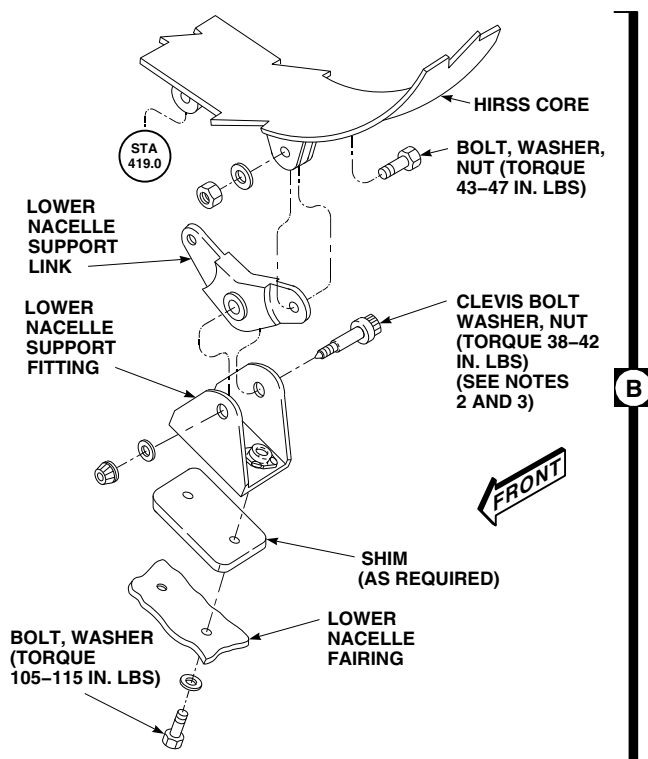
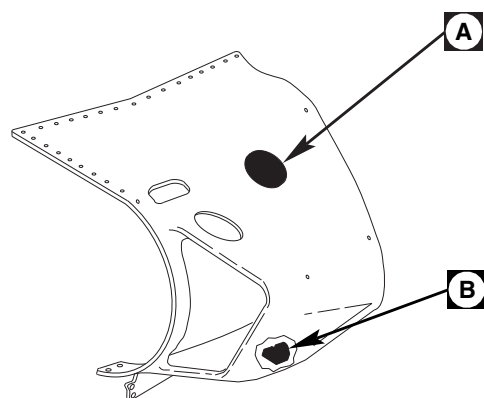
- (3) Secure upper nacelle support fitting to upper nacelle fairing with bolts, washers, and nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**

NOTE

Gap under bolthead is acceptable.

- (4) If HIRSS nacelle fairing is installed, install clevis bolt securing upper support fitting to upper support clevis. **TORQUE BOLT TO 38 - 42 INCH-POUNDS.**

- c. If necessary, install HIRSS nacelle fairing (PARA 4-4-13).



NOTES

1. MAXIMUM OF THREE SHIMS MAY BE USED UNDER UPPER NACELLE SUPPORT FITTING. TWO DIFFERENT SHIMS ARE AVAILABLE, A SOLID SHIM AND A LAMINATED SHIM. LAMINATED SHIM MAY APPEAR TO BE TWO SHIMS BUT SHOULD BE COUNTED AS ONE. ONE SHIM IS APPROXIMATELY 0.180-INCH THICK.
2. GAP UNDER BOLTHEAD IS ACCEPTABLE.
3. DIRECTION OF BOLTHEAD IS OPTIONAL (FRONT OR REAR).

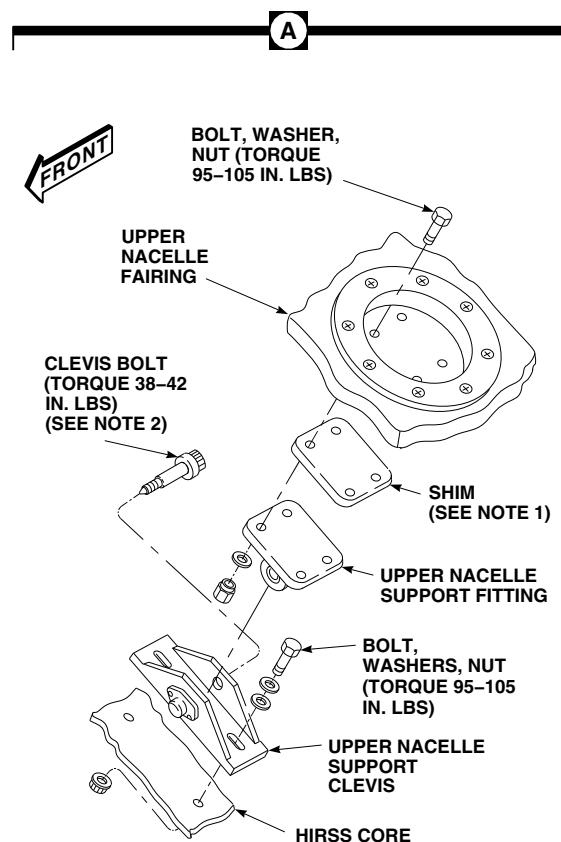


FIGURE 4-4-12-1. REPLACE HIRSS NACELLE FAIRING SUPPORT MOUNTS

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4-4-13 HIRSS NACELLE FAIRING.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
4-4-13.1	Replace HIRSS Nacelle Fairing	4-4-13-1	4-4-13.2	Repair HIRSS Nacelle Fairing	4-4-13-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Bit Set
- Drill, Rotary
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-6-1
- PARA 2-6-3
- PARA 2-6-5
- PARA 4-4-9
- PARA 4-4-12
- PARA 4-4-15

4-4-13.1 Replace HIRSS Nacelle Fairing.

NOTE

Replacement of left and right HIRSS nacelle fairing and attaching components is the same, except as noted.

4-4-13.1.1 Remove.

- Remove HIRSS extender (PARA 4-4-15).
- Remove HIRSS (PARA 4-4-9).
- Remove screws and washers securing nacelle fairing and splice strap to bulkhead (Figure 4-4-13-1, Sheet 1, Detail A).
- Remove clevis bolt from upper nacelle support fitting (Figure 4-4-13-1, Sheet 2, Detail B).
- Remove clevis bolt, washer, and nut from lower nacelle support fitting (Figure 4-4-13-1, Sheet 2, Detail C).

- Carefully lift nacelle fairing from HIRSS core.
- Remove bolts and washers from IPS collar and remove collar and suppressor module IPS blower duct from bulkhead (Figure 4-4-13-1, Sheet 2, Detail D).
- On right side only, remove bolts and washers securing suppressor module IPS blower duct to rear panel.

4-4-13.1.2 Inspect.

- Check nacelles for cracks. **NO CRACKS ALLOWED WITHIN 2-INCHES OF CORE SUPPORT OR NACELLE TO AIRFRAME ATTACHMENTS. ANY NUMBER OF CRACKS UP TO 1-INCH LONG ARE ALLOWED IN ALL OTHER AREAS.** If required, repair fiberglass (PARA 2-6-3).
- Check nacelle fairing for loose, broken, or missing hoist point nutplates. **NONE ALLOWED.** If required, replace nutplates.

- c. Check nutplates on firewall and bulkhead attachment areas. **TWO MISSING, LOOSE, OR BROKEN NUTPLATES ARE ALLOWED WITH AT LEAST TWO GOOD NUTPLATES BETWEEN DAMAGED NUTPLATES.** If damaged, replace nutplates.

4-4-13.1.3 Install.

- a. Install inlet particle separator (IPS) collar and suppressor module IPS blower duct on bulkhead with bolts and washers (Figure 4-4-13-1, Sheet 2, Detail D).
- b. On right side only, install bolts and washers securing suppressor module IPS blower duct to rear panel. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**
- c. With aid of assistant, carefully lift nacelle fairing into place on suppressor core and loosely install screws and washers securing nacelle fairing and splice strap to bulkhead (Figure 4-4-13-1, Sheet 1, Detail A).

NOTE

Gap under bolthead is acceptable.

- d. Install clevis bolt in upper support clevis and upper nacelle support fitting (Figure 4-4-13-1, Sheet 2, Detail B). **TORQUE BOLT TO 38 - 42 INCH-POUNDS.**

NOTE

Gap under bolthead is acceptable.

- e. Install clevis bolt, washer, and nut in lower nacelle support link and lower nacelle support fitting (Figure 4-4-13-1, Sheet 2, Detail C). **TORQUE BOLT TO 38 - 42 INCH-POUNDS.**
- f. Tighten screws securing nacelle fairing to bulkhead.
- g. Install HIRSS (PARA 4-4-9).
- h. Install HIRSS extender (PARA 4-4-15).

4-4-13.2 Repair HIRSS Nacelle Fairing.

NOTE

Repair of left and right nacelle fairing is the same, except as noted.

4-4-13.2.1 Disassemble.

- a. Remove upper and lower nacelle support fittings and shims (PARA 4-4-12).
- b. Remove suppressor module IPS blower duct components (PARA 4-4-13.1).
- c. Remove screws, washers, nuts, and support panel from nacelle fairing (Figure 4-4-13-2).
- d. Remove handhold by drilling out rivets using rotary drill and drill bit set.
- e. On left side, remove screws, washers, nuts, and rear panel support angle from nacelle fairing.
- f. Remove hoist point nutplates by drilling out rivets (Figure 4-4-13-2, Detail A).
- g. Remove screws securing fairing plate to nacelle fairing (Figure 4-4-13-2, Detail B).

4-4-13.2.2 Repair.

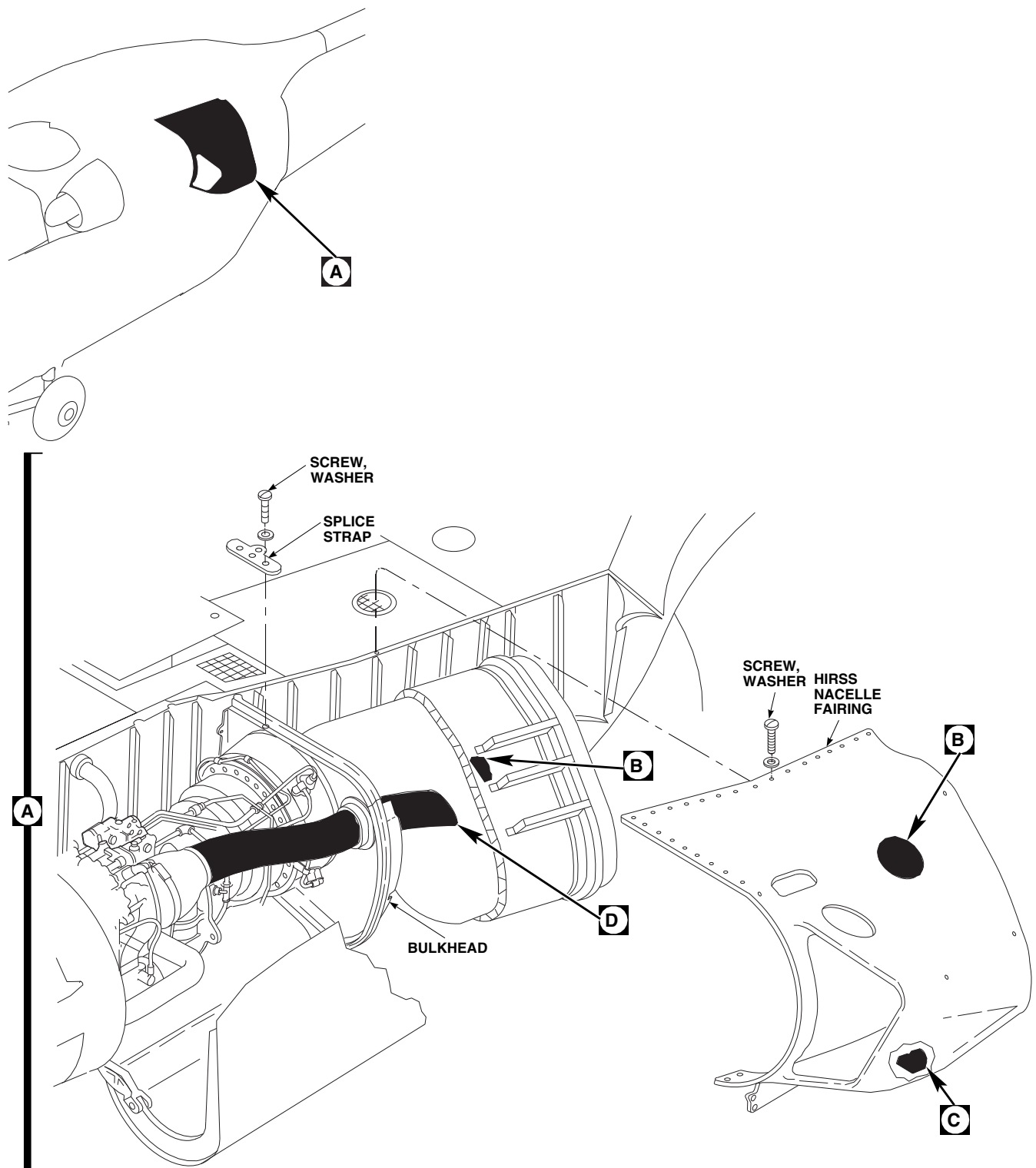
NOTE

Nacelle fairing is constructed of a fiberglass outer skin and an inner core of aluminum alloy 2024/F40 honeycomb.

- a. Evaluate damage to fairing skin and core (PARA 4-4-13.1 and PARA 2-6-1).
- b. If required, repair nacelle fairing outer skin (PARA 2-6-3).

4-4-13.2.3 Assemble.

- a. Install fairing plate on nacelle fairing with screws (Figure 4-4-13-2, Detail B).
- b. Install nutplates for hoist points with rivets (Figure 4-4-13-2, Detail A).



NOTES

1. GAP UNDER BOLTHEAD IS ACCEPTABLE.
2. DIRECTION OF BOLTHEAD IS OPTIONAL (FRONT OR REAR).

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FIGURE 4-4-13-1. REPLACE HIRSS NACELLE FAIRING (SHEET 1 OF 2)

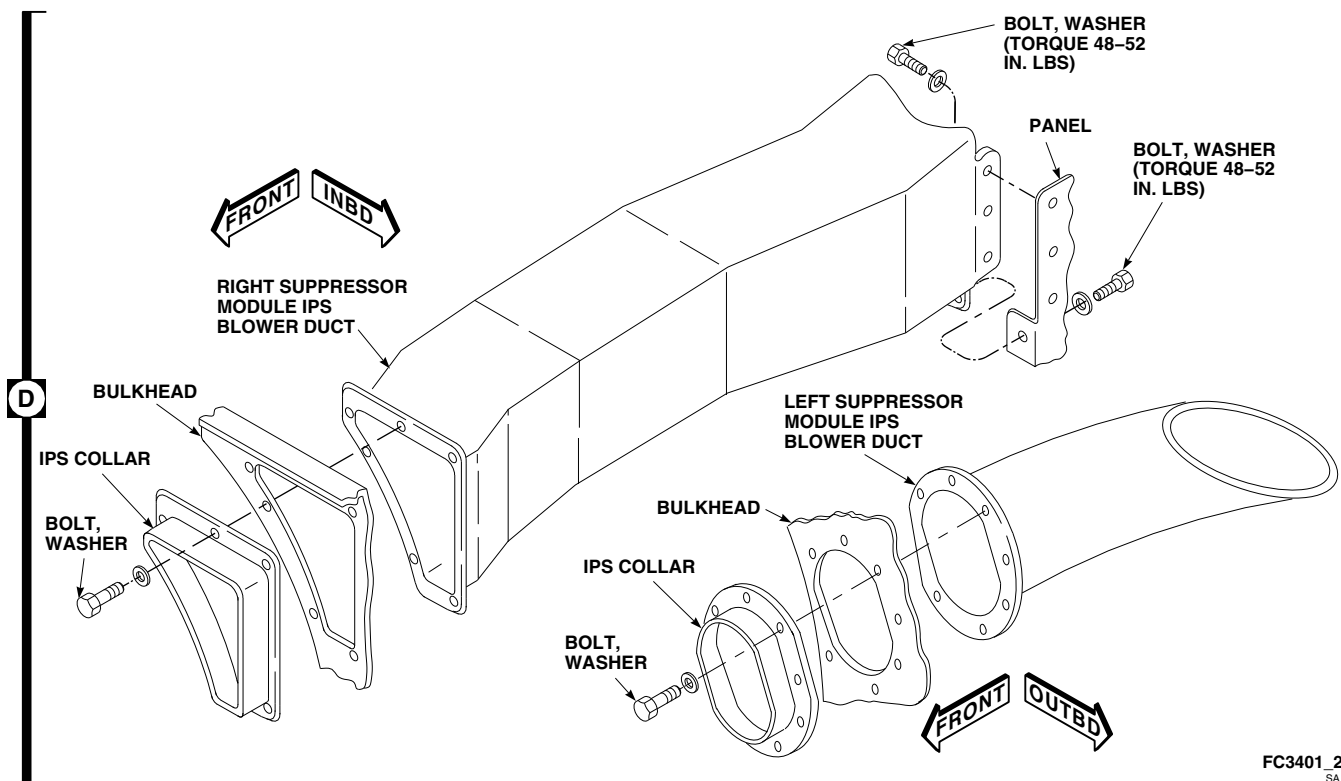
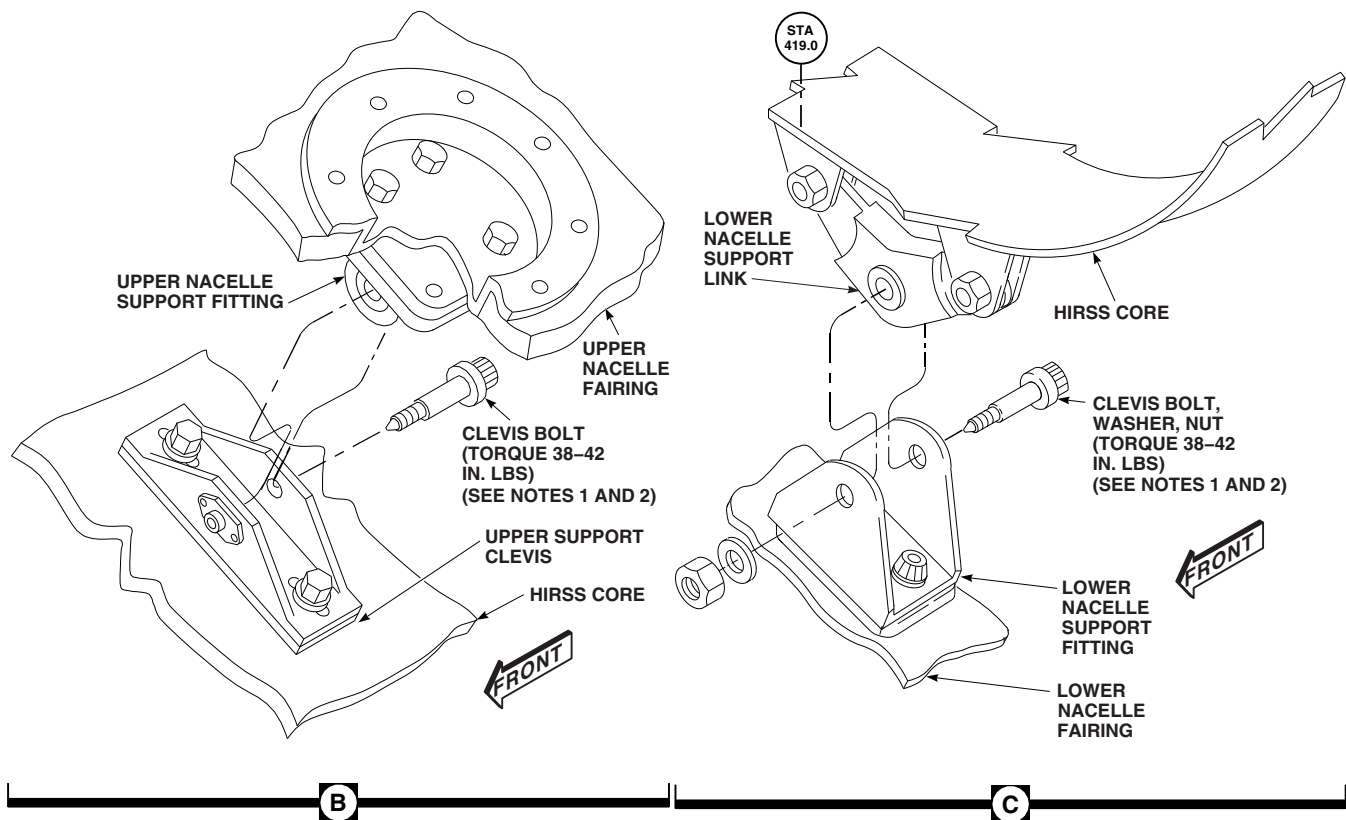


FIGURE 4-4-13-1. REPLACE HIRSS NACELLE FAIRING (SHEET 2 OF 2)

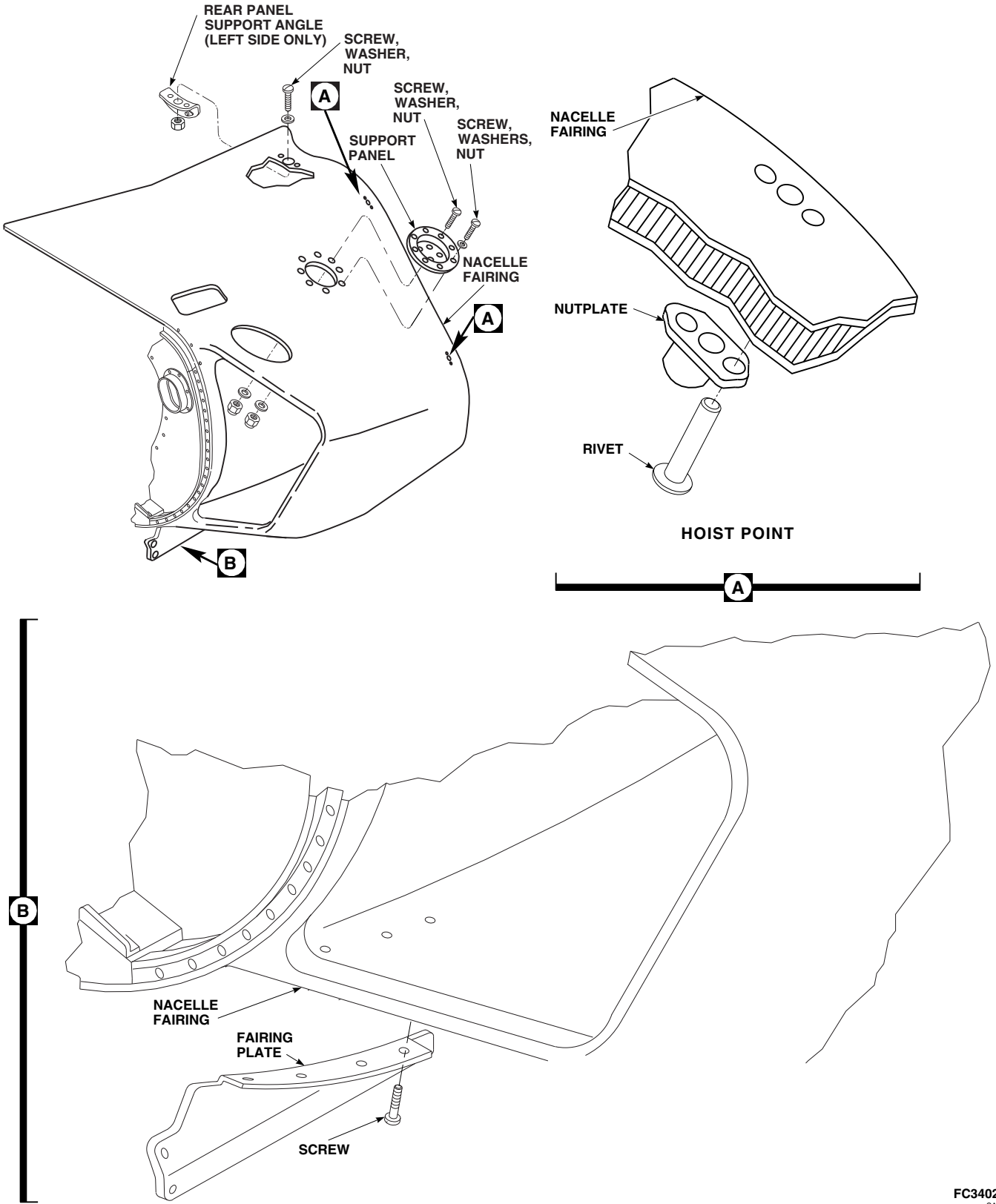


FIGURE 4-4-13-2. REPAIR HIRSS NACELLE FAIRING

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- c. Install handhold with rivets.
- d. Install suppressor module IPS ducting (PARA 4-4-13.1).
- e. On left side, install rear panel support angle on nacelle fairing with screws, washers, and nuts (Figure 4-4-13-2).
- f. Install support panel in opening on top of nacelle fairing with screws, washers, and nuts.
- g. Install upper and lower nacelle support fittings and shims (PARA 4-4-12).
- h. Touch up paint in repaired areas as required (PARA 2-6-5).

4-4-14 HIRSS SUPPRESSOR CORE AND BAFFLE.

PARAGRAPH BREAKDOWN

	PAGE
4-4-14.1 Replace HIRSS Suppressor Core and Baffle	4-4-14-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Fluorescent Penetrant Inspection Kit
- Light Source, Strong
- Plastic Mallet
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 4-4-13
- PARA 4-4-17
- PARA 4-5-8

4-4-14.1 Replace HIRSS Suppressor Core and Baffle.

NOTE

Replacement of left and right HIRSS suppressor core and baffles is the same, except as noted.

- On right bulkhead only, remove bolts and washers from inlet particle separator (IPS) collar and remove collar and IPS blower duct (PARA 4-4-13).
- Remove screws, washers, and nuts and remove bulkhead from bulkhead support flange on suppressor core (Figure 4-4-14-1, Detail A).

4-4-14.1.1 Remove.

- Remove HIRSS nacelle fairing (PARA 4-4-13).
- Remove bolts and washers, and slide baffle out of suppressor core (Figure 4-4-14-1).

4-4-14.1.2 Inspect.

- Table 4-4-14-1 defines allowable damage and wear to suppressor core (Figure 4-4-14-2, Sheet 1 and Sheet 2).
- Table 4-4-14-2 defines allowable damage and wear to baffle (Figure 4-4-14-3).

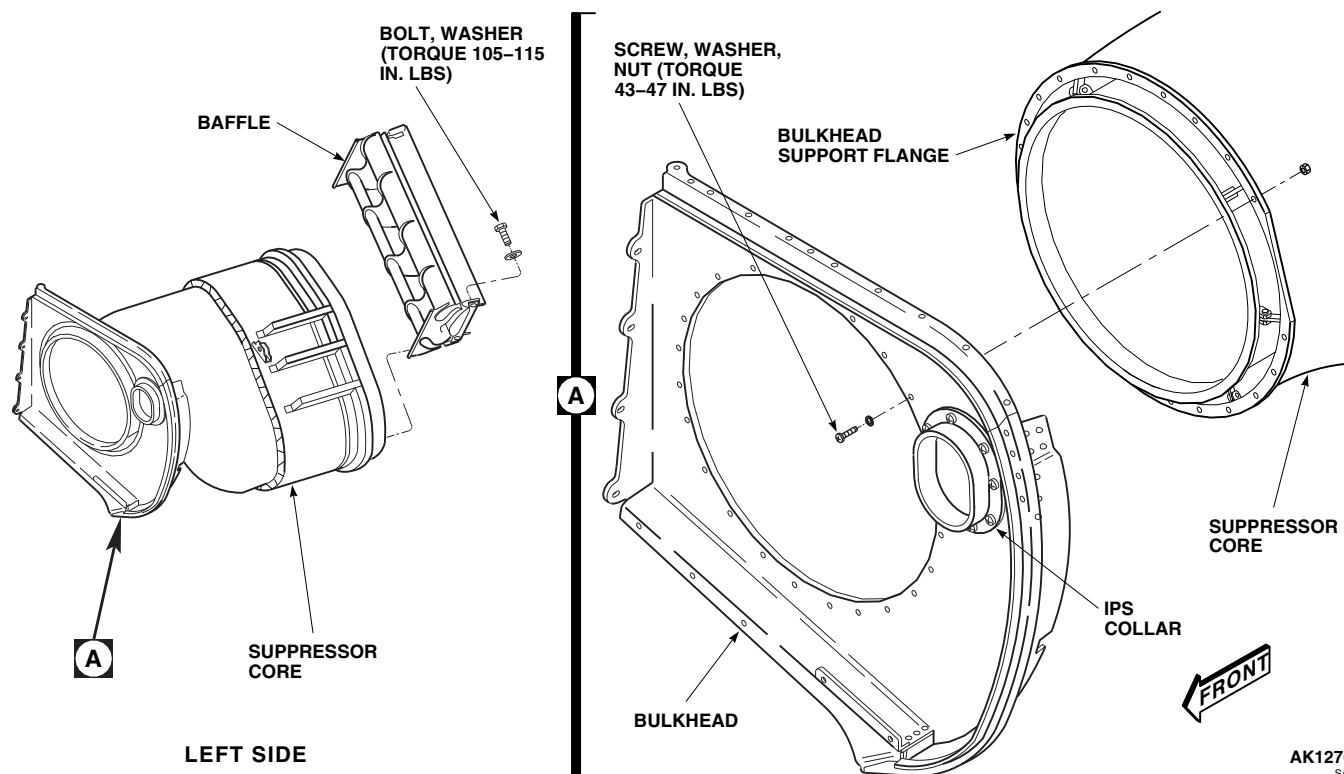


FIGURE 4-4-14-1. REPLACE HIRSS SUPPRESSOR CORE AND BAFFLE

Table 4-4-14-1. HIRSS Suppressor Core Inspection

Component	Type and Size	Repair Action
Overall Assembly	Discoloration	
	- - Any amount.	No repair required.
	Nicks	
	- - Any amount.	No repair required.
	Scratches	
	- - Any amount.	No repair required.
	Dents	
	- - Up to 2-inches in diameter and 1/4-inch deep.	No repair required. Touchup emissive coating, as required (PARA 4-4-17).

Table 4-4-14-1. HIRSS Suppressor Core Inspection (Cont)

Component	Type and Size	Repair Action
	- - Dents larger than specified limits.	Straighten dents by lightly tapping with plastic mallet and if necessary, touch up emissive coating (PARA 4-4-17). If dents cannot be straightened, replace core.
	Cracks (See Note)	
	- - Within 2-inches of mounts or stage-to-stage attachment locations, except stage 2 duct.	Unacceptable. Replace core.
	Corrosion	
	- - Any amount that does not penetrate component.	No repair required.
	- - Any corrosion which causes holes or breaks in component.	Unacceptable. Replace core.
	Warping	
	- - With baffle installed, view core from rear to see if any first stage inner duct surface is visible around line-of-sight shields.	Unacceptable. Replace core.
	- - With strong light source placed in first stage duct cavity and baffle installed behind light, view core from rear to see if any direct light is visible from first stage inner duct.	Unacceptable. Replace core.
	Missing or damaged emissive coating	
	- - Up to 2-square inches.	No repair required.
	- - More than 2-square inches.	Touch up emissive coating (PARA 4-4-17).
	Loose rivets	

Table 4-4-14-1. HIRSS Suppressor Core Inspection (Cont)

Component	Type and Size	Repair Action
Bulkhead	- - Up to 10% of rivets in any joint other than nutplates.	No repair required.
	- - Greater than above limit or any nutplate rivets.	Replace rivets.
	Missing rivets	
	- - Any rivets.	Replace rivets.
	Cracks (See Note)	
Finger Seals	- - Within 2-inches of bulkhead where it attaches to support flange on core.	Unacceptable. Replace bulkhead.
	- - In all other areas, any number of cracks up to 1-inch long which do not intersect or will not cause a piece to break off bulkhead.	No repair required.
	Cracks (See Note)	
	- - Breaking into rivet holes in AMS 5510 bracket or bracket holder.	Unacceptable. Replace inlet seal (PARA 4-5-8).
	- - In all other areas of bracket, any number of cracks up to 0.50-inch long which do not intersect or will not cause a piece to break off bracket.	No repair required.
	- - Any crack greater than 0.25-inch long in INCO 718 finger seal and brace.	Unacceptable. Replace inlet seal (PARA 4-5-8).
	- - Cracks in above component not more than 0.25-inch long.	No repair required.
I	Bends or deformation	
	- - Fingers bent or deformed out of plane $\pm 1/16$ -inch.	Bend back into shape. Inspect for cracks. If cracked, replace inlet seal (PARA 4-5-8).

Table 4-4-14-1. HIRSS Suppressor Core Inspection (Cont)

Component	Type and Size	Repair Action
	- - Greater than above limits.	Unacceptable. Replace inlet seal (PARA 4-5-8).
	Missing seal fingers	Up to three fingers may be missing provided spacing between missing fingers is greater than 60°.
Stage 1 Inner Duct Assy	Cracks (See Note)	
	- - In turning vanes, three cracks per vane up to 0.50-inch long not in same axial plane and do not intersect.	No repair required.
	- - In any other area of duct skin or turning vane supports, any number of cracks up to 0.50-inch long which do not intersect or will not cause a piece to break off duct.	No repair required.
	- - Greater than above limits or breaking into forward link supports.	Unacceptable. Replace core.
	Vane bending or distortion	
	- - Up to 0.50-inch as long as front/rear and/or up/down movement of front vanes is not more than 0.25-inch or bending/twisting is not enough to disconnect vanes.	No repair required.
	- - Greater than above limits.	Unacceptable. Replace core.
Stage 1 Outer Duct Assy	Cracks (See Note)	
	- - In duct skin or line-of-sight shields, any number of cracks up to 1-inch long that do not intersect or will not cause a piece to break off component.	No repair required.
	- - In wiggle strips, two cracks per leg up to 0.25-inch long which do not intersect.	No repair required.

Table 4-4-14-1. HIRSS Suppressor Core Inspection (Cont)

Component	Type and Size	Repair Action
Stage 2 Duct	- - Greater than above limits.	Unacceptable. Replace core.
	- - Within 2-inches of bulkhead support flange or breaking into forward link supports or rivet holes.	Unacceptable. Replace core.
	Cracks (See Note)	
	- - Breaking into baffle guide pin or slot.	Unacceptable. Replace core.
	- - Within 2-inches of mounts or stage-to-stage attachment locations at 8 or more locations with a length of 2-inches each or longer, within 1 1/4-inches of another crack with individual cracks having no more than one connecting perpendicular segment not more than 1 1/2-inches long.	Unacceptable. Replace core.
	- - In all other areas, any number of cracks up to 1-inch long which do not intersect or will not cause a piece to break off duct.	No repair required.
	- - Greater than above limits.	Unacceptable. Replace core.
	Missing or damaged emissive coating	
	- - Up to 6-square inches.	No repair required.
	- - More than 6-square inches.	Touch up emissive coating (PARA 4-4-17).
Baffle Guide Pin	Wear	
	- - Diameter of pin not less than 0.218-inch.	No repair required.
	- - Diameter of pin less than 0.218-inch.	Unacceptable. Replace core.
	Missing or loose	Unacceptable. Replace core.

Table 4-4-14-1. HIRSS Suppressor Core Inspection (Cont)

Component	Type and Size	Repair Action
Baffle Rails	Bends or distortion	Unacceptable. Replace core.
Second/Third Stage Wiggle Strips	Cracks (See Note)	
	- - Two per leg, not more than 0.25-inch long, which do not intersect.	No repair required.
	- - Greater than above limits.	Unacceptable. Replace core.
	Distortion	
	- - Any amount as long as gap dimension between second and third stage remains same.	No repair required.
Stage 3 Duct	- - Any distortion which reduces gap.	Unacceptable. Replace core.
	Cracks (See Note)	
	- - Any size against or breaking into mount plates or brackets.	Unacceptable. Replace core.
	- - In all other areas, any number of cracks up to 1-inch long that do not intersect or will not cause a piece to break off duct.	No repair required.

NOTE:

Fluorescent penetrant inspect, Type I, Method C, using fluorescent penetrant inspection kit (ASTM E1417).

Table 4-4-14-2. HIRSS Baffle Inspection

Component	Type and Size	Repair Action
Baffle Guide Pin	Wear	
	- - Diameter of pin not less than 0.218-inch.	No repair required.

Table 4-4-14-2. HIRSS Baffle Inspection (Cont)

Component	Type and Size	Repair Action
Baffle Guide Slot	- - Diameter of pin less than 0.218-inch.	Unacceptable. Replace baffle.
	Missing or loose	Unacceptable. Replace baffle.
	Cracks (See Note) or bends	Unacceptable. Replace baffle.
	Wear	
	- - Width of slot not more than 0.385-inch.	No repair required.
Endplates	- - Greater than above limit.	Unacceptable. Replace baffle.
	Cracks (See Note)	
	- - Breaking into mounting bolt-holes.	Unacceptable. Replace baffle.
	- - All other areas, any number of cracks not more than 0.50-inch long.	No repair required.
	- - Greater than above limits.	Unacceptable. Replace baffle.
	Bending or twisting	
	- - Any amount as long as baffle fits properly in rails, guide pin slot, and forward fish mouth seal on stage 2 duct.	No repair required.
Splitter Plates and Support Brackets	- - Greater than above limits.	Unacceptable. Replace baffle.
	Cracks (See Note)	
	- - In parent metal, any number of cracks up to 0.50-inch long which do not intersect or will not cause a piece to break off plates or brackets.	No repair required.

Table 4-4-14-2. HIRSS Baffle Inspection (Cont)

Component	Type and Size	Repair Action
	- - In braze, any cracks not more than 1-inch long.	No repair required.
	- - Greater than above limits.	Unacceptable. Replace baffle.
Baffle Cooling Slots (six in small baffles and two in large baffles)	Distortion	
	- - Any amount as long as gap dimension of slot remains same.	No repair required.
	- - Any distortion which reduces gap.	Unacceptable. Replace baffle.
Small Baffle Cooling Slot Standoff Pins	Loose, bent, missing	Unacceptable. Replace baffle.
Baffles	Cracks (See Note)	
	- - Breaking into small baffle standoff pins or expansion slots.	Unacceptable. Replace baffle.
	- - Breaking into large baffle hinge support pivots.	Unacceptable. Replace baffle.
	- - All other areas on baffles, any number of cracks up to 1-inch long which do not intersect or will not cause a piece to break off baffles.	No repair required.
	- - Greater than above limits.	Unacceptable. Replace baffle.
Large Baffle Hinge Support Pivots	Braze	
	- - Up to 20% of any joint may be damaged or broken.	No repair required.
	- - Greater than above limits.	Unacceptable. Replace baffle.
Overall Baffle Assy	Distortion	

Table 4-4-14-2. HIRSS Baffle Inspection (Cont)

Component	Type and Size	Repair Action
	- - Up to 0.25-inch in any area beyond original baffle dimensions.	No repair required.
	- - Greater than above limits.	Unacceptable. Replace baffle.

NOTE:

Fluorescent penetrant inspect, Type I, Method C, using fluorescent penetrant inspection kit (ASTM E1417).

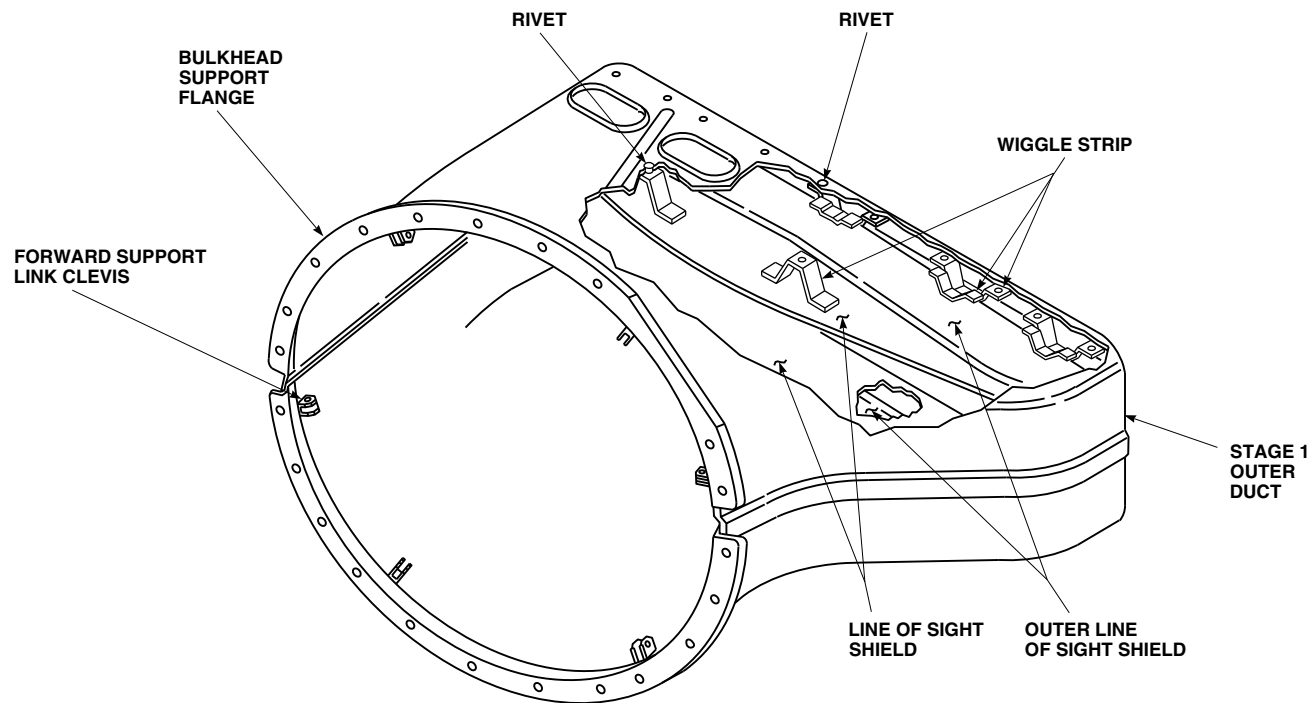
4-4-14.1.3 Install.

- a. Install bulkhead on suppressor core bulkhead support flange with screws, washers, and nuts (Figure 4-4-14-1, Detail A). **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
- b. On right bulkhead only, install inlet particle separator (IPS) collar and IPS blower duct , using bolts and washers (PARA 4-4-13).

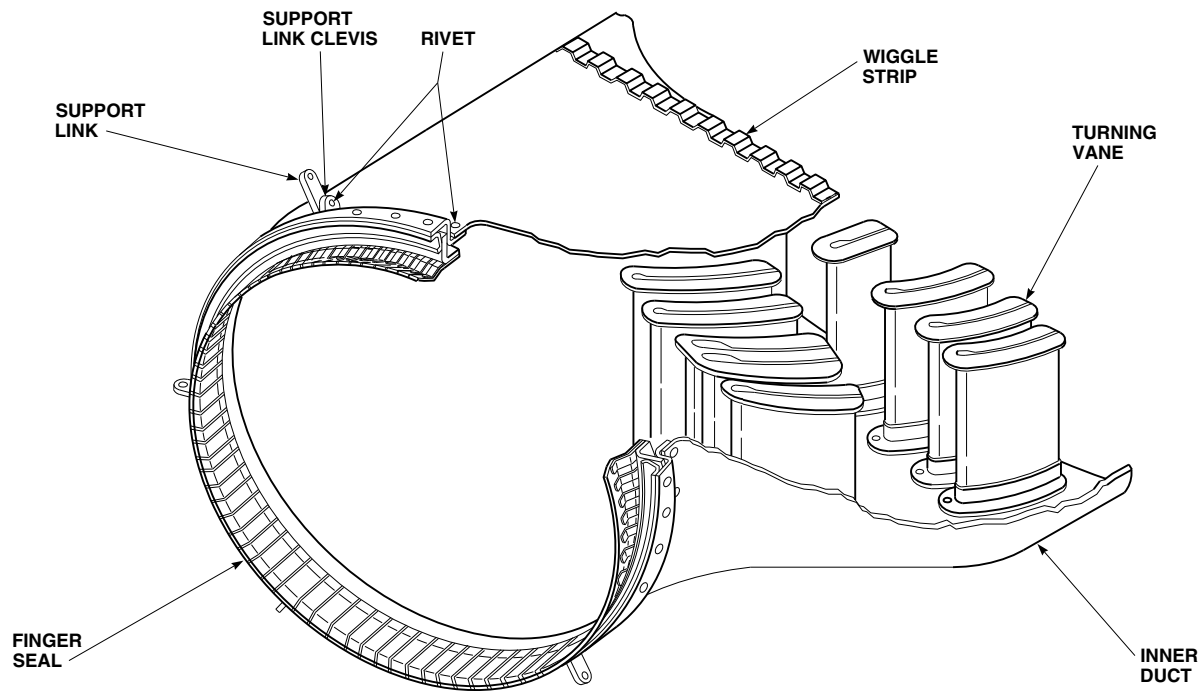
- c. Apply light coat of antiseize compound, Item 74, Appendix D, to threads of baffle installation bolts.
- d. Slide baffle into core. Install bolts and washers (Figure 4-4-14-1). **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**
- e. Install HIRSS nacelle fairing (PARA 4-4-13).

NOTE

If helicopter is not operated in a hostile environment, it is not necessary to install baffle.



STAGE 1 OUTER DUCT AND
LINE OF SIGHT SHIELDS



STAGE 1 INNER DUCT

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FIGURE 4-4-14-2. INSPECT HIRSS SUPPRESSOR CORE (SHEET 1 OF 2)

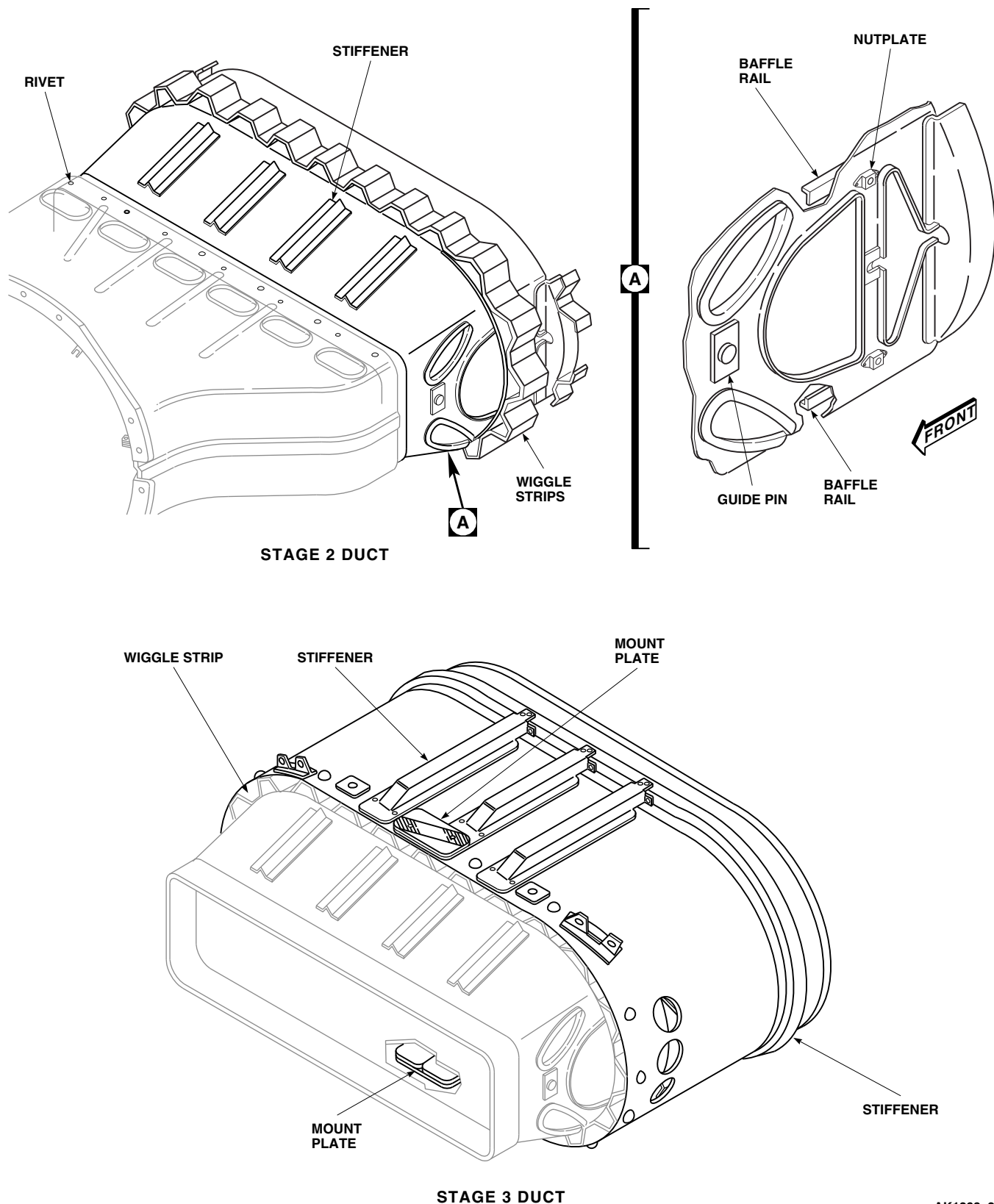


FIGURE 4-4-14-2. INSPECT HIRSS SUPPRESSOR CORE (SHEET 2 OF 2)

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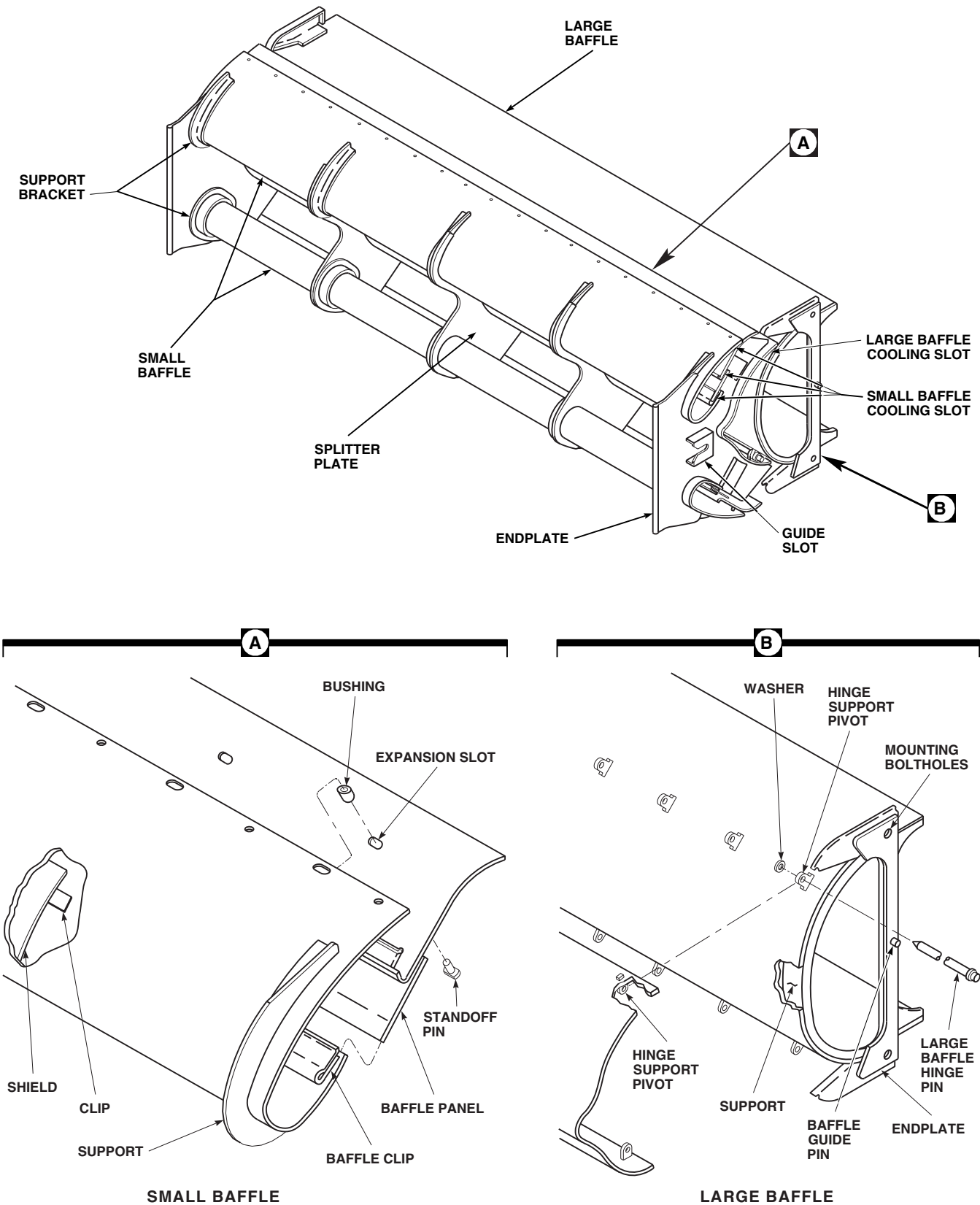


FIGURE 4-4-14-3. INSPECT HIRSS BAFFLE

4-4-15 HIRSS EXTENDER.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
4-4-15.1	Replace HIRSS Extender	4-4-15-1	4-4-15.2	Repair HIRSS Extender	4-4-15-5

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill
- Fluorescent Penetrant Inspection Kit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 55, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-2

4-4-15.1 Replace HIRSS Extender.

NOTE

Replacement of left and right HIRSS extenders is the same.

4-4-15.1.1 Remove.

NOTE

HIRSS extender must be supported during removal.

- With assistant supporting HIRSS extender, remove bolts and washers securing inboard support struts to rear panel and outboard sup-

port strut to fuselage (Figure 4-4-15-1, Sheet 2, Detail B and Detail C).

- Remove bolts, washers, and nuts securing inboard support struts to support strut angle and outboard support strut to angle.
- Remove bolts, washers, spacers, and nuts securing upper support channels and lower support channel to nacelle fairing. Remove HIRSS extender (Figure 4-4-15-1, Sheet 1, Detail A).

4-4-15.1.2 Inspect.

Table 4-4-15-1 defines allowable damage and wear to HIRSS extender.

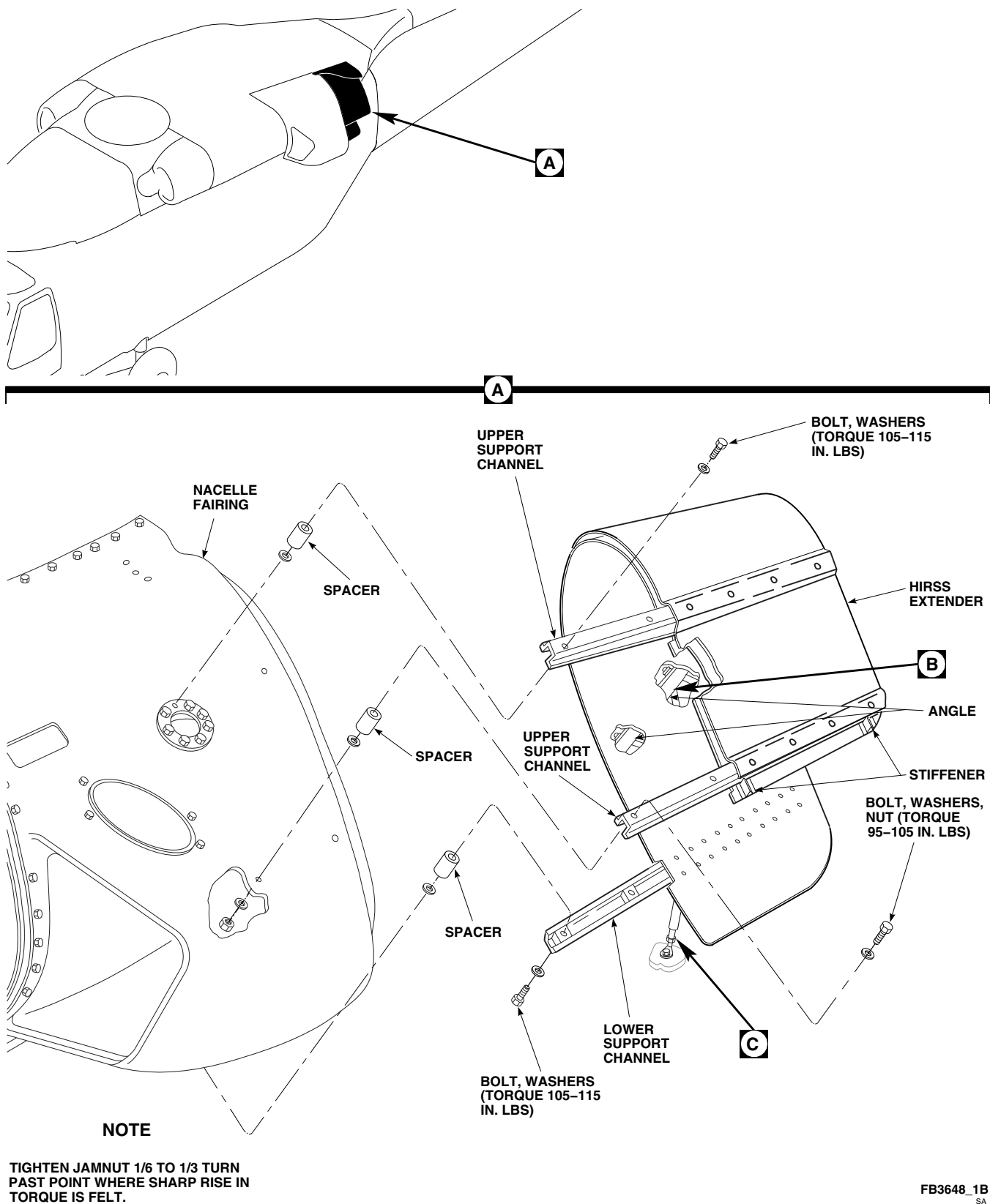


FIGURE 4-4-15-1. REPLACE HIRSS EXTENDER (SHEET 1 OF 2)

4-4-15-2

Change 10

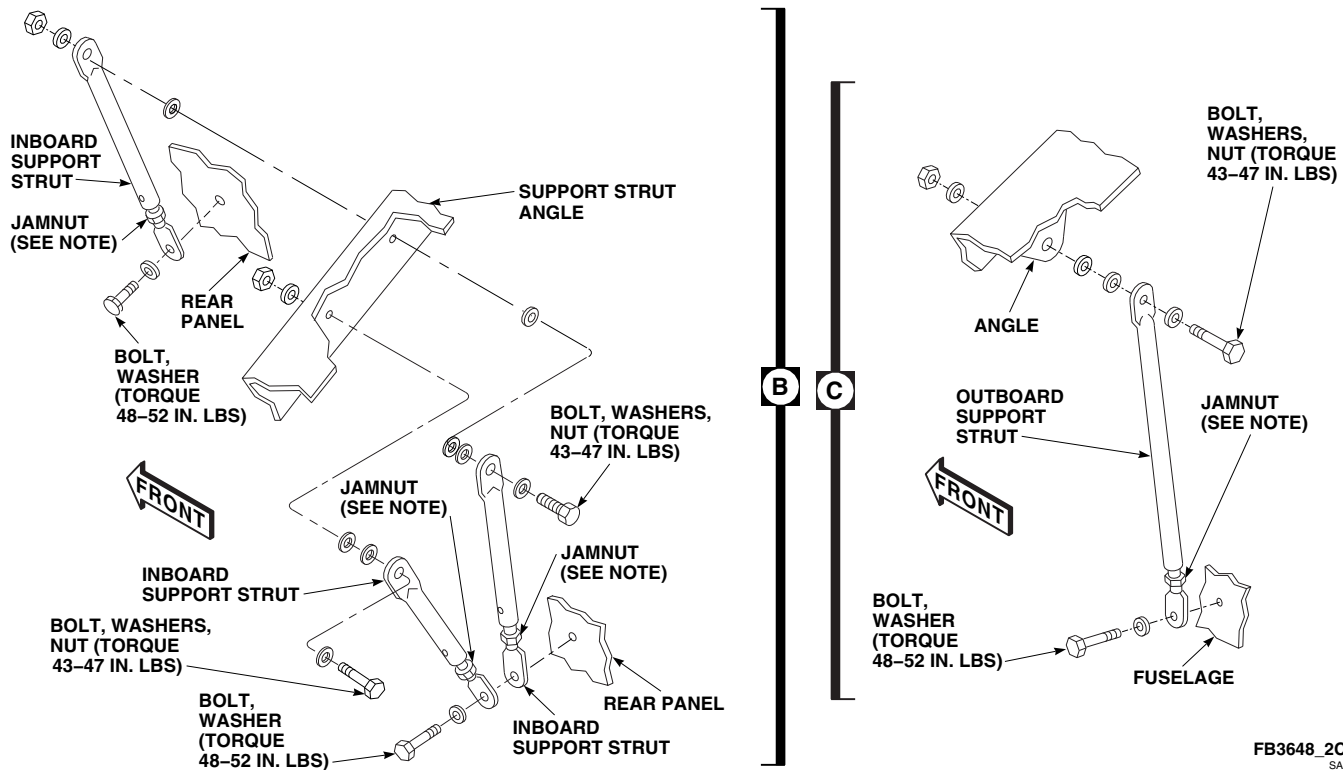


FIGURE 4-4-15-1. REPLACE HIRSS EXTENDER (SHEET 2 OF 2)

Table 4-4-15-1. HIRSS Extender Inspection

Component	Type and Size	Repair Action
Extender	Cracks (See Note)	
	- - In support strut angles.	Repair support strut angles (PARA 4-4-15.2.2).
	- - In front lower surface, at lower support channel attachment point, less than ½ the width of reinforcing doubler.	Install reinforcing doubler (PARA 4-4-15.2.2).
	- - More than ½ the width of reinforcing doubler.	Unacceptable. Replace HIRSS extender.
	- - In support beams, struts, within 3 inches of support beams, strut attach angle, or any lower surface angle or channel.	Unacceptable. Replace HIRSS extender.

Table 4-4-15-1. HIRSS Extender Inspection (Cont)

Component	Type and Size	Repair Action
	- - All other locations any amount not over 1 inch in length, that does not intersect or will not cause a piece of extender to fall off.	Repair weld (See Note).
	Loose, broken, or missing fasteners	
	- - None allowed in support beams or struts.	Unacceptable. Replace extender.
	- - All other areas up to 15% of fasteners damaged, with at least two good fasteners between damaged fasteners.	Replace fastener.

NOTE:

Using fluorescent penetrant inspection kit, fluorescent penetrant inspect per ASTM E1417, Type I, Method C.

4-4-15.1.3 Install.**NOTE**

HIRSS extender must be supported during installation.

- a. Slide HIRSS extender over nacelle fairing and loosely secure upper and lower support channels with bolts, washers, spacers, and nuts (Figure 4-4-15-1, Sheet 1, Detail A).

NOTE

One washer may be installed between strut assemblies and rear panels/fuselage to prevent chafing between HIRSS fairing and strut.

- b. With assistant supporting HIRSS extender, secure inboard support struts to rear panel and outboard support strut to fuselage with bolts and washers (Figure 4-4-15-1, Sheet 2, Detail B and Detail C). Do not tighten hardware now.
- c. Secure inboard support struts to support strut angle and outboard support strut to angle with bolts, washers, and nuts. Do not tighten hardware now.
- d. **TORQUE UPPER AND LOWER SUPPORT CHANNEL BOLTS TO 105 - 115 INCH-POUNDS** (Figure 4-4-15-1, Sheet 1, Detail A).
- e. **TORQUE UPPER SUPPORT CHANNEL NUT TO 95 - 105 INCH-POUNDS.**

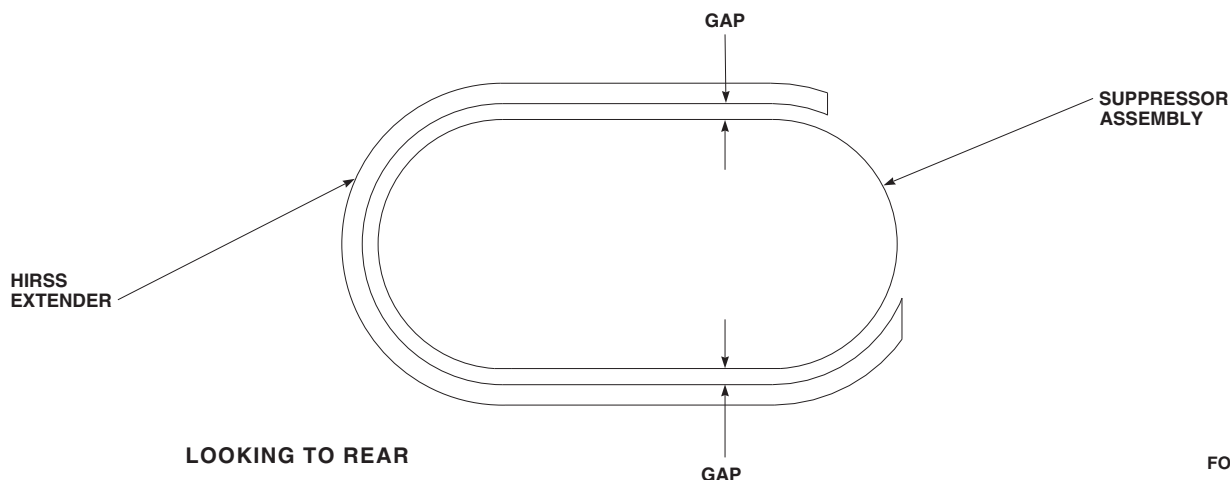


FIGURE 4-4-15-2. ADJUST HIRSS EXTENDER

NOTE

- Adjust strut length using adjustable rod ends as necessary to allow strut installation.
 - Inboard strut, 70308-03611-050, may not install properly to inboard support strut angle due to radius block being in the way. Inboard strut may be trimmed a maximum of $\frac{3}{16}$ -inch in order to obtain proper installation. Trim minimum amount necessary to obtain proper installation.
- f. Adjust length of inboard and outboard support struts to allow for proper installation of struts. **TIGHTEN JAMNUT $\frac{1}{4}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT** (Figure 4-4-15-1, Sheet 2, Detail B and Detail C). Make sure clearance exists between HIRSS extender and suppressor assembly (Figure 4-4-15-2).
 - g. **TORQUE UPPER SUPPORT STRUT ASSEMBLY FITTING NUTS TO 43 - 47 INCH-POUNDS** (Figure 4-4-15-1, Sheet 2, Detail B and Detail C).
 - h. **TORQUE LOWER SUPPORT STRUT ASSEMBLY FITTING BOLTS TO 48 - 52 INCH-POUNDS.**
 - i. Make sure area is clean and free of foreign material.

4-4-15.2 Repair HIRSS Extender.

NOTE

Repair of left and right HIRSS extender is the same.

4-4-15.2.1 Remove.

Remove HIRSS extender (PARA 4-4-15.1.1).

4-4-15.2.2 Repair.

NOTE

- Repair of HIRSS extender is limited to replacement of inboard support strut angle and addition of reinforcing doubler for lower support channel.
 - If repair kits are unavailable, use locally made parts and materials and install (PARA 2-6-2).
- a. Evaluate damage to HIRSS extender (PARA 4-4-15.1.2).
 - b. Replace inboard support strut angle as follows:
 - (1) Using drill, drill out rivets securing inboard support strut angle to HIRSS extender (Figure 4-4-15-3, Detail B).
 - (2) Install replacement inboard support strut angle, radius block, filler plate, and angle

clip and secure to extender with pins, collars, and rivets.

c. Add reinforcing doubler as follows:

- (1) Using drill, drill out rivets securing front end of lower support angle to HIRSS extender (Figure 4-4-15-3, Detail A).

- (2) Install doubler and fillers on HIRSS extender and secure using rivets wet with adhesive, Item 55, Appendix D.

4-4-15.2.3 Install.

Install HIRSS extender (PARA 4-4-15.1.3).

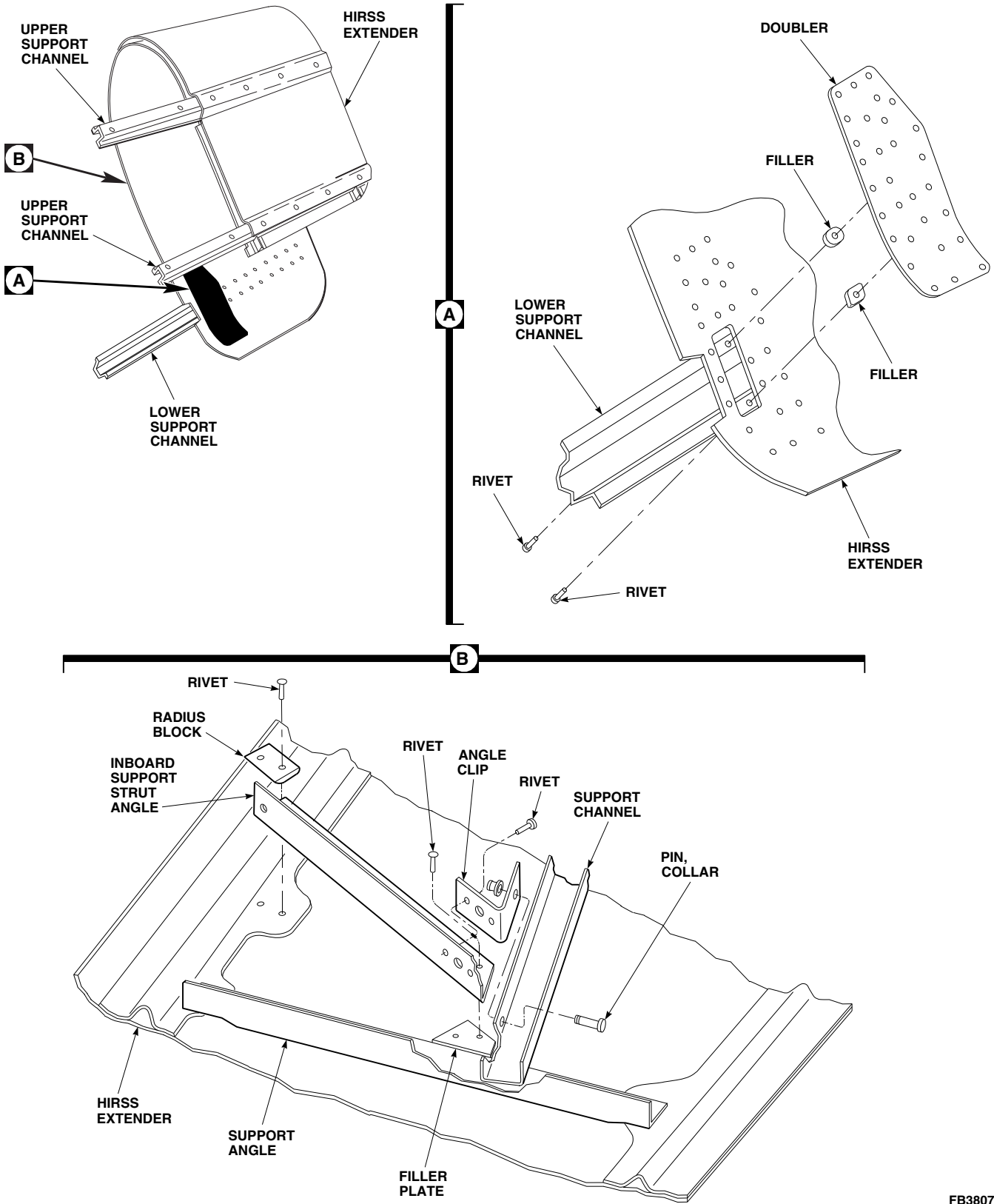


FIGURE 4-4-15-3. REPAIR HIRSS EXTENDER

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4-4-16 HIRSS BULKHEAD REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
4-4-16.1 Repair HIRSS Bulkhead	4-4-16-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit

References

- PARA 4-4-9
- PARA 4-4-14

4-4-16.1 Repair HIRSS Bulkhead.

NOTE

Repair of left and right HIRSS bulkhead is the same.

4-4-16.1.1. Remove.

Remove firewall support angle from bulkhead (PARA 4-4-9).

4-4-16.1.2. Repair.

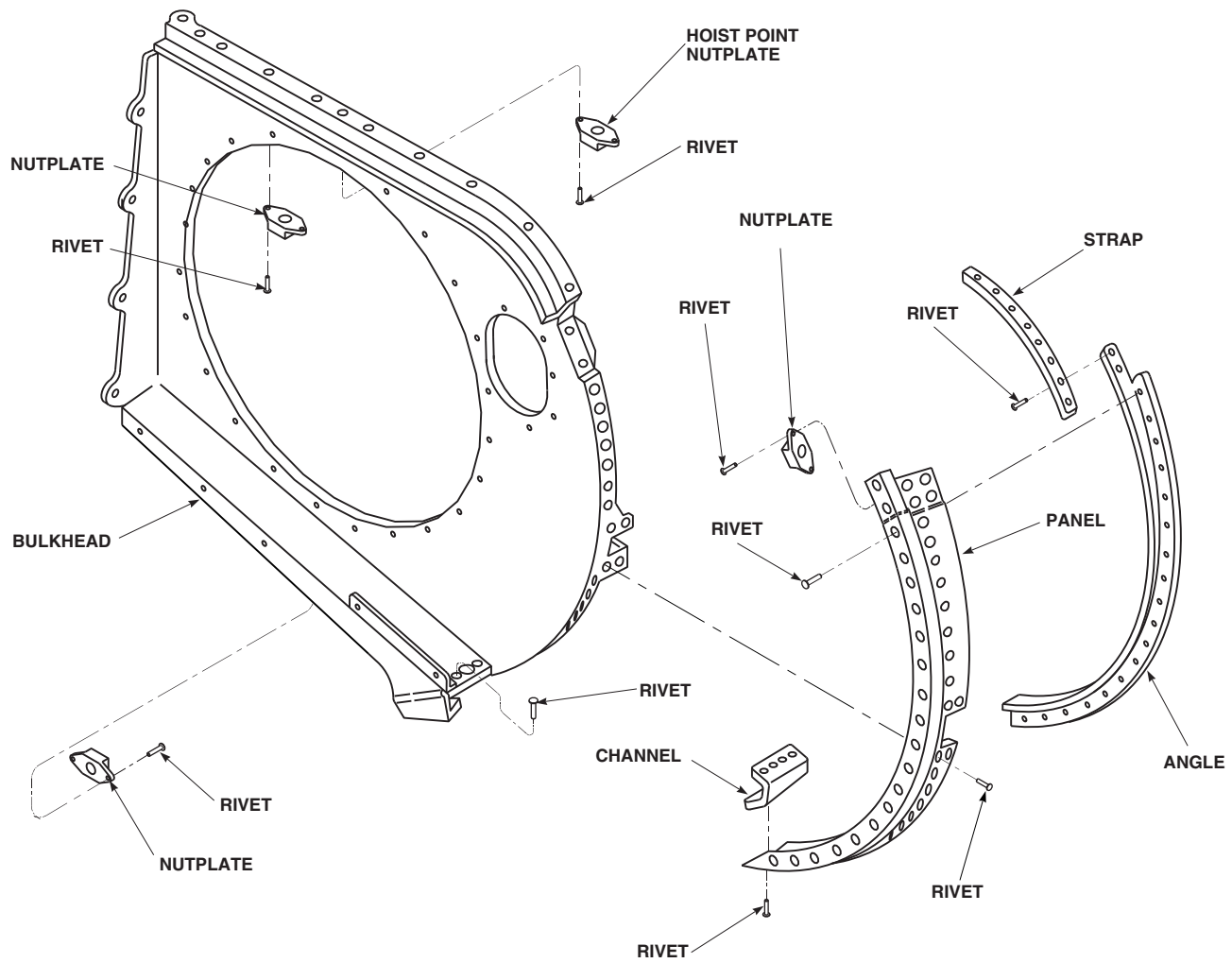
NOTE

Bulkhead repair is limited to replacement of nutplates and riveted components.

- Evaluate damage to bulkhead (PARA 4-4-14).
- Repair or replace components as required (Figure 4-4-16-1).
- Replace loose, broken, or missing rivets as required.

4-4-16.1.3. Install.

Install firewall support angle on bulkhead (PARA 4-4-9).

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4-4-17 HIRSS EMISSIVE COATING REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
4-4-17.1 Repair HIRSS Emissive Coating	4-4-17-1

INITIAL SETUP

Tools

- Heat Gun
- Nonmetallic Scraper, Locally-Made
- Wire Brush

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Brush, Item 83, Appendix D

- Dry-Cleaning Solvent, Item 124, Appendix D
- Paint, Heat Resisting, Item 221B, Appendix D

References

- Appendix D
- PARA 4-4-14

4-4-17.1 Repair HIRSS Emissive Coating.

NOTE

- Coating repair should be performed after all other maintenance or repairs are done to suppressor, either on or off helicopter.
 - Inspection areas and damage limits for emissive coating repairs are listed in PARA 4-4-14.
- Examine coating for areas of erosion, looseness, cracking, or flaking.
 - Using wire brush or nonmetallic scraper, remove damaged coating by brushing or careful scraping.
 - Using abrasive cloth, Item 1A, Appendix D, clean area down to parent metal, and feather edges around area to be recoated.



Only dry-cleaning solvent, Item 124, Appendix D, may be used. Solvent shall be new and clean. Do not use other cleaning compounds.

NOTE

- Allow at least 30 minutes for solvent to evaporate.
 - Make sure that all oily residue is removed by solvent wash.
- Clean areas to be coated by washing with dry-cleaning solvent, Item 124, Appendix D. Repeat cleaning until all oily residue is removed.

NOTE

Heat resisting paint must be used within 24 hours after mixing.

- e. Mix heat resisting paint, Item 221B, Appendix D, per manufacturer's instructions.
- f. Using brush, Item 83, Appendix D, brush on a smooth, wet coat using broad, even strokes.
- g. Allow coating to air-dry until it becomes matte gray.

- h. Cure coating using heat gun at 600°F (316°C) for at least 30 minutes.

NOTE

Complete curing of coating can be done by running engine at idle speed for 5 minutes.

- i. Inspect coating after curing. Repeat application procedure if coating is blistering, cracking, flaking, or if coverage is incomplete.

4-4-18 **LOAD-DEMAND ROTARY INPUT.**

PARAGRAPH BREAKDOWN

	PAGE
4-4-18.1 Replace Load-Demand Rotary Input	4-4-18-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill Rod, 1⁄8-inch diameter, 10-inches long
- Hex Wrench, 3⁄8-inch
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 193, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 4-3-3

4-4-18.1 **Replace Load-Demand Rotary Input.**

NOTE

Replacement of No. 1 and No. 2 engine load-demand rotary inputs is the same.

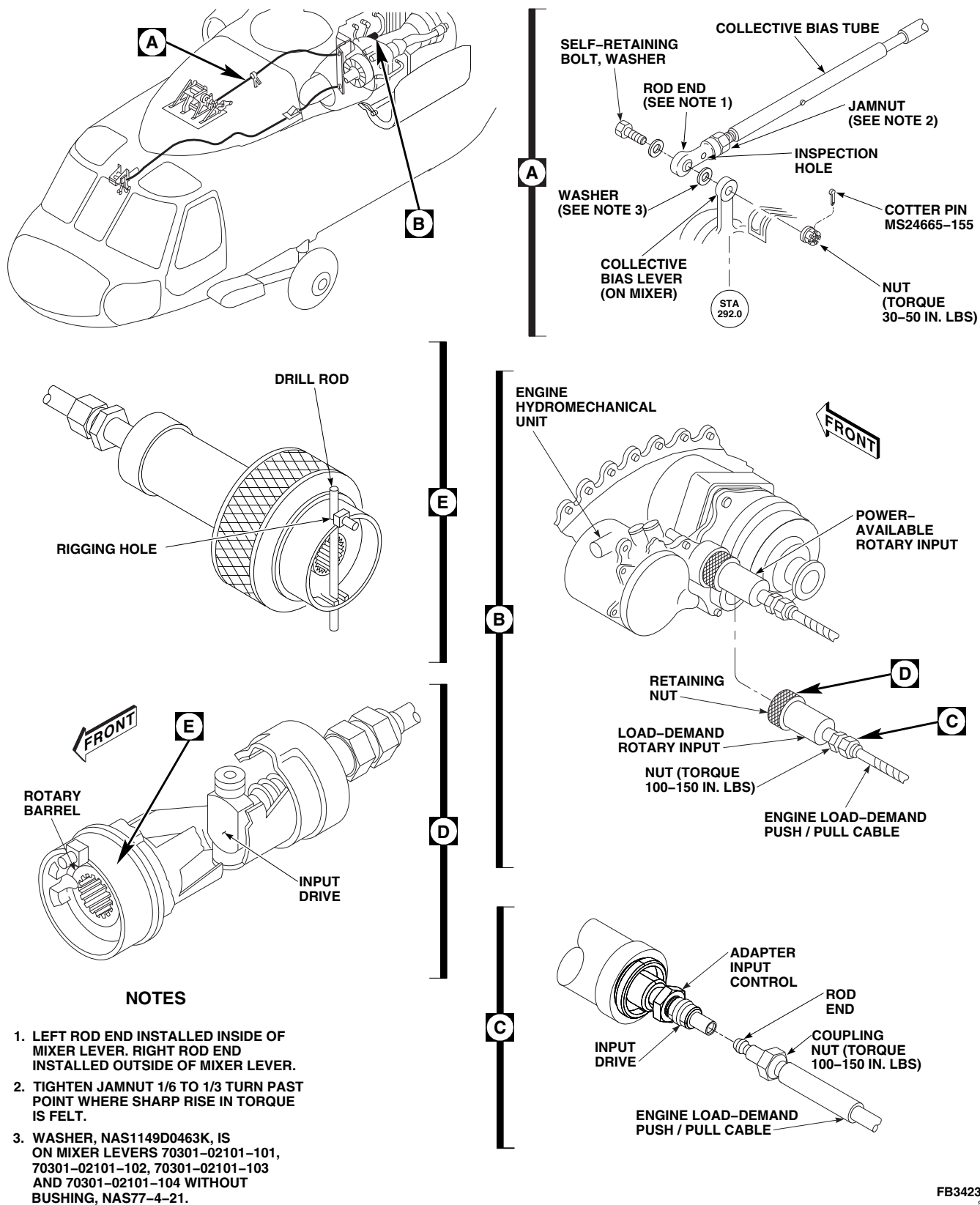
4-4-18.1.1 **Remove.**

- Open engine cowling.
- Open main rotor pylon sliding cover.
- Loosen coupling nut securing engine load-demand push/pull cable to adapter input control (Figure 4-4-18-1, Detail C).
- Disconnect load-demand rotary input from engine hydromechanical unit (Figure 4-4-18-1, Detail B).



Injury to personnel and damage to equipment will result if engine load-demand rotary input spring is not compressed and released carefully. Slowly compress and release spring to avoid sudden release of spring tension.

- Compress spring on engine load-demand rotary input by inserting 3⁄8-inch hex wrench through splines in front end of rotary barrel (Figure 4-4-18-1, Detail D). Push against front of input drive until rear of input drive extends beyond adapter.
- Unhook rear of input drive from engine load-demand push/pull cable rod end (Figure 4-4-18-1, Detail C). Slowly release pressure on hex

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wrench until input drive is in full front position.

- g. Remove retaining nut from rotary input (Figure 4-4-18-1, Detail B).
- h. Perform engine load-demand push/pull cable friction test (PARA 4-3-3).

4-4-18.1.2 Install.

- a. Install retaining nut on rotary input (Figure 4-4-18-1, Detail B).

WARNING

Injury to personnel and damage to equipment will result if engine load-demand rotary input spring is not compressed and released carefully. Slowly compress and release spring to avoid sudden release of spring tension.

- b. Compress spring of engine load-demand rotary input by inserting $\frac{3}{8}$ -inch hex wrench through splines in front end of rotary barrel (Figure 4-4-18-1, Detail D). Push against front of input drive until rear of input drive extends beyond adapter.

NOTE

When connecting engine load-demand push/pull cable to rotary input, make sure internal connection is made.

- c. Hook engine load-demand push/pull cable rod end on exposed end of input drive (Figure 4-4-18-1, Detail C). Slowly release pressure on hex wrench, and remove hex wrench from rotary adapter.
- d. Thread coupling nut on adapter input control. Do not torque coupling nut now.
- e. Connect external hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

CAUTION

Damage to collective bias tube spring pin will result if collective stick is moved with engine load-demand push/pull cable attached to collective bias lever and drill rod installed in rotary input. Do not move collective stick with cable attached and drill rod installed.

- f. Turn on backup hydraulic pump and hold collective stick in low stick position.
- g. Make sure $\frac{1}{8}$ -inch diameter drill rod in rotary input slides freely in and out of rigging hole after installing bolt in rod end and collective bias lever (Figure 4-4-18-1, Detail E). If drill rod does not slide freely, readjust collective bias tube rod end as follows:
 - (1) Loosen jamnut on collective bias tube rod end (Figure 4-4-18-1, Detail A).
 - (2) Remove bolt, washers, and nut securing collective bias tube rod end to collective bias lever.
 - (3) Turn collective bias tube rod end a half-turn at a time until $\frac{1}{8}$ -inch diameter drill rod slides freely in and out of rigging hole.
 - (4) Insert 0.020-inch lockwire, Item 193, Appendix D, into inspection hole in rod end. Adjusted rod end must prevent wire from passing through inspection hole.
 - (5) Install collective bias tube rod end on mixer with self-retaining bolt, washers, and nut.
 - (6) Tighten jamnut on collective bias tube rod end. **TIGHTEN JAMNUT $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
 - (7) Recheck that $\frac{1}{8}$ -inch diameter drill rod slides freely in and out of rigging hole when collective stick is in low position.

NOTE

Perform the next step only if collective bias tube rod end has been removed from collective bias lever.

until it bottoms in support bracket. Slight axial/radial play is normal. **TORQUE NUT CONNECTING LOAD-DEMAND PUSH/PULL CABLE TO ROTARY INPUT TO 100 - 150 INCH-POUNDS.**

- h. **TORQUE NUT SECURING COLLECTIVE BIAS TUBE ROD END TO COLLECTIVE BIAS LEVER TO 30 - 50 INCH-POUNDS.**

Install cotter pin.

- i. Remove 1/8-inch diameter drill rod from load-demand rotary input.

- j. Connect load-demand rotary input to engine hydromechanical unit (HMU) (Figure 4-4-18-1, Detail B).

- k. Make sure rotary input alignment pin engages slot on engine HMU. Tighten retaining nut

- l. Disconnect external hydraulic and electrical (PARA 1-6-15 and PARA 1-6-16).

- m. Make sure area is clean and free of foreign material.

- n. Close main rotor pylon sliding cover.

- o. Close engine cowling.

- p. Perform maintenance test flight (Appropriate MTF).

4-4-19 POWER-AVAILABLE ROTARY INPUT.

PARAGRAPH BREAKDOWN

		PAGE
4-4-19.1	Replace Power-Available Rotary Input	4-4-19-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill Rod, 1⁄8-inch diameter, 10-inches long
- Torque Wrench, 30 - 150 in. lbs

References

- Appropriate MTF
- PARA 4-4-33

4-4-19.1 Replace Power-Available Rotary Input.

4-4-19.1.2 Install.

NOTE

The following procedures apply to both No. 1 and No. 2 engine power-available rotary inputs.

4-4-19.1.1 Remove.

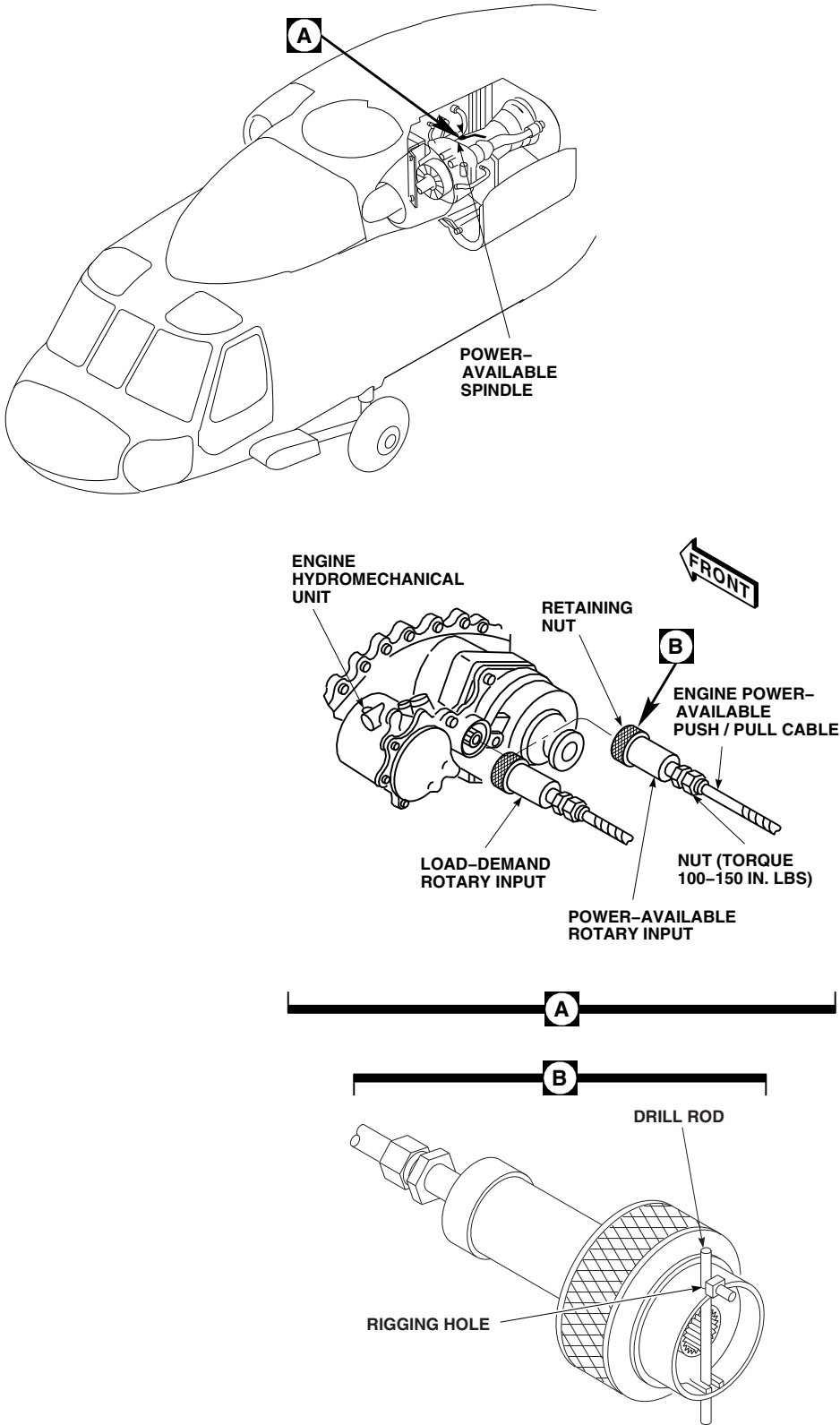
- Open engine cowling.
- Place quadrant ENG POWER CONT lever OFF.
- Loosen retaining nut and disconnect power-available rotary input from engine hydromechanical unit (Figure 4-4-19-1, Detail A).
- Loosen nut and disconnect power-available rotary input from engine power-available push/pull cable. Remove retaining nut from rotary input.

NOTE

When connecting engine power-available push/pull cable to rotary input, make sure internal connection is made.

- Install retaining nut on rotary input and connect rotary input to engine power-available push/pull cable. Do not torque nut connecting power-available push/pull cable to rotary input now (Figure 4-4-19-1, Detail A).
- Place ENG POWER CONT lever on quadrant in IDLE detent.
- Insert 1⁄8-inch diameter drill rod in rotary input (Figure 4-4-19-1, Detail B). If rod can be inserted, remove rod and secure rotary input to engine. **TORQUE NUT CONNECTING ENGINE POWER-AVAILABLE PUSH/PULL CABLE TO ROTARY INPUT TO 100 - 150 INCH-POUNDS** (Figure 4-4-19-1, Detail A).

- d. If rod cannot be inserted, rig engine power-available system (PARA 4-4-33).
- e. When all rigging has been completed, place quadrant ENG POWER CONT lever OFF.
- f. Make sure area is clean and free of foreign material.
- g. Close engine cowling.
- h. Perform maintenance test flight (Appropriate MTF).



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FIGURE 4-4-19-1. REPLACE POWER-AVAILABLE ROTARY INPUT

4-4-20 ENGINE SPEED CONTROL POTENTIOMETER.

PARAGRAPH BREAKDOWN

	PAGE
4-4-20.1 Replace Engine Speed Control Potentiometer	4-4-20-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 4-2-5

4-4-20.1 Replace Engine Speed Control Potentiometer.

4-4-20.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove soundproofing panels as required to access engine speed control potentiometer (PARA 2-4-69).
- Disconnect electrical connector, P901, from engine speed control potentiometer (Figure 4-4-20-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove screws, washers, and lockwashers securing engine speed control potentiometer to supports. Remove potentiometer.

4-4-20.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install engine speed control potentiometer on supports with screws, washers, and lockwashers (Figure 4-4-20-1, Detail A).
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P901, to engine speed control potentiometer.
- Make sure area is clean and free of foreign material.
- Install soundproofing panel in overhead cabin (PARA 2-4-69).
- Perform operational check of engine speed trim system (PARA 4-2-5).

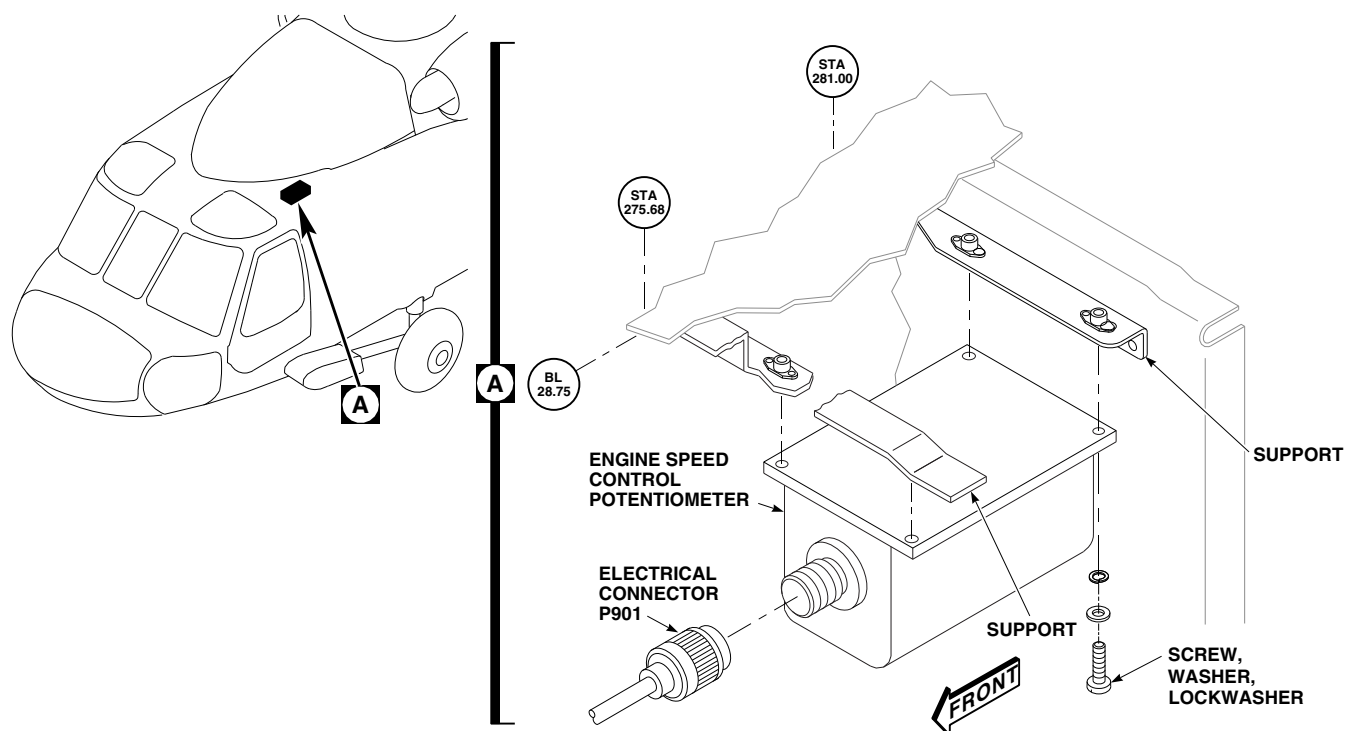
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FIGURE 4-4-20-1. REPLACE ENGINE SPEED CONTROL POTENTIOMETER

4-4-21 ENGINE CONTROL QUADRANT.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
4-4-21.1	Replace Engine Control Quadrant	4-4-21-1	4-4-21.2	Adjust Secondary Stop Blocks	4-4-21-5

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill Rod, 0.125-inch diameter, 5 inches long
- Feeler Gage
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 193, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 4-4-24
- PARA 4-4-33
- PARA 4-5-12

4-4-21.1 Replace Engine Control Quadrant.

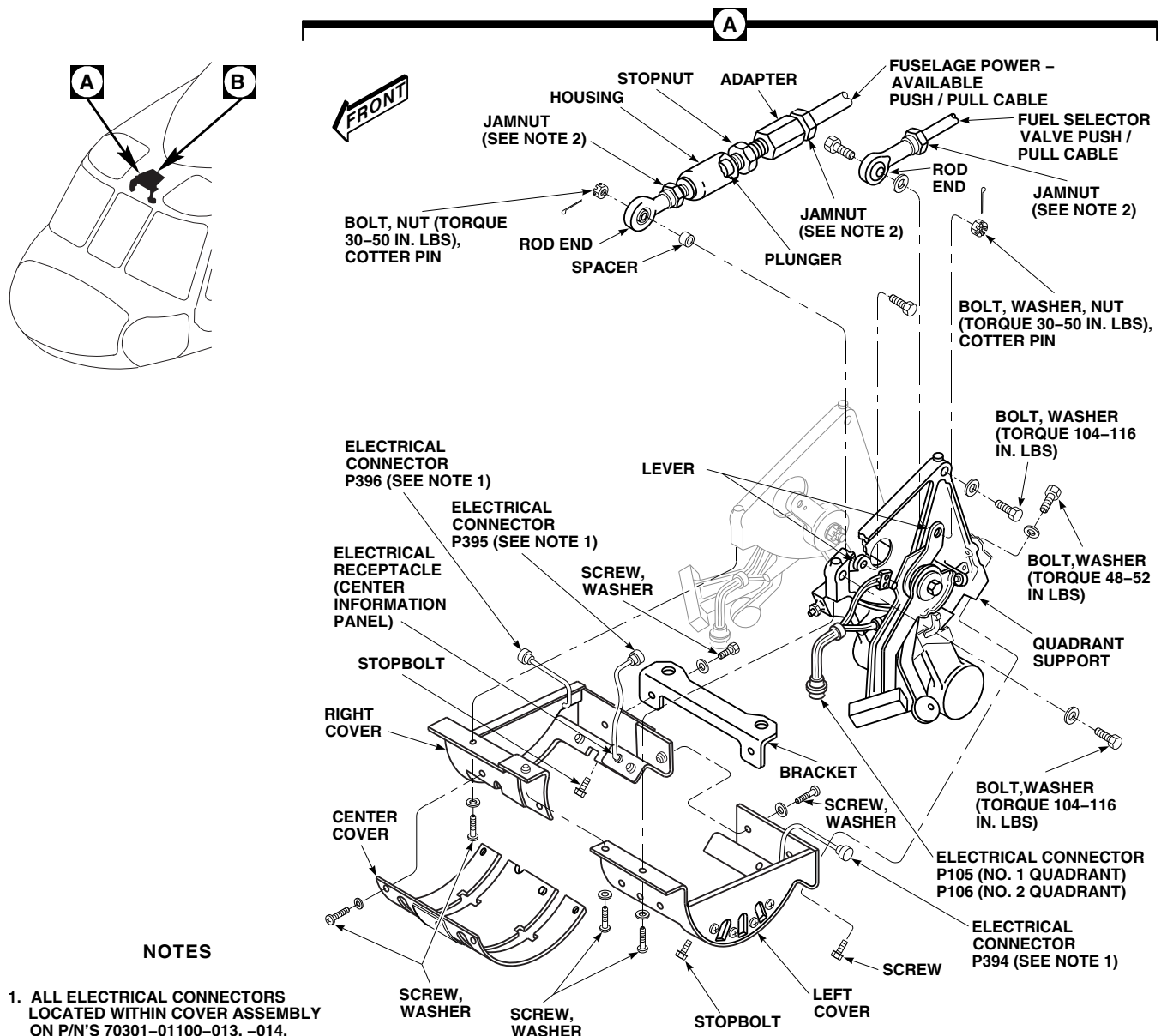
NOTE

The following procedures apply to both No. 1 and No. 2 quadrants.

4-4-21.1.1. Remove.

- Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).
- Move fuel selector levers forward.
- Unlock fastener studs holding upper console to cockpit overhead, and hinge down upper console.
- Remove screws, washers, and bracket at rear of quadrant (Figure 4-4-21-1, Sheet 1, Detail A or Sheet 2, Detail B).

- On quadrant, 70301-01100-011, remove center and side covers from quadrant. Remove stop-bolt (Figure 4-4-21-1, Sheet 1, Detail A).
- On quadrants, 70301-01100-013, -014, and 70301-85100-011 and -012, remove quadrant covers as follows (Figure 4-4-21-1, Sheet 1, Detail A and Sheet 2, Detail B):
 - Remove screws and washers from center cover. Remove cover before disconnecting electrical connector from receptacle on right cover.
 - Install protective caps and plugs on electrical connector and receptacle.
 - Remove screws and washers from left cover. Lower cover and disconnect electrical connector, P394.



NOTES

1. ALL ELECTRICAL CONNECTORS LOCATED WITHIN COVER ASSEMBLY ON P/N'S 70301-01100-013, -014, 70301-85100-011, -012 QUADRANTS. ELECTRICAL CONNECTORS ARE NOT INSTALLED ON P/N 70301-01100-011 QUADRANTS.
2. TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.
3. ALL ELECTRICAL CONNECTORS LOCATED WITHIN COVER ASSEMBLY ON P/N'S 70301-01100-013, -014, 70301-85100-011, -012 QUADRANTS.
4. ROTOR BRAKE INTERLOCK AND SOLENOID INSTALLED ON P/N 70303-85100-011 AND -012 QUADRANTS.

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FIGURE 4-4-21-1. REPLACE ENGINE CONTROL QUADRANT (SHEET 1 OF 2)

4-4-21-2

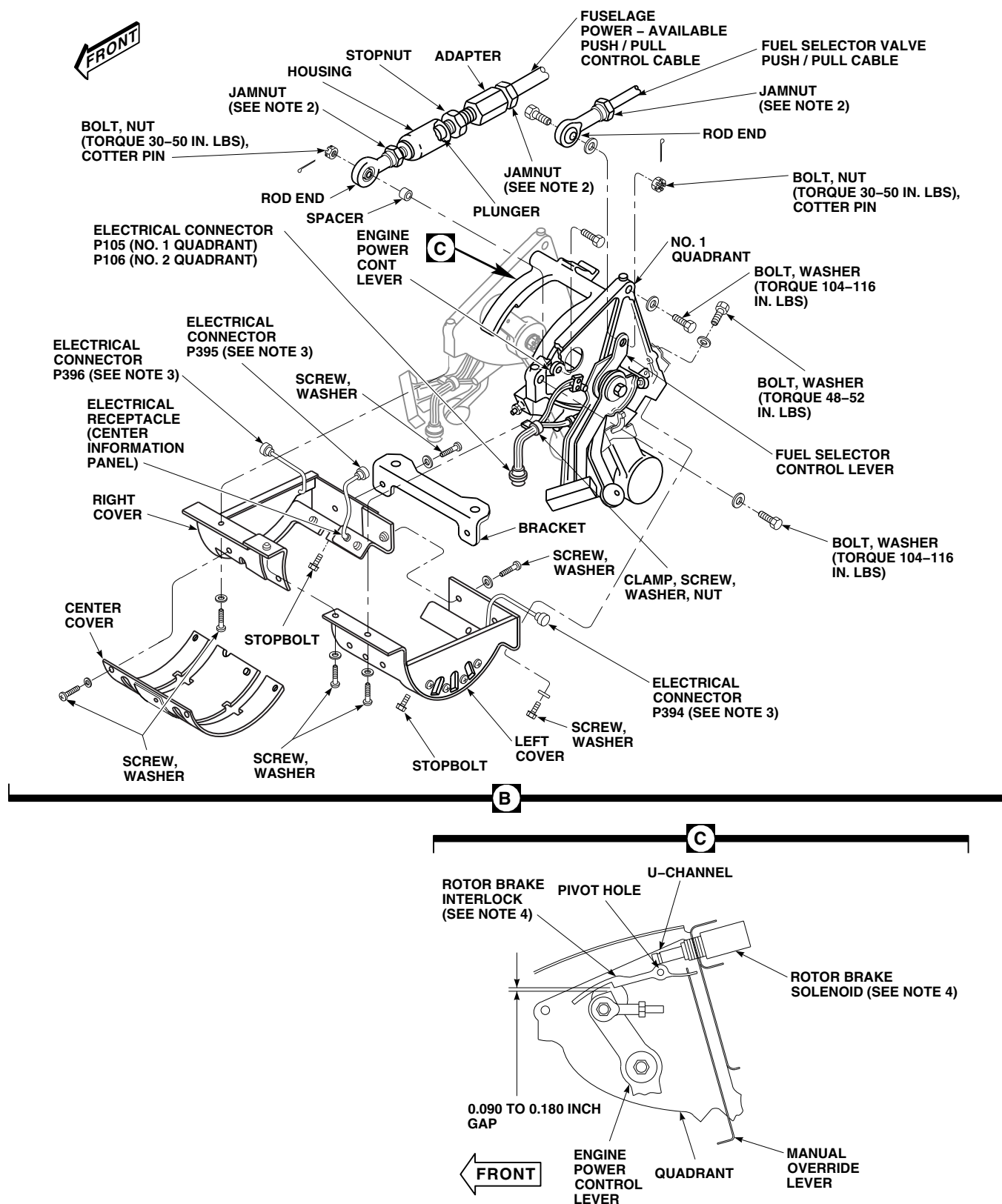


FIGURE 4-4-21-1. REPLACE ENGINE CONTROL QUADRANT (SHEET 2 OF 2)

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(4) Install protective caps and plugs on electrical connectors.

(5) Remove stopbolt.

(6) Remove screws and washers from right cover. Lower right cover and disconnect electrical connectors, P395 and P396.

(7) Install protective caps and plugs on electrical connectors.

(8) Remove stopbolt.

g. **RTR BRAKE** Remove manual override lever (Figure 4-4-21, Sheet 2, Detail C).◀

h. Remove bolts, washers, spacers, and nuts holding fuselage power-available push/pull cable and fuel selector valve push/pull cables to quadrant levers (Figure 4-4-21-1, Sheet 1, Detail A and Sheet 2, Detail B).

i. Remove clamp and disconnect electrical connector, P105 (No. 1 quadrant) or P106 (No. 2 quadrant).

j. Install protective caps and plugs on electrical connectors.

k. **RTR BRAKE** Note position of rotor brake interlock (Figure 4-4-21-1, Sheet 2, Detail C).◀

l. Remove bolt and washers holding quadrant to cockpit overhead. Remove quadrant support (Figure 4-4-21-1, Sheet 1, Detail A and Sheet 2, Detail B).

m. **RTR BRAKE** Remove rotor brake interlock (Figure 4-4-21-1, Sheet 2, Detail C).◀

4-4-21.1.2. Inspect.

a. Check all threaded inserts for proper engagement of support and broken or cracked threads.

b. **IF INSERT IS LOOSE, BROKEN, OR CRACKED, REPLACE INSERT** (PARA 4-5-12).

4-4-21.1.3. Install.

CAUTION

Damage to starter switch actuators will result if engine power control lever is moved from FLY or IDLE to OFF, while pressing starter button. Do not press starter button while moving engine power control lever.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. **RTR BRAKE** Position rotor brake interlock in pivot hole in quadrant (Figure 4-4-21, Sheet 2, Detail C).◀

c. Position quadrant on cockpit overhead and install bolts and washers. **TORQUE OUTBOARD FACING BOLTS TO 104 - 116 INCH-POUNDS. TORQUE REAR FACING BOLTS TO 48 - 52 INCH-POUNDS** (Figure 4-4-21-1, Sheet 1, Detail A and Sheet 2, Detail B).

d. **RTR BRAKE** Make sure rotor brake interlock is in hole in quadrant and engages rotor brake solenoid. Install manual override lever in slot of rotor brake interlock end and rotate lever 90° so tang in lever faces forward (Figure 4-4-21-1, Sheet 2, Detail C).◀

e. Remove protective caps and plugs from electrical connectors.

f. Inspect and treat electrical connectors (PARA 1-7-20).

g. Connect electrical connector, P105 (No. 1 quadrant), or P106 (No. 2 quadrant) (Figure 4-4-21-1, Sheet 1, Detail A or Sheet 2, Detail B). Install clamp around connector and secure with screw, washer, and nut.

h. **RTR BRAKE** Measure gap between power control lever and rotor brake interlock using feeler gage. **MAXIMUM GAP ALLOWED IS 0.090 TO 0.180-INCH.** If gap is not within tolerance, adjust rotor brake solenoid (Figure 4-4-21-1, Sheet 2, Detail C) (PARA 4-4-24).◀

4-4-21-4

- i. Move engine power control lever so rigging holes in lever and quadrant support line up. Insert 0.125-inch diameter x 5 inches long drill rod through rigging holes in lever and quadrant support.
- j. Insert 0.020-inch lockwire, Item 193, Appendix D, into inspection hole in rod end. Adjusted rod end must prevent wire from passing through inspection hole.

CAUTION

To prevent the power control lever from bending, make sure the inboard side of the lever is supported when inserting control cable to power lever bolt.

- k. Adjust rod end on fuselage power-available push/pull control cable until rod end bearing and quadrant engine power control lever holes line up. **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Secure quadrant lever to control cable rod end using bolt, spacer, and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin (Figure 4-4-21-1, Sheet 1, Detail A or Sheet 2, Detail B).
- l. Move fuel selector lever forward. Install fuel selector valve push/pull cable to lever with bolt, washer and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.
- m. Rig engine power-available system (PARA 4-4-33).
- n. Make sure area is clean and free of foreign material.
- o. Remove protective caps and plugs from electrical connectors and receptacles.
- p. Inspect and treat electrical connectors and receptacles (PARA 1-7-20).

CAUTION

Damage to harness will result if wiring is pinched when installing covers. Be

careful to avoid pinching wiring harness when installing covers.

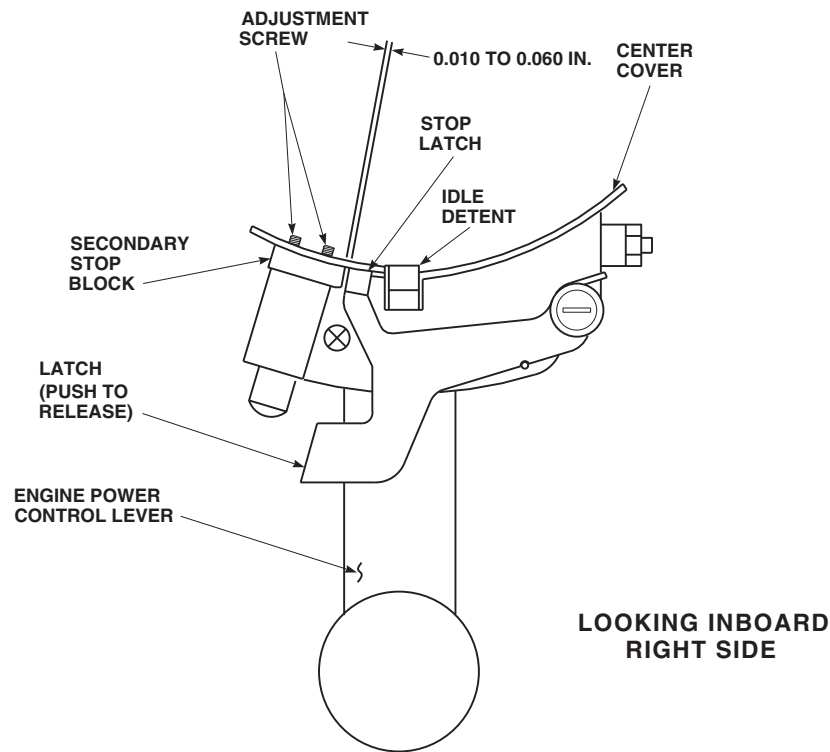
- q. Install quadrant covers as follow:
 - (1) On quadrant, 70301-01100-011, install center and side covers on quadrant with washers and screws. Install stopbolt (Figure 4-4-21-1, Sheet 1, Detail A).
 - (2) On quadrant, 70301-01100-013, -014, and 70301-85100-011, -012, install quadrant covers as follows (Figure 4-4-21-1, Sheet 1, Detail A and Sheet 2, Detail B):
 - (a) Connect electrical connectors, J395 and J396, and install right cover on quadrant with screws and washers. Install stopbolt.
 - (b) Connect electrical connector, J394, and install left cover on quadrant with screws and washers. Install stopbolt.
 - (c) Insert electrical connector on center cover into receptacle on right cover, and secure center cover with screws and washers.
 - (3) Install bracket at rear of quadrant with screws and washers.
- r. Hinge up upper console and lock fastener studs holding upper console to cockpit overhead.
- s. Move fuel selector levers back to the off position.

4-4-21.2 Adjust Secondary Stop Blocks.

NOTE

The following adjustment is for both No. 1 and No. 2 secondary stop blocks.

- a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).
- b. With engine power control lever in the IDLE position and fully engaged in the ground idle position detent located in the center cover as-

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sembly, using feeler gage, measure the gap between the stop latch and the secondary stop block located on the center access cover (Figure 4-4-21-2).

- c. Adjust gap to 0.010 to 0.060-inch by loosening the two adjustment screws on the stop block.
- d. Tighten adjustment screws. Place engine power control lever in the OFF position.

4-4-22 ENGINE CONTROL QUADRANT CANOPY PLUNGER.

PARAGRAPH BREAKDOWN

	PAGE
4-4-22.1 Replace/Adjust Engine Control Quadrant Canopy Plunger	4-4-22-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Spring Scale

References

- PARA 1-6-16
- PARA 4-4-21
- PARA 4-5-12

4-4-22.1 Replace/Adjust Engine Control Quadrant Canopy Plunger.

4-4-22.1.1 Remove.

- Remove quadrant covers (PARA 4-4-21).
- Remove jamnut and washer from canopy plunger (Figure 4-4-22-1, Detail A).
- Remove canopy plunger from fitting on support.

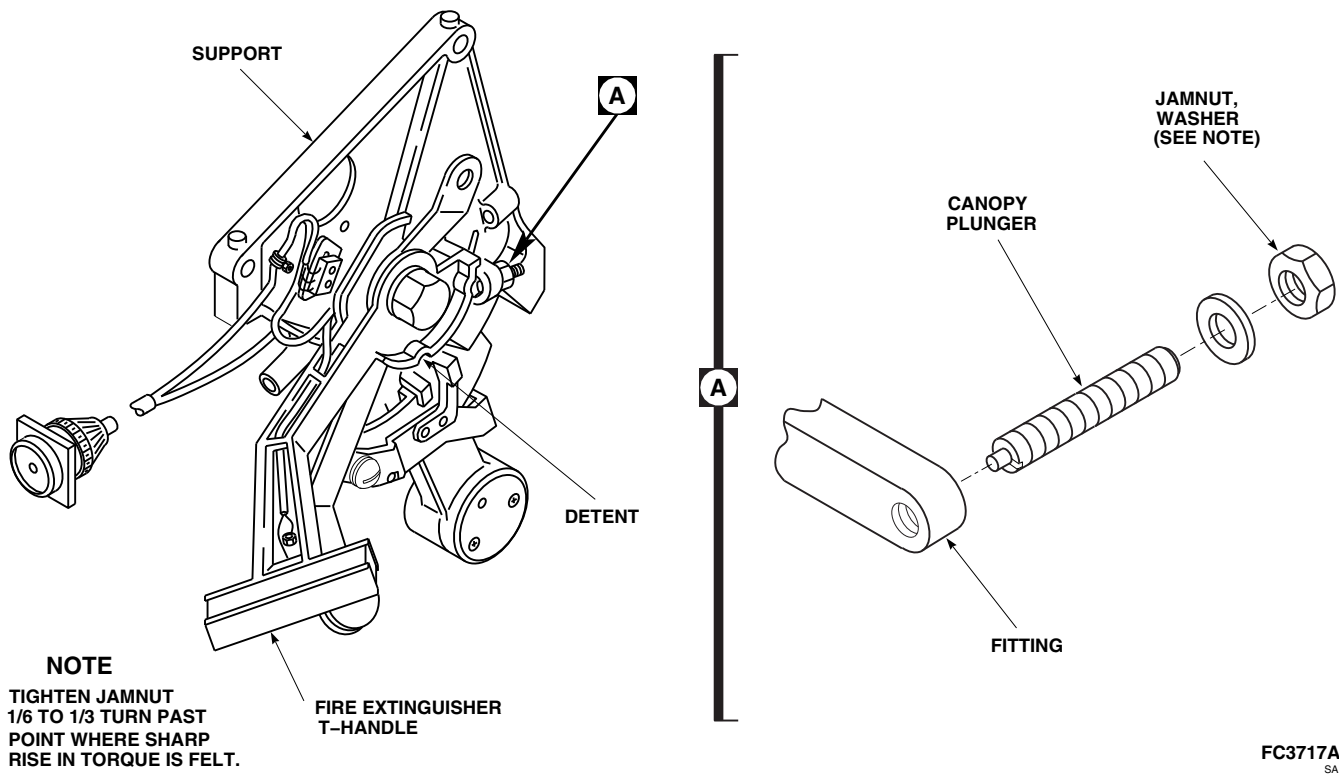
4-4-22.1.2 Inspect.

Check all threaded inserts for proper engagement of support and broken or cracked threads. **NO DISENGAGEMENT ALLOWED. NO BROKEN OR CRACKED THREADS ALLOWED.** If damage is found, replace insert (PARA 4-5-12).

4-4-22.1.3 Install.

- Position fire extinguisher T-handle so detent is lined up with hole in fitting (Figure 4-4-22-1, Detail A).

- Thread canopy plunger into fitting until plunger is lightly touching detent in handle.
- Install jamnut and washer on canopy plunger. Do not tighten jamnut now.
- Using spring scale attached to T intersection of handle, move handle through full range of travel while adjusting canopy plunger. **ADJUST PLUNGER UNTIL A FORCE OF 5 - 7 POUNDS IS OBTAINED TO MOVE PLUNGER FROM DETENTS.**
- TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- Install quadrant covers (PARA 4-4-21).



4-4-23 ENGINE STARTER SWITCH ADJUSTMENT.

PARAGRAPH BREAKDOWN

		PAGE
4-4-23.1	Adjust Engine Starter Switch	4-4-23-1

INITIAL SETUP

- PARA 4-4-21

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 1-6-16
- PARA 4-2-8

4-4-23.1 Adjust Engine Starter Switch.



Damage to starter switch actuator will result if ENG POWER CONT lever is moved from FLY or IDLE to OFF, while pressing starter button. Do not press starter button while moving ENG POWER CONT lever.

NOTE

Do not adjust starter switch on bench. Adjust switch after quadrant is installed in helicopter, and power-available push/pull cable is rigged.

- a. Turn off all electrical power (PARA 1-6-16).
 b. Move fuel selector levers forward.
 c. Unlock fastener studs holding upper console to cockpit overhead, and hinge down upper console.
- d. Remove quadrant covers (PARA 4-4-21).
 e. Place ENG POWER CONT lever on quadrant against off stop.
 f. Loosen screws holding starter switch to quadrant support, just enough to move switch (Figure 4-4-23-1, Detail A).
 g. With starter button fully depressed, move starter switch until arm is against starter button, and switch clicks. Tighten screws to hold switch in place.
 h. Move ENG POWER CONT lever back and forth several times, then place lever against off stop again.
 i. Make sure that starter switch cannot be actuated by moving lever against off stop.
 j. Press starter button and check for click of switch. Readjust as necessary.
 k. Make sure area is clean and free of foreign material.

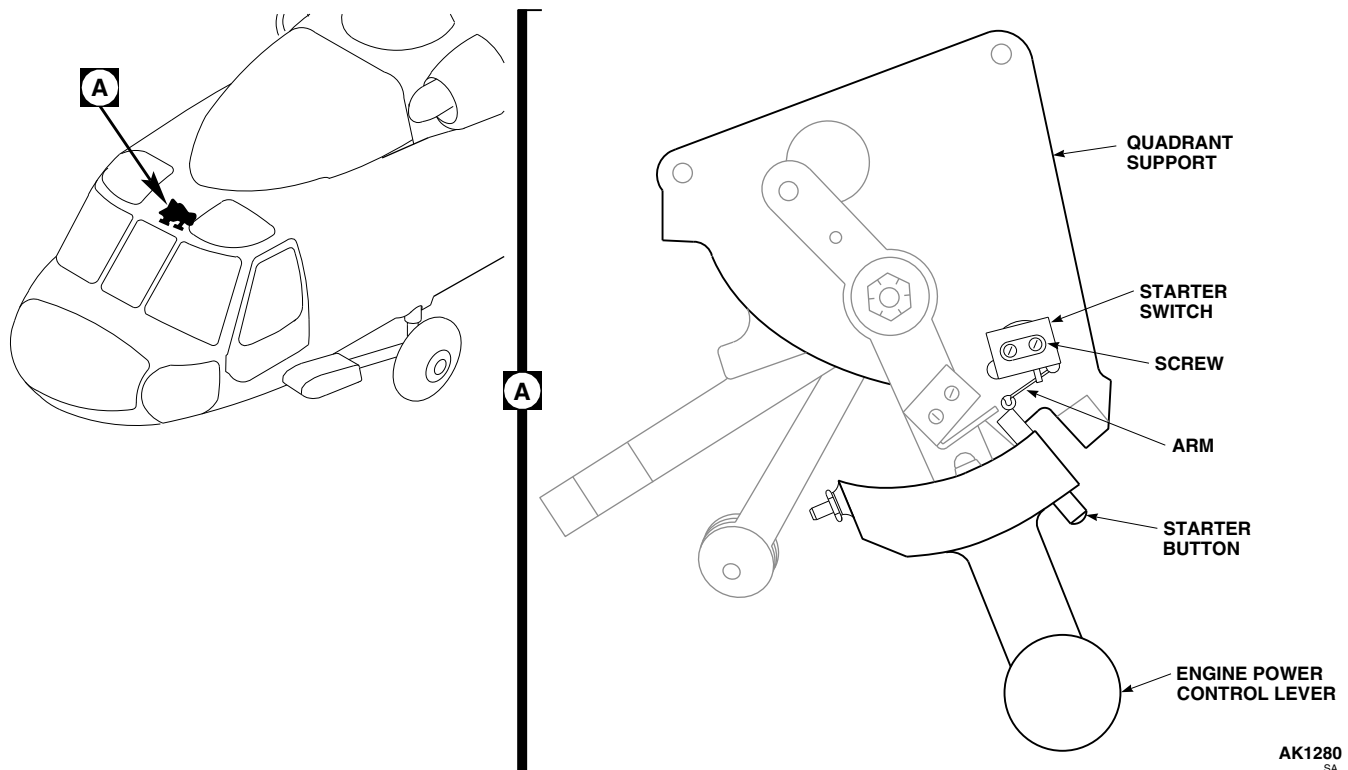


FIGURE 4-4-23-1. ADJUST ENGINE STARTER SWITCH

- l. Install quadrant covers (PARA 4-4-21).
- m. Hinge up and fasten upper console.
- n. Move fuel selector levers back to the off position.
- o. Perform operational check of engine start and ignition system (PARA 4-2-8).

4-4-24 ROTOR BRAKE SOLENOID.

PARAGRAPH BREAKDOWN

	PAGE
4-4-24.1 Replace Rotor Brake Solenoid	4-4-24-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage

Materials/Parts

- Protective Caps and Plugs
- Washers, NAS1149D0763K

References

- PARA 1-6-16
- PARA 1-6-18
- PARA 1-7-20
- PARA 4-4-21
- PARA 4-4-33

4-4-24.1 Replace Rotor Brake Solenoid.

4-4-24.1.1 Remove.

- Remove engine control quadrant (left or right) (PARA 4-4-21).
- Disconnect electrical connector, P165 (Figure 4-4-24-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove jamnut and lockwasher and remove rotor brake solenoid from quadrant support. Remove washers from solenoid and retain for installation.

4-4-24.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install washers on rotor brake solenoid.
- Position rotor brake solenoid in quadrant support.

- Install lockwasher and jamnut on solenoid and engage solenoid shaft in U-channel on rotor brake interlock (Figure 4-4-24-1, Detail A). **TIGHTEN JAMNUT ¼ TO ⅓ TURN PAST POINT UNTIL A SHARP RISE IN TORQUE IS FELT.**
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P165.
- Install engine control quadrant (PARA 4-4-21).

4-4-24.1.3 Adjust.

- Connect external electrical power (PARA 1-6-16).
- If engaged, disengage rotor brake (PARA 1-6-18).

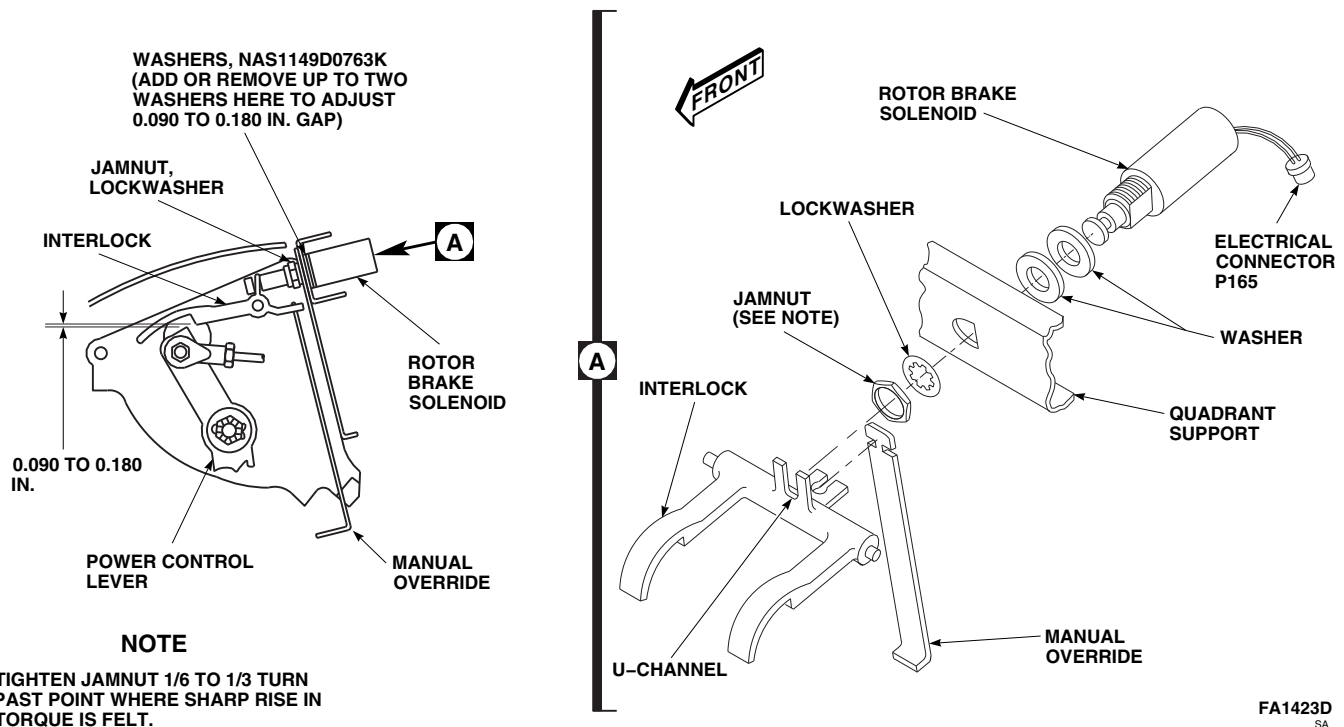


FIGURE 4-4-24-1. REPLACE ROTOR BRAKE SOLENOID

- c. Using feeler gage, measure gap between rotor brake interlock and engine power control lever. **GAP SHALL BE 0.090 - 0.180-INCH** (Figure 4-4-24-1, Detail A). If gap is not within limits, perform the following:
 - (1) Engage rotor brake (PARA 1-6-18).
 - (2) Turn off all electrical power (PARA 1-6-16).
 - (3) Remove engine control quadrant (PARA 4-4-21).
 - (4) Remove rotor brake solenoid (PARA 4-4-24.1.1).
 - (5) Add up to two washers, NAS1149D0763K, between quadrant support and solenoid to increase gap or remove washers to reduce gap.
 - (6) Install rotor brake solenoid (PARA 4-4-24.1.2).
 - (7) Repeat step a. through step c. until gap is within limits.
- d. If not already done, engage rotor brake (PARA 1-6-18).
- e. Disconnect external power (PARA 1-6-16).
- f. Loosen fasteners that secure upper console and carefully lower.
- g. Install manual override (Figure 4-4-24-1, Detail A).
- h. Make sure area is clean and free of foreign material.
- i. Carefully raise upper console and secure with fasteners.
- j. Rig engine power-available system (PARA 4-4-33).

4-4-25 ENGINE STARTER ABORT SWITCH ADJUSTMENT.

PARAGRAPH BREAKDOWN

		PAGE
4-4-25.1	Adjust Engine Starter Abort Switch	4-4-25-1

INITIAL SETUP

Tools

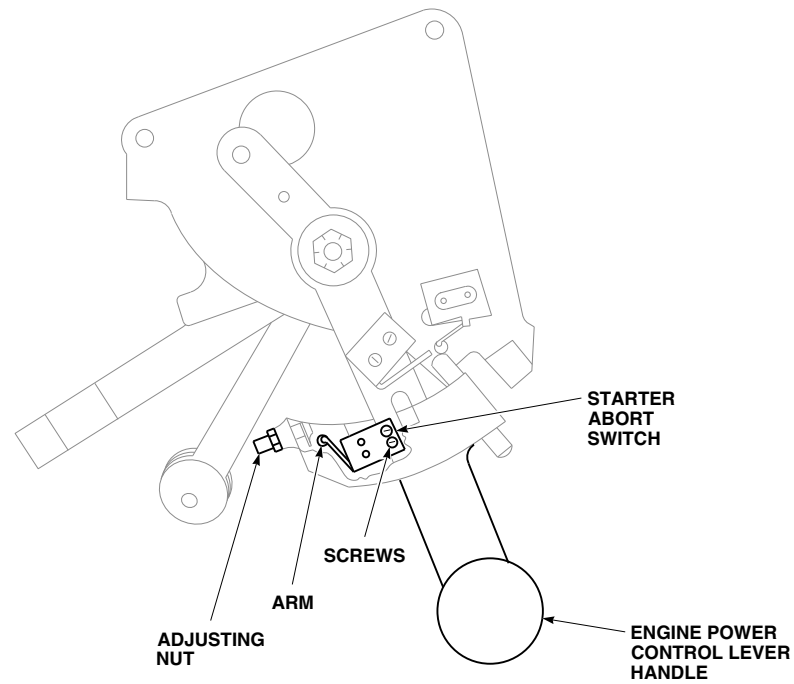
- Aircraft Mechanic’s Toolkit
- Pencil

References

- PARA 4-2-9

4-4-25.1 Adjust Engine Starter Abort Switch.

- a. Turn quadrant adjusting nut fully clockwise (Figure 4-4-25-1).
- b. Pull ENG POWER CONT lever handle down as far as it will go and hold. Using pencil, mark a line on lever 0.060-inch above top of handle.
- c. With handle down as far as it will go, turn adjusting nut counterclockwise until switch clicks. Release handle.
- d. While pulling ENG POWER CONT lever handle down, look at lever to see where top of handle is when switch clicks.
- e. If top of handle is then below pencil marking, no further adjustment is required.
- f. If there is no click with handle fully down, turn adjusting nut counterclockwise in quarter-turn units until a click is heard.
- g. If top of handle is covering pencil marking, turn adjusting nut clockwise in quarter-turn units until switch clicks within 0.060-inch spread range.
- h. Perform operational check of engine controls quadrant (PARA 4-2-9).

AA9699
SA**FIGURE 4-4-25-1. ADJUST ENGINE STARTER ABORT SWITCH**

4-4-26 ENGINE STARTER OVERRIDE SWITCH ADJUSTMENT.

PARAGRAPH BREAKDOWN

		PAGE
4-4-26.1	Adjust Engine Starter Override Switch	4-4-26-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

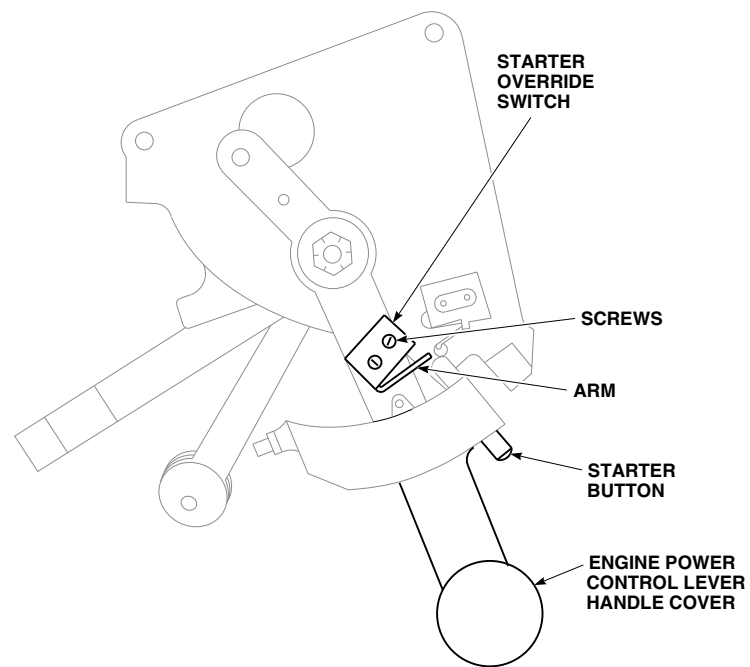
- PARA 4-2-9
- PARA 4-4-21

4-4-26.1 Adjust Engine Starter Override Switch.

- Remove screws, washers, and center cover of quadrant (PARA 4-4-21).
- Loosen screws holding starter override switch to ENG POWER CONT lever, just enough to move switch (Figure 4-4-26-1).
- With starter button fully pressed, move starter override switch until arm is against starter but-

ton, and switch clicks. Tighten screws holding switch to ENG POWER CONT lever bracket.

- Make sure area is clean and free of foreign material.
- Install center quadrant cover with screws and washers (PARA 4-4-21).
- Perform operational check of engine controls quadrant (PARA 4-2-9).

AA9698
SA**FIGURE 4-4-26-1. ADJUST ENGINE STARTER OVERRIDE SWITCH**

4-4-27 ENGINE LOAD-DEMAND PUSH/PULL CABLE.

PARAGRAPH BREAKDOWN

	PAGE
4-4-27.1 Replace Engine Load-Demand Push/Pull Cable	4-4-27-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill Rods, 1⁄8-inch diameter, 3-inches long
- Powerplant Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Corrosion Compound, Item 108, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lockwire, Item 196, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Scouring Pad, Item 270, Appendix D

- Sealing Compound, Item 292, Appendix D
- Spiral Wrap, Item 315, Appendix D
- Tape, Item 317C, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 4-3-2
- PARA 4-3-3
- PARA 4-4-4

4-4-27.1 Replace Engine Load-Demand Push/Pull Cable.

NOTE

Replacement of left and right load-demand push/pull cables is the same.

4-4-27.1.1 Remove.

- Open engine cowling.
- Remove engine air inlet (PARA 4-4-4).
- Open main rotor pylon sliding cover.
- Connect external electrical power (PARA 1-6-16).
- Turn on backup hydraulic power (PARA 1-6-15).
- Move collective stick in cockpit to low stick position.
- Disconnect load-demand rotary input from hydromechanical unit (Figure 4-4-27-1, Sheet 1, Detail A). Insert 1⁄8-inch diameter drill rod through rigging hole in outer housing and inner rotary driver of rotary input to hold spring

compressed (Figure 4-4-27-1, Sheet 2, Detail G).

CAUTION

Shearing of spring pin in collective bias tube will result if collective stick is moved while drill rod is installed. Do not move collective stick when drill rod is installed.

- h. Disconnect engine load-demand push/pull cable from load-demand rotary input. Make sure retaining nut remains on load-demand rotary input (Figure 4-4-27-1, Sheet 1, Detail B).
- i. Turn off backup hydraulic power (PARA 1-6-15).
- j. Unclamp cable from firewall.
- k. Remove screws, washers, seal retainers, and seal from firewall and cable (Figure 4-4-27-1, Sheet 1, Detail C).
- l. Remove tape and remove spiral wrap from cable.
- m. Remove bolt, washers, and nut securing rod end of collective bias tube to collective bias lever on mixer. Remove rod end (Figure 4-4-27-1, Sheet 1, Detail D).
- n. Remove rod end and jamnut from collective bias tube. Inspect rod end for radial and axial play. **NONE ALLOWED.** If play is found, replace rod end (Figure 4-4-27-1, Sheet 1, Detail E).
- o. Remove screws, washers, and nuts and slide retainer away from support (Figure 4-4-27-1, Sheet 1, Detail F).
- p. Remove cable from support and helicopter. Remove retainer from cable.
- q. If collective bias tube and cable need to be separated, perform the following:

- (1) Remove spring pin from collective bias tube and cable (Figure 4-4-27-1, Sheet 1, Detail E).
- (2) Slide cable from collective bias tube.
- (3) Connect new cable to collective bias tube with spring pin.

CAUTION

Spring pin is part of a shear section. Only use one wrap (loop) when lockwiring this pin.

NOTE

There are two spring pin holes in collective bias tube. Either hole can be used until final adjustment at installation.

- (4) Using only one wrap (loop), lockwire, Item 196, Appendix D, spring pin to collective bias tube.

4-4-27.1.2 Repair.

- a. Using scouring pad, Item 270, Appendix D, remove surface corrosion from rod end of collective bias tube.
- b. Clean surface with corrosion compound, Item 108, Appendix D.
- c. Using epoxy primer coating kit, Item 135, Appendix D, apply two coats of epoxy primer to surfaces where plating was disturbed.

4-4-27.1.3 Install.

- a. Slide retainer onto cable and collective bias tube and insert load-demand push/pull cable and bias tube through support (Figure 4-4-27-1, Sheet 1, Detail F). Install jamnut and rod end on collective bias tube (Figure 4-4-27-1, Sheet 1, Detail E).
- b. Route cable from collective bias lever at STA 292.0 on upper front deck through engine compartment firewall.

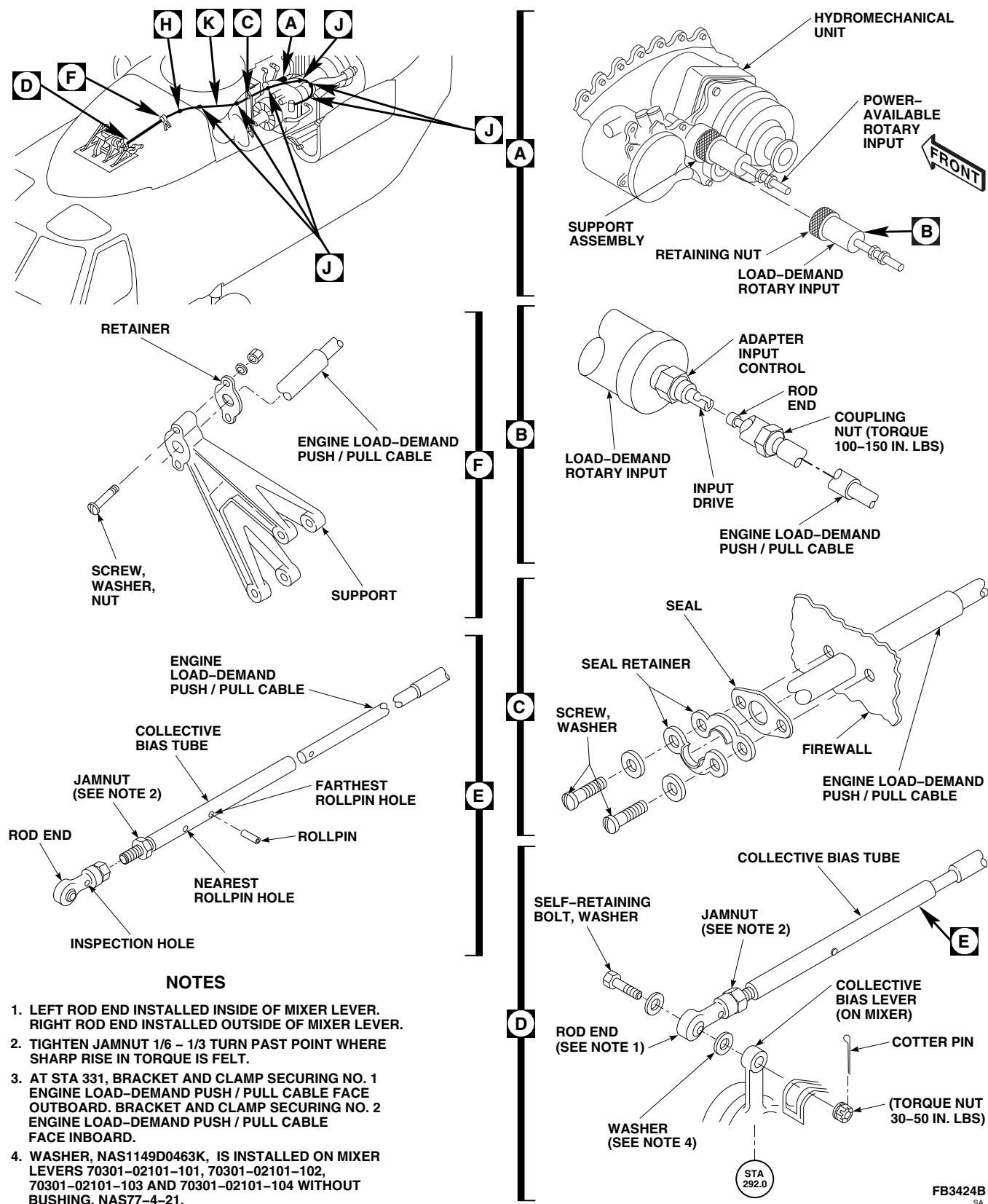
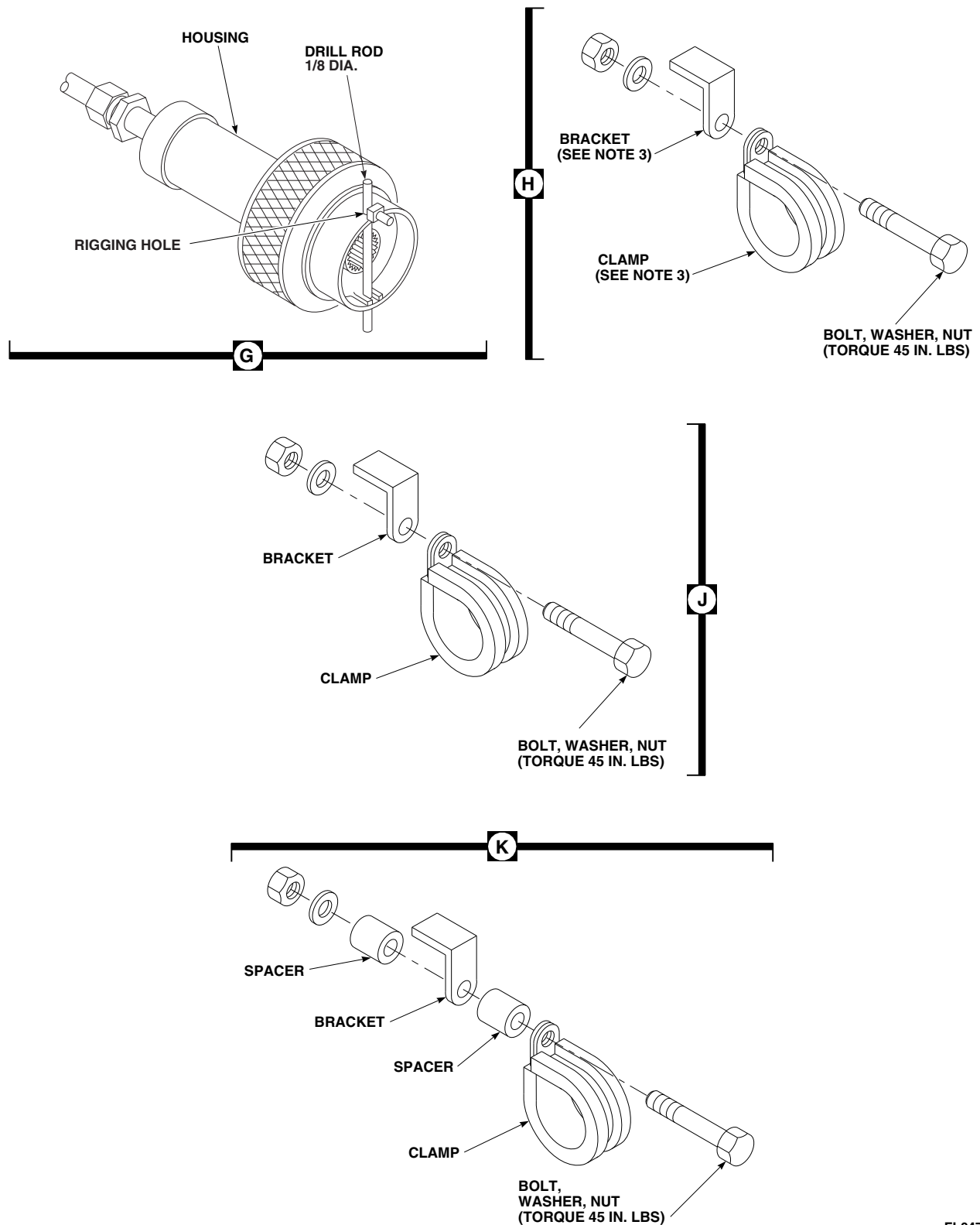


FIGURE 4-4-27-1. REPLACE ENGINE LOAD-DEMAND PUSH/PULL CABLE (SHEET 1 OF 2)



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FIGURE 4-4-27-1. REPLACE ENGINE LOAD-DEMAND PUSH/PULL CABLE (SHEET 2 OF 2)

- c. Add spiral wrap, Item 315, Appendix D, to cable from STA 325.0 through STA 330.0, and from STA 353.0 through STA 368.0. Secure spiral wrap in place with tape, Item 317C, Appendix D.
- d. Secure retainer on support with screws, washers, and nuts (Figure 4-4-27-1, Sheet 1, Detail F).

NOTE

Make sure retaining nut is installed on load-demand rotary input before connecting to engine load-demand push/pull cable.

- e. Compress spring of load-demand rotary input and insert 1/8-inch diameter drill rod (Figure 4-4-27-1, Sheet 2, Detail G).

NOTE

When connecting cable to rotary input, make sure internal connection is made.

- f. Connect cable to load-demand rotary input and loosely install coupling nut. Do not torque coupling nut now (Figure 4-4-27-1, Sheet 1, Detail B).
- g. Install seal and seal retainers around push/pull cable and secure on firewall with screws and washers (Figure 4-4-27-1, Sheet 1, Detail C).

CAUTION

Failure to properly orient engine load-demand push/pull cable brackets and clamps at STA 331.0 can cause chafing between rotating swashplate and cables. Make sure bracket and clamp securing No. 1 cable at STA 331.0 face outboard, and bracket and clamp securing No. 2 cable at STA 331.0 face inboard (Figure 4-4-27-1, Sheet 2, Detail H).

- h. Secure engine load-demand push/pull cable at STA 331.0 with bolts, washers, nuts, brackets (left outboard and right inboard) and clamps

(left outboard and right inboard). **TORQUE NUTS TO 45 INCH-POUNDS.** Perform clearance check as follows:

- (1) Turn on electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).
- (2) Center stick and yaw pedals.
- (3) Place collective stick in full down position.
- (4) With aid of assistant, turn main rotor head and inspect clearance between rotating swashplate lug and left and right engine load-demand push/pull cable (Figure 4-4-27-2, Detail A). **CLEARANCE SHALL NOT BE LESS THAN 1.25-INCHES.** If not within limits, reposition cables within clamps and recheck.

NOTE

When clamping cable, make sure adequate clearance exists between cable and surrounding structure and components.

- i. Clamp engine load-demand push/pull cable to every rib along firewall.
- j. Check for minimum clearance of 1/4-inch between engine load-demand push/pull cable and engine fuel filter head. Reposition cable or add clamps as necessary.
- k. Install engine load-demand push/pull cable with bolts, washers, spacers, nuts, and brackets (Figure 4-4-27-1, Sheet 2, Detail J and Detail K). **TORQUE NUTS TO 45 INCH-POUNDS.**

NOTE

Cyclic stick position has no effect on rigging of load-demand push/pull cable.

1. Turn on backup hydraulic power (PARA 1-6-15). Hold collective stick down against stop and adjust rod end bearing to line up with hole in collective bias lever at STA 292.0 on upper front deck as follows (Figure 4-4-27-1, Sheet 1, Detail D):

- (1) Thread rod end on or off collective bias tube to line up holes (Figure 4-4-27-1, Sheet 1, Detail E).
- (2) If rod end bottoms on collective bias tube before boltholes line up, remove spring pin, line up hole in load-demand push/pull cable with hole in collective bias tube nearest to rod end, and install spring pin. Using only one wrap (loop), lockwire, Item 196, Appendix D, spring pin to tube. Adjust rod end again.
- (3) If boltholes line up, check that lockwire, Item 196, Appendix D, cannot pass through inspection hole in rod end. If it does, remove spring pin, line up hole in load-demand push/pull cable with hole in collective bias tube farthest from rod end, and install spring pin. Using only one wrap (loop), lockwire, Item 196, Appendix D, spring pin to tube. Adjust rod end again.
- (4) **TIGHTEN JAMNUT $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

CAUTION

Shearing of spring pin in collective bias tube will result if collective stick is moved while drill rod is installed. Do not move collective stick with push/pull cable attached to collective bias lever and drill rod installed in rotary input.

NOTE

- Right load-demand push/pull cable rod end is connected to outside of collective bias lever. Left load-demand push/pull cable rod end is connected to inside of collective bias lever.
- On helicopter with collective bias lever 70301-02101-041 through -044 installed, do not install washer, NAS1149D0463K, between collective

bias tube rod end and collective bias lever.

- Make sure washer, SS4409-060, or equivalent, is installed between bolthead and collective bias tube rod end. Install washer, NAS1149D0463K, between collective bias lever and collective bias tube rod end only if collective bias lever, 70301-02101-101, 70301-02101-102, 70301-02101-103, or 70301-02101-104, without bushing installed.
 - Adjust rod end as required to allow $\frac{1}{8}$ -inch diameter drill rod in rotary input to slide freely before rod end is connected at mixer lever.
- m. Connect rod end to collective bias lever with self-retaining bolt, washers, and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin (Figure 4-4-27-1, Sheet 1, Detail D).
 - n. Make sure $\frac{1}{8}$ -inch diameter drill rod in rotary input slides freely in and out of rigging hole after installing bolt in rod end and collective bias lever (Figure 4-4-27-1, Sheet 2, Detail G). If necessary, adjust collective bias tube rod end as follows (Figure 4-4-27-1, Sheet 1, Detail D):
 - (1) Loosen jamnut on collective bias tube rod end.
 - (2) Remove bolt, washers, and nut securing collective bias tube rod end to collective bias lever.
 - (3) Turn collective bias tube rod end a half-turn at a time until $\frac{1}{8}$ -inch diameter drill rod slides freely in and out of rigging hole.
 - (4) Install collective bias tube rod end on mixer with self-retaining bolt, washers, and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.
 - (5) Tighten jamnut on collective bias tube rod end. **TIGHTEN JAMNUT $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

- (6) Recheck that 1/8-inch diameter drill rod slides freely in and out of rigging hole when collective stick is in low position (Figure 4-4-27-1, Sheet 2, Detail G).

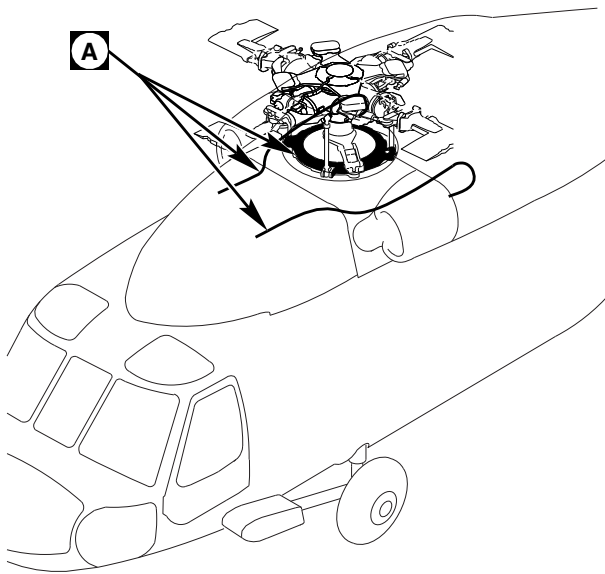
NOTE

Slight axial and radial play from load-demand rotary input is acceptable when retaining nut is correctly installed against support assembly.

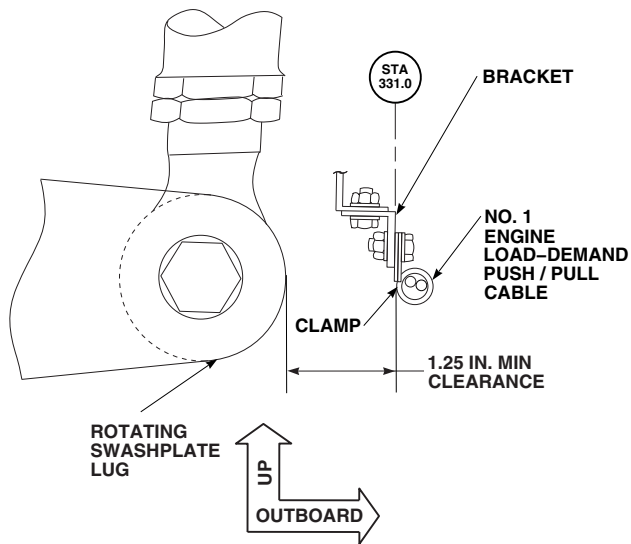
- o. Remove 1/8-inch diameter drill rod from load-demand rotary input.
- p. Connect load-demand rotary input to hydromechanical unit (HMU). Make sure rotary input alignment pin engages in slot on HMU. **TIGHTEN RETAINING NUT UNTIL IT BOTTOMS IN SUPPORT ASSEMBLY** (Figure 4-4-27-1, Sheet 1, Detail A).
- q. Coat threads of engine load-demand push/pull cable rod end with lubricating oil, Item 205, Appendix D. **TORQUE NUT CONNECTING CABLE TO ROTARY INPUT TO 100 - 150**

INCH-POUNDS (Figure 4-4-27-1, Sheet 1, Detail B).

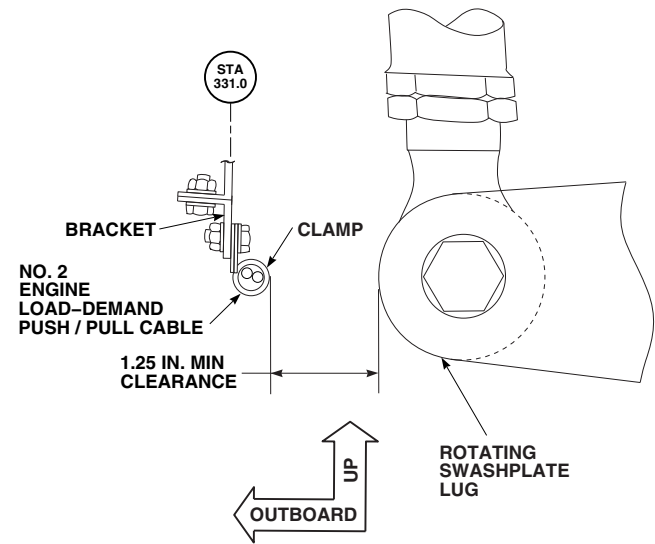
- r. Perform engine control push/pull cable inspection (PARA 4-3-2).
- s. Perform engine load-demand push/pull cable friction check (PARA 4-3-3).
- t. After installing engine load-demand push/pull cable rod end, seal threads extending past nut with sealing compound, Item 292, Appendix D.
- u. Close and latch main rotor pylon sliding cover.
- v. Install engine air inlet (PARA 4-4-4).
- w. Close engine cowling and latch.
- x. Disconnect backup hydraulic power (PARA 1-6-15) and external electrical power (PARA 1-6-16).
- y. Perform maintenance test flight (Appropriate MTF).



A



LEFT SIDE



RIGHT SIDE

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FIGURE 4-4-27-2. ENGINE LOAD-DEMAND PUSH/PULL CABLE CLEARANCE

4-4-28 ENGINE POWER-AVAILABLE PUSH/PULL CABLE.

PARAGRAPH BREAKDOWN

	PAGE
4-4-28.1 Replace Engine Power-Available Push/Pull Cable	4-4-28-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Powerplant Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Spiral Wrap, Item 315, Appendix D
- Tape, Item 317C, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 4-4-4
- PARA 4-4-33

4-4-28.1 Replace Engine Power-Available Push/Pull Cable.

NOTE

Replacement of both No. 1 and No. 2 engine power-available push/pull cables is the same.

4-4-28.1.1 Remove.

- Open engine cowling.
- Remove engine air inlet (PARA 4-4-4).
- Place quadrant ENG POWER CONT lever OFF.
- Disconnect engine power-available rotary input from hydromechanical unit (Figure 4-4-28-1, Detail B).

- Disconnect fuselage power-available push/pull cable from engine power-available push/pull cable at firewall (Figure 4-4-28-1, Detail A).
- Remove nut securing engine power-available push/pull cable and remove cable from firewall. Keep nut for installation of new cable.
- Unclamp engine power-available push/pull cable from firewall. Remove nut from cable. Keep nut for installation of new cable.
- Remove tape and remove spiral wrap from cable.
- Disconnect engine power-available push/pull cable from power-available rotary input. Make sure retaining nut stays on power-available rotary input (Figure 4-4-28-1, Detail B).

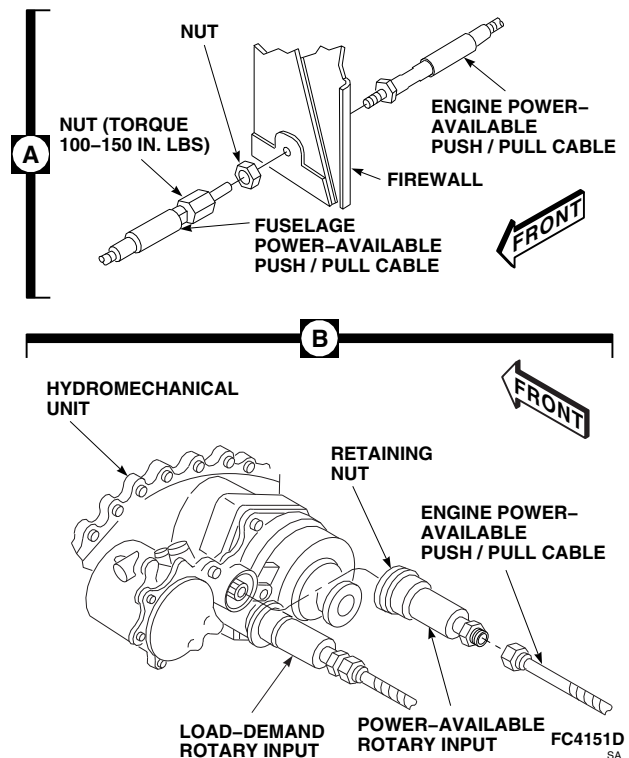
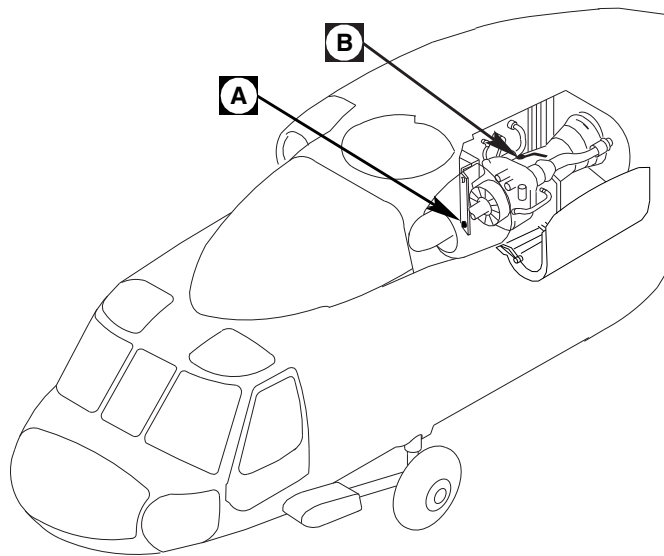


FIGURE 4-4-28-1. REPLACE ENGINE POWER-AVAILABLE PUSH/PULL CABLE

4-4-28.1.2 Install.

NOTE

- Make sure retaining nut is installed on power-available rotary input before connecting to engine power-available push/pull cable.
- When connecting engine power-available push/pull cable to rotary input, make sure internal connection is made.

- a. Connect engine power-available push/pull cable to power-available rotary input. Do not torque nut now (Figure 4-4-28-1, Detail B).
- b. Install nut on end of engine power-available push/pull cable.
- c. Add spiral wrap, Item 315, Appendix D, as necessary from firewall to power-available rotary input in areas where cable may chafe. Secure spiral wrap in place with tape, Item 317C, Appendix D.

- d. Place end of engine power-available push/pull cable through firewall. Install nut securing cable to firewall (Figure 4-4-28-1, Detail A).

NOTE

When clamping cable, make sure there is adequate clearance between cable and surrounding structure and components.

- e. Clamp engine power-available push/pull cable to firewall.

NOTE

When connecting engine power-available push/pull cables, make sure internal connection is made.

- f. Connect fuselage power-available push/pull cable to engine power-available push/pull cable and clamp to every rib along firewall. **TORQUE NUT CONNECTING PUSH/PULL CABLES TO 100 - 150 INCH-POUNDS.**
- g. Rig engine power-available system (PARA 4-4-33).

- h. Install engine air inlet (PARA 4-4-4).
- i. Close and latch engine cowling.
- j. Perform maintenance test flight (Appropriate MTF).

4-4-29 FUSELAGE POWER-AVAILABLE PUSH/PULL CABLE.

PARAGRAPH BREAKDOWN

	PAGE
4-4-29.1 Replace Fuselage Power-Available Push/Pull Cable	4-4-29-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Powerplant Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 9, Appendix D
- Acetone, Item 25, Appendix D
- Adhesive, Item 49, Appendix D
- Machinery Towel, Item 211, Appendix D

- Sealing Compound, Item 292, Appendix D
- Spiral Wrap, Item 315, Appendix D
- Tape, Item 317C, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 1-6-16
- PARA 2-4-69
- PARA 4-4-4
- PARA 4-4-21
- PARA 4-4-33

4-4-29.1 Replace Fuselage Power-Available Push/Pull Cable.

NOTE

Replacement of both left and right side fuselage power-available push/pull cables is the same.

4-4-29.1.1 Remove.

- Remove engine control quadrant covers (PARA 4-4-21).
- Move quadrant ENG POWER CONT lever in IDLE detent.

- Open engine cowlings.
- Disconnect power-available rotary input from hydromechanical unit (Figure 4-4-29-1, Detail C).
- Remove bolt, spacer, and nut securing rod end of fuselage power-available push/pull cable to quadrant ENG POWER CONT lever (Figure 4-4-29-1, Detail A).
- Remove bolts, washers, and bracket halves securing fuselage power-available push/pull cable and fuel selector valve push/pull cable to cockpit ceiling.

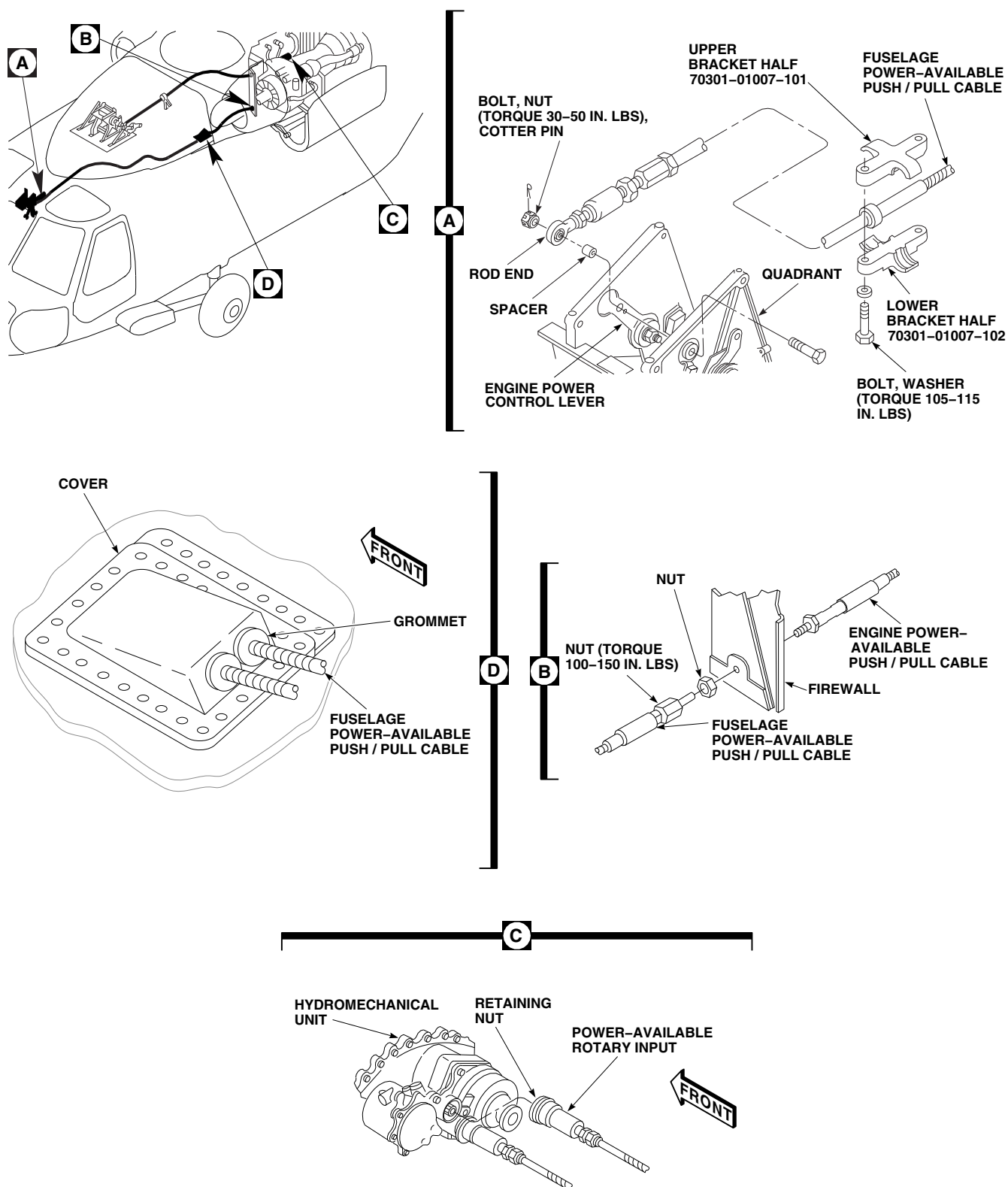
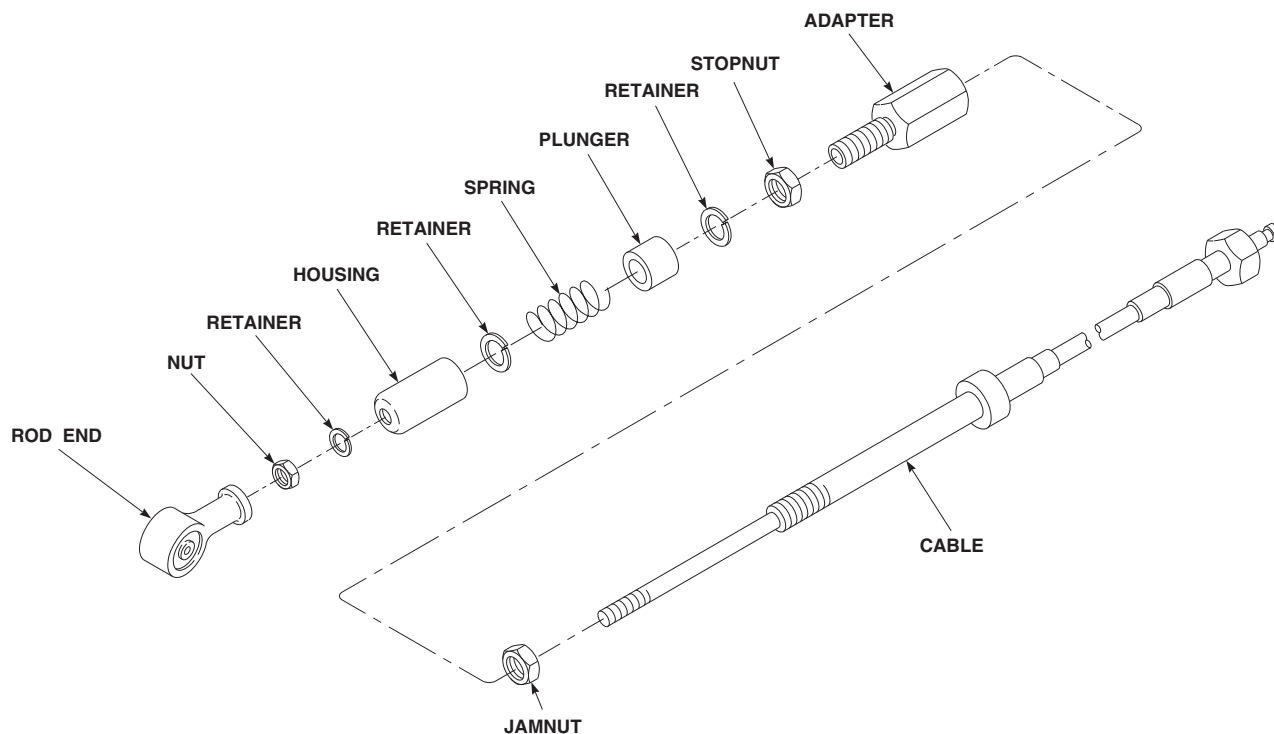
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FIGURE 4-4-29-1. REPLACE FUSELAGE POWER-AVAILABLE PUSH/PULL CABLE



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FIGURE 4-4-29-2. DISASSEMBLE/ASSEMBLE FUSELAGE POWER-AVAILABLE PUSH/PULL CABLE

- g. Remove engine air inlet (PARA 4-4-4).
- h. Remove soundproofing panels from cabin ceiling as necessary (PARA 2-4-69).
- i. Disconnect fuselage power-available push/pull cable from engine power-available push/pull cable at firewall (Figure 4-4-29-1, Detail B).
- j. Unclamp fuselage power-available push/pull cable from helicopter.
- k. Remove tape and remove spiral wrap from cable.
- l. Carefully pull cable from grommet and remove fuselage power-available push/pull cable from helicopter (Figure 4-4-29-1, Detail D).
- m. Remove rod end, nut, retainer, housing, retainer, spring, plunger, retainer, stopnut, adapter, jamnut, and cable (Figure 4-4-29-2).

4-4-29.1.2 Install.

- a. Inspect grommet in cover at STA 338.0 for security and wear. If grommet is loose or worn

through, replace grommet as follows (Figure 4-4-29-1, Detail D):

- (1) Make 45° cut through edge of replacement grommet.
- (2) Using nonmetallic scraper and machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D, remove old grommet and clean adhesive from cover. Do not allow solvent to dry on cover. Keep turning cloth after each wipe.
- (3) Using abrasive cloth, Item 9, Appendix D, lightly scuff inside of outer groove of grommet.
- (4) Apply light coat of adhesive, Item 49, Appendix D, to groove of grommet and edge of hole in cover. Allow adhesive to air-dry for 10 minutes or until tacky.
- (5) Install grommet in hole in cover. For best results, allow adhesive to cure for 24 hours.

- b. Install jamnut, adapter, stopnut, retainer, plunger, spring, retainer, housing, retainer, nut, and rod end on replacement cable (Figure 4-4-29-2).
- c. Route fuselage power-available push/pull cable from quadrant ENG POWER CONT lever in cockpit ceiling to engine compartment firewall.
- d. Carefully slide fuselage power-available push/pull cable through grommet (Figure 4-4-29-1, Detail D).
- e. Apply sealing compound, Item 292, Appendix D, to edge of grommet and cover, at 45° slice and around cable.
- f. Add spiral wrap, Item 315, Appendix D, 5-inches long, where cable passes through each fuselage frame. Secure spiral wrap in place with tape, Item 317C, Appendix D.

NOTE

When connecting push/pull cable, make sure internal connection is made.

- g. Connect fuselage power-available push/pull cable to engine power-available push/pull cable at firewall (Figure 4-4-29-1, Detail B). **TORQUE NUT TO 100 - 150 INCH-POUNDS.**
- h. Clamp fuselage power-available push/pull cable to fuel selector valve push/pull cable every 6 inches from ENG POWER CONT lever to STA 340.0.



Damage to equipment will result if bracket halves are not installed properly. Bracket halves are not interchangeable. Bracket half, 70301-01007-101, is upper bracket half; bracket half, 70301-01007-102, is lower bracket half.

- i. Secure fuselage power-available push/pull cable and fuel selector valve push/pull cable to cockpit ceiling with bolts, washers, and bracket halves (Figure 4-4-29-1, Detail A). **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**
- j. Secure rod end of power-available push/pull cable to quadrant ENG POWER CONT lever with bolt, spacer, and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin in nut.
- k. Clamp fuselage power-available push/pull cable to structure.
- l. Rig engine power-available system (PARA 4-4-33).
- m. Install engine air inlet (PARA 4-4-4).
- n. Install soundproofing panels in cabin ceiling (PARA 2-4-69).
- o. Close and latch engine cowling.
- p. Perform maintenance test flight (Appropriate MTF).

4-4-30 FUEL SELECTOR VALVE PUSH/PULL CABLE.

PARAGRAPH BREAKDOWN

	PAGE
4-4-30.1 Replace Fuel Selector Valve Push/Pull Cable	4-4-30-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Powerplant Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 9, Appendix D
- Acetone, Item 25, Appendix D
- Adhesive, Item 49, Appendix D

- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 292, Appendix D
- Spiral Wrap, Item 315, Appendix D
- Tape, Item 317C, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-69
- PARA 4-4-4
- PARA 4-4-21

4-4-30.1 Replace Fuel Selector Valve Push/Pull Cable.

NOTE

Replacement of both left and right side fuel selector valve push/pull cables is the same.

4-4-30.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Move ENG FUEL SYS lever OFF.
- Unlock fastener studs securing upper console to cockpit overhead, and hinge down.

- Remove quadrant covers (PARA 4-4-21).
- Remove bolt, washer, and nut securing fuel selector valve push/pull cable rod end to quadrant fuel selector lever. Remove rod end and nut from cable (Figure 4-4-30-1, Detail A).
- Inspect rod end for axial and radial play. **NONE ALLOWED.**
- Remove bolts, washers, and bracket halves securing fuel selector valve push/pull cable and fuselage power-available push/pull cable to cockpit ceiling.
- Open engine cowling.
- Open oil cooler compartment access doors.

- j. Remove engine air inlet (PARA 4-4-4).
- k. Remove soundproofing panels from cabin ceiling (PARA 2-4-69).
- l. Disconnect fuel selector valve push/pull cable from fuel selector valve control box (Figure 4-4-30-1, Detail B).
- m. Unclamp fuel selector valve push/pull cable from firewall and structure.
- n. Remove tape and remove spiral wrap from fuel selector valve push/pull cable.
- o. Carefully pull fuel selector valve push/pull cable from grommet and remove cable from helicopter (Figure 4-4-30-1, Detail D).
- b. Route fuel selector valve push/pull cable from quadrant fuel selector lever to oil cooler compartment firewall.
- c. Carefully slide fuel selector valve push/pull cable through grommet.
- d. Apply sealing compound, Item 292, Appendix D, to edge of grommet and cover at 45° slice and around fuel selector valve push/pull cable.
- e. Connect fuel selector valve push/pull cable to fuel selector valve control box. **TORQUE COUPLING NUT TO 100 - 150 INCH-POUNDS** (Figure 4-4-30-1, Detail B and Detail C).
- f. Add spiral wrap, Item 315, Appendix D, to fuel selector valve push/pull cable at these locations:

4-4-30.1.2 Install.

- a. Inspect grommet in cover at STA 338.0 for security and wear. If grommet is loose or worn through, perform the following (Figure 4-4-30-1, Detail D):
 - (1) Make a 45° cut through edge of replacement grommet.
 - (2) Using nonmetallic scraper and machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D, remove old grommet and clean adhesive from cover. Do not allow solvent to dry on airframe. Keep turning towel after each wipe.
 - (3) Using abrasive cloth, Item 9, Appendix D, lightly scuff insides of outer groove of grommet.
 - (4) Apply light coat of adhesive, Item 49, Appendix D, to groove of grommet and edge of hole in cover. Allow adhesive to air-dry for 10 minutes or until tacky.
 - (5) Install grommet in hole in cover. For best results, allow adhesive to cure for 24 hours.
- (1) From STA 245.0 through STA 253.0.
- (2) From STA 257.0 through STA 263.0.
- (3) From STA 270.0 through STA 275.0.
- (4) From STA 278.0 through STA 283.0 at BL 5.0.
- (5) From BL 8.0 through BL 12.0 at STA 283.0.
- (6) From STA 290.0 through STA 294.0.
- (7) From STA 297.0 through STA 303.0.
- (8) From STA 313.0 through STA 318.0.
- (9) From STA 325.0 through STA 330.0.
- (10) From STA 350.0 through STA 360.0.
- g. Secure spiral wrap in place with tape, Item 317C, Appendix D.
- h. Clamp fuel selector valve push/pull cable to fuselage power-available push/pull cable every 6-inches from ENG POWER CONT lever to STA 340.0.

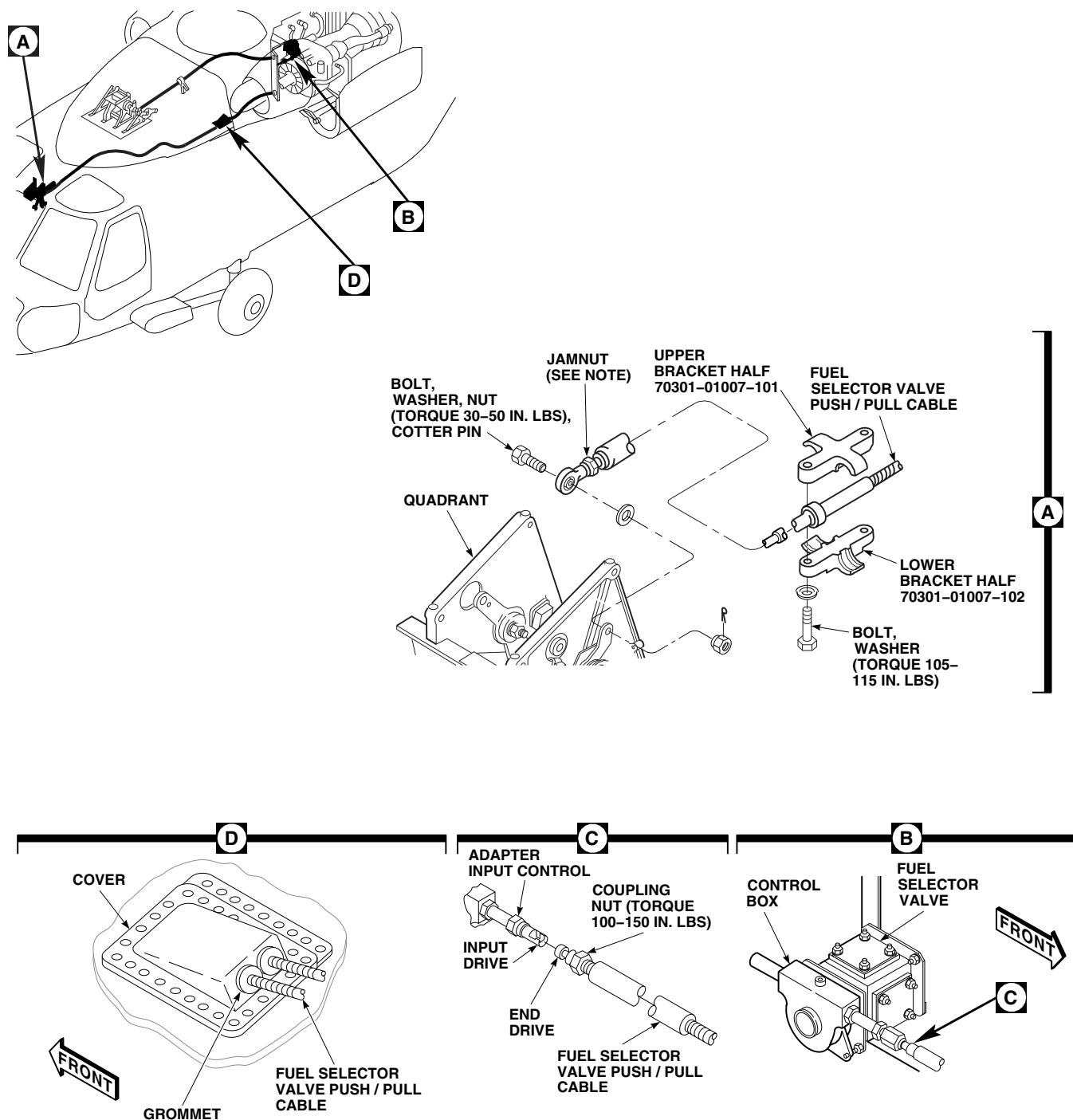


FIGURE 4-4-30-1. REPLACE FUEL SELECTOR VALVE PUSH/PULL CABLE

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CAUTION

Damage to equipment will result if bracket halves are not installed properly. Bracket halves are not interchangeable. Bracket half, 70301-01007-101, is upper bracket half; 70301-01007-102, is lower bracket half.

NOTE

Keyway on fuel selector valve and control box should be OFF.

- i. Install fuel selector valve push/pull cable and fuselage power-available push/pull cable to cockpit ceiling with bolts, washers, and bracket halves. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS** (Figure 4-4-30-1, Detail A).
- j. Clamp fuel selector valve push/pull cable to firewall and structure.

NOTE

Make sure ENG FUEL SYS lever is OFF.

- k. Thread jamnut and rod end onto fuel selector valve push/pull cable until rod end bearing and

fuel selector lever holes line up. **TIGHTEN JAMNUT $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

- l. Bolt fuel selector valve push/pull cable rod end to quadrant fuel selector lever.
- m. Move fuel selector lever through DIRECT and X FEED positions and check by feel for fuel selector control box detenting in each position.
- n. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.
- o. Install quadrant covers (PARA 4-4-21).
- p. Move fuel selector levers back to the off position.
- q. Hinge up and lock upper console to cockpit overhead.
- r. Install soundproofing panels in cabin ceiling (PARA 2-4-69).
- s. Close oil cooler compartment access doors.
- t. Install engine air inlet (PARA 4-4-4).
- u. Close and latch engine cowling.

4-4-31 FUEL SELECTOR VALVE CONTROL BOX.

PARAGRAPH BREAKDOWN

		PAGE
4-4-31.1	Replace Fuel Selector Valve Control Box	4-4-31-1

INITIAL SETUP

- Torque Wrench, 30 - 150 in. lbs

Tools

- Aircraft Mechanic’s Toolkit
- Powerplant Repairer’s Toolkit

4-4-31.1 Replace Fuel Selector Valve Control Box.

NOTE

Replacement of both left and right side fuel selector control boxes is the same.

4-4-31.1.1. Remove.

- Place ENG FUEL SYS lever OFF.
- Open engine cowling.
- Open oil cooler compartment access doors.
- Disconnect fuel selector cable from fuel selector valve control box (Figure 4-4-31-1, Detail A).
- Remove lockring and fuel selector valve control box from fuel selector valve.

4-4-31.1.2. Install.

- Line up keyway on fuel selector valve and control box at OFF. Install control box on fuel selector valve with slot in ear over torque stud. Install lockring (Figure 4-4-31-1, Detail A).
- Connect fuel selector cable to control box. **TORQUE NUT TO 100 - 150 INCH-POUNDS.**
- Move ENG FUEL SYS lever to DIR, XFD, and OFF while an assistant checks to make sure shaft of fuel selector valve turns.
- Make sure area is clean and free of foreign material.
- Close oil cooler compartment access doors.
- Close and latch engine cowling.

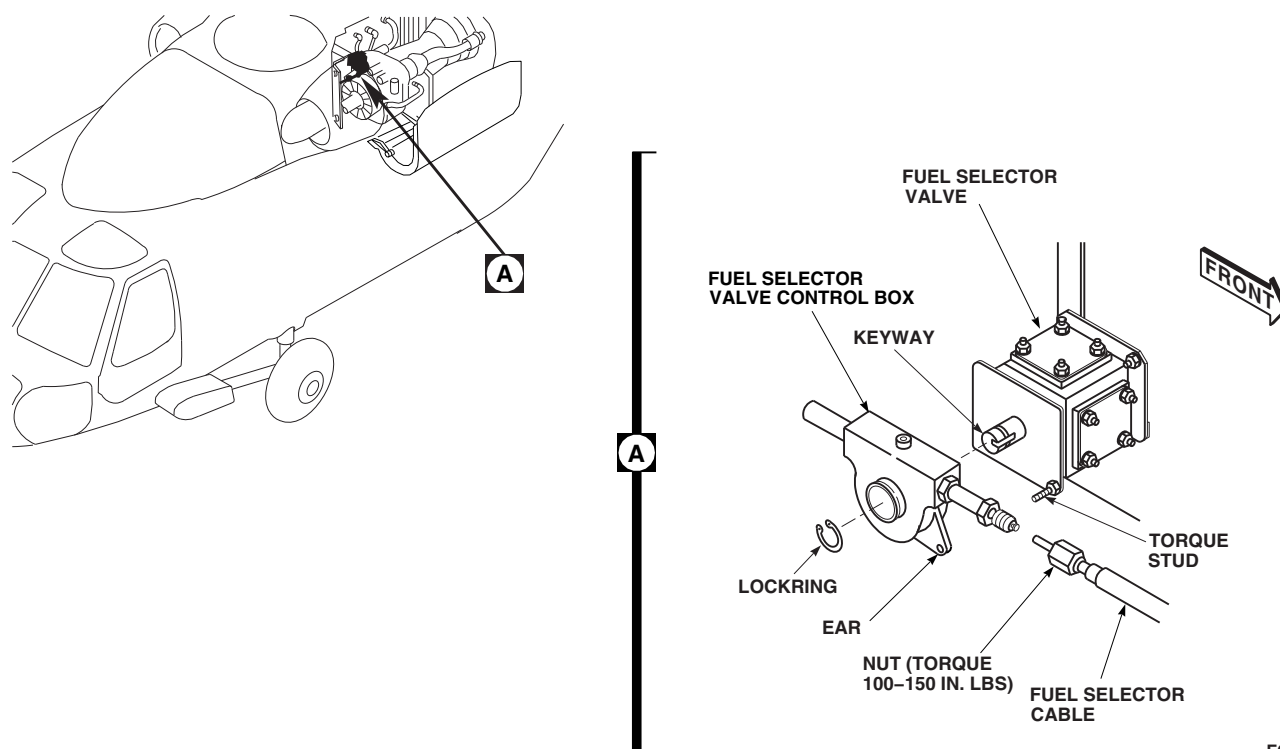
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FIGURE 4-4-31-1. REPLACE FUEL SELECTOR VALVE CONTROL BOX

4-4-31-2

4-4-32 ENGINE LOAD-DEMAND PUSH/PULL CABLE SUPPORTS.

PARAGRAPH BREAKDOWN

	PAGE
4-4-32.1 Replace Engine Load-Demand Push/Pull Cable Supports	4-4-32-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Fluorescent Penetrant Inspection Kit
- Powerplant Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Primer, Item 258, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 4-3-3
- PARA 4-4-27

4-4-32.1 Replace Engine Load-Demand Push/Pull Cable Supports.

NOTE

Replacement of left and right engine control cable supports is the same.

4-4-32.1.1 Remove.

- Open main rotor pylon sliding cover.
- Disconnect engine load-demand push/pull cable from engine control cable support (PARA 4-4-27).
- Remove bolts, washers, and nuts securing engine control cable support to accessory module (Figure 4-4-32-1, Detail A). Remove cable support.

4-4-32.1.2 Inspect.

Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect cable support for cracks. **NONE ALLOWED.** If cracked, replace cable support.

4-4-32.1.3 Install.

- Using primer, Item 258, Appendix D, coat boltholes of engine load-demand push/pull cable supports on accessory modules.

NOTE

Install cable support with short arm at top and recess facing rear.

- Position cable support on accessory module. Install washers between lugs as necessary to fill gaps. Install bolts, washers, and nuts securing

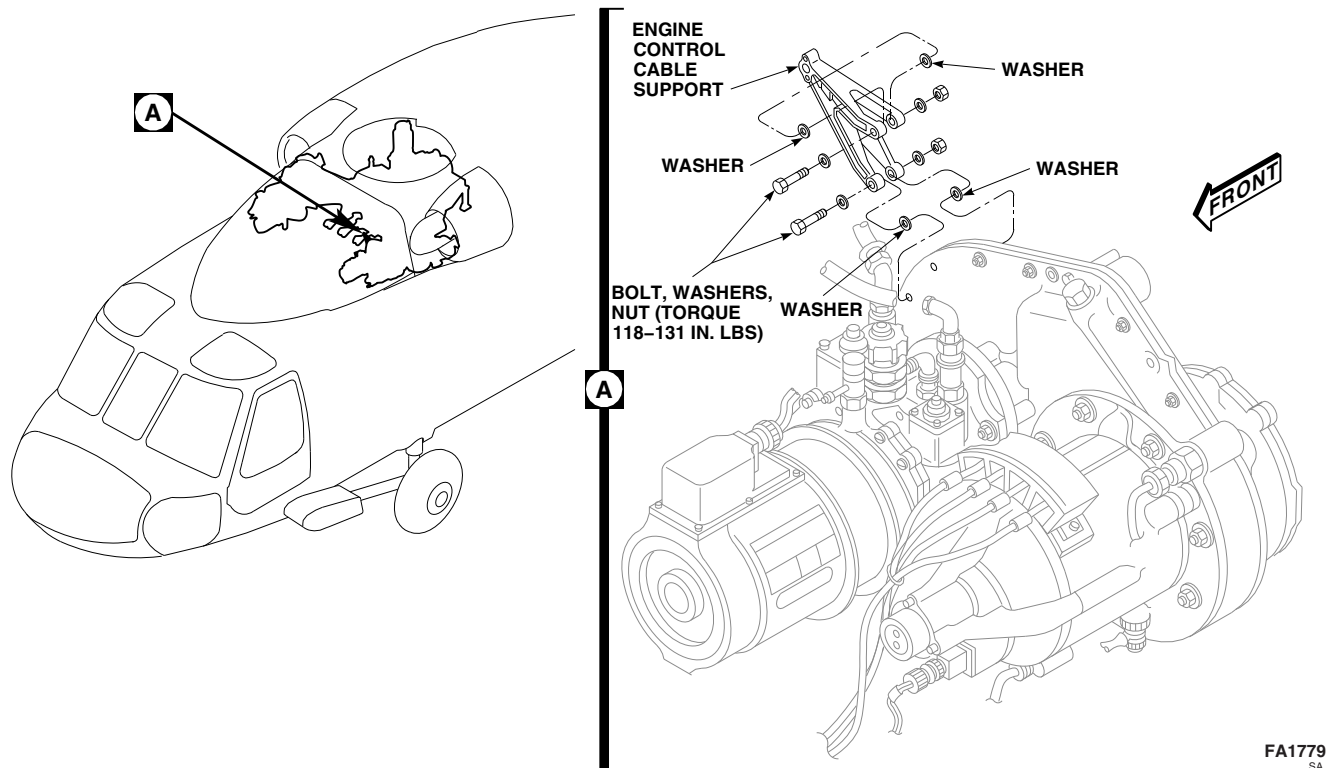
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FIGURE 4-4-32-1. REPLACE ENGINE LOAD-DEMAND PUSH/PULL CABLE SUPPORTS

cable support to accessory module. **TORQUE NUTS TO 118 - 131 INCH-POUNDS** (Figure 4-4-32-1, Detail A).

- c. Connect engine load-demand push/pull cable to cable support (PARA 4-4-27).

- d. Perform engine load-demand push/pull cable friction check (PARA 4-3-3).
- e. Make sure area is clean and free of foreign material.
- f. Close main rotor pylon sliding cover.

4-4-33 ENGINE POWER-AVAILABLE SYSTEM RIGGING AND ADJUSTMENT.

PARAGRAPH BREAKDOWN

	PAGE
4-4-33.1 Rig and Adjust Engine Power-Available System	4-4-33-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill Rod, 1⁄8-inch diameter, 3-inches long
- Drill Rod, 1⁄8-inch diameter, 10-inches long
- Electrical Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

References

- Appropriate MTF
- PARA 1-6-16

- PARA 4-4-19
- PARA 4-4-21
- PARA 4-4-23
- PARA 4-4-25
- PARA 4-4-26
- PARA 4-4-28
- PARA 4-4-29
- PARA 4-5-16

4-4-33.1 Rig and Adjust Engine Power-Available System.

NOTE

The following procedure applies to both No. 1 and No. 2 systems.

- Turn off all electrical power (PARA 1-6-16).
- Move fuel selector levers toward front.
- Unlock fastener studs securing upper console to cockpit overhead, and hinge down.
- Remove quadrant covers (PARA 4-4-21).
- Move ENG POWER CONT lever to OFF.

- Open engine cowling.
- Disconnect power-available rotary input from engine hydromechanical unit (HMU) (Figure 4-4-33-1, Detail A).
- Move quadrant ENG POWER CONT lever to IDLE detent and insert 10-inch, 1⁄8-inch diameter drill rod through rigging holes in lever and quadrant support.
- Position rotary input to approximate installed position. Insert 3-inch, 1⁄8-inch diameter drill rod through rigging hole in outer housing and inner rotary driver of rotary input (Figure 4-4-33-1, Detail C).
- If drill rod slides in and out freely, or with slight friction, proceed to step 1.

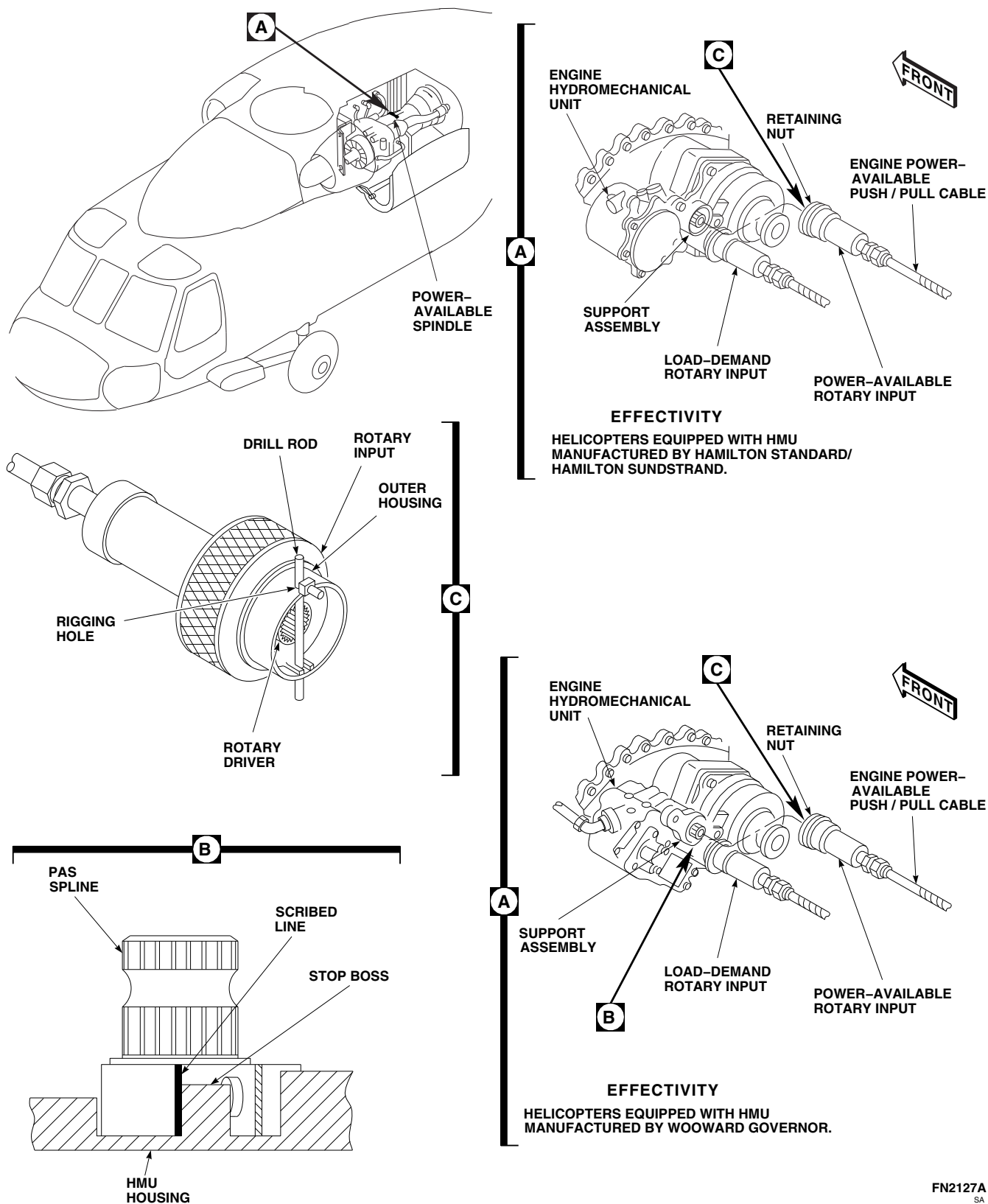


FIGURE 4-4-33-1. RIG ENGINE POWER-AVAILABLE SYSTEM

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- k. If drill rod cannot be inserted in rotary input, perform the following:

- (1) Remove bolt, spacer, and nut securing rod end of power-available push/pull cable to ENG POWER CONT lever (Figure 4-4-33-2, Detail A).
- (2) Position rotary input barrel to align holes and insert drill rod (Figure 4-4-33-1, Detail C).
- (3) Pull rod end bearing forward to remove slack in cable (Figure 4-4-33-2). Loosen jamnut and adjust rod end until rod end bearing and quadrant POWER CONT lever holes line up.
- (4) **TIGHTEN JAMNUT $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- (5) Install fuselage power-available push/pull cable rod on quadrant ENG POWER CONT lever, with bolt, spacer, and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

- l. Remove drill rods from rigging holes in quadrant support, lever, and rotary input (Figure 4-4-33-1, Detail C).

- m. Position rotary input at HMU mounting location and check for any twisting or rotation required to make connection (Figure 4-4-33-1, Detail A). If twisting or rotation is required, remove by adjusting connection of rotary input and power-available cable (PARA 4-4-19).

NOTE

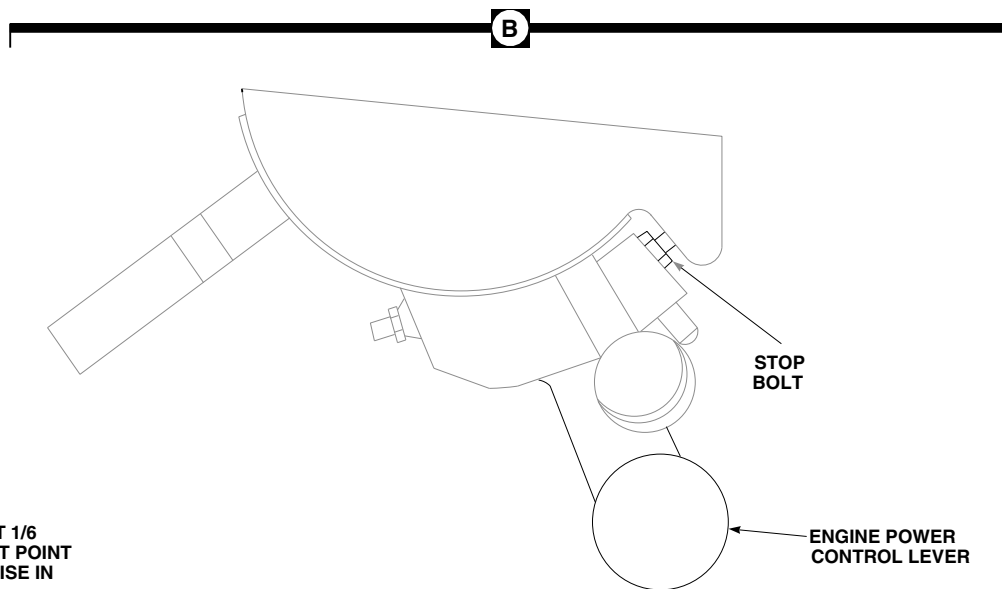
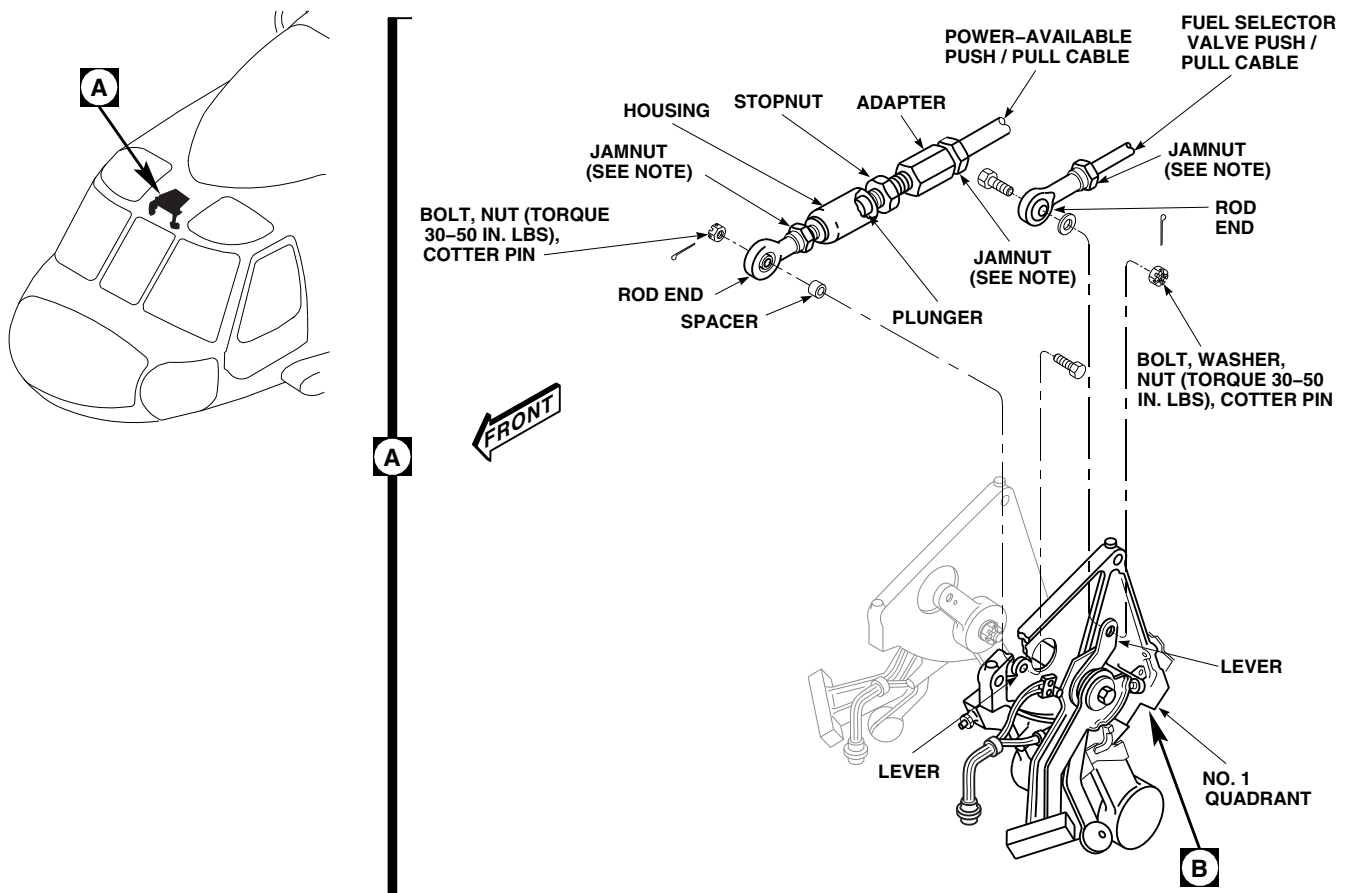
Slight axial and radial play from the power-available rotary input is acceptable when retaining nut is correctly installed against support assembly.

- n. Connect power-available rotary input to HMU. Make sure that rotary input alignment pin engages slot on engine HMU. **TIGHTEN RETAINING NUT UNTIL IT BOTTOMS IN SUPPORT ASSEMBLY.**

- o. Install quadrant covers (PARA 4-4-21).
- p. Move quadrant ENG POWER CONT lever to FLY detent. If stopnut contacts housing before adapter contacts plunger, rotate stopnut away from housing.
- q. Adjust adapter on fuselage power-available push/pull cable to contact face of plunger. Turn adapter one complete turn against plunger face. **TIGHTEN JAMNUT $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT** (Figure 4-4-33-2, Detail A).

NOTE

- There are two types of HMUs. HMUs are identified by manufacturer information on data plate and external appearance.
 - HMUs are manufactured by Hamilton Standard/Hamilton Sundstrand and Woodward Governor.
 - HMUs manufactured by Hamilton Standard/Hamilton Sundstrand utilize an audible click to note when DEC lockout is reached.
 - HMUs manufactured by Woodward Governor do not utilize an audible click when DEC lockout is reached. DEC lockout is indicated by alignment of scribed line on power-available spindle with stop boss on HMU housing.
- r. Inspect data plate on HMU to determine which HMU is installed.
- s. On HMUs manufactured by Hamilton Standard/Hamilton Sundstrand, rig for DEC lockout as follows:
- (1) Move quadrant ENG POWER CONT lever from FLY detent towards LOCKOUT and listen at engine HMU for a click to indicate DEC lockout has occurred. Stop moving lever as soon as click is heard.
 - (2) Adjust stopnut against housing. Turn stopnut back two full turns.



NOTE
TIGHTEN JAMNUT 1/6
TO 1/3 TURN PAST POINT
WHERE SHARP RISE IN
TORQUE IS FELT.

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FIGURE 4-4-33-2. ADJUST ENGINE POWER-AVAILABLE SYSTEM

- (3) Move ENG POWER CONT lever to IDLE to reset DEC.
 - (4) Move ENG POWER CONT lever to LOCKOUT and listen at HMU for click. Click must be heard before stopnut contacts housing.
- t. On HMUs manufactured by Woodward Governor, rig for DEC lockout as follows:
- (1) Move quadrant ENG POWER CONT lever from FLY detent towards LOCKOUT. Stop moving lever when scribed line on power-available spindle is aligned with stop boss on HMU housing (Figure 4-4-33-1, Detail B). Scribed line must be completely visible.
 - (2) Adjust stopnut in quadrant against housing. Turn stopnut back two full turns.
 - (3) Move quadrant ENG POWER CONT lever to IDLE to reset DEC.
 - (4) Move quadrant ENG POWER CONT lever towards LOCK OUT and check that scribed line on power-available spindle is aligned with stop boss on HMU housing and is completely visible. Alignment must occur before stopnut contacts housing.
- u. If click is not heard before housing contacts stopnut, or if housing contacts stopnut before alignment occurs, repeat step r. and step s. or step r. and step t.
- v. If attempt to rig is still unsuccessful, repeat step h. through step s. or step h. through step t.
- w. If attempt to rig is still unsuccessful, replace engine power-available push/pull cable and fuselage power-available push/pull cable (PARA 4-4-28 and PARA 4-4-29).
- x. Adjust OFF stop as follows:
- (1) Turn stop bolt in so push/pull cable internal stop will limit travel when quadrant ENG POWER CONT lever is moved to rear (Figure 4-4-33-2, Detail B).
 - (2) Turn stop bolt out until it contacts face of quadrant ENG POWER CONT lever when lever limits have been reached.
 - (3) Release quadrant ENG POWER CONT lever and turn stop bolt out one full turn to make sure OFF stop is reached before push/pull cable internal stop or engine stop.
- y. Adjust quadrant ENG FUEL SYS lever as follows:
- (1) Place quadrant ENG FUEL SYS lever in line with OFF on quadrant side cover.
 - (2) Adjust rod end on fuel selector valve push/pull cable so rod end bearing and lever holes line up. **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT** (Figure 4-4-33-2, Detail A).
 - (3) Install fuel selector valve push/pull cable to quadrant ENG FUEL SYS lever with bolt, washer, and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.
- z. Adjust engine starter switch (PARA 4-4-23).
- aa. Adjust starter override switch (PARA 4-4-26).
- ab. Adjust starter abort switch (PARA 4-4-25).
- ac. Adjust fire extinguisher arming switch (PARA 4-5-16).
- ad. Adjust secondary stop blocks (PARA 4-4-21).
- ae. Make sure area is clean and free of foreign material.
- af. Move fuel selector levers back to OFF.
- ag. Hinge up and lock upper console to cockpit overhead.
- ah. Close engine cowling.
- ai. Perform maintenance test flight (Appropriate MTF).

4-4-34 ENGINE MOUNT RESTRAINER RINGS.

PARAGRAPH BREAKDOWN

	PAGE
4-4-34.1 Replace Engine Mount Restrainer Rings	4-4-34-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Clecos
- Drill Bit, Letter D (0.2460-inch diameter)
- Drill Bit, Letter G (0.2610-inch diameter)
- Drill Bit, No. 4 (0.2090-inch diameter)
- Drill Bit, 15/64-inch (0.2344-inch diameter)
- Drill Bit, ¼-inch (0.2500-inch diameter)
- Drill, Rotary
- Fluorescent Penetrant Inspection Kit
- Grease Pencil

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Fastener, MS90354-0604
- Hi-Lok Collar, HL1087APBW-8
- Hi-Lok Pin, HL20PB-8-5

References

- Appendix D
- ASTM E1417
- PARA 2-6-5
- PARA 4-4-6

4-4-34.1 Replace Engine Mount Restrainer Rings.

NOTE

Replacement of front and rear restrainer rings in No. 1 and No. 2 forward support tubes is the same.

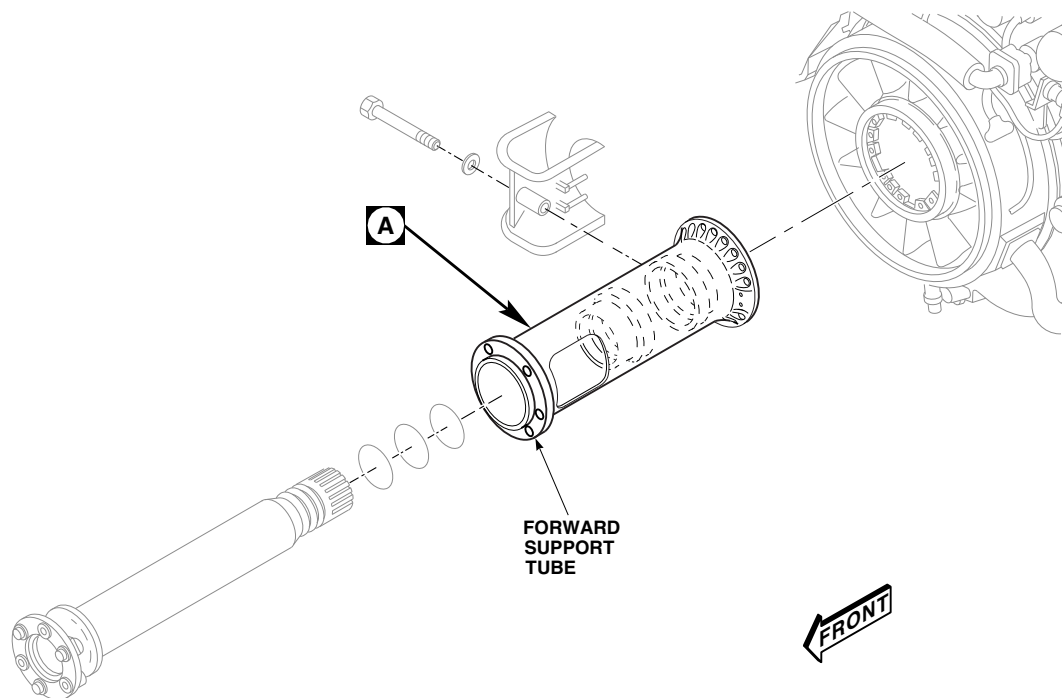
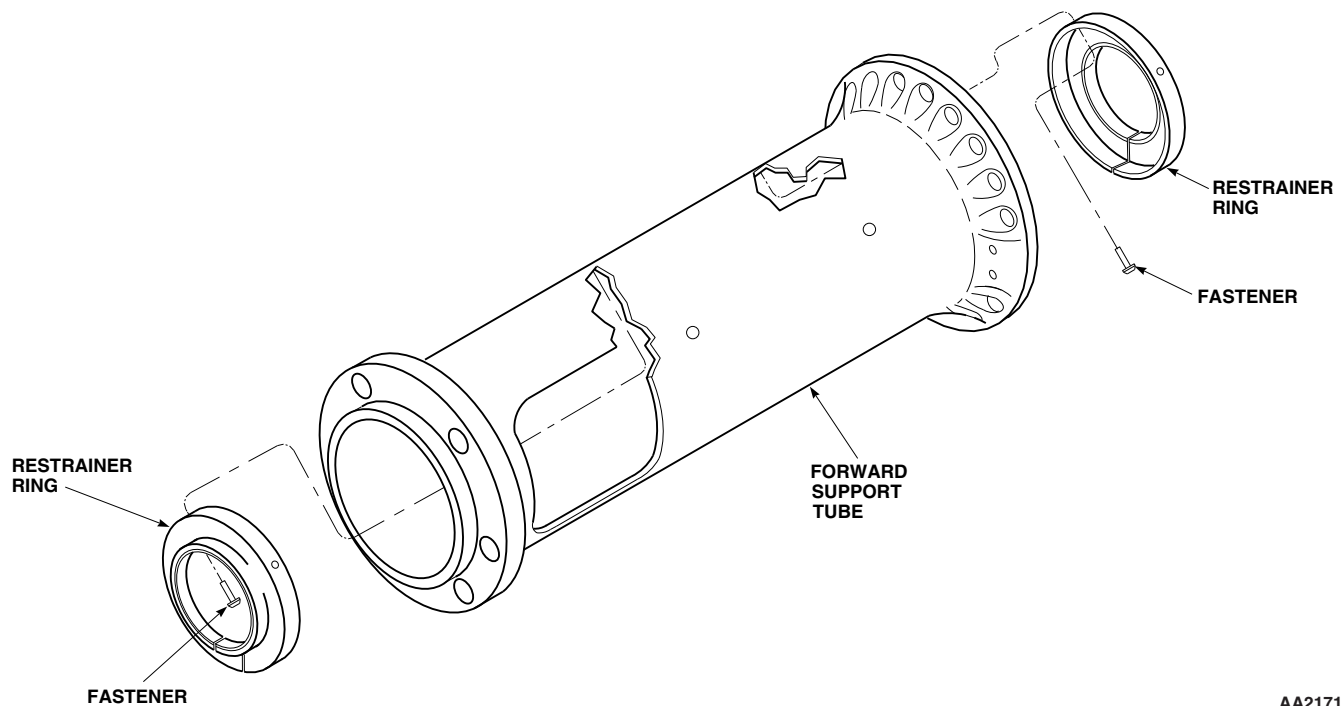
4-4-34.1.1 Remove.

- Remove forward support tube/engine output shaft (PARA 4-4-6).

- Using drill with No. 4 drill bit, remove four fasteners securing restrainer ring to forward support tube (Figure 4-4-34-1, Detail A). Do not allow rivet to spin in hole. Remove ring from tube.
- Using abrasive cloth, Item 10, Appendix D, remove any remaining primer from area around holes.

4-4-34.1.2 Install.

- Using three clecos, install restrainer ring to inside of tube with slot aligned vertically.

**A**AA2171
SA**FIGURE 4-4-34-1. REPLACE ENGINE MOUNT RESTRAINER RINGS**

Leave hole closest to slot on window side of tube open (Figure 4-4-34-1, Detail A).

- b. Using drill with 1/4-inch drill bit, predrill open hole.
- c. Using drill with letter G drill bit, enlarge open hole. Install cleco to open hole.
- d. Remove cleco from adjacent hole on window side of tube and repeat step b. and step c. for this hole.
- e. Remove cleco from next hole on non-window side of tube and using drill with 15/64-inch drill bit, predrill hole.
- f. Using drill with letter D drill bit, enlarge open hole. Install cleco to open hole.
- g. Remove cleco from remaining hole and repeat step e. and step f. for this hole.
- h. Using grease pencil, mark location of slot and remove restrainer ring from tube.
- i. Deburr holes in restrainer ring and forward support tube.
- j. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect area around holes for cracks (ASTM E1417). **NONE ALLOWED.**
- k. Touch up area around hole with corrosion resistant coating, Item 118, Appendix D.

NOTE

Restrainer rings must be installed with slots 180° opposed.

- l. Using three clecos, install restrainer ring to forward support tube with slot aligned with mark. Leave hole closest to slot on window side of tube open.
- m. Install fastener wet with epoxy primer from epoxy primer coating kit, Item 135, Appendix D, to open hole.
- n. Remove cleco from adjacent hole on window side of tube and install fastener wet with epoxy primer from epoxy primer coating kit, Item 135, Appendix D.
- o. Remove cleco from next hole on non-window side of tube and install Hi-Lok pin, and Hi-Lok collar, wet with epoxy primer from epoxy primer coating kit, Item 135, Appendix D.
- p. Remove cleco from remaining hole and install Hi-Lok pin, and Hi-Lok collar, wet with epoxy primer from epoxy primer coating kit, Item 135, Appendix D.
- q. Apply epoxy primer from epoxy primer coating kit, Item 135, Appendix D, to fastener heads.
- r. Touch up paint to reworked areas (PARA 2-6-5).
- s. Install forward support tube/engine output shaft (PARA 4-4-6).

SECTION 4-5

MAINTENANCE PROCEDURES (BENCH)

SECTION OVERVIEW

PARAGRAPH	TITLE
4-5-1	Aft Engine Mount Struts (BENCH)
4-5-2	Aft Engine Mount Fittings (BENCH)
4-5-3	Aft Engine Mount Links (BENCH)
4-5-4	Aft Engine Mount Supports (BENCH)
4-5-5	Engine Mount Crotch Assembly (BENCH)
4-5-6	Engine Exhaust Ejector and Fairings (BENCH)
4-5-7	Engine Exhaust Module Duct Assembly (BENCH)
4-5-8	HIRSS Inlet Seal (BENCH)
4-5-9	Rotary Input (BENCH)
4-5-10	Engine Control Quadrant Power Lever (BENCH)
4-5-11	Engine Control Quadrant T-Handle and Fuel Lever (BENCH)
4-5-12	Engine Control Quadrant Threaded Inserts (BENCH)
4-5-13	Engine Control Quadrant Engine Starter Switch (BENCH)
4-5-14	Engine Control Quadrant Starter Abort Switch (BENCH)
4-5-15	Engine Control Quadrant Starter Override Switch (BENCH)
4-5-16	Engine Control Quadrant Fire Extinguisher Arming Switch (BENCH)
4-5-17	Engine Control Quadrant Front Cover Information Plate (BENCH)
4-5-18	Engine Control Quadrant Left and Right Cover Information Plates (BENCH)

PARAGRAPH

TITLE

4-5-19

Engine Control Quadrant Front Cover Information Plate Wiring
Harness (BENCH)

4-5-1 AFT ENGINE MOUNT STRUTS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
4-5-1.1	Repair Aft Engine Mount Struts (BENCH)	4-5-1-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Arbor Press
- Bushing Installation and Removal Kit, 65700-20002-041
- Drift
- Drill Press
- Feeler Gage, 0.005-inch
- Fluorescent Penetrant Inspection Kit
- Fly Cutter
- Protection, Skin and Breathing
- Reamer Set
- Roller Staking Tool

Materials/Parts

- Abrasive Cloth, Item 5, Appendix D
- Abrasive Cloth, Item 8, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Adhesive, Item 53, Appendix D
- Corrosion Compound, Item 108, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Petrolatum, Item 234, Appendix D
- Primer, Item 258, Appendix D
- Scouring Pad, Item 270, Appendix D

References

- Appendix D
- ASTM E1417

4-5-1.1 Repair Aft Engine Mount Struts (BENCH).

NOTE

Strut repair consists of replacement of bearings and bushings, inspecting, and refinishing.

4-5-1.1.1 Remove Bearing/Bushings.

- a. Remove bearing as follows (Figure 4-5-1-1):

WARNING

- Bushings installed in aft engine mount struts are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.
- No minor repair or corrosion removal allowed on beryllium copper components.

CAUTION

Damage to strut will occur if staked lip of bearing is improperly removed. Be careful not to damage strut when machining off staked lip of bearing.

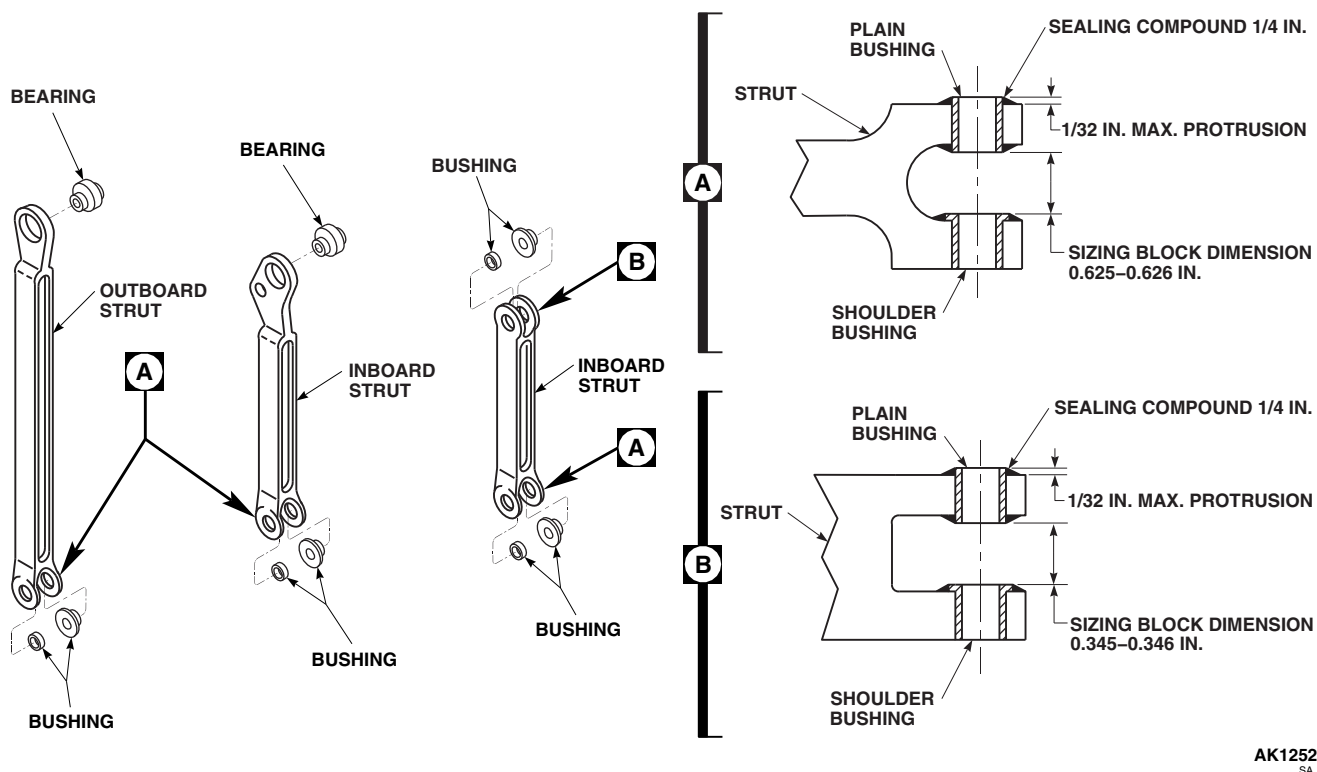
- (1) Clamp strut to baseplate of drill press. Using fly cutter, machine off part of bearing outer race swaged over chamfer on strut.
 - (2) Press bearing out from strut using arbor press and suitable drift.
- b. Remove bushing(s) from strut, using arbor press and bushing installation and removal kit.

4-5-1.1.2 Inspect.

- Inspect bearing for corrosion and damage. **PITS SHALL NOT BE VISIBLE IN 0.250-INCH DIAMETER OR EXCEED 0.010-INCH DEEP.** If pits are visible in 0.250-inch diameter or limit is exceeded, replace bearing.
- Inspect bearing for axial play. **AXIAL PLAY SHALL NOT EXCEED 0.020-INCH.** If limit is exceeded, replace bearing.
- Inspect bearing for radial play. **RADIAL PLAY SHALL NOT EXCEED 0.005-INCH.** If limit is exceeded, replace bearing.
- Inspect bearing bore for scratches. **SCRATCHES SHALL NOT EXCEED 0.500-INCH LONG AND 0.005-INCH DEEP.** If limit is exceeded, replace bearing.
- Remove minor corrosion and scratches from bearings using scouring pad, Item 270, Appendix D.
- Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect strut for cracks. **NONE ALLOWED.** If cracked, replace strut.
- Inspect strut for nicks, gouges, scratches, and pitting. **DAMAGE SHALL NOT EXCEED 0.250-INCH LONG AND 0.032-INCH DEEP.** If limit is exceeded, replace strut.
- Measure bearing hole wear per applicable dimension in Table 4-5-1-1.
- Measure bushing hole wear per applicable dimension in Table 4-5-1-1.
- Measure inner mount strut to strut mount bolt-hole wear per applicable dimension in Table 4-5-1-1.

4-5-1.1.3 Repair Engine Mounts.

- a. Repair engine mount struts as follows:
- (1) Blend out nicks, gouges, scratches, and pitting using 1/2-inch abrasive wheel, Item



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FIGURE 4-5-1-1. REPAIR AFT ENGINE MOUNT STRUTS (BENCH)

22, Appendix D. **BLENDED AREA SHALL NOT EXCEED 0.250-INCH LONG AND 0.032-INCH DEEP.** If limits are exceeded, replace strut.

ply two coats of epoxy primer from epoxy primer coating kit, Item 135, Appendix D, to surfaces where plating was disturbed.

(2) Repair damage to nickel plating as follows:

- (a) Using abrasive cloth, Item 5, Appendix D, blend out nickel plating in discrepant area. Blend discrepant area, removing only plating. Do not remove parent material.
- (b) Using abrasive cloth, Item 10, Appendix D, polish blended area to remove abrasive marks and match area smoothness.
- (c) Fluorescent penetrant inspect strut using fluorescent penetrant inspection kit (ASTM E1417). **NONE ALLOWED.** If cracked, replace strut.
- (d) Clean surface with corrosion compound, Item 108, Appendix D. Ap-

(3) Repair damage to parent metal as follows:

- (a) Using abrasive cloth, Item 8, Appendix D, blend damaged area removing least amount of material.
- (b) Using abrasive cloth, Item 10, Appendix D, polish blended area to remove abrasive marks and match surface smoothness.
- (c) Clean reworked area with dry-cleaning solvent, Item 124, Appendix D.
- (d) Clean surface with corrosion compound, Item 108, Appendix D. Apply two coats of epoxy primer from epoxy primer coating kit, Item 135, Appendix D, to surfaces where plating was disturbed.

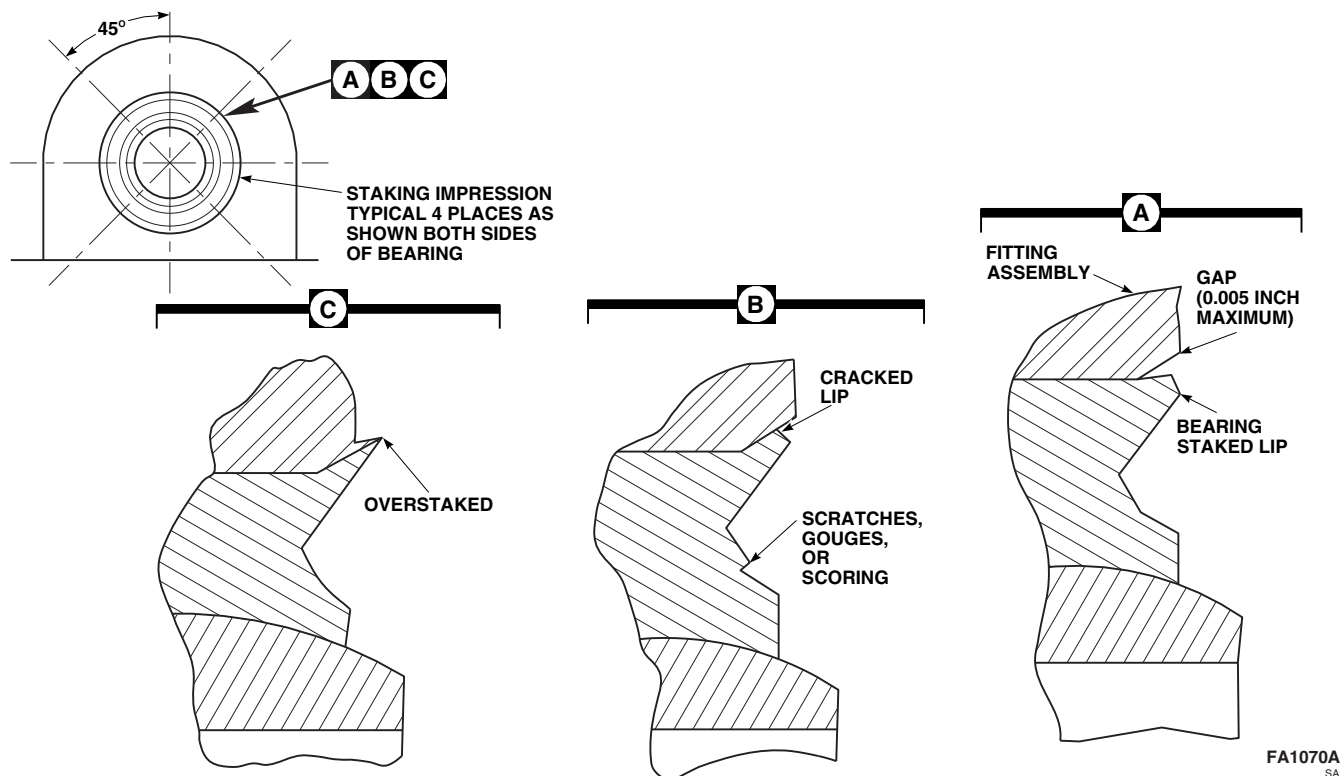


FIGURE 4-5-1-2. BEARING STAKING ACCEPTANCE

4-5-1.1.4 Install Bearing/Bushing.

- a. Install bearing as follows (Figure 4-5-1-1):

WARNING

- Bushings installed in aft engine mount struts are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.
- No minor repair or corrosion removal allowed on beryllium copper components.

- (1) Using arbor press, press bearing into strut.

- (2) Apply light coat of petrolatum, Item 234, Appendix D, to bearing groove to be staked and to roller staking tool.

- (3) Stake bearing in strut using roller staking tool. Staking flare is complete when 0.005-inch feeler gage cannot be inserted between flare lip and strut.

- (4) Repeat step a. (2) and step a. (3) for other side of bearing.

- b. Inspect bearing installation for the following. If installation does not meet the following criteria (Figure 4-5-1-2, Detail A, Detail B, and Detail C), replace bearing:

- (1) **ANY GAP BETWEEN BEARING STAKED LIP AND FITTING ASSEMBLY SHALL NOT EXCEED 0.005-INCH.**
- (2) **NO CRACKS ALLOWED IN STAKED LIP.**

- (3) **SCRATCHES, GOUGES, OR SCORING CAUSED BY STAKING TOOL CONTACTING UNSTAKED PORTION OF BEARING STAKE GROOVE NOT ALLOWED.**
 - (4) **STAKED LIP SHALL NOT BE OVER-STAKED, FORMING A FEATHERED OR KNIFE EDGE BEYOND FACE OF FITTING ASSEMBLY.**
 - (5) **VISUAL DEFECTS SUCH AS IMPRESSIONS, SCRATCHES, OR SCORING AND ENTRAPMENT OF CHIPS CAUSED BY STAKING OPERATION, IN THE STAKED RETENTION GROOVE OF BEARING, ARE ACCEPTABLE.**
- c. Install bushings as follows:
- (1) Coat bushings and strut bushing holes with primer, Item 258, Appendix D.
- NOTE**
- Flange of shoulder bushing to be located on inside of strut.
- (2) Press shoulder bushing into strut using arbor press and bushing installation and removal kit (Figure 4-5-1-1, Detail A or Detail B). Remove extra primer.
 - (3) Place sizing block on shoulder bushing and press plain bushing into strut using arbor press and bushing installation and removal kit. Remove extra primer.
 - (4) Measure distance between bushings as follows:
 - (a) On struts, 70305-02105-043 and 70305-02106-043, distance shall be 0.625 - 0.626-inch.
 - (b) On strut, 70305-02119-042, distance between upper bushings shall be 0.345 - 0.346-inch. Distance between lower bushings shall be 0.625 - 0.626-inch.
 - (5) Using reamer set, ream bushings to 0.3122 - 0.3132-inch diameter.
 - (6) Apply adhesive, Item 53, Appendix D, around bushings.

Table 4-5-1-1. Maximum Allowable Bore Dimensions For Engine Strut, Support, Links, and Bushings

Basic Assy Part No./Bore	Applicable Bushing Dash Number	
	-101	-102
70305-02105/bushing	0.4376-in.	0.4376-in.
70305-02106/bushing	0.4376-in.	0.4376-in.
70305-02119/bushing	0.4376-in.	0.4376-in.
Basic Bushing Part No./Diameter	Applicable Bushing Dash Number	
	-101	-102
70305-22112/Inside	(See Notes)	(See Notes)
70305-22112/Outside	0.4396-in.	0.4396-in.
70305-02112/Inside	(See Notes)	(See Notes)
70305-02112/Outside	0.4396-in.	0.4396-in.
Basic Assy Part No./Bore	Maximum Diameter	
	(See Notes)	
70305-02105/bushing not installed	(See Notes)	
70305-02105/bearing	0.9063-in.	
70305-02106/bearing	0.9063-in.	
70305-02110/bearing	0.6878-in.	
70305-02111/bearing	0.6878-in.	
70305-02115/bushing not installed	0.6370-in.	
Link Assy Part No.	Large Bore	Small Bore
70305-02107-101	0.5040-in.	0.3757-in.
70305-02107-102	0.5035-in.	0.3772-in.

NOTES:

**Table 4-5-1-1. Maximum Allowable Bore Dimensions For Engine Strut, Support, Links, and Bushings
(Cont)**

0.3158-in. when using bolts, NAS464P5-17 and NAS464P5-12.

0.3165-in. when using bolts, NAS6705DU14 and NAS6705DU18.

4-5-2 AFT ENGINE MOUNT FITTINGS (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
4-5-2.1 Repair Aft Engine Mount Fittings (BENCH)	4-5-2-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Engine Mount Staking Tool, STD1850.451-005
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Wheel, Item 22, Appendix D
- Corrosion Compound, Item 108, Appendix D

- Epoxy Primer Coating Kit, Item 135, Appendix D
- Paint Remover, Item 230, Appendix D
- Scouring Pad, Item 270, Appendix D

References

- Appendix D
- ASTM E1417

4-5-2.1 Repair Aft Engine Mount Fittings (BENCH).

NOTE

Repair of inner and outer aft engine mount fittings is the same.

4-5-2.1.1 Remove Bearing.

Remove bearing from aft engine mount fitting (Figure 4-5-2-1).

4-5-2.1.2 Inspect.

- Inspect bearing for corrosion and damage. **PITS SHALL NOT BE VISIBLE IN 0.250-INCH DIAMETER OR EXCEED 0.010-INCH DEEP.** If pits are visible in 0.250-inch diameter or limit is exceeded, replace bearing.

- Inspect bearing for axial play. **AXIAL PLAY SHALL NOT EXCEED 0.020-INCH.** If limit is exceeded, replace bearing.
- Inspect bearing for radial play. **RADIAL PLAY SHALL NOT EXCEED 0.005-INCH.** If limit is exceeded, replace bearing.
- Inspect bearing bore for scratches. **SCRATCHES SHALL NOT EXCEED 0.500-INCH LONG AND 0.005-INCH DEEP.** If limits are exceeded, replace bearing.
- Remove minor corrosion and scratches using scouring pad, Item 270, Appendix D.
- Remove paint from fitting using paint remover, Item 230, Appendix D.
- Using fluorescent penetrant inspection kit, fluorescent penetrant inspect fitting for cracks

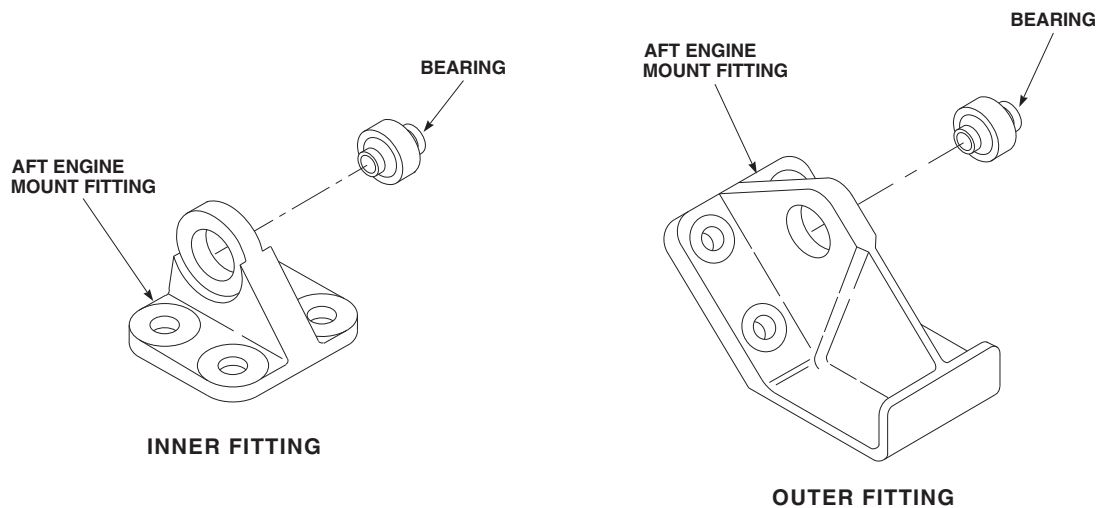
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FIGURE 4-5-2-1. REPAIR AFT ENGINE MOUNT FITTINGS (BENCH)

(ASTM E1417). **NONE ALLOWED.** If cracked, replace fitting.

- h. Inspect fitting for nicks, gouges, scratches, and pitting. **DAMAGE SHALL NOT EXCEED 0.250-INCH LONG AND 0.032-INCH DEEP.** If limit is exceeded, replace fitting.
- i. Blend out nicks, gouges, scratches, and pitting on inboard fitting using 1/2-inch-radius abrasive wheel, Item 22, Appendix D. **BLENDED AREA SHALL NOT EXCEED 0.250-INCH LONG AND 0.032-INCH DEEP.** If limits are exceeded, replace fitting.
- j. Blend out nicks, gouges, scratches, and pitting on outboard fitting using 1/2-inch-radius abrasive wheel, Item 22, Appendix D. **BLENDED AREA SHALL NOT EXCEED**

0.500-INCH LONG AND 0.032-INCH DEEP. If limits are exceeded, replace fitting.

- k. Blend out pits on inboard and outboard fitting using scouring pad, Item 270, Appendix D. Apply corrosion compound, Item 108, Appendix D.
- l. Clean surface and apply epoxy primer from epoxy primer coating kit, Item 135, Appendix D, to surfaces where plating was disturbed.

4-5-2.1.3 Install Bearing.

- a. Install bearing in aft engine mount fitting (Figure 4-5-2-1).
- b. Stake bearing in housing using engine mount staking tool.

4-5-3 AFT ENGINE MOUNT LINKS (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
4-5-3.1 Repair Aft Engine Mount Links (BENCH)	4-5-3-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Wheel, Item 22, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 4-5-1

4-5-3.1 Repair Aft Engine Mount Links (BENCH).

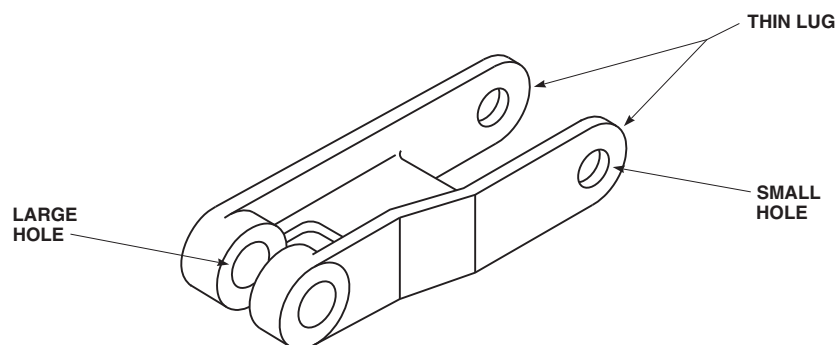
NOTE

Repair of all aft engine mount links is the same.

- Measure for wear in holes at both ends of link (PARA 4-5-1).
- Inspect link in area of thin lugs for nicks, gouges, scratches, and pitting. **DAMAGE SHALL NOT EXCEED 0.250-INCH LONG AND 0.020-INCH DEEP.** If limit is exceeded, replace link (Figure 4-5-3-1).
- Inspect remaining area of link for nicks, gouges, scratches, and pitting. **DAMAGE SHALL NOT EXCEED 0.500-INCH LONG**

AND 0.032-INCH DEEP. If limits are exceeded, replace link.

- Using fluorescent penetrant inspection kit, fluorescent penetrant inspect link for cracks (ASTM E1417). **NONE ALLOWED.** If cracked, replace link.
- Blend out nicks, gouges, scratches, and pitting in area of thin lugs using 1/2-inch-radius abrasive wheel, Item 22, Appendix D. **BLENDED AREA SHALL NOT EXCEED 0.250-INCH LONG AND 0.020-INCH DEEP.** If limits are exceeded, replace link.
- Blend out nicks, gouges, scratches, and pitting in remaining area of link 1/2-inch-radius abrasive wheel, Item 22, Appendix D. **BLENDED AREA SHALL NOT EXCEED 0.500-INCH LONG AND 0.032-INCH DEEP.** If limits are exceeded, replace link.

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SA**FIGURE 4-5-3-1. REPAIR AFT ENGINE MOUNT LINKS (BENCH)**

4-5-4 AFT ENGINE MOUNT SUPPORTS (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
4-5-4.1 Repair Aft Engine Mount Supports (BENCH)	4-5-4-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Fluorescent Penetrant Inspection Kit

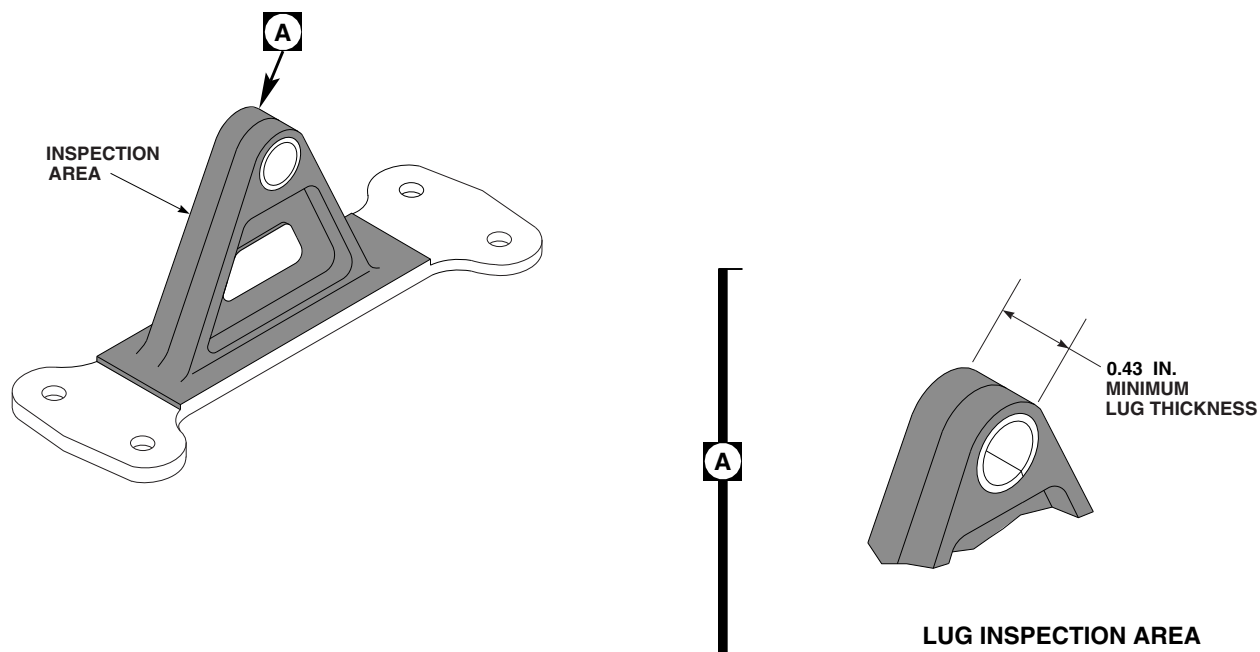
References

- ASTM E1417
- PARA 4-5-1

4-5-4.1 Repair Aft Engine Mount Supports (BENCH).

- a. Inspect lug inspection area of aft engine mount support for nicks, gouges, scratches, and pitting (Figure 4-5-4-1, Detail A). **DAMAGE SHALL NOT EXCEED 0.032-INCH DEEP AND MINIMUM LUG THICKNESS OF 0.43-INCH MUST BE MAINTAINED.** If limits are exceeded, replace aft engine mount support.
- b. Inspect remaining inspection area of aft engine mount support for nicks, gouges, scratches, and pitting (Figure 4-5-4-1). **DAMAGE SHALL NOT EXCEED 0.032-INCH DEEP OVER ENTIRE SHADED REGION ON BOTH FRONT AND REAR SURFACES OF FITTING.** If limits are exceeded, replace aft engine mount support.
- c. Inspect aft engine mount support mount bolt bore wear per applicable dimension (PARA 4-5-1).

- d. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect support for cracks (ASTM E1417). **NONE ALLOWED.** If cracked, replace aft engine mount support.
- e. Blend out nicks, gouges, scratches, and pitting on lug inspection area. **AREA SHALL NOT EXCEED 0.032-INCH DEEP AND MINIMUM LUG THICKNESS OF 0.43-INCH MUST BE MAINTAINED AFTER BLENDING.** If limits are exceeded, replace aft engine mount support.
- f. Blend out nicks, gouges, scratches, and pitting on remaining inspection area. **AREA SHALL NOT EXCEED 0.032-INCH DEEP OVER ENTIRE SHADED REGION ON BOTH FRONT AND REAR SURFACES OF FITTING AFTER BLENDING.** If limit is exceeded, replace aft engine mount support.

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SA**FIGURE 4-5-4-1. REPAIR AFT ENGINE MOUNT SUPPORTS (BENCH)**

4-5-5 ENGINE MOUNT CROTCH ASSEMBLY (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
4-5-5.1 Repair Engine Mount Crotch Assembly (BENCH)	4-5-5-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Backing Strip, Locally-Made, Figure H-130, Appendix H
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry Ice, Item 125, Appendix D

- Engine Inlet Firewall Seal, Locally-Made, Figure H-65, Appendix H
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Sealing Compound, Item 273, Appendix D

References

- Appendix D
- Appendix H
- ASTM E1417
- PARA 2-6-5

4-5-5.1 Repair Engine Mount Crotch Assembly (BENCH).

4-5-5.1.1 Replace Pins.

- Remove loose or damaged pins (Figure 4-5-5-1).
- Inspect pin bores for wear. **PIN BORES MUST NOT EXCEED 0.1275-INCH.** If limit is exceeded for either pin bore, replace crotch assembly.
- Place replacement pin(s) in dry ice, Item 125, Appendix D, for minimum of 30 minutes.
- Coat pin bores with sealing compound, Item 273, Appendix D, and insert pins in bores.

- Allow sealing compound to dry for 1 hour.

4-5-5.1.2 Replace Crotch Components.

NOTE

Individual parts of seal assembly may be replaced after seal is removed from crotch assembly and disassembled.

- Remove rivets securing seal assembly and segment ring to crotch assembly (Figure 4-5-5-1).
- Remove rivets securing heat shield, backing strip, seal, and firewall plate. Replace parts as required (Figure 4-5-5-1, Detail A).
- Inspect crotch assembly as follows:

- (1) Inspect crotch, in areas other than the crotch flange, for missing or damaged anodized finish. **NONE ALLOWED.** If anodized finish is missing or damaged, replace crotch.
- (2) Inspect crotch flange for scratches, gouges, wear grooves, and rub marks. If damage is found, repair as follows (Figure 4-5-5-1, Detail B):

NOTE

- When repairing damage found on outside corners of crotch flange, do not exceed 0.125-inch maximum.
- Rework not allowed on inside corners of crotch flange.

- (a) Blend out scratches, gouges, wear grooves, and rub marks using abrasive cloth, Item 10, Appendix D. **FLANGE HEIGHT TO BE NO LESS THAN 0.135-INCH AFTER LOCALIZED REPAIR. MAXIMUM ALLOWABLE DEPTH OF REPAIR ALONG FLANGE WIDTH TO BE 0.125-INCH. REWORK NOT ALLOWED IN**

AREA INDICATED IN DETAIL B.
If limits are exceeded, replace crotch.

- (b) Using fluorescent penetrant inspection kit, perform fluorescent penetrant inspection of repaired areas (ASTM E1417).
 - (c) Apply corrosion resistant coating, Item 118, Appendix D, to repaired areas.
 - (d) Apply epoxy primer from epoxy primer coating kit, Item 135, Appendix D, to repaired areas (PARA 2-6-5).
- d. If replacement seal is required, locally make engine inlet firewall seal (Figure H-65, Appendix H).
 - e. If replacement backing strip is required, locally make backing strip (Figure H-130, Appendix H).
 - f. Assemble heat shield, backing strip, seal, and firewall plate and secure with rivets (Figure 4-5-5-1, Detail A).
 - g. Secure seal assembly and segment ring to crotch assembly with rivets (Figure 4-5-5-1). **█**

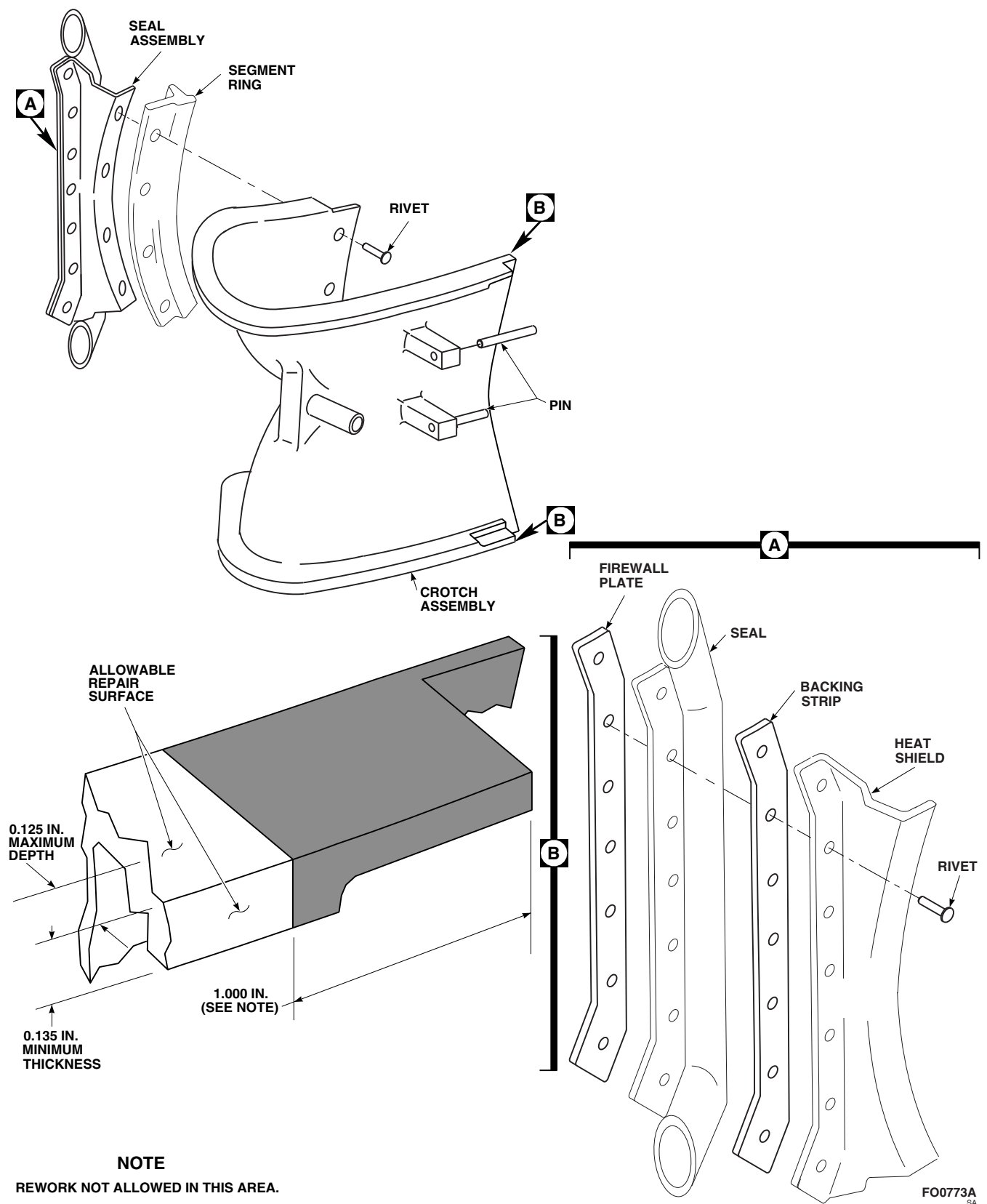


FIGURE 4-5-5-1. REPAIR ENGINE MOUNT CROTCH ASSEMBLY (BENCH)

4-5-6 ENGINE EXHAUST EJECTOR AND FAIRINGS (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
4-5-6.1 Repair Engine Exhaust Ejector and Fairings (BENCH)	4-5-6-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Fluorescent Penetrant Inspection Kit
- Welding Shop Set
- White Light Source, 200 foot-candles minimum

Materials/Parts

- Filler Wire, AMS 4951
- Filler Wire, 347, MIL-R-5031, Class 5A

References

- AMS 2689
- AMS 4951
- ASTM E1417
- MIL-R-5031
- MIL-W-8611
- PARA 4-4-8

4-5-6.1 Repair Engine Exhaust Ejector and Fairings (BENCH).

4-5-6.1.1 Repair.

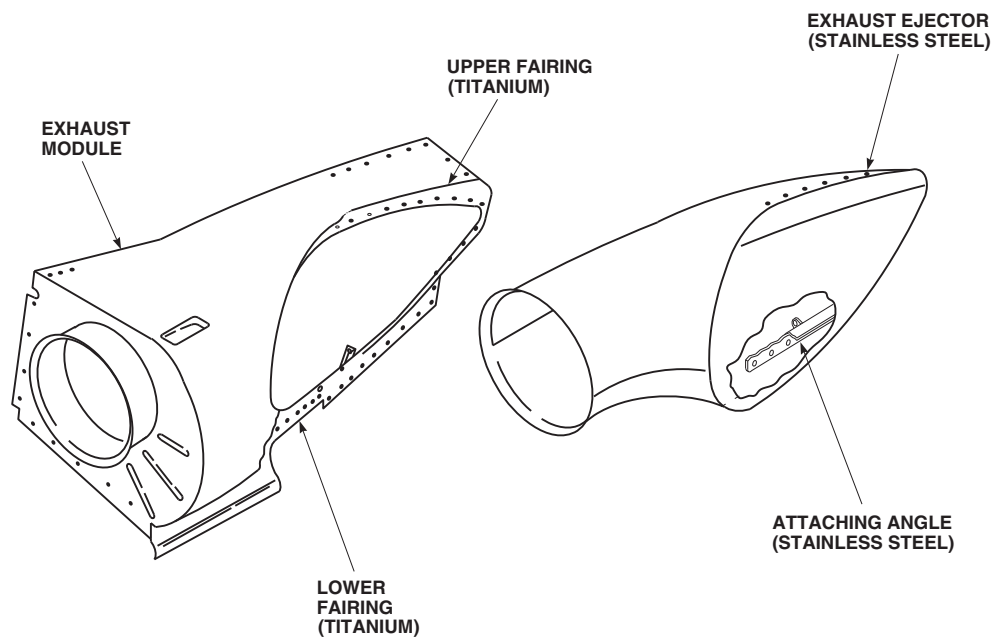
- Remove exhaust ejector from exhaust module (PARA 4-4-8).
- Weld stainless steel exhaust ejector and attaching angle by gas tungsten (GTA) method per MIL-W-8611, using 347 filler wire, MIL-R-5031, Class 5A (Figure 4-5-6-1).
- Weld titanium upper and lower fairings per AMS 2689, using filler wire, AMS 4951.

4-5-6.1.2 Inspect.

- Using fluorescent penetrant inspection kit, fluorescent penetrant inspect repaired area (ASTM E1417), with the following exceptions:

- Inspector shall be dark-conditioned for a minimum of 5 minutes each time dark area is entered from white light area.
- Inspector shall be dark-conditioned for a minimum of 5 minutes after looking directly into beam of black light or after using white light in inspection booth.
- White light source shall provide a minimum of 200 foot-candles of illumination, at part surface, for inspection of fluorescent penetrant indications.

- Permanently mark repaired component "PS-004". Marking shall be in same style and location as part number.

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SA**FIGURE 4-5-6-1. REPAIR ENGINE EXHAUST EJECTOR AND FAIRINGS (BENCH)**

4-5-7 ENGINE EXHAUST MODULE DUCT ASSEMBLY (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
4-5-7.1	Replace Engine Exhaust Module Duct Assembly (BENCH)	4-5-7-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Drill Bit Set
- Drill, Rotary

- Exhaust Fairing Bulkhead Seal, Locally-Made, Figure H-115, Appendix H

- Rivet Gun

References

- Appendix H

4-5-7.1 Replace Engine Exhaust Module Duct Assembly (BENCH).

4-5-7.1.1. Remove.

- a. Remove bulkhead from exhaust module by drilling out rivets on outside edge of bulkhead (Figure 4-5-7-1).
- b. Drill out rivets holding duct assembly to bulkhead and remove duct assembly (Figure 4-5-7-1, Detail A).

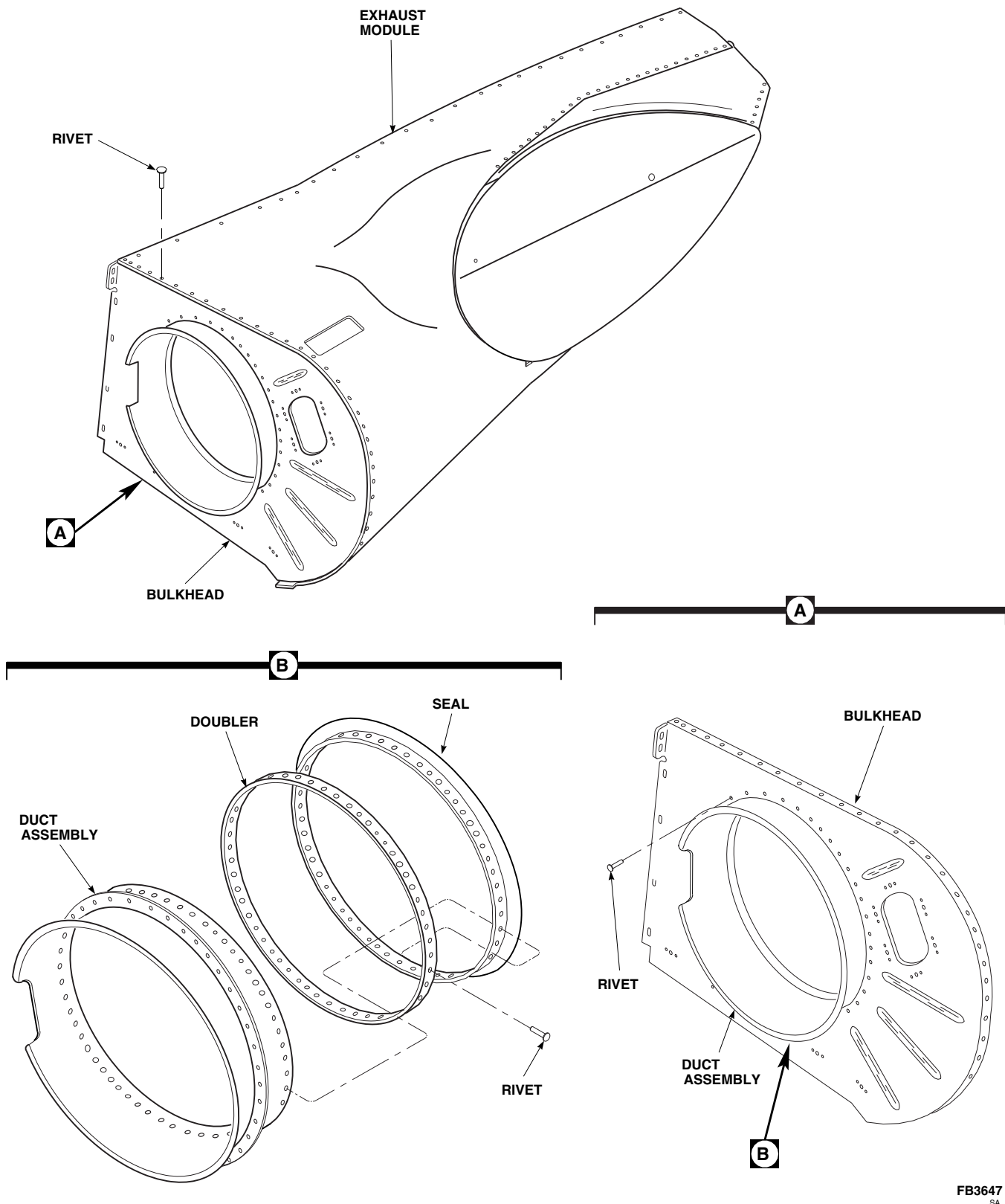
- b. Locally-make exhaust fairing bulkhead seal (Figure H-115, Appendix H).
- c. Assemble seal and doubler on duct assembly and secure with rivets.

4-5-7.1.3. Install.

4-5-7.1.2. Replace Seal.

- a. Drill out rivets holding doubler and seal to duct assembly (Figure 4-5-7-1, Detail B).

- a. Install duct assembly on bulkhead and secure with rivets (Figure 4-5-7-1, Detail A).
- b. Install bulkhead on exhaust module and secure outside edge with rivets (Figure 4-5-7-1).



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FIGURE 4-5-7-1. REPLACE ENGINE EXHAUST MODULE DUCT ASSEMBLY (BENCH)

4-5-7-2

4-5-8 HIRSS INLET SEAL (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
4-5-8.1 Replace HIRSS Inlet Seal (BENCH)	4-5-8-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Bit Set

- Drill, Rotary
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 4-4-9
- PARA 4-4-13

4-5-8.1 Replace HIRSS Inlet Seal (BENCH).

4-5-8.1.1. Remove.

- Remove HIRSS module (PARA 4-4-9).
- Remove nacelle fairing (PARA 4-4-13).
- Remove bulkhead from bulkhead support flange by removing screws, washers, and nuts (Figure 4-5-8-1, Detail A).
- Using rotary drill, drill out rivets securing inlet seal to stage 1 inner duct (Figure 4-5-8-1, Detail B).
- Remove inlet seal from stage 1 inner duct of suppressor core.

4-5-8.1.2. Install.

- Install new inlet seal into stage 1 inner duct of suppressor core (Figure 4-5-8-1, Detail B and Detail C).
- Position bracket of inlet seal 0.883 to 0.963-inch from stage 1 outer duct flange (Figure 4-5-8-1, Detail C).
- Install inlet seal to stage 1 inner duct with rivets (Figure 4-5-8-1, Detail B).
- Install bulkhead on bulkhead support flange of suppressor core and attach with screws, washers and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS** (Figure 4-5-8-1, Detail A).
- Install nacelle fairing (PARA 4-4-13).
- Install HIRSS module (PARA 4-4-9).

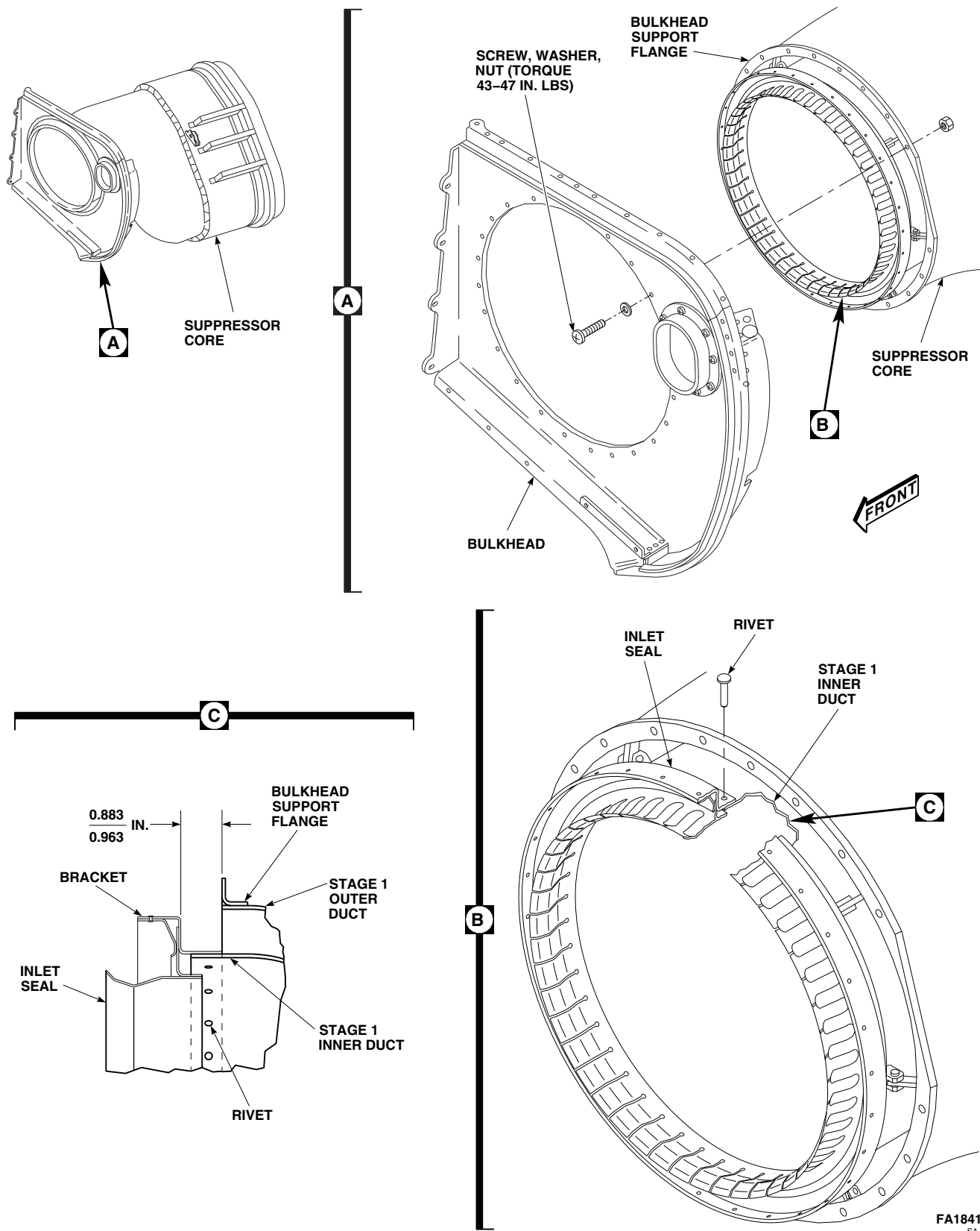


FIGURE 4-5-8-1. REPLACE HIRSS INLET SEAL (BENCH)

4-5-8-2

4-5-9 ROTARY INPUT (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
4-5-9.1 Repair Rotary Input (BENCH)	4-5-9-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Fluorescent Penetrant Inspection Kit
- Inside Micrometer
- Mallet
- Micrometer
- Oven
- Phenolic Drift
- Rotary Input Bearing Removal Tube, Locally-Made, Figure H-211, Appendix H
- Spring Tester
- Torque Wrench, 30 - 150 in. lbs

- Vise, Padded

Materials/Parts

- Abrasive Cloth, Item 7, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Grease, Item 150, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 277, Appendix D
- Trichloroethane 1,1,1, Item 341, Appendix D

References

- Appendix D
- Appendix H
- ASTM E1417

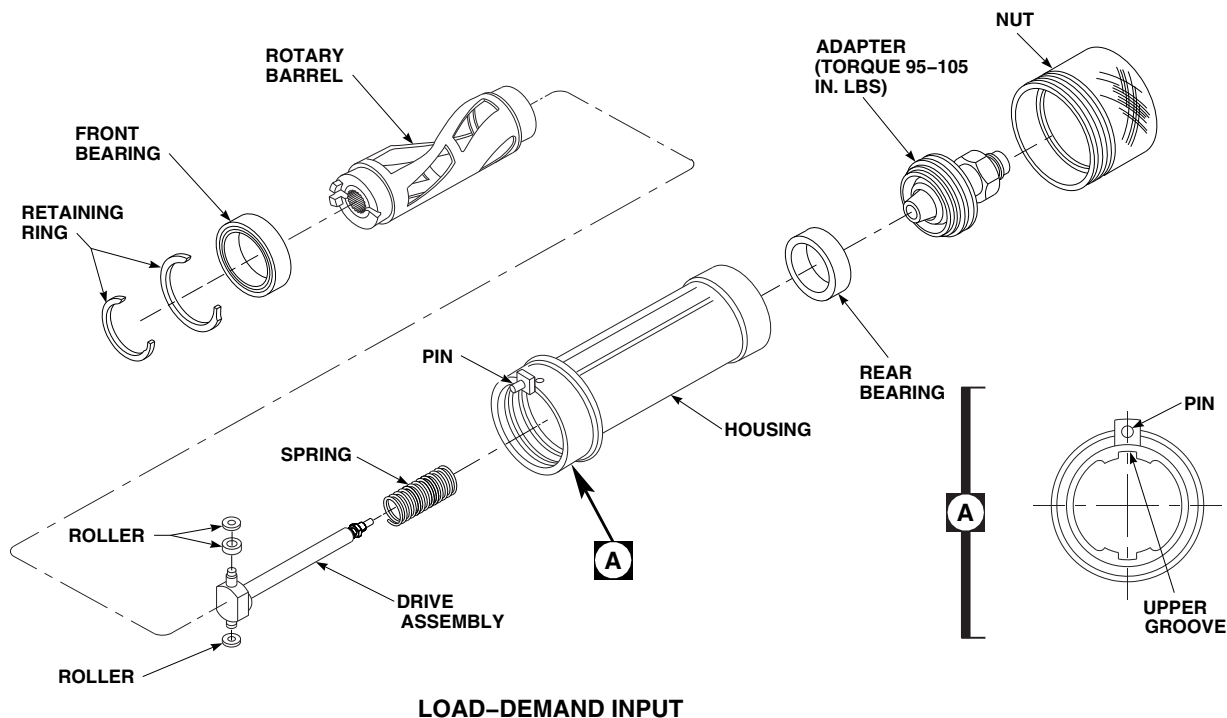
4-5-9.1 Repair Rotary Input (BENCH).

NOTE

- The following procedure applies to both power-available and load-demand rotary

inputs. When maintaining these rotary inputs, keep parts separated.

- Repair of both rotary inputs is the same, except as noted.

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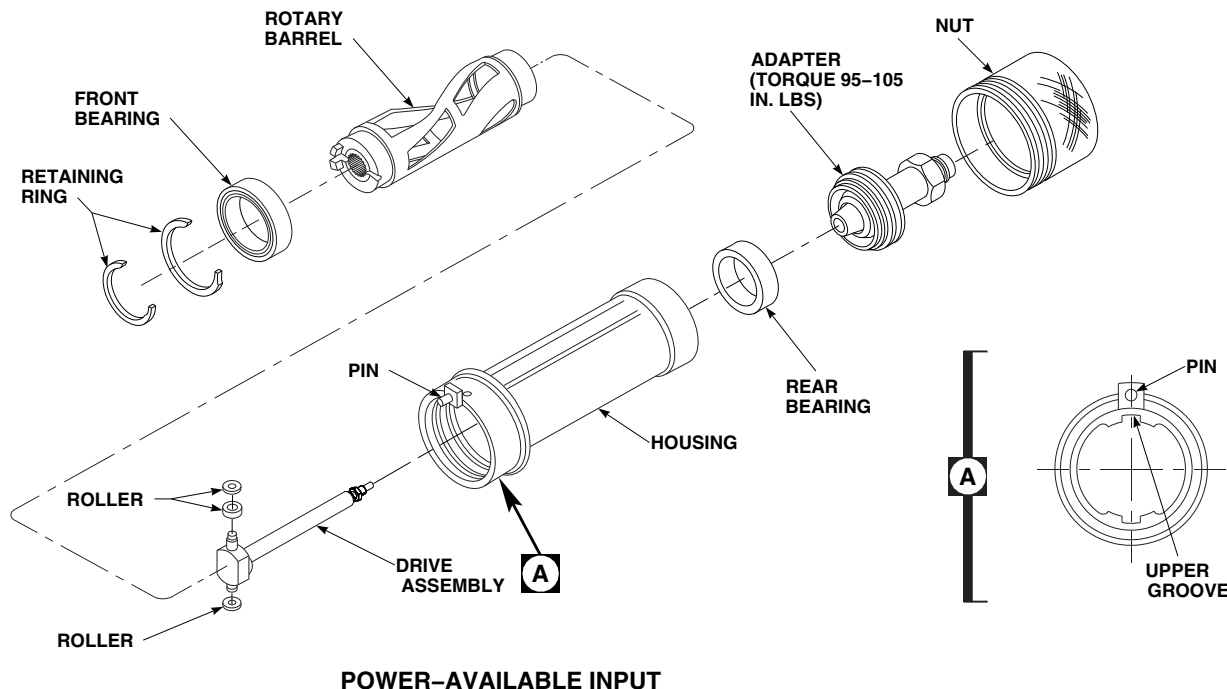
- a. Place rotary input in padded vise.

NOTE

Heating of rotary input may be necessary in order to weaken sealing compound bond between adapter and housing.

- b. Remove adapter from housing (Figure 4-5-9-1 and Figure 4-5-9-2). If adapter cannot be removed from housing, perform the following:
 - (1) Place rotary input in 250°F (121°C) preheated oven for 15 - 20 minutes.
 - (2) Repeat step a. and step b.
- c. On load-demand input only, remove spring from adapter (Figure 4-5-9-1).
- d. Remove large retaining ring from housing (Figure 4-5-9-1 and Figure 4-5-9-2).
- e. Remove small retaining ring from end of rotary barrel.
- f. Remove front bearing from housing. If bearing is seated tightly in housing, perform the following:
 - (1) Locally make rotary input bearing removal tube (Figure H-211, Appendix H).
 - (2) Place tube in rear of housing over drive assembly.
 - (3) Lightly tap tube to unseat bearing from housing.
- g. Remove rotary barrel and drive assembly from housing.
- h. Remove drive assembly from rotary barrel.
- i. Remove rollers from drive assembly.
- j. Remove rear bearing from housing. If necessary, lightly tap bearing with phenolic drift to unseat it from housing.

4-5-9-2 Change 11



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FIGURE 4-5-9-2. POWER-AVAILABLE INPUT REPAIR (BENCH)

4-5-9.1.2 Inspect.

CAUTION

Damage to bearings will result if contact with solvents occur. Do not allow solvents to contact bearings.

- a. Clean housing bearing bores using cleaning cloth, Item 102, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D (Figure 4-5-9-1 and Figure 4-5-9-2).
- b. Clean remaining metal parts with dry-cleaning solvent, Item 124, Appendix D.
- c. Inspect bearings for damage and wear.
- d. Check load-demand input spring with spring tester. **COMPRESS LOAD-DEMAND INPUT SPRING TO 1 1/4-INCHES. SPRING PRESSURE MUST BE AT LEAST 24 POUNDS, BUT NOT EXCEED 26 POUNDS.** If not within limits, replace spring.
- e. Check retaining rings for damage. **NONE ALLOWED.** If damaged, replace retaining rings.
- f. Inspect rotary barrel as follows:
 - (1) Visually inspect rotary barrel for nicks, scratches, and corrosion. **NONE ALLOWED.** If damage is found, replace rotary barrel.
 - (2) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type 1, Method C, rotary barrel for cracks and fractures (ASTM E1417). **NONE ALLOWED.** If cracked or fractured, replace rotary barrel.
 - (3) Inspect slots of rotary barrel for wear. **WIDTH OF NARROW SLOT SHALL NOT EXCEED 0.2530-INCH. OUTSIDE DIAMETER OF SLOTTED END SHALL NOT BE LESS THAN 1.0615-INCHES. OUTSIDE DIAMETER OF NONSLOTTED END SHALL NOT BE LESS THAN 0.9995-INCH. INSIDE DIAMETER OF NONSLOTTED END**

SHALL NOT EXCEED 0.870-INCH. If not within limits, replace rotary barrel.

Method C, adapter for cracks and fractures (ASTM E1417). **NONE ALLOWED.** If cracked or fractured, replace adapter.

g. Inspect drive assembly as follows:

- (1) Inspect drive assembly for nicks, scratches, and corrosion. **NONE ALLOWED.** If damaged, replace drive assembly.
- (2) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type 1, Method C, drive assembly for cracks and fractures (ASTM E1417). **NONE ALLOWED.** If cracked or fractured, replace drive assembly.

h. Inspect rollers as follows:

- (1) Inspect rollers for nicks, scratches, and corrosion. **NONE ALLOWED.** If damaged, replace roller.
- (2) Using micrometer and inside micrometer, measure rollers for wear. **OUTSIDE DIAMETER OF ROLLERS, 70301-02117-101 AND -102, SHALL NOT BE LESS THAN 0.249-INCH. INSIDE DIAMETER OF ROLLER, 70301-02117-101, SHALL NOT EXCEED 0.153-INCH. INSIDE DIAMETER OF ROLLER, 70301-02117-102, SHALL NOT EXCEED 1/8-INCH.** If not within limits, replace roller.

i. Inspect adapter as follows:

- (1) Inspect all external surfaces of adapter except threads for nicks, scratches, and corrosion. **DEPTH OF EXTERNAL DAMAGE SHALL NOT EXCEED 0.010-INCH DEEP.** If limit is exceeded, replace adapter. If within limits, clean external surface corrosion using abrasive cloth, Item 7, Appendix D.
- (2) Inspect internal surfaces and all threads of adapter for damage. **NONE ALLOWED.** If damage is found, replace adapter.
- (3) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type 1,

- j. Inspect pin at rear of housing for looseness. **NONE ALLOWED.** If pin is loose or missing, replace housing.

k. Inspect housing as follows:

- (1) Inspect all external surfaces of housing except threads for nicks, scratches, and corrosion. **DEPTH OF EXTERNAL DAMAGE SHALL NOT EXCEED 0.010-INCH DEEP.** If limit is exceeded, replace housing. If within limits, clean external surface corrosion using abrasive cloth, Item 7, Appendix D.
- (2) Inspect internal surfaces and all threads of housing for damage. **NONE ALLOWED.** If damage is found, replace housing.
- (3) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type 1, Method C, housing for cracks and fractures (ASTM E1417). **NONE ALLOWED.** If cracked or fractured, replace housing.
- (4) Using inside micrometer, inspect front bearing bore of housing for wear. **INSIDE DIAMETER OF FRONT BEARING BORE SHALL NOT EXCEED 1.501-INCHES.** If limit is exceeded, replace housing.
- (5) Using inside micrometer, inspect rear bearing bore of housing for wear. **INSIDE DIAMETER OF REAR BEARING BORE SHALL NOT EXCEED 1.2505-INCHES.** If limit is exceeded, replace housing.

l. Inspect nut as follows:

- (1) Inspect all external surfaces of nut except threads for nicks, scratches, and corrosion. **DEPTH OF EXTERNAL DAMAGE SHALL NOT EXCEED 0.010-**

INCH DEEP. If limit is exceeded, replace nut. If within limits, clean up external surface corrosion using abrasive cloth, Item 7, Appendix D.

- (2) Inspect internal surfaces and all threads of nut for damage. **NONE ALLOWED.** If damage is found, replace nut.

NOTE

False indications of cracking are possible due to knurling on nut. If cracks are detected in knurled area, reinspect to verify crack.

- (3) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type 1, Method C, nut for cracks and fractures (ASTM E1417). **NONE ALLOWED.** If cracked or fractured, replace nut.

4-5-9.1.3 Assemble.

- a. Apply thin coat of primer, Item 258, Appendix D, to inside diameter of front bearing and outside diameter of rear bearing.
- b. Install rear bearing in housing while primer is wet (Figure 4-5-9-1 and Figure 4-5-9-2).
- c. Install front bearing onto rotary barrel and secure with small retaining ring.
- d. Lubricate all moving parts with grease, Item 150, Appendix D.
- e. Install rollers on drive assembly.

NOTE

Side of drive assembly with two rollers fits into narrow slot in rotary barrel.

- f. Install drive assembly in rotary barrel.
- g. Apply thin coat of primer, Item 258, Appendix D, to outside diameter of front bearing.
- h. Install rotary barrel and drive assembly into housing. Make sure side of drive assembly with two rollers is in upper groove (Figure 4-5-9-1, Detail A and Figure 4-5-9-2, Detail A).
- i. If necessary, reseal rear bearing by lightly tapping with mallet.
- j. Clean retaining ring bore in housing using cleaning cloth, Item 102, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.
- k. Install large retaining ring in groove in housing assembly (Figure 4-5-9-1 and Figure 4-5-9-2).
- l. On load-demand rotary input only, push drive assembly to rear of housing assembly. Position spring onto drive assembly (Figure 4-5-9-1).
- m. Apply sealing compound, Item 277, Appendix D, to threads of adapter and housing (Figure 4-5-9-1 and Figure 4-5-9-2).
- n. Position adapter over end of drive assembly. Thread adapter into housing. **TORQUE ADAPTER TO 95 - 105 INCH-POUNDS.**
- o. Check assembled rotary input for smooth operation.

4-5-10 ENGINE CONTROL QUADRANT POWER LEVER (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
4-5-10.1 Replace Engine Control Quadrant Power Lever (BENCH)	4-5-10-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Arbor Press

Materials/Parts

- Washer, NAS1149F0663P

References

- PARA 4-5-14
- PARA 4-5-15

4-5-10.1 Replace Engine Control Quadrant Power Lever (BENCH).

4-5-10.1.1 Remove.

- Remove override and abort switches (PARA 4-5-15 and PARA 4-5-14).
- Remove nut and washer(s) from bolt (Figure 4-5-10-1).
- Slide engine control quadrant power lever from bolt.
- Remove bushing and laminated shim from power lever.

4-5-10.1.2 Disassemble.

- On quadrant, 70301-01100-014, remove nut, washers, bolt, spacers, torsion spring, and latch stop from engine power control lever (Figure 4-5-10-1).

NOTE

There are two small compressed springs and a latch in engine control quadrant power lever. Be careful when removing cover to avoid losing parts.

- Remove screws and cover from lever (Figure 4-5-10-2).
- Remove lever from handle.
- Remove heavy spring, light spring, and latch from lever.
- Remove nut and washer securing abort switch screw and spring to handle. Remove screw and spring.
- Remove pin from starter button and remove button and spring from plunger.
- Remove plunger from handle.

4-5-10.1.3 Inspect/Repair.

- a. Inspect bushing for wear (Figure 4-5-10-1). **WEAR OF BUSHING SHALL NOT EXCEED INSIDE DIAMETER OF 0.375-INCH.** If limit is exceeded, replace bushing.
- b. Inspect bearing for damage and wear (Figure 4-5-10-2). **NONE ALLOWED.** If required, remove bearing from lever, using arbor press.

4-5-10.1.4 Assemble.

- a. If bearing was removed, install new bearing in lever using arbor press (Figure 4-5-10-2).
- b. Position plunger in handle.
- c. Install starter button and spring on plunger, using pin.
- d. Position abort switch screw and spring in handle.
- e. Install nut and washer to secure abort switch screw.

NOTE

- Light spring refers to C0240-022-0810M.
- Heavy spring refers to C0240-038-1000M.
- f. Position latch in light spring and install latch and spring, as an assembly, in upper slot of lever.
- g. Install heavy spring in lower slot of lever.
- h. Position lever in handle.
- i. Install cover on lever, using screws.
- j. On quadrant, 70301-01100-014, install latch stop assembly as follows:
 - (1) Position torsion spring on engine power control lever (Figure 4-5-10-1).
 - (2) Install latch stop, then engage torsion spring.

NOTE

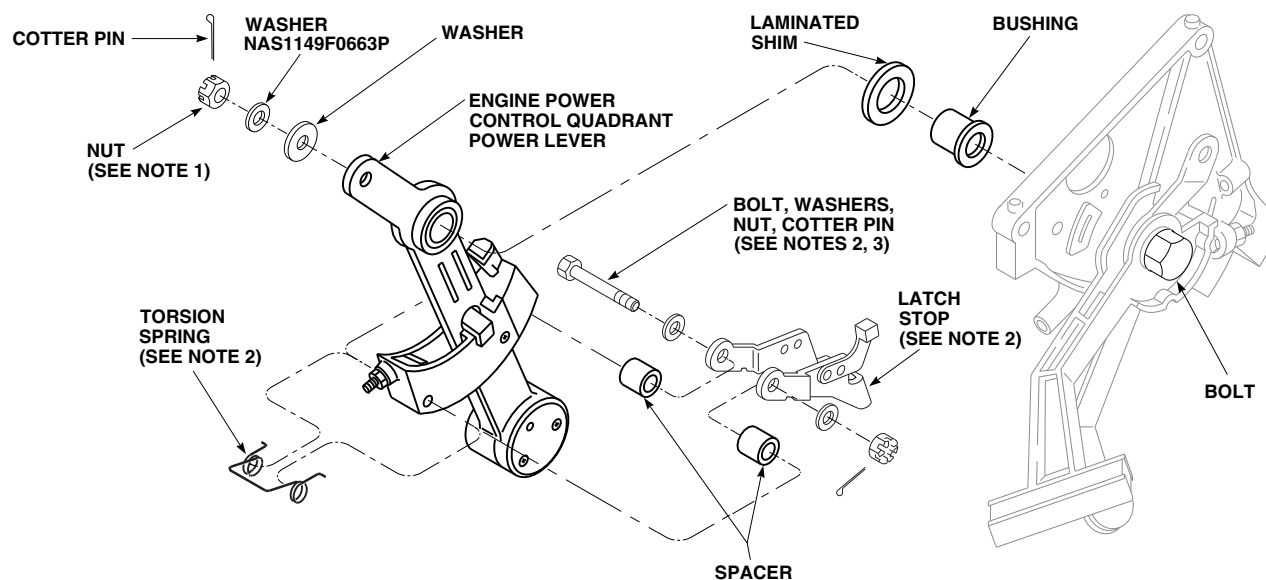
Make sure bolt securing latch stop is installed with bolthead facing outboard.

- (3) Install bolt, washers, spacers, and nut holding latch stop in place.
- (4) Install cotter pin in nut.

4-5-10.1.5 Install.**NOTE**

If a new laminated shim is being installed, peel laminations as required to obtain minimum clearance between parts, with control lever moving freely, without drag.

- a. Install laminated shim, with stainless steel side facing engine power control lever, on bushing (Figure 4-5-10-1).
- b. Install bushing and laminated shim in engine power control lever.
- c. Install engine power control lever on bolt with washer and nut. If necessary, shim with washers, NAS1149F0663P, as required to properly install cotter pin.
- d. **TIGHTEN NUT UNTIL HOLE IN BOLT LINES UP WITH SLOT IN NUT, THEN TIGHTEN ONE MORE CASTELLATION.**
- e. Move engine power control lever, fire extinguisher T-handle, and fuel control lever, and check for binding or looseness.
- f. If handles or levers bind or are too loose, perform the following:
 - (1) Remove nut from bolt.
 - (2) Add or remove shims as required.
 - (3) Install nut. Repeat step d. through step f. until no binding or looseness exists.
- g. Install cotter pin in nut.

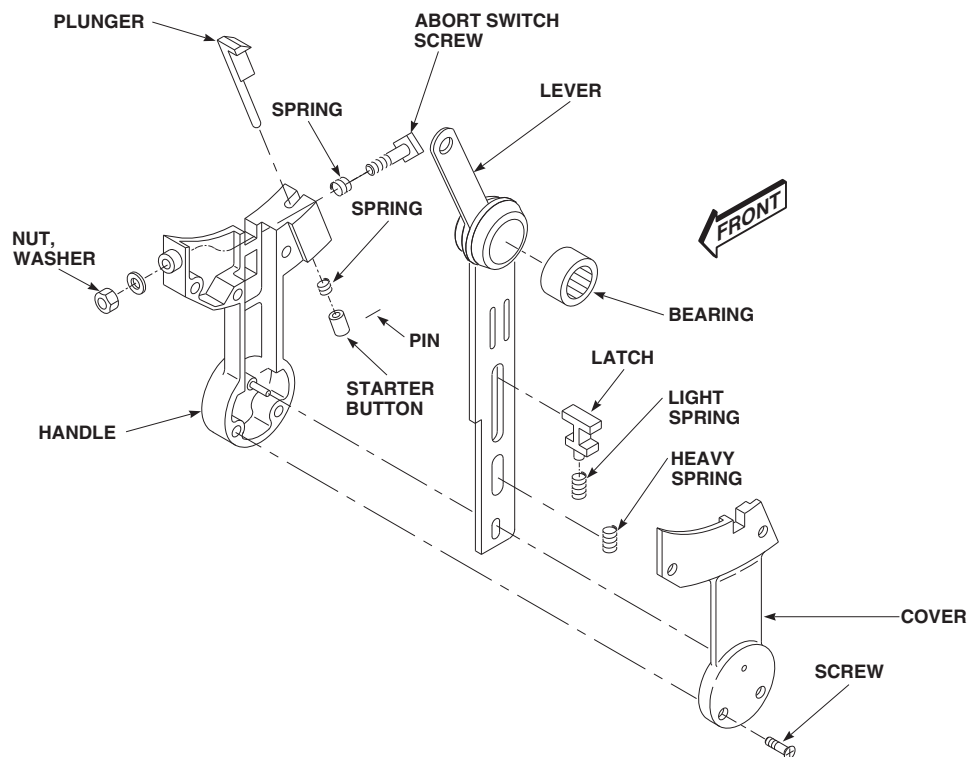


NOTES

1. TIGHTEN NUT UNTIL HOLE IN BOLT LINES UP WITH SLOT IN NUT. THEN TIGHTEN ONE MORE CASTELLATION.
2. ON QUADRANT, 70301-01100-014.
3. INSTALL WITH BOLTHEAD FACING OUTBOARD.

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FIGURE 4-5-10-1. REPLACE ENGINE CONTROL QUADRANT POWER LEVER (BENCH)



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FIGURE 4-5-10-2. REPAIR ENGINE CONTROL QUADRANT POWER LEVER (BENCH)

- h. Install override and abort switches (PARA 4-5-15 and PARA 4-5-14).

4-5-11 ENGINE CONTROL QUADRANT T-HANDLE AND FUEL LEVER (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
4-5-11.1 Replace Engine Control Quadrant T-Handle and Fuel Lever (BENCH)	4-5-11-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Protective Caps and Plugs
- Tags

References

- PARA 1-7-20
- PARA 4-5-10
- PARA 4-5-13
- PARA 4-5-16

4-5-11.1 Replace Engine Control Quadrant T-Handle and Fuel Lever (BENCH).

4-5-11.1.1. Remove.

- Remove engine starter and fire extinguisher arming switches (PARA 4-5-13 and PARA 4-5-16).
- Remove engine power control lever (PARA 4-5-10).
- Slide bolt, fire extinguisher T-handle, and fuel control lever, as an assembly, from quadrant support (Figure 4-5-11-1).
- Remove bolt and washer from fire extinguisher T-handle and fuel control lever.
- Remove bushing from fire extinguisher T-handle and fuel control lever.
- Remove laminated shim from bushing.

- Remove fuel control lever, washer, and laminated shim from fire extinguisher T-handle. Tag shim for assembly.
- Disconnect electrical connectors.
- Install protective caps and plugs on electrical connectors.
- Remove electrical connector from electrical wires.
- Carefully remove electrical tubing insulation from wire bundle.
- Remove fire extinguisher T-handle wire from wire bundle.

4-5-11.1.2. Inspect/Repair.

Check inside diameter of fire extinguisher T-handle bushing for maximum wear of 0.3765-inch. **IF WEAR IS BEYOND TOLERANCE, REPLACE BUSHING.**

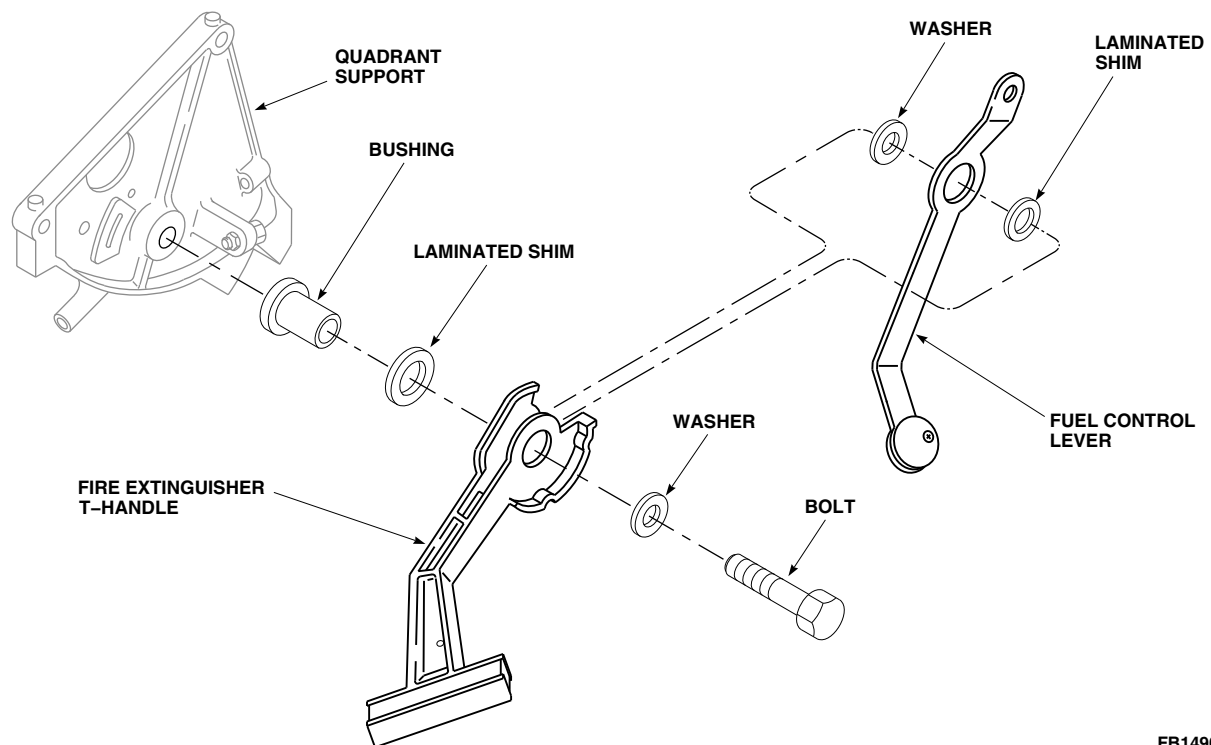
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FIGURE 4-5-11-1. REPLACE ENGINE CONTROL QUADRANT T-HANDLE AND FUEL LEVER (BENCH)

4-5-11.1.3. Install.

- a. Position wire from fire extinguisher T-handle in wire bundle.
- b. Remove protective caps and plugs from electrical connectors.
- c. Inspect and treat electrical connectors (PARA 1-7-20).
- d. Measure and cut electrical wire to correct length.
- e. Install electrical tubing insulation on wire bundle.
- f. Connect electrical wire to electrical connector.
- g. Remove tag from laminated shim.
- h. If a new laminated shim is being installed, peel laminations as required to obtain minimum clearance between parts.
- i. Position fuel control lever, washer, and laminated shim, with stainless steel side of shim against lever, between ears of fire extinguisher T-handle (Figure 4-5-11-1).
- j. Install laminated shim with stainless steel side facing fire extinguisher T-handle on bushing.
- k. Install bushing through fire extinguisher T-handle, washer, fuel control lever, and shim.
- l. Install bolt and washer through fire extinguisher T-handle and fuel control lever. Install shim on bolt.
- m. Slide bolt, fire extinguisher T-handle, and fuel control lever, as an assembly, in quadrant support.
- n. Install engine power control lever (PARA 4-5-10).
- o. Install fire extinguisher and starter switches (PARA 4-5-13 and PARA 4-5-16).

4-5-12 ENGINE CONTROL QUADRANT THREADED INSERTS (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
4-5-12.1 Replace Engine Control Quadrant Threaded Inserts (BENCH)	4-5-12-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
 - Airframe Repairer’s Toolkit
 - Drill Press
 - Protection, Eye
- Insert Repair Kit, K4-162-162L
 - Insert Repair Kit, K4-192-192L
 - Insert Repair Kit, K4-258-258L
 - Installation Wrench
 - Removal Bit
 - Swage Tool

4-5-12.1 Replace Engine Control Quadrant Threaded Inserts (BENCH).

NOTE

- Replacement of all threaded inserts is the same, except as noted.
- Use tool kit, K4-162-162L, for insert, SR162L, tool kit, K4-192-192L, for insert, SR192L, and tool kit, K4-258-258L, for insert, SR258L.

4-5-12.1.1 Remove.

- Install required removal bit from tool kit in drill press.
- For insert, SR162L, using drill press and removal bit, drill to depth of 0.090-inch.
- For inserts, SR192L or SR258L, using drill press and removal bit, drill to depth of 0.107-inch.

- Using installation wrench, turn insert counter-clockwise and remove insert (Figure 4-5-12-1).

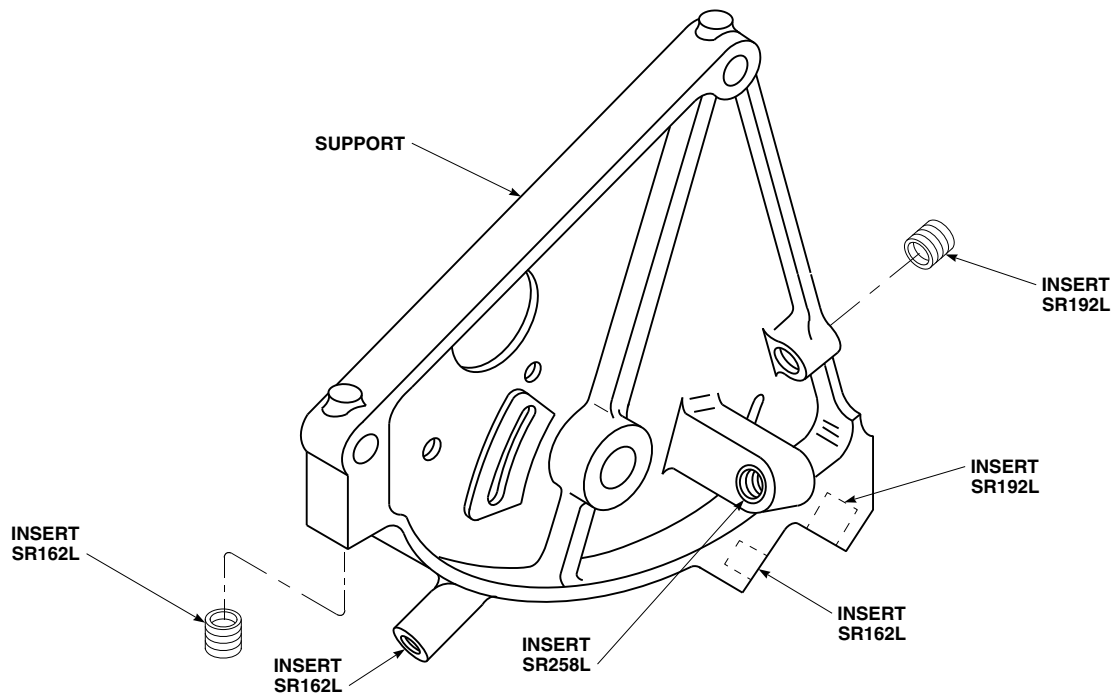
WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- Use clean, dry compressed air to remove any loose chips from hole.

4-5-12.1.2 Install.

- For insert, SR162L, using installation wrench, thread insert into hole until top of insert is

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0.010 - 0.020-inch below boss surface of support (Figure 4-5-12-1).

- b. For inserts, SR192L or SR258L, using installation wrench, thread insert into hole until top of insert is 0.015 - 0.025-inch below boss surface of support.

- c. Insert swage tool in insert and apply downward pressure until shoulder of tool is bottomed against boss surface. Make sure shoulder is bottomed against boss surface.

4-5-13 ENGINE CONTROL QUADRANT ENGINE STARTER SWITCH (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
4-5-13.1	Replace Engine Control Quadrant Engine Starter Switch (BENCH)	4-5-13-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Gun, Electric
- Tags

Materials/Parts

- Adhesive, Item 55, Appendix D
- Solder, Item 313, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 4-4-21
- PARA 4-4-23

4-5-13.1 Replace Engine Control Quadrant Engine Starter Switch (BENCH).

4-5-13.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Lower overhead console.
- Remove quadrant covers (PARA 4-4-21).
- Remove screws, lockwashers, nutplate, and spacer plate holding starter switch to quadrant support (Figure 4-5-13-1, Detail A).
- Place identification tags on starter switch wires.
- Remove sealing compound to expose solder connections on engine starter switch.



Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Using electric soldering gun, unsolder wires from starter switch.

4-5-13.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).

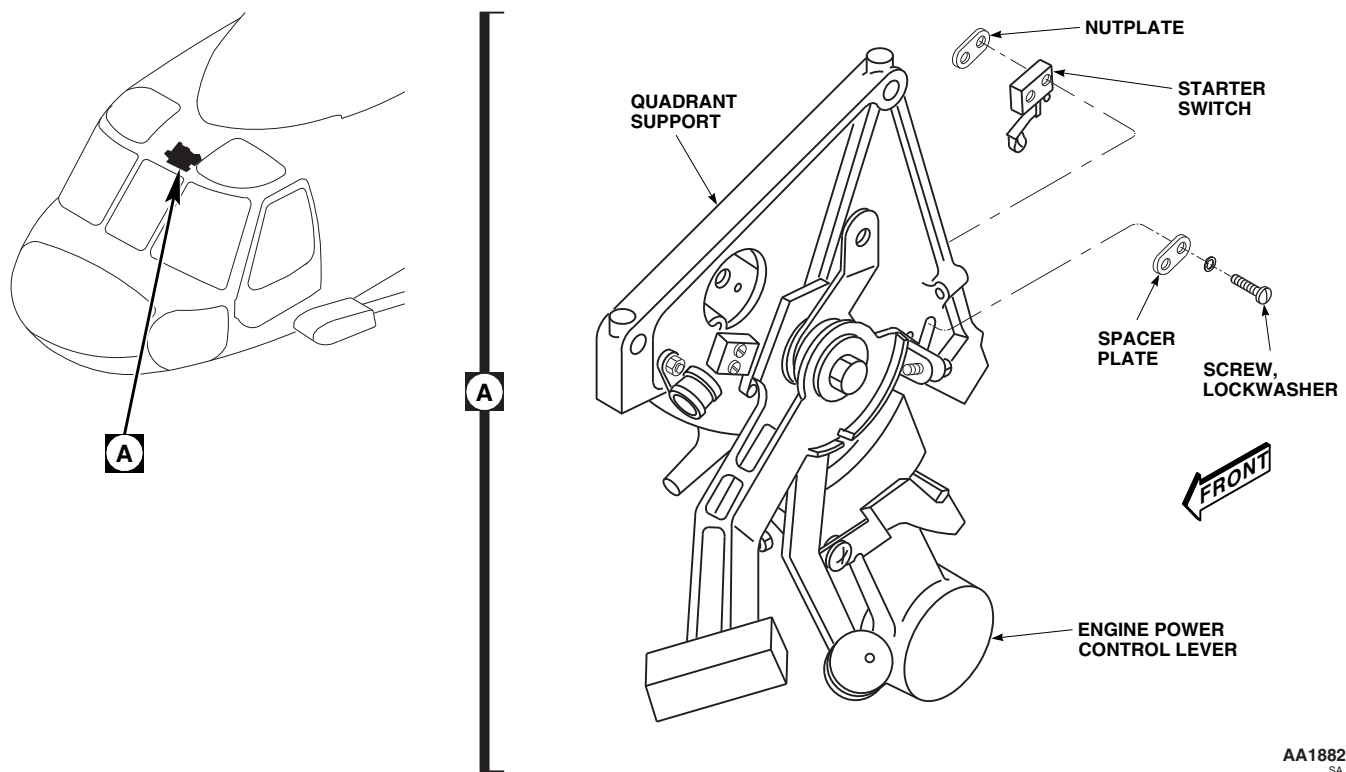


FIGURE 4-5-13-1. REPLACE ENGINE CONTROL QUADRANT ENGINE STARTER SWITCH (BENCH)

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- b. Identify and solder, Item 313, Appendix D, wires to starter switch using electric soldering gun. Remove tags.
- c. Apply adhesive, Item 55, Appendix D, to cover soldered connections.
- d. Install starter switch on quadrant support with screws, lockwashers, nutplate, and spacer plate. Do not tighten screws now (Figure 4-5-13-1, Detail A).
- e. Adjust engine starter switch (PARA 4-4-23).
- f. Install quadrant covers (PARA 4-4-21).
- g. Hinge up and secure upper console.

4-5-14 ENGINE CONTROL QUADRANT STARTER ABORT SWITCH (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
4-5-14.1	Replace Engine Control Quadrant Starter Abort Switch (BENCH)	4-5-14-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Gun, Electric
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Adhesive, Item 56, Appendix D
- Grommet, Nonmetallic, Item 152A, Appendix D

- Insulating Compound, Item 159, Appendix D
- Solder, Item 313, Appendix D
- Tags
- Tape, Item 317C, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 4-4-21
- PARA 4-4-25

4-5-14.1 Replace Engine Control Quadrant Starter Abort Switch (BENCH).

4-5-14.1.1 Remove.

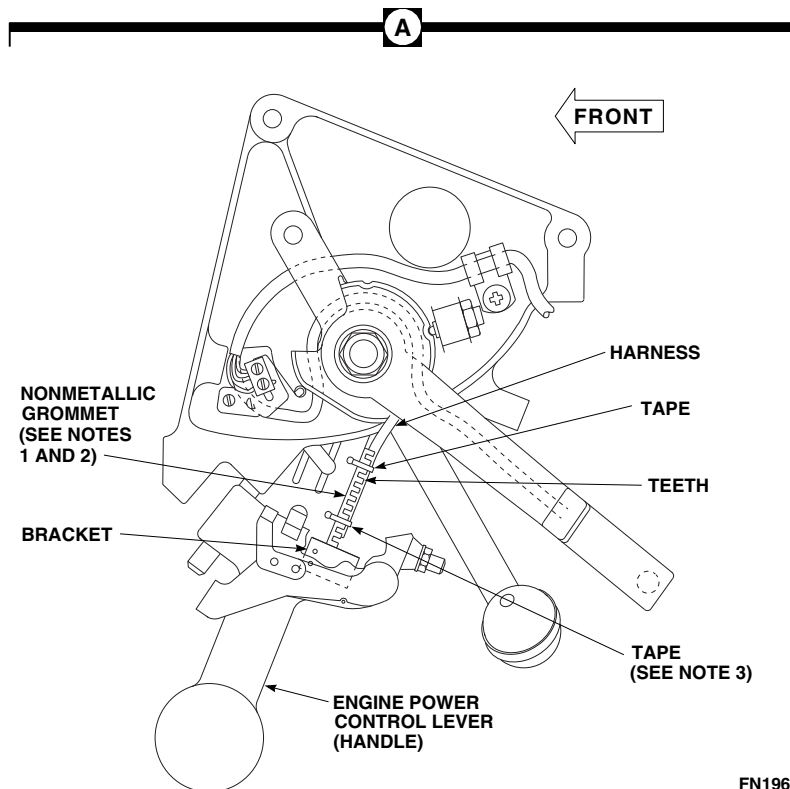
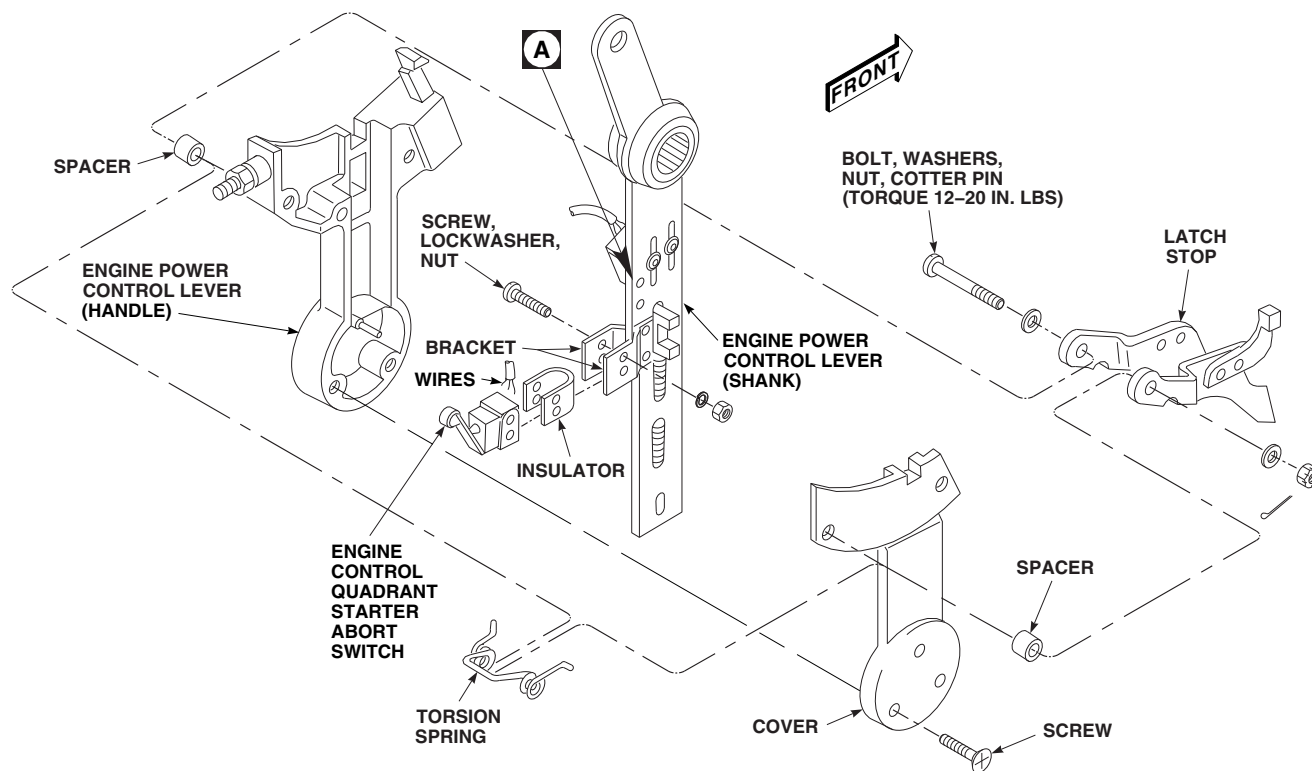
- Remove quadrant covers (PARA 4-4-21).
- On quadrant, 70301-01100-014, remove bolt, washers, spacers, torsion spring, nut, and latch stop from engine power control lever (Figure 4-5-14-1).

NOTE

There are two small compressed springs and a latch in engine power control lever

assembly. Be careful when removing cover to avoid losing parts.

- Remove screws securing cover to handle portion of engine power control lever.
- Remove screws, lockwashers, and nuts securing insulator and engine control quadrant starter abort switch to brackets on shank of engine power control lever. Remove insulator.
- Remove tape securing nonmetallic grommet and harness to shank of engine power control lever (Figure 4-5-14-1, Detail A). Discard nonmetallic grommet.



NOTES

1. WHEN FABRICATING NONMETALLIC GROMMET, MAKE SURE THERE ARE 11 FULL TEETH SQUARE AND FLUSH AT EACH END.
2. POSITION NONMETALLIC GROMMET AS CLOSE AS POSSIBLE TO BRACKET.
3. WHEN SECURING NONMETALLIC GROMMET AT LOWER HOLE, MAKE SURE TAPE IS SECURED BETWEEN THIRD AND FOURTH TOOTH FROM BOTTOM.

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FIGURE 4-5-14-1. REPLACE ENGINE CONTROL QUADRANT STARTER ABORT SWITCH (BENCH)

- f. Remove screw, lockwashers, and nuts securing engine control quadrant starter abort switch and insulator to engine power control lever. Remove switch and insulator from lever (Figure 4-5-14-1).
- g. Tag wires on engine control quadrant starter abort switch.
- h. Remove adhesive or insulating compound from wires to expose soldered connections on engine control quadrant starter abort switch.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- i. Using electric soldering gun, unsolder wires from engine control quadrant starter abort switch.

4-5-14.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- b. Using solder, Item 313, Appendix D, and electric soldering gun, solder wires to engine control quadrant starter abort switch (Figure 4-5-14-1). Remove tags.
- c. Apply adhesive, Item 56, Appendix D, or insulating compound, Item 159, Appendix D, to cover soldered connections.

- d. Secure engine control quadrant starter abort switch to insulator and position between brackets on shank of engine power control lever and secure with screws, lockwashers, and nuts.
- e. Fabricate new nonmetallic grommet by trimming nonmetallic grommet, Item 152A, Appendix D, from raw material, making sure there are eleven full teeth, and is square and flush at each end (approximately 1 $\frac{3}{8}$ -inch) (Figure 4-5-14-1, Detail A).

NOTE

Depending on wire dressing of engine control quadrant starter abort switch and adhesive or insulation compound application, nonmetallic grommet may not reach brackets. In this case, position nonmetallic grommet as low as possible to brackets.

- f. Position nonmetallic grommet against shank of engine power control lever and brackets.

NOTE

Pyramid-like condition would prevent chaff condition on grommet.

- g. Prior to installing starter abort switch harness, make sure to finger-squeeze starter abort switch harness to create a pyramid-like condition with internal wires.
- h. Position harness in teeth of nonmetallic grommet.

CAUTION

Damage to tape may result when securing nonmetallic grommet at lower hole. Chafing condition may exist when quadrant covers are installed if tape is secured in lower two teeth. When securing nonmetallic grommet at lower hole, make sure tape is secured between third and fourth tooth to prevent chafing.

NOTE

When securing harness to shank of engine power control lever, make sure tape is secured between teeth of nonmetallic grommet to prevent slippage.

- i. Using tape, Item 317C, Appendix D, secure harness and nonmetallic grommet to holes in shank of engine power control lever. Make sure tape is trapped between teeth of nonmetallic grommet. For lower hole, make sure tape is secured between third and fourth tooth.
- j. Trim ends of tape as close as possible to knot.
- k. Make sure area is clean and free of foreign material.

CAUTION

Damage to wiring will result if caution is not used when installing covers on engine power control lever. Carefully position wires and install covers to avoid pinching and chafing conditions.

- l. Secure cover on handle of engine power control lever with screws (Figure 4-5-14-1).
- m. On quadrant, 70301-01100-014, install latch stop as follows:
 - (1) Position torsion spring on engine power control lever.
 - (2) Install latch stop on engine power control lever and engage torsion spring.

NOTE

Make sure latch stop is installed with bolthead outboard.

- (3) Install bolt, washers, spacers, and nut securing latch stop in place. **TORQUE NUT TO 12 - 20 INCH-POUNDS.**
 - (4) Install cotter pin in nut.
- n. Install quadrant covers (PARA 4-4-21).
- o. Adjust engine starter abort switch (PARA 4-4-25).

4-5-15 ENGINE CONTROL QUADRANT STARTER OVERRIDE SWITCH (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
4-5-15.1	Replace Engine Control Quadrant Starter Override Switch (BENCH)	4-5-15-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Gun, Electric

Materials/Parts

- Adhesive, Item 55, Appendix D
- Solder, Item 313, Appendix D

- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 4-4-21
- PARA 4-4-26

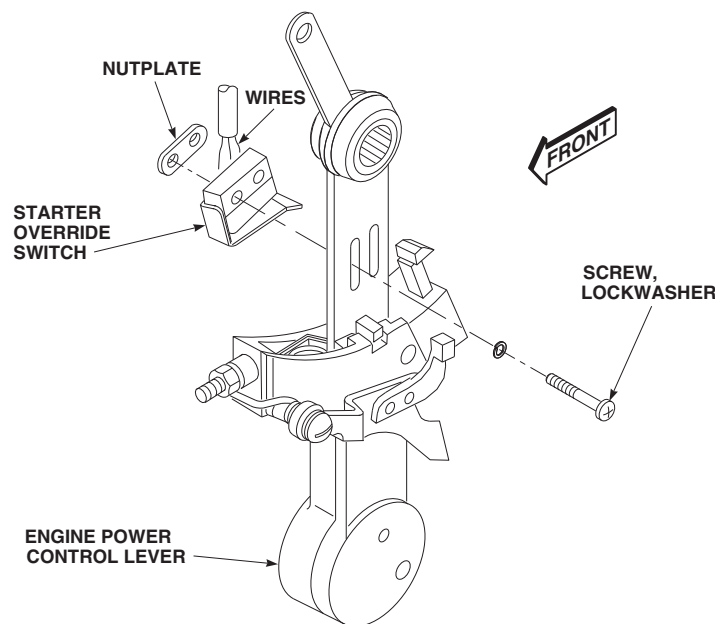
4-5-15.1 Replace Engine Control Quadrant Starter Override Switch (BENCH).

4-5-15.1.1 Remove.

- Remove quadrant covers (PARA 4-4-21).
- Remove screws, lockwashers, and nutplate securing starter override switch to engine power control lever (Figure 4-5-15-1).
- Place identification tags on starter override switch wires.
- Remove sealing compound to expose soldered connections on starter override switch.



- Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.
- Using electric soldering gun, unsolder wires from starter override switch.

**NOTE**

ENGINE POWER CONTROL LEVER
SHOWN REMOVED FOR CLARITY.

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FIGURE 4-5-15-1. REPLACE ENGINE CONTROL QUADRANT STARTER OVERRIDE SWITCH (BENCH)

4-5-15.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- b. Identify and using solder, Item 313, Appendix D, solder wires to starter override switch using electric soldering gun. Remove tags.
- c. Apply adhesive, Item 55, Appendix D, to cover soldered connections.
- d. Install starter override switch on engine power control lever with screws, lockwashers, and nutplate. Do not tighten screws now (Figure 4-5-15-1).
- e. Adjust starter override switch (PARA 4-4-26).
- f. Install quadrant covers (PARA 4-4-21).

4-5-16 ENGINE CONTROL QUADRANT FIRE EXTINGUISHER ARMING SWITCH (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
4-5-16.1	Replace Engine Control Quadrant Fire Extinguis-her Arming Switch (BENCH)	4-5-16-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Gun, Electric
- Spring Scale

Materials/Parts

- Adhesive, Item 56, Appendix D
- Insulating Compound, Item 159, Appendix D

- Solder, Item 313, Appendix D
- Tags

References

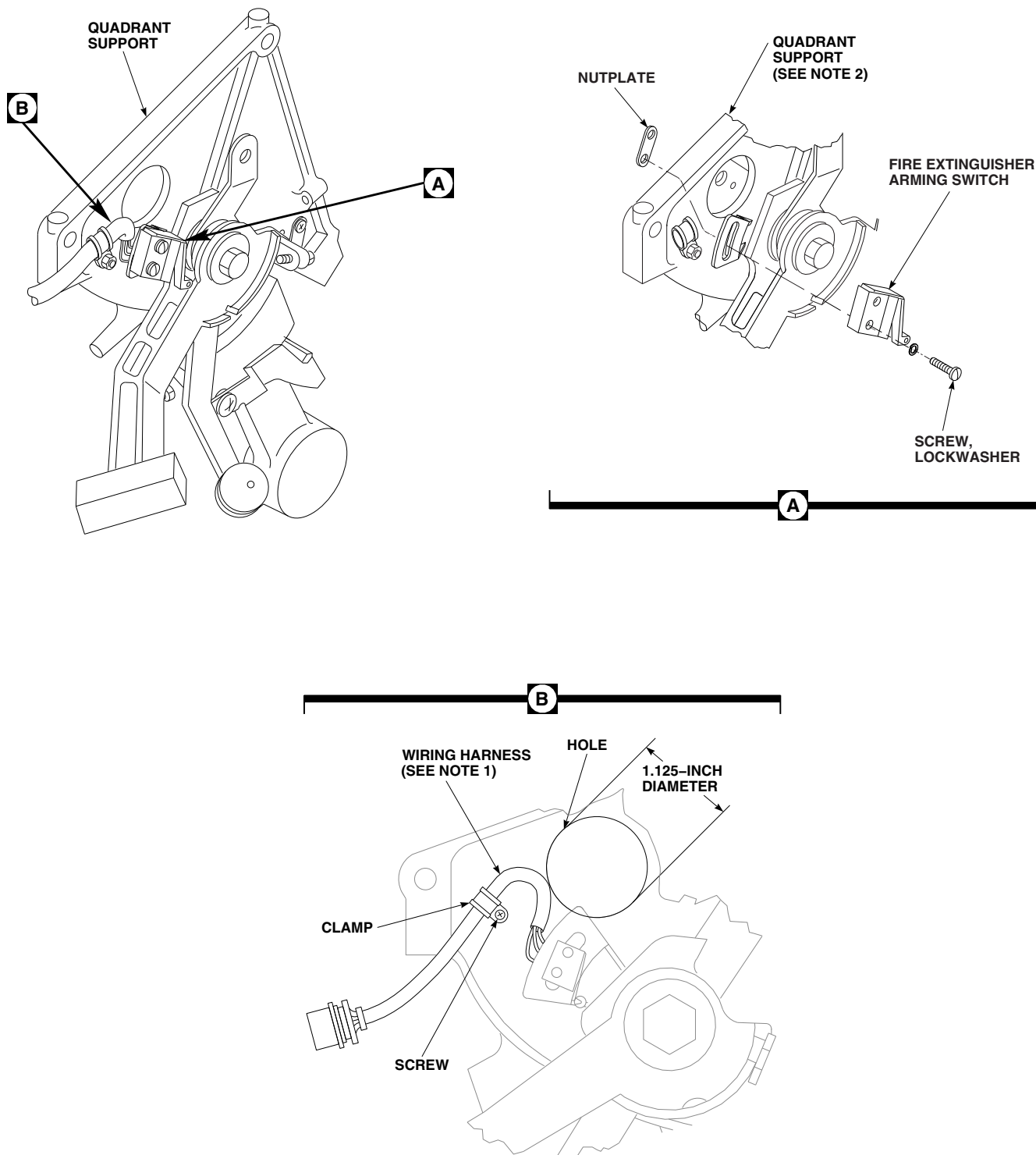
- Appendix D
- PARA 1-6-16
- PARA 4-2-9
- PARA 4-4-21

4-5-16.1 Replace Engine Control Quadrant Fire Extinguisher Arming Switch (BENCH).

4-5-16.1.1 Remove.

- Remove quadrant covers (PARA 4-4-21).

- Remove screws, lockwashers, and nutplate securing fire extinguisher arming switch to quadrant support (Figure 4-5-16-1, Detail A).
- Place identification tags on fire extinguisher arming switch wires.



NOTES

1. WIRING HARNESS NOT TO PROTRUDE PAST 1.125-INCH DIAMETER HOLE AS SHOWN.
2. HARNESS SHOWN REMOVED FOR CLARITY.

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FIGURE 4-5-16-1. REPLACE ENGINE CONTROL QUADRANT FIRE EXTINGUISHER ARMING SWITCH (BENCH)

- d. Remove sealing compound to expose soldered connections on fire extinguisher arming switch.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- e. Using electric soldering gun, unsolder wires from fire extinguisher arming switch (Figure 4-5-16-1, Detail B).

4-5-16.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- b. Identify and using solder, Item 313, Appendix D, solder wires to fire extinguisher arming switch using electric soldering gun (Figure 4-5-16-1, Detail B). Remove tags.
- c. Apply adhesive, Item 56, Appendix D, or insulating compound, Item 159, Appendix D, to cover soldered connections.
- d. Install fire extinguisher arming switch on quadrant support with screws, lockwashers, and nutplate (Figure 4-5-16-1, Detail A). Do not tighten screws now.
- e. Adjust fire extinguisher arming switch (PARA 4-5-16.1.3).

- f. If required, loosen screw on wiring harness clamp and adjust service loop so no part of wiring harness extends into 1 1/8-inch diameter hole (Figure 4-5-16-1, Detail B). Tighten screw.

- g. Install quadrant covers (PARA 4-4-21).

4-5-16.1.3 Adjust.

- a. If required, remove quadrant covers (PARA 4-4-21).
- b. Move fire extinguisher T-handle forward until plunger engages in stowed detent (Figure 4-5-16-2, Sheet 1).
- c. Secure spring scale to apex of fire extinguisher T-handle.
- d. Using spring scale, adjust plunger to release from detent position when breakout force of 3 - 4 pounds is applied at T intersection of lever assembly.
- e. Loosen screws securing fire extinguisher arming switch to support just enough to move switch.
- f. Move fire extinguisher arming switch until arm is against fire extinguisher lever and switch clicks. Tighten screws to secure switch.
- g. Move fire extinguisher T-handle rear as far as travel permits. Check that switch is off.
- h. If actuator has been previously adjusted, perform operational check of fire extinguisher arming switch (PARA 4-2-9).
- i. If fire extinguisher arming switch fails to operate properly, adjust actuator as follows (Figure 4-5-16-2, Sheet 2, Detail A):
 - (1) Bend roller end of actuator arm until actuator makes contact with fire extinguisher T-handle cam when in ON (armed) position.
 - (2) Verify actuator releases fire extinguisher arming switch when T-handle is in OFF (non-armed) position.

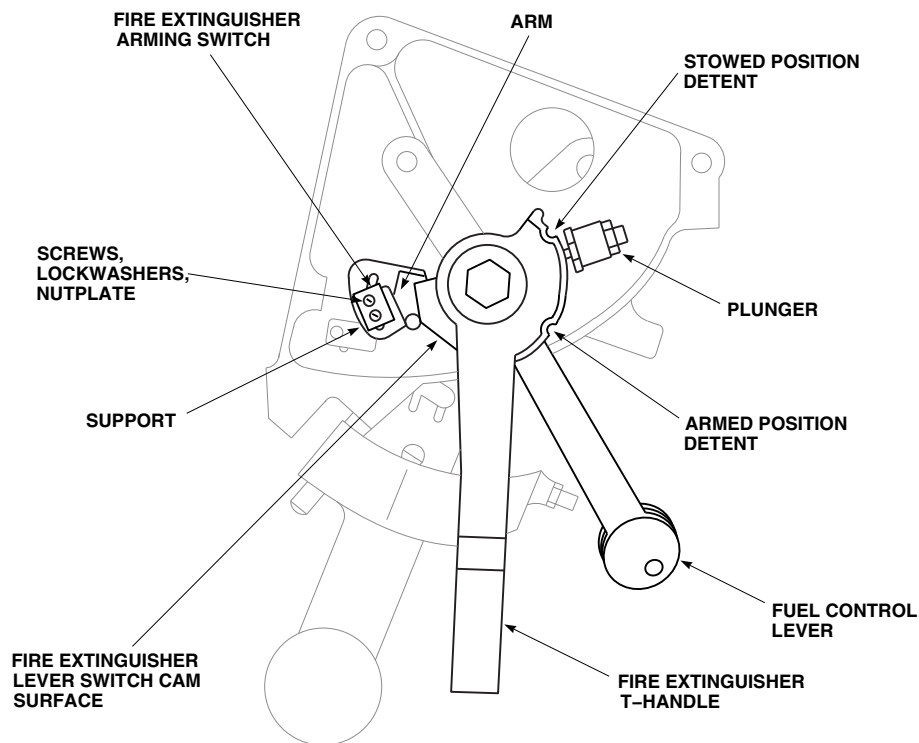
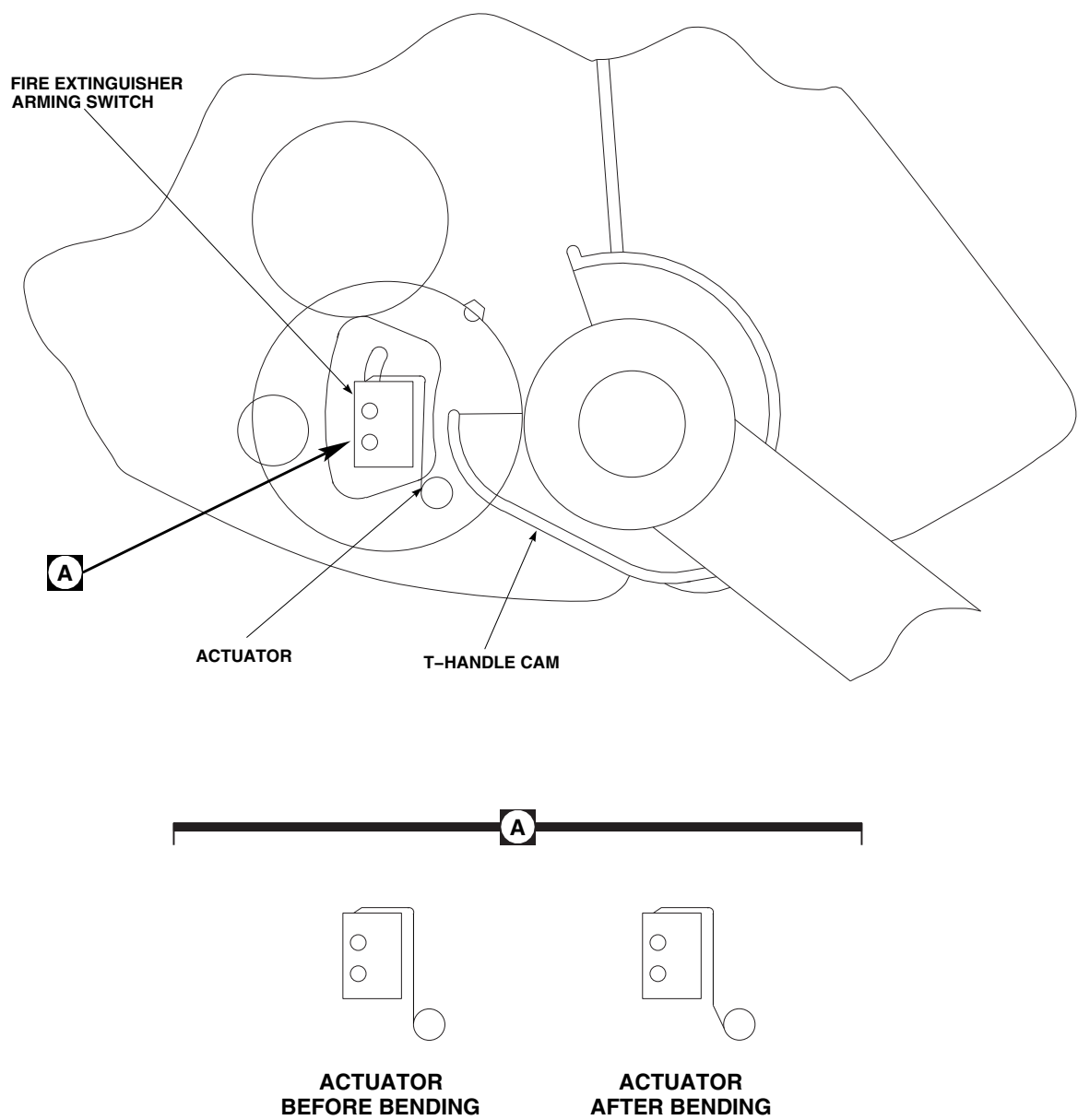
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FIGURE 4-5-16-2. ADJUST ENGINE CONTROL QUADRANT FIRE EXTINGUISHER ARMING SWITCH (BENCH) (SHEET 1 OF 2)

- (3) Check for proper switch operation.
- j. If fire extinguisher arming switch fails to operate after adjustment of actuator, replace actuator as follows:
 - (1) Remove screws, lockwashers, and nutplate securing fire extinguisher arming switch to support (Figure 4-5-16-2, Sheet 1). Remove fire extinguisher arming switch.
 - (2) Remove actuator from fire extinguisher arming switch.
 - (3) Install replacement actuator on fire extinguisher arming switch.
- (4) Secure fire extinguisher arming switch on support and secure using screws, lockwashers, and nutplate.
- (5) Check for proper switch operation.
- k. If fire extinguisher arming switch fails to operate after replacement of actuator, adjust actuator per step i.
 - l. If switch fails to operate properly after adjustment of actuator, replace fire extinguisher arming switch.
- m. Move fire extinguisher T-handle from stowed position detent to rear several times and check for proper switch operation.



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FIGURE 4-5-16-2. ADJUST ENGINE CONTROL QUADRANT FIRE EXTINGUISHER ARMING SWITCH (BENCH) (SHEET 2 OF 2)

4-5-17 ENGINE CONTROL QUADRANT FRONT COVER INFORMATION PLATE (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
4-5-17.1	Replace Engine Control Quadrant Front Cover Information Plate (BENCH)	4-5-17-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 9-2-5
- PARA 9-2-25

4-5-17.1 Replace Engine Control Quadrant Front Cover Information Plate (BENCH).

4-5-17.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove screws and washers and disconnect front information plate from quadrant center cover (Figure 4-5-17-1).
- Install protective caps and plugs on electrical connectors.

4-5-17.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).

- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Gently position front information plate on quadrant center cover and push into electrical connector.
- Install front information plate to center cover with screws and washers (Figure 4-5-17-1).
- Perform operational check of upper console lights (PARA 9-2-25 and PARA 9-2-5).

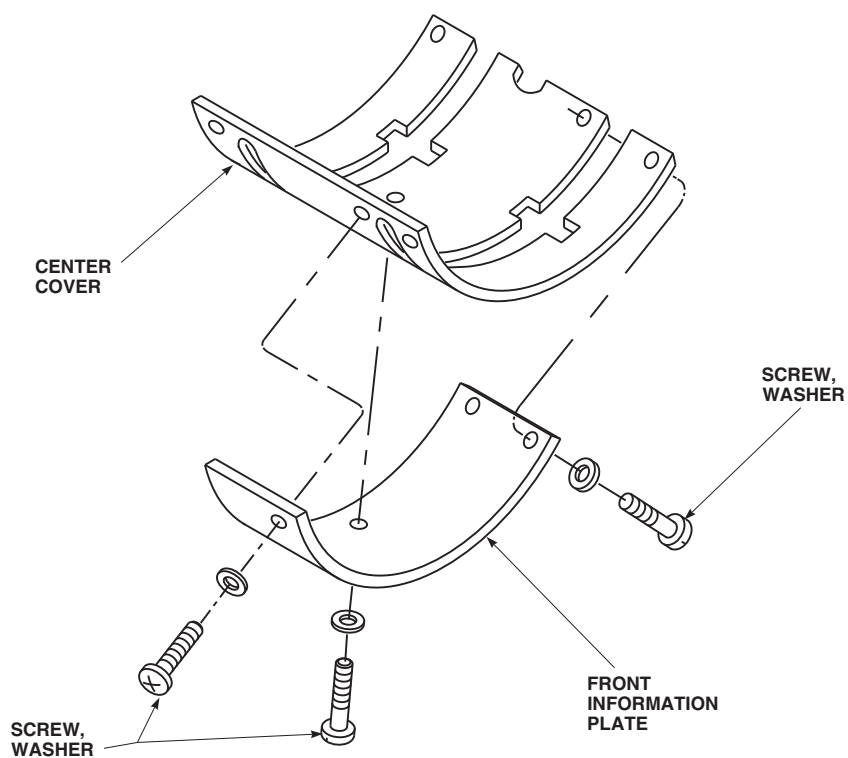
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FIGURE 4-5-17-1. REPLACE ENGINE CONTROL QUADRANT FRONT COVER INFORMATION PLATE (BENCH)

4-5-18 ENGINE CONTROL QUADRANT LEFT AND RIGHT COVER INFORMATION PLATES (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
4-5-18.1	Replace Engine Control Quadrant Left and Right Cover Information Plates (BENCH)	4-5-18-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Crimp Tool
- Electrical Repairer’s Toolkit
- Heat Gun
- Insertion-Extraction Tool

Materials/Parts

- Protective Caps and Plugs
- Insulation Sleeving, Boot, Item 160B, Appendix D

- Insulation Sleeving, Heat-Shrinkable, Item 164, Appendix D

References

- Appendix D
- PARA 1-7-20
- PARA 4-4-21
- PARA 9-2-5
- PARA 9-2-25

4-5-18.1 Replace Engine Control Quadrant Left and Right Cover Information Plates (BENCH).

NOTE

Replacement of the left and right information plate is the same, except as noted.

4-5-18.1.1. Remove.

- Remove engine quadrant cover (PARA 4-4-21).
- Remove screws holding information plate to side cover (Figure 4-5-18-1).

- Disconnect electrical connector, P394, left information plate, or P396, right information plate.
- Install protective caps and plugs on electrical connectors.
- Remove wires and pins from connector using insertion-extraction tool and save connector for replacement information plate.

4-5-18.1.2. Install.

- Remove protective caps and plugs on electrical connectors.

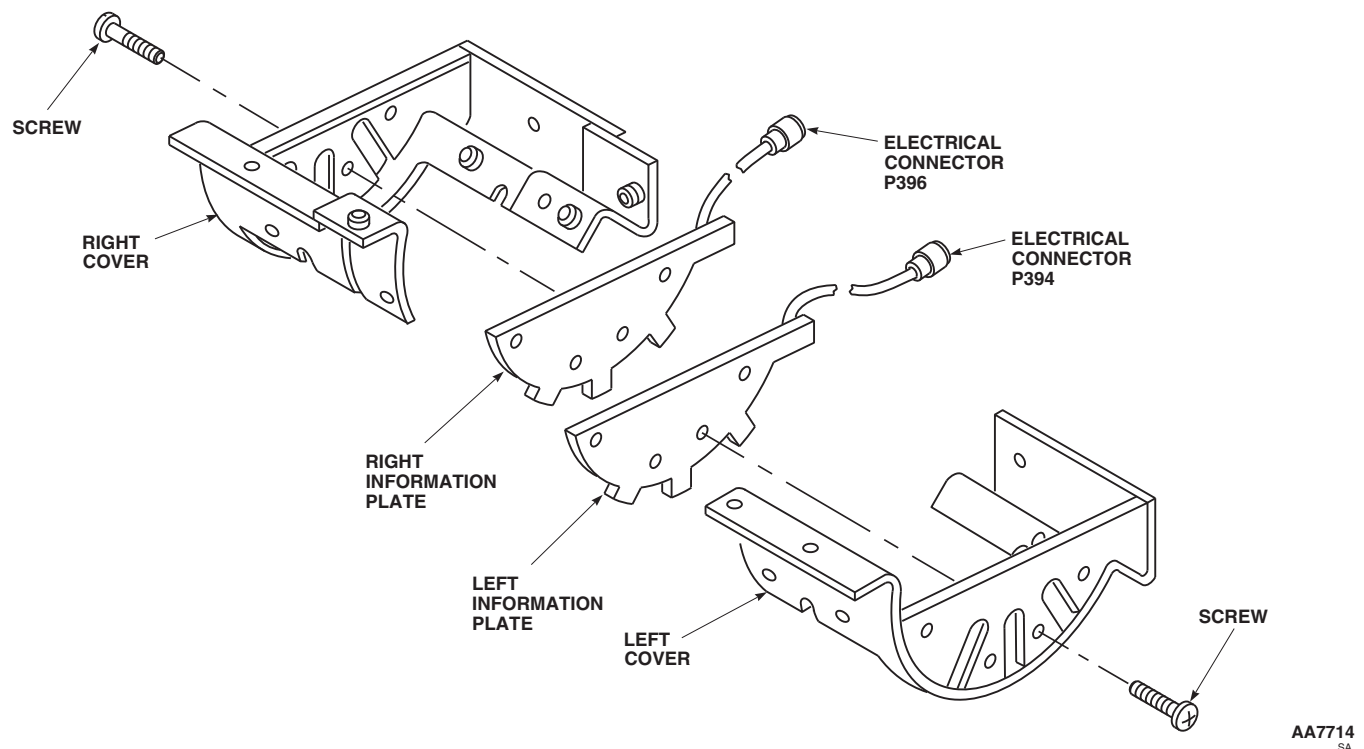


FIGURE 4-5-18-1. REPLACE ENGINE CONTROL QUADRANT LEFT AND RIGHT COVER INFORMATION PLATES (BENCH)

- b. Inspect and treat electrical connectors (PARA 1-7-20).
- c. Install heat-shrinkable insulation sleeving, Item 164, Appendix D, and boot insulation sleeving, Item 160B, Appendix D, and slide back. Do not heat shrink now.
- d. Install pins on information plate electrical wires using crimp tool.
- e. Feed wires through backshell and insert pins into connector using insertion-extraction tool as follows:

Wire No.	Connector Pin No.
A-20	1
B-20N	2
- f. Slide boot over backshell and using heat gun apply heat to insulation boot and sleeving to shrink fit. Do not overheat.
- g. Install information plate to inside of side cover with screws (Figure 4-5-18-1).
- h. Connect electrical connector, P394, left information plate, or P396, right information plate.
- i. Install engine quadrant cover (PARA 4-4-21).
- j. Perform operational check of upper console lights (PARA 9-2-25 and PARA 9-2-5).

4-5-19 ENGINE CONTROL QUADRANT FRONT COVER INFORMATION PLATE WIRING HARNESS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
4-5-19.1	Replace Engine Control Quadrant Front Cover Information Plate Wiring Harness (BENCH)	4-5-19-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- Protection, Eye and Skin
- Soldering Gun, Electric

Materials/Parts

- Protective Caps and Plugs
- Solder, Item 313, Appendix D

- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 4-4-21
- PARA 9-2-5
- PARA 9-2-25

4-5-19.1 Replace Engine Control Quadrant Front Cover Information Plate Wiring Har-ness (BENCH).

4-5-19.1.1 Remove.

- Remove center and right side covers from quadrant assembly (PARA 4-4-21).
- Disconnect electrical connector, P395 (Figure 4-5-19-1).
- Install protective caps and plugs on electrical connectors.



Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- Tag wires. Using electric soldering gun, unsolder wires from connector on quadrant right side

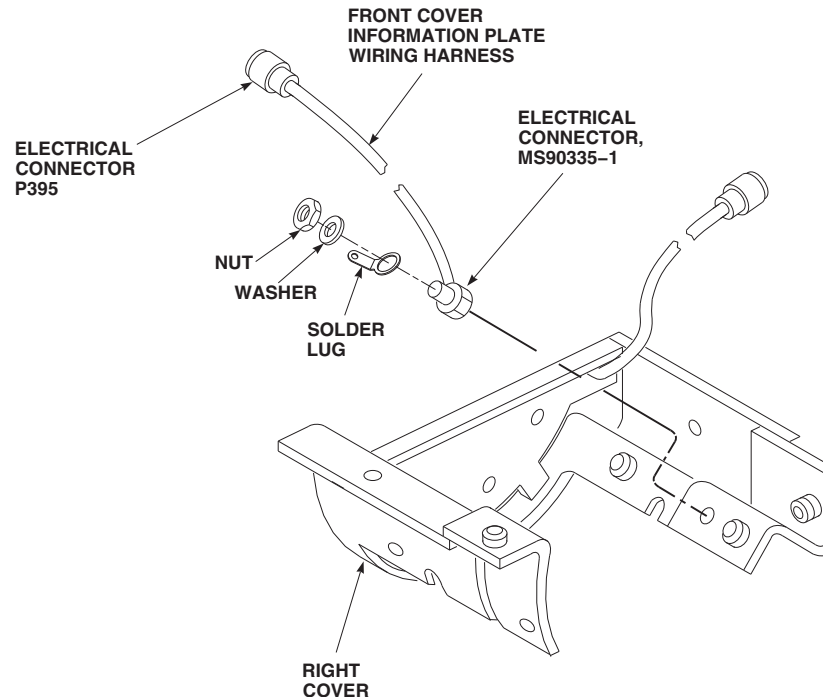
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FIGURE 4-5-19-1. REPLACE ENGINE CONTROL QUADRANT FRONT COVER INFORMATION PLATE WIRING HARNESS (BENCH)

cover assembly and remove front cover information wiring harness.

- e. Remove nut, washer, solder lug, and electrical connector.

4-5-19.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install electrical connector, MS90335-1, on quadrant cover assembly and secure with nut, washer, and solder lug (Figure 4-5-19-1).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective

clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

- c. Solder, Item 313, Appendix D, wiring harness to terminals using electric soldering gun. Untag wires.
- d. Remove protective caps and plugs from electrical connectors.
- e. Inspect and treat electrical connectors (PARA 1-7-20).
- f. Connect electrical connector, P395.
- g. Install right side and center covers on quadrant assembly (PARA 4-4-21).
- h. Perform operational check of upper console lights (PARA 9-2-25 and PARA 9-2-5).